Riemann Zeta Function: Literature and Local Corpus Working Synthesis

Living working document — September 12, 2026

Contributing-model provenance. The final corpus consolidation and current working synthesis were performed with GPT-5.6 Sol Ultra. The local research-note inputs were produced primarily with ChatGPT 5.4 Pro mode and ChatGPT 5.5 Pro mode, with occasional ChatGPT 5.6 Pro mode contributions. Some portions may also contain exploratory inputs from Claude Opus 4.6 and Fable 5. Those two systems were used predominantly for broad-scope conversation and exploratory notes rather than systematic proof audit; any particular mathematical contribution from them remains subject to claim-level provenance checking. Models are recorded here as contributing tools, not as legal authors. Human authorship and attribution are governed by the claim-level audit of this living corpus.

Abstract

This is a living research synthesis, not a claim of a completed paper or endpoint theorem. It consolidates the relevant literature and a modern local corpus at claim level, preserves exact hypotheses and coordinates, audits proofs, searches for explicit morphisms, and records derived results only after literature, novelty, and verification gates are satisfied.

Contents

1 Scope, evidence, and admission rules

The research object is the Riemann zeta function and mathematically evidenced adjacent structures represented in the authorized literature and modern local corpus. The Riemann hypothesis may motivate or receive consequences from the work, but it is not a forced deliverable. Every statement is framed by what is actually proved.

For each mathematical object, the mandatory work cycle is:

1. query the existing machine-readable index;
2. read the relevant human-authored primary literature at content level;
3. state the object, hypotheses, coordinate conventions, and known results;
4. read and audit the maintained local material;
5. calculate, prove, formalize, and compare;
6. propagate each admitted result through every affected dependency and satellite.

Local AI-generated or AI-assisted documents are routing and proof-candidate artifacts, not independent scholarly authority. Their citations and mathematical dependencies are followed recursively to primary or canonical public sources. A genuinely novel local result may be attributed to Kokuno Yumeto only after literature and novelty searches and a complete proof audit; the proof must be reproduced here rather than outsourced to a private note.

No simplification or normalization is permitted. Presentations are related by explicit typed morphisms or explicit obstructions while both original objects remain visible. No theorem is admitted merely because it appears in a note, generated index, prior task, numerical experiment, or certificate filename.

Current admission status. Three bounded local results now have complete checkable proofs and stable theorem records: the exact finite $U(3)$ derivative moments in theorem 2.31, the source-block dimension formula in theorem 2.33, and the fixed-coordinate block bijection in theorem 2.34. The first has two independent exact computational replays under `CERT-N3-CUE-20260825-0001`; their Lean replays remain queued as `FORMAL-20260825-0007` and `FORMAL-20260825-0008`; both replays now pass with hash-pinned sources, compiled outputs, and measured peaks below 2 GB. They are admitted only at those finite computational and coordinate scopes. No CUE-to-block, CUE-to-Lie, or quotient/lift morphism has been proved, and no larger local programme theorem is admitted by these status changes.

2 Literature foundations

2.1 Status and source editions

This section is the first bounded content-level literature tranche. It does not treat a search hit, a title, or a bibliography entry as mathematical evidence. Fourteen human-authored sources now have content-level records: Bogomolny's complete review, Goldston's complete lecture notes, bounded analytic/random-matrix portions of the Schumayer–Hutchinson and Connes surveys, the complete Keating–Snaith, Montgomery, Grover–Mezzadri–Simm, Bourgade–Hughes–Nikeghbali–Yor, and Bourgade–Nikeghbali–Rouault primary papers, Guinand's complete journal article, Weil's complete 1952 article, the bounded Fourier-analysis chapters of Rudin used in the exact transform proofs, Bourgade's complete conditional-Haar paper, and the complete fifth chapter of Bourgade's thesis. No OCR was used. The sources are cited conventionally throughout this chapter and collected in the bibliography [13, 14, 15, 16, 23, 24, 25, 26, 27, 28, 36, 37, 38, 39]. The fourth identity was checked against the native first page and PDF metadata as well as the public record. The identifier `2501.06560`, previously attached to it by a generated bibliographic graph, belongs to a different work cited inside the survey.

Weil was read first through the native text of Denise Vella–Chemla's public 2020 French transcription. Every one of the fourteen pages of the original collected-works reprint was then obtained as a separate public scan leaf and checked visually. The original scan controls conjugation bars, signs, kernels, and denominators; the transcription remains a distinct edition. Guinand was read visually from all thirteen pages of the public journal-volume scan.

The sources do not all have the same evidentiary role. Goldston gives a detailed analytic derivation and carefully separates theorems from conjectures. Bogomolny supplies an especially useful map of the quantum-chaos and random-matrix interfaces, while repeatedly identifying the semiclassical calculations as heuristic. The other two sources are routing surveys whose theorem-level assertions must be checked against their primary references. Keating–Snaith is a primary source: its finite circular-ensemble integrals, cumulant calculations, and random-matrix limits are theorem content, while its transfers to zeta moments and prime-factor models are explicitly conjectural or heuristic. Montgomery, Guinand, Weil, and Grover–Mezzadri–Simm are primary sources; their different hypotheses, parameter spaces, limits, and theorem/conjecture boundaries are retained term by term below. The three Bourgade-led papers and the thesis chapter supply the direct human coordinate and probability precedents used in the derivative-state crosswalk below.

2.2 Explicit-formula presentations

The following presentations are deliberately retained as separate objects. Their common subject does not make their test spaces, transforms, convergence rules, or boundary terms identical.

2.2.1 A sharp Chebyshev presentation

Let

$$\psi(x) = \sum_{p^m \le x} \log p, \qquad \psi_0(x) = \frac{\psi(x+0) + \psi(x-0)}{2}.$$

Goldston records, for $x > 1$ [14, eqs. (3.1), (3.3), and (3.4)],

$$\psi_0(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}), \tag{1}$$

where conjugate zeros are grouped because the zero sum is not absolutely convergent. A truncated presentation retains additional variables and jump data:

$$\psi(x) = x - \sum_{|\gamma| \le T} \frac{x^\rho}{\rho} + O\Big(\frac{x}{T}(\log xT)^2\Big) + O\bigg((\log x)\min\Big(1, \frac{x}{T\|x\|}\Big)\bigg), \tag{2}$$

where $\|x\|$ is the distance to the nearest integer. The final term records the jumps at prime powers; it is not disposable presentation noise.

2.2.2 An additive even-test/Fourier presentation

Bogomolny takes a function h analytic in $|\operatorname{Im} r| \le \frac{1}{2} + \delta$, even, and satisfying $|h(r)| \le A(1+|r|)^{-2-\delta}$, with transform [13, explicit-formula discussion]

$$g(u) = \frac{1}{2\pi} \int_{\mathbb{R}} h(r) e^{-iru}\, dr. \tag{3}$$

The displayed explicit formula has an archimedean Γ'/Γ integral, a prime-power sum weighted by $g(n \log p)$, two pole terms, and $-g(0)\log \pi$. The source writes the zero-side summand as $h(s_n)$ while defining h in an ordinate variable. Until the $s_n \leftrightarrow r_n$ shift is checked against the primary formula, this is a recorded presentation rather than an admitted theorem statement. In particular, the later formal substitution $h = \delta_E$, which the source introduces only after explicitly setting aside convergence, is not licensed by the stated test space.

2.2.3 Guinand's primary half-line cosine/Hankel presentation

Guinand assumes RH throughout the 1948 paper and writes every nontrivial zero as $\frac{1}{2} + i\gamma$, with multiplicity retained. His cosine-transform convention is [25, pp. 107–108, eq. (1.1)]

$$g(y) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^\infty f(x)\cos(xy)\, dx. \tag{4}$$

For Theorem 3, f is an integral, tends to zero at infinity, and $xf'(x) \in L^p(0,\infty)$ for some $1 < p \le 2$. Put

$$\mathcal{P}_f(T) = \sideset{}{'}\sum_{0 < m\log p < T} \frac{\log p}{p^{m/2}} f(m \log p), \qquad \mathcal{P}_1(T) = \sideset{}{'}\sum_{0 < m\log p < T} \frac{\log p}{p^{m/2}},$$

$$\mathcal{A}_f(T) = \int_0^T f(x) e^{x/2}\, dx, \qquad \mathcal{H}_f = \frac{1}{2}\int_0^\infty f(x)\left(\frac{1}{x} - \frac{e^{-x/2}}{\sinh x}\right) dx,$$

$$\mathcal{Z}_g(T) = \sum_{0<\gamma<T} g(\gamma), \qquad \mathcal{D}_g(T) = \frac{1}{2\pi}\int_0^T g(x) \log\frac{x}{2\pi}\, dx.$$

The dash halves a prime-power term exactly on the cutoff. Guinand's literal theorem is [25, pp. 112–115, Thms. 3 and 3′]

$$\lim_{T\to\infty}\Big\{\mathcal{P}_f(T)-\mathcal{A}_f(T)-f(T)\big(\mathcal{P}_1(T)-2e^{T/2}\big)\Big\}-\mathcal{H}_f=-\sqrt{2\pi}\lim_{T\to\infty}\big\{\mathcal{Z}_g(T)-\mathcal{D}_g(T)\big\}. \tag{5}$$

The boundary term is part of the theorem. Theorem 3′ permits its omission if either

$$\sum_{n\le N}\Lambda(n)-N=O\Big(N^{1/2}\log^{\alpha}N\Big)\quad\text{with }\alpha=1/p,$$

or $T^{1/2}f(T)\to 0$. It is not omitted merely because a later presentation looks cleaner.

The paper also supplies a distinct order-3/2 Hankel-transform pair. With

$$\mathcal{F}(x)=\frac{1}{x}\left\{N(x)-\Big(\frac{x}{2\pi}\log\frac{x}{2\pi}-\frac{x}{2\pi}\Big)\right\},$$

the explicitly displayed prime-power function $\mathcal{G}(x)$ in Guinand's equation (2.9) is the order-3/2 Hankel transform of $\mathcal{F}$. The inversion lemmas retain their L^p hypotheses and recover the averaged value $\{\mathcal{F}(y+0)+\mathcal{F}(y-0)\}/2$. The resulting Theorem 2 is an exact formula for $\{N(T+0)+N(T-0)\}/2$; neither averaging operation is erased here [25, pp. 108–112, Thms. 1–2].

Guinand records the multiplicative change of variables rather than leaving it implicit [25, p. 115, immediately after Thm. 3′]:

$$t=e^{x},\qquad U=e^{T},\qquad dx=\frac{dt}{t},\qquad e^{x/2}dx=t^{-1/2}dt, \tag{6}$$

so the prime/main-term expression becomes

$$\lim_{U\to\infty}\left\{\sum_{n<U}\frac{\Lambda(n)}{n^{1/2}}f(\log n)-\int_1^U f(\log t)t^{-1/2}\,dt\right\}. \tag{7}$$

Finally, the compact cosine cutoff in Theorem 4 produces a term $-X+O(1)$ precisely when y is a positive zero ordinate and $O(1)$ otherwise. The inverse resonance is

$$\sum_{0<\gamma<X}\cos(y\gamma)=\begin{cases}-\dfrac{X}{2\pi}\dfrac{\log p}{p^{m/2}}+O(\log X), & y=m\log p,\\ O(\log X), & y\ne m\log p.\end{cases}$$

These are RH-conditional summation theorems, not an operator construction [25, pp. 115–119, Thms. 4–5].

2.2.4 A multiplicative Mellin/local-place presentation

Connes presents a function $f:\mathbb{R}_{>0}\to\mathbb{C}$ with piecewise C^1 regularity, averaged one-sided values at its finitely many jumps, and, for some $\delta>0$, [16, explicit-formula presentation]

$$f(x)=O(x^{\delta})\quad(x\to 0+),\qquad f(x)=O(x^{-1-\delta})\quad(x\to\infty).$$

Set

$$\widetilde{f}(s)=\int_0^\infty f(x)x^{s-1}\,dx,\qquad f^{\sharp}(x)=x^{-1}f(x^{-1}). \tag{8}$$

The survey attributes to Guinand and Weil the presentation

$$\sum_{\rho} \widetilde{f}(\rho) = \widetilde{f}(0) + \widetilde{f}(1) - \sum_{v} \mathcal{W}_v(f), \tag{9}$$

with finite-place distributions

$$\mathcal{W}_p(f) = (\log p) \sum_{m \geq 1} \big(f(p^m) + f^\sharp(p^m)\big) \tag{10}$$

and

$$\mathcal{W}_{\mathbb{R}}(f) = (\log(4\pi) + \gamma_E) f(1) + \int_1^\infty \left(f(x) + f^\sharp(x) - \frac{2f(1)}{x}\right) \frac{dx}{x - x^{-1}}. \tag{11}$$

The zero sum is again generally conditional. This later survey specialization is useful, but the exact 1952 source is the next, separately retained object.

2.2.5 Weil's primary Hecke/idèle-class presentation

Let k be a number field of degree d, I_k its idèle group, P_k the principal idèles, and $C_k = I_k/P_k$. Weil fixes a character χ of C_k, its conductor $\mathfrak{f}$, and archimedean data $\eta_\lambda \in \{1, 2\}$, f_λ, and φ_λ, choosing the associated representative with $\sum_\lambda \eta_\lambda \varphi_\lambda = 0$ [26, Collected Papers II, pp. 48–52, eqs. (1)–(7)]. For

$$A = (2\pi)^{-d} |\Delta| N\mathfrak{f},$$

the completed Hecke function $\Lambda(s) = G(s)L(s)$ satisfies the source's conjugated functional equation

$$\Lambda(1 - \bar{s}) = \kappa\, \overline{\Lambda(s)}, \qquad |\kappa| = 1.$$

Weil's additive logarithmic test is $F : \mathbb{R} \to \mathbb{C}$ with

$$\Phi(s) = \int_{-\infty}^{\infty} F(x) e^{(s-1/2)x}\, dx. \tag{12}$$

Condition (A) requires F to be continuous and continuously differentiable away from finitely many first-kind jumps of F and F', with the value at each jump equal to the average of its one-sided limits. Condition (B) requires, for some $b > 0$,

$$F(x), F'(x) = O\Big(e^{-(1/2+b)|x|}\Big).$$

The singular archimedean functional uses Weil's literal finite-part convention. If $\beta(x) = |x| a(x)$, then

$$\mathrm{PF} \int_{\mathbb{R}} a(x)\, dx = \lim_{\lambda \to \infty} \left\{ \int_{\mathbb{R}} (1 - e^{-\lambda |x|}) a(x)\, dx - 2\beta(0) \log \lambda \right\}. \tag{13}$$

It differs from Schwartz's Pf by a Dirac multiple and is not replaced by an ordinary integral. Define

$$K_{1,f}(x) = \frac{e^{(1/2-f)|x|}}{|e^x - e^{-x}|}, \qquad K_{2,f}(x) = \frac{e^{-f|x|/2}}{|e^{x/2} - e^{-x/2}|}. \tag{14}$$

If $\omega = \beta + i\gamma$ runs over the zeros of Λ with multiplicity, Weil's formula (11) is [26, Collected Papers II, pp. 57–58, eq. (11)]

$$\begin{aligned}\lim_{T\to\infty} \sum_{|\gamma|<T} \Phi(\omega) = \delta_\chi \int_{\mathbb{R}} F(x)(e^{x/2} + e^{-x/2})\,dx + F(0)\log A \\ - \sum_{\mathfrak{p}} \sum_{n\geq 1} \frac{\log N\mathfrak{p}}{(N\mathfrak{p})^{n/2}} \left\{\chi(\mathfrak{p})^n F(\log N\mathfrak{p}^n) + \chi(\mathfrak{p})^{-n} F(\log N\mathfrak{p}^{-n})\right\} \\ - \sum_{\lambda} \mathrm{PF} \int_{\mathbb{R}} F(x) e^{-i\varphi_\lambda x} K_{\eta_\lambda, f_\lambda}(x)\,dx. \end{aligned} \tag{15}$$

The cutoff is symmetric in γ; the finite-place orientations $\pm n \log N\mathfrak{p}$ both remain present.

Weil then proves a positivity equivalence [26, Collected Papers II, pp. 58–59]. For

$$F(x) = \int_{\mathbb{R}} F_0(x+t)\overline{F_0(t)}\,dt, \qquad \Phi(s) = \Phi_0(s)\overline{\Phi_0(1-\bar{s})}, \tag{16}$$

$L(s)$ satisfies RH if and only if the common value of (15) is nonnegative for every F_0 satisfying (A) and (B). The converse is constructive. From an off-line zero $\omega_0 = \beta_0 + i\gamma_0$, set

$$z = i(s - \tfrac{1}{2} - i\gamma_0), \qquad \eta = i(\omega - \tfrac{1}{2} - i\gamma_0).$$

Choose a real polynomial Q vanishing at the bounded competing η's but not at the selected pair, put $P(z) = zQ(z)Q(-z)$, and use $\Psi_0(z) = P(z)e^{-Az^2}$. For sufficiently large A, the explicitly displayed negative selected-zero contribution dominates the Gaussian tail. Thus the converse is not merely an abstract separation assertion.

The global form fixes the Haar normalizations rather than suppressing them:

$$d^\times x = \frac{dx}{2|x|} \quad (k_v = \mathbb{R}), \qquad d^\times x = \frac{dr\,d\theta}{\pi r} \quad (k_v = \mathbb{C}),$$

and a nonarchimedean unit group has measure $\log N\mathfrak{p}$. For the positive-type idèle-class function P, formula (12) is [26, Collected Papers II, pp. 59–61, eqs. (12)–(13)]

$$\begin{aligned} D(P) = \rho P(1) + \int_{C_k} P(\xi)(\|\xi\|^{1/2} + \|\xi\|^{-1/2})\,d\xi \\ - \sum_v \int_{k_v^*} \frac{P(x) - P(1)\theta_v(\|x\|)}{\|x-1\|\,\|x\|^{-1/2}}\,d^\times x, \end{aligned} \tag{17}$$

where

$$\rho = \log|\Delta| + \log(2\pi)^{-d} - \sum_\lambda \mathrm{PF} \int_{\mathbb{R}} \theta_\lambda(e^{-x}) K_{\eta_\lambda, 0}(x)\,dx. \tag{18}$$

D is independent of the permitted θ_v choices and is of positive type if and only if all Hecke L-functions over k satisfy RH. In the one-dimensional function-field case, $\rho = (2g-2)\log q$, and positivity holds because the corresponding theorem is known.

2.2.6 Montgomery's Poisson-weighted presentation

Montgomery's 1973 paper assumes RH and writes every nontrivial zero as $\rho = \frac{1}{2} + i\gamma$ with $\gamma \in \mathbb{R}$. For $1 < \sigma < 2$, $x \geq 1$, and $\tau = |t|+2$, its retained explicit formula is [24, pp. 185–187, eqs. (18)–(22)]

$$\begin{aligned}
&(2\sigma - 1)\sum_{\gamma} \frac{x^{i\gamma}}{(\sigma - \frac{1}{2})^2 + (t-\gamma)^2} \\
&= -x^{-1/2}\left(\sum_{n\leq x} \Lambda(n)(x/n)^{1-\sigma+it} + \sum_{n>x} \Lambda(n)(x/n)^{\sigma+it}\right) \\
&\quad - \frac{\zeta'}{\zeta}(1-\sigma+it)x^{1/2-\sigma+it} + \frac{x^{1/2}(2\sigma-1)}{(\sigma-1+it)(\sigma-it)} \\
&\quad - x^{-1/2}\sum_{m\geq 1} \frac{(2\sigma-1)x^{-2m}}{(\sigma-1-it-2m)(\sigma+it+2m)}.
\end{aligned} \tag{19}$$

The two prime ranges, logarithmic-derivative term, pole term, and trivial-zero series are distinct terms. The derivation first excludes $x = 1$ and prime powers in Landau's identity; continuity after subtraction restores every $x \geq 1$. The functional equation gives

$$\frac{\zeta'}{\zeta}(1-\sigma+it) = -\frac{\zeta'}{\zeta}(\sigma-it) - \log\tau + O_\sigma(1),$$

so (19) also yields the same two prime sums together with

$$x^{1/2-\sigma+it}\big(\log\tau + O_\sigma(1)\big) + O_\sigma(x^{1/2}\tau^{-2}) + O_\sigma(x^{-2}\tau^{-1}).$$

For the later $\sigma = \frac{3}{2}$ coordinate presentation, define

$$a_n(x) = \min\Big((n/x)^{1/2}, (x/n)^{3/2}\Big). \tag{20}$$

Multiplying the specialization of (19) by $x^{1/2-it}$ gives Goldston's algebraically equivalent form [14, eqs. (3.11)–(3.14)]

$$\begin{aligned}
2x^{1/2-it}\sum_{\gamma} \frac{x^{i\gamma}}{1+(t-\gamma)^2} &= -\sum_{n\geq 1} \frac{\Lambda(n)a_n(x)}{n^{it}} + \frac{2x^{1-it}}{(\frac{1}{2}+it)(\frac{3}{2}-it)} \\
&\quad + x^{-1/2}\big(\log(|t|+2) + O(1)\big) + O\left(\frac{x^{-2}}{|t|+2}\right).
\end{aligned} \tag{21}$$

Goldston separately observes that one can write every zero as $\rho = \frac{1}{2} + i\gamma$ with complex γ to obtain an unconditional identity. That complex-coordinate convention cannot be substituted into later real-ordinate estimates without rechecking them.

2.3 Exact logarithmic-coordinate morphisms

Proposition 2.1 (Additive–multiplicative coordinate isomorphism)**.** *The map*

$$E : (\mathbb{R}, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \qquad E(u) = e^u,$$

is a real-analytic group isomorphism with inverse $L(x) = \log x$. It transports Haar coordinates by $dx/x = du$.

Proof. $E(u+v) = E(u)E(v)$. Strict monotonicity and the range of the real exponential give a singleton fibre over each $x > 0$ and the two-sided inverse L. Differentiating $x = e^u$ gives $dx = e^u du = x du$. $\square$

Fix $c \in \mathbb{R}$ and put $g_c(u) = e^{cu} f(e^u)$. Whenever the displayed integrals are absolutely convergent, the same coordinate map gives

$$\widetilde{f}(c+it) = \int_{\mathbb{R}} g_c(u) e^{itu}\, du. \tag{22}$$

Thus, for the convention $\mathcal{F}_-[g](r) = \int_{\mathbb{R}} g(u)e^{-iru}\, du$,

$$\widetilde{f}(c+it) = \mathcal{F}_-[g_c](-t). \tag{23}$$

The sign and the $1/(2\pi)$ in (3) are therefore convention data, not factors to be silently removed. The inverse coordinate formula is $f(x) = x^{-c} g_c(\log x)$; Fourier inversion has its own regularity hypotheses.

There is also an exact convolution morphism. For

$$(f *_\times h)(x) = \int_0^\infty f(y)h(x/y)\frac{dy}{y}, \qquad U(f)(u) = f(e^u), \tag{24}$$

absolute convergence and $y = e^v$ give

$$U(f *_\times h)(u) = \int_{\mathbb{R}} U(f)(v)U(h)(u-v)\, dv = (U(f) *_+ U(h))(u). \tag{25}$$

For Weil's normalization [26, Collected Papers II, pp. 52–58] there is a particularly exact specialization. Put

$$\mathcal{B}(F)(v) = v^{-1/2}F(\log v), \qquad \mathcal{B}^{-1}(\phi)(x) = e^{x/2}\phi(e^x). \tag{26}$$

These are two-sided inverses and

$$\int_0^\infty \mathcal{B}(F)(v)v^{s-1}\, dv = \Phi(s).$$

Condition (B) transports exactly to $\mathcal{B}(F)(v) = O(v^b)$ at $0+$ and $O(v^{-1-b})$ at infinity; jump locations a_i map bijectively to e^{a_i}. No fibre is merged.

Now restrict to half-line functions for which the following even extension satisfies Weil's conditions:

$$E(f)(x) = \frac{f(|x|)}{\sqrt{2\pi}}. \tag{27}$$

The inverse on the even subspace is $f(t) = \sqrt{2\pi}E(f)(t)$ for $t \geq 0$, and

$$\Phi_{E(f)}(\tfrac{1}{2} + iy) = \int_{\mathbb{R}} E(f)(x)e^{iyx}\, dx = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos(xy)\, dx = g(y). \tag{28}$$

This proves the transform factor rather than normalizing it away. Weil's exponential decay also gives $T^{1/2}f(T) \to 0$, so Guinand's boundary term [25, pp. 112–115, Thm. 3′] vanishes on this explicit intersection subspace by Theorem 3′.

For $k = \mathbb{Q}$ and the trivial character,

$$d = 1, \quad \Delta = 1, \quad N\mathfrak{f} = 1, \quad A = (2\pi)^{-1}, \quad (\eta_1, f_1, \varphi_1) = (1, 0, 0).$$

Consequently, the finite-place and pole coordinates specialize term by term as

$$\sum_p \sum_{m\ge 1} \frac{\log p}{p^{m/2}}\{E(f)(m\log p)+E(f)(-m\log p)\} = \sqrt{\frac{2}{\pi}}\sum_p\sum_{m\ge 1}\frac{\log p}{p^{m/2}} f(m\log p), \tag{29}$$

$$\int_{\mathbb{R}} E(f)(x)(e^{x/2}+e^{-x/2})\,dx = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)(e^{x/2}+e^{-x/2})\,dx. \tag{30}$$

Under RH, evenness pairs γ and $-\gamma$. There is no central zeta zero: the alternating eta series at $1/2$ has strictly positive paired terms, while $1-\sqrt{2}<0$, so $\zeta(1/2)<0$. Hence

$$\sum_{|\gamma|<T} \Phi_{E(f)}(\tfrac{1}{2}+i\gamma) = 2\sum_{0<\gamma<T} g(\gamma). \tag{31}$$

For a different self-dual function with a central zero, its multiplicity times $g(0)$ would remain as an exceptional term.

Subtracting the two primary formulae only after (29)–(31) gives an exact residual archimedean identity. With

$$K(x) = \frac{e^{|x|/2}}{|e^x-e^{-x}|}, \qquad L_g = \lim_{T\to\infty}\int_0^T g(t)\log\frac{t}{2\pi}\,dt,$$

one obtains

$$\begin{aligned} 2\int_0^\infty f(x)e^{-x/2}\,dx - \mathrm{PF}\int_{\mathbb{R}} f(|x|)K(x)\,dx - f(0)\log(2\pi) \\ = 2\mathcal{H}_f + \sqrt{\frac{2}{\pi}}\,L_g. \end{aligned} \tag{32}$$

This is proved on the stated common subspace as an algebraic corollary of the two primary theorems. A direct distribution-level proof of (32), independent of subtracting them, remains a formalized analytic obligation.

There is also an exact candidate for Montgomery's Poisson kernel, using the retained Montgomery kernel [24, pp. 185–187, eqs. (18)–(22)]. For

$$a=\sigma-\tfrac{1}{2}, \qquad u=\log x, \qquad F_{a,t,u}(v) = e^{-a|v-u|}e^{-it(v-u)},$$

where $1<\sigma<2$ and $x\ge 1$, condition (B) holds and direct integration gives

$$\Phi_{a,t,u}(s) = \frac{2a\,x^{s-1/2}}{a^2-(s-1/2-it)^2}. \tag{33}$$

Therefore

$$\Phi_{a,t,u}(\tfrac{1}{2}+i\gamma) = \frac{(2\sigma-1)x^{i\gamma}}{(\sigma-\frac{1}{2})^2+(t-\gamma)^2}, \tag{34}$$

exactly Montgomery's zero kernel. At a prime power n,

$$n^{-1/2}F(\log n) = \begin{cases} x^{1/2-\sigma+it}n^{\sigma-1-it}, & n\le x,\\ x^{\sigma-1/2+it}n^{-\sigma-it}, & n>x,\end{cases}$$

$$n^{-1/2}F(-\log n) = x^{1/2-\sigma+it}n^{-\sigma+it},$$

so both of Montgomery's prime ranges and the $\zeta'/\zeta(\sigma - it)$ series are recovered without approximation. Moreover,

$$\Phi_{a,t,u}(1) = \frac{x^{1/2}(2\sigma - 1)}{(\sigma - 1 + it)(\sigma - it)},$$

which is exactly his rational pole term. What is still open is one explicit commutator: $\Phi(0)$, $F(0)\log A$, Weil's real-place PF term, and the negative-log prime series must be shown from the zeta functional equation and the gamma partial fractions to equal Montgomery's $-x^{1/2-\sigma+it}\zeta'/\zeta(1-\sigma + it)$ plus his complete trivial-zero series. No term is discarded while that identity remains open.

2.4 The pair-correlation hierarchy

Assume RH for this subsection, as Montgomery does throughout the 1973 paper [24, pp. 181–192], and count zeros with multiplicity. Put

$$w(u) = \frac{4}{4 + u^2}, \qquad c_T = \frac{T}{2\pi}\log T, \qquad T \geq 2, \tag{35}$$

and retain both of the following presentations:

$$F(x, T) = \sum_{0<\gamma,\gamma'\leq T} x^{i(\gamma-\gamma')} w(\gamma - \gamma'), \qquad x > 0, \tag{36}$$

and Montgomery's normalized form

$$F_M(\alpha, T) = c_T^{-1} \sum_{0<\gamma,\gamma'\leq T} T^{i\alpha(\gamma-\gamma')} w(\gamma - \gamma'). \tag{37}$$

For each retained $T > 1$, $x = T^\alpha$ is a bijection $\mathbb{R} \to \mathbb{R}_{>0}$ with inverse $\alpha = (\log x)/(\log T)$, and

$$F_M(\alpha, T) = c_T^{-1} F(T^\alpha, T).$$

This coordinate bijection does not erase either presentation.

Theorem 2.2 (Montgomery's theorem as printed in 1973)**.** *[24, pp. 181 and 187–191, eqs. (2) and (18)–(34)] The function $F_M(\alpha, T)$ is real and even. For every $\varepsilon > 0$ there is $T_0(\varepsilon)$ such that*

$$F_M(\alpha, T) \geq -\varepsilon \qquad (\alpha \in \mathbb{R},\ T > T_0(\varepsilon)).$$

For fixed $\varepsilon > 0$, uniformly for $0 \leq \alpha \leq 1 - \varepsilon$,

$$F_M(\alpha, T) = \alpha + o(1) + (1 + o(1))T^{-2\alpha}\log T \qquad (T \to \infty). \tag{38}$$

The complete primary paper has now been content-read [24]. It says that more effort could extend the uniform estimate through $\alpha = 1$, but its printed proof establishes (38) on $[0, 1 - \varepsilon]$.

Two later refinements must remain later refinements. Goldston's exact identity [14, eqs. (4.1)–(4.11)]

$$F(x, T) = \frac{2}{\pi} \int_{\mathbb{R}} \left| \sum_{0<\gamma\leq T} \frac{x^{i\gamma}}{1 + (t - \gamma)^2} \right|^2 dt \tag{39}$$

proves $F(x, T) \geq 0$ and $F(x, T) = F(x^{-1}, T)$; a sieve refinement recorded by Goldston extends the asymptotic range through $\alpha = 1$. Neither result is silently retrojected into the weaker literal theorem statement above.

2.4.1 Fourier conventions, pair measure, and support loss

Montgomery uses [24, pp. 181–182, eqs. (3)–(6)]

$$\widehat{r}_M(\alpha) = \int_{\mathbb{R}} r(u)e^{-2\pi i\alpha u}\,du, \qquad r(u) = \int_{\mathbb{R}} \widehat{r}_M(\alpha)e^{2\pi i\alpha u}\,d\alpha,$$

whereas Goldston uses [14, Sec. 5]

$$\widehat{r}_G(\alpha) = \int_{\mathbb{R}} r(u)e^{2\pi i\alpha u}\,du, \qquad r(u) = \int_{\mathbb{R}} \widehat{r}_G(\alpha)e^{-2\pi i\alpha u}\,d\alpha.$$

The convention morphism is the involution

$$\widehat{r}_G(\alpha) = \widehat{r}_M(-\alpha). \tag{40}$$

Because F_M is even, changing α to $-\alpha$ proves that the two resulting pairings agree. Under Montgomery's convention the exact bridge is

$$\sum_{0<\gamma,\gamma'\le T} r\left((\gamma-\gamma')\frac{\log T}{2\pi}\right) w(\gamma-\gamma') = c_T \int_{\mathbb{R}} \widehat{r}_M(\alpha)F_M(\alpha,T)\,d\alpha. \tag{41}$$

The primary theorem mainly permits $\operatorname{supp}\widehat{r}_M \subseteq [-1+\delta, 1-\delta]$; Goldston's endpoint refinement permits support through $[-1,1]$.

This boundary can be stated as an exact measure map. Define

$$\mu_T = c_T^{-1} \sum_{0<\gamma,\gamma'\le T} w(\gamma-\gamma')\,\delta_{(\gamma-\gamma')\log T/(2\pi)}. \tag{42}$$

Then

$$F_M(\alpha,T) = \int_{\mathbb{R}} e^{2\pi i\alpha u}\,d\mu_T(u), \qquad \int_{\mathbb{R}} r(u)\,d\mu_T(u) = \int_{\mathbb{R}} F_M(\alpha,T)\widehat{r}_M(\alpha)\,d\alpha. \tag{43}$$

The full Fourier–Stieltjes transform is injective on finite measures, but restriction to $[-1,1]$ is not injective on finite signed measures. Indeed, choose a nonzero even real $\phi \in C_c^\infty((-3,-2)\cup(2,3))$, let g be its inverse Fourier transform, and put $d\nu = g(u)\,du$. Then g is real Schwartz, $\nu \neq 0$, and $\widehat{\nu} = \phi$ vanishes on $[-1,1]$. Thus μ and $\mu+\nu$ have identical restricted transforms in the signed-measure category. No stronger fibre claim inside the positive-measure cone is being made. In particular, the support-limited theorem does not determine arbitrary interval indicators or the full pair measure.

2.4.2 The two finite-height spacing coordinates

Montgomery's test coordinate [24] and the local mean-density coordinate used in the random-matrix comparison [23] are respectively

$$u_M = \frac{\log T}{2\pi}(\gamma-\gamma'), \qquad u_D = \frac{\log(T/(2\pi))}{2\pi}(\gamma-\gamma'). \tag{44}$$

For $T \neq 2\pi$ their exact transition and inverse are

$$u_D = \frac{\log(T/(2\pi))}{\log T}\,u_M, \qquad u_M = \frac{\log T}{\log(T/(2\pi))}\,u_D. \tag{45}$$

The fibres are singletons for fixed $T \ge 2$, $T \neq 2\pi$. At $T = 2\pi$, u_D collapses every difference to zero while u_M remains bijective. For $2 \le T < 2\pi$ the transition reverses orientation; for $T > 2\pi$ it preserves orientation. Its factor tends to one as $T \to \infty$, but it is not replaced by one at finite height.

2.4.3 Conjectures, orientations, and the diagonal

Montgomery's literal signed-interval conjecture is: for fixed real $a < b$ [24, pp. 183–184, eqs. (11)–(15)],

$$\sum_{\substack{0<\gamma,\gamma'\leq T\\ 2\pi a/\log T\leq\gamma-\gamma'\leq 2\pi b/\log T}} 1 \sim \left[\int_a^b \left(1-\left(\frac{\sin\pi u}{\pi u}\right)^2\right) du + \delta(a,b)\right] c_T, \tag{46}$$

where $\delta(a,b)=1$ when $0\in[a,b]$ and is zero otherwise. The extra term is exactly the diagonal $\gamma=\gamma'$. The paper separately proposes $F_M(\alpha,T)=1+o(1)$ for $\alpha\geq 1$, uniformly in bounded intervals, and calls the two assertions essentially equivalent; that phrase is not a printed biconditional proof.

Goldston later separates the following two conjectures [14, Sec. 6, eqs. (6.6)–(6.9)]:

Strong pair correlation (SPC). For each fixed bounded M, $F_M(\alpha,T)=1+o(1)$ for $1\leq\alpha\leq M$. A strong Hardy–Littlewood prime-pair input motivates a bounded subrange but does not turn the full statement into the preceding theorem.

Interval pair correlation (PCC). For fixed $\beta>0$, the scaled count of pairs with $0<\gamma'-\gamma\leq 2\pi\beta/\log T$ is conjectured to tend to

$$\int_0^\beta \left(1-\left(\frac{\sin\pi u}{\pi u}\right)^2\right) du. \tag{47}$$

Goldston proves, under the real-ordinate hypothesis, that SPC implies PCC together with the corresponding average-multiplicity statement. This implication does not identify SPC and PCC. For $0<a<b$, swapping (γ,γ') is an exact bijection between the two opposite signed orientations. If the interval contains zero, the diagonal term must instead be retained or explicitly subtracted. Montgomery's primary paper itself therefore confirms that Schumayer–Hutchinson's pair-correlation discussion collapses a conjecture into the proved theorem [15, pair-correlation discussion]. The conditional simple-zero consequence remains the support-limited bound $(2/3-o(1))N(T)$; the primary gap calculation explicitly permits the constant 0.68, whereas Goldston's later exposition reports $0.669\ldots$.

2.5 GUE, CUE, and the maps that are still missing

In the bulk, after scaling a GUE spectrum to unit mean density and sending the matrix size to infinity, the correlation functions have the determinantal presentation [13, random-matrix correlation discussion]

$$R_n(E_1,\ldots,E_n)=\det\big[K(E_i,E_j)\big]_{i,j=1}^n, \qquad K(E_i,E_j)=\frac{\sin\pi(E_i-E_j)}{\pi(E_i-E_j)}. \tag{48}$$

In particular, $R_2(u)=1-(\sin(\pi u)/(\pi u))^2$. This explains the comparison with (47) and (46). Montgomery reports that Dyson identified this same pair function for large random complex Hermitian or unitary matrices, and then proposes the higher-correlation determinant in (48) [24, pp. 184–185, eqs. (15)–(17)]. This records the historical bridge from the primary zeta paper; the cited Mehta random-matrix source has not yet been independently content-read. The bridge supplies no map from an individual zeta zero to an individual matrix eigenvalue and constructs no fixed operator.

For a retained height $T > 2\pi$, the exact affine scaling map is

$$U_T(\gamma) = \frac{\log(T/(2\pi))}{2\pi}(\gamma - T), \qquad U_T^{-1}(u) = T + \frac{2\pi}{\log(T/(2\pi))}u. \tag{49}$$

For fixed T this map has singleton fibres. If T is forgotten, or if only differences are retained, absolute height and a common translation are lost. At $T = 2\pi$ the scale vanishes. The asymptotic convergence of the resulting point configurations to GUE statistics is a separate conjectural statement. Conversely, the height dependence of (49) does not prove that a fixed spectral operator cannot exist.

Finite CUE theorem. CUE enters through a different finite object. For $U \in U(N)$ with normalized Haar measure and $\theta \in \mathbb{R}$, put

$$Z_N(U;\theta) = \det(I_N - Ue^{-i\theta}). \tag{50}$$

Keating–Snaith prove, for $\Re s > -1$ [23, pp. 57–64, eqs. (1), (6), and (20)–(26)],

$$M_N(s) := \mathbb{E}_{U(N)}|Z_N(U;\theta)|^s = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+s)}{\Gamma(j+s/2)^2}. \tag{51}$$

Their proof inserts the CUE joint eigenphase density and evaluates the resulting N-fold integral by Selberg's integral. The product has meromorphic continuation in s; outside $\Re s > -1$ that continuation is not automatically the original Haar expectation. The exact parameter morphism $s = 2\lambda$ has inverse $\lambda = s/2$ and transports the moment domain to $\Re\lambda > -1/2$. Thus (51) becomes

$$\mathbb{E}_{U(N)}|Z_N(U;\theta)|^{2\lambda} = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+2\lambda)}{\Gamma(j+\lambda)^2}. \tag{52}$$

They also retain the branch data needed for the phase-ratio moment and obtain [23, eqs. (7) and (27)–(33)]

$$L_N(s) := \mathbb{E}_{U(N)}\left(\frac{Z_N(U;\theta)}{Z_N(U;\theta)^*}\right)^{s/2} = \prod_{j=1}^{N} \frac{\Gamma(j)^2}{\Gamma(j+s/2)\Gamma(j-s/2)}. \tag{53}$$

The branch is defined by continuous variation along $\theta - i\varepsilon$ away from eigenphases. Erasing this prescription changes the object.

Large-N theorem and log-characteristic-polynomial limit. For fixed λ in the initial moment domain [23, eqs. (15)–(16) and (80)–(88)],

$$\lim_{N\to\infty} N^{-\lambda^2}\mathbb{E}_{U(N)}|Z_N(U;\theta)|^{2\lambda} = \frac{G(1+\lambda)^2}{G(1+2\lambda)}. \tag{54}$$

For integers $k \geq 1$, this coefficient is

$$f_{\mathrm{CUE}}(k) = \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}, \tag{55}$$

giving 1, 1/12, 42/9!, and 24024/16! at $k = 1, 2, 3, 4$. The same primary paper proves that [23, eqs. (9)–(14) and (34)–(79)]

$$\left(\frac{\Re \log Z_N}{\sqrt{\frac{1}{2} \log N}}, \frac{\Im \log Z_N}{\sqrt{\frac{1}{2} \log N}} \right) \Longrightarrow (X, Y), \tag{56}$$

where X and Y are independent standard normal variables. The proof differentiates the exact generating products, computes polygamma cumulants, and shows that every normalized higher pure or mixed cumulant vanishes. At finite N, independence is not asserted.

For $\beta = 1, 2, 4$, corresponding respectively to COE, CUE, and CSE, the same Selberg calculation gives the retained family [23, eqs. (109)–(111)]

$$M_N^{(\beta)}(s) = \prod_{j=0}^{N-1} \frac{\Gamma(1 + j\beta/2)\Gamma(1 + s + j\beta/2)}{\Gamma(1 + s/2 + j\beta/2)^2}. \tag{57}$$

At $\beta = 2$, the index change $k = j+1$ recovers (51). This compatibility does not assign an L-function family to the other two finite ensembles.

Exact spacing-parameter map and conjectural zeta transfer. Keating–Snaith equate the CUE mean eigenphase spacing $2\pi/N$ with the high-height zeta-zero spacing $2\pi/\log(T/(2\pi))$. This gives the exact continuous parameter map [23, eqs. (8) and (92)]

$$D : (2\pi, \infty) \longrightarrow (0, \infty), \qquad D(T) = \log \frac{T}{2\pi}, \qquad D^{-1}(n) = 2\pi e^n. \tag{58}$$

It has singleton fibres and satisfies $2\pi/D(T) = 2\pi/\log(T/(2\pi))$. Actual matrix size is integer; applying a floor, ceiling, or nearest-integer map is an additional lossy operation with interval fibres. Even before that projection, (58) maps scale parameters only: it maps neither individual zeros to eigenphases nor zeta functions to characteristic polynomials.

For real λ where the factors are defined, set

$$a(\lambda) = \prod_p (1 - p^{-1})^{\lambda^2} \sum_{m=0}^{\infty} \left(\frac{(\lambda)_m}{m!} \right)^2 p^{-m}. \tag{59}$$

The Keating–Snaith conjecture is [23, eqs. (4)–(5), (19), and (93)–(96)]

$$\frac{1}{T} \int_0^T \left| \zeta\left(\frac{1}{2} + it \right) \right|^{2\lambda} dt \sim a(\lambda) \frac{G(1+\lambda)^2}{G(1+2\lambda)} (\log T)^{\lambda^2}. \tag{60}$$

The Barnes-G factor is supplied by the proved CUE limit; the arithmetic factor is motivated by a truncated Euler product and an independence model for prime contributions. Those steps, the transfer through (58), and (60) are not proved by the finite Haar integral. The paper explicitly distinguishes local random-matrix statistics from long-range prime corrections. This primary reading also resolves the survey defects: the ensemble is CUE, 1/12 is $f_{\mathrm{CUE}}(2)$, the cited classical author is A. E. Ingham, and the retained scale is $\log(T/(2\pi))$, not merely "of order $\log T$."

Finally, Bogomolny's finite-height two-point formula is a valuable heuristic object but not a theorem. It uses smoothing, a diagonal approximation, the Hardy–Littlewood prime-pair conjecture, $\log((n + r)/n) \approx r/n$, and a sum-to-integral passage. Its diagonal and off-diagonal terms retain arithmetic Euler factors and approach the GUE expression after the height-dependent scaling. The source itself describes the error as supposed and concludes that these random-matrix results are not rigorously proved. The equation label `exact_R` therefore cannot be used as an evidence status [13, finite-height two-point calculation].

2.6 Higher derivative CUE moments: exact exponents and the missing Lie arrow

Grover, Mezzadri, and Simm work with Haar probability measure on $\mathrm{U}(N)$ and the characteristic polynomial

$$\Lambda_N(z) = \det(1 - zU^\dagger). \tag{61}$$

For finite lists of nonnegative integers $\mu = (\mu_1, \ldots, \mu_K)$ and $\nu = (\nu_1, \ldots, \nu_L)$, their joint derivative moment is

$$M_{\mu,\nu}(z, N) = \mathbb{E}\left[\prod_{i=1}^{K} \Lambda_N^{(\mu_i)}(z) \prod_{j=1}^{L} \overline{\Lambda_N^{(\nu_j)}(z)}\right]. \tag{62}$$

Here $|\mu| = \sum_i \mu_i$, $\mu! = \prod_i \mu_i!$, and similarly for ν [28, eq. (1.1)]. The companion Easy Riders exposition identifies this as the channel owner's recent paper written with a PhD supervisor and collaborator, describes the work as primarily random matrix theory, and gives the author's geometric account of the global, mesoscopic, and microscopic probe-point regimes. In that account, the microscopic scale moves the probe toward the unit circle at the eigenvalue-spacing scale and is the regime compared with the zeta function near its critical-line zeros [40, 0:25–0:35, 3:03–3:08, and 7:44–10:52]. A later video records a separate open random-matrix problem concerning a one-mode/two-mode transition and the search for a human-readable analytic mechanism rather than an opaque machine certificate [41, 3:19–6:15 and 7:20–12:08]. These videos are author-side motivation and geometric exposition. The theorem statements, hypotheses, and proofs below are taken from the paper, while the later first-peeled $N = 3$ coordinates are independently derived in theorems 2.37 and 2.38; neither video states those coordinates. For $\boldsymbol{z} = (z_1, \ldots, z_K)$ and $\boldsymbol{w} = (w_1, \ldots, w_L)$, retain the source's differential and coalescence notation

$$\mathcal{D}_{\boldsymbol{z}}^{\mu} = \prod_{i=1}^{K} \frac{\partial^{\mu_i}}{\partial z_i^{\mu_i}}, \qquad F(\boldsymbol{z})|_{\boldsymbol{z}=z} = \lim_{z_1,\ldots,z_K \to z} F(\boldsymbol{z}), \tag{63}$$

where coalescence occurs after differentiation. Let $\mathcal{M}_{\mu,\cdot}$ be the set of $K \times L$ nonnegative-integer matrices with row sums μ, and let $\mathcal{M}_{\cdot,\nu}$ be the set of such matrices with column sums ν. Define

$$p_{n,m}(z, \overline{z}) = \sum_{r=0}^{\min(n,m)} \binom{n}{r}\binom{m}{r} z^{m-r}\overline{z}^{n-r}. \tag{64}$$

Then for each fixed $\alpha \in (0, 1)$, uniformly for $|z| < 1 - N^{-\alpha}$ as $N \to \infty$,

$$M_{\mu,\nu}(z, N) = \frac{\mu!\nu!}{(1 - |z|^2)^{KL+|\mu|+|\nu|}} \sum_{\substack{Q \in \mathcal{M}_{\mu,\cdot} \\ R \in \mathcal{M}_{\cdot,\nu}}} \prod_{i=1}^{K}\prod_{j=1}^{L} p_{Q_{ij},R_{ij}}(z, \overline{z}) + O\Big(e^{-N^{\delta}}\Big) \tag{65}$$

for some $\delta > 0$ [28, Thm. 1.1]. At $z = 0$ the polynomial in (64) is nonzero precisely when its two indices agree. Consequently

$$\frac{1}{\mu!\nu!} \lim_{N\to\infty} M_{\mu,\nu}(0, N) = \#\big(\mathcal{M}_{\mu,\cdot} \cap \mathcal{M}_{\cdot,\nu}\big), \tag{66}$$

the exact contingency-table count recorded by the source.

The equal-length microscopic theorem has substantially more content than its exponent. Fix $K = L = s$, put

$$z_N = 1 - \frac{c}{N}, \qquad \tau = c + \overline{c}, \tag{67}$$

and, for partitions λ, ρ of lengths at most s, put

$$\alpha_i(\lambda) = \lambda_i + s - i, \qquad \beta_j(\rho) = \rho_j + s - j, \qquad \lambda!_{(s)} = \prod_{i=1}^{s} \alpha_i(\lambda)!, \tag{68}$$

with the analogous definition of $\rho!_{(s)}$, and

$$I_r(\tau) = \int_0^1 x^r e^{-\tau x}\, dx. \tag{69}$$

When $K_{\lambda\mu}$ or $K_{\rho\nu}$ is formed, the source first rearranges the relevant list in weakly decreasing order and discards zero entries; this Kostka-content convention is not a change to the list lengths K and L in (62). Then, uniformly for c in compact subsets of $\mathbb{C}$,

$$\lim_{N\to\infty} \frac{M_{\mu,\nu}(z_N, N)}{N^{|\mu|+|\nu|+s^2}} = \mu!\nu! \sum_{\substack{\lambda\vdash|\mu|,\ \ell(\lambda)\le s\\ \rho\vdash|\nu|,\ \ell(\rho)\le s}} \frac{K_{\lambda\mu}K_{\rho\nu}}{\lambda!_{(s)}\rho!_{(s)}} \det\Big(I_{\alpha_i(\lambda)+\beta_j(\rho)}(\tau)\Big)_{i,j=1}^{s}. \tag{70}$$

At $c = 0$, this is exactly

$$\lim_{N\to\infty} \frac{M_{\mu,\nu}(1, N)}{N^{|\mu|+|\nu|+s^2}} = \mu!\nu! \sum_{\substack{\lambda\vdash|\mu|,\ \ell(\lambda)\le s\\ \rho\vdash|\nu|,\ \ell(\rho)\le s}} \frac{K_{\lambda\mu}K_{\rho\nu}}{\lambda!_{(s)}\rho!_{(s)}} \det\left(\frac{1}{\lambda_i+\rho_j+2s-i-j+1}\right)_{i,j=1}^{s}. \tag{71}$$

Both right-hand sides are strictly positive [28, Thm. 1.2]. The proof passes through an exact finite-N Kostka/determinant formula, a confluent determinant identity, the microscopic derivative asymptotic, and an Andréief–Schur positivity argument [28, Sec. 2].

The unequal-length coordinates show where the quadratic term comes from. They also expose a sign defect in version 1 of the source. If $K \ge L$, put $d = K-L$, set $z_i = 1-a_i/N$ and $w_j = 1-b_j/N$, and let Ω be the $K\times K$ matrix whose ith row is

$$\Big(1, a_i, \dots, a_i^{K-L-1}, I(a_i+b_1), \dots, I(a_i+b_L)\Big), \qquad I(x) = \frac{1-e^{-x}}{x}. \tag{72}$$

With the source's convention $\Delta(\boldsymbol{a}) = \prod_{i<j}(a_j - a_i)$ and with the excess columns displayed before the kernel columns, the corrected formula is

$$\lim_{N\to\infty} \frac{M_{\mu,\nu}(1-c/N, N)}{N^{|\mu|+|\nu|+KL}} = (-1)^{|\mu|+|\nu|+L(K-L)}\ \mathcal{D}^{\mu}_{\boldsymbol{a}}\mathcal{D}^{\nu}_{\boldsymbol{b}} \frac{\det\Omega}{\Delta(\boldsymbol{a})\Delta(\boldsymbol{b})}\bigg|_{\boldsymbol{a}=c,\ \boldsymbol{b}=\bar{c}}. \tag{73}$$

Version 1 prints the same display without $(-1)^{L(K-L)}$ [28, Sec. 2, theorem labelled `micro`]. The missing parity is not a matter of normalization: the exact $K = 2, L = 1, N = 1$ Haar integral has the opposite sign from the printed determinant quotient. The proof-bearing audit below derives the finite-N factor by padding the equal-length determinant with zero coordinates, proves (73) coordinate by coordinate, and records the counterexample. All equal-length formulas have $K-L = 0$ and are therefore unchanged. The power of N is still obtained term by term: the derivative change of variables contributes $|\mu| + |\nu|$, and the two Vandermonde factors, the L kernel columns, and the $K - L$ excess columns contribute KL.

The number-theoretic statement is separate. Assuming the Lindelöf hypothesis, the authors prove, as $\sigma \downarrow \frac{1}{2}$,

$$\lim_{T\to\infty} \frac{1}{T}\int_1^T \prod_{i=1}^{K} \zeta^{(\mu_i)}(\sigma+it) \prod_{j=1}^{L} \overline{\zeta^{(\nu_j)}(\sigma+it)}\, dt \sim \frac{(-1)^{|\mu|+|\nu|} a_{\mu,\nu} h_{\mu,\nu}}{(2\sigma-1)^{KL+|\mu|+|\nu|}}, \tag{74}$$

where

$$a_{\mu,\nu} = \prod_p \left\{ (1-p^{-1})^{KL} \sum_{m\geq 0} \frac{\Gamma(K+m)}{m!\Gamma(K)} \frac{\Gamma(L+m)}{m!\Gamma(L)} p^{-m} \right\}, \tag{75}$$

$$h_{\mu,\nu} = \mu!\nu! \sum_{\substack{Q\in\mathcal{M}_{\mu,\cdot} \\ R\in\mathcal{M}_{\cdot,\nu}}} \prod_{i=1}^{K} \prod_{j=1}^{L} \binom{Q_{ij}+R_{ij}}{Q_{ij}}. \tag{76}$$

This is unconditional when $K, L \leq 2$. In particular, for every positive integer k the source proves unconditionally that

$$\lim_{T\to\infty} \frac{1}{T} \int_1^T \left|\zeta^{(k)}(\sigma+it)\right|^4 \, dt \sim \frac{6}{\pi^2} \frac{h_k}{(2\sigma-1)^{4(k+1)}}, \qquad \sigma \downarrow \tfrac{1}{2}, \tag{77}$$

$$h_k = (k!)^4(k+1) \left\{ \binom{4k+3}{2k+2} - 2\binom{2k+1}{k}^2 \right\}. \tag{78}$$

By contrast, replacing the shift by $\sigma = \frac{1}{2} + c/\log T$ and transporting the coefficient from (70) is explicitly proposed as a reasonable conjecture, not proved [28, Prop. 1.3, Thm. 1.4, and the paragraph following eq. (1.22)].

2.6.1 Proof-bearing audit of the Grover–Mezzadri–Simm tranche

The preceding subsection records the statements. This subsection records the proof chain at the level needed to reproduce it. The primary object is version 1 of the paper together with four locally pinned source dependencies used in its proofs: the characteristic-polynomial determinant of Akemann–Vernizzi [29, Thm. 1 and its proof], Krattenthaler's divided-difference reduction [30, Lemma 22], Balantekin's determinant Taylor expansion [31, App. B], and the finite-kernel microscopic estimate of Simm–Wei [32, Sec. 5.1]. The formulas and deductions below are a reconstruction with those human sources cited; author-side videos are not used as theorem evidence. Macdonald's Schur-function identities, Hua's deeper provenance for the determinant expansion, and the Titchmarsh mean-value estimates remain separately marked imported boundaries wherever they enter.

Exact finite-N generating determinant. For pairwise distinct coordinates $\mathbf{z} = (z_1, \ldots, z_K)$ and $\mathbf{w} = (w_1, \ldots, w_L)$, define

$$\mathcal{F}_{N;K,L}(\mathbf{z},\mathbf{w}) = \mathbb{E}\left[\prod_{i=1}^{K} \Lambda_N(z_i) \prod_{j=1}^{L} \overline{\Lambda_N(\overline{w_j})} \right], \qquad \mathcal{K}_{N+L}(r) = \sum_{q=0}^{N+L-1} r^q. \tag{79}$$

The bar in the second product has a precise role: $\overline{\Lambda_N(\overline{w})} = \det(I - wU)$, whereas $\Lambda_N(z) = \det(I - zU^\dagger)$. Thus z and w are independent holomorphic coordinates until the final coalescence $w = \overline{z}$.

Theorem 2.3 (Corrected Akemann–Vernizzi determinant in the CUE coordinates)**.** *Suppose $K \geq L \geq 1$ and put $d = K - L$. Let $\Omega_N^{\mathrm{ker}|\mathrm{exc}}(\mathbf{z},\mathbf{w})$ be the $K \times K$ matrix whose ith row is*

$$\left(z_i^{K-L}\mathcal{K}_{N+L}(z_i w_1), \ldots, z_i^{K-L}\mathcal{K}_{N+L}(z_i w_L), 1, z_i, \ldots, z_i^{K-L-1}\right). \tag{80}$$

Let $\Omega_N^{\mathrm{exc}|\mathrm{ker}}$ be obtained by moving the last d columns to the front, without changing the order inside either block. Then

$$\mathcal{F}_{N;K,L}(\mathbf{z},\mathbf{w}) = (-1)^{Ld}\frac{\det\Omega_N^{\mathrm{ker}|\mathrm{exc}}(\mathbf{z},\mathbf{w})}{\Delta(\mathbf{z})\Delta(\mathbf{w})} = \frac{\det\Omega_N^{\mathrm{exc}|\mathrm{ker}}(\mathbf{z},\mathbf{w})}{\Delta(\mathbf{z})\Delta(\mathbf{w})}. \tag{81}$$

Both sides extend holomorphically across every collision of z-coordinates or w-coordinates.

Proof. We spell out the specialization because every exponent used later comes from it. Under the Weyl integration formula, the eigenvalues $x_r = e^{i\theta_r}$ of a Haar unitary have joint density

$$\frac{1}{N!(2\pi)^N}\,|\Delta(x_1,\ldots,x_N)|^2\prod_{r=1}^{N} d\theta_r. \tag{82}$$

Multiplication by $\prod_{r,i}(1-z_ix_r^{-1})\prod_{r,j}(1-w_jx_r)$ appends the external coordinates to the two Vandermonde determinants. Replacing monomials by the monic orthogonal polynomials does not change a Vandermonde determinant. For Haar measure on the unit circle these polynomials are the monomials and all norms are one. Expanding both determinants and applying orthogonality

$$\frac{1}{2\pi}\int_0^{2\pi} x^p\overline{x^q}\,d\theta = \delta_{pq} \tag{83}$$

integrates the N eigenvalue rows. Equivalently, one applies the projection identity

$$\frac{1}{2\pi}\int_0^{2\pi}\mathcal{K}_m(vx^{-1})\mathcal{K}_m(xu)\,d\theta = \mathcal{K}_m(vu), \qquad \mathcal{K}_m(r) = \sum_{q=0}^{m-1} r^q, \tag{84}$$

successively; the factors $N, N-1,\ldots,1$ cancel the $N!$ in (82). When $K = L$, the surviving external block is $\big(\mathcal{K}_{N+K}(z_iw_j)\big)_{i,j=1}^K$, giving

$$\mathcal{F}_{N;K,K} = \frac{\det(\mathcal{K}_{N+K}(z_iw_j))_{i,j=1}^K}{\Delta(\mathbf{z})\Delta(\mathbf{w})}. \tag{85}$$

For $K > L$, introduce $d = K - L$ auxiliary coordinates $\boldsymbol{\varepsilon} = (\varepsilon_1,\ldots,\varepsilon_d)$ and apply (85) to the ordered K-tuple $(w_1,\ldots,w_L,\varepsilon_1,\ldots,\varepsilon_d)$. Since $\Lambda_N(0) = 1$, its left-hand side tends to $\mathcal{F}_{N;K,L}$ as $\boldsymbol{\varepsilon}\to\mathbf{0}$. The denominator factors exactly as

$$\Delta(\mathbf{w},\boldsymbol{\varepsilon}) = \Delta(\mathbf{w})\Delta(\boldsymbol{\varepsilon})\prod_{j=1}^{L}\prod_{r=1}^{d}(\varepsilon_r - w_j), \tag{86}$$

and hence

$$\lim_{\boldsymbol{\varepsilon}\to\mathbf{0}}\frac{\Delta(\mathbf{w},\boldsymbol{\varepsilon})}{\Delta(\mathbf{w})\Delta(\boldsymbol{\varepsilon})} = (-1)^{Ld}\prod_{j=1}^{L} w_j^d. \tag{87}$$

In the numerator, division by $\Delta(\boldsymbol{\varepsilon})$ and confluence at zero replace the auxiliary columns by

$$\frac{1}{r!}\frac{\partial^r}{\partial\varepsilon^r}\mathcal{K}_{N+K}(z_i\varepsilon)\bigg|_{\varepsilon=0} = z_i^r, \qquad 0\le r<d. \tag{88}$$

For each remaining kernel column j, subtract the linear combination $\sum_{r=0}^{d-1} w_j^r (z_i^r)_i$. This does not change the determinant and leaves

$$\mathcal{K}_{N+K}(z_i w_j) - \sum_{r=0}^{d-1} (z_i w_j)^r = (z_i w_j)^d \mathcal{K}_{N+L}(z_i w_j). \tag{89}$$

Factoring w_j^d from each of the L kernel columns cancels $\prod_j w_j^d$ in (87); division by the remaining $(-1)^{Ld}$ gives the first equality in (81). Moving the L kernel columns past the d excess columns contributes a second $(-1)^{Ld}$ and gives the excess-first equality with no prefactor.

This padding proof is also the exact CUE-coordinate specialization of the two-block determinant proved by Akemann–Vernizzi [29, pp. 8–12, especially the induction displayed in eq. (2.12)]. Their determinant in variables v_i, u_j is correct. The parity in (81) enters when the reciprocal coordinates used for $\det(I - zU^*)$ are converted to the ascending excess columns displayed in (80).

The numerator is alternating separately in $\mathbf{z}$ and $\mathbf{w}$. Successive column subtractions, or Newton divided differences, factor every $z_j - z_i$ and every $w_j - w_i$. The quotient therefore extends holomorphically; at a collision its coordinates are the corresponding derivatives divided by factorials. This is the confluent specialization of Krattenthaler's explicit divided-difference identity [30, Lemma 22]. □

The factor in (81) is forced already in the smallest unequal case. Take $K = 2$, $L = 1$, $N = 1$ and write the Haar eigenvalue as $e^{\mathrm{i}\theta}$. Direct extraction of the constant Fourier coefficient gives

$$\begin{aligned}\mathcal{F}_{1;2,1}(z_1, z_2; w) &= \frac{1}{2\pi} \int_0^{2\pi} (1 - z_1 e^{-\mathrm{i}\theta})(1 - z_2 e^{-\mathrm{i}\theta})(1 - w e^{\mathrm{i}\theta})\, d\theta \\ &= 1 + w(z_1 + z_2).\end{aligned} \tag{90}$$

But the kernel-first numerator printed in version 1 is

$$\det \begin{pmatrix} z_1(1 + z_1 w) & 1 \\ z_2(1 + z_2 w) & 1 \end{pmatrix} = (z_1 - z_2)\big(1 + w(z_1 + z_2)\big), \tag{91}$$

whereas $\Delta(z_1, z_2) = z_2 - z_1$. Its uncorrected quotient is therefore the negative of (90); here $(-1)^{L(K-L)} = -1$ repairs it. This is a finite exact counterexample to the sign as printed, not a numerical asymptotic test. The cross-multiplied identity (90)–(91) and the doubled fixed-exponent polynomial identity are replayed by `GMSUnequalSign.lean`.

The companion receipt `FORMAL-20260827-0022.receipt.json` records Lean 4.32.0, mathlib commit beginning `81a5d257c8e4` (and stores the complete hash), exit code zero, and a peak process-tree working set below 1.41 GB. That certificate intentionally does not formalize Haar integration, the general confluent limit, or the large-N argument.

Mesoscopic regime: uniform kernel control and exact coefficient extraction. The proof of (65) has two logically separate parts. The first removes the finite kernel with a uniform error; the second evaluates every derivative of the limiting Cauchy product.

Lemma 2.4 (Uniform limiting product). *Fix $\alpha \in (0, 1)$ and the lists μ, ν. There is $\delta > 0$ such that, uniformly for $|z| < 1 - N^{-\alpha}$,*

$$M_{\mu,\nu}(z, N) = \mathcal{D}_{\mathbf{z}}^{\mu} \mathcal{D}_{\mathbf{w}}^{\nu} \prod_{i=1}^{K} \prod_{j=1}^{L} (1 - z_i w_j)^{-1} \Bigg|_{\mathbf{z}=z,\, \mathbf{w}=\overline{z}} + O(e^{-N^{\delta}}). \tag{92}$$

Proof. First take $K = L$. From (85),

$$\mathcal{K}_{N+K}(r) = \frac{1 - r^{N+K}}{1 - r}. \tag{93}$$

The Cauchy determinant identity gives

$$\frac{\det((1 - z_i w_j)^{-1})_{i,j=1}^{K}}{\Delta(\mathbf{z})\Delta(\mathbf{w})} = \prod_{i,j=1}^{K} (1 - z_i w_j)^{-1}. \tag{94}$$

For $d_N = 1 - |z| \geq N^{-\alpha}$, take each Cauchy differentiation contour to have radius $d_N/8$. On their product, $|z_i w_j| \leq 1 - c_0 d_N$ for an absolute $c_0 > 0$, and hence

$$\left| \frac{(z_i w_j)^{N+K}}{1 - z_i w_j} \right| \leq C d_N^{-1} e^{-c_0 N d_N} \leq C N^{\alpha} e^{-c_0 N^{1-\alpha}}. \tag{95}$$

The determinant has fixed size. Divided differences express its Vandermonde quotient as a finite determinant of iterated divided differences, so the same bound holds there up to a fixed power of d_N^{-1}. Cauchy's integral formula for the fixed total derivative order $|\mu| + |\nu|$ contributes another fixed power. Every such polynomial in N is absorbed by $e^{-c_0 N^{1-\alpha}}$; for example one may take any $0 < \delta < 1 - \alpha$. This proves the claimed error for $K = L$. If $K \neq L$, pad the shorter product with coordinates equal to zero and use $\Lambda_N(0) = 1$; the same estimate survives before differentiation. This proves (92). □

Lemma 2.5 (Coordinate extraction)**.** *For $|z| < 1$, the derivative in* (92) *is exactly*

$$\frac{\mu!\nu!}{(1 - |z|^2)^{KL+|\mu|+|\nu|}} \sum_{\substack{Q \in \mathcal{M}_{\mu,\cdot} \\ R \in \mathcal{M}_{\cdot,\nu}}} \prod_{i=1}^{K} \prod_{j=1}^{L} p_{Q_{ij},R_{ij}}(z, \overline{z}). \tag{96}$$

Proof. Make the invertible affine coordinate change

$$z_i = z + (1 - |z|^2)t_i, \qquad w_j = \overline{z} + (1 - |z|^2)u_j. \tag{97}$$

No normalization is suppressed: its inverse is $t_i = (z_i - z)/(1 - |z|^2)$ and $u_j = (w_j - \overline{z})/(1 - |z|^2)$, and its Jacobian contributes $(1 - |z|^2)^{-|\mu|-|\nu|}$ to the derivative. Direct multiplication gives

$$1 - z_i w_j = (1 - |z|^2)\big((1 - \overline{z}t_i)(1 - zu_j) - t_i u_j\big). \tag{98}$$

Therefore the derivative equals

$$\frac{\mu!\nu!}{(1 - |z|^2)^{KL+|\mu|+|\nu|}} [\mathbf{t}^{\mu}\mathbf{u}^{\nu}] \prod_{i,j} \frac{1}{(1 - \overline{z}t_i)(1 - zu_j) - t_i u_j}. \tag{99}$$

For each cell (i, j) the geometric and negative-binomial expansions give

$$\frac{1}{(1 - \overline{z}t)(1 - zu) - tu} = \sum_{r \geq 0} \frac{t^r u^r}{(1 - \overline{z}t)^{r+1}(1 - zu)^{r+1}}, \tag{100}$$

$$[t^q u^s] \frac{1}{(1 - \overline{z}t)(1 - zu) - tu} = \sum_{r=0}^{\min(q,s)} \binom{q}{r}\binom{s}{r} \overline{z}^{q-r} z^{s-r} = p_{q,s}(z, \overline{z}). \tag{101}$$

Distributing the exponent of each t_i among the L cells in row i gives precisely a matrix Q with row sums μ; distributing each u_j among column j gives precisely a matrix R with column sums ν. Multiplication of the independent cell coefficients in (101) proves (96). Combining the two lemmas proves (65).

At $z = 0$, (101) is 1 if $q = s$ and 0 otherwise. Consequently only $Q = R$ survives, and the simultaneous row and column constraints give (66) without an additional limit argument. $\square$

Lemma 2.6 (The 2×2 finite binomial sum)**.** *For $k \in \mathbb{N}_0$,*

$$\sum_{n,m=0}^{k} \binom{n+m+k+1}{k}\binom{3k-n-m+1}{k} = (k+1)\left\{\binom{4k+3}{2k+2} - 2\binom{2k+1}{k}^2\right\}. \tag{102}$$

Proof. Introduce independent coefficient variables x, y. Two finite geometric sums give the exact coefficient representation

$$\begin{aligned}
&\sum_{n,m=0}^{k} \binom{n+m+k+1}{k}\binom{3k-n-m+1}{k} \\
&\quad = [x^k y^k](1+x)^{k+1}(1+y)^{3k+1}\left(\sum_{r=0}^{k}\left(\frac{1+x}{1+y}\right)^r\right)^2 \\
&\quad = [x^k y^k](1+x)^{k+1}(1+y)^{k+1}\left(\frac{(1+y)^{k+1}-(1+x)^{k+1}}{y-x}\right)^2.
\end{aligned} \tag{103}$$

The quotient is the finite polynomial $\sum_{r=0}^{k}(1+y)^{k-r}(1+x)^r$. Expanding it, grouping terms by the total exponent, and applying $\sum_j \binom{A}{j}\binom{B}{R-j} = \binom{A+B}{R}$ yields respectively the full Vandermonde coefficient $\binom{4k+3}{2k+2}$ and the two equal boundary coefficients $\binom{2k+1}{k}^2$. The conversion $(r+1)\binom{r+k+1}{k} = (k+1)\binom{r+k+1}{k+1}$ supplies the factor $k+1$. Substitution in (103) gives (102). $\square$

For $\mu = \nu = (k, k)$, write

$$Q = \begin{pmatrix} n & k-n \\ m & k-m \end{pmatrix}, \qquad R = \begin{pmatrix} k-p & k-q \\ p & q \end{pmatrix}. \tag{104}$$

Two applications of finite Vandermonde convolution first sum p, q and give the left side of (102). The lemma therefore proves the closed formula (78); no asymptotic or numerical fitting enters that constant.

Microscopic regime with unequal list lengths. We next prove the coordinate limit (73), including the power of N and the signs which are hidden if the two determinant presentations are simply identified. Assume $K \geq L \geq 1$ and put $d = K - L \geq 0$; the case $K < L$ follows by conjugating the moment and exchanging the two coordinate blocks. Use the excess-first finite determinant in (81), and put

$$z_i = 1 - \frac{a_i}{N}, \qquad w_j = 1 - \frac{b_j}{N}. \tag{105}$$

The coordinate map is affine and invertible for fixed N; hence

$$\mathcal{D}^{\mu}_{\mathbf{z}}\mathcal{D}^{\nu}_{\mathbf{w}} = (-N)^{|\mu|+|\nu|}\mathcal{D}^{\mu}_{\mathbf{a}}\mathcal{D}^{\nu}_{\mathbf{b}}. \tag{106}$$

The two Vandermonde factors transform exactly as

$$\Delta(\mathbf{z}) = (-N)^{-K(K-1)/2}\Delta(\mathbf{a}), \qquad \Delta(\mathbf{w}) = (-N)^{-L(L-1)/2}\Delta(\mathbf{b}). \tag{107}$$

For a kernel entry, uniformly on compact subsets of $(a,b) \in \mathbb{C}^2$,

$$\begin{aligned}\frac{1}{N}\mathcal{K}_{N+L}\left(\left(1-\frac{a}{N}\right)\left(1-\frac{b}{N}\right)\right) &= \frac{1}{N}\sum_{q=0}^{N+L-1}\left(1-\frac{a+b}{N}+\frac{ab}{N^2}\right)^q \\ &\longrightarrow \int_0^1 e^{-(a+b)x}\,dx = I(a+b).\end{aligned} \tag{108}$$

This is a uniform Riemann-sum limit; the value at $a+b=0$ is the removable value 1.

The d excess columns have entries $(1-a_i/N)^r$, $0 \le r < d$. The binomial identity

$$\left(1-\frac{a}{N}\right)^r = \sum_{q=0}^{r}\binom{r}{q}\left(-\frac{1}{N}\right)^q a^q \tag{109}$$

is an upper-triangular change from the monomial columns $1, a, \ldots, a^{d-1}$. Its determinant is

$$(-1)^{d(d-1)/2}N^{-d(d-1)/2}. \tag{110}$$

The L kernel columns contribute N^L. The reciprocal Vandermondes contribute

$$N^{K(K-1)/2+L(L-1)/2}. \tag{111}$$

Thus the net exponent is

$$\frac{K(K-1)}{2}+\frac{L(L-1)}{2}-\frac{d(d-1)}{2}+L = KL. \tag{112}$$

Because the finite determinant is now in excess-first order, no further block permutation is made. Together with the two reciprocal Vandermonde signs, (110) gives the fixed sign

$$(-1)^{K(K-1)/2+L(L-1)/2+d(d-1)/2} = (-1)^{Ld}, \tag{113}$$

because the exponent equals $Ld+L(L-1)+d(d-1)$ and the last two summands are even. Combining this with the derivative sign $(-1)^{|\mu|+|\nu|}$ in (106) gives exactly $(-1)^{|\mu|+|\nu|+L(K-L)}$. Substituting (108) into the exact determinant gives exactly the matrix Ω in (73).

It remains to justify differentiating after the limit and then coalescing the coordinates. For each N, the quotient in (81) is a polynomial. After division by the power N^{KL} just computed, its divided-difference form converges uniformly on every compact set to $\det\Omega/(\Delta(\mathbf{a})\Delta(\mathbf{b}))$. The latter quotient is entire, because its numerator is alternating in each coordinate block. The Weierstrass theorem and Cauchy's formula therefore give locally uniform convergence of every fixed derivative. This proves (73) without assuming the coalesced expression in advance. For an independent coalesced check, take $K=2, L=1$, zero derivative lists, and $c=0$. The exact excess-first determinant gives

$$\mathcal{F}_{N;2,1}(1,1;1) = \frac{d}{dz}\big(z\mathcal{K}_{N+1}(z)\big)\Big|_{z=1} = \frac{(N+1)(N+2)}{2}, \tag{114}$$

so $N^{-2}\mathcal{F}_{N;2,1} \to 1/2$. On the limiting side $\det\begin{pmatrix}1 & I(a_1+b)\\ 1 & I(a_2+b)\end{pmatrix}/(a_2-a_1) \to I'(0) = -1/2$; the factor $(-1)^{L(K-L)} = -1$ is therefore again forced. When $K=L$, $d=0$, so every equal-length microscopic formula is unchanged.

The explicit fourth moment at microscopic distance. For $K = L = 2$, before differentiation and coalescence the limiting quotient is

$$f(a_1, a_2, b_1, b_2) = \frac{I(a_1 + b_1)I(a_2 + b_2) - I(a_1 + b_2)I(a_2 + b_1)}{(a_2 - a_1)(b_2 - b_1)}. \tag{115}$$

Let $a(u) = a_1 + u(a_2 - a_1)$ and $b(v) = b_1 + v(b_2 - b_1)$. Applying the fundamental theorem of calculus once in each divided-difference coordinate gives the denominator-free identity

$$f = -\int_0^1 \int_0^1 \big[I'(a(u) + b_1)I'(a_2 + b(v)) - I''(a(u) + b(v))I(a_2 + b_1)\big]\, du\, dv. \tag{116}$$

Indeed, this follows by applying $-\partial_u \partial_v$ to

$$\frac{I(a(u) + b_1)I(a_2 + b(v)) - I(a(u) + b(v))I(a_2 + b_1)}{(a_2 - a_1)(b_2 - b_1)} \tag{117}$$

and evaluating its four boundary values. Since $I^{(r)}(x) = (-1)^r I_r(x)$, Leibniz's rule followed by $a_1 = a_2 = b_1 = b_2 = c \in \mathbb{R}$ turns $\partial_{a_1}^k \partial_{a_2}^k \partial_{b_1}^k \partial_{b_2}^k f$ into

$$\begin{aligned}-(1-u)^k v^k \sum_{i,j=0}^{k} \binom{k}{i}\binom{k}{j} \Big[& u^{k-i}(1-v)^j I_{3k-i-j+1}(2c) I_{k+i+j+1}(2c) \\ &- u^{k-i}(1-v)^{k-j} I_{4k-i-j+2}(2c) I_{i+j}(2c)\Big].\end{aligned} \tag{118}$$

All factors are continuous on the compact square, so differentiation under the integral is valid. The required beta integrals are explicit:

$$\int_0^1 u^{k-i}(1-u)^k du = \frac{(k-i)!k!}{(2k-i+1)!}, \tag{119}$$

$$\int_0^1 v^k (1-v)^{k-j} dv = \frac{k!(k-j)!}{(2k-j+1)!}, \tag{120}$$

$$\int_0^1 v^k (1-v)^j dv = \frac{k!j!}{(k+j+1)!}. \tag{121}$$

Consequently

$$C_{ij}^{(1)}(k) = \binom{k}{i}\binom{k}{j} \frac{(k-i)!k!}{(2k-i+1)!} \frac{k!(k-j)!}{(2k-j+1)!}, \tag{122}$$

$$C_{ij}^{(2)}(k) = \binom{k}{i}\binom{k}{j} \frac{(k-i)!k!}{(2k-i+1)!} \frac{k!j!}{(k+j+1)!}. \tag{123}$$

This is the source's coefficient pair after rearranging factorials. We have therefore proved the full formula

$$\begin{aligned}\lim_{N\to\infty} \frac{\mathbb{E}|\Lambda_N^{(k)}(1 - c/N)|^4}{N^{4k+4}} = \sum_{i,j=0}^{k} \big[& C_{ij}^{(1)}(k) I_{4k-i-j+2}(2c) I_{i+j}(2c) \\ &- C_{ij}^{(2)}(k) I_{3k-i-j+1}(2c) I_{k+i+j+1}(2c)\big].\end{aligned} \tag{124}$$

At $c = 0$, $I_r(0) = 1/(r+1)$ gives the source's double finite sum on the unit circle. Direct substitution of $k = 1$ reduces that sum to $61/10080$; this is a check of the coordinate and sign bookkeeping, not an additional asymptotic assumption.

The confluent Kostka identity. The exact finite-N theorem rests on the following identity. We include its proof so that the two coalescence operations are not black boxes.

Lemma 2.7 (Confluent determinant with prescribed derivatives)**.** *Let $f_1, \dots, f_s$ be analytic near z, and let $\mu = (\mu_1, \dots, \mu_s) \in \mathbb{N}_0^s$. With $\alpha_i(\lambda) = \lambda_i + s - i$,*

$$\mathcal{D}_{\mathbf{x}}^{\mu} \frac{\det(f_j(x_i))_{i,j=1}^s}{\Delta(\mathbf{x})}\Bigg|_{x_1=\cdots=x_s=z} = (-1)^{s(s-1)/2}\mu! \sum_{\substack{\lambda\vdash|\mu| \\ \ell(\lambda)\le s}} \frac{K_{\lambda\mu}}{\lambda!_{(s)}} \det(f_j^{(\alpha_i(\lambda))}(z))_{i,j=1}^s. \tag{125}$$

Proof. Write $x_i = z + y_i$. Taylor expansion of each row and the Cauchy–Binet formula give

$$\det(f_j(z+y_i))_{i,j=1}^s = \sum_{0\le n_1<\cdots<n_s} \frac{\det(f_j^{(n_i)}(z))_{i,j=1}^s}{n_1!\cdots n_s!} \det(y_i^{n_j})_{i,j=1}^s. \tag{126}$$

For completeness, (126) follows by factoring the infinite matrices $A_{i,n} = y_i^n/n!$ and $B_{n,j} = f_j^{(n)}(z)$ and summing the determinants of all s-column minors; analyticity gives absolute convergence in a sufficiently small polydisc. This is the identity recorded by Balantekin, with its earlier provenance to Hua [31, App. B].

Every strictly increasing sequence has a unique partition coordinate

$$n_i = \lambda_{s+1-i} + i - 1. \tag{127}$$

Reversing the derivative rows gives

$$\det(f_j^{(n_i)}(z)) = (-1)^{s(s-1)/2} \det(f_j^{(\alpha_i(\lambda))}(z)), \tag{128}$$

while the alternant formula for the Schur polynomial gives

$$\det(y_i^{n_j}) = \Delta(\mathbf{y}) s_\lambda(\mathbf{y}), \qquad n_1!\cdots n_s! = \lambda!_{(s)}. \tag{129}$$

After division by $\Delta(\mathbf{y}) = \Delta(\mathbf{x})$, use

$$s_\lambda(\mathbf{y}) = \sum_{\eta\vdash|\lambda|} K_{\lambda\eta} m_\eta(\mathbf{y}), \qquad \mathcal{D}_{\mathbf{y}}^{\mu} s_\lambda(\mathbf{y})\big|_{\mathbf{y}=0} = \mu! K_{\lambda\mu}. \tag{130}$$

Only $|\lambda| = |\mu|$ survives. Substitution proves (125); the symmetric-function identities used here are in Macdonald [33, Ch. I]. □

Theorem 2.8 (Exact finite-N derivative moment)**.** *Let μ, ν both have length s, set $w = |z|^2$, and set $K_{N,s}(w) = \sum_{m=0}^{N+s-1} w^m$. Then*

$$M_{\mu,\nu}(z,N) = \mu!\nu! \sum_{\substack{\lambda\vdash|\mu|,\ \ell(\lambda)\le s \\ \rho\vdash|\nu|,\ \ell(\rho)\le s}} \frac{K_{\lambda\mu}K_{\rho\nu}}{\lambda!_{(s)}\rho!_{(s)}} z^{|\rho|-|\lambda|} \times \det\left(\left(w^{\alpha_i(\lambda)} K_{N,s}^{(\alpha_i(\lambda))}(w)\right)^{(\beta_j(\rho))}\right)_{i,j=1}^s. \tag{131}$$

The right side has no singularity at $z = 0$.

Proof. Apply theorem 2.7 to the $\mathbf{z}$ block of (85), with $f_j(u) = K_{N,s}(uw_j)$. Since $f_j^{(r)}(u) = w_j^r K_{N,s}^{(r)}(uw_j)$, this gives

$$\mathcal{D}^{\mu}_{\mathbf{z}} \frac{\det(K_{N,s}(z_i w_j))}{\Delta(\mathbf{z})}\bigg|_{\mathbf{z}=z}$$
$$= (-1)^{s(s-1)/2}\mu! \sum_{\lambda\vdash|\mu|} \frac{K_{\lambda\mu}}{\lambda!_{(s)}} \det\Big(w_j^{\alpha_i(\lambda)} K_{N,s}^{(\alpha_i(\lambda))}(zw_j)\Big)_{i,j=1}^{s}. \tag{132}$$

Terms with more than s parts are understood to be absent. Divide by $\Delta(\mathbf{w})$ and apply the same lemma to

$$g_i(v) = v^{\alpha_i(\lambda)} K_{N,s}^{(\alpha_i(\lambda))}(zv). \tag{133}$$

The two confluent signs cancel. If $H_i(u) = u^{\alpha_i(\lambda)} K_{N,s}^{(\alpha_i(\lambda))}(u)$, the chain rule gives

$$g_i^{(\beta_j(\rho))}(\overline{z}) = z^{\beta_j(\rho)-\alpha_i(\lambda)} H_i^{(\beta_j(\rho))}(|z|^2). \tag{134}$$

Factoring the powers of z from the rows and columns gives $z^{\sum_j \beta_j - \sum_i \alpha_i} = z^{|\rho|-|\lambda|}$, proving (131).

The apparently negative powers in (134) do not define the value at zero. The unfactored coordinate is the polynomial

$$g_i^{(\beta)}(\overline{z}) = \sum_{m=\max(\alpha_i,\beta)}^{N+s-1} (m)_{\alpha_i}(m)_\beta z^{m-\alpha_i}\overline{z}^{m-\beta}, \tag{135}$$

where $(m)_r = m!/(m-r)!$. Thus the determinant, and hence the factored expression by continuation, is regular at $z = 0$. □

For $\mu = \nu = (1^s)$, $K_{\lambda,(1^s)}$ is the number f_λ of standard tableaux, $|\lambda| = |\rho| = s$, and the explicit power of z is one. Therefore theorem 2.8 specializes term by term to the exact Simm–Wei first-derivative formula [32, Thm. 1.5]; this is a proved specialization, not an analogy.

Equal-length microscopic limit and strict positivity. Let $z_N = 1 - c/N$, $w_N = |z_N|^2$, and $\tau = c + \overline{c}$. For fixed nonnegative integers a, b,

$$\Big(u^a K_{N,s}^{(a)}(u)\Big)^{(b)}\bigg|_{u=w_N} \sim N^{a+b+1} I_{a+b}(\tau), \tag{136}$$

uniformly for c in compact subsets of $\mathbb{C}$. To prove it, use the exact integral

$$K_{N,s}(u) = (N+s)\int_0^1 (1 - x(1-u))^{N+s-1}dx. \tag{137}$$

After a derivatives,

$$u^a K_{N,s}^{(a)}(u) = \frac{(N+s)!}{(N+s-a-1)!} u^a \int_0^1 x^a (1 - x(1-u))^{N+s-a-1}dx. \tag{138}$$

In the subsequent b derivatives, a term with q derivatives falling on u^a is of order $N^{a+b-q+1}$. The unique leading term has $q = 0$. Since $w_N = 1-\tau/N+O(N^{-2})$, dominated convergence on $[0, 1]$ gives $(1 - x(1 - w_N))^{N+O(1)} \to e^{-\tau x}$ uniformly on compact c-sets. This proves (136); it also reconstructs the estimate used from Simm–Wei [32, eqs. (5.22)–(5.25)].

For the (i,j) determinant entry of (131), take $a = \alpha_i(\lambda)$ and $b = \beta_j(\rho)$. Row and column factorization gives the exact total exponent

$$\begin{aligned}\sum_i \alpha_i(\lambda) + \sum_j \beta_j(\rho) + s &= |\lambda| + |\rho| + s(s-1) + s \\ &= |\mu| + |\nu| + s^2. \end{aligned} \tag{139}$$

Since $z_N^{|\rho|-|\lambda|} \to 1$, substitution in the exact finite-N formula proves (70). Setting $c = 0$ gives (71), because $I_r(0) = 1/(r+1)$.

Finally, Andréief's identity [32, Sec. 5.1, lemma labelled "Andréief identity"] transforms the determinant sum into

$$\frac{\mu!\nu!}{s!} \int_{[0,1]^s} \left(\prod_{q=1}^{s} e^{-\tau x_q} dx_q \right) \left(\sum_{\substack{\lambda \vdash |\mu| \\ \ell(\lambda) \le s}} \frac{K_{\lambda\mu}}{\lambda!_{(s)}} s_\lambda(\mathbf{x}) \right) \left(\sum_{\substack{\rho \vdash |\nu| \\ \ell(\rho) \le s}} \frac{K_{\rho\nu}}{\rho!_{(s)}} s_\rho(\mathbf{x}) \right) \Delta(\mathbf{x})^2. \tag{140}$$

Here $\tau = 2\Re c$ is real, so the exponential weight is strictly positive. Kostka numbers are nonnegative and Schur polynomials are positive on $(0,1)^s$. The squared Vandermonde is positive away from the collision diagonals, a set of full measure. Hence the integral is strictly positive. This establishes that the displayed power of N is the actual leading order rather than only an upper bound.

Shifted zeta moments: the Dirichlet-series and Euler-product proof. We now reconstruct the number-theoretic argument and keep its conditional and unconditional parts separate. Throughout this paragraph $K, L \ge 1$. For small shift vectors $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$, define

$$\sigma_{\boldsymbol{\alpha}}(m) = \sum_{n_1 \cdots n_K = m} n_1^{-\alpha_1} \cdots n_K^{-\alpha_K}, \tag{141}$$

$$A_\mu(m) = (-1)^{|\mu|} \, \partial_{\boldsymbol{\alpha}}^{\mu} \sigma_{\boldsymbol{\alpha}}(m)\big|_{\boldsymbol{\alpha}=0} = \sum_{n_1 \cdots n_K = m} \prod_{i=1}^{K} (\log n_i)^{\mu_i}. \tag{142}$$

Absolute convergence for $\Re s > 1$ gives, by termwise shift differentiation,

$$\prod_{i=1}^{K} \zeta^{(\mu_i)}(s) = (-1)^{|\mu|} \sum_{m \ge 1} \frac{A_\mu(m)}{m^s}. \tag{143}$$

The elementary bound

$$|A_\mu(m)| \le (\log m)^{|\mu|} d_K(m) \tag{144}$$

will control every smoothing and limiting step.

Lemma 2.9 (Long mean at fixed σ)**.** *Assume the Lindelöf hypothesis. For every fixed $\sigma > 1/2$,*

$$\begin{aligned}\lim_{T \to \infty} \frac{1}{T} \int_1^T \prod_{i=1}^{K} \zeta^{(\mu_i)}(\sigma + it) \prod_{j=1}^{L} \overline{\zeta^{(\nu_j)}(\sigma + it)} \, dt \\ = (-1)^{|\mu|+|\nu|} \sum_{m \ge 1} \frac{A_\mu(m) A_\nu(m)}{m^{2\sigma}}. \end{aligned} \tag{145}$$

Source boundary and checked algebraic reduction. Under Lindelöf, Cauchy's integral formula on a circle contained in $\Re s > 1/2$ gives, for every fixed derivative order and every $\varepsilon > 0$,

$$\zeta^{(r)}(\sigma + it) = O_{r,\sigma,\varepsilon}((1 + |t|)^{\varepsilon}). \tag{146}$$

Grover–Mezzadri–Simm state that this is the derivative-list modification of Titchmarsh's proof of Theorem 7.9; Simm–Wei write out the repeated-first-derivative case [32, Sec. 3, especially the proof following their eq. (3.20)]. The exact coefficient substitution can be checked without importing any estimate. If $\delta > 0$ and $\eta > \max\{0, 1 - \Re z\}$, Mellin inversion and absolute convergence on $\Re w = \Re z + \eta > 1$ give

$$(-1)^{|\mu|} \sum_{m \geq 1} \frac{A_{\mu}(m)}{m^{z}} e^{-\delta m} = \frac{1}{2\pi i} \int_{\Re w = \Re z + \eta} \Gamma(w - z) \prod_{i=1}^{K} \zeta^{(\mu_i)}(w)\, \delta^{z-w}\, dw. \tag{147}$$

Here the residue at the pole $w = z$ of $\Gamma(w - z)$ is the unsmoothed zeta product. A valid contour displacement goes to $\Re w = \Re z - \eta'$ with $0 < \eta' < \min\{1, \Re z - 1/2\}$; it cannot go "to the line through z". The displacement must retain the $w = z$ residue and, when it lies between the two lines, the pole at $w = 1$ of the zeta factors. Gamma decay controls the latter for large $|\Im z|$, but the complete uniform estimates needed to average the remainder, let $\delta \downarrow 0$, and retain only the diagonal $m = n$ are precisely the Titchmarsh input [34, Chs. VII and XIII]. Since that book has not yet been content-read in the authorized corpus, this paragraph verifies the coordinate substitution and correct contour geometry but does not claim an independent replacement for the cited mean-value proof. Equation (145) remains a human-sourced published theorem. $\square$

Lemma 2.10 (Exact Euler factorization)**.** *For shifts in a sufficiently small neighbourhood of zero and $\sigma > 1/2$,*

$$\sum_{m \geq 1} \frac{\sigma_{\boldsymbol{\alpha}}(m)\sigma_{\boldsymbol{\beta}}(m)}{m^{2\sigma}} = H_{K,L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma) \prod_{i=1}^{K} \prod_{j=1}^{L} \zeta(2\sigma + \alpha_i + \beta_j), \tag{148}$$

where $H_{K,L}$ is analytic near $(\mathbf{0}, \mathbf{0}, \frac{1}{2})$ and

$$\begin{aligned} H_{K,L}(\mathbf{0}, \mathbf{0}, \tfrac{1}{2}) &= a_{\mu,\nu} \\ &= \prod_{p} (1 - p^{-1})^{KL} \sum_{r \geq 0} \binom{K + r - 1}{r} \binom{L + r - 1}{r} p^{-r}. \end{aligned} \tag{149}$$

Only the lengths K, L, not the derivative entries, enter this arithmetic factor.

Proof. Both coefficient functions are multiplicative. At a prime p, writing $a_1 + \cdots + a_K = r = b_1 + \cdots + b_L$ gives the local factor

$$S_p = 1 + \sum_{r \geq 1} p^{-2r\sigma} \sum_{\substack{a_1 + \cdots + a_K = r \\ b_1 + \cdots + b_L = r}} p^{-\sum_i a_i \alpha_i - \sum_j b_j \beta_j}. \tag{150}$$

The inverse of the proposed polar product has local expansion

$$\prod_{i,j} (1 - p^{-2\sigma - \alpha_i - \beta_j}) = 1 - \sum_{i,j} p^{-2\sigma - \alpha_i - \beta_j} + O(p^{-4\sigma + O(\|\boldsymbol{\alpha}\| + \|\boldsymbol{\beta}\|)}). \tag{151}$$

The $r = 1$ term in (150) is exactly the positive double sum in (151); it cancels. Therefore the product of the two local factors is $1 + O(p^{-4\sigma+o(1)})$, uniformly in a small shift polydisc. The prime product converges normally whenever the polydisc is chosen inside the region $4\sigma - o(1) > 1$, proving analyticity near $\sigma = 1/2$.

At zero shifts, the number of weak K-compositions of r is $d_K(p^r) = \binom{K+r-1}{r}$, and similarly for L. Setting $\sigma = 1/2$ in the normally convergent residual product yields (149). □

Differentiate (148) by $\partial^\mu_{\boldsymbol{\alpha}} \partial^\nu_{\boldsymbol{\beta}}$ at zero. Because $H_{K,L}$ and all of its derivatives are bounded near $\sigma = 1/2$, the largest pole is the term in which every derivative lands on the KL zeta factors. If even one derivative lands on $H_{K,L}$, the total pole order decreases by at least one. Taylor expansion at the shift origin gives the exact leading term

$$\partial^\mu_{\boldsymbol{\alpha}} \partial^\nu_{\boldsymbol{\beta}} \prod_{i,j} \zeta(2\sigma + \alpha_i + \beta_j)\Bigg|_{\boldsymbol{\alpha}=\boldsymbol{\beta}=0} = \mu!\nu! \sum_{\substack{Q \in \mathcal{M}_{\mu,\cdot} \\ R \in \mathcal{M}_{\cdot,\nu}}} \prod_{i,j} \frac{\zeta^{(Q_{ij}+R_{ij})}(2\sigma)}{Q_{ij}! R_{ij}!}. \tag{152}$$

Since

$$\zeta^{(r)}(2\sigma) \sim \frac{(-1)^r r!}{(2\sigma - 1)^{r+1}}, \tag{153}$$

the total denominator exponent in every summand is $KL + |\mu| + |\nu|$, and

$$\frac{(Q_{ij} + R_{ij})!}{Q_{ij}! R_{ij}!} = \binom{Q_{ij} + R_{ij}}{Q_{ij}}. \tag{154}$$

Combining theorems 2.9 and 2.10 therefore proves (74), with exactly the coefficient $h_{\mu,\nu}$ displayed in the preceding sourced statement.

The unconditional length-at-most-two cases. For fixed $\sigma > 1/2$ and sufficiently small shifts, define

$$\mathcal{G}_{K,L}(\boldsymbol{\alpha}, \boldsymbol{\beta}; \sigma) = \begin{cases} \zeta(2\sigma + \alpha_1 + \beta_1), & (K, L) = (1, 1), \\ \prod_{j=1}^2 \zeta(2\sigma + \alpha_1 + \beta_j), & (K, L) = (1, 2), \\ \prod_{i=1}^2 \zeta(2\sigma + \alpha_i + \beta_1), & (K, L) = (2, 1), \\ \dfrac{\prod_{i,j=1}^2 \zeta(2\sigma + \alpha_i + \beta_j)}{\zeta(4\sigma + \alpha_1 + \alpha_2 + \beta_1 + \beta_2)}, & (K, L) = (2, 2). \end{cases} \tag{155}$$

For $(1, 1)$, Titchmarsh's mean-value theorem for Dirichlet series pairs equal indices. For $(1, 2)$, the coefficient of n in the second product is the sum over $m_1 m_2 = n$; the diagonal pairing then separates into two factors, and $(2, 1)$ follows by conjugation. This is the algebraic reduction in the published proof. When $1/2 < \sigma \le 1$, the displayed Dirichlet series are not pointwise absolutely convergent, so the pairing must be justified by the L^2 limiting statement in Titchmarsh's Theorem 7.1 rather than by a formal multiplication of infinite series. With that imported mean-value theorem, the first three cases give

$$\lim_{T \to \infty} \frac{1}{T} \int_1^T \prod_i \zeta^{(\mu_i)}(\sigma + it) \prod_j \overline{\zeta^{(\nu_j)}(\sigma + it)}\, dt = \partial^\mu_{\boldsymbol{\alpha}} \partial^\nu_{\boldsymbol{\beta}} \mathcal{G}_{K,L}\big|_0 \tag{156}$$

for the first three cases without a hypothesis.

For $(2,2)$ and $s=\sigma+it$ with $0<\sigma<1$, the approximate functional equation

$$\zeta(s)=\sum_{n\le x} n^{-s}+\chi(s)\sum_{n\le y} n^{s-1}+O\Big(x^{-\sigma}+|t|^{1/2-\sigma}y^{\sigma-1}\Big),\qquad x,y\ge 1,\qquad xy=\frac{t}{2\pi},\tag{157}$$

is differentiated by Cauchy's formula on a fixed circle contained in $0<\Re s<1$. With $x=y=\sqrt{t/(2\pi)}$ this choice is valid for $t\ge 2\pi$; the bounded initial interval makes no contribution after division by T. The derivative remainder is the displayed error times a fixed power of $\log t$. The case $\sigma\ge 1$ is handled separately by the convergent Dirichlet-series mean-value argument. On averaging the product of four main sums for $1/2<\sigma<1$, the diagonal $n_1n_2=n_3n_4$ survives. Its Euler product is exactly

$$\begin{aligned}\sum_{n_1n_2=n_3n_4} & n_1^{-\sigma-\alpha_1}n_2^{-\sigma-\alpha_2}n_3^{-\sigma-\beta_1}n_4^{-\sigma-\beta_2}\\ &=\frac{\prod_{i,j=1}^{2}\zeta(2\sigma+\alpha_i+\beta_j)}{\zeta(4\sigma+\alpha_1+\alpha_2+\beta_1+\beta_2)}.\end{aligned}\tag{158}$$

At each prime this is the finite rational identity obtained by summing the four geometric exponents subject to $a_1+a_2=b_1+b_2$. The differentiated off-diagonal and AFE remainders are $O_{\varepsilon,\mu,\nu}(T^{1-2\sigma+\varepsilon}(\log T)^C)$ after division by T. Choosing $0<\varepsilon<2\sigma-1$ makes it vanish. This is the exact analytic estimate imported by the paper from Titchmarsh's Theorem 7.5 and reproduced for the first-derivative case by Simm–Wei [32, Sec. 3, Lemma 3.1]. It proves the last case of (156).

As $\sigma\downarrow 1/2$, the denominator in the $(2,2)$ case tends to $\zeta(2)$; hence its reciprocal is $6/\pi^2$. Derivatives falling on this regular denominator are lower order. For $\mu=\nu=(k,k)$, the numerator calculation is exactly the 2×2 contingency-table calculation in theorem 2.6. This proves the unconditional fourth-moment formula (77) and its constant (78).

Proof obligation 2.11 (Classical off-diagonal source boundary)**.** The authorized indexed corpus did not return a local copy of Titchmarsh's book during this tranche. The audit above content-reads the pinned Grover–Mezzadri–Simm and Simm–Wei ranges, but both sources import classical estimates. It therefore does not falsely claim a fresh line-by-line derivation of Titchmarsh's Theorems 7.1, 7.5, and 7.9, or of the Chapter XIII mean-value equivalence invoked by Grover–Mezzadri–Simm. A later content-level reading of an authorized copy must bind the exact hypotheses and pages of all four inputs to this obligation. Until then, the fixed-shift results remain correctly attributed published theorems with explicit imported analytic lemmas, not independently certified theorems of this corpus.

For the real parameter $\sigma>1/2$, the substitution $\sigma=1/2+c/\log T$ requires fixed real $c>0$. It changes the order of limits: the fixed-σ long mean used above is no longer uniform by anything proved here. Therefore the proposed transfer of (70) to that shift remains a conjecture. None of the CUE or shifted-mean formulas above supplies a map to individual zeta zeros.

Exact crosswalk to the already derived CUE coordinates. For a fixed $A\in\mathrm{U}(N)$ with eigenvalues $z_1,\dots,z_N$, define

$$\Phi_N(s)=\det(sI-A)=\prod_{j=1}^{N}(s-z_j),\qquad \chi_A=(-1)^N\det(A)^{-1}.\tag{159}$$

Because $z_j^{-1} = \overline{z_j}$, the GMS/CRS polynomial is not merely analogous to this monic polynomial: the exact phase morphism is

$$\Lambda_A(s) = \det(I - sA^*) = \chi_A \Phi_N(s). \tag{160}$$

It is invertible for fixed A, with inverse multiplication by χ_A^{-1}, and it loses no derivative data. If

$$V_N = \Phi_N(1), \qquad F_N = \frac{1}{N}\Phi_N'(1), \tag{161}$$

then

$$\Lambda_A(1) = \chi_A V_N, \qquad \Lambda_A'(1) = \chi_A N F_N. \tag{162}$$

Consequently, for $\mu = \nu = (1^s)$,

$$M_{(1^s),(1^s)}(1, N) = \mathbb{E}|\Lambda_A'(1)|^{2s} = N^{2s}\mathbb{E}|F_N|^{2s}. \tag{163}$$

Thus the exact finite-N GMS theorem specializes to both the Simm–Wei determinant and the raw CRS first-derivative moment already used later in this manuscript.

The completed derivative is a coordinate obtained by the explicitly proved triangular first-jet map later in this file tree. After choosing the eigenangle lifts, the corresponding phase ρ_A and square-root branch exactly as in (325),

$$\mathcal{Z}_A'(1) = \rho_A N \left(F_N - \frac{1}{2} V_N \right). \tag{164}$$

Equation (164) states the morphism, rather than declaring the raw and completed derivatives categorically different. Its inverse requires both the completed derivative and the value coordinate. For derivative orders above one, the first-jet map is insufficient: the full Leibniz-triangular map among $(\Phi_N^{(r)}(1))_{0 \le r \le m}$ and the corresponding completed jet must be written and proved before higher GMS moments can be transported to completed coordinates. No such higher-jet transport, and no endpoint statement about individual zeros, is claimed here.

2.6.2 Complete source-level audit of Simm–Wei

This subsection records a complete content-level reading of Simm and Wei, *On moments of the derivative of CUE characteristic polynomials and the Riemann zeta function*, arXiv:2409.03687v2 [32]. The source TeX, its compiled bibliography, and the arXiv source archive were read as distinct pinned objects. We distinguish results proved in Simm–Wei, imported from an identified source, independently checked here, and not proved in this version; the last status is strictly source-relative. An exact bounded certificate, `CERT-SIMM-WEI-20260828-0001`, independently replays the finite polynomial checks stated below. It is not used as a replacement for any analytic argument.

Objects, coordinates, and the three limits. Let $U \in U(N)$ have normalized Haar law and put

$$\Lambda_N(z) = \det(I - zU^*), \qquad G_N(z) = \log \Lambda_N(z), \qquad |z| < 1. \tag{165}$$

The logarithm is the analytic branch fixed by $G_N(0) = 0$. If $e^{\mathrm{i}\theta_1}, \ldots, e^{\mathrm{i}\theta_N}$ are the eigenvalues of U, then

$$G_N(z) = \sum_{j=1}^{N} \log(1 - z e^{-\mathrm{i}\theta_j}) = -\sum_{k \ge 1} \frac{\mathrm{Tr}(U^{-k})}{k} z^k, \tag{166}$$

$$G_N'(z) = -\sum_{j=1}^{N} \frac{e^{-\mathrm{i}\theta_j}}{1 - z e^{-\mathrm{i}\theta_j}} = -\sum_{k \ge 1} \mathrm{Tr}(U^{-k}) z^{k-1}. \tag{167}$$

The first equality in (167) corrects the source's summation-letter typo: the summand is indexed by j, not by k. Haar rotation shows that $\mathbb{E}|\Lambda_N'(z)|^{2s}$ depends on z only through $u = |z|^2$. The source keeps three different limiting problems:

$$\begin{array}{c|c|c} \text{regime} & u & \text{distance } 1-u \\ \hline \text{global} & u \in [0,1) \text{ fixed} & 1-u \\ \text{mesoscopic} & 1-N^{-\alpha},\ 0<\alpha<1 & N^{-\alpha} \\ \text{microscopic} & 1-c/N,\ c \in \mathbb{R} \text{ fixed} & c/N. \end{array} \tag{168}$$

No one of these coordinate prescriptions is silently substituted for another.

The Gaussian field and the global moment. Let $(\mathcal{N}_k)_{k\geq 1}$ be independent standard complex Gaussian variables, so their real and imaginary parts have variance $1/2$, and define on the open disc

$$G(z) = \sum_{k\geq 1} \frac{\mathcal{N}_k}{\sqrt{k}} z^k, \qquad G'(z) = \sum_{k\geq 1} \sqrt{k}\,\mathcal{N}_k z^{k-1}, \qquad \Lambda(z) = e^{G(z)}. \tag{169}$$

The trace central limit theorem and strong Szegő machinery give, for fixed points in the disc [60, 61, 62],

$$(G_N(z_1), G_N'(z_2)) \Longrightarrow (G(z_1), G'(z_2)). \tag{170}$$

Direct summation of the power-series covariances gives, without a change of coordinates,

$$\Gamma_{11} = -\log(1-|z_1|^2), \qquad \Gamma_{12} = \frac{z_1}{1-z_1\overline{z_2}}, \qquad \Gamma_{22} = \frac{1}{(1-|z_2|^2)^2}. \tag{171}$$

Here $\Gamma_{ab} = \mathbb{E}(Q_a\overline{Q_b})$ for $Q = (G(z_1), G'(z_2))$, and the relation matrix $\mathbb{E}(Q_aQ_b)$ is zero.

Theorem 2.12 (Simm–Wei global joint moment)**.** *Fix $|z_1|, |z_2| < 1$. For $s, h \in \mathbb{C}$ with $\operatorname{Re} h > -1$, interpret positive-real complex powers using the real logarithm. Then*

$$\lim_{N\to\infty} \mathbb{E}\left[\left|\frac{\Lambda_N'(z_2)}{\Lambda_N(z_2)}\right|^{2h} |\Lambda_N(z_1)|^{2s}\right] = \frac{e^{-s^2\rho}\Gamma(h+1)\,{}_1F_1(h+1,1;s^2\rho)}{(1-|z_2|^2)^{2h}(1-|z_1|^2)^{s^2}}, \tag{172}$$

where

$$\rho = \frac{|z_1|^2(1-|z_2|^2)^2}{|1-z_1\overline{z_2}|^2}. \tag{173}$$

In particular, for fixed $|z|<1$ and $\operatorname{Re} s > -1$,

$$\lim_{N\to\infty} \mathbb{E}|\Lambda_N'(z)|^{2s} = \frac{e^{-s^2|z|^2}\Gamma(s+1)\,{}_1F_1(s+1,1;s^2|z|^2)}{(1-|z|^2)^{s^2+2s}}. \tag{174}$$

For $s \in \mathbb{N}$ this is

$$\frac{s!L_s(-s^2|z|^2)}{(1-|z|^2)^{s^2+2s}}. \tag{175}$$

Gaussian calculation and analytic dependency chain. Put $D = \det\Gamma$. The density of $Q = (w_1, w_2)$ is

$$\frac{1}{\pi^2 D}\exp\left[-\frac{\Gamma_{22}|w_1|^2 - \Gamma_{12}\overline{w_1}w_2 - \overline{\Gamma_{12}}w_1\overline{w_2} + \Gamma_{11}|w_2|^2}{D}\right]. \tag{176}$$

For real s and $\operatorname{Re} h > -1$, complete the square in w_1 in the integral of $|w_2|^{2h} e^{s(w_1+\overline{w_1})}$. After scaling $w_2 \mapsto \sqrt{\Gamma_{22}}w_2$ one obtains

$$\Gamma_{22}^h e^{s^2 D/\Gamma_{22}} \frac{1}{\pi} \int_{\mathbb{C}} |w|^{2h} e^{-|w|^2 + s\xi w + s\overline{\xi w}}\, d^2 w, \qquad \xi = \frac{\Gamma_{12}}{\sqrt{\Gamma_{22}}}. \tag{177}$$

Angular integration kills unequal holomorphic and antiholomorphic powers. Radial integration then gives

$$\begin{aligned} \frac{1}{\pi} \int_{\mathbb{C}} |w|^{2h} e^{-|w|^2 + s\xi w + s\overline{\xi w}}\, d^2 w &= \sum_{k \geq 0} \frac{\Gamma(h+k+1)}{(k!)^2} s^{2k} |\xi|^{2k} \\ &= \Gamma(h+1) {}_1F_1(h+1, 1; s^2 |\xi|^2). \end{aligned} \tag{178}$$

The covariance entries imply

$$|\xi|^2 = \rho, \qquad \Gamma_{22}^h e^{s^2 D/\Gamma_{22}} = (1-|z_2|^2)^{-2h} (1-|z_1|^2)^{-s^2} e^{-s^2 \rho}, \tag{179}$$

which proves the formula for real s; both sides are entire in s, giving the stated extension. Kummer's transformation yields $e^{-x} {}_1F_1(s+1, 1; x) = L_s(-x)$ for integer s.

It remains to justify passage from convergence in law to expectations. For nonnegative $\operatorname{Re} h$, Cauchy–Schwarz and a sufficiently high integer moment reduce the claim to bounded integer moments of Λ_N' and positive or negative moments of Λ_N; the latter are the interior-disc bounds of Forrester–Keating [63, Cor. 2]. For $-1 < \operatorname{Re} h < 0$, the negative-moment estimate proved next supplies the missing uniform integrability. Thus the Gaussian calculation, the trace limit, and the uniform-integrability theorem together prove (172); none alone suffices. □

For integer s, the numerator in (175) also follows by differentiating the exact Gaussian product

$$\mathbb{E} \prod_{i=1}^{s} \Lambda(z_i) \prod_{j=1}^{s} \overline{\Lambda(\overline{w_j})} = \prod_{i,j=1}^{s} (1 - z_i w_j)^{-1}. \tag{180}$$

After all z_i derivatives are merged, the product is $F(z, \mathbf{w}) H(z, \mathbf{w})$ with

$$F(z, \mathbf{w}) = \prod_{j=1}^{s} (1 - z w_j)^{-s}, \qquad H(z, \mathbf{w}) = \left(\sum_{j=1}^{s} \frac{w_j}{1 - z w_j} \right)^{s}. \tag{181}$$

If exactly k of the w derivatives strike F, the merged contribution is

$$\binom{s}{k}^2 (s-k)! (s|z|)^{2k} (1 - |z|^2)^{-s^2 - 2s}, \tag{182}$$

and summing $0 \leq k \leq s$ gives $s! L_s(-s^2 |z|^2)$.

Negative moments: small balls in two real coordinates. Set

$$f_r(\theta) = -\frac{e^{-\mathrm{i}\theta}}{1 - r e^{-\mathrm{i}\theta}}, \qquad X_N(f_r) = \sum_{j=1}^{N} f_r(\theta_j) = G_N'(r). \tag{183}$$

Theorem 2.13 (Simm–Wei negative-moment bound). *For fixed $0 \le r < 1$, $N > 4$, and $0 \le a < 2$, a constant $C_{a,r}$ independent of N satisfies*

$$\mathbb{E}|G_N'(r)|^{-a} \le C_{a,r}, \qquad \mathbb{E}|\Lambda_N'(r)|^{-a} \le C_{a,r}. \tag{184}$$

Proof with the uniform stationary-point calculation made explicit. Integration by parts reduces the first estimate to the two-dimensional small-ball bound

$$\mathbb{P}(|X_N(f_r)| < y) \le C_r y^2, \qquad 0 < y < 1, \tag{185}$$

because $\int_0^1 y^{1-a} dy < \infty$ exactly for $a < 2$. Halász's concentration inequality [64, Sec. 5] gives

$$\mathbb{P}(|X_N(f_r)| \le y) \le y^2 \int_{|\xi_1|<y^{-1}} \int_{|\xi_2|<y^{-1}} |\psi_N(\xi_1, \xi_2)|\, d\xi_1\, d\xi_2, \tag{186}$$

where $\xi = |\xi_1| + |\xi_2|$, $\eta_j = \xi_j/\xi$, and

$$\psi_N = \mathbb{E}e^{\mathrm{i}\xi X_N(g)}, \qquad g(\theta) = \frac{\eta_1(r - \cos\theta) + \eta_2 \sin\theta}{1 + r^2 - 2r\cos\theta}. \tag{187}$$

The characteristic function is the Toeplitz determinant of $h(\theta) = e^{\mathrm{i}\xi g(\theta)}$. Hadamard's inequality and stationary phase give, for $\xi > N^8$,

$$|\psi_N(\xi_1, \xi_2)| \le C_r^N N^{-N/2} \xi^{-N/4}. \tag{188}$$

The uniformity needed here is visible in coordinates. If

$$A = \eta_2(1 + r^2), \quad B = \eta_1(1 - r^2), \quad C = 2r\eta_2, \tag{189}$$

then a zero of g' satisfies $A\cos\theta + B\sin\theta = C$, while the derivative of this numerator in the tangent direction has square

$$(-A\sin\theta + B\cos\theta)^2 = A^2 + B^2 - C^2 = (1 - r^2)^2(\eta_1^2 + \eta_2^2). \tag{190}$$

Because $|\eta_1| + |\eta_2| = 1$, the last parenthesis is at least $1/2$. The denominator in (187) lies between $(1 - r)^2$ and $(1 + r)^2$. Hence all zeros are uniformly nondegenerate. This also covers $\eta_2 = 0$ directly, and, by compactness and the implicit function theorem, covers the perturbed phases $\xi g - k\theta$ occurring in the Fourier coefficients when $|k| \le N$ and $\xi > N^8$. Thus the invoked stationary-phase estimate [65, Ch. 6] has the required uniform constants; checking only one fixed parameter value would not have done so.

For the low and intermediate ranges, Johansson's exponential estimate [62, proof of Prop. 2.8] applies to the smooth family (187). Its quantities are uniformly controlled because that family is compact in every C^m norm for fixed $r < 1$. In particular

$$b = \frac{1}{2\pi} \int_0^{2\pi} (g')^2 = \frac{(\eta_1^2 + \eta_2^2)(1 + r^2)}{2(1 - r)^3(1 + r)^3} \tag{191}$$

is uniformly positive, and one obtains

$$|\psi_N| \le e^{-c_r(\xi_1^2+\xi_2^2)} \quad (\xi < N), \qquad |\psi_N| \le e^{-c_r N^2} \quad (N \le \xi < N^8). \tag{192}$$

Integrating (188) and (192) over the three regions gives a constant independent of N in (186), proving (185). The finitely many sizes $5 \le N \le 8$ are absorbed by the same Fourier estimate with an N-dependent constant and then a maximum over this finite set.

Finally $\Lambda_N' = G_N' \Lambda_N$. Choose conjugate Hölder exponents $q, \ell > 1$ with $aq < 2$. Then

$$\mathbb{E}|\Lambda_N'(r)|^{-a} \le (\mathbb{E}|G_N'(r)|^{-aq})^{1/q} (\mathbb{E}|\Lambda_N(r)|^{-a\ell})^{1/\ell}. \tag{193}$$

The second factor is bounded by Forrester–Keating for large N [63, Cor. 2]. For the remaining finite sizes, $|\Lambda_N(r)| \ge (1 - r)^N$ pointwise. This completes both bounds. $\square$

Jensen's formula and derivative zeros. Let $n_N(r)$ count zeros of Λ_N' in $|z| < r$, with multiplicity. Jensen's formula, the preceding negative moments, and moment bounds for the smooth linear statistics G_N', G_N'', and G_N give uniform integrability. The required cumulant bounds are imported from Soshnikov [66]. Consequently, uniformly for r in a compact subset of $[0, 1)$,

$$\lim_{N\to\infty} \int_0^r \frac{\mathbb{E}n_N(t)}{t}, dt = -\log(1 - r^2), \qquad \lim_{N\to\infty} \mathbb{E}n_N(r) = \frac{2r^2}{1 - r^2}. \tag{194}$$

Indeed,

$$\int_0^r \frac{\mathbb{E}n_N(t)}{t}, dt = \mathbb{E}\log|\Lambda_N'(r)| - \mathbb{E}\log|\Lambda_N'(0)|, \tag{195}$$

and differentiation gives

$$\mathbb{E}n_N(r) = r\,\mathbb{E}\,\mathrm{Re}\,\frac{\Lambda_N''(r)}{\Lambda_N'(r)}. \tag{196}$$

The convergence required for the last quotient is joint convergence of G_N', G_N'', and G_N, followed by the identity $\Lambda_N''/\Lambda_N' = G_N''/G_N' + G_N'$. The source sentence naming $r(d/dr)\log|G_N|$ is therefore read as a variable-name slip, not as the actual target.

Exact finite-N determinant. For a partition $\lambda \vdash s$, pad λ to length s, let f_λ be the number of standard Young tableaux of shape λ, and retain the source's nonstandard factorial

$$\lambda! := \prod_{i=1}^{s} (\lambda_i + s - i)!. \tag{197}$$

Put

$$p_i = \lambda_i + s - i, \qquad q_j = \mu_j + s - j, \qquad K_N(u) = \sum_{k=0}^{N+s-1} u^k. \tag{198}$$

Theorem 2.14 (Simm–Wei finite determinant)**.** *For $N, s \in \mathbb{N}$ and $z \in \mathbb{C}$,*

$$\mathbb{E}|\Lambda_N'(z)|^{2s} = \sum_{\lambda,\mu\vdash s} \frac{f_\lambda f_\mu}{\lambda!\mu!} \det\Big(\Big[u^{p_i} K_N^{(p_i)}(u)\Big]^{(q_j)}\Big)_{i,j=1}^{s}, \qquad u = |z|^2. \tag{199}$$

Proof. The Akemann–Vernizzi characteristic-polynomial identity gives

$$\mathbb{E}\prod_{i=1}^{s} \det(I - z_i U) det(I - w_i U^*) = \frac{\det(K_N(z_i w_j))_{i,j=1}^{s}}{\Delta(\mathbf{z})\Delta(\mathbf{w})} \tag{200}$$

[29, Thm. 1]. The numerator is alternating in each list. For an alternating polynomial $P(x_1, \ldots, x_s)$, the confluent derivative identity used by Simm–Wei is

$$\prod_{i=1}^{s} \partial_{x_i} \frac{P(\mathbf{x})}{\Delta(\mathbf{x})}\Bigg|_{\mathbf{x}=x} = \sum_{\lambda\vdash s} \frac{f_\lambda}{\prod_i (\lambda_i + s - i)!} \prod_i \partial_{x_i}^{\lambda_i+s-i} P(\mathbf{x})\Bigg|_{\mathbf{x}=x}, \tag{201}$$

with the harmless common sign determined by the chosen orientation of Δ; applying it once in each variable list cancels that sign. Its recursion deletes one corner of a Young diagram, which is exactly the standard-tableau recursion, hence its coefficient is f_λ.

Apply (201) to both lists in (200). Expanding the determinant by permutations allows the derivatives to be taken entrywise. At $z_i = w_j = r$, differentiation of $K_N(z_i w_j)$ gives a factor $r^{q_j - p_i}$

multiplying $[u^{p_i}K_N^{(p_i)}(u)]^{(q_j)}$. The determinant-wide factor is $r^{\sum_j q_j-\sum_i p_i} = 1$, since both sums equal $s(s+1)/2$. This proves (199) for $r \neq 0$, and polynomial continuation gives $r = 0$. The source's displayed induction has one denominator running to n where the preceding and following expressions require $n-1$; the corrected range is used here. $\square$

The independent certificate evaluates the right side of (199) and a separate Weyl-density Laurent constant term. Their exact common values are

N	s	$\mathbb{E}\lvert\Lambda_N'(z)\rvert^{2s}$
1	$1,2,3$	1
2	1	$1+4u$
2	2	$2(1+8u+8u^2)$
3	1	$1+4u+9u^2$
3	2	$2+32u+116u^2+144u^3+81u^4$.

(202)

These are bounded checks of the general identity, not a finite-case proof of it.

Mesoscopic limit: the discarded tail is exponentially small.

Theorem 2.15 (Simm–Wei mesoscopic derivative moment)**.** *For fixed $s \in \mathbb{N}$, $0<\alpha<1$, and $u_N = 1-N^{-\alpha}$,*

$$\mathbb{E}|\Lambda_N'(z_N)|^{2s} \sim N^{\alpha(s^2+2s)}s!L_s(-s^2), \qquad |z_N|^2 = u_N. \tag{203}$$

Proof. Write the finite kernel with its tail retained:

$$K_N(u) = \frac{1}{1-u} - \frac{u^{N+s}}{1-u}. \tag{204}$$

All derivative orders in (199) are bounded in terms of the fixed s. After any fixed number m of differentiations, the second term at u_N is bounded by a fixed power of N times

$$u_N^{N+s-m} \le \exp(-(N+s-m)N^{-\alpha}) = \exp(-N^{1-\alpha}+O(N^{-\alpha})). \tag{205}$$

The determinant has fixed size, so this error is negligible compared with every power of N. Thus the finite kernel may, in this regime and after this estimate, be replaced by $\widetilde{K}(u) = (1-u)^{-1}$. The resulting partition determinant is not newly evaluated: it is exactly the global integer identity (175). Therefore

$$\begin{aligned}\mathbb{E}|\Lambda_N'(z_N)|^{2s} &\sim (1-u_N)^{-s^2-2s}s!L_s(-s^2u_N)\\ &\sim N^{\alpha(s^2+2s)}s!L_s(-s^2),\end{aligned} \tag{206}$$

which proves the claim. This supplies the uniform estimate implicit in the source's short replacement of K_N by $\widetilde{K}$. $\square$

Microscopic limit and positivity. For $p \ge 0$ the finite kernel has the exact beta-integral derivative

$$u^pK_N^{(p)}(u) = \frac{(N+s)!}{(N+s-p-1)!}u^p\int_0^1 x^p(1-x(1-u))^{N+s-p-1}, dx. \tag{207}$$

This formula retains the factorial and the finite endpoint; neither is normalized away.

Theorem 2.16 (Simm–Wei microscopic derivative moment)**.** *Fix $c \in \mathbb{R}$, $s \in \mathbb{N}$, and set $u_N = 1 - c/N$ whenever this is nonnegative. Then*

$$\mathbb{E}|\Lambda_N'(z_N)|^{2s} \sim N^{s^2+2s} b_s(c), \tag{208}$$

where

$$b_s(c) = \sum_{\lambda,\mu \vdash s} \frac{f_\lambda f_\mu}{\lambda!\mu!} \det\left(\int_0^1 x^{p_i+q_j} e^{-cx}\, dx\right)_{i,j=1}^{s} > 0. \tag{209}$$

Equivalently, if

$$F_c(v,w) = \int_0^1 J_0(2\sqrt{vx}) J_0(2\sqrt{wx}) e^{-cx}\, dx, \tag{210}$$

then

$$b_s(c) = \partial_v^s \partial_w^s \det\left(\partial_v^{i-1}\partial_w^{j-1} F_c(v,w)\right)_{i,j=1}^{s}\Big|_{v=w=0}. \tag{211}$$

Proof. Differentiate (207) q times. The summand in which every derivative strikes the integral has the largest factorial power. Since p, q range over a fixed finite set and c is fixed, dominated convergence gives the entrywise expansion

$$\left[u^p K_N^{(p)}(u)\right]^{(q)}\Big|_{u=1-c/N} = N^{p+q+1}\left(\int_0^1 x^{p+q} e^{-cx}\, dx + O_{p,q,c}(N^{-1})\right). \tag{212}$$

Factor N^{p_i+1} from row i and N^{q_j} from column j. Since

$$\sum_i p_i = \sum_j q_j = s + \frac{s(s-1)}{2} = \frac{s(s+1)}{2}, \tag{213}$$

the total power is N^{s^2+2s}. Substitution in (199) yields (209).

For positivity, Andréief's identity rewrites each moment determinant as an integral of two alternants. Summing the two partition lists separately gives the square

$$b_s(c) = \frac{1}{s!}\int_{[0,1]^s} \left(\sum_{\lambda \vdash s} \frac{f_\lambda}{\lambda!} s_\lambda(\mathbf{x})\right)^2 e^{-c\sum_j x_j} \Delta(\mathbf{x})^2\, d\mathbf{x}. \tag{214}$$

Every factor is nonnegative for $\mathbf{x} \in [0,1]^s$, and the displayed polynomial is not identically zero, so the integral is strictly positive. This nonvanishing also licenses the asymptotic equivalence sign after the entrywise limit.

To obtain (211), use the hook-content relation between f_λ, $s_\lambda(1,\dots,1)$, and (197); introduce one degree-marking variable for the condition $|\lambda| = s$; and apply the Schur–Bessel determinant expansion

$$\sum_{\lambda:\, \ell(\lambda)\le s} \frac{s_\lambda(\mathbf{x}) s_\lambda(-\mathbf{v})}{(\lambda!)^2} = \frac{\det(J_0(2\sqrt{x_i v_j}))_{i,j=1}^s}{\Delta(\mathbf{x})\Delta(\mathbf{v})}. \tag{215}$$

This is the character expansion used by Simm–Wei, with provenance in Balantekin [31, App. B]. Applying Andréief again and then the confluent determinant identity

$$\frac{\det(B(v_i,w_j))}{\Delta(\mathbf{v})\Delta(\mathbf{w})}\Bigg|_{\mathbf{v}=v,\mathbf{w}=w} = \left(\prod_{k=0}^{s-1}(k!)^{-2}\right) \det(\partial_v^{i-1}\partial_w^{j-1} B(v,w)) \tag{216}$$

leaves exactly (211). □

The integral definition (210) is correct as printed. One subsequent displayed Christoffel–Darboux identity in v2 has both an undefined z and an extra factor 2. The correct $c = 0$ identity is

$$\int_0^1 J_0(2\sqrt{vx})J_0(2\sqrt{wx})\,dx = \frac{\sqrt{w}J_0(2\sqrt{v})J_0'(2\sqrt{w}) - \sqrt{v}J_0'(2\sqrt{v})J_0(2\sqrt{w})}{v-w}, \tag{217}$$

continued across $v = w$. It follows from $x = t^2$ and the standard Bessel product integral. At $w = 0$, termwise integration gives

$$\sum_{m\geq 0} \frac{(-v)^m}{(m!)^2(m+1)}, \tag{218}$$

whose constant term is 1; the printed right side has constant term 2. The certificate checks the first eight exact coefficients. Because the microscopic theorem uses (210), the defect in the later identification does not alter (208) or (211).

The zeta-side results: theorem, conditional theorem, and conjecture. For $\sigma > 1/2$, define, when the limit exists,

$$\mathcal{M}_s'(\sigma) = \lim_{T\to\infty} \frac{1}{T}\int_1^T |\zeta'(\sigma + \mathrm{i}t)|^{2s}\,dt. \tag{219}$$

Simm–Wei propose for complex s with $\operatorname{Re} s > 0$ the asymptotic

$$\mathcal{M}_s'(\sigma) \sim \frac{a_s h_s}{(2\sigma-1)^{s^2+2s}}, \qquad h_s = e^{-s^2}\Gamma(s+1)\,{}_1F_1(s+1,1;s^2), \tag{220}$$

as $\sigma \downarrow 1/2$, with the Keating–Snaith arithmetic factor

$$a_s = \prod_p (1-p^{-1})^{s^2} \sum_{m\geq 0} \left(\frac{\Gamma(s+m)}{\Gamma(m+1)\Gamma(s)}\right)^2 p^{-m}. \tag{221}$$

Equation (220) is a conjectural transfer from the global CUE boundary coefficient. It is not a map from individual CUE eigenvalues or derivative zeros to individual zeta zeros.

Theorem 2.17 (Unconditional fourth derivative moment in the fixed-σ mean)**.** *As $\sigma \downarrow 1/2$,*

$$\mathcal{M}_2'(\sigma) \sim \frac{6}{\pi^2}\frac{34}{(2\sigma-1)^8}. \tag{222}$$

Proof and independent pole extraction. Introduce four independent shifts. The fixed-σ mean-value calculation gives

$$\mathcal{M}_2'(\sigma) = \partial_{\alpha_1}\partial_{\alpha_2}\partial_{\beta_1}\partial_{\beta_2} \left.\frac{\prod_{i,j=1}^2 \zeta(2\sigma + \alpha_i + \beta_j)}{\zeta(4\sigma + \alpha_1 + \alpha_2 + \beta_1 + \beta_2)}\right|_{\boldsymbol{\alpha}=\boldsymbol{\beta}=0}. \tag{223}$$

The identity follows by the approximate functional equation, differentiation by Cauchy's formula, the diagonal condition $n_1 n_2 = n_3 n_4$, and Ramanujan's local Euler-factor identity; this is the argument of Simm–Wei [32, Sec. 3]. Their symmetric choice $x = y = \sqrt{t/(2\pi)}$ satisfies the stated $x, y \geq 1$ hypothesis only for $t \geq 2\pi$. Starting the argument there repairs the boundary; the omitted finite interval is $o(T)$ after division by T.

Put $\delta = 2\sigma - 1$. The leading singular part of the numerator of (223) is

$$P(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{i,j=1}^{2} (\delta + \alpha_i + \beta_j)^{-1}. \tag{224}$$

Exact differentiation gives

$$\delta^8 \partial_{\alpha_1} \partial_{\alpha_2} \partial_{\beta_1} \partial_{\beta_2} P\Big|_0 = 34. \tag{225}$$

The denominator in (223) tends to $\zeta(2) = \pi^2/6$; any derivative that strikes that regular factor has lower pole order. This proves (222). Equation (225) is also replayed symbolically by the bounded certificate. □

Let $\ell_s = \log * \cdots * \log$ be the s-fold Dirichlet convolution, so

$$(\zeta'(w))^s = (-1)^s \sum_{n \geq 1} \frac{\ell_s(n)}{n^w} \qquad (\operatorname{Re} w > 1). \tag{226}$$

Theorem 2.18 (Simm–Wei, conditional on Lindelöf)**.** *Assume the Lindelöf hypothesis. For every integer $s \geq 3$ and every $\sigma > 1/2$,*

$$\mathcal{M}'_s(\sigma) = \sum_{n \geq 1} \frac{\ell_s(n)^2}{n^{2\sigma}}, \tag{227}$$

and, as $\sigma \downarrow 1/2$,

$$\mathcal{M}'_s(\sigma) \sim a_s \frac{s! L_s(-s^2)}{(2\sigma - 1)^{s^2+2s}}. \tag{228}$$

Factorization after the mean-value input. The passage from Lindelöf to (227) is an adaptation of Titchmarsh's mean-value argument: its smoothed Mellin integral replaces $\zeta(w)^s$ by $(\zeta'(w))^s$, and Cauchy's formula transfers the Lindelöf bound to ζ' [34, Chs. 7 and 13]. The present audit has not independently replayed that identified human-source dependency from a local copy.

For the algebraic part, define

$$\sigma_{\boldsymbol{\alpha}}(m) = \sum_{n_1 \cdots n_s = m} n_1^{-\alpha_1} \cdots n_s^{-\alpha_s}. \tag{229}$$

Near the zero shift, absolute convergence permits termwise differentiation, and

$$\sum_{m \geq 1} \frac{\sigma_{\boldsymbol{\alpha}}(m) \sigma_{\boldsymbol{\beta}}(m)}{m^{2\sigma}} = H_\sigma(\boldsymbol{\alpha}, \boldsymbol{\beta}) \prod_{i,j=1}^{s} \zeta(2\sigma + \alpha_i + \beta_j). \tag{230}$$

Cancellation of every first-order prime term shows that H_σ is holomorphic in a fixed neighbourhood of the origin for σ near $1/2$; hence it and all derivatives needed here are locally bounded by the several-variable Cauchy formula. This is the precise boundedness used in the proof. No global uniform bound on the entire half-plane $\operatorname{Re} w > 1/2$ is required. On the diagonal shift,

$$H_\sigma(\delta_0, \ldots, \delta_0; \delta_0, \ldots, \delta_0) = \zeta(2\sigma + 2\delta_0)^{-s^2} \sum_{n \geq 1} \frac{d_s(n)^2}{n^{2\sigma + 2\delta_0}}. \tag{231}$$

The source's lower limit $n = 0$ in this display is a typographical error; Dirichlet series begin at 1. It follows that $H_\sigma(0, 0) \to a_s$.

In differentiating (230), any derivative falling on H_σ leaves at least one fewer derivative on the s^2 simple pole factors and therefore produces a strictly smaller pole order. The leading term is $H_\sigma(0,0)$ times all $2s$ derivatives of the pole product. Exact product-rule counting gives

$$\prod_i \partial_{\alpha_i} \prod_j \partial_{\beta_j} \prod_{i,j} (\delta + \alpha_i + \beta_j)^{-1} \Bigg|_0 = \frac{\sum_{k=0}^{s} \binom{s}{k}^2 (s-k)! s^{2k}}{\delta^{s^2+2s}} = \frac{s! L_s(-s^2)}{\delta^{s^2+2s}}. \tag{232}$$

Taking $\delta = 2\sigma - 1$ proves (228). $\square$

The logical boundary is now exact. The fourth moment is unconditional within the fixed-σ mean-value theorem used in (223); the integer moments $s \geq 3$ use Lindelöf through (227); and the noninteger statement (220) remains conjectural in this source. None of these statements supplies an individual-zero correspondence.

Alternative finite-N structure: exact status of the source proof. The notation in the introduction of v2 calls A a block matrix and then divides A by Vandermondes, although every later use requires a scalar determinant. The typed objects are therefore

$$\mathcal{A}(\mathbf{z}, \mathbf{w}) = \begin{pmatrix} V_{\mathbf{z}} & Y_{\mathbf{z}} \\ Y_{\mathbf{w}} & V_{\mathbf{w}} \end{pmatrix}, \qquad A(\mathbf{z}, \mathbf{w}) = \det \mathcal{A}(\mathbf{z}, \mathbf{w}), \qquad G(\mathbf{z}, \mathbf{w}) = \frac{A(\mathbf{z}, \mathbf{w})}{\Delta(\mathbf{z})\Delta(\mathbf{w})}, \tag{233}$$

where $(V_{\mathbf{z}})_{ij} = z_i^{j-1}$ and $(Y_{\mathbf{z}})_{ij} = z_i^{N+2s-j}$, and similarly for $\mathbf{w}$. Put also

$$F(\mathbf{z}, \mathbf{w}) = \prod_{i,j=1}^{s} (1 - z_i w_j)^{-1}. \tag{234}$$

The Conrey–Farmer–Keating–Rubinstein–Snaith Schur determinant gives the exact factorization of the characteristic-polynomial correlation as $F(-\mathbf{w}, -\mathbf{z})G(-\mathbf{w}, -\mathbf{z})$ [67].

For $0 \leq h_1 \leq h_2 \leq s$, Simm–Wei compute the derivatives of F explicitly as

$$\begin{aligned} &\binom{s}{h_1}\binom{s}{h_2} \prod_{j=1}^{s-h_2} \partial_{w_j} \prod_{i=1}^{s-h_1} \partial_{z_i} F(\mathbf{z}, \mathbf{w}) \Bigg|_{\mathbf{z}=\mathbf{w}=-r} \\ &= \frac{(-sr)^{h_2-h_1}}{(1-r^2)^{s^2+2s-h_1-h_2}} \frac{e^{-s^2 r^2}}{h_1! h_2! \Gamma(h_2 - h_1 + 1)} \frac{\Gamma(s+1)^2}{\Gamma(s - h_2 + 1)} {}_1F_1(s + 1 - h_1, h_2 - h_1 + 1; s^2 r^2). \end{aligned} \tag{235}$$

They then define

$$b_{h_1,h_2}(N, r) = (-sr)^{h_2-h_1} \prod_{j=1}^{h_2} \partial_{w_j} \prod_{i=1}^{h_1} \partial_{z_i} G(\mathbf{z}, \mathbf{w}) \Bigg|_{\mathbf{z}=\mathbf{w}=-r} \tag{236}$$

and group the Leibniz expansion by $h = h_1 + h_2$ into

$$\mathbb{E}|\Lambda_N'(z)|^{2s} = \sum_{h \geq 0} \frac{C_h(N, r)}{(1-r^2)^{s^2+2s-h}}. \tag{237}$$

The displayed formula for C_h is the sum over $h_1 < h_2$, doubled by symmetry, plus the diagonal $h_1 = h_2$ term. This finite Leibniz decomposition follows from (235) once the derivatives of G are known.

The source then states a general coefficient proposition expressing those derivatives of G as finite sums

$$\prod_{j=1}^{h_2} \partial_{w_j} \prod_{i=1}^{h_1} \partial_{z_i} G \Bigg|_{-r,-r} = (-1)^{h_1+h_2} \sum_{m\geq 0} \sum_{\ell=1}^{s} d_{m,\ell,s}(h_1,h_2) r^{2N\ell-(s^2-s+h_1+h_2)+2m}. \tag{238}$$

For $(h_1,h_2) = (0,0)$ an active Laplace-expansion proof is present. For general $(h_1,h_2) \neq (0,0)$, the long derivation immediately following the proposition is commented out in arXiv:2409.03687v2 and no replacement active proof occurs. Consequently (238), and the portions of the alternative structure theorem that depend on it, are not proved in this source version. Generalized Vandermonde determinants, cited by the source at this point, have independent human provenance in Heineman [68]; that citation does not by itself supply the commented coefficient derivation.

There is, however, one exact incompatibility in the theorem as printed. For positive integer s,

$$\frac{1}{\Gamma(s-h_2+1)} = 0 \qquad (h_2 > s). \tag{239}$$

Hence every term contributing to C_h vanishes when $h > 2s$, so

$$C_h(N,r) = 0 \qquad (h > 2s). \tag{240}$$

The simultaneous printed assertion that, for every $h \geq 0$, C_h has degree $Ns + s^2 + s - h$ cannot hold for these zero polynomials. A corrected statement must at least restrict any degree claim to the nonzero range $0 \leq h \leq 2s$ and separately establish that the displayed leading coefficient does not vanish. This is an exact internal contradiction, not an inference from a missing proof.

For $s = 1$ the entire structure can be retained independently, for every N. From

$$M_N(u) := \mathbb{E}|\Lambda'_N(z)|^2 = \sum_{j=1}^{N} j^2 u^{j-1} = \frac{d}{du}\left(u \frac{d}{du} \frac{1-u^{N+1}}{1-u}\right), \tag{241}$$

direct differentiation and collection by powers of $(1-u)^{-1}$ gives

$$M_N(u) = \frac{C_0}{(1-u)^3} + \frac{C_1}{(1-u)^2} + \frac{C_2}{1-u}, \tag{242}$$

with

$$C_0 = (1+u)(1-u^{N+1}), \qquad C_1 = -2(N+1)u^{N+1}, \qquad C_2 = -(N+1)^2 u^N. \tag{243}$$

This all-N identity is proved by (241); the certificate additionally replays it for $1 \leq N \leq 8$.

Non-derivative CUE appendix and Painlevé interfaces. Simm–Wei's appendix keeps the related but distinct moment

$$M_N(s,z) = \int_{U(N)} |\Lambda_N(z)|^{2s}\, d\mu. \tag{244}$$

On $|z| = 1$, the Keating–Snaith product and Barnes-G asymptotic are

$$M_N(s,e^{\mathrm{i}\theta}) = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+2s)}{\Gamma(j+s)^2}, \qquad M_N(s,e^{\mathrm{i}\theta}) \sim N^{s^2} \frac{G(1+s)^2}{G(1+2s)}, \tag{245}$$

for $\operatorname{Re} s > -1/2$ [23]. For integer $s > 0$ and arbitrary z, the appendix imports from the non-Hermitian characteristic-polynomial literature the Jacobi integral

$$M_N(s,z) = \frac{1}{c_{s,N}} \int_{[0,1]^s} \prod_{j=1}^{s} (1 - t_j(1-|z|^2))^N \Delta(\mathbf{t})^2 \, d\mathbf{t}, \tag{246}$$

where

$$c_{s,N} = \prod_{j=0}^{s-1} \frac{\Gamma(N+j+1)\Gamma(j+1)\Gamma(j+2)}{\Gamma(N+s+j+1)} \tag{247}$$

[69, 70]. The change of variables is retained exactly: for $|z| < 1$,

$$M_N(s,z) = (1-|z|^2)^{-s^2} \mathbb{P}\Big(t_{\max}^{(0,N)} \le 1 - |z|^2\Big). \tag{248}$$

This is the map to the Jacobi largest-eigenvalue distribution; it is not merely a similarity of exponents.

The integer microscopic and global/mesoscopic consequences of (246) are, respectively,

$$M_N(s,z_N) \sim N^{s^2} \prod_{j=1}^{s} \frac{1}{\Gamma(j)\Gamma(j+1)} \int_{[0,1]^s} e^{-c\sum t_j} \Delta(\mathbf{t})^2 \, d\mathbf{t}, \qquad |z_N|^2 = 1 - c/N, \tag{249}$$

$$M_N(s,z) \longrightarrow (1-|z|^2)^{-s^2}, \qquad |z| < 1 \text{ fixed}, \tag{250}$$

$$M_N(s,z_N) \sim N^{\alpha s^2}, \qquad |z_N|^2 = 1 - N^{-\alpha}. \tag{251}$$

The finite-N Painlevé VI representation is obtained through the Jacobi largest-eigenvalue law and Forrester–Witte [72]; the microscopic Painlevé V expression uses the Laguerre law and Tracy–Widom/Forrester–Witte [73, 71]. The extension in complex s is not inferred from the appearance of the formulas: it is imported from the uniform Toeplitz asymptotics of Claeys–Its–Krasovsky [80] and the cited Painlevé identifications.

The source's theorem saying that the Painlevé V microscopic formula holds for every $\operatorname{Re} s > -1/2$ omits in its standalone wording that this is the $c > 0$ scaling used immediately before it. The formula contains $(N/c)^{s^2}$ and an integral from c to ∞, so $c > 0$ and the stated branch conventions are part of the typed result. This is a missing local hypothesis, not evidence against the cited Toeplitz theorem.

Source corrections, dependencies, and nonclaims. The following record separates editorial corrections from mathematical status.

1. In the logarithmic derivative, the finite eigenvalue sum is over $j = 1, \dots, N$; the source prints k while retaining θ_j.
2. Two small-ball displays contain an extra parenthesis inside $\mathbb{P}(|X_N(f)| < y)$. The event used in the proof is the one just written.
3. The symmetric approximate-functional-equation choice begins at $t = 2\pi$, not at $t = 1$; the finite initial interval has been retained and then shown negligible.
4. The diagonal Dirichlet series in (231) starts at $n = 1$.
5. The Euler factor needed near the origin is holomorphic and locally bounded. The argument invokes no estimate uniform on the entire half-plane toward $\operatorname{Re} w = 1/2$; this audit makes no claim about whether such an estimate is proved elsewhere.

6. In the confluent induction, the final denominator product is through $n-1$.
7. In the mixed derivative of A, the derivative orders use μ_i with the matching w_i; one source display prints a mismatched index.
8. The Laplace expansion is indexed by $S \in \binom{[2s]}{s}$, not by a subset of the set of all such subsets.
9. The mesoscopic determinant has a missing closing parenthesis around $u^p \widetilde{K}^{(p)}(u)$ before the qth derivative.
10. The Jensen quotient estimate uses $|G_N'(r)|^{-q(1+\varepsilon)}$; one absolute-value delimiter is missing in the source.
11. The Bessel quotient is corrected by (217).
12. In the final combinatorial appendix, one product indexed by the length-ℓ list is printed through s, and a later factorial contains an undefined p where the surrounding indices require s. Those appendix coefficient formulas are not used as independent evidence here.

The strong Szegő/trace limit, Halász inequality, stationary phase, Johansson bound, Forrester–Keating interior moments, Soshnikov cumulants, Akemann–Vernizzi determinant, Balantekin character expansion, classical mean-value theorem, and Painlevé/Toeplitz identifications remain separately attributed dependencies. The present audit supplies the coordinate substitutions and repairs above, the all-N $s=1$ identity, the pole-coefficient extraction, the Bessel factor diagnostic, and bounded finite determinant checks. It does not claim a proof of the conjectural noninteger zeta transfer, a proof of the commented general G-coefficient proposition, a new individual-zero map, or any endpoint statement about the zero set of the zeta function.

2.6.3 Exact CMV/Verblunsky coordinates behind the derivative peel

The local peeling notes preserve two different source layers that must not be conflated. Grover–Mezzadri–Simm supply the derivative-moment theorems above. Killip–Nenciu supply the finite Haar spectral-measure coordinates: the Szegő recurrences, the independent Verblunsky laws, and the CMV realization of the circular ensemble [35, eqs. (2.1)–(2.2), Prop. 3.3, Thm. 1, and App. B]. The formulas denoted T_n and $\eta_n^{(r)}$ in the local notes were attributed there to video screenshots. The following calculation derives the actual Haar ladder from the Killip–Nenciu coordinates and therefore makes the mathematical dependency independently checkable. It does not assert that Killip–Nenciu printed the final T_n notation.

Theorem 2.19 (Exact Haar–Verblunsky peel and telescoping derivative)**.** *Let $N \geq 1$. For a spectral measure of a Haar-distributed $U \in \mathrm{U}(N)$, let $\Phi_0 = 1$ and let Φ_n be the monic orthogonal polynomials in the Killip–Nenciu convention*

$$\Phi_{n+1}(z) = z\Phi_n(z) - \overline{\alpha_n}\Phi_n^*(z), \tag{252}$$

$$\Phi_{n+1}^*(z) = \Phi_n^*(z) - \alpha_n z \Phi_n(z). \tag{253}$$

Almost surely,

$$\alpha_n \sim \Theta_{2N-2n-1} \quad (0 \leq n \leq N-2), \qquad \alpha_{N-1} \sim \mathrm{Unif}\{z : |z| = 1\}, \tag{254}$$

and these variables are independent. Define, for $0 \leq n \leq N-1$,

$$A_n = \Phi_n(1), \qquad u_n = \frac{\Phi_n'(1)}{\Phi_n(1)}, \qquad \beta_n = \overline{\alpha_n}\frac{\overline{A_n}}{A_n}. \tag{255}$$

Then the following statements hold.

1. *The triangular coordinate map*

$$\mathcal{K}_N : (\alpha_0, \dots, \alpha_{N-1}) \longmapsto (\beta_0, \dots, \beta_{N-1}) \tag{256}$$

is a bijection of $\mathbb{D}^{N-1} \times \{z : |z| = 1\}$*. Its inverse is explicit:*

$$A_0 = 1, \qquad A_n = \prod_{j=0}^{n-1} (1 - \beta_j), \qquad \alpha_n = \overline{\beta_n} \frac{\overline{A_n}}{A_n}. \tag{257}$$

It transports the product law in (254) *to independent variables*

$$d\mu_{N-n-1}(\beta_n) = \frac{N-n-1}{\pi} (1 - |\beta_n|^2)^{N-n-2} d^2\beta_n \quad (0 \le n \le N-2), \tag{258}$$

with β_{N-1} *uniform on the unit circle. Thus the local measure* μ_m *is literally* Θ_{2m+1}*, not an asymptotic or normalized surrogate.*

2. *For* $0 \le n \le N-2$*, the logarithmic-derivative state obeys exactly*

$$u_{n+1} = T_n(\beta_n; u_n) := \frac{1 + u_n - \beta_n(n - \overline{u_n})}{1 - \beta_n}. \tag{259}$$

If $w_n = 2u_n - n$*, then*

$$\Re w_{n+1} = \frac{1 - |\beta_n|^2}{|1 - \beta_n|^2} \bigl(1 + \Re w_n\bigr). \tag{260}$$

Consequently $\Re w_0 = 0$ *and* $\Re w_n > 0$ *for* $1 \le n \le N-1$*.*

3. *On the actual size-*N *ladder, put*

$$\eta_n(u) = \eta_n^{(N-n-1)}(u) := \frac{N - 1 - \overline{u}}{N - n + u}. \tag{261}$$

Every denominator in (261) *is nonzero and* $|\eta_n(u_n)| < 1$*. The exact one-step identity is*

$$(1 - \beta_n)\bigl(N - n - 1 + u_{n+1}\bigr) = (N - n + u_n)\bigl(1 - \beta_n \eta_n(u_n)\bigr). \tag{262}$$

4. *The complete product telescopes to the characteristic-polynomial derivative:*

$$\frac{\Phi_N'(1)}{N} = \prod_{n=0}^{N-1} \bigl(1 - \beta_n \eta_n(u_n)\bigr). \tag{263}$$

Therefore, pathwise,

$$\frac{|\Phi_N'(1)|^2}{N^2} = \prod_{n=0}^{N-1} |1 - \beta_n \eta_n(u_n)|^2, \tag{264}$$

where the last factor is the boundary-circle peel and the preceding factors are the independent disk peels.

Finally, the Grover–Mezzadri–Simm and local characteristic polynomial conventions are related without loss of information by

$$\Lambda_N(z) = \det(1 - zU^\dagger) = (-1)^N \overline{\det U}\, \det(zI - U) = (-1)^N \overline{\det U}\, \Phi_N(z). \tag{265}$$

The factor has modulus one and is independent of z, so the derivative magnitudes in the two conventions agree exactly.

Proof. Killip–Nenciu define Θ_ν by the disk density

$$\frac{\nu - 1}{2\pi}(1 - |z|^2)^{(\nu-3)/2} d^2 z. \tag{266}$$

Putting $\nu = 2N - 2n - 1 = 2(N - n - 1) + 1$ gives (258) coefficient by coefficient and exponent by exponent. It remains to prove that the phase change in (255) preserves the product law.

First, $A_0 = 1$. Evaluating (252) at $z = 1$ gives

$$A_{n+1} = A_n - \overline{\alpha_n}\,\overline{A_n} = A_n(1 - \beta_n). \tag{267}$$

If $n \le N - 2$, then $|\beta_n| = |\alpha_n| < 1$, so induction proves $A_n \ne 0$ for every $n \le N - 1$ and proves the product formula in (257). Solving $\beta_n = \overline{\alpha_n}\overline{A_n}/A_n$ for α_n gives the displayed inverse, including the terminal circle coordinate. This proves bijectivity. Conditionally on $\alpha_0, \dots, \alpha_{n-1}$, the multiplier $\overline{A_n}/A_n$ is fixed and unimodular. Complex conjugation followed by multiplication by that phase preserves every radial Θ_{2m+1} law, and it preserves the terminal uniform circle law. Iterated conditioning therefore proves both the stated marginal laws and independence of all the β_n.

Write $D_n = \Phi_n'(1)$. The coefficient definition of reversal gives

$$(\Phi_n^*)'(1) = n\overline{A_n} - \overline{D_n} = \overline{A_n}(n - \overline{u_n}). \tag{268}$$

Differentiating (252) at 1 now yields

$$\begin{aligned} D_{n+1} &= A_n + D_n - \overline{\alpha_n}\big(n\overline{A_n} - \overline{D_n}\big) \\ &= A_n\big(1 + u_n - \beta_n(n - \overline{u_n})\big). \end{aligned} \tag{269}$$

Division by (267) proves (259). With $w_n = 2u_n - n$, direct substitution into that transition gives

$$w_{n+1} = \frac{w_n + 1 + \beta_n(\overline{w_n} + 1)}{1 - \beta_n}. \tag{270}$$

Multiplication by $1 - \overline{\beta_n}$ and taking real parts gives (260); no inequality has been discarded.

For $u = u_n$ one has

$$\begin{aligned} |N - n + u|^2 - |N - 1 - \overline{u}|^2 &= (2N - n - 1)(1 - n + 2\Re u) \\ &= (2N - n - 1)(1 + \Re w_n) > 0. \end{aligned} \tag{271}$$

This proves simultaneously that the denominator of (261) is nonzero and that its modulus is strictly less than one. Substituting (259) and collecting the two terms containing β_n gives

$$\begin{aligned} &(1 - \beta_n)(N - n - 1 + u_{n+1}) \\ &\quad = N - n + u_n - \beta_n(N - 1 - \overline{u_n}) = (N - n + u_n)(1 - \beta_n \eta_n(u_n)), \end{aligned} \tag{272}$$

which is (262). Multiplying it for $0 \le n \le N-2$, using $u_0 = 0$, and cancelling only the displayed nonzero intermediate factors gives

$$\prod_{n=0}^{N-2} (1 - \beta_n \eta_n(u_n)) = \frac{A_{N-1}(1 + u_{N-1})}{N}. \tag{273}$$

The terminal instance of (269), for which no division by $1 - \beta_{N-1}$ is needed, is

$$D_N = A_{N-1}(1 + u_{N-1})\big(1 - \beta_{N-1}\eta_{N-1}(u_{N-1})\big). \tag{274}$$

Together with (273), this proves (263) and its squared-modulus form.

For a spectral measure supported at the N eigenvalues, the monic Φ_N is their product polynomial, hence $\Phi_N(z) = \det(zI - U)$ almost surely [35, finite-support discussion and App. B]. Finally, $1 - zU^\dagger = -U^\dagger(zI - U)$ proves (265) by taking determinants. □

There is one literal source inconsistency outside the argument just used. In the Killip–Nenciu proposition transferring the Jacobi measure to real Verblunsky coordinates, the final parameter is printed as $\alpha_{2n-1} \equiv 1$, whereas the same section's setup, the theorem it proves, and the determinant calculation use $\alpha_{2n-1} = -1$ [35, Prop. 5.3 and the surrounding discussion]. The line is retained as a source defect; no part of theorem 2.19 uses it.

2.6.4 Human prior art and the exact deformed-coordinate derivative crosswalk

The finite derivative peel must be situated against two directly relevant human-authored probability constructions before any novelty language is used. Bourgade–Hughes–Nikeghbali–Yor derive a recursive Haar construction and prove, for Haar $V_N \in \mathrm{U}(N)$,

$$\det(I_N - e^{\mathrm{i}\theta} V_N) \overset{\text{law}}{=} \prod_{k=1}^{N} \left(1 + e^{\mathrm{i}\theta_k}\sqrt{B_{1,k-1}}\right), \tag{275}$$

where the phases are independent uniform variables and the $B_{1,k-1}$ are independent beta variables (with $B_{1,0} = 1$), and they derive the corresponding Mellin–Fourier transform [36, Prop. 2.1, Prop. 2.2, eqs. (1.1), (2.6)]. Their introduction explicitly places this random characteristic-polynomial model beside the Keating–Snaith model for $\zeta(1/2 + it)$; this is a sourced zeta edge, not an endpoint theorem. They also prove the joint independent sum/product representation

$$(\Im \log Z_N, |Z_N|) \overset{\text{law}}{=} \left(\sum_{j=1}^{N} W_j, \prod_{j=1}^{N} B_{j,j-1}\, 2\cos W_j\right), \tag{276}$$

with the explicit cosine-power densities for W_j [36, Thm. 3.1, eqs. (3.7)–(3.8)]. These are determinant-value and logarithmic probability laws; neither theorem differentiates the Szegő recurrence at $z = 1$.

Bourgade–Nikeghbali–Rouault supply a second, closer coordinate precedent. For finite OPUC data they define

$$\gamma_k(z) = z - \frac{\Phi_{k+1}(z)}{\Phi_k(z)}, \qquad \gamma_k = \gamma_k(1), \tag{277}$$

and prove the recursive identities

$$\gamma_k = \overline{\alpha_k} \prod_{j<k} \frac{1 - \overline{\gamma_j}}{1 - \gamma_j}, \qquad |\gamma_k| = |\alpha_k|, \tag{278}$$

with a bijection between the finite α and γ coordinates. They then prove

$$\Phi_N(1) = \det(I - U) = \prod_{k=0}^{N-1} (1 - \gamma_k) \tag{279}$$

and, under circular-Jacobi sampling, independence of the γ_k together with explicit disk/circle densities and the Mellin–Fourier product formula [37, Defs. (3.1)–(3.3), Prop. 3.2, Thm. 3.1, Prop. 4.1].

The apparent coordinate gap can in fact be closed exactly. Write $A_k = \Phi_k(1)$ as in theorem 2.19, and retain the inverse Schur iterate $b_k(z) = \Phi_k(z)/\Phi_k^*(z)$ from the source.

Theorem 2.20 (Literal γ–β identity and lossless derivative-state lift)**.** *Let $N \geq 1$, and put*

$$\mathcal{G}_N = \mathbb{D}^{N-1} \times \mathbb{T}, \qquad \mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}.$$

For finite Szegő data $(\alpha_0, \ldots, \alpha_{N-1}) \in \mathcal{G}_N$, let γ_k be the Bourgade–Nikeghbali–Rouault deformed coefficient and let β_k be the phase-rotated coefficient in (255)*. Then:*

1. *For every $0 \leq k \leq N-1$,*

$$\gamma_k = \beta_k = \overline{\alpha_k}\frac{\overline{A_k}}{A_k}. \tag{280}$$

Thus the sourced triangular $\alpha \mapsto \gamma$ map and the independently reconstructed $\alpha \mapsto \beta$ map are the same coordinate map, not merely two equidistributed maps. The inverse is

$$A_0 = 1, \qquad A_k = \prod_{j<k}(1 - \gamma_j), \qquad \alpha_k = \overline{\gamma_k}\frac{\overline{A_k}}{A_k}. \tag{281}$$

Its image is all of $\mathcal{G}_N$, every fibre is a singleton, and there is no exceptional locus on $\mathcal{G}_N$.

2. *Define the retained state graph*

$$\begin{aligned}\mathcal{S}_N = \{(\gamma_0, \ldots, \gamma_{N-1}; u_0, \ldots, u_{N-1}) : \gamma \in \mathcal{G}_N,\ u_0 = 0,\\ u_{k+1} = T_k(\gamma_k; u_k) \quad (0 \leq k \leq N-2)\}.\end{aligned} \tag{282}$$

The map

$$\mathcal{J}_N : \mathcal{G}_N \longrightarrow \mathcal{S}_N, \qquad \gamma \longmapsto (\gamma; u(\gamma)) \tag{283}$$

is a bijection. Its inverse is the projection $(\gamma; u) \mapsto \gamma$; hence its fibres are singletons and it loses no information. Every denominator in the recursion and in

$$\eta_k(u_k) = \frac{N - 1 - \overline{u_k}}{N - k + u_k} \tag{284}$$

is nonzero on this graph.

3. *Let $m_{\mathbb{T}}$ be normalized Haar measure on $\mathbb{T}$ and put*

$$\Pi_N = \left(\bigotimes_{k=0}^{N-2} \mu_{N-k-1}\right) \otimes m_{\mathbb{T}}. \tag{285}$$

Under Haar measure on $\mathrm{U}(N)$, the γ coordinate has law Π_N. Consequently the full lifted state has the exact push-forward law

$$(\mathcal{J}_N)_*\Pi_N = \delta_0(du_0) \prod_{k=0}^{N-2} \left[\mu_{N-k-1}(d\gamma_k)\delta_{T_k(\gamma_k;u_k)}(du_{k+1})\right] m_{\mathbb{T}}(d\gamma_{N-1}). \tag{286}$$

In particular, the one-step conditional kernel is explicitly

$$K_k(u, d\gamma\, du') = \mu_{N-k-1}(d\gamma)\, \delta_{T_k(\gamma;u)}(du') \qquad (0 \le k \le N-2). \tag{287}$$

4. *The determinant value and its derivative are two explicitly different functions on the same retained coordinate space:*

$$V_N(\gamma) = \prod_{k=0}^{N-1} (1-\gamma_k) = \Phi_N(1), \tag{288}$$

$$F_N(\gamma) = \prod_{k=0}^{N-1} (1-\gamma_k \eta_k(u_k(\gamma))) = \frac{\Phi_N'(1)}{N}. \tag{289}$$

Therefore the exact derivative law is $(F_N)_*\Pi_N$. Although the coordinates γ_k are independent, the displayed derivative factors are not asserted to be independent: u_k is the deterministic function of the complete prefix $(\gamma_0, \dots, \gamma_{k-1})$ specified above.

Proof. By the source convention, $\Phi_k^*(z) = z^k \overline{\Phi_k(\overline{z}^{-1})}$. Hence

$$\Phi_k^*(1) = \overline{\Phi_k(1)} = \overline{A_k}, \qquad b_k(1) = \frac{A_k}{\overline{A_k}}. \tag{290}$$

For $k \le N-1$, the product formula $A_k = \prod_{j<k}(1-\beta_j)$ contains only disk coordinates, so $A_k \ne 0$. Substitution into the sourced definition $\gamma_k(1) = \overline{\alpha_k}/b_k(1)$ gives

$$\gamma_k = \frac{\overline{\alpha_k}}{A_k/\overline{A_k}} = \overline{\alpha_k}\frac{\overline{A_k}}{A_k} = \beta_k, \tag{291}$$

which proves (280). The inverse, image, fibres, and absence of an exceptional locus now follow from the already proved triangular inverse (257); no coordinate has been discarded.

Given γ, the equations defining (282) determine u_0, then u_1, and so on through u_{N-1}. Induction proves uniqueness. Projection onto the retained γ coordinates is therefore a two-sided inverse to $\mathcal{J}_N$. The denominator and strict-modulus statements are precisely the ordered identities (260) and (271), now with $\gamma_k = \beta_k$.

Specialize the Bourgade–Nikeghbali–Rouault circular-Jacobi theorem to its Dyson parameter $\beta_{\mathrm{D}} = 2$ and singularity parameter $\delta = 0$. Their interior density becomes, with $m = N-k-1$,

$$\frac{m}{\pi}(1-|z|^2)^{m-1} d^2 z = d\mu_m(z), \tag{292}$$

and their terminal density is $m_{\mathbb{T}}$. Their theorem supplies independence, so the law is exactly (285). Since $u_{k+1} = T_k(\gamma_k; u_k)$ is a deterministic Borel map and γ_k is independent of its prefix, disintegration one coordinate at a time gives (286); equivalently, testing both sides against any bounded Borel function and applying Fubini gives identical iterated integrals. Finally (288) is the sourced determinant factorization, while (289) is (263) after (280). This proves every asserted map and push-forward formula. □

Bourgade's conditional-Haar paper gives a second derivative product, but on a different measure. Its relation to the preceding coordinates is also exact rather than categorical.

Proposition 2.21 (Conditional-Haar residual map, coordinate reversal, and determinant tilt)**.** *Let $n \ge 1$ and $p \in \mathbb{Z}_{\ge 0}$. Endow a unitary matrix of size $n+p$ with Bourgade's Haar measure conditioned to have p eigenvalues equal to 1. Write its remaining eigenvalues as $e^{i\theta_1}, \ldots, e^{i\theta_n}$ and set*

$$Z_U(z) = \det(zI - U) = (z-1)^p Q_n(z), \qquad Q_n(z) = \prod_{j=1}^{n} (z - e^{i\theta_j}). \tag{293}$$

Then:

1. *The residual spectrum has the circular-Jacobi law $\mathrm{CJ}_{p,2}^{(n)}$. Its deformed coefficients $(\gamma_0, \ldots, \gamma_{n-1})$ are independent and*

$$\frac{Z_U^{(p)}(1)}{p!} = Q_n(1) = \prod_{k=0}^{n-1} (1 - \gamma_k). \tag{294}$$

2. *For $0 \le k \le n-2$, with $m = n-k-1$, γ_k has density*

$$c_{m,p}(1-|z|^2)^{m-1}|1-z|^{2p}\, d^2 z, \qquad c_{m,p} = \frac{\Gamma(m+p+1)^2}{\pi\Gamma(m)\Gamma(m+2p+1)}. \tag{295}$$

   The terminal coordinate has density

$$\frac{\Gamma(p+1)^2}{\Gamma(2p+1)}|1-z|^{2p}\, dm_{\mathbb{T}}(z). \tag{296}$$

3. *Put $\mathcal{X}_n = \mathbb{T} \times \mathbb{D}^{n-1}$, ordered as $(X_1, \ldots, X_n)$, and define the coordinate reversal*

$$\mathcal{R}_n : \mathcal{G}_n \longrightarrow \mathcal{X}_n, \qquad X_\ell = (\mathcal{R}_n \gamma)_\ell = \gamma_{n-\ell} \qquad (1 \le \ell \le n). \tag{297}$$

   This is a bijection with inverse $\gamma_k = X_{n-k}$. Its push-forward of $\Pi_{n,p}$ is exactly the product law of the independent variables in Bourgade's derivative-product theorem. This is a law-preserving coordinate identification; it does not silently identify two source constructions pointwise on an unstated common sample space. Thus X_1 is the sampled circle coordinate, while for $\ell \ge 2$ the unsampled radial variable satisfies $|X_\ell|^2 \sim \mathrm{Beta}(1, \ell-1)$.

4. *Let $\Pi_{n,p}$ be the product law in (295)–(296) and let $\Pi_{n,0}$ be the CUE product law. The exact change of measure is*

$$\frac{d\Pi_{n,p}}{d\Pi_{n,0}}(\gamma) = \frac{|V_n(\gamma)|^{2p}}{\mathcal{Z}_{n,p}}, \qquad \mathcal{Z}_{n,p} = \prod_{m=0}^{n-1} \frac{\Gamma(m+1)\Gamma(m+2p+1)}{\Gamma(m+p+1)^2}. \tag{298}$$

   Hence the conditional first-nonzero derivative and the unconditioned derivative F_n from (289) are explicitly placed on the same coordinate space, with their observable maps and measures retained. No equality of those two random variables is asserted.

Proof. The residual eigenvalue density is

$$\mathrm{const} \cdot \prod_{j<\ell} |e^{i\theta_j} - e^{i\theta_\ell}|^2 \prod_{j=1}^{n} |1 - e^{i\theta_j}|^{2p}, d\theta_1 \cdots d\theta_n, \tag{299}$$

which is exactly $\mathrm{CJ}^{(n)}_{p,2}$ in the source coordinates [38, eq. (1.2), Def. 2.5, Cor. 4.1]. Differentiating (293) exactly p times at 1 leaves only the term in which all p derivatives hit $(z-1)^p$, proving $Z_U^{(p)}(1)/p! = Q_n(1)$. The sourced deformed-coefficient factorization proves the last equality in (294).

In the circular-Jacobi density theorem, set $\delta = p$, $\beta_{\mathrm{D}} = 2$, and $m = n-k-1$. The source constant becomes $c_{m,p}$ and gives (295); its terminal density gives (296) [37, Thm. 3.1 and eqs. (3.1)–(3.3)]. For $m \geq 1$, under the CUE base law write $z = e^{\mathrm{i}\Theta}\sqrt{B}$ with Θ uniform and $B \sim \mathrm{Beta}(1,m)$. Expanding both integer powers and integrating the phase gives

$$\begin{aligned}\mathbb{E}|1-z|^{2p} &= \sum_{j=0}^{p} \binom{p}{j}^2 \mathbb{E}B^j \\ &= \sum_{j=0}^{p} \binom{p}{j}^2 \frac{j!\,\Gamma(m+1)}{\Gamma(m+j+1)} = \frac{\Gamma(m+1)\Gamma(m+2p+1)}{\Gamma(m+p+1)^2}.\end{aligned} \tag{300}$$

The last finite sum is the Chu–Vandermonde evaluation of ${}_2F_1(-p,-p;m+1;1)$. For $m=0$, phase integration on the circle gives the same formula through $\sum_{j=0}^{p}\binom{p}{j}^2 = \binom{2p}{p}$. Taking reciprocals of these normalizing factors gives exactly the constants in (295) and (296).

For the source variable X_ℓ, the unsampled radial square has $\mathrm{Beta}(1,\ell-1)$ law. The coordinate γ_k has the same base radial law when $m = n-k-1 = \ell-1$, which is precisely $k = n-\ell$; the $|1-z|^{2p}$ sampling is identical. Independence on both product spaces therefore proves $(\mathcal{R}_n)_*\Pi_{n,p} = \mathrm{Law}(X_1,\ldots,X_n)$ coordinate by coordinate. The displayed inverse proves bijectivity, including the terminal $m=0$ coordinate, without asserting an unproved pointwise coupling of the reflection and OPUC realizations. Finally, multiplication of the independent one-factor density ratios yields

$$\prod_{k=0}^{n-1}|1-\gamma_k|^{2p} = \left|\prod_{k=0}^{n-1}(1-\gamma_k)\right|^{2p} = |V_n(\gamma)|^{2p}, \tag{301}$$

and multiplication of (300) for $m = 0,\ldots,n-1$ gives $\mathcal{Z}_{n,p}$. This proves (298) and the complete crosswalk. $\square$

The paper's zeta edge is the Keating–Snaith value model and Selberg-type logarithmic limit, not a statement about zeros. Bourgade's thesis repeats the conditional derivative theorem as Theorem 5.1 and then retains several further results that do not occur in the paper: the trace central limit theorem under finitely pinned eigenvalues, the martingale realization of the circular-β_{D} trace limit, and the joint convergence

$$\left(\frac{\log\det(I-u_n)}{\sqrt{\log n}}, \operatorname{Tr} u_n\right) \implies \sqrt{\frac{2}{\beta_{\mathrm{D}}}}(X_1,X_2), \tag{302}$$

where X_1, X_2 are independent standard complex normal variables [39, Chap. 5, Thms. 5.1, 5.11, 5.14 and Cor. 5.13]. The proof of the joint limit uses the same independent deformed coefficients but splits a finite prefix controlling the trace from a tail controlling the normalized logarithm, then uses the L^2 tail estimate and Slutsky's lemma. This is human prior art for a state-dependent use of the independent coordinates; it does not make the derivative factors in (289) independent. The exact $\gamma = \beta$ identity and state uniqueness are also replayed in bounded Lean file `GammaBetaCrosswalk.lean`; the probability disintegration above remains a standard human proof rather than an axiom in the formal kernel.

2.6.5 The Winn–Forrester Cauchy trace as the terminal deformed-coordinate state

Winn's finite-N argument and Forrester's circular-β_{D} extension use the stereographic variables $x_j = \cot(\theta_j/2)$ to turn a completed-characteristic- polynomial derivative into the trace of a symmetric Cauchy ensemble [43, §3.2][42, Prop. 2.1]. The following result proves the missing coordinate map from that trace all the way back to the retained γ-state. In particular, the Cauchy trace is not merely analogous to the terminal state in theorem 2.20: it is an explicit rational observable of that state on its exact open domain.

Theorem 2.22 (Stereographic, trace, and terminal-state factorization)**.** *Let $N \geq 1$. On the ordered eigenangle chart*

$$\Theta_N^\circ = (0, 2\pi)^N$$

define

$$\mathrm{St}_N : \Theta_N^\circ \longrightarrow \mathbb{R}^N, \qquad \mathrm{St}_N(\theta_1, \ldots, \theta_N) = \left(\cot\frac{\theta_1}{2}, \ldots, \cot\frac{\theta_N}{2}\right). \tag{303}$$

Put $x_j = \cot(\theta_j/2)$ and

$$T = \Sigma_N(\mathbf{x}) := \sum_{j=1}^{N} x_j.$$

Then the following assertions hold.

1. *The map* (303) *is a bijection, with inverse*

$$\mathrm{St}_N^{-1}(x_1, \ldots, x_N) = (2\,\mathrm{arccot}\, x_1, \ldots, 2\,\mathrm{arccot}\, x_N), \qquad \mathrm{arccot}\, x \in (0, \pi). \tag{304}$$

It is equivariant for simultaneous permutation of coordinates and therefore descends to a bijection of the corresponding unordered configuration spaces. Coordinatewise,

$$e^{\mathrm{i}\theta_j} = \frac{x_j + \mathrm{i}}{x_j - \mathrm{i}}, \qquad |1 - e^{\mathrm{i}\theta_j}| = \frac{2}{\sqrt{1 + x_j^2}}, \tag{305}$$

$$|e^{\mathrm{i}\theta_k} - e^{\mathrm{i}\theta_j}| = \frac{2|x_k - x_j|}{\sqrt{1 + x_k^2}\sqrt{1 + x_j^2}}, \qquad |d\theta_j| = \frac{2\,dx_j}{1 + x_j^2}. \tag{306}$$

The excluded locus is exactly an eigenangle 0 *modulo* 2π*, equivalently an eigenvalue* 1*; it lies at the infinite boundary of this finite x-chart.*

The trace map $\Sigma_N : \mathbb{R}^N \to \mathbb{R}$ is surjective, has the explicit right inverse $j_N(t) = (t, 0, \ldots, 0)$, and has fibres

$$\Sigma_N^{-1}(t) = j_N(t) + \ker \Sigma_N = \{\mathbf{x} \in \mathbb{R}^N : x_1 + \cdots + x_N = t\}. \tag{307}$$

Thus it loses no coordinate when $N = 1$ and identifies precisely the affine translates of the $(N-1)$-dimensional hyperplane $\ker \Sigma_N$ *when $N > 1$.*

2. *Let*

$$\Phi_N(z) = \prod_{j=1}^{N}(z - e^{\mathrm{i}\theta_j}), \qquad u_N = \frac{\Phi_N'(1)}{\Phi_N(1)}.$$

On Θ_N° one has the exact terminal identity

$$u_N = \sum_{j=1}^{N} \frac{1}{1 - e^{\mathrm{i}\theta_j}} = \frac{N}{2} + \frac{\mathrm{i}}{2} T, \qquad 2u_N - N = \mathrm{i}T. \tag{308}$$

In the deformed coordinates of theorem 2.20, put

$$\mathcal{G}_N^\circ = \{\gamma \in \mathcal{G}_N : V_N(\gamma) \neq 0\} = \{\gamma \in \mathcal{G}_N : \gamma_{N-1} \neq 1\}. \tag{309}$$

The Cauchy trace is the explicit observable

$$\mathcal{T}_N : \mathcal{G}_N^\circ \longrightarrow \mathbb{R}, \qquad \mathcal{T}_N(\gamma) = -\mathrm{i}\left(2N \frac{F_N(\gamma)}{V_N(\gamma)} - N\right). \tag{310}$$

Its complete fibre over $t \in \mathbb{R}$ is

$$\mathcal{T}_N^{-1}(t) = \{\gamma \in \mathcal{G}_N^\circ : 2N F_N(\gamma) = (N + \mathrm{i}t) V_N(\gamma)\}. \tag{311}$$

Hence two retained states are identified by this observable exactly when their terminal ratios F_N/V_N agree; no stronger independence or injectivity statement is being made.

3. *Define Winn's real completed characteristic polynomial by*

$$\mathcal{V}_U(\vartheta) = \exp\left(\frac{\mathrm{i}N(\vartheta + \pi)}{2} - \frac{\mathrm{i}}{2}\sum_{j=1}^{N} \theta_j\right) \prod_{j=1}^{N} \left(1 - e^{\mathrm{i}(\theta_j - \vartheta)}\right). \tag{312}$$

Then $|\mathcal{V}_U(0)| = |V_N|$ and

$$\frac{\mathcal{V}_U'(0)}{\mathcal{V}_U(0)} = \mathrm{i}\left(u_N - \frac{N}{2}\right) = -\frac{T}{2}. \tag{313}$$

For $s \in \mathbb{Z}_{\geq 0}$ and real $-\frac{1}{2} < h < s + \frac{1}{2}$, let

$$\mathcal{F}_N(s, h) = \int_{\mathrm{U}(N)} |\mathcal{V}_U(0)|^{2(s-h)} |\mathcal{V}_U'(0)|^{2h} \, d\mu_N^{\mathrm{Haar}}(U). \tag{314}$$

With $\Pi_{N,s}$ and $\mathcal{Z}_{N,s}$ as in (298), the exact factorization is

$$\mathcal{F}_N(s, h) = 2^{-2h} \mathcal{Z}_{N,s} \mathbb{E}_{\Pi_{N,s}}\left[|\mathcal{T}_N|^{2h}\right], \qquad \mathcal{Z}_{N,s} = \mathcal{F}_N(s, 0). \tag{315}$$

The locus removed in (309) has zero $\Pi_{N,s}$-measure, so no choice of a value for $\mathcal{T}_N$ there changes the expectation.

4. *The push-forward of the determinant-tilted CUE eigenangle law under St_N is the symmetric Cauchy unitary ensemble with the literal density*

$$d\nu_{N,s}(\mathbf{x}) = \frac{2^{N^2 + 2sN}}{(2\pi)^N N! \, \mathcal{Z}_{N,s}} \prod_{1 \leq j < k \leq N} (x_k - x_j)^2 \prod_{j=1}^{N} (1 + x_j^2)^{-(N+s)} \, d\mathbf{x}. \tag{316}$$

Consequently

$$\mathbb{E}_{\Pi_{N,s}}[|\mathcal{T}_N|^{2h}] = \int_{\mathbb{R}^N} \left|\sum_{j=1}^{N} x_j\right|^{2h} d\nu_{N,s}(\mathbf{x}). \tag{317}$$

This is an equality of push-forward laws and observables. It does not identify a particular Haar matrix, a particular deformed-coordinate sample, and a particular Cauchy matrix pointwise without specifying a coupling.

Proof. For $x = \cot(\theta/2)$ with $0 < \theta < 2\pi$, elementary half-angle algebra gives

$$e^{\mathrm{i}\theta} = \frac{x+\mathrm{i}}{x-\mathrm{i}}, \qquad \frac{dx}{d\theta} = -\frac{1}{2}(1+x^2).$$

The first equation and its inverse prove the coordinatewise bijection and (304); applying it to two coordinates and taking absolute values proves (305)– (306). Coordinatewise application also proves permutation equivariance. The statement about the excluded locus follows from the endpoint behaviour of $\cot(\theta/2)$. The assertions about Σ_N follow by direct substitution of j_N, and subtracting $j_N(t)$ from the equation $\Sigma_N(\mathbf{x}) = t$ proves the full fibre formula (307).

Logarithmic differentiation of Φ_N gives the first sum in (308). For each coordinate,

$$\frac{1}{1-e^{\mathrm{i}\theta}} = \frac{1}{2} + \frac{\mathrm{i}}{2}\cot\frac{\theta}{2}.$$

Summing proves the rest of that equation. By (288)–(289),

$$u_N = \frac{\Phi_N'(1)}{\Phi_N(1)} = N\frac{F_N(\gamma)}{V_N(\gamma)}.$$

All nonterminal factors in $V_N = \prod_k(1-\gamma_k)$ are nonzero because $\gamma_k \in \mathbb{D}$ for $k < N-1$; this proves both equalities in (309). Substitution in (308) proves (310), including that its value is real on the stated domain. Multiplying the equation $\mathcal{T}_N(\gamma) = t$ by $\mathrm{i}V_N(\gamma)$, and using $V_N \neq 0$, gives exactly (311) in both directions.

Let $Z_U(\vartheta) = \prod_j(1-e^{\mathrm{i}(\theta_j-\vartheta)})$. At zero, $Z_U(0) = \Phi_N(1) = V_N$. Direct differentiation gives

$$\frac{Z_U'(0)}{Z_U(0)} = \mathrm{i}\sum_{j=1}^{N}\frac{e^{\mathrm{i}\theta_j}}{1-e^{\mathrm{i}\theta_j}} = \mathrm{i}(u_N - N).$$

The derivative of the exponential prefactor in (312) contributes $\mathrm{i}N/2$. Therefore

$$\frac{\mathcal{V}_U'(0)}{\mathcal{V}_U(0)} = \mathrm{i}(u_N - N/2) = -T/2,$$

which proves (313). Its absolute-value form turns the integrand of (314) into

$$2^{-2h}|V_N|^{2s}|T|^{2h}.$$

Now apply the exact Radon–Nikodym identity (298); its normalizing factor at $p = s$ is $\mathcal{Z}_{N,s}$. This proves (315). The terminal circle coordinate has a density relative to Haar measure, so its singleton value 1 has probability zero.

Finally Weyl's eigenangle density is

$$\frac{1}{(2\pi)^N N!}\prod_{j<k}|e^{\mathrm{i}\theta_k} - e^{\mathrm{i}\theta_j}|^2\, d\boldsymbol{\theta}.$$

Multiplying the pair factors and Jacobians in (306) gives

$$2^{N(N-1)}2^N\prod_{j<k}(x_k-x_j)^2\prod_j(1+x_j^2)^{-N} = 2^{N^2}\prod_{j<k}(x_k-x_j)^2\prod_j(1+x_j^2)^{-N}.$$

Moreover $|V_N|^{2s} = 2^{2sN}\prod_j(1+x_j^2)^{-s}$ by (305). Division by the exact tilt normalizer $\mathcal{Z}_{N,s}$ proves (316); applying the trace map Σ_N proves (317). $\square$

For completeness, Winn's finite-N theorem can now be stated without hiding the coordinate change. For a partition λ, write h_λ for its hook-length product and $[a]_\lambda$ for the generalized Pochhammer symbol, and set

$$\mathcal{C}_N(p,s) = (-2)^p \sum_{\lambda \vdash_s p} \frac{[s]_\lambda [-N]_\lambda}{[2s]_\lambda h_\lambda^2}. \tag{318}$$

If $h = (2m-1)/2$, $m \in \mathbb{N}$, and $s \in \mathbb{N}$ with $s > h - 1/2$, then Winn proves

$$\begin{aligned}\mathcal{F}_N(s,h) = \frac{2(-1)^{h+1/2}}{2^{2h}\pi}\mathcal{F}_N(s,0)\Bigg\{ \sum_{p=1}^{2h}\sum_{\ell=1}^{p} \binom{2h}{p-\ell}\frac{(-1)^\ell}{\ell}(-N)^{2h-p} p!\,\mathcal{C}_N(p,s) \\ + \sum_{p=2h+1}^{sN} \frac{(2h)!(p-2h-1)!}{N^{p-2h}}\mathcal{C}_N(p,s)\Bigg\}. \end{aligned}\tag{319}$$

This is Theorem 2.2 of the source. Its proof is not a formal manipulation of the final answer: Winn derives the regularized Fourier kernel, proves the exact Cauchy integral, evaluates its characteristic function by Laguerre Wronskians and Jack hypergeometric functions, establishes the required small-regularizer limits, and only then collects the partition coefficients in Sections 3–6 [43, Thm. 2.2 and §§3–6].

Forrester retains the same Cauchy trace but replaces the Vandermonde square by the Dyson power $\beta_{\mathrm{D}} > 0$. For $s \in \mathbb{Z}_{\geq 0}$ and $-1/2 < \operatorname{Re} h < s + 1/2$, his large-$N$ theorem is

$$\begin{aligned}\lim_{N\to\infty} N^{-2s^2/\beta_{\mathrm{D}}-2h}\mathcal{F}_{N,\beta_{\mathrm{D}}}(s,h) = \frac{1}{2^{2h}}\prod_{j=1}^{s}\frac{\Gamma(2j/\beta_{\mathrm{D}})}{\Gamma(2(s+j)/\beta_{\mathrm{D}})} \\ \times \frac{1}{\cos(\pi h)}\sum_{\kappa}\frac{(-2h)_{|\kappa|}}{|\kappa|!\,[4s/\beta_{\mathrm{D}}]_\kappa^{(\beta_{\mathrm{D}}/2)}}\left(\frac{4}{\beta_{\mathrm{D}}}\right)^{|\kappa|} C_\kappa^{(\beta_{\mathrm{D}}/2)}((1)^s), \end{aligned}\tag{320}$$

with limiting values taken at removable half-integer singularities. He also gives the limiting trace density as an explicit inverse Fourier–Laplace integral of a Jack ${}_0F_1^{(\beta_{\mathrm{D}}/2)}$ function [42, Thms. 1.1–1.2]. Equation (310) supplies the exact $\beta_{\mathrm{D}} = 2$ deformed-coordinate lift of that trace. No $\beta_{\mathrm{D}} \neq 2$ lift to the present γ-state has been constructed here, and none of these moment identities is a statement about individual zeta zeros.

2.6.6 The Conrey–Rubinstein–Snaith first jet and its exact place in the Cauchy trace

Conrey, Rubinstein, and Snaith use a different characteristic-polynomial coordinate from the one above. The conversion is an everywhere invertible phase and an explicit triangular first-jet map; writing it out fixes every constant in their two derivative moments [44, §§1–4].

Theorem 2.23 (Exact CRS polynomial and first-jet crosswalk)**.** *Let $A \in \mathrm{U}(N)$ have ordered eigenvalues $z_j = e^{\mathrm{i}\theta_j}$, and put*

$$\delta_A = \det A = \prod_{j=1}^{N} z_j, \qquad \Phi_N(s) = \prod_{j=1}^{N}(s - z_j), \qquad V_N = \Phi_N(1), \quad F_N = \frac{1}{N}\Phi_N'(1).$$

The CRS characteristic polynomial is

$$\Lambda_A(s) = \det(I - sA^*) = \prod_{j=1}^{N}(1 - s z_j^{-1}).$$

Choose the eigenangle lifts and the branch of $s^{-N/2}$ used in their completed function, and define

$$\chi_A = (-1)^N \delta_A^{-1}, \qquad \rho_A = e^{-\pi \mathrm{i} N/2} e^{\frac{\mathrm{i}}{2}\sum_j \theta_j} \chi_A.$$

Both χ_A and ρ_A have modulus one. Then:

1. *For every $s \in \mathbb{C}$,*
$$\Lambda_A(s) = \chi_A \Phi_N(s). \tag{321}$$
For fixed A this is a linear bijection of polynomial spaces, with inverse $\Phi_N = \chi_A^{-1}\Lambda_A$, singleton fibres, empty exceptional locus, and no information loss.

2. *If*
$$\mathcal{Z}_A(s) = e^{-\pi \mathrm{i} N/2} e^{\frac{\mathrm{i}}{2}\sum_j \theta_j} s^{-N/2} \Lambda_A(s),$$
then the value–derivative jets at $s = 1$ satisfy
$$\begin{pmatrix} \mathcal{Z}_A(1) \\ \mathcal{Z}'_A(1) \end{pmatrix} = \rho_A \begin{pmatrix} 1 & 0 \\ -N/2 & 1 \end{pmatrix} \begin{pmatrix} V_N \\ \Phi'_N(1) \end{pmatrix}. \tag{322}$$
The inverse is
$$\begin{pmatrix} V_N \\ \Phi'_N(1) \end{pmatrix} = \rho_A^{-1} \begin{pmatrix} 1 & 0 \\ N/2 & 1 \end{pmatrix} \begin{pmatrix} \mathcal{Z}_A(1) \\ \mathcal{Z}'_A(1) \end{pmatrix}. \tag{323}$$
Thus the jet map has singleton fibres, empty exceptional locus, and zero information loss. Changing one eigenangle lift by 2π changes the displayed square-root phase by a sign, exactly as it changes the chosen completed function; all absolute-value identities below are lift-independent.

3. *Pointwise on the whole eigenangle space, including $V_N = 0$,*
$$\Lambda'_A(1) = \chi_A \, N F_N, \tag{324}$$
$$\mathcal{Z}'_A(1) = \rho_A \, N \left(F_N - \frac{1}{2} V_N \right). \tag{325}$$
On the open set $\mathcal{G}_N^\circ$ of (309), this becomes
$$\mathcal{Z}'_A(1) = \frac{\mathrm{i}\rho_A}{2} V_N \mathcal{T}_N. \tag{326}$$
The centered polynomial coordinate $D_N := F_N - \frac{1}{2}V_N$ remains defined at $V_N = 0$; only the quotient $\mathcal{T}_N = -\mathrm{i}(2NF_N/V_N - N)$ fails there. Equivalently,
$$\frac{\mathrm{i}}{2} V_N \mathcal{T}_N = N \left(F_N - \frac{1}{2} V_N \right)$$
has the unique polynomial continuation displayed on the right.

4. *Let $\Pi_{N,0}$ be the CUE γ-coordinate law and, for an integer $k \geq 1$, define*
$$M^{\Lambda}_{N,k} = \int_{\mathrm{U}(N)} |\Lambda'_A(1)|^{2k} \, dA, \qquad M^{\mathcal{Z}}_{N,k} = \int_{\mathrm{U}(N)} |\mathcal{Z}'_A(1)|^{2k} \, dA.$$

Then the exact finite-N identities are

$$M^{\Lambda}_{N,k} = N^{2k}\mathbb{E}_{\Pi_{N,0}}|F_N|^{2k}, \tag{327}$$

$$\begin{aligned} M^{\mathcal{Z}}_{N,k} &= N^{2k}\mathbb{E}_{\Pi_{N,0}}\left|F_N - \frac{1}{2}V_N\right|^{2k} = \mathcal{F}_N(k,k) \\ &= 2^{-2k}\mathcal{Z}_{N,k}\mathbb{E}_{\Pi_{N,k}}|\mathcal{T}_N|^{2k}. \end{aligned} \tag{328}$$

Consequently the CRS b'_k asymptotic is exactly the $s = h = k$ member of Winn's joint-moment family, after the explicit jet and tilt maps above.

Proof. For each j,

$$s - z_j = -z_j(1 - sz_j^{-1}).$$

Multiplication gives $\Phi_N(s) = (-1)^N\delta_A\Lambda_A(s)$, proving (321) and its inverse. Multiplication by the nonzero scalar χ_A proves the image, fibre, and exceptional-locus assertions.

Substitution into the definition of $\mathcal{Z}_A$ gives $\mathcal{Z}_A(s) = \rho_A s^{-N/2}\Phi_N(s)$. Evaluation and differentiation at $s = 1$ give

$$\mathcal{Z}_A(1) = \rho_A V_N, \qquad \mathcal{Z}'_A(1) = \rho_A\left(\Phi'_N(1) - \frac{N}{2}V_N\right),$$

which is (322). The two triangular matrices in (322)–(323) multiply to the identity; because $\rho_A \neq 0$, the inverse and fibre assertions follow. Differentiating (321) and using $\Phi'_N(1) = NF_N$ proves (324)–(325). Equation (310) gives

$$\frac{\mathrm{i}}{2}V_N\mathcal{T}_N = \frac{1}{2}(2NF_N - NV_N) = N\left(F_N - \frac{1}{2}V_N\right)$$

when $V_N \neq 0$, proving (326) and its stated polynomial continuation.

Taking absolute values in (324)–(325), using $|\chi_A| = |\rho_A| = 1$, and pushing Haar measure through the proved CUE-to-γ coordinate map gives the first equalities in (327)–(328). The pointwise identity $|\mathcal{Z}'_A(1)| = |V_N||\mathcal{T}_N|/2$ also equals the absolute value of Winn's completed derivative at zero by (313). Hence $M^{\mathcal{Z}}_{N,k} = \mathcal{F}_N(k,k)$, and (315) with $s = h = k$ proves the final equality. □

The finite triangular inverse and the identity $\frac{\mathrm{i}}{2}V_N\mathcal{T}_N = N(F_N - \frac{1}{2}V_N)$ are independently replayed by `FORMAL-20260826-0019`. The Lean file proves both compositions of the jet map and its inverse, the trace factorization under the explicit hypothesis $V_N \neq 0$, and the polynomial value of the centered coordinate at $V_N = 0$. It introduces no asymptotic, measure-theoretic, or number-theoretic axiom.

The source's asymptotic constants can now be recorded without suppressing their normalization. Set

$$\mathcal{D}_k(x) = x^{-k^2/2}\det_{1\le i,j\le k}\left(I_{i+j-1}(2\sqrt{x})\right);$$

the leading power of the determinant cancels the displayed power at zero. For fixed $k \in \mathbb{N}$, Conrey–Rubinstein–Snaith prove

$$M^{\Lambda}_{N,k} = b_k N^{k^2+2k} + O_k(N^{k^2+2k-1}), \tag{329}$$

$$M^{\mathcal{Z}}_{N,k} = b'_k N^{k^2+2k} + O_k(N^{k^2+2k-1}), \tag{330}$$

where

$$b_k = (-1)^{k(k+1)/2} \sum_{h=0}^{k} \binom{k}{h} \left(\frac{d}{dx}\right)^{k+h} \left(e^{-x}\mathcal{D}_k(x)\right)\Big|_{x=0}, \tag{331}$$

$$b'_k = (-1)^{k(k+1)/2} \left(\frac{d}{dx}\right)^{2k} \left(e^{-x/2}\mathcal{D}_k(x)\right)\Big|_{x=0}. \tag{332}$$

They also prove the independent combinatorial evaluation

$$b'_k = (-1)^{k(k+1)/2} \sum_{\substack{m_0+\cdots+m_k=2k \\ m_i\geq 0}} \binom{2k}{m_0,\ldots,m_k} \left(-\frac{1}{2}\right)^{m_0} \prod_{i=1}^{k} \frac{1}{(2k-i+m_i)!} \prod_{1\leq i<j\leq k} (m_j - m_i + i - j). \tag{333}$$

These are Theorems 1–3 of the source [44, Thms. 1–3].

Proof mechanism for (329)–(333). The source begins with the shifted CUE product formula and converts its $\binom{2k}{k}$ shuffle sum into a k-fold residue integral. For shifts $\alpha_j = O(N^{-1})$, replacing $z(w) = 1/(1-e^{-w})$ by $1/w$ produces the leading contour integral with error $O_k(N^{k^2-1})$. Differentiating all $2k$ shifts gives the $O_k(N^{k^2+2k-1})$ remainders above.

After moving the squared Vandermonde outside as $\Delta^2(d/dL)$, the completed derivative gives the leading operator

$$\frac{(-1)^{k(k+1)/2}}{k!}\Delta^2\left(\frac{d}{dL}\right)\left(\frac{d}{dt}\right)^{2k}\left[e^{-Nt/2}\prod_{r=1}^{k}\frac{1}{2\pi \mathrm{i}}\oint \frac{e^{L_r w+t/w}}{w^{2k}}\,dw\right]_{\substack{L_r=N\\ t=0}}. \tag{334}$$

For the uncompleted derivative, the exact reciprocal-power identity

$$\left(\sum_r w_r^{-1} - N\right)^k \left(\sum_r w_r^{-1}\right)^k = \sum_{h=0}^{k}\binom{k}{h} N^{k-h}\left(\sum_r w_r^{-1} - N\right)^{k+h}$$

gives

$$\frac{(-1)^{k(k+1)/2}}{k!}\sum_{h=0}^{k}\binom{k}{h}N^{k-h}\Delta^2\left(\frac{d}{dL}\right)\left(\frac{d}{dt}\right)^{k+h}\left[e^{-Nt}\prod_{r=1}^{k}\frac{1}{2\pi \mathrm{i}}\oint \frac{e^{L_r w+t/w}}{w^{2k}}\,dw\right]_{\substack{L_r=N\\ t=0}}. \tag{335}$$

The one-variable integral is checked coefficientwise:

$$\frac{1}{2\pi \mathrm{i}}\oint \frac{e^{Lw+t/w}}{w^{2k}}\,dw = \frac{L^{2k-1}I_{2k-1}(2\sqrt{Lt})}{(Lt)^{k-1/2}}.$$

For

$$f_t(L) = \sum_{r=0}^{\infty}\frac{t^r L^{2k-1+r}}{r!(2k-1+r)!},$$

direct expansion of both Vandermonde determinants gives

$$\Delta^2\left(\frac{d}{dL}\right)\prod_{r=1}^{k} f_t(L_r)\Big|_{L_r=N} = k!\det_{1\leq i,j\leq k}\left(f_t^{(i+j-2)}(N)\right).$$

Substitution of the Bessel series followed by $x = Nt$ extracts N^{k^2+2k} and yields (331)–(332), including the stated error terms.

For the second form of b'_k, the source instead uses the elementary Laplace-convolution identity

$$\frac{1}{2\pi\mathrm{i}} \oint \frac{e^{Lw}}{w \prod_{j=1}^{2k}(w - \alpha_j)}\, dw = \int_{\substack{x_j \geq 0 \\ \sum_j x_j \leq L}} e^{\sum_j \alpha_j x_j}\, d\mathbf{x}$$

and the simplex monomial evaluation

$$\int_{\substack{x_j \geq 0 \\ \sum_j x_j \leq L}} x_1 \cdots x_n \, d\mathbf{x} = \frac{L^{2k+n}}{(2k+n)!}.$$

Differentiating the $2k$ shifts, expanding by multiplicities $m_0, \ldots, m_k$, and applying the same squared-Vandermonde determinant produces a row-polynomial determinant. Its top-degree term is Vandermonde and its vanishing hyperplanes are $m_j - m_i = j - i$; therefore it equals $\prod_{i<j}(m_j - m_i + i - j)$. Factoring $1/(2k - i + m_i)!$ from row i gives (333). $\square$

The source then makes a separate number-theoretic conjecture. With

$$a_k = \prod_p \left(1 - \frac{1}{p}\right)^{k^2} \sum_{m=0}^{\infty} \left(\frac{\Gamma(m+k)}{m!\Gamma(k)}\right)^2 p^{-m}, \tag{336}$$

it conjectures, for fixed positive integer k,

$$\frac{1}{T}\int_0^T \left|\zeta'\left(\frac{1}{2} + \mathrm{i}t\right)\right|^{2k} dt \sim a_k b_k (\log T)^{k^2+2k}, \tag{337}$$

$$\frac{1}{T}\int_0^T |Z'(t)|^{2k}\, dt \sim a_k b'_k (\log T)^{k^2+2k}. \tag{338}$$

Equations (337)– (338) are Conjecture 1, not consequences of the proved random-matrix asymptotics [44, Conj. 1 and Remarks 1–7]. The authors state that they do not solve the complex-r moment problem, treat only leading terms, do not prove b_k or b'_k nonzero, and do not make their numerical heuristic $b_k \sim 4^{-k} \prod_{j=0}^{k-1} j!/(k+j)!$ rigorous. They also attribute the broader mixed completed moment to Hughes's thesis; that dependency remains unread until its own source is inspected. None of these statements supplies an individual-zero conclusion.

2.6.7 Dehaye's joint-moment programme and the exact angular–radial jet

Dehaye studies the same value–derivative data in an angular coordinate and then passes through Schur functions, Frobenius coordinates, and finite determinant sums [45]. None of the phases or derivative conventions may be silently identified: the angular derivative contains a translation by N, whereas the completed angular derivative is the pull-back of the CRS radial derivative. The following theorem records the complete coordinate map before any asymptotic or meromorphic continuation is used.

Theorem 2.24 (Dehaye angular jet, completed pull-back, and moment transport)**.** *Let $U \in \mathrm{U}(N)$ have lifted eigenangles $\theta_1, \ldots, \theta_N$ and put $z_j = e^{\mathrm{i}\theta_j}$. Retain*

$$\Phi_N(s) = \prod_{j=1}^{N}(s - z_j), \qquad V_N = \Phi_N(1), \qquad F_N = \frac{1}{N}\Phi'_N(1),$$

and let ρ_U be the phase in (322)*. Define Dehaye's angular functions by*

$$Z_U^{\mathrm{D}}(\vartheta) = \prod_{j=1}^{N} \left(1 - e^{\mathrm{i}(\theta_j - \vartheta)}\right), \tag{339}$$

$$\mathcal{V}_U^{\mathrm{D}}(\vartheta) = e^{\mathrm{i}N(\vartheta+\pi)/2} e^{-\mathrm{i}\sum_j \theta_j/2} Z_U^{\mathrm{D}}(\vartheta). \tag{340}$$

Then:

1. *On the lifted angular cover $s = e^{\mathrm{i}\vartheta}$,*

$$Z_U^{\mathrm{D}}(\vartheta) = e^{-\mathrm{i}N\vartheta} \Phi_N(e^{\mathrm{i}\vartheta}). \tag{341}$$

Consequently the value–angular-derivative jet is

$$\begin{pmatrix} Z_U^{\mathrm{D}}(0) \\ (Z_U^{\mathrm{D}})'(0) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\mathrm{i}N & \mathrm{i} \end{pmatrix} \begin{pmatrix} V_N \\ \Phi_N'(1) \end{pmatrix}, \qquad \begin{pmatrix} V_N \\ \Phi_N'(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ N & -\mathrm{i} \end{pmatrix} \begin{pmatrix} Z_U^{\mathrm{D}}(0) \\ (Z_U^{\mathrm{D}})'(0) \end{pmatrix}. \tag{342}$$

This is an everywhere invertible linear map of $\mathbb{C}^2$, with singleton fibres, empty exceptional locus, and no information loss.

2. *The phase is not discarded. From $\chi_U = (-1)^N (\det U)^{-1}$ one obtains exactly*

$$\rho_U = e^{\mathrm{i}N\pi/2} e^{-\mathrm{i}\sum_j \theta_j/2}.$$

With the branch of $s^{-N/2}$ transported along the same lifted angle,

$$\mathcal{V}_U^{\mathrm{D}}(\vartheta) = \mathcal{Z}_U(e^{\mathrm{i}\vartheta}). \tag{343}$$

Hence the two completed first jets satisfy

$$\begin{pmatrix} \mathcal{V}_U^{\mathrm{D}}(0) \\ (\mathcal{V}_U^{\mathrm{D}})'(0) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix} \begin{pmatrix} \mathcal{Z}_U(1) \\ \mathcal{Z}_U'(1) \end{pmatrix}, \qquad \begin{pmatrix} \mathcal{Z}_U(1) \\ \mathcal{Z}_U'(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -\mathrm{i} \end{pmatrix} \begin{pmatrix} \mathcal{V}_U^{\mathrm{D}}(0) \\ (\mathcal{V}_U^{\mathrm{D}})'(0) \end{pmatrix}. \tag{344}$$

These are again bijections with singleton fibres and no exceptional locus. The global map $\vartheta \mapsto e^{\mathrm{i}\vartheta}$ has fibres $\vartheta + 2\pi\mathbb{Z}$ on the unlifted circle; the completed square-root phase changes by $(-1)^N$ around one turn. Equation (343) *therefore lives on the retained lifted cover, not on a quotient which forgets that sign.*

3. *On $V_N \neq 0$, put $u_N = \Phi_N'(1)/\Phi_N(1)$ and retain the Cauchy trace $\mathcal{T}_N$ from* (310)*. Then*

$$\frac{(Z_U^{\mathrm{D}})'(0)}{Z_U^{\mathrm{D}}(0)} = \mathrm{i}(u_N - N) = -\frac{\mathcal{T}_N + \mathrm{i}N}{2}, \tag{345}$$

$$\frac{(\mathcal{V}_U^{\mathrm{D}})'(0)}{\mathcal{V}_U^{\mathrm{D}}(0)} = \mathrm{i}\left(u_N - \frac{N}{2}\right) = -\frac{\mathcal{T}_N}{2}. \tag{346}$$

The quotient exceptional locus is exactly $V_N = 0$; the underlying angular and completed jets in the first two items remain polynomial there.

4. Preserve Dehaye's three finite-N moment symbols:

$$|\mathcal{M}|_N^{\mathrm{D}}(2k,r) = \int_{\mathrm{U}(N)} |Z_U^{\mathrm{D}}(0)|^{2k} \left| \frac{(Z_U^{\mathrm{D}})'(0)}{Z_U^{\mathrm{D}}(0)} \right|^r dU, \tag{347}$$

$$(\mathcal{M})_N^{\mathrm{D}}(2k,r) = \int_{\mathrm{U}(N)} |Z_U^{\mathrm{D}}(0)|^{2k} \left(\frac{(Z_U^{\mathrm{D}})'(0)}{Z_U^{\mathrm{D}}(0)} \right)^r dU, \tag{348}$$

$$|\mathcal{V}|_N^{\mathrm{D}}(2k,r) = \int_{\mathrm{U}(N)} |\mathcal{V}_U^{\mathrm{D}}(0)|^{2k} \left| \frac{(\mathcal{V}_U^{\mathrm{D}})'(0)}{\mathcal{V}_U^{\mathrm{D}}(0)} \right|^r dU. \tag{349}$$

For $k \in \mathbb{Z}_{\geq 0}$ and real $r \geq 0$ with $2k - r > -1$, the exact determinant-tilt transports are

$$|\mathcal{M}|_N^{\mathrm{D}}(2k,r) = 2^{-r} \mathcal{Z}_{N,k} \mathbb{E}_{\Pi_{N,k}} (\mathcal{T}_N^2 + N^2)^{r/2}, \tag{350}$$

$$|\mathcal{V}|_N^{\mathrm{D}}(2k,r) = 2^{-r} \mathcal{Z}_{N,k} \mathbb{E}_{\Pi_{N,k}} |\mathcal{T}_N|^r. \tag{351}$$

For $r \in \mathbb{Z}_{\geq 0}$ in the same integrability range,

$$(\mathcal{M})_N^{\mathrm{D}}(2k,r) = \left(-\frac{1}{2}\right)^r \mathcal{Z}_{N,k} \mathbb{E}_{\Pi_{N,k}} (\mathcal{T}_N + \mathrm{i}N)^r. \tag{352}$$

In particular, for $r = 2h$, $|\mathcal{V}|_N^{\mathrm{D}}(2k,2h) = \mathcal{F}_N(k,h)$ exactly. The expectation map is a scalar observable, not an injective map on probability laws: its fibre consists of all laws having the displayed moment. On finite signed measures its kernel is the explicit hyperplane on which the integral of the displayed observable is zero.

Proof. For every j,

$$e^{-\mathrm{i}\vartheta}(e^{\mathrm{i}\vartheta} - z_j) = 1 - z_j e^{-\mathrm{i}\vartheta},$$

so multiplication proves (341). Evaluation and the chain rule give

$$Z_U^{\mathrm{D}}(0) = V_N, \qquad (Z_U^{\mathrm{D}})'(0) = \mathrm{i}\big(\Phi_N'(1) - NV_N\big),$$

which is the first matrix in (342). Direct matrix multiplication proves the displayed inverse.

Using $\det U = e^{\mathrm{i}\sum_j \theta_j}$ in the definition of ρ_U gives the retained phase formula. Substitution of (341) into (340) then gives

$$\mathcal{V}_U^{\mathrm{D}}(\vartheta) = \rho_U e^{-\mathrm{i}N\vartheta/2} \Phi_N(e^{\mathrm{i}\vartheta}) = \mathcal{Z}_U(e^{\mathrm{i}\vartheta}),$$

including the branch on the lifted cover. The chain rule at zero gives the diagonal jet map and its inverse.

Division of the angular derivative identity by V_N gives $\mathrm{i}(u_N - N)$. Equation (308) says $u_N = N/2 + \mathrm{i}\mathcal{T}_N/2$, proving (345). The completed identity follows either by differentiating its phase or from the diagonal jet map and (326).

Finally $|Z_U^{\mathrm{D}}(0)| = |V_N|$ and the exact Radon–Nikodym formula (298) changes Haar integration weighted by $|V_N|^{2k}$ into $\mathcal{Z}_{N,k}\Pi_{N,k}$. Substitution of the two logarithmic derivatives proves (350)–(352). The source's local integrability calculation at the divisor $V_N = 0$ is precisely $2k - r > -1$. $\square$

The finite algebra above immediately reproduces Dehaye's four triangular relations. For $h \in \mathbb{Z}_{\geq 0}$, the two finite-$N$ relations are

$$|\mathcal{M}|_N^{\mathrm{D}}(2k,2h) = \sum_{j=0}^{h} (\mathrm{i}N)^{h-j} \binom{h}{j} (\mathcal{M})_N^{\mathrm{D}}(2k, h+j), \tag{353}$$

$$|\mathcal{V}|_N^{\mathrm{D}}(2k,2h) = \sum_{j=0}^{h} \binom{h}{j} \left(-\frac{N^2}{4}\right)^{h-j} |\mathcal{M}|_N^{\mathrm{D}}(2k,2j). \tag{354}$$

After division by N^{k^2+2h} and passage to the corresponding existing limits, the remaining two relations are literally

$$|\mathcal{M}|^{\mathrm{D}}(2k,2h)=\sum_{j=0}^{h}\mathrm{i}^{h-j}\binom{h}{j}(\mathcal{M})^{\mathrm{D}}(2k,h+j), \tag{355}$$

$$|\mathcal{V}|^{\mathrm{D}}(2k,2h)=\sum_{j=0}^{h}\binom{h}{j}\left(-\frac{1}{4}\right)^{h-j}|\mathcal{M}|^{\mathrm{D}}(2k,2j). \tag{356}$$

All four are finite binomial transforms at integer h. Dehaye also records the separate integer identity

$$|\mathcal{V}|^{\mathrm{D}}(2k,2h)=\sum_{j=0}^{2h}\binom{2h}{j}\left(\frac{\mathrm{i}}{2}\right)^{j}(\mathcal{M})^{\mathrm{D}}(2k,2h-j), \tag{357}$$

but explicitly warns that its right side is not the analytic continuation in h of the left side: its derivation uses $|x|^{2h}=x^{2h}$ for real x and integer h [45, §2].

We now record the source theorem without merging proved random-matrix statements with its number-theoretic proposals. Put

$$g(k)=\frac{G(k+1)^2}{G(2k+1)},\qquad Y_r(u)=\prod_{\substack{1\le a\le r-1\\ a\ \mathrm{odd}}}(u^2-a^2)^{\alpha_a(r)},\qquad \alpha_a(r)=\left\lfloor\frac{-a+\sqrt{a^2+4r}}{2}\right\rfloor. \tag{358}$$

For those real parameters for which the integrals exist, define the scaled limits by retaining Dehaye's normalization:

$$\mathcal{A}(2k,r)=\lim_{N\to\infty}N^{-k^2-r}\mathcal{A}_N(2k,r),\qquad \mathcal{A}\in\{|\mathcal{M}|^{\mathrm{D}},(\mathcal{M})^{\mathrm{D}},|\mathcal{V}|^{\mathrm{D}}\}.$$

Theorem 2.25 (Dehaye's sourced rationality and determinant formulas)**.** *For $r\in\mathbb{Z}_{\ge 0}$, Dehaye proves, as a meromorphic function of k,*

$$(\mathcal{M})^{\mathrm{D}}(2k,r)=\left(-\frac{\mathrm{i}}{2}\right)^{r}g(k)\frac{X_r(2k)}{Y_r(2k)}, \tag{359}$$

where X_r and Y_r are even monic polynomials in $\mathbb{Z}[u]$ of equal degree and Y_r is the literal product in (358)*.*

For $h\in\mathbb{Z}_{\ge 0}$, he also proves the existence of integer polynomials $\widehat{X}_{2h}$ and $\widetilde{X}_{2h}$ and constants $\widehat{C}(h),\widetilde{C}(h)$ such that

$$|\mathcal{M}|^{\mathrm{D}}(2k,2h)=\widehat{C}(h)g(k)\frac{\widehat{X}_{2h}(2k)}{Y_{2h}(2k)}, \tag{360}$$

$$|\mathcal{V}|^{\mathrm{D}}(2k,2h)=\widetilde{C}(h)g(k)\frac{\widetilde{X}_{2h}(2k)}{Y_{2h}(2k)}, \tag{361}$$

with $\deg\widehat{X}_{2h}=\deg X_{2h}>\deg\widetilde{X}_{2h}$. The further assertions

$$\widehat{C}(h)=2^{-2h},\qquad \widetilde{C}(h)=\frac{(2h)!}{h!2^{3h}},\qquad \deg X_{2h}-\deg\widetilde{X}_{2h}=2h,$$

with both displayed numerators monic, are labelled conjectural in the source and are not part of the proved theorem.

Here is the complete algebraic-combinatorial route to that statement. If $\lambda = \left\{ \begin{smallmatrix} \vec{\mathbf{s}} \\ \vec{\mathbf{t}} \end{smallmatrix} \right\}$ has rank d, Dehaye uses the reduced El–Samra–King determinant

$$s_\lambda(1^R) = \det_{1 \le i,j \le d} \left[\frac{(R+s_i)!}{(R-t_j-1)!s_i!t_j!(1+s_i+t_j)} \right] \tag{362}$$

and its Cauchy-product expansion. For vectors with repeated coordinates the alternating extension is zero; thus no division by an informal ordering choice is made. The rectangle insertion is the exact identity

$$s_{\langle N^k \rangle \cup \left\{ \begin{smallmatrix} \overline{\mathbf{s}} \\ \overline{\mathbf{t}} \end{smallmatrix} \right\}}(1^{2k}) = s_{\langle N^k \rangle}(1^{2k}) s_{\left\{ \begin{smallmatrix} \overline{\mathbf{s}} \\ \overline{\mathbf{t}} \end{smallmatrix} \right\}}(1^{2k}) \times \prod_{i=1}^{d} \frac{(N-s_i)^{(k)}(k-t_i)^{(k)}}{(s_i+k+1)^{(k)}(N+t_i+1)^{(k)}}. \tag{363}$$

The dual Cauchy identity, Abel-regularized logarithmic derivative, Murnaghan– Nakayama rule, and $\mathrm{U}(N)$ character orthogonality first give

$$(\mathcal{M})_N^{\mathrm{D}}(2k,r) = (-\mathrm{i})^r \sum_{\boldsymbol{\mu} \in \mathbb{N}_+^r} \sum_{\lambda \subseteq \langle N^k \rangle} \chi_{\boldsymbol{\mu}}^{\lambda} s_{\langle N^k \rangle \cup \lambda}(1^{2k}). \tag{364}$$

The sum is finite after character orthogonality: only partitions between the $k \times N$ and $2k \times N$ rectangles survive. This finiteness also justifies removing the Abel parameter by dominated convergence precisely in the range $2k - r > -1$.

For completeness, the filled-structure count used to evaluate (364) is

$$\#T(r,d,\overline{\mathbf{p}},\overline{\mathbf{q}},v) = \sum_{\substack{\overline{\mathbf{s}},\overline{\mathbf{t}} \in \mathbb{N}^d \\ \sum_i t_i = v,\ d + \sum_i (s_i+t_i) = r}} \frac{r! \prod_i \binom{p_i}{s_i}\binom{q_i}{t_i}}{\prod_i s_i! t_i! \prod_{i=1}^d \left(d+1-i+\sum_{j=i}^d (s_j+t_j) \right)}. \tag{365}$$

Antisymmetrizing the unsorted Frobenius coordinates uses the exact determinant identity

$$\sum_{\sigma,\tau \in \mathfrak{S}_d} \frac{\operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)}{\prod_{i=1}^d \left(d+1-i+\sum_{j=i}^d (s_{\sigma(j)}+t_{\tau(j)}) \right)} = \prod_{i<j} (s_i-s_j)(t_i-t_j) \prod_{i,j=1}^{d} \frac{1}{1+s_i+t_j}. \tag{366}$$

The two sides have the same alternating zeros, the same poles, and the same degree; the proportionality constant is fixed to one by sending one s coordinate to infinity.

Define the finite generating series $\mathcal{E}_{N,k}(z) = \sum_{r>0} (\mathcal{M})_N^{\mathrm{D}}(2k,r)(\mathrm{i}z)^r/r!$. Dehaye's exact finite-$N$ determinant formula states that, coefficientwise for $r < 2k+1$,

$$\mathcal{E}_{N,k}(z) = s_{\langle N^k \rangle}(1^{2k}) \sum_{d \ge 1} \sum_{\vec{\mathbf{s}},\vec{\mathbf{t}} \in \mathbb{N}^d} \det_{i,j} \left[\frac{z^{1+s_i+t_j}}{s_i! t_j! (1+s_i+t_j)} \right] \times \sum_{\substack{\overline{\mathbf{p}} \in [0,N-1]^d \\ \overline{\mathbf{q}} \in [0,k-1]^d}} \det_{i,j} \left[\frac{k \binom{p_i}{s_i}\binom{q_j}{t_j}\binom{k+p_i}{p_i}\binom{k-1}{q_j}(N-p_i)^{(k)}(-1)^{q_j}}{(N+q_j+1)^{(k)}(1+p_i+q_j)} \right]. \tag{367}$$

Every coefficient is a finite sum because $s_i \le p_i$ and $t_j \le q_j$.

The single kernel remaining in the large-N limit is

$$H^{N,k,s,t} = s!t! \sum_{\substack{0\le p<N \\ 0\le q<k}} \frac{k(N-p)^{(k)}(-1)^q}{(N+q+1)^{(k)}(1+p+q)} \binom{k+p}{p}\binom{k-1}{q}\binom{p}{s}\binom{q}{t}, \tag{368}$$

$$H^{k,s,t} = \lim_{N\to\infty} N^{-1-s-t} H^{N,k,s,t} = \frac{1}{1+s+t}\frac{\Gamma(2k-t)\Gamma(k+s+1)}{\Gamma(k-t)\Gamma(2k+s+1)} \tag{369}$$

$$= \frac{1}{1+s+t}\prod_{a=-t}^{s}\frac{k+a}{2k+a}. \tag{370}$$

Initially k is an integer with $k > t$. The proof expands the q dependence and uses

$$\sum_{j=0}^{k-1}\sum_{q=0}^{k-1}(-1)^q\binom{k-1}{q}q^jX^j = (-1)^{k+1}(k-1)!X^{k-1};$$

all lower powers cancel. The remaining p sum is a Riemann sum for a beta integral. The resulting terminating sum is evaluated by the stated Saalschützian ${}_3F_2(1)$ identity, giving (369); gamma recurrence gives the product form.

Substitution into (367), together with $s_{\langle N^k\rangle}(1^{2k}) \sim g(k)N^{k^2}$, gives the coefficientwise asymptotic series

$$\sum_{r>0}(\mathcal{M})^{\mathrm{D}}(2k,r)\frac{(\mathrm{i}z)^r}{r!} = g(k)\sum_{d\ge1}\sum_{\vec{\mathbf{s}},\vec{\mathbf{t}}\in\mathbb{N}^d}\det_{i,j}\left[\frac{1}{s_i!t_j!(1+s_i+t_j)}\right]^2$$
$$\times\prod_{i,j=1}^{d}\frac{\Gamma(2k-t_j)\Gamma(k+s_i+1)}{\Gamma(k-t_j)\Gamma(2k+s_i+1)}z^{d+\sum_i(s_i+t_i)}. \tag{371}$$

At integer k this initially covers coefficients $r < 2k+1$. For each fixed r, the source invokes a Pochhammer-contour continuation on the integral side and Carlson's theorem on the difference to continue meromorphically in k. The growth verification is attributed also to Dehaye's thesis; it remains a human-source analytic proof boundary, not an axiom in the finite coordinate certificate.

Equation (370) makes every coefficient a rational function of k, satisfies $H^{k,s,t} = H^{-k,t,s}$, and exposes possible poles only at the odd coordinates $2k = \pm a$. Minimizing $r = d + \sum_i(s_i+t_i)$ subject to distinct Frobenius coordinates gives exactly the exponent $\alpha_a(r)$ in (358). The source fixes the monic normalization through

$$\sum_{\substack{d\ge1,\ \vec{\mathbf{s}},\vec{\mathbf{t}}\in\mathbb{N}^d \\ d+\sum_i(s_i+t_i)=r}}\det_{i,j}\left[\frac{1}{s_i!t_j!(1+s_i+t_j)}\right]^2 2^{-r} = \frac{1}{r!2^r}. \tag{372}$$

The final equality is left to the reader in the source. It is part of the published proof of the general theorem; independently, the attached coefficient certificate below verifies every resulting monicity and transformation through the complete provided range, not for arbitrary r.

There is a second, lossless presentation of the same coefficient data. Define Macdonald's ninth variation by

$$\widetilde{h}_m^{(R)} = \frac{(R-1)!}{(R+m-1)!m!}, \qquad \widetilde{s}_\lambda^{(R)} = \det\left[\widetilde{h}_{\lambda_i-i+j}^{(R-j+1)}\right].$$

Dehaye proves the Giambelli form

$$\widetilde{s}_\lambda^{(R)} = \det_{1\le i,j\le d}\left[\frac{\Gamma(R-t_j)/\Gamma(R+s_i+1)}{s_i!t_j!(1+s_i+t_j)}\right] \tag{373}$$

by an induction on hook partitions followed by the standard Giambelli exercise. Consequently

$$\sum_{r>0}(\mathcal{M})^{\mathrm{D}}(2k,r)\frac{(\mathrm{i}z)^r}{r!} = g(k)\sum_\lambda \widetilde{s}_\lambda^{(2k)} s_\lambda(1^k)z^{|\lambda|} \tag{374}$$

coefficientwise for fixed r and sufficiently large integer k, followed by the same meromorphic-continuation boundary. An attempted Gessel–Cauchy reduction ends with a matrix which is not Toeplitz, so the proposed Szegő step does not apply; the source records this as a failed simplification, not a theorem.

The arXiv source package contains the algorithm and all extended numerator data. The deterministic certificate `CERT-DEHAYE-2008-20260826-0001` parses every coefficient of

$$X_r\ (1\le r\le 60), \qquad \widehat{X}_{2h}\ (1\le h\le 30), \qquad \widetilde{X}_{2h}\ (1\le h\le 30)$$

over $\mathbb{Z}[u]$. It verifies that all 120 polynomials are even and monic, checks all sixty exact denominator-preserving transforms between the three moment families, matches all forty-five embedded table rows, and uses exact root isolation after $y=u^2$ to reproduce all thirty real-root counts. In particular $\widetilde{X}_{42}$ is the first supplied completed numerator not having all roots real: it has degree 44, forty real roots, and four nonreal roots. This is a finite exact statement about an attached polynomial, not a theorem that its observed root pattern persists.

Finally, the source's number-theoretic statements remain separate. With the same arithmetic factor $a(k)$ as in (336), it proposes

$$\mathcal{I}(2k,r) = a(k)|\mathcal{M}|^{\mathrm{D}}(2k,r), \tag{375}$$

$$\mathcal{J}(2k,r) = a(k)|\mathcal{V}|^{\mathrm{D}}(2k,r), \tag{376}$$

for the correspondingly normalized zeta and Hardy integrals. These are conjectural model transfers. The source reports a conditional number-theory value for $\mathcal{J}(2,1)$ and therefore proposes a value for $|\mathcal{V}|^{\mathrm{D}}(2,1)$, but explicitly fails to continue its integer-h formulas to that point. Its $k=1$ proposition only gives the horizontal meromorphic continuation of $(\mathcal{M})^{\mathrm{D}}(2k,r)$:

$$\begin{aligned}\sum_{r>0}(\mathcal{M})^{\mathrm{D}}(2,r)\frac{(\mathrm{i}z)^r}{r!} &= \sum_{d\ge1}\sum_{\vec{\mathbf{s}},\vec{\mathbf{t}}\in\mathbb{N}^d}\det_{i,j}\left[\frac{1}{s_i!t_j!(1+s_i+t_j)}\right]^2\\ &\quad\times\prod_{i,j=1}^d\left[2^{-\nu(t_j)}\frac{1-t_j}{2+s_i}\right]z^{d+\sum_i(s_i+t_i)}, \qquad \nu(0)=0, \quad \nu(t)=1\ (t\ge2),\end{aligned} \tag{377}$$

where $\nu(1)$ is immaterial because $1-t_j=0$. The vertical continuation in h remains missing. The source also does not solve noninteger r, does not obtain a full formula for $|\mathcal{V}|^{\mathrm{D}}(2k,r)$, and does not supply an individual-zero or endpoint conclusion.

2.6.8 Shifted-Hurwitz source boundary and exact finite $U(2)$ coordinates

The standard analytic input is narrower than the local consequences drawn from it. For $\Re a>0$, the Hurwitz function $\zeta(s,a)$ is meromorphic in s, with its only singularity a simple pole at $s=1$ of residue 1, and is analytic in a for fixed $s \neq 1$. Moreover

$$\zeta(0,a) = \frac{1}{2} - a, \qquad \zeta(s,a) = \sum_{n=0}^{\infty} \frac{(s)_n}{n!} \zeta(s+n)(1-a)^n \quad (|a-1| < 1), \tag{378}$$

where the second identity is an identity of meromorphic functions of s away from its displayed singular coordinates. These are Apostol's equations 25.11.1, 25.11.10, 25.11.13, and 25.11.17 in DLMF Chapter 25 [51, §25.11]. Substituting $a = 1 + ty$ in (378) gives the local shifted expansion with the literal factor $(-t)^n$; if $0 \le y \le L$ and $|t|L < 1$, its convergence is uniform in y on compact s-sets avoiding the resulting poles. Integration against a compact probability measure then gives the moment coefficients without changing variables or deleting any endpoint.

Fekih-Ahmed gives additional Hurwitz representations, but one proof in that source does not establish its stated full parameter range $0 < a \le 1$: the final domination writes $e^{-at} = e^{-t/2}e^{-(a-1/2)t}$ and evaluates the latter gamma integral, which requires $a > 1/2$ [55, proof of Thm. 1, eqs. (20)–(23)]. No statement below uses that estimate. The pole, value, parameter derivative, and Taylor coordinates are taken from the Apostol source just cited; the compact-measure and zero-deformation consequences remain independent local arguments.

Theorem 2.26 (Completed compact shifted-Hurwitz principal part)**.** *Let μ be a probability measure supported in $[0, L]$, let*

$$m_n = \int_0^L y^n \, d\mu(y), \qquad Z_{\mu,t}(s) = \int_0^L \zeta(s, 1+ty) \, d\mu(y),$$

and assume $1 + ty > 0$ on the support. Define

$$\Lambda_{\mu,t}(s) = \pi^{-s/2}\Gamma(s/2)Z_{\mu,t}(s), \qquad u = s(s-1), \quad v = 2s-1, \quad q = \frac{1}{u}, \quad \ell = \frac{v}{u}.$$

Then

$$\operatorname*{Res}_{s=1} \Lambda_{\mu,t}(s) = 1, \tag{379}$$

$$\operatorname*{Res}_{s=0} \Lambda_{\mu,t}(s) = -1 - 2tm_1, \tag{380}$$

$$\mathrm{PP}_{\{0,1\}} \Lambda_{\mu,t} = (1 + tm_1)q - tm_1\ell. \tag{381}$$

The sum of the two finite endpoint residues is $-2tm_1$.

Proof. The Hurwitz pole at $s = 1$ has residue 1, independent of its positive second argument. Because μ has total mass one, $\operatorname{Res}_{s=1} Z_{\mu,t} = 1$. The completing factor at $s = 1$ is $\pi^{-1/2}\Gamma(1/2) = 1$, giving the first residue.

At $s = 0$, (378) gives

$$Z_{\mu,t}(0) = \int_0^L \left(\frac{1}{2} - (1+ty)\right) d\mu(y) = -\frac{1}{2} - tm_1.$$

Since $\Gamma(s/2) = 2/s + O(1)$ and $\pi^{-s/2} = 1 + O(s)$, the residue of the completion is twice this value, proving (380).

Write the endpoint principal part as $a/s + b/(s-1)$. The two residues just computed give $a = -1 - 2tm_1$ and $b = 1$. The retained quotient and lift coordinates satisfy

$$q = \frac{1}{s-1} - \frac{1}{s}, \qquad \ell = \frac{1}{s-1} + \frac{1}{s}.$$

Thus $Aq + B\ell$ has endpoint residues $(-A+B, A+B)$. The coordinate map

$$(A, B) \longmapsto (-A+B, A+B)$$

has inverse $(a, b) \mapsto ((b-a)/2, (a+b)/2)$, zero kernel, full image, and singleton fibres. Substituting $(a, b) = (-1-2tm_1, 1)$ gives $A = 1+tm_1$ and $B = -tm_1$, which is (381). Adding the two residues proves the final statement. $\square$

For the zero-deformation calculation one needs the full moment lattice, not only m_1. When $|t|L < 1$, the uniform substitution into (378) gives

$$Z_{\mu,t}(s) = \sum_{n=0}^{\infty} (-t)^n \frac{(s)_n}{n!} m_n \zeta(s+n), \tag{382}$$

first in the common defining half-plane and then meromorphically in s.

Theorem 2.27 (All-order motion of a simple zeta zero)**.** *Let ρ be a simple nontrivial zero of ζ, and put*

$$F_n(s) = (-1)^n \frac{(s)_n}{n!} m_n \zeta(s+n), \qquad a_{n,j} = \frac{1}{j!} F_n^{(j)}(\rho).$$

For sufficiently small t, there is a unique holomorphic branch

$$\rho_\mu(t) = \rho + \sum_{N=1}^{\infty} c_N t^N$$

with $Z_{\mu,t}(\rho_\mu(t)) = 0$. Its coefficients are determined without discarded orders by

$$c_N = -\frac{1}{\zeta'(\rho)} \sum_{\substack{n,j \geq 0 \\ (n,j) \neq (0,1)}} a_{n,j} [t^N] \left[t^n \left(\sum_{m=1}^{N-1} c_m t^m \right)^j \right]. \tag{383}$$

Only finitely many summands contribute to a fixed N. In particular,

$$c_1 = m_1 \frac{\rho \zeta(\rho+1)}{\zeta'(\rho)}, \tag{384}$$

$$c_2 = -\frac{\frac{1}{2}\zeta''(\rho) c_1^2 + F_1'(\rho) c_1 + F_2(\rho)}{\zeta'(\rho)}, \tag{385}$$

where

$$F_1(s) = -s\, m_1 \zeta(s+1), \qquad F_2(s) = \frac{s(s+1)}{2} m_2 \zeta(s+2).$$

Proof. No shifted term $\rho+n$ equals 1. Hence (382) is holomorphic in a neighborhood of $(s, t) = (\rho, 0)$. At $t = 0$ it is $\zeta(s)$, and its s-derivative at ρ is the nonzero number $\zeta'(\rho)$. The analytic implicit-function theorem gives the unique holomorphic branch.

Set $w(t) = \rho_\mu(t) - \rho$. Taylor expansion in the retained s coordinate gives

$$0 = Z_{\mu,t}(\rho + w(t)) = \sum_{n,j \geq 0} a_{n,j} t^n w(t)^j.$$

Here $a_{0,0} = \zeta(\rho) = 0$ and $a_{0,1} = \zeta'(\rho)$. In the coefficient of t^N, the single term $(n, j) = (0, 1)$ contributes $\zeta'(\rho) c_N$. Every other term depends only on $c_1, \ldots, c_{N-1}$: if $n \geq 1$, the leading t^n lowers the required coefficient of w^j, while if $n = 0$ and $j \geq 2$, a monomial containing $c_N t^N$ also contains at least one further positive power of t. Thus w may be replaced by its truncation through order $N-1$ in the remaining sum. Solving the coefficient equation for c_N proves (383). The bounds $n \leq N$ and $j \leq N$ make the sum finite. Collecting orders one and two gives (385). $\square$

The theorem follows each simple zero only in its local implicit branch. It does not assert that the moment sequence is recovered injectively from the branches, that different branches remain disjoint globally, or that any line or region is preserved under the deformation.

The weak-measure reconstruction used by the local shifted-Hurwitz programme also has a terminal classical source. Bernstein's original two-page proof introduces

$$B_m f(x) = \sum_{j=0}^{m} f(j/m) \binom{m}{j} x^j (1-x)^{m-j}$$

and proves $B_m f \to f$ uniformly for every $f \in C([0,1])$ [48, eqs. (1)–(5)]. Retaining all measure coordinates gives the following exact consequence.

Proposition 2.28 (Bernstein pushforward reconstruction)**.** *Let ν be a probability measure on $[0,1]$ and set*

$$\lambda_{m,j} = \binom{m}{j} \int_0^1 x^j (1-x)^{m-j} \, d\nu(x), \qquad \nu_m = \sum_{j=0}^{m} \lambda_{m,j} \delta_{j/m}.$$

Then every $\lambda_{m,j}$ is nonnegative, $\sum_{j=0}^{m} \lambda_{m,j} = 1$, and $\nu_m \Rightarrow \nu$.

Proof. Nonnegativity is pointwise, and the binomial identity gives

$$\sum_{j=0}^{m} \lambda_{m,j} = \int_0^1 \sum_{j=0}^{m} \binom{m}{j} x^j (1-x)^{m-j} \, d\nu(x) = \int_0^1 1 \, d\nu = 1.$$

For every $f \in C([0,1])$,

$$\int f \, d\nu_m = \int_0^1 B_m f(x) \, d\nu(x).$$

Hence

$$\left| \int f \, d\nu_m - \int f \, d\nu \right| \leq \|B_m f - f\|_\infty \longrightarrow 0$$

by Bernstein's uniform approximation theorem. This is exactly weak convergence on the compact interval. $\square$

We next close the proof omitted by the local exact-$N = 2$ statement. Put $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, let U be Haar distributed in $U(2)$, and set

$$\Phi_2(z) = \det(zI - U), \qquad X_2 = |\Phi_2'(1)|^2.$$

Theorem 2.29 (Exact Haar $U(2)$ derivative law)**.** *There are independent coordinates*

$$R \sim \mathrm{Beta}\left(\frac{1}{2}, \frac{3}{2}\right), \qquad \phi \sim \mathrm{Unif}(\mathbb{T}),$$

for which

$$X_2 = 4\bigl(1 + R - 2\sqrt{R}\cos\phi\bigr) = 4\left|1 - \sqrt{R}\, e^{i\phi}\right|^2. \tag{386}$$

The density of R is

$$f_R(r) = \frac{2}{\pi} r^{-1/2}(1-r)^{1/2}, \qquad 0 < r < 1. \tag{387}$$

The folding map used to obtain (R, ϕ) has two-point fibres in the interior and single boundary fibres after the periodic endpoint is identified. The support of X_2 is $[0, 16]$, and $\mathbb{E}X_2 = 5$.

Proof. Write the eigenvalues as $z_j = e^{i\theta_j}$. The normalized Haar eigenangle density is

$$\frac{1}{2!(2\pi)^2} |e^{i\theta_1} - e^{i\theta_2}|^2 \, d\theta_1 \, d\theta_2 \tag{388}$$

by the Weyl integration formula [42, eq. (1.3)]. The torus automorphism

$$(\theta_1, \theta_2) \longmapsto (\alpha, \beta) = (\theta_1 - \theta_2, \theta_2)$$

has integer matrix $\left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$, determinant 1, inverse $(\alpha, \beta) \mapsto (\alpha + \beta, \beta)$, zero kernel, full image, and singleton fibres. Therefore it preserves torus Lebesgue measure. Since $|e^{i\alpha} - 1|^2 = 4\sin^2(\alpha/2)$, the joint density becomes

$$\frac{\sin^2(\alpha/2)}{2\pi^2} \, d\alpha \, d\beta.$$

Choose the half-open representative $\alpha \in [0, 2\pi)$ and put

$$\delta = \frac{\alpha}{2} \in [0, \pi), \qquad \phi_0 = \beta + \delta \pmod{2\pi}.$$

For fixed δ, the second coordinate is a circle translation, so its Jacobian is one and its inverse is $\beta = \phi_0 - \delta$. Thus δ and ϕ_0 are independent, ϕ_0 is uniform, and

$$f_\delta(\delta) = \frac{2}{\pi} \sin^2 \delta. \tag{389}$$

The two eigenvalues retain the exact centre-and-difference coordinates

$$z_1 = e^{i(\phi_0 + \delta)}, \qquad z_2 = e^{i(\phi_0 - \delta)}.$$

Consequently

$$\Phi_2'(1) = 2 - z_1 - z_2 = 2 - 2e^{i\phi_0} \cos \delta, \tag{390}$$

$$X_2 = 4\big(1 + \cos^2 \delta - 2 \cos \delta \cos \phi_0\big). \tag{391}$$

Now retain the sign lost by squaring the cosine in a translated phase coordinate:

$$r = \cos^2 \delta, \qquad \phi = \begin{cases} \phi_0, & 0 \le \delta \le \pi/2, \\ \phi_0 + \pi \pmod{2\pi}, & \pi/2 < \delta < \pi. \end{cases} \tag{392}$$

For $0 < r < 1$, the two inverse branches are

$$(\delta, \phi_0) = \left(\arccos \sqrt{r}, \phi\right), \qquad \left(\pi - \arccos \sqrt{r}, \phi - \pi\right), \tag{393}$$

and on either branch

$$\left|\frac{dr}{d\delta}\right| = 2\sqrt{r}\sqrt{1-r}.$$

Both branches have the density $(2/\pi)(1-r)$ inherited from (389). Summing their Jacobian contributions gives

$$2\frac{(2/\pi)(1-r)}{2\sqrt{r}\sqrt{1-r}} = \frac{2}{\pi}r^{-1/2}(1-r)^{1/2}.$$

This integrates to one because $B(1/2,3/2) = \Gamma(1/2)\Gamma(3/2)/\Gamma(2) = \pi/2$. The phase change on each branch is a circle translation. More explicitly, the joint pushforward density is

$$\frac{2}{\pi}r^{-1/2}(1-r)^{1/2}\,dr\,\frac{d\phi}{2\pi},$$

which proves both the beta law and independence. At $r = 0$ the two formulas in (393) meet at $\delta = \pi/2$; at $r = 1$, the representatives $\delta = 0$ and $\delta = \pi$ are the same periodic α-point. These null boundary coordinates therefore have single fibres on the torus, while every interior fibre has the two points displayed in (393).

By construction, $\cos\delta\cos\phi_0 = \sqrt{r}\cos\phi$. Substitution into (391) proves (386). The product measure has full topological support $[0,1]\times\mathbb{T}$, and at $r = 1$ the expression $8(1-\cos\phi)$ ranges over $[0,16]$, proving the support statement. Finally,

$$\mathbb{E}R = \frac{1/2}{1/2+3/2} = \frac{1}{4}, \qquad \mathbb{E}\cos\phi = 0,$$

so (386) gives $\mathbb{E}X_2 = 4(1+1/4) = 5$. $\square$

Corollary 2.30 (Exact $U(2)$ shifted-Hurwitz transform)**.** *For $t \geq 0$, define*

$$Z_{2,t}(s) = \mathbb{E}\,\zeta(s, 1+tX_2).$$

Then

$$Z_{2,t}(s) = \frac{2}{\pi}\int_0^1 r^{-1/2}(1-r)^{1/2}\left[\frac{1}{2\pi}\int_0^{2\pi}\zeta\big(s, 1+4t(1+r-2\sqrt{r}\cos\phi)\big)\,d\phi\right]dr. \tag{394}$$

For $\Re s > 1$, this equals

$$\frac{2}{\pi}\sum_{n=0}^{\infty}\int_0^1 r^{-1/2}(1-r)^{1/2}A_n(r)^{-s}{}_2F_1\left(\frac{s}{2},\frac{s+1}{2};1;\frac{B(r)^2}{A_n(r)^2}\right)dr, \tag{395}$$

where

$$A_n(r) = n+1+4t(1+r), \qquad B(r) = 8t\sqrt{r}. \tag{396}$$

Proof. Equation (394) is the pushforward formula from Theorem 2.29. If $\sigma = \Re s > 1$, then

$$\sum_{n=0}^{\infty}\mathbb{E}\left|(n+1+tX_2)^{-s}\right| \leq \sum_{n=0}^{\infty}(n+1)^{-\sigma} < \infty.$$

Fubini's theorem therefore permits the Hurwitz series to pass through both probability integrals. Its nth denominator is

$$n+1+tX_2 = A_n(r) - B(r)\cos\phi,$$

and

$$A_n(r) - B(r) = n+1+4t(1-\sqrt{r})^2 > 0, \tag{397}$$

so $0 \leq B(r)/A_n(r) < 1$.

For arbitrary complex s and $0 \le q < 1$, the binomial series is uniformly absolutely convergent in ϕ:

$$(1 - q\cos\phi)^{-s} = \sum_{k=0}^{\infty} \frac{(s)_k}{k!} q^k \cos^k \phi.$$

The odd powers average to zero, while

$$\frac{1}{2\pi} \int_0^{2\pi} \cos^{2m} \phi \, d\phi = \frac{(2m)!}{4^m (m!)^2}.$$

Using the duplication identity $(s)_{2m} = 4^m (s/2)_m ((s+1)/2)_m$ gives

$$\frac{1}{2\pi} \int_0^{2\pi} (A - B\cos\phi)^{-s} \, d\phi = A^{-s} \sum_{m=0}^{\infty} \frac{(s/2)_m ((s+1)/2)_m}{(m!)^2} \left(\frac{B^2}{A^2}\right)^m \tag{398}$$

$$= A^{-s} {}_2F_1\left(\frac{s}{2}, \frac{s+1}{2}; 1; \frac{B^2}{A^2}\right). \tag{399}$$

This also fixes the beta and Gauss-hypergeometric conventions used by (395); compare Forrester's equations (J1), (J1x), and (J2) [42]. Applying (399) term by term proves (395). □

Forrester's theorem is a scaled large-N joint-moment theorem. The source supplies (388) and the special-function conventions, but it does not state (386) or (395). The latter two are the independent finite-N pushforward and integration calculation just proved.

2.6.9 An exact finite $U(3)$ derivative-moment certificate

The following finite calculation is independent of the preceding large-N limits. Let $z_j = e^{i\theta_j}$, let e_r be the rth elementary symmetric polynomial in z_1, z_2, z_3, and put

$$p = 3 - 2e_1 + e_2 = \Phi_3'(1), \qquad p^\iota = p(z_1^{-1}, z_2^{-1}, z_3^{-1}). \tag{400}$$

Keating and Snaith give the normalized Haar $U(N)$ eigenangle density explicitly [23, Sec. 2.1, eq. (21)]. For a Laurent polynomial f at $N = 3$, integration of each torus character therefore gives the exact coordinate identity

$$\mathbb{E}_{U(3)} f = \frac{1}{3!} \operatorname{CT}\left[f \prod_{1 \le i < j \le 3} (1 - z_i/z_j)(1 - z_j/z_i) \right]. \tag{401}$$

Here CT is the coefficient of $z_1^0 z_2^0 z_3^0$; no limit or numerical quadrature occurs.

Proposition 2.31 (Exact first five $U(3)$ derivative moments)**.** *For $q = 1, 2, 3, 4, 5$,*

$$\begin{array}{c|ccccc} q & 1 & 2 & 3 & 4 & 5 \\ \hline \mathbb{E}_{U(3)} |\Phi_3'(1)|^{2q} & 14 & 375 & 15510 & 847280 & 55331424. \end{array} \tag{402}$$

Proof. By (400), complex conjugation on the eigenvalue torus sends p to p^ι. Thus the qth entry of (402) is exactly

$$\frac{1}{6} \operatorname{CT}\left[p^q (p^\iota)^q \prod_{1 \le i < j \le 3} (1 - z_i/z_j)(1 - z_j/z_i) \right]. \tag{403}$$

The certificate represents an element of $\mathbb{Z}[z_1^{\pm 1}, z_2^{\pm 1}, z_3^{\pm 1}]$ by its finite coefficient map $\mathbb{Z}^3 \to \mathbb{Z}$. Multiplication is the exact convolution

$$(a * b)_\gamma = \sum_{\alpha+\beta=\gamma} a_\alpha b_\beta, \qquad (p^\iota)_\alpha = p_{-\alpha}. \tag{404}$$

It constructs every factor in (403), extracts the zero exponent, and checks exact divisibility by 6. A second implementation expands the same expression independently with SymPy after multiplication by $(z_1 z_2 z_3)^{q+2}$; since every exponent is at least $-(q+2)$, the coefficient of $z_1^{q+2} z_2^{q+2} z_3^{q+2}$ is precisely the original Laurent constant term. Both exact integer implementations return the displayed five values. The $q = 1$ value also equals $1^2 + 2^2 + 3^2 = 14$, the character-orthogonality check.

The replay command, interpreter and SymPy versions, source and script hashes, complete values, resource observations, and nonclaims are recorded in `CERT-N3-CUE-20260825-0001`. Its executable is `certificates/n3_cue_constant_terms/verify_n3_cue_constant_terms.py`, with SHA–256 `adbebd6540f886dbae17aaa6e85bff5a62ede96a23d7aa56902128c3d38d02fd`. □

The proposition certifies only the finite $U(3)$ moments. It does not certify the later local quadratic trace algebra, a quotient/lift comparison, or any statement about zeta zeros.

2.6.10 Finite Weyl operator coordinates with an explicit inverse

Schwinger constructs the cyclic operators $N^{-1/2} U^m V^n$ and proves their completeness and Hilbert–Schmidt orthonormality [46, pp. 575–576]. Tolar restates that cyclic result and classifies finite quantum kinematics by the elementary-divisor tensor decomposition of a finite abelian configuration group [47, Secs. 2–3]. The following direct coordinate theorem retains an arbitrary finite abelian group and supplies the inverse map, fibres, and twisted product explicitly.

Let A be a finite abelian group of order r, let $\widehat{A} = \mathrm{Hom}(A, \mathbb{C}^\times)$, and let $H_A = \mathbb{C}[A]$ with basis $(e_x)_{x \in A}$. Define

$$T_a e_x = e_{a+x}, \qquad M_\chi e_x = \chi(x) e_x, \qquad W_{a,\chi} = T_a M_\chi. \tag{405}$$

Let $\mathbb{C}_\sigma[A \times \widehat{A}]$ denote the vector space with basis $[a, \chi]$ and the retained twisted product

$$[a, \chi] \star [b, \psi] = \chi(b)[a + b, \chi\psi]. \tag{406}$$

Proposition 2.32 (Finite Weyl coordinate isomorphism)**.** *The linear map*

$$\mathcal{W} : \mathbb{C}_\sigma[A \times \widehat{A}] \longrightarrow \mathrm{End}_{\mathbb{C}}(H_A), \qquad \mathcal{W}([a, \chi]) = W_{a,\chi}, \tag{407}$$

is an algebra isomorphism. Its inverse is

$$\mathcal{W}^{-1}(X) = \frac{1}{r} \sum_{a \in A} \sum_{\chi \in \widehat{A}} \mathrm{Tr}(W_{a,\chi}^* X)[a, \chi]. \tag{408}$$

In particular, $\ker \mathcal{W} = 0$*, its image is the entire endomorphism algebra, and every fibre is a singleton. The restriction to the coordinate hyperplane with zero* $[0, 1]$*-coefficient is a linear isomorphism onto* $\mathfrak{sl}(H_A)$*.*

Proof. For every $x \in A$,

$$M_\chi T_b e_x = \chi(b + x) e_{b+x} = \chi(b) T_b M_\chi e_x.$$

Hence

$$W_{a,\chi} W_{b,\psi} = \chi(b) W_{a+b,\chi\psi}, \tag{409}$$

which proves multiplicativity of (407). It also proves associativity of the displayed twisted product without deleting its cocycle: both bracketings of three basis coordinates acquire the scalar $\chi(b)\chi(c)\psi(c)$.

The Hilbert–Schmidt product can be computed on the retained basis:

$$\operatorname{Tr}(W_{a,\chi}^{*}W_{b,\psi}) = \sum_{x\in A}\langle W_{a,\chi}e_x, W_{b,\psi}e_x\rangle \tag{410}$$

$$= \delta_{a,b}\sum_{x\in A}\overline{\chi(x)}\psi(x) = r\,\delta_{a,b}\delta_{\chi,\psi}. \tag{411}$$

The last equality follows directly from character orthogonality: if $\eta = \psi\chi^{-1}$ is nontrivial and $\eta(y) \neq 1$, then translation by y gives

$$\sum_{x\in A}\eta(x) = \sum_{x\in A}\eta(x+y) = \eta(y)\sum_{x\in A}\eta(x),$$

so the sum is zero; for $\eta = 1$ it is r. Thus the r^2 operators $W_{a,\chi}$ are nonzero and mutually orthogonal. Since $|\widehat{A}| = |A| = r$ and $\dim_{\mathbb{C}}\operatorname{End}(H_A) = r^2$, they form a basis.

For an arbitrary $X \in \operatorname{End}(H_A)$, orthogonal projection onto that basis gives

$$X = \frac{1}{r}\sum_{a,\chi}\operatorname{Tr}(W_{a,\chi}^{*}X)W_{a,\chi},$$

which proves both inverse equations for (408). Those equations give the asserted kernel, image, and fibres.

Finally, $W_{0,1} = I$, while

$$\operatorname{Tr}(W_{a,\chi}) = r\,\delta_{a,0}\delta_{\chi,1}.$$

Therefore the $[0,1]$ coefficient of $\mathcal{W}^{-1}(X)$ is $r^{-1}\operatorname{Tr}(X)$. It vanishes exactly when X is traceless, proving the final restricted isomorphism. $\square$

The theorem is not a claim that the ordinary group algebra of $A\times\widehat{A}$ has been identified with the endomorphism algebra: the cocycle $\chi(b)$ in (406) is part of the domain. Nor is the arbitrary-A inverse (408) attributed to Schwinger or Tolar; it is proved above. Tolar's displayed Weyl composition line literally prints the first output coordinate as $\rho+\rho$ rather than an expression involving both inputs. That source line is not silently repaired and is not used in this proof. No map from (407) to a CUE measure, shifted-Hurwitz transform, or Kostka-labelled summand has been constructed.

2.6.11 Audited local source-block coordinates

The statement that the microscopic power is $|\mu| + |\nu| + s^2$ is correct under the exact hypotheses of (70). A local construction places the same integer in a separate coordinate problem. Its original TeX has a malformed second subscript in the display defining the block; the following is an explicit emendation to ν, not a silent normalization.

Let $W = \mathbb{C}^s$ and define, with every direct-sum coordinate retained,

$$J_\mu = \bigoplus_{i=1}^{s}\mathbb{C}^{\mu_i}, \qquad J_\nu^\vee = \bigoplus_{j=1}^{s}(\mathbb{C}^{\nu_j})^\vee, \qquad \mathfrak{B}_{\mu,\nu} = \operatorname{End}_{\mathbb{C}}(W)\oplus J_\mu \oplus J_\nu^\vee. \tag{412}$$

Proposition 2.33 (Exact source-block dimension)**.** *For $\ell(\mu) = \ell(\nu) = s$,*

$$\dim_{\mathbb{C}}\mathfrak{B}_{\mu,\nu} = s^2 + |\mu| + |\nu|. \tag{413}$$

Proof. The three displayed summands in (412) have respective dimensions s^2, $\sum_{i=1}^{s} \mu_i = |\mu|$, and $\sum_{j=1}^{s} \nu_j = |\nu|$. Dimension is additive on this retained finite direct sum, which gives (413). $\square$

For $\mu = \nu = (1^s)$, use the standard summand-coordinate identifications $J_\mu \cong W$ and $J_\nu^\vee \cong W^\vee$. They preserve the chosen decomposition and give the linear map

$$\Phi : \mathrm{End}(W) \oplus W \oplus W^\vee \longrightarrow \mathfrak{sl}(W \oplus \mathbb{C}), \qquad \Phi(A, u, \varphi) = \begin{pmatrix} A & u \\ \varphi & -\operatorname{Tr}(A) \end{pmatrix}. \tag{414}$$

Proposition 2.34 (Block-coordinate bijection)**.** *The map Φ in* (414) *is a complex-linear bijection. Its kernel is zero, its image is the whole codomain, and every fibre is a singleton. Pulling the matrix commutator back along Φ makes it a Lie-algebra isomorphism. In the fixed standard coordinates $W = \mathbb{C}^s$, the codomain is $\mathfrak{sl}_{s+1}(\mathbb{C})$.*

Proof. The trace of the displayed block matrix is $\operatorname{Tr}(A) - \operatorname{Tr}(A) = 0$, so Φ has the stated codomain. Every $X \in \mathfrak{sl}(W \oplus \mathbb{C})$ has a unique block decomposition

$$X = \begin{pmatrix} B & v \\ \psi & \lambda \end{pmatrix}, \qquad B \in \mathrm{End}(W), \quad v \in W, \quad \psi \in W^\vee, \quad \lambda \in \mathbb{C}.$$

The traceless equation is $\operatorname{Tr}(B) + \lambda = 0$, hence $\lambda = -\operatorname{Tr}(B)$. Therefore

$$\Psi(X) = (B, v, \psi) \tag{415}$$

is a two-sided inverse: $\Psi\Phi = \mathrm{id}$ and $\Phi\Psi = \mathrm{id}$. These equations give zero kernel, full image, and singleton fibres without a dimension argument. Finally define $[x, y]_{\mathfrak{B}} = \Psi([\Phi(x), \Phi(y)]_{\mathrm{mat}})$. The Lie identities and bracket preservation follow through the two-sided inverse, while the direct-sum coordinates remain recorded rather than being discarded. $\square$

This proves only the numerical dimension equality and the explicit block-coordinate bijection. Grover–Mezzadri–Simm construct neither (412) nor a map from $M_{\mu,\nu}$, its finite-N determinant, its Kostka-labelled summands, its Haar law, or its microscopic limit to either side of Φ. Those domain/codomain arrows and their compatibility equations remain open; equality of dimensions is not such an arrow.

The neighbouring split-zero construction uses the standard language of semirings and semimodules. As terminal human provenance for that language, Golan defines a basis and a free semimodule, proves the free universal property, proves that every semimodule is a quotient of a free one, and proves that projectivity is equivalent to being a retract of a free semimodule [56, Chap. 17, pp. 191–196, Props. 17.11, 17.12, 17.14, and 17.16]. These citations support exactly those definitions and results. The chapter contains neither the local semiring $G(R)$, its support character, a support-diagram classification, nor a zeta/CUE interface. The local object-level reconstruction and its functorial extension therefore remain separate proof and provenance obligations.

2.6.12 Exact deck and residue coordinates from the next sibling

The next independently content-read sibling uses

$$u = s(s-1), \qquad v = 2s - 1, \tag{416}$$

but its later quadratic-trace section reuses the letter v for $|c|^2$. The two occurrences are retained as different typed variables. In this subsection v means only the deck coordinate in (416).

Proposition 2.35 (Affine deck-coordinate isomorphism)**.** *Let*

$$\mathcal{C} = \{(u, v) \in \mathbb{C}^2 : v^2 = 1 + 4u\}.$$

The polynomial map

$$F : \mathbb{A}^1_s \longrightarrow \mathcal{C}, \qquad F(s) = \big(s(s-1), 2s-1\big)$$

is an isomorphism with inverse $G(u, v) = (v+1)/2$. It has zero information loss and singleton fibres. The involution $\sigma(s) = 1 - s$ is transported to $(u, v) \mapsto (u, -v)$. By contrast, the quotient coordinate $\pi(s) = u = s(s-1)$ has the two-point fibre $\{s, 1-s\}$ away from $s = 1/2$, and the singleton branch fibre $\{1/2\}$ over $u = -1/4$.

Proof. For every s,

$$(2s-1)^2 = 4s^2 - 4s + 1 = 1 + 4s(s-1),$$

so F lands in $\mathcal{C}$. Conversely, if $(u, v) \in \mathcal{C}$ and $s = (v+1)/2$, then

$$2s - 1 = v, \qquad s(s-1) = \frac{(v+1)(v-1)}{4} = \frac{v^2-1}{4} = u.$$

Thus $GF = \mathrm{id}$ and $FG = \mathrm{id}$, proving every assertion about F. Also

$$F(1-s) = \big(s(s-1), -(2s-1)\big).$$

Finally $\pi(s) = \pi(t)$ is equivalent to $(s-t)(s+t-1) = 0$, so $t = s$ or $t = 1-s$. These values coincide exactly when $s = 1/2$, whose image is $-1/4$. $\square$

Let $\mathcal{P}$ be the two-dimensional space of endpoint principal parts

$$\mathcal{P} = \left\{ \frac{a}{s} + \frac{b}{s-1} : a, b \in \mathbb{C} \right\}, \qquad q = \frac{1}{u}, \qquad \ell = \frac{v}{u}.$$

Proposition 2.36 (Complete endpoint-residue coordinate map)**.** *The ordered basis (q, ℓ) is related to the ordered endpoint-residue coordinates (a, b) by*

$$Aq + B\ell = \frac{-A+B}{s} + \frac{A+B}{s-1}. \tag{417}$$

Hence

$$(A, B) \longmapsto (a, b) = (-A+B, A+B)$$

is a linear isomorphism with inverse

$$A = \frac{b-a}{2}, \qquad B = \frac{a+b}{2}.$$

Its kernel is zero, its image is all of $\mathbb{C}^2$, and all fibres are singletons. The endpoint-residue vectors of $q\,ds$ and $\ell\,ds$ are respectively $(-1, 1)$ and $(1, 1)$. Moreover

$$\ell\,ds = \frac{du}{u} = d\log u$$

has residues $1, 1, -2$ at $s = 0, 1, \infty$, so its complete projective residue is zero.

Proof. Partial fractions give

$$q = \frac{1}{s-1} - \frac{1}{s}, \qquad \ell = \frac{1}{s-1} + \frac{1}{s},$$

which proves (417); solving its two scalar equations gives the displayed inverse. The inverse equations prove the kernel, image, and fibre statements. The same partial fractions give the two residue vectors. Since $du = (2s-1)ds = v\,ds$, one has $\ell\,ds = du/u$. The rational function $u = s(s-1)$ has simple zeros at 0 and 1 and a double pole at infinity. The residue of $d\log u$ is the corresponding order, giving $1, 1, -2$. □

These two propositions provide the exact coordinate morphisms inside the quotient/lift presentation. They do not map the split-zero semiring, a CUE measure, a Weyl operator, or the later quadratic trace cover to that presentation.

The same sibling states that the supported zero $e = 0_R$ is divisibility-prime for every nonzero commutative ring R. That generality is false. For $R = \mathbb{Z}/6\mathbb{Z}$, take $x = 2$ and $y = 3$. Then $xy = e$, so $e \mid xy$; but for every $g \in G(R)$, eg is e when g is supported and is τ when $g = \tau$. Therefore $e \nmid 2$ and $e \nmid 3$. The statement becomes correct after adding the hypothesis that R is an integral domain. The separate ideal identity $I^G J^G = (IJ)^G$ is not affected by this counterexample.

2.7 Exact first-peeled $N = 3$ quotient and physical trace cover

The following calculation closes the previously open input equations for the local first-peeled $N = 3$ construction. The Grover–Mezzadri–Simm paper and the accompanying author exposition supply the CUE setting and its scale regimes [28, 40]; they do not state the coordinates below. Accordingly, every displayed identity is derived here from the explicit local parametrization.

For $\beta_0 \in \mathbb{D} = \{\beta \in \mathbb{C} : |\beta| < 1\}$ define

$$\lambda = \frac{(1-2\overline{\beta_0})(1-\beta_0)}{(3-2\beta_0)(1-\overline{\beta_0})}, \qquad c = \frac{\overline{\beta_0}(1-\beta_0)}{(3-2\beta_0)(1-\overline{\beta_0})}, \qquad d = \frac{2-\beta_0}{3-2\beta_0}. \tag{418}$$

Put

$$u = |\lambda|^2, \qquad v = |c|^2, \qquad w = |d|^2. \tag{419}$$

The letter v in this subsection is precisely the nonnegative modulus coordinate $|c|^2$; no identification with the deck coordinate in (416) is being made.

Theorem 2.37 (Exact base quotient and fibres)**.** *Write*

$$\beta_0 = x + \mathrm{i}y, \qquad q_0 = x^2 + y^2, \qquad D_0 = 9 - 12x + 4q_0.$$

Then

$$u = \frac{1 - 4x + 4q_0}{D_0}, \qquad v = \frac{q_0}{D_0}, \qquad w = \frac{4 - 4x + q_0}{D_0} = \frac{1 - u + 2v}{2}. \tag{420}$$

Define

$$h(u,v) = 1 - 3u + 8v, \tag{421}$$
$$g(u,v) = 1 - 3u + 2v, \tag{422}$$
$$\Delta_3(u,v) = 18u + 32v - 1 - (9u - 16v)^2, \tag{423}$$

and the semialgebraic set

$$\Omega_3 = \left\{(u,v) \in \mathbb{R}^2 : u \ge 0,\ v \ge 0,\ g(u,v) > 0,\ \Delta_3(u,v) \ge 0\right\}. \tag{424}$$

The map

$$\pi_3 : \mathbb{D} \longrightarrow \Omega_3, \qquad \beta_0 \longmapsto (u(\beta_0), v(\beta_0))$$

is surjective. Its complete inverse-fibre coordinates are

$$x = \frac{1 - 9u + 32v}{4h}, \qquad q_0 = \frac{6v}{h}, \qquad y = \pm\frac{\sqrt{\Delta_3}}{4h}. \tag{425}$$

If $\Delta_3 > 0$, the fibre consists of the conjugate pair $x \pm \mathrm{i}\sqrt{\Delta_3}/(4h)$. If $\Delta_3 = 0$, the fibre is the single real point x. Thus π_3 loses exactly the sign of y away from the branch locus. Moreover $u < 1$ everywhere on Ω_3.

Proof. Since $|\beta_0| < 1$, neither $3 - 2\beta_0$ nor $1 - \overline{\beta_0}$ vanishes. Taking absolute values in (418) and cancelling only the equal nonzero factors $|1 - \beta_0| = |1 - \overline{\beta_0}|$ gives

$$|\lambda|^2 = \frac{|1 - 2\beta_0|^2}{|3 - 2\beta_0|^2}, \qquad |c|^2 = \frac{|\beta_0|^2}{|3 - 2\beta_0|^2}, \qquad |d|^2 = \frac{|2 - \beta_0|^2}{|3 - 2\beta_0|^2}.$$

Expanding the four squared moduli gives the first three fractions in (420). If $A_0 = 1 - 4x + 4q_0$, then

$$D_0 - A_0 + 2q_0 = 8 - 8x + 2q_0 = 2(4 - 4x + q_0),$$

which proves the formula for w without deleting any coordinate.

Substitution of $u = A_0/D_0$ and $v = q_0/D_0$ gives the exact identities

$$h = \frac{6}{D_0}, \qquad g = \frac{6(1 - q_0)}{D_0}, \qquad \Delta_3 = 16h^2y^2. \tag{426}$$

Here $D_0 = |3 - 2\beta_0|^2 > 0$ and $q_0 < 1$. Consequently $u, v \geq 0$, $g > 0$, and $\Delta_3 \geq 0$, so π_3 maps into Ω_3.

Conversely, let $(u, v) \in \Omega_3$. Since

$$h = g + 6v > 0,$$

all expressions in (425) are defined. Set

$$x = \frac{1 - 9u + 32v}{4h}, \qquad q_0 = \frac{6v}{h}, \qquad y^2 = \frac{\Delta_3}{16h^2}.$$

The following polynomial identity proves that the proposed radial and Cartesian coordinates agree:

$$96vh - (1 - 9u + 32v)^2 - \Delta_3 = 0, \tag{427}$$

because division by $16h^2$ yields $q_0 - x^2 - y^2 = 0$. Direct substitution also gives

$$9 - 12x + 4q_0 = \frac{6}{h}, \qquad 1 - 4x + 4q_0 = \frac{6u}{h}, \qquad 1 - q_0 = \frac{g}{h}. \tag{428}$$

The last identity and $g, h > 0$ show that $q_0 < 1$. Hence both choices $\beta_0 = x \pm \mathrm{i}y$ lie in $\mathbb{D}$, and the first two identities in (428) reconstruct the prescribed u and v through (420). This proves surjectivity. The same inverse formulas show that no other points occur in a fibre and give the stated branch behaviour.

Finally,

$$1 - u = \frac{D_0 - A_0}{D_0} = \frac{8(1 - x)}{D_0} > 0,$$

because $x < 1$ for every $x^2 + y^2 < 1$. Thus $u < 1$ on the entire image. $\square$

Define the residual phase coordinate

$$Z = c\,\overline{d}\,\overline{\lambda}. \tag{429}$$

Theorem 2.38 (Exact trace equations and lifted physical cover)**.** *For the coordinates above,*

$$Z + \overline{Z} = R(u,v) := \frac{9u^2 - 16uv - 10u - 16v + 1}{16}, \tag{430}$$

$$Z\overline{Z} = P(u,v) := uv\frac{1-u+2v}{2}. \tag{431}$$

Consequently Z satisfies

$$Z^2 - RZ + P = 0. \tag{432}$$

Put

$$X = 9u - 16v + 1, \qquad \Delta = 36u - X^2, \qquad S = P - \frac{R^2}{4}, \qquad J = Z - \frac{R}{2}. \tag{433}$$

Then

$$\Delta = \Delta_3, \qquad S = \frac{(u-1)^2\Delta}{1024}, \qquad \overline{J} = -J, \qquad J^2 = -S. \tag{434}$$

Let

$$\widetilde{\Omega}_3 = \left\{(u,v,J) \in \mathbb{R}^2 \times \mathrm{i}\mathbb{R} : (u,v) \in \Omega_3,\ J^2 + S(u,v) = 0\right\}. \tag{435}$$

The lifted map

$$\widetilde{\pi}_3 : \mathbb{D} \longrightarrow \widetilde{\Omega}_3, \qquad \beta_0 \longmapsto (u,v,J)$$

is a bijection. Its inverse is

$$\beta_0 = \frac{1 - 9u + 32v}{4h} + \frac{8J}{(u-1)h}, \tag{436}$$

where the second summand is purely imaginary; equivalently,

$$x = \frac{1 - 9u + 32v}{4h}, \qquad y = -\frac{8\mathrm{i}J}{(u-1)h}. \tag{437}$$

Complex conjugation on $\mathbb{D}$ is transported exactly to $(u,v,J) \mapsto (u,v,-J)$. The lifted map has singleton fibres and zero information loss.

Proof. Using (418), cancel the same nonzero $1-\beta_0$ and $1-\overline{\beta_0}$ factors as in the preceding proof and rationalize the remaining denominator. This gives

$$Z = \frac{N}{D_0^2}, \qquad N = \overline{\beta_0}(2-\overline{\beta_0})(1-2\beta_0)(3-2\beta_0). \tag{438}$$

With $\beta_0 = x + \mathrm{i}y$ and $q_0 = x^2 + y^2$, direct expansion gives

$$\operatorname{Re} N = -4q_0^2 + 16q_0x - 13q_0 - 6x^2 + 6x, \tag{439}$$

$$\operatorname{Im} N = 6y(x-1). \tag{440}$$

The trace numerator is fixed by the exact polynomial identity

$$9A_0^2 - 16A_0q_0 - 10A_0D_0 - 16q_0D_0 + D_0^2 = 32\operatorname{Re} N. \tag{441}$$

Substituting $u = A_0/D_0$ and $v = q_0/D_0$ into the definition of R therefore yields

$$R = \frac{32\,\mathrm{Re}\,N}{16D_0^2} = 2\,\mathrm{Re}\,Z = Z + \overline{Z},$$

which proves (430). The norm equation does not require an expansion:

$$Z\overline{Z} = |c|^2|d|^2|\lambda|^2 = uvw = uv\frac{1-u+2v}{2}.$$

Equation (432) follows from $Z^2 - (Z+\overline{Z})Z + Z\overline{Z} = 0$.

The base coordinate change has the explicit affine inverse

$$X = 9u - 16v + 1, \qquad v = \frac{9u+1-X}{16}. \tag{442}$$

If $Y = 9u - 16v$, then

$$\Delta_3 = 18u + 32v - 1 - Y^2 = 36u - (Y+1)^2 = \Delta.$$

Substitution of the inverse in (442) gives, without discarding v,

$$R = \frac{X(u+1) - 20u}{16}, \qquad P = \frac{u(u+9-X)(9u+1-X)}{256}. \tag{443}$$

Hence

$$4P - R^2 = \frac{(u-1)^2(36u - X^2)}{256}, \tag{444}$$

which is precisely the displayed factorization of $S = P - R^2/4$.

By (430), $J = (Z - \overline{Z})/2$ and therefore $\overline{J} = -J$. Equations (438) and (440) give

$$J = \mathrm{i}\,\frac{6y(x-1)}{D_0^2}.$$

Using $h = 6/D_0$ and

$$u - 1 = \frac{A_0 - D_0}{D_0} = \frac{8(x-1)}{D_0},$$

this becomes the coordinate formula

$$J = \frac{\mathrm{i}\,y(u-1)h}{8}. \tag{445}$$

Together with $y^2 = \Delta/(16h^2)$ from (425), it yields

$$J^2 = -\frac{y^2(u-1)^2h^2}{64} = -\frac{(u-1)^2\Delta}{1024} = -S.$$

Thus $\widetilde{\pi}_3$ lands in $\widetilde{\Omega}_3$.

For the inverse direction, take $(u, v, J) \in \widetilde{\Omega}_3$. The coordinate J is purely imaginary, and theorem 2.37 gives $h > 0$ and $u < 1$, so (437) is defined and has real y. Modulo $J^2 = -S$,

$$y^2 = \frac{-64J^2}{(u-1)^2h^2} = \frac{64S}{(u-1)^2h^2} = \frac{\Delta}{16h^2}.$$

The base inverse proof now shows that $\beta_0 = x + \mathrm{i}y$ lies in $\mathbb{D}$ and reconstructs u, v. Finally,

$$\frac{\mathrm{i}\,y(u-1)h}{8} = \frac{\mathrm{i}(-8\mathrm{i}J)}{8} = J,$$

so it also reconstructs the sheet coordinate. This proves both inverse compositions, hence bijectivity and singleton fibres. Replacing y by $-y$ in (445) fixes u, v and replaces J by $-J$, proving the conjugation statement. $\square$

To state the interface without identifying two coordinates that happen to use the same letters, let

$$\mathcal{C}_{a,b} = \{(a,b) \in \mathbb{C}^2 : b^2 = 1 + 4a\}, \qquad F_{a,b}(s) = \big(s(s-1), 2s-1\big).$$

This is a typed coordinate copy of the curve $\mathcal{C}$ in theorem 2.35. The renaming map

$$\rho_{a,b} : \mathcal{C} \longrightarrow \mathcal{C}_{a,b}, \qquad (u_{\mathrm{deck}}, v_{\mathrm{deck}}) \longmapsto (a,b) = (u_{\mathrm{deck}}, v_{\mathrm{deck}})$$

is an affine isomorphism with the displayed assignment reversed as its inverse. Thus the new letters prevent a type collision; they delete no coordinate and impose no normalization. Write

$$\kappa(u,v,J) = (u,v,-J), \qquad \tau(a,b) = (a,-b)$$

for the physical conjugation involution and the deck involution, respectively.

Theorem 2.39 (Exact carrier, equivariant sheet map, and fixed-locus obstruction)**.** *For $(u,v,J) \in \widetilde{\Omega}_3$, retain*

$$h = 1 - 3u + 8v$$

and define the recovered disc coordinate

$$\mathfrak{b}(u,v,J) = \frac{1 - 9u + 32v}{4h} + \frac{8J}{(u-1)h}. \tag{446}$$

The following statements hold.

1. *The carrier map*

$$\mathcal{M}_{\mathrm{car}} : \widetilde{\Omega}_3 \longrightarrow \mathcal{C}_{a,b}, \qquad \mathcal{M}_{\mathrm{car}}(u,v,J) = \big(\mathfrak{b}(\mathfrak{b}-1), 2\mathfrak{b}-1\big) \tag{447}$$

is a real-rational semialgebraic isomorphism onto

$$\mathcal{C}_{\mathbb{D}} := F_{a,b}(\mathbb{D}).$$

It has singleton fibres and zero information loss. Explicitly, for $(a,b) \in \mathcal{C}_{\mathbb{D}}$ set

$$\begin{aligned} \beta &= \frac{b+1}{2}, & D_\beta &= (3-2\beta)(3-2\overline{\beta}), & & \\ u_\beta &= \frac{(1-2\beta)(1-2\overline{\beta})}{D_\beta}, & v_\beta &= \frac{\beta\overline{\beta}}{D_\beta}, & h_\beta &= \frac{6}{D_\beta}, \\ J_\beta &= \frac{(\beta-\overline{\beta})(u_\beta - 1)h_\beta}{16}. & & & & \end{aligned} \tag{448}$$

Then $(a,b) \mapsto (u_\beta, v_\beta, J_\beta)$ is the complete inverse of (447).

The carrier does not intertwine the two involutions everywhere. Its exact intertwining locus is

$$\mathcal{M}_{\mathrm{car}}(\kappa(u,v,J)) = \tau(\mathcal{M}_{\mathrm{car}}(u,v,J)) \iff \operatorname{Re}\mathfrak{b}(u,v,J) = \frac{1}{2}. \tag{449}$$

2. *Define the retained imaginary coordinate and the sheet parameter by*

$$y_* = -\frac{8\mathrm{i}J}{(u-1)h}, \qquad s_{\mathrm{sh}} = \frac{1}{2} + \mathrm{i}y_* = \frac{1}{2} + \frac{8J}{(u-1)h}. \tag{450}$$

The sheet map

$$\begin{aligned}\mathcal{M}_{\mathrm{sh}} : \widetilde{\Omega}_3 &\longrightarrow \mathcal{C}_{a,b}, & \mathcal{M}_{\mathrm{sh}}(u,v,J) = F_{a,b}(s_{\mathrm{sh}})\\ &= \left(\frac{64J^2}{(u-1)^2h^2} - \frac{1}{4}, \frac{16J}{(u-1)h}\right) \\ &= \left(-\frac{1}{4} - y_*^2, 2\mathrm{i}y_*\right) \end{aligned} \tag{451}$$

is real-rational and exactly equivariant:

$$\mathcal{M}_{\mathrm{sh}} \circ \kappa = \tau \circ \mathcal{M}_{\mathrm{sh}}. \tag{452}$$

Its image and every fibre are

$$\operatorname{im} \mathcal{M}_{\mathrm{sh}} = \left\{ \left(-\frac{1}{4} - t^2, 2\mathrm{i}t\right) : -1 < t < 1 \right\}, \tag{453}$$

$$\mathcal{M}_{\mathrm{sh}}^{-1}\left(-\frac{1}{4} - t^2, 2\mathrm{i}t\right) = \left\{ \widetilde{\pi}_3(x + \mathrm{i}t) : -\sqrt{1-t^2} < x < \sqrt{1-t^2} \right\}. \tag{454}$$

Thus this map retains the sign and magnitude of the physical sheet coordinate y_* and loses exactly the real coordinate x along the displayed horizontal open interval. In particular, the fibre at $t = 0$ is the whole real diameter.

3. *Let*

$$p_3 : \widetilde{\Omega}_3 \to \Omega_3, \quad p_3(u,v,J) = (u,v), \qquad p_a : \mathcal{C}_{a,b} \to \mathbb{A}^1_a, \quad p_a(a,b) = a.$$

The induced base map is the explicit rational function

$$m_{\mathrm{sh}} : \Omega_3 \longrightarrow \mathbb{A}^1_a, \qquad m_{\mathrm{sh}}(u,v) = -\frac{1}{4} - \frac{\Delta_3(u,v)}{16h(u,v)^2}, \tag{455}$$

and the square commutes exactly:

$$\begin{tikzcd} \widetilde{\Omega}_3 \arrow[r, "\mathcal{M}_{\mathrm{sh}}"] \arrow[d, "p_3"'] & \mathcal{C}_{a,b} \arrow[d, "p_a"] \\ \Omega_3 \arrow[r, "m_{\mathrm{sh}}"] & \mathbb{A}^1_a. \end{tikzcd}$$

Its image is $(-5/4, -1/4]$. If $0 \le r < 1$, its fibre over $a = -1/4 - r$ is

$$m_{\mathrm{sh}}^{-1}\left(-\frac{1}{4} - r\right) = \left\{ \pi_3(x + \mathrm{i}\sqrt{r}) : -\sqrt{1-r} < x < \sqrt{1-r} \right\}, \tag{456}$$

where $\pi_3(x + \mathrm{i}\sqrt{r}) = \pi_3(x - \mathrm{i}\sqrt{r})$ records the already proved conjugation quotient rather than deleting that identification.

4. *There is no injective set map*

$$\mathcal{M} : \widetilde{\Omega}_3 \longrightarrow \mathcal{C}_{a,b}$$

that satisfies $\mathcal{M} \circ \kappa = \tau \circ \mathcal{M}$. Hence, in particular, there is no involution-equivariant semialgebraic, continuous, real-analytic, or algebraic isomorphism between these two specified involutive spaces.

Proof. By theorem 2.37, $h > 0$ and $u < 1$ throughout Ω_3. Every denominator in (446)– (455) is therefore nonzero. The inverse formula (436) says exactly that $\mathfrak{b}(u, v, J) = \beta_0$. Since $\widetilde{\pi}_3$ is a bijection and $F_{a,b}$ is the coordinate-renamed isomorphism proved in theorem 2.35, their composition $\mathcal{M}_{\mathrm{car}} = F_{a,b} \circ \widetilde{\pi}_3^{-1}$ is a bijection onto $F_{a,b}(\mathbb{D})$. The first two formulas in (448) are the forward coordinates (420), while the last is (445) written with $\beta - \overline{\beta} = 2\mathrm{i}\,\mathrm{Im}\,\beta$. They therefore give both inverse compositions explicitly.

The involution κ replaces $\mathfrak{b}$ by $\overline{\mathfrak{b}}$, whereas τ replaces the parameter of $F_{a,b}$ by $1 - \mathfrak{b}$. Because $F_{a,b}$ is injective, the two outputs agree precisely when

$$\overline{\mathfrak{b}} = 1 - \mathfrak{b},$$

which is equivalent to $2\,\mathrm{Re}\,\mathfrak{b} = 1$. This proves (449) without asserting global equivariance.

Equation (437) gives $y_* = y$. Therefore $-1 < y_* < 1$, and direct multiplication gives

$$s_{\mathrm{sh}}(s_{\mathrm{sh}} - 1) = -\frac{1}{4} - y_*^2, \qquad 2s_{\mathrm{sh}} - 1 = 2\mathrm{i}y_*.$$

This proves that (451) lands in the curve and gives its image. Under κ, y_* changes to $-y_*$, so s_{sh} changes to $1 - s_{\mathrm{sh}}$; applying $F_{a,b}$ gives (452). Since $F_{a,b}$ is injective, two source points have the same image exactly when they have the same y_*. The bijection $\widetilde{\pi}_3 : \mathbb{D} \to \widetilde{\Omega}_3$ then gives the complete fibres (454).

From (425),

$$y_*^2 = \frac{\Delta_3(u, v)}{16h(u, v)^2}.$$

Consequently the first coordinate of (451) is exactly (455), which proves the commuting square. The disc inequality $x^2 + y_*^2 < 1$ gives $0 \le y_*^2 < 1$, the stated image interval, and the complete base fibres (456).

It remains to prove the obstruction rather than infer it from the different presentations. The fixed locus of the physical involution is

$$\mathrm{Fix}(\kappa) = \{\widetilde{\pi}_3(x) : -1 < x < 1\},$$

because κ corresponds bijectively to conjugation on $\mathbb{D}$. The fixed locus of the target involution is

$$\mathrm{Fix}(\tau) = \{(-1/4, 0)\},$$

because $b = -b$ forces $b = 0$ and the curve equation then forces $a = -1/4$. For any equivariant set map $\mathcal{M}$, every point fixed by κ must map to a point fixed by τ. Thus $\mathcal{M}$ sends the entire open real diameter to the single point $(-1/4, 0)$ and cannot be injective. This proves the last assertion in every listed subcategory. □

Remark 2.40 (Exact scope of the interface). The carrier map is a lossless composition through the fully recovered β_0 coordinate; it is not asserted to encode CUE dynamics, Haar measure, or the derivative-moment recursion. The equivariant sheet map proves a genuine compatibility of involutions, but its exact fibres prove the accompanying loss of x. The fixed-locus argument proves that no alternative equivariant map can also be injective. In particular, the target coordinate b has been constructed explicitly and is not identified with the source modulus coordinate $v = |c|^2$. These maps and the obstruction establish the precise relation between the two presentations; they imply no critical-line or zeta-zero statement.

2.7.1 Complete reconstruction of the CUE–Mellin peeling programme

This subsection reconstructs the complete mathematical content of the three modern local CUE–Mellin TeX versions. Their exact source identities are:

> maintained source (1,591 lines):
> `4d4ed40c9968cc8cdfd28ebe129316ee04abcef47cef6683c01c47bd972daa4f`
> predecessor (1,032 lines):
> `21e450e24691c845c2d7d60b74e43555a854ccaf33407cbbe56d7dcc5a46b4a9`
> predecessor (877 lines):
> `7666d8a685914f91f0de96cb5e420c989001150a30a910bc33bb82249d61acea`
> separate 17-page PDF:
> `78eb0f5a545fa5c05b459aed0f973ae6fedcccd5615f84a914a5cd8179f47441`.

They are versions of one programme, but deletion from a later version is not treated as a mathematical retraction. In particular, the earlier auxiliary Gauss law, ratio-lattice count, and cubic trace calculation are audited below rather than lost. The PDF was compared through native PDF text with a fresh two-pass compilation of the maintained TeX. The mathematical body agrees; the 28 one-sided comparison lines are the two versions of fourteen table-of-contents, pagination, and date lines. Thus it is a rendered snapshot, not an additional mathematical source. No OCR is involved.

The human-source boundary is fixed first. Haar CUE, Weyl integration, and the derivative moment problem come from the published random-matrix literature [42, 43, 44, 45, 28, 32]. The exact independent Verblunsky variables come from Killip–Nenciu, with the derivative telescope derived in theorem 2.19 from their coordinates [35]. Bernstein's original construction supplies the approximation operator used in the measure recovery [48]; Hausdorff is the historical source for the finite-difference moment criterion [49]. The local formulas below are then either consequences proved here from those inputs or independently derived finite identities. No novelty priority is asserted.

The fixed-rank Mellin object and its exact support. For normalized Haar measure on $U(N)$, retain

$$\Phi_N(z) = \det(zI - U), \qquad X_N(U) = |\Phi_N'(1)|^2, \qquad \mathsf{M}_N(k) = \mathbb{E}X_N^k \tag{457}$$

whenever the last integral is finite. Equation (265) gives the exact convention map to $\Lambda_N(z) = \det(1 - zU^*)$: multiplication by the z-independent phase $(-1)^N\overline{\det U}$ has inverse $(-1)^N \det U$, zero kernel, singleton fibres, and preserves every derivative magnitude. Consequently, for $s \in \mathbb{Z}_{>0}$,

$$\mathsf{M}_N(s) = M_{(1^s),(1^s)}(1, N) \tag{458}$$

in the notation of Grover–Mezzadri–Simm [28, eq. (1.1)].

Theorem 2.41 (Exact fixed-rank support and Mellin representation)**.** *For every $N \geq 2$, the push-forward probability*

$$\nu_N = (X_N)_* \operatorname{Haar}_{U(N)}$$

has exact support

$$\operatorname{supp} \nu_N = [0, L_N], \qquad L_N = N^2 4^{N-1}. \tag{459}$$

For every $k \geq 0$,

$$\mathsf{M}_N(k) = \int_0^{L_N} t^k \, d\nu_N(t). \tag{460}$$

The same identity is tautological for any other real k for which the integral is finite; the theorem makes no additional negative-moment assertion.

Proof. If $z_1, \dots, z_N \in \mathbb{T}$ are the eigenvalues, then

$$\Phi_N'(1) = \sum_{m=1}^{N} \prod_{j \neq m} (1 - z_j).$$

Every factor has modulus at most 2, so $|\Phi_N'(1)| \leq N2^{N-1}$ and $0 \leq X_N \leq L_N$. At $U = I_N$, $\Phi_N(z) = (z-1)^N$ and $X_N = 0$; at $U = -I_N$, $\Phi_N(z) = (z+1)^N$ and $X_N = L_N$. The continuous image of the compact connected group $U(N)$ in $\mathbb{R}$ is a compact interval, hence it is exactly $[0, L_N]$.

It remains to distinguish image from measure support. If $y = X_N(U_0)$ and V is any open interval containing y, continuity makes $X_N^{-1}(V)$ an open neighbourhood of U_0. Normalized Haar measure has full support, so this neighbourhood has positive measure. Thus every image point belongs to the pushforward support. The reverse inclusion was already proved by the bound. Equation (460) is the definition of a pushforward integral. □

The complete moment cone and a constructive inverse. Push ν_N forward by $t \mapsto t/L_N$ and write the resulting probability as $\widetilde{\nu}_N$. Its integer moments are

$$c_{N,n} = \int_0^1 x^n \, d\widetilde{\nu}_N(x) = \frac{\mathsf{M}_N(n)}{L_N^n}. \tag{461}$$

Use the forward difference $\Delta c_{N,n} = c_{N,n+1} - c_{N,n}$ without changing this sign convention.

Theorem 2.42 (Finite differences, positivity, and constructive recovery)**.** *For all $n, r \geq 0$,*

$$(-1)^r \Delta^r c_{N,n} = \int_0^1 x^n (1-x)^r \, d\widetilde{\nu}_N(x) \geq 0. \tag{462}$$

For $m \geq 0$ and $0 \leq j \leq m$, define

$$\begin{aligned} \lambda_{N,m,j} &= \binom{m}{j} (-1)^{m-j} \Delta^{m-j} c_{N,j} \\ &= \binom{m}{j} \int_0^1 x^j (1-x)^{m-j} \, d\widetilde{\nu}_N(x), \end{aligned} \tag{463}$$

$$\widetilde{\nu}_{N,m} = \sum_{j=0}^{m} \lambda_{N,m,j} \delta_{j/m}. \tag{464}$$

Then the weights are nonnegative and sum to one, and

$$\widetilde{\nu}_{N,m} \Rightarrow \widetilde{\nu}_N. \tag{465}$$

Equivalently, the integer sequence in (461) *constructively recovers the complete compact measure and therefore every nonnegative real moment:*

$$\frac{\mathsf{M}_N(k)}{L_N^k} = \lim_{m \to \infty} \sum_{j=0}^{m} \left(\frac{j}{m} \right)^k \lambda_{N,m,j}, \qquad k \geq 0. \tag{466}$$

Proof. Expanding the difference gives

$$(-1)^r \Delta^r c_{N,n} = \sum_{q=0}^{r} \binom{r}{q} (-1)^q c_{N,n+q} = \int_0^1 x^n \sum_{q=0}^{r} \binom{r}{q} (-x)^q \, d\widetilde{\nu}_N(x),$$

and the binomial sum is $(1-x)^r$. This proves (462) and both expressions in (463). The weights are therefore nonnegative, while

$$\sum_{j=0}^{m} \lambda_{N,m,j} = \int_0^1 \sum_{j=0}^{m} \binom{m}{j} x^j (1-x)^{m-j}\, d\widetilde{\nu}_N(x) = 1.$$

For $f \in C([0,1])$,

$$\int f\, d\widetilde{\nu}_{N,m} = \int_0^1 \sum_{j=0}^{m} f(j/m) \binom{m}{j} x^j (1-x)^{m-j}\, d\widetilde{\nu}_N(x).$$

The inner sum is Bernstein's polynomial $B_m f$, and $\|B_m f - f\|_\infty \to 0$ by Bernstein's theorem [48, eqs. (1)–(5)]. This proves weak convergence. Finally take $f(x) = x^k$, which is continuous for $k \geq 0$. This is the specialization of the general coordinate-preserving reconstruction in theorem 2.28, now written explicitly in the CUE moments. □

Two positivity consequences retain useful information that the scalar moments alone can hide. For every $(a_0, \ldots, a_m) \in \mathbb{C}^{m+1}$,

$$\sum_{i,j=0}^{m} a_i \overline{a_j} \mathsf{M}_N(i+j) = \int_0^{L_N} \left| \sum_{i=0}^{m} a_i t^i \right|^2 d\nu_N(t) \geq 0, \tag{467}$$

so every moment Hankel matrix is positive semidefinite. Hölder's inequality gives

$$\mathsf{M}_N(\theta k_1 + (1-\theta)k_0) \leq \mathsf{M}_N(k_1)^\theta \mathsf{M}_N(k_0)^{1-\theta} \tag{468}$$

whenever both endpoint moments are finite and $0 \leq \theta \leq 1$.

Proposition 2.43 (Stieltjes generating function with exact radius)**.** *The series*

$$G_N(\tau) = \sum_{n \geq 0} \mathsf{M}_N(n) \tau^n$$

has radius L_N^{-1} and, for $|\tau| < L_N^{-1}$,

$$G_N(\tau) = \int_0^{L_N} \frac{d\nu_N(t)}{1 - \tau t}. \tag{469}$$

Proof. The upper support bound gives $\mathsf{M}_N(n) \leq L_N^n$. Conversely, for every $\varepsilon > 0$, exact support gives $\nu_N([L_N - \varepsilon, L_N]) > 0$ and hence

$$\mathsf{M}_N(n) \geq \nu_N([L_N - \varepsilon, L_N])(L_N - \varepsilon)^n.$$

Therefore $\lim_n \mathsf{M}_N(n)^{1/n} = L_N$. Cauchy–Hadamard proves the radius, and uniform convergence of the geometric series on $[0, L_N]$ proves (469). □

Several large-rank moment scales. For each fixed positive integer s, Grover–Mezzadri–Simm prove

$$\mathsf{M}_N(s) \sim b_s N^{s^2+2s}, \qquad b_s > 0 \tag{470}$$

as the $(1^s), (1^s)$ instance of their microscopic theorem [28, Thm. 1.2 and Sec. 2]; the complete source-level determinant audit is in section 2.6.1.

Theorem 2.44 (No deterministic scale carries two convergent nonzero moments)**.** *If $r, s \in \mathbb{Z}_{>0}$ and $r \neq s$, there are no $a_N > 0$ and $c_r, c_s \in (0,\infty)$ such that*

$$\frac{\mathsf{M}_N(r)}{a_N^r} \longrightarrow c_r, \qquad \frac{\mathsf{M}_N(s)}{a_N^s} \longrightarrow c_s. \tag{471}$$

For every fixed $m \geq 1$ and $\alpha > 0$,

$$\mathbb{P}(X_N \geq N^\alpha) \leq N^{m^2+2m-\alpha m+o(1)}. \tag{472}$$

For $0 < \theta < 1$,

$$\mathbb{P}\Big(X_N \geq (\theta b_m)^{1/m} N^{m+2}(1+o(1))\Big) \geq (1-\theta)^2 \frac{b_m^2}{b_{2m}} N^{-2m^2+o(1)}. \tag{473}$$

Proof. If (471) held, then $a_N^r \sim (b_r/c_r)N^{r^2+2r}$ and $a_N^s \sim (b_s/c_s)N^{s^2+2s}$. Taking logarithms and dividing by $\log N$ would make the same sequence $\log a_N/\log N$ converge to both $r+2$ and $s+2$.

Markov's inequality applied to X_N^m gives

$$\mathbb{P}(X_N \geq N^\alpha) \leq \mathsf{M}_N(m)N^{-\alpha m},$$

which becomes (472). For completeness, the inequality used in the other direction is obtained directly. If $Z \geq 0$, then

$$\mathbb{E}Z \leq \theta\mathbb{E}Z + \mathbb{E}[Z\mathbf{1}_{\{Z\geq\theta\mathbb{E}Z\}}] \leq \theta\mathbb{E}Z + (\mathbb{E}Z^2)^{1/2}\mathbb{P}(Z \geq \theta\mathbb{E}Z)^{1/2}.$$

After rearrangement this is the Paley–Zygmund bound [50]. Put $Z = X_N^m$ and use (470) at m and $2m$ to obtain (473). □

The maximal-support coordinate illustrates the same multiscale behaviour without an asymptotic theorem. Schur-character orthogonality gives

$$\mathsf{M}_N(1) = \sum_{j=0}^{N-1}(N-j)^2 = \frac{N(N+1)(2N+1)}{6}. \tag{474}$$

Indeed, $\Phi_N(z) = \sum_{j=0}^{N}(-1)^j e_j(U)z^{N-j}$ and the exterior-power characters satisfy $\mathbb{E}[e_j(U)\overline{e_\ell(U)}] = \delta_{j\ell}$; this is the same Schur-coordinate orthogonality used throughout the derivative-moment literature [33, 45, 28]. Hence $\mathbb{E}(X_N/L_N) \to 0$. Since $0 \leq X_N/L_N \leq 1$, Markov's inequality proves $X_N/L_N \to 0$ in probability and therefore weakly. This conclusion concerns the retained maximal-support scale L_N; it does not replace the distinct microscopic regimes in Grover–Mezzadri–Simm or Simm–Wei.

The measure-valued Verblunsky recursion. Define the state domain

$$\mathcal{A}_j = \{u \in \mathbb{C} : 1 - j + 2\operatorname{Re} u > 0\}, \qquad 0 \leq j \leq N-1. \tag{475}$$

It contains $u_0 = 0$. For $u \in \mathcal{A}_j$ retain, without changing variables,

$$T_j(\beta; u) = \frac{1 + u - \beta(j - \overline{u})}{1-\beta}, \tag{476}$$

$$\eta_j(u) = \frac{N-1-\overline{u}}{N-j+u}, \tag{477}$$

$$\gamma_j(\beta, u) = |1 - \beta\eta_j(u)|^2. \tag{478}$$

For $0 \le j \le N-2$, let

$$d\mu_{N-j-1}(\beta) = \frac{N-j-1}{\pi}(1-|\beta|^2)^{N-j-2}\mathbf{1}_{\mathbb{D}}(\beta)d^2\beta, \tag{479}$$

and take the terminal β_{N-1} uniformly on $\mathbb{T}$. These are exactly the independent pushforward laws proved in theorem 2.19, not fitted approximations.

Theorem 2.45 (State kernel, boundary law, and measure peel)**.** *The map $T_j(\beta;\cdot)$ sends $\mathcal{A}_j$ into $\mathcal{A}_{j+1}$ for every $\beta \in \mathbb{D}$, and $|\eta_j(u)| < 1$ on $\mathcal{A}_j$. Starting at a state $u \in \mathcal{A}_j$, define the future product recursively by*

$$Y_{N-1}(u) = |1 - e^{-\mathrm{i}\Theta}\eta_{N-1}(u)|^2, \tag{480}$$

$$Y_j(u) = \gamma_j(\beta_j, u)Y_{j+1}(T_j(\beta_j; u)), \qquad j \le N-2, \tag{481}$$

where Θ is uniform on $[0, 2\pi]$ and all future variables are independent. Let $\mu_{j,u}$ be the law of $Y_j(u)$. Then

$$\mu_{j,u} = \int_{\mathbb{D}} \sigma_{\gamma_j(\beta,u)}\mu_{j+1,T_j(\beta;u)}\, d\mu_{N-j-1}(\beta), \tag{482}$$

where $\sigma_a\mu = (x \mapsto ax)_*\mu$. This equality means equality against every bounded continuous test function and therefore specifies the map, its domain, and its codomain. Moreover,

$$\mathsf{M}_N(k) = N^{2k}\int y^k\, d\mu_{0,0}(y), \qquad k \ge 0. \tag{483}$$

At the boundary put $q = |\eta_{N-1}(u)|$. If $q > 0$, $\mu_{N-1,u}$ has density

$$\frac{\mathbf{1}_{[(1-q)^2,(1+q)^2]}(y)}{\pi\sqrt{(y-(1-q)^2)((1+q)^2-y)}}\, dy; \tag{484}$$

if $q = 0$, it is the atom δ_1. For every $k \ge 0$,

$$\int y^k\, d\mu_{N-1,u}(y) = {}_2F_1(-k,-k;1;q^2). \tag{485}$$

Proof. Put $w = 2u - j$. Direct substitution in (476) gives

$$1 + \mathrm{Re}(2T_j(\beta;u) - (j+1)) = \frac{1-|\beta|^2}{|1-\beta|^2}(1 + \mathrm{Re}\, w) > 0, \tag{486}$$

so the state domain is preserved. A second direct calculation gives

$$|N-j+u|^2 - |N-1-\overline{u}|^2 = (2N-j-1)(1-j+2\,\mathrm{Re}\, u) > 0, \tag{487}$$

which proves both nonvanishing of the denominator and $|\eta_j(u)| < 1$.

Conditioning (481) on $\beta_j = \beta$ gives, for bounded continuous f,

$$\mathbb{E}[f(Y_j(u)) \mid \beta_j = \beta] = \int f(\gamma_j(\beta,u)y)\, d\mu_{j+1,T_j(\beta;u)}(y).$$

Integration in β proves (482). The pathwise product in (264) proves (483); no moment recursion has been substituted for the underlying random variable.

At the boundary, $Y = 1+q^2-2q\cos\Theta$. The pushforward of uniform angle under $x = \cos\Theta$ has density $(\pi\sqrt{1-x^2})^{-1}$ on $(-1,1)$. The affine map $y = 1+q^2-2qx$ has inverse $x = (1+q^2-y)/(2q)$ and derivative magnitude $(2q)^{-1}$; substitution gives (484). If $q = 0$, the map is constantly one. Finally, $q < 1$ and the two binomial series converge absolutely:

$$\frac{1}{2\pi}\int_0^{2\pi}(1-qe^{-\mathrm{i}\theta})^k(1-qe^{\mathrm{i}\theta})^k\,d\theta = \sum_{n\geq 0}\frac{(-k)_n^2}{(n!)^2}q^{2n},$$

because angular integration retains exactly the equal Fourier indices. This is (485); $q = 0$ is included by direct evaluation. □

The exact transform carrier. Let $\mathcal{C}_c$ be the real vector space of finite signed compactly supported Borel measures on $[0,\infty)$, with additive convolution $*$ and unit δ_0. For $\mu \in \mathcal{C}_c$, $t \geq 0$, and $\operatorname{Re} s > 1$, define

$$\mathcal{L}[\mu](t) = \int e^{-tx}\,d\mu(x), \tag{488}$$

$$\mathcal{Z}_s[\mu](t) = \int \zeta(s, 1+tx)\,d\mu(x), \tag{489}$$

$$\mathcal{D}[\mu](t) = \int \psi(1+tx)\,d\mu(x). \tag{490}$$

Theorem 2.46 (Transported convolution algebra and Bose coordinate)**.** *The Laplace map is an injective algebra homomorphism*

$$\mathcal{L}[\mu * \nu] = \mathcal{L}[\mu]\mathcal{L}[\nu]. \tag{491}$$

For

$$(\mathcal{B}_s F)(t) = \frac{1}{\Gamma(s)}\int_0^\infty \frac{x^{s-1}}{e^x-1}F(tx)\,dx, \tag{492}$$

one has

$$\mathcal{B}_s\mathcal{L}[\mu] = \mathcal{Z}_s[\mu], \qquad -\operatorname{FP}_{s=1}\mathcal{Z}_s[\mu] = \mathcal{D}[\mu]. \tag{493}$$

Put

$$\mathfrak{C}_s = \{(\mathcal{L}[\mu], \mathcal{Z}_s[\mu], \mathcal{D}[\mu]) : \mu \in \mathcal{C}_c\}.$$

Then

$$\mathbf{T}_s : \mathcal{C}_c \longrightarrow \mathfrak{C}_s, \qquad \mu \longmapsto (\mathcal{L}[\mu], \mathcal{Z}_s[\mu], \mathcal{D}[\mu]), \tag{494}$$

is a bijection onto its displayed image, with zero kernel and singleton fibres. Transporting $*$ through this bijection makes $\mathfrak{C}_s$ a commutative unital algebra.

For $a \geq 0$ let $S_aF(t) = F(at)$. Then

$$\mathbf{T}_s[\sigma_a\mu] = S_a\mathbf{T}_s[\mu], \qquad \mathcal{B}_sS_a = S_a\mathcal{B}_s, \tag{495}$$

and the measure peel becomes the exact linear recursion

$$\mathbf{T}_s[\mu_{j,u}] = \int_{\mathbb{D}} S_{\gamma_j(\beta,u)}\mathbf{T}_s[\mu_{j+1,T_j(\beta;u)}]\,d\mu_{N-j-1}(\beta). \tag{496}$$

This is an averaging of dilation operators on a transported convolution algebra; it is not being identified with convolution multiplication.

Proof. Fubini gives

$$\mathcal{L}[\mu * \nu](t) = \iint e^{-t(x+y)}\, d\mu(x) d\nu(y) = \mathcal{L}[\mu](t)\mathcal{L}[\nu](t).$$

If $\mathcal{L}[\mu]$ vanishes identically and $\operatorname{supp}\mu \subset [0, R]$, then all derivatives at zero vanish, so $\int x^n\, d\mu(x) = 0$ for every n. Polynomial density in $C([0, R])$ and the Riesz representation theorem imply $\mu = 0$; equivalently this is the compact-support instance of Fourier–Stieltjes uniqueness [27]. Thus the first component of (494) already recovers μ.

The classical Mellin integral

$$\zeta(s, a) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1} e^{-(a-1)x}}{e^x - 1}\, dx, \qquad \operatorname{Re} s > 1, a > 0, \tag{497}$$

and the simple pole at $s = 1$ with residue one are recorded in the canonical Hurwitz source [51, §25.11(i),(vii)]. The constant term needed here can be derived without importing an unstated formula. For $a > 0$,

$$D(s, a) := \zeta(s, a) - \zeta(s) = \sum_{n=0}^{\infty} \big((n + a)^{-s} - (n + 1)^{-s}\big).$$

The series and all of its s-derivatives converge locally uniformly for $\operatorname{Re} s > 0$: on a compact a-interval the mean-value theorem bounds the rth derivative of the nth difference by $C_r(1 + \log^{r+1}(n + 1))(n + 1)^{-\operatorname{Re} s-1}$. Hence D is holomorphic near $s = 1$. Using the Laurent expansion $\zeta(s) = (s - 1)^{-1} + \gamma + O(s - 1)$ and the digamma series [53, eq. (5.7.6)],

$$D(1, a) = \sum_{n=0}^{\infty} \left(\frac{1}{n + a} - \frac{1}{n + 1} \right) = -\psi(a) - \gamma,$$

so indeed $\zeta(s, a) = (s - 1)^{-1} - \psi(a) + O(s - 1)$, locally uniformly for $a > 0$. Substitute $a = 1 + ty$ in (497), integrate in y, and use compact support to justify Fubini; this proves the first identity in (493). Integrating the Laurent expansion uniformly for a in the resulting compact interval gives

$$\mathcal{Z}_s[\mu](t) = \frac{\mu([0, \infty))}{s - 1} - \mathcal{D}[\mu](t) + O(s - 1),$$

which proves the finite-part identity. Bijectivity onto the displayed image and the transported product now follow from injectivity of $\mathcal{L}$.

Finally, every transform in (488)–(490) satisfies $F_{\sigma_a \mu}(t) = F_\mu(at)$. This proves the first dilation equation; direct substitution in (492) proves the second. Apply these identities to (482) to obtain (496). □

The complete noninteger $U(2)$ moment. The exact Haar coordinate theorem theorem 2.29 gives independent

$$R \sim \operatorname{Beta}\left(\frac{1}{2}, \frac{3}{2}\right), \qquad \phi \sim \operatorname{Unif}(\mathbb{R}/2\pi\mathbb{Z}), \qquad X_2 = 4|1 - \sqrt{R} e^{i\phi}|^2, \tag{498}$$

including the two-sheet folding map, its boundary fibres, and the exact support $[0, 16]$.

Corollary 2.47 (Closed $U(2)$ Mellin moment)**.** *For every real $k \geq 0$,*

$$\mathsf{M}_2(k) = 4^k\,{}_3F_2\left(\begin{matrix}-k,-k,\frac{1}{2}\\1,2\end{matrix};1\right). \tag{499}$$

Proof. Conditioning on $R = r$ and integrating the uniform phase gives

$$\frac{1}{2\pi}\int_0^{2\pi} |1-\sqrt{r}e^{\mathrm{i}\phi}|^{2k}\,d\phi = {}_2F_1(-k,-k;1;r). \tag{500}$$

Indeed, expand both binomial factors; the angular integral deletes precisely the unequal indices. The beta density in (387) therefore yields

$$\mathsf{M}_2(k) = \frac{2^{2k+1}}{\pi}\int_0^1 r^{-1/2}(1-r)^{1/2}{}_2F_1(-k,-k;1;r)\,dr. \tag{501}$$

For $k \geq 0$ the hypergeometric coefficients are nonnegative, so Tonelli applies. Termwise beta integration gives

$$\int_0^1 r^{n-1/2}(1-r)^{1/2}\,dr = B\left(n+\frac{1}{2},\frac{3}{2}\right) = \frac{\pi}{2}\frac{(1/2)_n}{(2)_n}.$$

Substitution in (501) is exactly the defining ${}_3F_2$ series in (499); the hypergeometric conventions and the terminating Chu–Vandermonde identity are recorded in the canonical reference [52]. Independently, the finite certificate below compares the terminating right-hand side with Weyl constant terms for $k = 0,\ldots,7$, obtaining

$$1, 5, 34, 285, 2766, 29666, 340740, 4108509.$$

□

The first peeled $U(3)$ compact law. Let $\beta_0 \in \mathbb{D}$ and retain the exact rational coordinates

$$\lambda = \frac{(1-2\overline{\beta_0})(1-\beta_0)}{(3-2\beta_0)(1-\overline{\beta_0})}, \qquad c = \frac{\overline{\beta_0}(1-\beta_0)}{(3-2\beta_0)(1-\overline{\beta_0})}, \qquad d = \frac{2-\beta_0}{3-2\beta_0}. \tag{502}$$

These are the same coordinates as (418); their base quotient, inverse, fibres, branch locus, and physical J-sheet are proved without omissions in theorems 2.37 and 2.38.

The connection to the state recursion is also exact. Starting with $u_0 = 0$ gives

$$u_1 = T_0(\beta_0;0) = \frac{1}{1-\beta_0}, \qquad \eta_1(u_1) = \lambda. \tag{503}$$

For the next disk variable $\beta \in \mathbb{D}$, put $u_2 = T_1(\beta;u_1)$. Substitution in (476) and (477) gives the retained phase relation

$$\eta_2(u_2) = -\frac{\beta-1}{\overline{\beta}-1}\frac{c+\overline{\beta}d}{1-\lambda\beta}. \tag{504}$$

The first factor on the right has modulus one. Hence, with

$$A_{\beta_0}(\beta) = |1-\lambda\beta|^2, \qquad B_{\beta_0}(\beta) = |c+\overline{\beta}d|^2, \tag{505}$$

one has $B/A = |\eta_2(u_2)|^2 < 1$. This proves the precise state-space arrow behind the local inner kernel; it is not an analogy between two formulas.

Theorem 2.48 (Local and global $U(3)$ compact measures)**.** *For $\phi \in [0, \pi]$ define*

$$\Theta_{\beta_0}(\beta, \phi) = A_{\beta_0}(\beta) + B_{\beta_0}(\beta) + 2\sqrt{A_{\beta_0}(\beta) B_{\beta_0}(\beta)} \cos \phi. \tag{506}$$

Then, for $k \geq 0$, the local inner kernel is the moment

$$\Psi_k(\beta_0) = \frac{1}{\pi} \int_{\mathbb{D}} A_{\beta_0}(\beta)^k {}_2F_1\left(-k, -k; 1; \frac{B_{\beta_0}(\beta)}{A_{\beta_0}(\beta)}\right) d^2\beta \tag{507}$$

$$= \frac{1}{\pi^2} \int_{\mathbb{D}} \int_0^{\pi} \Theta_{\beta_0}(\beta, \phi)^k \, d\phi \, d^2\beta. \tag{508}$$

For fixed β_0, β, if $B > 0$ the conditional pushforward in t has density

$$\frac{\mathbf{1}_{[(\sqrt{A}-\sqrt{B})^2, (\sqrt{A}+\sqrt{B})^2]}(t)}{\pi\sqrt{(t - (\sqrt{A} - \sqrt{B})^2)((\sqrt{A} + \sqrt{B})^2 - t)}} \, dt; \tag{509}$$

if $B = 0$ it is δ_A.

Define

$$\Xi(\beta_0, \beta, \phi) = |3 - 2\beta_0|^2 \Theta_{\beta_0}(\beta, \phi) \tag{510}$$

and push the probability measure

$$\frac{2}{\pi}(1 - |\beta_0|^2) d^2\beta_0 \otimes \frac{d^2\beta}{\pi} \otimes \frac{d\phi}{\pi} \tag{511}$$

forward under Ξ. The result is exactly ν_3 from theorem 2.41; in particular

$$\mathsf{M}_3(k) = \int_0^{144} t^k \, d\nu_3(t), \qquad \operatorname{supp} \nu_3 = [0, 144]. \tag{512}$$

Proof. Fix β_0, β and abbreviate $A = A_{\beta_0}(\beta) > 0$ and $r = \sqrt{B/A} < 1$. Replacing a uniform angle by its translate if necessary,

$$A + B + 2\sqrt{AB} \cos \phi = A|1 + re^{\mathrm{i}\phi}|^2.$$

The same diagonal Fourier calculation as in (500) gives

$$\frac{1}{\pi} \int_0^{\pi} (A + B + 2\sqrt{AB} \cos \phi)^k \, d\phi = A^k {}_2F_1(-k, -k; 1; B/A),$$

proving (508). The change of variables from $x = \cos \phi$ to $t = A + B + 2\sqrt{AB}x$ gives (509); when $B = 0$ the map is constant.

For the outer step, the first factor of the pathwise derivative peel is $|1 - (2/3)\beta_0|^2$. Multiplying by the retained 3^2 in (483) gives $|3 - 2\beta_0|^2$. The next disk law is uniform on $\mathbb{D}$, and the last circle law is precisely the angle just integrated. Therefore the pushforward of (511) is the law of X_3, proving the first assertion in (512). Its exact support now follows from theorem 2.41 with $L_3 = 9 \cdot 16 = 144$; the local source's direct inequality $\Xi \leq 144$ is consistent with, but weaker than, this equality. □

Integer kernels on the retained phase cover. Put

$$u=|\lambda|^2,\qquad v=|c|^2,\qquad w=|d|^2=\frac{1-u+2v}{2},\qquad Z=c\overline{d}\,\overline{\lambda}. \tag{513}$$

The complete (u,v,J) cover, including the inverse to β_0, is already proved in theorem 2.38; here the retained Z coordinate obeys

$$Z+\overline{Z}=R(u,v)=\frac{9u^2-16uv-10u-16v+1}{16},\qquad Z\overline{Z}=P(u,v)=uvw. \tag{514}$$

Choose $a=\sqrt{u}$, $b=\sqrt{v}$, and $\rho=\sqrt{w}$. If $ab\rho>0$, choose δ modulo 2π by $Z=ab\rho e^{-\mathrm{i}\delta}$; if $ab\rho=0$, $Z=0$ and δ is an immaterial fibre coordinate. For $s\in\mathbb{Z}_{\geq 0}$ and $0\leq m\leq s$, set

$$P_{s,m}(z)=(1-az)^{s-m}(b+\rho e^{-\mathrm{i}\delta}z)^m=\sum_{n=0}^{s}p_{s,m,n}z^n, \tag{515}$$

$$p_{s,m,n}=\sum_{j=0}^{\min(m,n)}(-a)^{n-j}b^{m-j}\rho^j e^{-\mathrm{i}j\delta}\binom{s-m}{n-j}\binom{m}{j}. \tag{516}$$

Theorem 2.49 (Hermitian square and complete trace descent)**.** *For every integer $s\geq 0$,*

$$\Psi_s=\sum_{m=0}^{s}\binom{s}{m}^2\sum_{n=0}^{s}\frac{|p_{s,m,n}|^2}{n+1}. \tag{517}$$

Define

$$T_0=2,\qquad T_1=R,\qquad T_{\ell+1}=RT_\ell-PT_{\ell-1}. \tag{518}$$

Then $T_\ell=Z^\ell+\overline{Z}^\ell$ and the phase-invariant kernel is the explicit polynomial

$$\Psi_s^\sharp(u,v)=\sum_{m=0}^{s}\binom{s}{m}^2\left(A_{s,m}+\sum_{\ell=1}^{\min(m,s-m)}(-1)^\ell T_\ell B_{s,m,\ell}\right), \tag{519}$$

$$A_{s,m}=\sum_{p=0}^{s-m}\sum_{q=0}^{m}\frac{\binom{s-m}{p}^2\binom{m}{q}^2}{p+q+1}u^p v^{m-q}w^q, \tag{520}$$

$$B_{s,m,\ell}=\sum_{p=0}^{s-m-\ell}\sum_{q=0}^{m-\ell}\frac{\binom{s-m}{p+\ell}\binom{s-m}{p}\binom{m}{q+\ell}\binom{m}{q}}{p+q+\ell+1}u^p v^{m-q-\ell}w^q. \tag{521}$$

It belongs to $\mathbb{Q}[u,v]$, has total degree exactly s, and

$$[v^s]\Psi_s^\sharp(u,v)=\frac{1}{s+1}\binom{2s+1}{s}. \tag{522}$$

Proof. For integer s, ${}_2F_1(-s,-s;1;x)=\sum_{m=0}^{s}\binom{s}{m}^2x^m$. Inserting this terminating identity in (507) turns the mth summand into

$$|1-a\beta|^{2(s-m)}|b+\rho e^{\mathrm{i}\delta}\overline{\beta}|^{2m}=|P_{s,m}(\beta)|^2.$$

The exact disk orthogonality

$$\frac{1}{\pi}\int_{\mathbb{D}}\beta^n\overline{\beta}^r\,d^2\beta=\frac{\delta_{nr}}{n+1}$$

proves (517) after substituting (516).

The quadratic equation $Z^2 - RZ + P = 0$ and its conjugate imply $Z^{\ell+1} + \overline{Z}^{\ell+1} = R(Z^\ell + \overline{Z}^\ell) - P(Z^{\ell-1} + \overline{Z}^{\ell-1})$; induction proves the trace recurrence. Now expand $|p_{s,m,n}|^2$. Terms with equal phase indices give (520). If the two indices differ by $\ell > 0$, the term and its conjugate combine to $(-1)^\ell(Z^\ell + \overline{Z}^\ell)$ times the coefficient in (521). Summing all allowed indices proves (519); because $w, R, P \in \mathbb{Q}[u,v]$, the recurrence puts every T_ℓ and therefore $\Psi_s^\sharp$ in $\mathbb{Q}[u,v]$.

For the degree bound, $A_{s,m}$ has degree at most s. Induction in (518) gives $\deg T_\ell \leq 2\ell$, while $\deg B_{s,m,\ell} \leq s - 2\ell$; hence every off-diagonal term also has degree at most s. To isolate the pure v^s coefficient, set $u = 0$. Then $w = v + 1/2$, $P = 0$, $R = 1/16 - v$, and $T_\ell = R^\ell$ for $\ell \geq 1$. At a fixed m, both the diagonal and off-diagonal contributions have v-degree at most m. Thus only $m = s$ can contribute to v^s, and its coefficient is

$$\sum_{q=0}^{s} \frac{\binom{s}{q}^2}{q+1} = \frac{1}{s+1} \sum_{q=0}^{s} \binom{s}{q}\binom{s+1}{q+1} = \frac{1}{s+1}\binom{2s+1}{s},$$

by Vandermonde's identity. It is nonzero, proving both (522) and exact degree s. □

The first four resulting polynomials, retained rather than referred to by a private source, are

$$\Psi_1^\sharp = \frac{5 + u + 6v}{4}, \tag{523}$$

$$\Psi_2^\sharp = \frac{47}{24} + \frac{11}{4}u + \frac{28}{3}v - \frac{11}{8}u^2 + 4uv + \frac{10}{3}v^2, \tag{524}$$

$$\begin{aligned} \Psi_3^\sharp = {} & \frac{105}{32} + \frac{465}{32}u + \frac{75}{2}v - \frac{225}{32}u^2 + \frac{195}{4}uv + \frac{405}{8}v^2 \\ & - \frac{65}{32}u^3 - \frac{15}{2}u^2v + \frac{225}{8}uv^2 + \frac{35}{4}v^3, \end{aligned} \tag{525}$$

$$\begin{aligned} \Psi_4^\sharp = {} & \frac{1}{320}\big(1759u^4 - 16224u^3v - 11812u^3 + 3024u^2v^2 - 12816u^2v - 2334u^2 \\ & + 50176uv^3 + 165984uv^2 + 117312uv + 18732u + 8064v^4 + 78848v^3 \\ & + 121296v^2 + 40752v + 1719\big). \end{aligned} \tag{526}$$

Proposition 2.50 (Exact sum-of-squares obstruction after descent)**.** *The polynomial $\Psi_2^\sharp$ is not a sum of squares in $\mathbb{R}[u,v]$, although its pullback to the retained phase cover is the Hermitian square sum* (517)*.*

Proof. If a real polynomial of degree at most two is a sum of polynomial squares, each summand has degree at most one: otherwise the sum of the nonzero squares of their top homogeneous parts would be the zero polynomial. Thus any such representation of $\Psi_2^\sharp$ would have a positive-semidefinite affine Gram matrix. Coefficient comparison in the fixed ordered basis $(1, u, v)$ gives the unique symmetric matrix

$$Q = \begin{pmatrix} 47/24 & 11/8 & 14/3 \\ 11/8 & -11/8 & 2 \\ 14/3 & 2 & 10/3 \end{pmatrix}, \qquad \Psi_2^\sharp = (1, u, v)Q(1, u, v)^\mathsf{T}.$$

Its leading 2×2 principal minor equals

$$\frac{47}{24}\left(-\frac{11}{8}\right) - \left(\frac{11}{8}\right)^2 = -\frac{55}{12} < 0.$$

Hence Q is not positive semidefinite. The conclusion concerns polynomial squares in the quotient coordinates; equation (517) records the exact map to the different Hermitian-square presentation. $\square$

The outer integral in (512), or independently the Weyl constant-term formula in theorem 2.31, gives

$$\mathsf{M}_3(1) = 14, \qquad \mathsf{M}_3(2) = 375, \qquad \mathsf{M}_3(3) = 15510, \qquad \mathsf{M}_3(4) = 847280, \qquad \mathsf{M}_3(5) = 55331424. \tag{527}$$

These five values have independent exact Python/SymPy and bounded Lean certificates recorded in `CERT-N3-CUE-20260825-0001` and `FORMAL-20260825-0010`.

The earlier Gauss coefficient law, with its moment domain repaired. For $m \in \mathbb{Z}_{\geq 1}$ and real $k > -1/2$, put

$$F_m(z) = {}_2F_1(-k, -k; m; z), \qquad a_n^{(m,k)} = \frac{(-k)_n^2}{(m)_n n!}. \tag{528}$$

Theorem 2.51 (Auxiliary exponent probability law)**.** *The numbers*

$$p_n^{(m,k)} = \frac{a_n^{(m,k)}}{F_m(1)}, \qquad n \geq 0, \tag{529}$$

form a probability distribution. If $Y_{m,k}$ has this law, then for $0 \leq \varepsilon \leq 1$,

$$\frac{F_m(1-\varepsilon)}{F_m(1)} = \mathbb{E}(1-\varepsilon)^{Y_{m,k}}. \tag{530}$$

If $k = s \in \mathbb{Z}_{\geq 0}$, the support is $\{0, \ldots, s\}$; if $k \notin \mathbb{Z}_{\geq 0}$, the support is infinite and

$$p_n^{(m,k)} \sim \frac{\Gamma(m)}{\Gamma(-k)^2 F_m(1)} n^{-(m+2k+1)}. \tag{531}$$

For $s \in \mathbb{Z}_{>0}$ the factorial moments are

$$\mathbb{E}Y_{m,s} = \frac{s^2}{m+2s-1}, \tag{532}$$

$$\mathbb{E}[Y_{m,s}(Y_{m,s}-1)] = \frac{s^2(s-1)^2}{(m+2s-2)(m+2s-1)}. \tag{533}$$

For $s = 0$, $Y_{m,0} = 0$ almost surely. If k is nonintegral, the same two formulas with s replaced by k hold respectively under the sharp convergence conditions $m + 2k > 1$ and $m + 2k > 2$.

Proof. The coefficients $a_n^{(m,k)}$ are nonnegative. Gauss' summation formula gives, because $m+2k > 0$,

$$F_m(1) = \sum_{n \geq 0} a_n^{(m,k)} = \frac{\Gamma(m)\Gamma(m+2k)}{\Gamma(m+k)^2} > 0 \tag{534}$$

[52, Gauss summation formula]. This proves normalization and (530). For integer s, $(-s)_n = 0$ exactly when $n > s$; for noninteger k no Pochhammer factor vanishes.

In the nonterminating case, gamma notation gives

$$p_n^{(m,k)} = \frac{\Gamma(m)}{\Gamma(-k)^2 F_m(1)} \frac{\Gamma(n-k)^2}{\Gamma(n+m)\Gamma(n+1)}.$$

The standard gamma-ratio asymptotic $\Gamma(n+a)/\Gamma(n+b) \sim n^{a-b}$ proves (531). Its power is strictly larger than 2 exactly when $m+2k>1$, and strictly larger than 3 exactly when $m+2k>2$, explaining the two nonintegral moment conditions.

Where the differentiated series converges at one,

$$F'_m(1) = \frac{k^2}{m}{}_2F_1(1-k, 1-k; m+1; 1),$$
$$F''_m(1) = \frac{k^2(k-1)^2}{m(m+1)}{}_2F_1(2-k, 2-k; m+2; 1).$$

Applying Gauss' formula to these two values and dividing by (534) gives

$$\frac{F'_m(1)}{F_m(1)} = \frac{k^2}{m+2k-1}, \qquad \frac{F''_m(1)}{F_m(1)} = \frac{k^2(k-1)^2}{(m+2k-2)(m+2k-1)}.$$

They are respectively $\mathbb{E}Y_{m,k}$ and $\mathbb{E}[Y_{m,k}(Y_{m,k}-1)]$. For positive integral s, the series terminates and the same identities follow from Chu–Vandermonde without convergence restrictions; $s=0$ is the separate point mass already stated. This repairs the earlier version's unrestricted display of the mean, without making any claim about the theorem's status outside that source. $\square$

The fixed-length multiplier set is a ratio-lattice image. For a finite real slice $\beta^{(1)}, \ldots, \beta^{(r)} \in (-1,1)$, set

$$q_i = \frac{1+\beta^{(i)}}{1-\beta^{(i)}} > 0, \qquad N_i = \begin{pmatrix} q_i & 1 \\ 0 & 1 \end{pmatrix}. \tag{535}$$

The top-left coordinate is a monoid morphism because

$$N_{i_1} \cdots N_{i_m} = \begin{pmatrix} \prod_{j=1}^m q_{i_j} & 1 + q_{i_1} + q_{i_1}q_{i_2} + \cdots + q_{i_1}\cdots q_{i_{m-1}} \\ 0 & 1 \end{pmatrix}. \tag{536}$$

Theorem 2.52 (Exact fixed-length multiplier rank)**.** *Let*

$$\mathcal{M}_m = \{q_{i_1} \cdots q_{i_m} : 1 \le i_j \le r\}, \qquad H = \left\langle \frac{q_1}{q_r}, \ldots, \frac{q_{r-1}}{q_r} \right\rangle \le \mathbb{R}^\times_{>0}, \tag{537}$$

and let $d = \operatorname{rank}_{\mathbb{Z}} H$. Then

$$|\mathcal{M}_m| = \Theta(m^d). \tag{538}$$

More precisely,

$$\mathcal{M}_m = q_r^m \left\{ \prod_{i=1}^{r-1} (q_i/q_r)^{n_i} : n_i \in \mathbb{Z}_{\ge 0}, \ \sum_{i=1}^{r-1} n_i \le m \right\}. \tag{539}$$

If $q_1, \ldots, q_r$ are multiplicatively independent, the displayed simplex map is injective and

$$|\mathcal{M}_m| = \binom{m+r-1}{r-1}. \tag{540}$$

For arbitrary positive q_i,

$$\operatorname{rank}_{\mathbb{Z}}\langle q_1, \ldots, q_r\rangle = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\log q_1, \ldots, \log q_r\}; \tag{541}$$

no algebraicity hypothesis is needed.

Proof. A word is determined at the scalar level by its Parikh vector $(n_1, \ldots, n_r)$ with $\sum_i n_i = m$. Replacing n_r by $m - \sum_{i<r} n_i$ gives (539). Since a finitely generated subgroup of $\mathbb{R}^\times_{>0}$ is torsion-free, $H \cong \mathbb{Z}^d$. Choose a basis of H. The exponent vector of each generator q_i/q_r is fixed, so every coordinate of every image of the integer simplex is bounded in absolute value by Cm. This gives $O(m^d)$ images.

Because the listed ratios generate a rank-d group, some d of their exponent vectors are linearly independent over $\mathbb{Q}$, hence over $\mathbb{Z}$. Restricting (539) to nonnegative powers of those d generators gives an injective set of $\binom{m+d}{d}$ products. This proves the lower bound, including the case $d = 0$. If all q_i are multiplicatively independent, equality of two fixed-length products forces equality of every Parikh coordinate, giving (540).

Finally, logarithm is an injective group homomorphism $\mathbb{R}^\times_{>0} \to (\mathbb{R}, +)$ with exponential as inverse on its image. Integral multiplicative relations among the q_i are therefore exactly integral additive relations among $\log q_i$; clearing denominators gives the same statement for rational relations. This proves (541). The earlier version counted with the rank of $\langle q_i \rangle$; fixed word length removes the common q_r^m direction, so the ratio lattice H is the exact object. $\square$

The cubic trace polynomial. Apply the invertible affine coordinate change already proved in (442):

$$X = 9u - 16v + 1, \qquad v = \frac{9u + 1 - X}{16}. \tag{542}$$

It has singleton fibres and deletes no coordinate.

Theorem 2.53 (Generic splitting group of the cubic descended kernel)**.** *Substitution of* (542) *into* (525) *gives*

$$\Psi_3^\sharp\left(u, \frac{9u + 1 - X}{16}\right) = \frac{5}{16384} Q_3(u, X), \tag{543}$$

where

$$\begin{aligned} Q_3(u, X) = &-7X^3 + (549u + 669)X^2 - (6645u^2 + 22746u + 8997)X \\ &+ 13783u^3 + 125949u^2 + 138933u + 19087. \end{aligned} \tag{544}$$

As a polynomial in X over $\mathbb{Q}(u)$, Q_3 is irreducible and its splitting field has Galois group S_3.

Proof. Equation (543) is direct coefficient collection after the displayed affine substitution. At $u = 0$,

$$Q_3(0, X) = -7X^3 + 669X^2 - 8997X + 19087.$$

Modulo 11, multiplication by the inverse of its nonzero leading scalar gives

$$f(X) = X^3 + 5X^2 + 3X - 5.$$

Its values at $0, 1, \ldots, 10$ are

$$6, 4, 7, 10, 8, 7, 2, 10, 4, 1, 7,$$

so it has no root in $\mathbb{F}_{11}$ and is irreducible there. Thus $Q_3(0, X)$ is irreducible over $\mathbb{Q}$. If $Q_3(u, X)$ factored over $\mathbb{Q}(u)$, Gauss' lemma would give a factorization in $\mathbb{Q}[u][X]$ after units were cleared. Since the leading X coefficient is the constant -7, the leading coefficients of both positive-X-degree factors would be nonzero constants; specialization at $u = 0$ would retain both degrees and contradict the preceding irreducibility. Therefore Q_3 is irreducible over $\mathbb{Q}(u)$.

For a cubic $aX^3 + bX^2 + cX + d$, substitution in $b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd$ gives

$$\operatorname{disc}_X Q_3 = 2^{18}3^3(290275u^6 + 731094u^5 + 2234709u^4 + 2884436u^3 + 3453789u^2 + 1714038u + 984779). \tag{545}$$

This discriminant is not a square in $\mathbb{Q}(u)$. Indeed, its nonzero specialization at $u = 0$ is $2^{18}3^3 \cdot 984779$; the final integer has digit sum 44 and is not divisible by 3, so the 3-adic valuation is the odd number 3. A rational-function square that is regular and nonzero at $u = 0$ specializes to a rational square there, which is impossible.

Irreducibility makes the Galois group a transitive subgroup of S_3. The discriminant is a square exactly when that group lies in A_3. Here it does not, so the transitive group is the full S_3. The exact substitution, modular values, fraction-field irreducibility, and discriminant expansion are independently replayed by the finite certificate below. □

Certificate and exact scope. The executable `certificates/cue_mellin_programme/verify_cue_mellin_finite_core.py` is recorded as `CERT-CUE-MELLIN-20260828-0001`. It uses exact integer, rational, Laurent-constant-term, and symbolic polynomial arithmetic. Two independent installed interpreters replay all 39 named checks: Python 3.13.9 with SymPy 1.13.1 and Python 3.13.1 with SymPy 1.13.3. It verifies the $U(2)$ Weyl/terminating-${}_3F_2$ equality through order seven; the exact $N = 3$ recursive-state arrow and its squared-modulus identity; the trace formula, exact degree, and leading coefficient through order eight; all four printed $\Psi_s^\sharp$; the Gram obstruction; the affine cubic transform; fraction-field irreducibility and the complete discriminant; and 42 terminating Gauss probability laws and their first two factorial moments.

The analytic and structural statements are not delegated to that bounded computation: theorems 2.41, 2.42, 2.44 to 2.49 and 2.51 to 2.53 contain their standard proofs and exact dependencies. The programme therefore supplies a fixed-rank compact Mellin measure, a constructive integer-to-real-moment inverse, an exact measure-valued derivative peel, a Laplace–Hurwitz–digamma transform carrier, complete $U(2)$ moments, and a phase-retaining $U(3)$ trace descent. The maps to the published CUE derivative framework are (458), (503), and (504); the map to the separate deck curve and its exact obstruction are already theorem 2.39. None of these finite-rank or transform statements supplies a map from an individual zeta zero to the CUE variables. Any such further implication remains a separately typed morphism and proof obligation.

2.7.2 An author-side CUE report: exact coordinates, proved overlap, and open obligations

The public research video [79] exposes three mathematically distinct workspaces. They concern, respectively, the joint law of the three zeros of Λ_4', a finite-rank OPUC–Painlevé treatment of the trace law and its Mellin transform, and an attempted analytic replacement of one certified inequality in a large-N bimodality argument. They are recorded separately below. The video is author-side evidence for the displayed claims and for the existence of the programmes; it is not used as a terminal theorem source. Twenty-five user-supplied frames were preserved in a private evidentiary cache with individual SHA-256 hashes. The images are not included in the public manuscript.

The $N = 4$ joint-law chart reported in the video. Use the convention

$$\Lambda_N(z) = \det(I - zU^*), \qquad U \sim \mathrm{Haar}(U(N)),$$

and let w_1, w_2, w_3 be the zeros of Λ_4', with whatever labelling is imposed by the unavailable source workspace. The displayed chart is

$$s_1 = |w_1|^2, \qquad s_2 = |w_2|^2, \qquad y_{12} = 2\,\mathrm{Re}(w_1\overline{w_2}), \tag{546}$$

and the third squared radius is reported to be recovered rationally as

$$s_3 = \sigma = -\frac{D_0}{D_1}. \tag{547}$$

The frame does not define D_0, D_1 or their domains. Therefore (547) is an exact transcription of a claimed coordinate map, not yet an evaluable definition in this manuscript.

The reported low-rank ladder is

$$\rho_2(s) = \frac{2}{\pi}\sqrt{\frac{1-s}{s}}, \qquad g_2(r) = \frac{4}{\pi}\sqrt{1-r^2},$$

followed at $N = 3$ by an algebraic law of degree four with the relative-angle coordinate reported as $y_{12} = 4st - s - t$. At $N = 4$ the claim is that the joint law is again a single algebraic density in the chart (546), while the branch sum in the three radii is only the fibre of the projection $(s_1, s_2, y_{12}) \mapsto (s_1, s_2, s_3)$.

The three displayed formulae are

$$|J| = \frac{s_3|\varepsilon|\,|C'(y)|\sqrt{4s_1s_2 - y^2}}{2|m|}, \tag{548}$$

$$p_4 = \frac{1}{3\pi^3}\sum \frac{|E_2|\,|\operatorname{Disc} Q|\,|\operatorname{Disc} P|^{1/2}}{|J|}, \tag{549}$$

$$f(s_1, s_2, y) = \frac{64}{3\pi^3}\frac{(s_1 + s_2 - y)V_{13}V_{23}}{D_1^4\sqrt{4s_1s_2 - y^2}}\sqrt{(1-s_1)(1-s_2)(-C_1)(1 + s_1s_2 - y)K_{13}K_{23}}. \tag{550}$$

No meanings are invented for $\varepsilon, C, m, E_2, P, Q, V_{13}, V_{23}, C_1, K_{13}, K_{23}, D_0, D_1$. Until their polynomial definitions, sign chambers, exceptional loci, and the source derivation are available, (548)–(550) remain candidate identities. In particular, the square roots in (550) require a specified real chamber or a branch convention.

The same report asserts that the support is the triangle-inequality region obtained from $|E_1| = 3|E_3|$, and distinguishes an internal discriminant wall, collinearity endpoints, and an excluded $E_2 = 0$ stratum. Across the internal wall the density is reported to remain bounded on the one-root side and to grow as

$$0.5235\,\epsilon^{-1/2}$$

on the three-root side. This numerical coefficient is not promoted to an exact constant. The report draws an obstruction to a radial algebraic or D-finite expression that omits the sign of the discriminant; that obstruction must be reproved from the local expansions on both sides of the wall before it is used.

The displayed validation used a 400,000-point Sturm-labelled calculation and a fresh 600,000-matrix exact-CUE$_4$ sample. It reports 256 bins per slab, slabwise L^1 discrepancies between 1.8% and 4.0% of slab mass, agreement of masses to 0.5%, no one of 1,300 bins outside 3.2 Poisson standard errors, and pooled-radial L^1 discrepancy 0.51% with reported theory mass 3.00025. These data identify the required future certificate. They do not replace the absent arrays, code, random seed, bin definitions, or exact arithmetic transcript. The video also reports two to four physical fibre points in the analogous $N = 5$ charts, so no persistence of the branch-free $N = 4$ mechanism is asserted.

The fixed-interior formulas that are already sourced. Let

$$M_{N,k}(z) = \mathbb{E}|\Lambda_N'(z)|^{2k}, \qquad 0 \le z < 1.$$

The video displays

$$\lim_{N\to\infty} M_{N,k}(z) = (1-z^2)^{-k^2-2k}\Gamma(k+1)\,{}_1F_1(-k;1;-k^2z^2), \qquad \operatorname{Re} k > -1. \tag{551}$$

This is not a new formula from the video. It is the Simm–Wei theorem already audited in section 2.6.2, written in Kummer coordinates [32, Thm. 1.1]. Indeed, Kummer's identity gives

$$e^{-k^2z^2}\,{}_1F_1(k+1;1;k^2z^2) = {}_1F_1(-k;1;-k^2z^2),$$

which is the exact morphism between the two presentations. For integral $k \ge 0$ this becomes $k!L_k(-k^2z^2)$. The passage $N \to \infty$ under the defining integral and the negative-moment range are justified by the uniform-integrability argument in the human source; they are not inferred from pointwise convergence.

Differentiation at $k = 0$ yields the next displayed formula. The linear term of $(-k^2-2k)\log(1-z^2)$ is $-2k\log(1-z^2)$, the hypergeometric factor has no linear term, and $\Gamma'(1) = -\gamma$. Hence

$$\lim_{N\to\infty} \mathbb{E}\log|\Lambda_N'(r)| = -\log(1-r^2) - \frac{\gamma}{2}. \tag{552}$$

The sourced uniform-integrability statement is again essential to the interchange. Applying the radial Jensen derivative gives

$$\lim_{N\to\infty} \mathbb{E}\mathcal{N}_N(r) = r\frac{d}{dr}\Big(-\log(1-r^2) - \frac{\gamma}{2}\Big) = \frac{2r^2}{1-r^2}. \tag{553}$$

Differentiating the count and dividing by the circumference $2\pi r$ gives the areal density

$$\varrho_\infty(r) = \frac{2}{\pi(1-r^2)^2}. \tag{554}$$

This is the zero intensity of the hyperbolic Gaussian analytic function with covariance $(1-z\overline{w})^{-2}$. In the audited Gaussian field convention $\Lambda_\infty' = \Lambda_\infty G'$; a video expression involving $\Lambda_\infty(-G)$ is not used until its G is explicitly defined.

The exact trace-convention map. For the convention fixed above,

$$\Lambda_N'(0) = -\operatorname{Tr}(U^*) = -\overline{\operatorname{Tr} U}, \tag{555}$$

whereas a frame writes $-\operatorname{Tr} U$. There is no contradiction and no silent normalization: the involution

$$\mathcal{C} : U(N) \longrightarrow U(N), \qquad U \longmapsto U^*,$$

is a measurable bijection with inverse $\mathcal{C}$, preserves Haar measure, and sends $\operatorname{Tr} U$ to $\overline{\operatorname{Tr} U}$. Thus it gives an exact law-preserving coordinate morphism, and in particular

$$|\Lambda_N'(0)| = |\operatorname{Tr} U| \circ \mathcal{C}.$$

Phase-sensitive identities still require the displayed map.

The OPUC–Painlevé system and its normalization. Tracy and Widom use the symbol

$$f_{\mathrm{TW}}(e^{i\theta};t) = \exp(t(e^{i\theta}+e^{-i\theta})) = \exp(2t\cos\theta)$$

and coefficients $U_n(t)$ [74, Secs. I and V]. The video uses $f_{\mathrm{vid}}(e^{i\theta};\tau) = \exp(\tau\cos\theta)$. The exact coordinate map is

$$t = \frac{\tau}{2}, \qquad U_n(t) = \alpha_{n-1}(\tau). \tag{556}$$

Tracy–Widom's discrete recurrence and equation (5.1), under (556) and the chain rule, become

$$\alpha_{n+1}+\alpha_{n-1} = -\frac{2(n+1)}{\tau}\frac{\alpha_n}{1-\alpha_n^2}, \tag{557}$$

$$\frac{d\alpha_n}{d\tau} = \frac{1}{2}(1-\alpha_n^2)(\alpha_{n+1}-\alpha_{n-1}). \tag{558}$$

This proves the coordinate equivalence of the two displayed relations. Analytic continuation to imaginary τ retains $1-\alpha_n^2$; replacing it by $1-|\alpha_n|^2$ would change the analytic system.

The video additionally reports, for $L = \log D_N$,

$$L'' = \frac{1}{2}(1-\alpha_{N-1}^2)(1-\alpha_{N-2}\alpha_N), \tag{559}$$

$$(\tau L')' = \tau(1-\alpha_{N-1}^2) = \tau\frac{D_{N+1}D_{N-1}}{D_N^2}, \tag{560}$$

$$a'' = -\frac{a(a')^2}{1-a^2} - \frac{a'}{\tau} + \frac{N^2 a}{\tau^2(1-a^2)} - a(1-a^2), \qquad a = \alpha_{N-1}, \tag{561}$$

$$(\tau\sigma''-\sigma')^2 = 4\sigma'(\tau\sigma'-2\sigma)(\tau-\sigma') + 4N^2(\tau-\sigma')^2, \qquad \sigma = \tau L'. \tag{562}$$

Equations (559)–(562) remain reported until their derivation is checked against the exact determinant normalization and initial germ. Every logarithmic equation is understood only where $D_N \neq 0$, or meromorphically after that qualification. The determinant size, Fourier sign, and value of α_{-1} must also be fixed before using the system.

The continuation of (562) can nevertheless be checked algebraically. Put $\tau = i\rho$ and $\Sigma(\rho) = \sigma(i\rho)$. Then

$$\sigma' = -i\Sigma', \qquad \sigma'' = -\Sigma'', \qquad \tau - \sigma' = i(\rho+\Sigma').$$

Substitution, followed only by multiplication by -1, gives precisely

$$(\rho\Sigma''-\Sigma')^2 + 4(\rho+\Sigma')\left[\Sigma'(\rho\Sigma'-2\Sigma) - N^2(\rho+\Sigma')\right] = 0. \tag{563}$$

The reported initial germ is

$$\Sigma_N(\rho) = \rho\,\partial_\rho \log\det[J_{j-\ell}(\rho)]_{j,\ell=0}^{N-1} = -\frac{\rho^2}{2} + O(\rho^4).$$

This Bessel Toeplitz determinant is the radial characteristic function of $\operatorname{Tr} U$; it is not the confluent microscopic Bessel kernel used in the Simm–Wei derivative-moment theorem, and no identification between them is made.

Trace-Mellin claims still requiring their proof artifact. The video reports a cutoff Mellin representation for $\mathbb{E}|\operatorname{Tr} U|^{2k}$, with convergence half-planes $\operatorname{Re} k > -1/2$ for $N = 2$ and $\operatorname{Re} k > -1$ for $N \geq 3$. It also reports

$$\mathbb{E} \log |\operatorname{Tr} U| = -\frac{1}{2} \quad (N = 2)$$

and, at $N = 3$,

$$\mathbb{E} \log |\operatorname{Tr} U| = \frac{\mathrm{Cl}_2(\pi/3)}{\pi} + \frac{7\sqrt{3}}{16\pi} - \frac{5}{6}. \tag{564}$$

The second value is described as PSLQ-identified and checked to 35 digits; without the one-dimensional integral reduction it remains a candidate exact evaluation.

The reported two-cluster stationary-phase expansion is

$$\sum_{n+m=N} c_{n,m} \rho^{-(n^2+m^2)/2} \exp\left(i \left[(n-m)\rho - \frac{\pi(n^2 - m^2)}{4} \right] \right) \left(1 + O(\rho^{-1})\right), \tag{565}$$

with

$$c_{n,m} = \frac{4^{nm} G(n+2) G(m+2)}{(2\pi)^{N/2} n! m!}. \tag{566}$$

It predicts exponents $N^2/4$ for even N and $(N^2+1)/4$ for odd N, and singular circles $|w| = N - 2m$ in the law of $\operatorname{Tr} U$, with local powers reported as $|r - (n-m)|^{(n^2+m^2)/2 - 3/2}$. These stationary-phase and singular-support claims are explicitly open in the displayed workspace. Neither the fixed-interior Simm–Wei decay estimate nor the boundary-derivative Mellin programme proves them, because those are different observables and phases.

Bothner–Wei provide a separate σ-Painlevé V to σ-Painlevé III$'$ interface for derivative-moment Hankel determinants [77]. Their object is not identified with the trace Toeplitz determinant above. Equality of the word "Painlevé" supplies no morphism.

The boundary bimodality problem. The third workspace uses the Mezzadri radius coordinate

$$x = (N-1)(1-r). \tag{567}$$

It reports a completed theorem that the corresponding radial density converges locally vaguely to q_∞ and that q_∞, and eventually every finite-N density, has at least two local maxima separated by a local minimum. The source proof is unavailable, so this endpoint is not yet a theorem of the present manuscript.

The coordinate must not be silently identified with the squared-radius Simm–Wei parameter c_{SW} defined by $r^2 = 1 - c_{\mathrm{SW}}/N$. Combining the two definitions gives the exact finite-N map

$$c_{\mathrm{SW}} = N(1 - r^2) = \frac{2N}{N-1} x - \frac{N}{(N-1)^2} x^2, \qquad c_{\mathrm{SW}} \longrightarrow 2x. \tag{568}$$

Moreover, the Simm–Wei theorem quoted above is uniform on compact subsets strictly inside the disc; it is not a boundary-uniform theorem in (567). Thus (568) is a coordinate bridge, not a transfer of convergence.

The sole machine-certified node in the reported proof is said to be

$$T_{1.5,0.4} > T_{2.2,0.2}, \qquad T_{2.7,0.3} > T_{2.2,0.2}. \tag{569}$$

The cited certificate uses 2,939 outward-rounded Arb determinant nodes on $0 \leq R \leq 20$, about forty bridge and tail rows extending to $R = 1600$, and reported margins greater than 1.27×10^{-4} and 7.9×10^{-3}. None of these numbers is independently certified here because the ledger is absent.

The exact analytic replacement requested in the video is the following problem.

Proof obligation 2.54 (Analytic persistence kernels)**.** Construct continuous compactly supported probability kernels K_E, K_V, K_O on $(0, \infty)$ whose support interiors are pairwise disjoint and ordered, and prove, without a machine-certified inequality,

$$\int_0^\infty K_E(x)q_\infty(x)\,dx > \int_0^\infty K_V(x)q_\infty(x)\,dx, \qquad \int_0^\infty K_O(x)q_\infty(x)\,dx > \int_0^\infty K_V(x)q_\infty(x)\,dx. \tag{570}$$

The five-kernel lemma and the transfer from q_∞ to large finite N are reported as already analytic, but they must be read before they are imported.

The model attempt shown in the video did not solve theorem 2.54. Its retained guidance is that the outer inequality appears to live in a cancelling Fourier tail beginning near $R = 5.5$, while the inner inequality is a near cancellation between the regions below and above $R = 5.5$. That observation is route selection, not a sign proof.

The primary literature gives a genuine but narrower boundary result. Mezzadri defines $I_p(x)$ as the fraction of derivative zeros in $1 - x/(N-1) \le |z| < 1$ and obtains the large-x asymptotic

$$I_p(x) \sim 1 - \frac{1}{x} \tag{571}$$

from his limiting density [75, Sec. 3]. He also gives the small-x heuristic

$$I_p(x) = \frac{8}{9\pi}x^{3/2} - \frac{64}{225\pi}x^{5/2} + \frac{128}{2205\pi}x^{7/2} + O(x^4). \tag{572}$$

Dueñez–Farmer–Froehlich–Hughes–Mezzadri–Phan prove the leading density law

$$Q(s) = \frac{4}{3\pi}s^{1/2} + O(s) \tag{573}$$

and prove the sharper close-pair contribution

$$\frac{4}{3\pi}s^{1/2} - \frac{82}{45\pi}s^{3/2} + O(s^{5/2}) \tag{574}$$

[76, Thm. 3.1 and Cor. 3.2]. The same paper displays the second bump experimentally for CUE, COE, independent unit-circle roots, and ζ', and explicitly asks for its underlying cause. It does not prove the later bimodality endpoint. This is the exact literature boundary: the small-s asymptotic is proved; the second-bump mechanism in that paper is observational; the later claimed persistence theorem remains to be audited from its own proof artifacts.

A newly routed critical-line source. Conrey's paper has now been acquired as a local, nonredistributable source [78]. With

$$\kappa = \liminf_{T\to\infty} \frac{N_0(T)}{N(T)}, \qquad \kappa^* = \liminf_{T\to\infty} \frac{N_0^*(T)}{N(T)},$$

where $N_0^*(T)$ counts simple critical-line zeros, its Theorem 1 states

$$\kappa \ge 0.4077, \qquad \kappa^* \ge 0.401. \tag{575}$$

Thus the title's two-fifths statement is weaker than the constants actually proved. The proof occupies the cited article and uses a mollifier of length $T^{4/7-\varepsilon}$ together with incomplete Kloosterman-sum estimates. The complete copyrighted article is retained locally for proof audit and citation; its protected bulk text is not reproduced in the public corpus.

Remark 2.55 (Admission boundary). Equations (551)–(554), the exact conjugation map (555), the Tracy–Widom coordinate map (556), and the primary-source statements (571)–(574) have human source chains or explicit derivations here. The $N = 4$ chart, the trace-Mellin continuation and stationary-phase tail, and the large-N bimodality endpoint remain typed candidate claims with exact missing obligations. No statement about individual Riemann zeros is deduced from any CUE claim in this subsection.

3 Bost–Connes–Marcolli systems: the zeta and character sectors

This section records a source-level bridge that was already present in the literature and was recalled in the local notes. The primary source is Eugene Ha and Frédéric Paugam, "Bost–Connes–Marcolli systems for Shimura varieties. I," arXiv:math/0507101v1 (2005) [11]. The exact local archive is the two-member source bundle. Its machine-readable receipt records the following identifiers:

```
archive SHA-256:
B553D565A1F2C6B0D4A81CD8FD8D4B
F4A3A108FAF20AC1490935902778F85628
canonical unit:  PUBUNIT-9E7A142AE210FF9C2665EF43
source TeX SHA-256:
7EFE5E2C38C48AEFAED921DB505B1454D3
6873913617BE28BF86B92EC738BF5C
bibliography SHA-256:
0E783DF3C62E8023A28AD2D0FDEB144D
850ECBE45035011D3F8DD81C14AACC44
```

3.1 The original partition function

For the rational Bost–Connes system let $(\varepsilon_n)_{n\geq 1}$ denote the standard energy basis and put

$$H_{\mathrm{BC}}\varepsilon_n = (\log n)\varepsilon_n. \tag{576}$$

For $\Re(s) > 1$ the trace is absolutely convergent and the source computes

$$Z_{\mathrm{BC}}(s) := \mathrm{Tr}(e^{-sH_{\mathrm{BC}}}) = \sum_{n\geq 1} e^{-s\log n} = \sum_{n\geq 1} n^{-s} = \zeta(s). \tag{577}$$

Thus the equality with the Riemann zeta function is an equality of a specified Gibbs trace with a Dirichlet series; it is not an assertion that the spectrum of H_{BC} is the set of non-trivial zeta zeros.

3.2 Number fields and the Dedekind sector

Let F be a number field, let $\widehat{\mathcal{O}}_F$ be the profinite completion of its ring of integers, and write

$$\widehat{\mathcal{O}}_F^{\natural} := \widehat{\mathcal{O}}_F \cap \mathbb{A}_{F,f}^{\times}.$$

The quotient $\mathcal{O}_F^{\times}\backslash\widehat{\mathcal{O}}_F^{\natural}$ is identified in the source with the semigroup I_F of non-zero integral ideals. The norm map therefore gives

$$\zeta_F(s) = \sum_{I\in I_F} \mathrm{Nm}(I)^{-s} = \sum_{x\in\mathcal{O}_F^{\times}\backslash\widehat{\mathcal{O}}_F^{\natural}} \mathrm{Nm}(x)^{-s}, \qquad \Re(s) > 1. \tag{578}$$

In the number-field BCM system the time evolution and Hamiltonian are, in the source's coordinates,

$$\sigma_t(f)(g, y) = \mathrm{Nm}(g)^{it} f(g, y), \qquad (H_y\xi)(g) = \log \mathrm{Nm}(g)\, \xi(g). \tag{579}$$

Substitution of (579) into the trace gives (578). This is the precise sense in which the rational system and its number-field analogue are the same construction with the norm semigroup changed.

3.3 Character twists and the Dirichlet L-series

Fix a conductor (ray modulus) $\mathfrak{m}$ and a finite class quotient through which a Hecke or Dirichlet character χ factors. The source uses the corresponding semigroup $K(\mathfrak{m})\backslash K^{\natural}(\mathfrak{m})$ and defines

$$L_F(s, \chi) := \sum_{x \in K(\mathfrak{m})\backslash K^{\natural}(\mathfrak{m})} \frac{\chi(x)}{\mathrm{Nm}(x)^s}. \tag{580}$$

On the Hilbert space of the BCM datum let a_χ be multiplication by χ and define the twisted trace by

$$\mathrm{Tr}_\chi(D) := \mathrm{Tr}(a_\chi D).$$

The source's lemma then gives the exact twisted partition identity

$$Z_{\mathfrak{m},\chi}(s) := \mathrm{Tr}_\chi(e^{-sH_y}) = L_F(s, \chi), \qquad \Re(s) > 1. \tag{581}$$

The proof is the direct basis trace: each basis vector indexed by x has energy $\log \mathrm{Nm}(x)$ and is multiplied by $\chi(x)$, so its trace contribution is exactly the summand in (580).

For clarity, suppose the chosen finite quotient is an abelian group G and the character decomposition is taken in the usual orthogonal basis $\widehat{G}$. Writing $w(I) = \mathrm{Nm}(I)^{-s}$, the typed components are

$$Z_\chi(s) = \sum_{I \in I_F} \chi([I])w(I), \qquad Z_{\mathbf{1}}(s) = \sum_{I \in I_F} w(I). \tag{582}$$

The second expression is the trivial-character component. In the rational case with the trivial class data, $Z_{\mathbf{1}}(s) = \zeta(s)$; for a general number field it is $\zeta_F(s)$. Non-trivial Z_χ are the corresponding Hecke/Dirichlet L-series. If an extension and its finite abelian character group have been fixed, the usual Euler-factor decomposition into those characters can be applied; without that extra finite quotient, one should not silently identify every Dedekind factorization with a single unspecified "first character."

3.4 The Artin–Hecke confirmation of the trivial sector

The representation-theoretic formulation is discussed in Cogdell's survey [12, pp. 7–8]. Fix notation with k the base field and K the extension. The trivial and regular representation sectors are

$$\begin{aligned} L(s, \mathbf{1}, K/k) = \zeta_k(s) = L(s, \mathbf{1}, k/k), \\ L(s, \mathrm{Ind}_{\{1\}}^{\mathrm{Gal}(K/k)} \mathbf{1}, K/k) = \zeta_K(s). \end{aligned} \tag{583}$$

Here the finite-prime Artin factors include the inertia invariants at ramified primes. To check the first identity, the trivial representation has one-dimensional inertia invariants and Frobenius acts by 1, giving $(1 - (N\mathfrak{p})^{-s})^{-1}$ at every prime of k. For the regular representation, the inertia-invariant permutation basis decomposes into Frobenius cycles indexed by primes $\mathfrak{P}$ of K over $\mathfrak{p}$, with cycle length $f(\mathfrak{P}/\mathfrak{p})$. The determinant is therefore

$$\prod_{\mathfrak{P}|\mathfrak{p}}(1-(N\mathfrak{p})^{-f(\mathfrak{P}/\mathfrak{p})s}),$$

which proves the second identity by Euler products for $\Re s > 1$ and then continuation. In the abelian case the regular representation is the sum of all characters, so $\zeta_K(s) = \prod_\chi L(s, \chi, K/k)$, whose trivial factor is $\zeta_k(s)$. This also corrects an earlier base/extension subscript mismatch in this working section. The untwisted number-field BCM trace is ζ_F for its own input field F; the Artin product records precisely how changing that field relates the character sectors.

Remark 3.1 (Scope of the bridge). Equations (577)–(582) are source-level identities in specified thermodynamic and character sectors. They formalize the recalled relation between the Riemann/Dedekind zeta functions and Dirichlet L-functions. Ha–Paugam also remark that adding a non-trivial infinite component requires a different groupoid construction; that remark is preserved as a source status, not upgraded here into a new theorem. No claim about zeta zeros, a spectral Hamiltonian for those zeros, or a universal identification of unrelated presentations is being made.

4 Split-zero support and shifted Dedekind sheets

This section reconstructs the common algebraic and analytic core of the overlapping split-zero and shifted-sheet notes. The semiring and semimodule terminology is standard in the sense of Golan [56, Chap. 17, pp. 191–202]. There is also a precise classical analogue: a strong semilattice of groups consists of groups connected by coherent structure homomorphisms, with multiplication transported to the meet index [57, §9.1, pp. 183–184, Thm. 9.1.6]. The theorem below is not quoted from either source. It is a join-oriented, R-linear refinement proved directly from the split-zero scalar action, including its morphisms and quasi-inverse constructions.

4.1 The split-zero semiring in literal coordinates

Let R be a nonzero commutative ring with identity. On the disjoint union

$$S = G(R) := R \sqcup \{\tau\}$$

define, for $r, r' \in R$,

$$r \oplus r' = r + r', \qquad r \oplus \tau = \tau \oplus r = r,$$

and

$$r \odot r' = rr', \qquad r \odot \tau = \tau \odot r = \tau, \qquad \tau \odot \tau = \tau.$$

Thus the semiring zero is τ, whereas

$$e := 0_R \in R \subset S$$

is a second, supported element lying over the arithmetic zero of R.

Write $\mathbb{B} = \{0, 1\}$ for the Boolean semiring and

$$D_R := \{(0_R, 0)\} \cup (R \times \{1\}) \subset R \times \mathbb{B}.$$

Proposition 4.1 (Coordinate model, projections, and the unique support character)**.** *The map*

$$\Theta_R : S \longrightarrow D_R, \qquad \Theta_R(\tau) = (0_R, 0), \qquad \Theta_R(r) = (r, 1),$$

is a semiring isomorphism. The coordinate projections

$$p_R : D_R \to R, \quad p_R(r, b) = r, \qquad \chi_R : D_R \to \mathbb{B}, \quad \chi_R(r, b) = b$$

are semiring homomorphisms, and $\chi_R\Theta_R$ is the only unital semiring homomorphism $S \to \mathbb{B}$. In particular,

$$p_R^{-1}(0_R) = \{\tau, e\}, \qquad \chi_R^{-1}(0) = \{\tau\}, \qquad \chi_R^{-1}(1) = R.$$

Proof. The map is bijective by the displayed disjoint decompositions. Coordinatewise addition in $R \times \mathbb{B}$ gives

$$(r, 1) + (r', 1) = (r + r', 1), \qquad (r, 1) + (0_R, 0) = (r, 1),$$

and coordinatewise multiplication gives

$$(r, 1)(r', 1) = (rr', 1), \qquad (r, 1)(0_R, 0) = (0_R, 0).$$

Hence D_R is closed under both operations and Θ_R preserves them. The two projections are therefore homomorphisms.

Let $f : S \to \mathbb{B}$ be a unital semiring homomorphism. It sends the semiring zero τ to 0 and 1_R to 1. The equality $e = 1_R \oplus (-1_R)$ in the supported copy of R gives

$$f(e) = 1 + f(-1_R) = 1$$

in the Boolean semiring. For every $r \in R$ one has $re = e$, so

$$f(r)f(e) = f(e) = 1.$$

Consequently $f(r) = 1$ for all supported r, while $f(\tau) = 0$; hence $f = \chi_R\Theta_R$. The three fibre identities now follow directly. $\square$

Proposition 4.2 (Products and an explicit multiplicative reconstruction)**.** *For nonzero commutative rings A, B, the map*

$$G(A \times B) \longrightarrow G(A) \times_{\mathbb{B}} G(B), \qquad \tau \longmapsto (\tau_A, \tau_B), \quad (a, b) \longmapsto (a, b),$$

is a semiring isomorphism.

Let M_R be the multiplicative monoid underlying $G(R)$ and let $\mathbb{N}[M_R]$ be its free commutative monoid semiring. If Q_R is the quotient by

$$[a] + [b] = [a + b] \quad (a, b \in R), \qquad [x] + [\tau] = [x] \quad (x \in M_R),$$

then the evaluation map $[x] \mapsto x$ induces a semiring isomorphism $Q_R \simeq G(R)$.

Proof. An element of the fibre product has equal Boolean supports in its two coordinates. It is therefore either (τ_A, τ_B) or a pair $(a, b) \in A \times B$. This gives the stated bijection, and the coordinate formulas in theorem 4.1 prove that it preserves both operations.

For the reconstruction, evaluation respects the defining relations and therefore induces $\bar{\pi} : Q_R \to G(R)$. Conversely define

$$\iota : G(R) \longrightarrow Q_R, \qquad \iota(\tau) = [\tau], \qquad \iota(r) = [r].$$

For supported r, r' the first relation gives $\iota(r \oplus r') = [r + r'] = [r] + [r']$; the second gives $\iota(r \oplus \tau) = [r] = [r] + [\tau]$. Multiplication is inherited from M_R, so ι is a semiring homomorphism in every remaining case as well. On generators,

$$\bar{\pi}\iota = \mathrm{id}_{G(R)}, \qquad \iota\bar{\pi}([x]) = [x].$$

Since the classes $[x]$ generate Q_R as a semiring, the second equality extends to $\iota\bar{\pi} = \mathrm{id}_{Q_R}$. Thus the two maps are explicit two-sided inverses; no unproved uniqueness-of-normal-form assertion is needed. □

4.2 All additive idempotents and the semimodule equivalence

Let M be a left S-semimodule and put

$$\varepsilon_M : M \to M, \qquad \varepsilon_M(m) = em.$$

Lemma 4.3 (The support skeleton is exactly the idempotent locus)**.** *For every $m \in M$,*

$$m + \varepsilon_M(m) = m.$$

Moreover

$$L_M := eM = \{y \in M : y + y = y\}.$$

With $l \vee l' := l + l'$, this set is a join-semilattice whose least element is the additive zero 0_M.

Proof. Because $1_R \oplus e = 1_R$ in S, distributivity gives

$$m + em = (1_R \oplus e)m = m.$$

If $l = em$, then

$$l + l = (e \oplus e)m = em = l,$$

so eM consists of additive idempotents. Conversely, suppose $y + y = y$. Since $e = 1_R \oplus (-1_R)$,

$$ey = y + (-1_R)y.$$

Using $y + y = y$ once gives

$$y + ey = (y + y) + (-1_R)y = y + (-1_R)y = ey.$$

The first displayed identity also says $y + ey = y$, hence $ey = y$. Thus every additive idempotent lies in eM. Closure under addition follows from $em + en = e(m + n)$, and the usual order $l \leq l'$ iff $l + l' = l'$ makes addition the join. Finally $0_M = e0_M$. □

For $l \in L_M$ define the fibre

$$M_l := \{m \in M : em = l\}.$$

Lemma 4.4 (Fibre modules and exact transport)**.** *Each M_l is canonically an R-module whose zero is l. If $l \leq l'$, then*

$$T_{l,l'} : M_l \to M_{l'}, \qquad T_{l,l'}(m) = m + l'$$

is R-linear, and

$$T_{l,l} = \mathrm{id}, \qquad T_{l',l''}T_{l,l'} = T_{l,l''} \quad (l \leq l' \leq l'').$$

Proof. For $m, n \in M_l$,

$$e(m+n) = em + en = l + l = l,$$

so the fibre is additively closed. By theorem 4.3, $m + l = m$. Furthermore

$$m + (-1_R)m = (1_R \oplus (-1_R))m = em = l,$$

so $(-1_R)m$ is the additive inverse of m inside the fibre. For $r \in R$,

$$e(rm) = (er)m = em = l,$$

because $er = e$. The restricted scalar action is therefore an R-module action, with $0_R m = em = l$.

If $l \leq l'$, then

$$e(m + l') = em + el' = l + l' = l',$$

so $T_{l,l'}$ is well-defined. Inside $M_{l'}$,

$$T_{l,l'}(m+n) = m + n + l' = (m + l') + (n + l')$$

because $l' + l' = l'$. Also every supported scalar fixes the support idempotents: $rl' = (re)l' = el' = l'$. Hence

$$T_{l,l'}(rm) = rm + l' = r(m + l').$$

Finally $m + l = m$, and for $l \leq l' \leq l''$,

$$(m + l') + l'' = m + (l' + l'') = m + l''.$$

These are the identity and composition equations. ☐

Define Diag_R as follows. An object is a join-semilattice L with least element 0, an R-module A_l for each $l \in L$, and R-linear maps

$$\rho_{l,l'} : A_l \to A_{l'} \qquad (l \leq l')$$

such that $\rho_{l,l} = \mathrm{id}$ and $\rho_{l',l''}\rho_{l,l'} = \rho_{l,l''}$. A morphism $(\phi, f) : (L, A, \rho) \to (L', A', \rho')$ consists of a zero- and join-preserving map $\phi : L \to L'$ and R-linear maps $f_l : A_l \to A'_{\phi(l)}$ satisfying

$$f_{l'}\rho_{l,l'} = \rho'_{\phi(l),\phi(l')} f_l.$$

Theorem 4.5 (Split-zero semimodules are R-linear join diagrams)**.** *There is an equivalence of categories*

$$G(R)\text{-}\mathrm{SMod} \simeq \mathrm{Diag}_R\,.$$

On an S-semimodule it sends M to $(L_M, \{M_l\}, \{T_{l,l'}\})$. In the opposite direction it sends $\mathcal{A} = (L, A, \rho)$ to the disjoint union

$$M(\mathcal{A}) = \bigsqcup_{l \in L} A_l$$

with, for $x \in A_l$ and $y \in A_m$,

$$\begin{aligned} x \boxplus y &= \rho_{l,l\vee m}(x) +_{A_{l\vee m}} \rho_{m,l\vee m}(y), \\ r \cdot x &= rx \in A_l \quad (r \in R), \\ \tau \cdot x &= 0_{A_0}. \end{aligned}$$

The additive zero is 0_{A_0}.

Proof. The forward construction is supplied by theorems 4.3 and 4.4. If $F : M \to N$ is S-linear, then

$$F(em) = eF(m),$$

so its restriction to L_M is a zero- and join-preserving map $\phi_F : L_M \to L_N$. Each restriction $F_l : M_l \to N_{\phi_F(l)}$ is R-linear, and

$$F_{l'}T_{l,l'}(m) = F(m + l') = F(m) + F(l') = T_{\phi_F(l),\phi_F(l')}F_l(m),$$

so these restrictions form a diagram morphism.

We next verify the reverse construction rather than appealing to an analogy. Commutativity of $\boxplus$ is immediate. If $x \in A_l$, $y \in A_m$, $z \in A_n$, and $u = l \vee m \vee n$, functoriality of the ρ's gives

$$(x \boxplus y) \boxplus z = \rho_{l,u}(x) + \rho_{m,u}(y) + \rho_{n,u}(z) = x \boxplus (y \boxplus z).$$

Also

$$x \boxplus 0_{A_0} = \rho_{l,l}(x) + \rho_{0,l}(0) = x.$$

For $r \in R$, linearity of transport proves $r(x \boxplus y) = rx \boxplus ry$ and $(r + r')x = rx \boxplus r'x$ within A_l. The cases involving τ are explicit:

$$(r \oplus \tau)x = rx = rx \boxplus 0_{A_0}, \qquad \tau(x \boxplus y) = 0_{A_0} = 0_{A_0} \boxplus 0_{A_0},$$

and $r(\tau x) = \tau(rx) = 0_{A_0}$. Associativity and the unit law for scalar multiplication reduce to the corresponding statements in each R-module. Hence $M(\mathcal{A})$ is an S-semimodule.

Given a diagram morphism (ϕ, f), define F on the component A_l to be f_l. For $x \in A_l, y \in A_m$, join preservation and naturality give

$$\begin{aligned} F(x \boxplus y) &= f_{l\vee m}\big(\rho_{l,l\vee m}(x) + \rho_{m,l\vee m}(y)\big) \\ &= \rho'_{\phi(l),\phi(l)\vee\phi(m)}f_l(x) + \rho'_{\phi(m),\phi(l)\vee\phi(m)}f_m(y) \\ &= F(x) \boxplus F(y). \end{aligned}$$

The scalar equations follow from R-linearity and $\phi(0) = 0$. Thus the reverse construction is functorial.

Starting from M, its fibres partition its underlying set. For $m \in M_l$ and $n \in M_{l'}$, put $u = l \vee l'$. Reconstructed addition is

$$(m + u) + (n + u) = m + n + u.$$

But $e(m + n) = u$, so theorem 4.3 gives $m + n + u = m + n$. Thus the reconstructed operation and scalar action are the original ones.

Starting from $\mathcal{A}$, one has

$$e \cdot x = 0_{A_l} \qquad (x \in A_l).$$

The support skeleton is therefore $\{0_{A_l} : l \in L\}$, with $0_{A_l} \boxplus 0_{A_m} = 0_{A_{l\vee m}}$; it is canonically L. The fibre above 0_{A_l} is exactly A_l, and for $l \le l'$ the reconstructed transport is

$$x \boxplus 0_{A_{l'}} = \rho_{l,l'}(x).$$

These canonical identifications are natural and prove that the functors are mutually quasi-inverse. $\square$

Corollary 4.6 (Cancellative and free-coordinate sectors)**.** *The full subcategory of additively cancellative $G(R)$-semimodules is equivalent to R-modules. For $F_n = G(R)^n$, the support skeleton is the Boolean lattice $\mathcal{P}(\{1,\dots,n\})$ under union; the fibre above I is R^I, and transport $I \subseteq J$ is extension by 0_R in $J \setminus I$.*

If M is generated by $m_1,\dots,m_n$, then L_M is join-generated by $em_1,\dots,em_n$, so $|L_M| \le 2^n$. For $l \in L_M$, the R-module M_l is generated by the transported $T_{em_i,l}(m_i)$ for which $em_i \le l$.

Proof. If M is cancellative, then $m+em = m = m+0_M$ implies $em = 0_M$ for every m, so its diagram has one support and is a single R-module. Conversely an R-module becomes an S-semimodule by letting τ act as zero; its additive monoid is cancellative.

For $x = (x_1,\dots,x_n) \in S^n$, the coordinate $(ex)_i$ is τ when $x_i = \tau$ and e when $x_i \in R$. Hence ex is identified with $I = \{i : x_i \in R\}$. The elements over I have arbitrary R-coordinates in I and τ elsewhere, giving R^I. Adding the support idempotent of J inserts 0_R in the coordinates $J \setminus I$.

Finally, every generated element is a finite sum of supported scalar multiples of the m_i; terms multiplied by τ are the global zero and may be removed. Supported scalars preserve support, while support of a sum is the join. This proves the join-generation and cardinality statements. If a sum has support l, each summand support is at most l; transporting every summand into M_l gives the asserted fibre generators. □

4.3 The complete ideal lattice and the exact zero-divisor boundary

For a ring ideal $I \lhd R$, put $I^G = I \sqcup \{\tau\} \subseteq G(R)$.

Theorem 4.7 (The ideal lattice has one additional bottom element)**.** *The assignments*

$$I \longmapsto I^G, \qquad J \longmapsto J \cap R$$

are mutually inverse inclusion-preserving bijections between ring ideals of R and the nonzero semiring ideals of $G(R)$, where "nonzero" means different from the semiring zero ideal $\{\tau\}$. Every semiring ideal is therefore exactly one of

$$\{\tau\} \quad \text{or} \quad I^G = I \sqcup \{\tau\} \quad (I \lhd R).$$

Thus the ideal lattice of $G(R)$ is the ideal lattice of R with one new bottom element adjoined strictly below

$$(0_R)^G = \{\tau, e\}.$$

On the lifted part the bijection preserves arbitrary intersections, generated joins, and products; in particular,

$$I^G J^G = (IJ)^G.$$

The extra bottom ideal is absorbing for ideal multiplication.

Proof. A semiring ideal J contains the semiring zero τ. If it contains no supported element, then $J = \{\tau\}$. Otherwise choose $x \in J \cap R$. Since J is closed under external multiplication,

$$ex = e \in J.$$

Consequently $J \cap R$ contains $0_R = e$. It is closed under ring addition and multiplication by elements of R; and if $y \in J \cap R$, then $(-1_R)y = -y \in J \cap R$. Hence it is a ring ideal and

$$J = (J \cap R) \sqcup \{\tau\}.$$

Conversely I^G is plainly closed under $\oplus$ and under multiplication by every element of $G(R)$. This proves the stated classification and the inverse bijection away from the extra bottom. Intersections and generated joins of lifted ideals follow from the inverse order isomorphism.

The product ideal $I^G J^G$ is generated by finite sums of products xy. A product with a τ factor is τ; a supported product is the ordinary product in R. Finite sums of the latter are precisely the elements of IJ, and τ is already present. Hence the product is $(IJ)^G$. Finally any product involving the ideal $\{\tau\}$ is generated only by τ, so that ideal is absorbing. $\square$

Proposition 4.8 (What divisibility by the supported zero detects)**.** *For $x \in G(R)$,*

$$e \mid x \iff x \in \{e, \tau\}.$$

Consequently the implication

$$e \mid xy \implies e \mid x \text{ or } e \mid y$$

holds for all $x, y \in G(R)$ if and only if R is an integral domain.

Proof. The image of multiplication by e is exactly $\{e, \tau\}$: for $r \in R$, $er = e$, while $e\tau = \tau$. This proves the first equivalence. If R is a domain and $e \mid xy$, then either one factor is τ, or $x, y \in R$ and $xy = 0_R$, which forces $x = 0_R$ or $y = 0_R$. In all cases e divides one factor. Conversely, if R has nonzero zero divisors a, b with $ab = 0_R$, then $e \mid ab$ but $a, b \notin \{e, \tau\}$, so e divides neither. For example, $2 \cdot 3 = 0$ in $\mathbb{Z}/6\mathbb{Z}$ gives the concrete failure. $\square$

4.4 Shifted Dirichlet flow and its exact endpoint boundary

Let

$$L(s) = \sum_{n \geq 1} a_n n^{-s}$$

converge absolutely in $\Re s > \sigma_a$. For real $t > -1$ define

$$L_t(s) = \sum_{n \geq 1} a_n (n+t)^{-s}.$$

For complex $|t| < 1$, the same notation uses the principal logarithm; every $n + t$ then lies in the open right half-plane.

Theorem 4.9 (Shift generator, all jets, and the convergent translation series)**.** *On $\Re s > \sigma_a$ the shifted series is locally normally convergent in its allowed t-domain and, for every integer $m \geq 0$,*

$$\partial_t^m L_t(s) = (-1)^m (s)_m L_t(s+m), \qquad (s)_m = s(s+1)\cdots(s+m-1). \tag{584}$$

For $|t| < 1$,

$$L_t(s) = \sum_{m \geq 0} (-1)^m \frac{(s)_m}{m!} t^m L(s+m), \tag{585}$$

locally normally on $\{\Re s > \sigma_a,\ |t| < 1\}$. Thus if D denotes the operator $(Df)(s) = sf(s+1)$, the notation $L_t = \exp(-tD)L$ is an equality represented by the convergent series (585) *on this domain, not merely a formal slogan.*

Proof. On a compact real interval $I \Subset (-1, \infty)$ there is $c_I > 0$ such that $n+t \geq c_I n$ for all $n \geq 1$ and $t \in I$. Hence the original absolute convergence majorizes the shifted series uniformly on compact subsets of $\Re s > \sigma_a$; the same argument applies after any fixed number of t-derivatives. Termwise differentiation gives

$$\partial_t^m (n+t)^{-s} = (-1)^m (s)_m (n+t)^{-s-m},$$

which proves (584). For complex t in a compact subdisc the estimate $|n+t| \geq n - |t|$ gives the same normal-convergence conclusion.

For each n and $|t| < 1$ the binomial theorem gives

$$(n+t)^{-s} = n^{-s} \sum_{m \geq 0} (-1)^m \frac{(s)_m}{m!} \left(\frac{t}{n}\right)^m.$$

To justify both interchanges, take a compact set on which $\Re s \geq \sigma > \sigma_a$, $|s| \leq A$, and $|t| \leq \rho < 1$. For $m \geq 1$,

$$\frac{|(s)_m|}{m!} \leq \frac{A}{m} \prod_{j=1}^{m-1} \left(1 + \frac{A}{j}\right) \leq C_A m^{A-1}.$$

Therefore the absolute double series is bounded by

$$\left(\sum_{n \geq 1} |a_n| n^{-\sigma}\right) \left(1 + C_A \sum_{m \geq 1} m^{A-1} \rho^m\right) < \infty.$$

Fubini's theorem for absolutely convergent series now gives (585), and the same majorant proves local normal convergence. The last statement is exactly the power-series definition of $\exp(-tD)$ applied term by term. □

Corollary 4.10 (Logarithmic shifted jets)**.** *On any open set where $L_t(s) \neq 0$, set*

$$H_m(t, s) = (-1)^m (s)_m \frac{L_t(s+m)}{L_t(s)}.$$

Then every $\partial_t^m \log L_t(s)$ is the universal logarithmic derivative polynomial in $H_1, \ldots, H_m$. In particular,

$$\begin{aligned}
\partial_t \log L_t(s) &= -s \frac{L_t(s+1)}{L_t(s)}, \\
\partial_t^2 \log L_t(s) &= s(s+1) \frac{L_t(s+2)}{L_t(s)} - s^2 \left(\frac{L_t(s+1)}{L_t(s)}\right)^2.
\end{aligned}$$

Proof. For a nonvanishing holomorphic function F, the identity $(\log F)' = F'/F$ and the quotient rule show inductively that $(\log F)^{(m)}$ is a polynomial in $F'/F, \ldots, F^{(m)}/F$ with integer coefficients. Substitute $F^{(j)}/F = H_j$ from (584). The first two quotient-rule computations are the displayed formulas. □

For the Riemann coefficients $a_n = 1$, one has

$$\zeta_t(s) := \sum_{n \geq 1} (n+t)^{-s} = \zeta(s, 1+t),$$

where the right side is the Hurwitz zeta function in the notation of DLMF 25.11.1 [51, Eq. 25.11.1].

Theorem 4.11 (A nonintegral shift is not an ordinary Dirichlet series)**.** *Let* $0 < t < 1$*. There is no sequence* $(b_m)_{m\geq 1}$ *whose ordinary Dirichlet series* $\sum_{m\geq 1} b_m m^{-s}$ *converges absolutely in a right half-plane and agrees there with* $\zeta_t(s)$*. At* $t = 1$*,*

$$\zeta_1(s) = \zeta(s) - 1 = \sum_{m\geq 2} m^{-s};$$

this ordinary Dirichlet series cannot have an Euler product all of whose local factors are normalized to have constant term 1*.*

Proof. For real $\sigma \to +\infty$, dominated convergence in the absolutely convergent positive series gives

$$(1+t)^{\sigma}\zeta_t(\sigma) = 1 + \sum_{n\geq 2}\left(\frac{1+t}{n+t}\right)^{\sigma} \longrightarrow 1.$$

Indeed, after fixing $\sigma_0 > 1$, the summands for $\sigma \geq \sigma_0$ are dominated by the summable sequence $(1+t)^{\sigma_0}(n+t)^{-\sigma_0}$.

Now let $F(s) = \sum b_m m^{-s}$ be a nonzero ordinary Dirichlet series absolutely convergent at σ_0, and let m_0 be its least nonzero coefficient index. Then

$$m_0^{\sigma}F(\sigma) = b_{m_0} + \sum_{m>m_0} b_m\left(\frac{m_0}{m}\right)^{\sigma} \longrightarrow b_{m_0}.$$

For the tail, dominated convergence applies after writing its absolute value term as

$$|b_m|m^{-\sigma_0}m_0^{\sigma_0}\left(\frac{m_0}{m}\right)^{\sigma-\sigma_0}.$$

If $F = \zeta_t$, comparison of the two nonzero leading exponential scales forces $m_0 = 1+t$, impossible because m_0 is an integer and $0 < t < 1$.

The $t = 1$ identity follows by the index change $m = n+1$. A convergent normalized Euler product

$$\prod_p \left(1 + \sum_{k\geq 1} c_{p,k}p^{-ks}\right)$$

has coefficient 1 at 1^{-s}: it is obtained by selecting the constant term from every local factor. The coefficient at 1^{-s} in $\zeta(s) - 1$ is 0, proving the final claim. □

For $R = \mathbb{Z}$, the ideals $n\mathbb{Z}$ give two different literal size coordinates:

$$|\mathbb{Z}/n\mathbb{Z}| = n, \qquad |G(\mathbb{Z}/n\mathbb{Z})| = n+1.$$

Thus the ordinary ideal sheet is $\zeta(s)$ while the split-size sheet is $\zeta(s)-1$. The change is exactly the retained unsupported element τ; it is not a normalization of the arithmetic-zero coordinate.

4.5 A lossless support-to-endpoint lattice morphism

The affine deck coordinates and their inverse were proved in theorem 2.35; the endpoint coefficient isomorphism was proved in theorem 2.36. We now construct the missing arrow from the rank-two split-zero support skeleton to that endpoint geometry.

Let $F_2 = G(R)^2$, with support coordinates labelled $0, 1$. Let

$$E = \mathbb{C}e_0 \oplus \mathbb{C}e_1, \qquad E_I = \operatorname{span}_{\mathbb{C}}\{e_i : i \in I\} \quad (I \subseteq \{0,1\}).$$

On the s-line put

$$u = s(s-1), \qquad v = 2s-1, \qquad q = \frac{1}{u}, \qquad \ell = \frac{v}{u},$$

and define the two-dimensional differential space

$$\mathcal{P}_\Omega = \mathrm{span}_{\mathbb{C}}\{q\,ds, \ell\,ds\}.$$

Theorem 4.12 (Boolean support cube to endpoint principal parts)**.** *The maps*

$$\begin{aligned} \Lambda : \mathcal{P}(\{0,1\}) &\longrightarrow \{0, \mathbb{C}e_0, \mathbb{C}e_1, E\}, & I &\longmapsto E_I, \\ \mathrm{Res}_{0,1} : \mathcal{P}_\Omega &\longrightarrow E, & Aq\,ds + B\ell\,ds &\longmapsto (-A+B)e_0 + (A+B)e_1 \end{aligned}$$

are lattice and vector-space isomorphisms, respectively. Their inverse data are

$$A = \frac{b-a}{2}, \qquad B = \frac{a+b}{2} \quad \text{for } ae_0 + be_1 \in E.$$

Consequently $I \mapsto \mathrm{Res}_{0,1}^{-1}(E_I)$ is the explicit lattice isomorphism

I	$\mathrm{Res}_{0,1}^{-1}(E_I)$
$\varnothing$	0
$\{0\}$	$\mathbb{C}\,ds/s$
$\{1\}$	$\mathbb{C}\,ds/(s-1)$
$\{0,1\}$	$\mathcal{P}_\Omega$.

The coordinate transposition on F_2 and the deck involution $\sigma(s) = 1-s$ act equivariantly on these lattices. The support projection

$$\pi_{\mathrm{supp}} : F_2 \to \mathcal{P}(\{0,1\})$$

has the complete fibre R^I over I; the two displayed isomorphisms have singleton fibres. Thus the composite from F_2 to endpoint subspaces loses exactly the R^I fibre coordinates and no support information.

Proof. Union and intersection of subsets map under $I \mapsto E_I$ to sum and intersection of coordinate subspaces, so Λ is a lattice isomorphism. Partial fractions give

$$q = \frac{1}{s-1} - \frac{1}{s}, \qquad \ell = \frac{1}{s-1} + \frac{1}{s}.$$

Hence the endpoint residues of $Aq\,ds + B\ell\,ds$ are $(a,b) = (-A+B, A+B)$. The coefficient matrix has determinant -2, and solving gives the displayed inverse. In particular,

$$\ell - q = \frac{2}{s}, \qquad q + \ell = \frac{2}{s-1},$$

which proves all four rows of the table.

Transposition sends a support subset I to the subset obtained by swapping 0 and 1. For differentials,

$$\sigma^*\left(\frac{ds}{s}\right) = \frac{ds}{s-1}, \qquad \sigma^*\left(\frac{ds}{s-1}\right) = \frac{ds}{s},$$

so the endpoint coordinate lines are swapped by the same permutation; 0 and the full space are fixed. This proves equivariance. The fibre statement for π_{supp} is theorem 4.6 with $n = 2$, while bijectivity gives singleton fibres for Λ and $\mathrm{Res}_{0,1}$. $\square$

This theorem is an exact morphism at the support-skeleton level. It does not identify the scalar semiring $G(R)$ with $\mathbb{C}$, nor does it discard the fibre coordinates while pretending that the resulting map is injective. The separate physical $N = 3$ cover and its two exact maps to the same deck curve, including the complete fibres and the fixed-locus obstruction, are already proved in theorem 2.39.

4.6 The endpoint Stone defect and its coprime kernel

Set

$$b(1) = 0, \qquad b(n) = 1 \quad (n \geq 2), \qquad \zeta(s) - 1 = \sum_{n\geq 1} b(n)n^{-s}.$$

Theorem 4.13 (Exact boundary defect and interior coprime series)**.** *For coprime positive integers m, n, define*

$$\delta_b(m,n) = b(mn) - b(m)b(n).$$

Then

$$\delta_b(m,n) = \begin{cases} 1, & \textit{exactly one of } m, n \textit{ is } 1, \\ 0, & \textit{otherwise.} \end{cases}$$

Consequently, for $\Re s, \Re t > 1$,

$$\sum_{(m,n)=1} \delta_b(m,n) m^{-s} n^{-t} = \zeta(s) + \zeta(t) - 2, \tag{586}$$

$$C(s,t) := \sum_{\substack{m,n\geq 2 \\ (m,n)=1}} m^{-s}n^{-t} = \frac{\zeta(s)\zeta(t)}{\zeta(s+t)} - \zeta(s) - \zeta(t) + 1. \tag{587}$$

Proof. If $m = n = 1$, both terms in δ_b vanish. If exactly one argument is 1, then $b(mn) = 1$ and $b(m)b(n) = 0$. If both are at least 2, both terms equal 1. This proves the pointwise formula, and summing its two boundary rays gives (586).

Absolute convergence in the stated domain permits an Euler factorization of the full coprime sum. For each prime p, at most one of its two exponents is positive, so

$$\begin{aligned} \sum_{(m,n)=1} m^{-s}n^{-t} &= \prod_p \left(1 + \sum_{j\geq 1} p^{-js} + \sum_{k\geq 1} p^{-kt}\right) \\ &= \prod_p \frac{1 - p^{-(s+t)}}{(1-p^{-s})(1-p^{-t})} = \frac{\zeta(s)\zeta(t)}{\zeta(s+t)}. \end{aligned}$$

Removing the row $m = 1$ and column $n = 1$, and restoring the twice-removed point $(1,1)$, gives (587). ☐

There is a literal two-point quotient behind the first formula. Let $q : \mathbb{N}_{\geq 1} \to \{\xi_1, \xi_*\}$ send 1 to ξ_1 and every $n \geq 2$ to ξ_*. Then $q(mn) = q(m)q(n)$ for the multiplication table

$$\xi_1^2 = \xi_1, \qquad \xi_1\xi_* = \xi_*\xi_1 = \xi_*^2 = \xi_*.$$

For the characteristic functions $u_0 = \mathbf{1}_{\{\xi_1\}}$ and $v_0 = \mathbf{1}_{\{\xi_*\}}$, pullback along multiplication gives

$$\Delta u_0 = u_0 \otimes u_0, \qquad \Delta v_0 = v_0 \otimes u_0 + u_0 \otimes v_0 + v_0 \otimes v_0.$$

Thus $\Delta v_0 - v_0 \otimes v_0$ is exactly the two boundary rays in (586), not merely a visual similarity.

4.7 Number-field shifts and the ideal-ratio kernel

Let K be a number field with ring of integers $\mathcal{O}_K$. Milne proves that the nonzero fractional ideals form the free abelian group on the nonzero prime ideals [58, Chap. 3, p. 53, Thm. 3.20], and that the numerical norm $N\mathfrak{a} = [\mathcal{O}_K : \mathfrak{a}]$ is multiplicative [58, Chap. 4, pp. 69–70, Prop. 4.2(a)]. Neukirch then defines

$$\zeta_K(s) = \sum_{0\neq\mathfrak{a}\lhd\mathcal{O}_K} N(\mathfrak{a})^{-s} = \sum_{n\geq 1} a_K(n)n^{-s}$$

and proves its absolutely convergent prime-ideal Euler product for $\Re s > 1$ [59, Chap. VII, §5, p. 457, Def. 5.1 and Prop. 5.2]. His Theorem 5.9 and Corollaries 5.10–5.11 prove the meromorphic continuation to $\mathbb{C}$, with the unique simple pole at $s = 1$ and the class-number-formula residue [59, pp. 465–467]. These are the analytic inputs used below; the shifted continuation is proved separately.

For $|t| < 1$, using the principal logarithm, define first on $\Re s > 1$

$$\zeta_{K,t}(s) := \sum_{n\geq 1} a_K(n)(n+t)^{-s} = \sum_{0\neq\mathfrak{a}\lhd\mathcal{O}_K} (N\mathfrak{a} + t)^{-s}.$$

Theorem 4.14 (Complete shifted Dedekind continuation)**.** *For $|t| < 1$ and $\Re s > 1$,*

$$\zeta_{K,t}(s) = \sum_{m\geq 0} (-1)^m \frac{(s)_m}{m!} t^m \zeta_K(s+m). \tag{588}$$

The part with $m \geq 1$ extends to an entire function of s, locally normally in $\mathbb{C} \times \{|t| < 1\}$. Hence (588) is the meromorphic continuation of $\zeta_{K,t}$ to the whole s-plane. Its only pole is the simple pole at $s = 1$, and its residue there equals $\operatorname{Res}_{s=1} \zeta_K(s)$.

Proof. The identity on $\Re s > 1$ is theorem 4.9 applied to the absolutely convergent Dirichlet series for ζ_K.

Fix $m \geq 1$. By the cited continuation theorem, $\zeta_K(s + m)$ is holomorphic except for a simple pole at $s = 1 - m$. The polynomial

$$(s)_m = \prod_{j=0}^{m-1} (s+j)$$

has a simple zero there: the factor with $j = m - 1$ vanishes and all other factors are nonzero. Therefore $(s)_m\zeta_K(s + m)$ is holomorphic at $1 - m$ and has no other singularity; it is entire.

It remains to prove convergence rather than merely say that continuation carries over termwise. Let $C \subset \mathbb{C}$ be compact and choose $A \geq 1$ with $|s| \leq A$ on C. For $m \geq A+2$, one has $\Re(s+m) \geq 2$ and the nonnegative Dirichlet coefficients give

$$|\zeta_K(s+m)| \leq \zeta_K(\Re(s+m)) \leq \zeta_K(2).$$

As in the proof of theorem 4.9,

$$\frac{|(s)_m|}{m!} \leq \frac{A}{m} \prod_{j=1}^{m-1} \left(1 + \frac{A}{j}\right) \leq C_A m^{A-1}.$$

On $|t| \leq \rho < 1$, the tail is therefore bounded uniformly on C by a constant times $\sum_m m^{A-1}\rho^m$. The finitely many earlier terms are entire by the cancellation just proved. The Weierstrass theorem now makes the $m \geq 1$ sum entire in s and locally normal in (s, t).

Thus the only possible pole in (588) comes from its $m = 0$ term, which is $\zeta_K(s)$. The remaining sum is entire, so neither the order nor the residue of that pole changes. Equality on the original half-plane and uniqueness of meromorphic continuation finish the proof. $\square$

Theorem 4.15 (The exact Dedekind ratio coefficients)**.** *For every integer $r \geq 1$ and $\Re s > 1$,*

$$Q_{K,r}(s) := \frac{\zeta_K(s+r)}{\zeta_K(s)} = \sum_{0 \neq \mathfrak{a} \lhd \mathcal{O}_K} \alpha_{K,r}(\mathfrak{a}) N(\mathfrak{a})^{-s}, \tag{589}$$

where the series is absolutely convergent, $\alpha_{K,r}$ is multiplicative, and

$$\alpha_{K,r}(\mathfrak{p}^k) = -\frac{N(\mathfrak{p})^r - 1}{N(\mathfrak{p})^{rk}} \qquad (k \geq 1), \tag{590}$$

$$\alpha_{K,r}(\mathfrak{a}) = (-1)^{\omega(\mathfrak{a})} \frac{\prod_{\mathfrak{p}|\mathfrak{a}} (N(\mathfrak{p})^r - 1)}{N(\mathfrak{a})^r}. \tag{591}$$

The meromorphic quotient $Q_{K,r}$ exists wherever the ratio is defined after continuation; (589) *is asserted only in its proved absolute-convergence half-plane.*

Proof. The sourced Euler product is nonzero on $\Re s > 1$, and there

$$Q_{K,r}(s) = \prod_{\mathfrak{p}} \frac{1 - N(\mathfrak{p})^{-s}}{1 - N(\mathfrak{p})^{-s-r}}.$$

For $q = N(\mathfrak{p})$ and $x = q^{-s}$, the local factor has the exact geometric expansion

$$\begin{aligned} \frac{1-x}{1-q^{-r}x} &= 1 + (q^{-r} - 1)x \sum_{j \geq 0} q^{-rj} x^j \\ &= 1 - \sum_{k \geq 1} (1 - q^{-r}) q^{-r(k-1)} x^k. \end{aligned}$$

Its $\mathfrak{p}^k$ coefficient is therefore

$$-(1 - q^{-r}) q^{-r(k-1)} = -\frac{q^r - 1}{q^{rk}},$$

which proves (590). Unique prime-ideal factorization then gives multiplicativity, and multiplying the prime-power values gives (591).

For $\sigma = \Re s > 1$, the absolute nonconstant mass of the local factor is

$$\sum_{k \geq 1} |\alpha_{K,r}(\mathfrak{p}^k)| q^{-k\sigma} = \frac{(1 - q^{-r}) q^{-\sigma}}{1 - q^{-(\sigma + r)}} \leq \frac{q^{-\sigma}}{1 - 2^{-(\sigma + r)}}.$$

The sum of $N(\mathfrak{p})^{-\sigma}$ over prime ideals is bounded by the corresponding positive sum over all nonzero ideals, which converges for $\sigma > 1$. Hence the product of absolute local series converges and its coefficient expansion is absolutely convergent. This proves (589) without extending the Dirichlet series beyond that domain. $\square$

Corollary 4.16 (Nonintegral Dedekind shifts)**.** *For real $0 < t < 1$, $\zeta_{K,t}$ is not an ordinary Dirichlet series in integer bases n^{-s} on any right half-plane of absolute convergence.*

Proof. Exactly one ideal, $\mathcal{O}_K$, has norm 1, so $a_K(1) = 1$. Since all $a_K(n)$ are nonnegative and the unshifted series converges absolutely for $\sigma > 1$, the same dominated convergence argument as in theorem 4.11 gives

$$(1+t)^\sigma \zeta_{K,t}(\sigma) \longrightarrow 1.$$

Every nonzero ordinary Dirichlet series has an integral leading base m_0 by the lemma proved inside theorem 4.11. Equality would force $m_0 = 1 + t$, which is impossible for $0 < t < 1$. $\square$

There is also a literal number-field version of the separated quotient size. For a nonzero ideal $\mathfrak{a}$,

$$|G(\mathcal{O}_K/\mathfrak{a})| = N(\mathfrak{a}) + 1.$$

By norm multiplicativity,

$$w(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b}) + 1,$$
$$w(\mathfrak{a})w(\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b}) + N(\mathfrak{a}) + N(\mathfrak{b}) + 1$$

for $w(\mathfrak{a}) = N(\mathfrak{a}) + 1$. Their difference is positive. Therefore the Dirichlet series

$$\sum_{0 \neq \mathfrak{a} \lhd \mathcal{O}_K} (N(\mathfrak{a}) + 1)^{-s}$$

cannot be reconstructed from prime-ideal factorization by treating w as a multiplicative ideal norm. This is the exact obstruction; it does not claim that every conceivable factorization of the resulting analytic function is impossible.

4.8 Executable finite certificate

The accompanying exact-arithmetic certificate `verify_split_zero_shifted_core.py` passes under Python/SymPy 3.13.9/1.13.1 and 3.13.1/1.13.3. Its 26 named checks exhaust the semiring laws, unique Boolean character, rank-two idempotent skeleton, ideal classification, generated ideal products, and supported-zero divisibility in the five models $R = \mathbb{Z}/n\mathbb{Z}$ for $n = 2, 3, 4, 5, 6$. It also checks the endpoint residue matrix and inverse, support-swap equivariance, the Stone defect through all coprime pairs up to 64, the exact local coprime Euler factor, shifted jets through order 8, Pochhammer pole cancellation through order 16, and the Dedekind local ratio coefficients over the stated finite test grid. This audit is what exposed the additional bottom ideal $\{\tau\}$ in theorem 4.7. The general category and convergence theorems remain the standard proofs above; finite replay is not substituted for them.

4.9 Interfaces retained and interfaces still requiring arrows

The finite CUE measure, Verblunsky recursion, Mellin–Hurwitz transform, $U(2)$ and $U(3)$ models, trace descent, finite phase-space algebra, and ratio-lattice carrier are proved in section 2.7.1. They are not repeated here. The present chapter adds three exact interfaces:

1. the full categorical equivalence $G(R)$-SMod $\simeq \mathrm{Diag}_R$, including all additive idempotents and morphisms;
2. the equivariant support-cube-to-endpoint-subspace map of theorem 4.12, with its exact fibres;
3. the shifted Dedekind continuation and ideal-ratio coefficient kernel of theorems 4.14 and 4.15.

The numerical equality

$$s^2 + 2s = \dim_{\mathbb{C}} \mathfrak{sl}_{s+1}(\mathbb{C})$$

and the explicit block bijection already proved in theorem 2.34 do not, by themselves, map a Kostka determinant, a Haar integral, or a microscopic asymptotic to a representation. Constructing such a functor remains an explicit interface problem. This is a statement about the arrows built in this corpus, not a claim about what has or has not been proved elsewhere.

5 Divisor sheaves, Artin chains, jet supports, and packet descent

This section reconstructs one programme from the local notes as a sequence of typed constructions. The classical inputs are deliberately narrow. Lee gives the local factorization of a meromorphic function on a Riemann surface, the associated divisor language, skyscraper sheaves, sheafification, and the acyclicity of flasque sheaves [81, pp. 93–95; Chap. 5, especially Thm. 5.9; pp. 184–188, Thm. 6.34]. Vakil gives a second formulation of sheafification by compatible germs and the one-dimensional regular-local-ring/DVR equivalences [82, §§2.2.9 and 2.4.4–2.4.7; §13.5]. All special constructions and all comparison maps below are proved here. In particular, neither classical source is being cited for the Boolean shadow or for the chain–cube adjunction.

5.1 The literal fixed pair at a zeta zero

Retain the split-zero semiring $S = G(\mathbb{C}) = \mathbb{C} \sqcup \{\tau\}$ of section 4. Its additive zero is τ. To avoid confusing that element with the ordinary complex number zero, write

$$e := 0_{\mathbb{C}} \in \mathbb{C} \subset S, \qquad A := \{\tau, e\} \subset S.$$

Proposition 5.1 (Fixed pair in literal coordinates)**.** *The operations inherited from S make A a unital semiring, with additive zero τ and multiplicative identity e. The map*

$$\beta : A \longrightarrow \mathbb{B} = \{0, 1\}, \qquad \beta(\tau) = 0, \quad \beta(e) = 1,$$

is a semiring isomorphism. If $u = -1 \in \mathbb{C} \subset S$, then

$$\mathrm{Fix}(u \cdot -) = A.$$

Consequently, at every zero ρ of ζ,

$$\zeta(\rho) - 1 = -1, \qquad \mathrm{Fix}\big((\zeta(\rho) - 1) \cdot -\big) = A.$$

Proof. The four sums and products in A are

$$\tau + \tau = \tau, \quad \tau + e = e + \tau = e, \quad e + e = e,$$

and

$$\tau\tau = \tau e = e\tau = \tau, \qquad e^2 = e.$$

These are exactly the Boolean tables under β. Multiplication by u fixes τ because τ is absorbing. For a supported element $z \in \mathbb{C} \subset S$, $uz = z$ is the ordinary equality $-z = z$, hence $2z = 0$ and therefore $z = 0_{\mathbb{C}} = e$. This proves the fixed-set identity. The final assertion is the direct substitution $\zeta(\rho) = 0$; it detects no further analytic property of ρ. □

The internal unit of A is e, whereas the unit of S is $1_{\mathbb{C}}$. Thus the inclusion $A \subset S$ is deliberately not being called a unit-preserving semiring map.

5.2 A product sheaf and the exact sheafification map

Let X be a Riemann surface and let

$$D = \sum_{x \in Z} m_x[x]$$

be an effective divisor with closed discrete support $Z \subset X$ and $m_x \in \mathbb{Z}_{\geq 1}$. For an open $U \subset X$, put

$$P_D(U) := \bigoplus_{x \in U \cap Z} A^{m_x}, \qquad \mathcal{V}_D(U) := \prod_{x \in U \cap Z} A^{m_x}, \qquad \mathcal{A}_D(U) := \prod_{x \in U \cap Z} A.$$

The direct sum means finite support relative to the basepoint $(\tau, \ldots, \tau)$; every restriction map forgets the coordinates outside the smaller open set.

Lemma 5.2 (Product sheaf on discrete support). *For any family $(M_x)_{x \in Z}$ of sets with fixed algebraic structure,*

$$\mathcal{F}_M(U) := \prod_{x \in U \cap Z} M_x$$

is a sheaf with that structure. Its stalk is M_x at $x \in Z$ and the terminal one-element object at $y \notin Z$.

Proof. For an open cover $U = \bigcup_i U_i$, compatible families on the U_i specify one and only one value at each $x \in U \cap Z$: choose any i containing x, and compatibility makes the choice irrelevant. These values form the unique global family. If $x \in Z$, discreteness gives a neighbourhood W with $W \cap Z = \{x\}$, so the directed system defining the stalk is eventually constant with value M_x. If $y \notin Z$, closedness gives a neighbourhood disjoint from Z, and the directed system is eventually terminal. □

Theorem 5.3 (Finite support sheafifies to arbitrary support). *The coordinate inclusion*

$$\eta_D : P_D \longrightarrow \mathcal{V}_D$$

exhibits $\mathcal{V}_D$ as the sheafification of P_D in sheaves of A-semimodules.

Proof. The target is a sheaf by theorem 5.2. Let $\mathcal{G}$ be a sheaf of A-semimodules and let $\varphi : P_D \to \mathcal{G}$ be a presheaf morphism. Fix $s = (s_x)_{x \in U \cap Z} \in \mathcal{V}_D(U)$. For every $x \in U \cap Z$, choose $U_x \subset U$ with $U_x \cap Z = \{x\}$, and put $U_0 = U \setminus Z$. The family

$$\{U_0\} \cup \{U_x : x \in U \cap Z\}$$

covers U. On U_x, the section s is the finite-support section s_x; on U_0 it is zero. If $x \neq y$, then $(U_x \cap U_y) \cap Z = \varnothing$, and the same is true of $U_x \cap U_0$. Hence

$$\varphi_{U_x}(s_x), \qquad \varphi_{U_0}(0)$$

agree on every overlap. They glue uniquely in $\mathcal{G}$; call the result $\widetilde{\varphi}_U(s)$. The construction commutes with restriction and with the A-semimodule operations because both statements hold on the displayed cover and sections of a sheaf are determined locally. Thus $\widetilde{\varphi} : \mathcal{V}_D \to \mathcal{G}$ is a sheaf morphism.

If s already lies in $P_D(U)$, the glued local images are precisely the restrictions of $\varphi_U(s)$, so $\widetilde{\varphi}\eta_D = \varphi$. If another extension has this property, it agrees with $\widetilde{\varphi}$ on every U_x and on U_0, hence everywhere. This is the universal property of sheafification. It also agrees with the compatible-germ construction in [82, §2.4.4–2.4.7]. □

5.3 Boolean coordinates, reduction, and lost information

For $m \geq 1$, let $[m] = \{1, \dots, m\}$ and $B_m = \mathcal{P}([m])$, ordered by inclusion and equipped with union as join. Define

$$\Theta_m : A^m \longrightarrow B_m, \qquad \Theta_m(a) = \{r : a_r = e\}.$$

Proposition 5.4 (Boolean cube and reduced retract)**.** *The map Θ_m is an isomorphism of A-semimodules, where the scalar action on B_m is*

$$\tau I = \varnothing, \qquad eI = I.$$

The maps

$$\Sigma_m : A^m \to A, \quad \Sigma_m(a) = a_1 + \cdots + a_m, \qquad \Delta_m : A \to A^m, \quad \Delta_m(a) = (a, \dots, a),$$

are A-linear and satisfy $\Sigma_m \Delta_m = \mathrm{id}_A$. In subset coordinates,

$$\Sigma_m(I) = \begin{cases} \tau, & I = \varnothing, \\ e, & I \neq \varnothing, \end{cases} \qquad \Delta_m(\tau) = \varnothing, \quad \Delta_m(e) = [m].$$

Thus the only nontrivial fibre of Σ_m is the set of all $2^m - 1$ nonempty subsets.

Proof. Coordinatewise addition in A^m is Boolean OR, so $\Theta_m(a + b) = \Theta_m(a) \cup \Theta_m(b)$; the inverse assigns e to the coordinates in a subset and τ to its complement. The formulas for Σ_m and Δ_m are then the nonempty-set detector and the inclusion of the two extreme vertices. They preserve joins and the scalar action, and their composite fixes both elements of A. The fibre count is immediate from the displayed formula. $\square$

Applying these maps independently at each x gives sheaf morphisms

$$\mathcal{V}_D \xrightarrow{\Sigma_D} \mathcal{A}_D \xrightarrow{\Delta_D} \mathcal{V}_D, \qquad \Sigma_D \Delta_D = \mathrm{id}_{\mathcal{A}_D}.$$

If $\mathcal{V}_D \cong \mathcal{A}_D$, then their stalks at x have respectively 2^{m_x} and 2 elements; hence $m_x = 1$. Conversely that condition makes the definitions identical. This proves, without erasing the retraction, that

$$\mathcal{V}_D \cong \mathcal{A}_D \quad \Longleftrightarrow \quad m_x = 1 \text{ for every } x \in Z.$$

5.4 The intrinsic Artin chain and both adjoints to the cube inclusion

Let $\mathfrak{m}_x$ be the maximal ideal of the analytic local ring $\mathcal{O}_{X,x}$ and define

$$Q_{x,m} := \mathcal{O}_{X,x} / \mathfrak{m}_x^m.$$

A centred holomorphic coordinate t identifies

$$\mathcal{O}_{X,x} \cong \mathbb{C}\{t\}, \qquad Q_{x,m} \cong \mathbb{C}\{t\}/(t^m) \cong \mathbb{C}[t]/(t^m).$$

Proposition 5.5 (Complete ideal lattice)**.** *The ideals of $Q_{x,m}$ are exactly*

$$I_k := \mathfrak{m}_x^k / \mathfrak{m}_x^m, \qquad 0 \leq k \leq m,$$

and

$$Q_{x,m} = I_0 \supset I_1 \supset \cdots \supset I_m = 0.$$

This chain is intrinsic: changing the centred coordinate changes its polynomial presentation but not any I_k.

Proof. Every nonzero convergent germ has a unique factorization $t^k u(t)$ with $k \geq 0$ and $u(0) \neq 0$. If $J \neq 0$ is an ideal of $\mathbb{C}\{t\}$, choose the least order k of a nonzero element of J. A least-order element is $t^k u$ with u a unit, so $t^k \in J$; every element of J has order at least k, so $J \subseteq (t^k)$. Hence $J = (t^k)$. Ideals of the quotient correspond to ideals of $\mathbb{C}\{t\}$ containing (t^m), which are precisely (t^k) for $0 \leq k \leq m$. Finally $(t) = \mathfrak{m}_x$, so the displayed ideals are the coordinate-free powers of the maximal ideal. This is the analytic instance of the DVR ideal classification recorded in [82, §13.5, especially Thm. 13.5.4]. ☐

Define terminal segments

$$T_r := \{r, r+1, \ldots, m\} \quad (1 \leq r \leq m), \qquad T_{m+1} := \varnothing,$$

and let $C_m = \{T_1, \ldots, T_{m+1}\} \subset B_m$. The intrinsic identification

$$\mathrm{Idl}(Q_{x,m}) \longrightarrow C_m, \qquad I_k \longmapsto T_{k+1},$$

preserves inclusion and joins. Let $i_m : C_m \hookrightarrow B_m$ be inclusion, and define

$$r_m(I) = \begin{cases} \varnothing, & I = \varnothing, \\ T_{\min I}, & I \neq \varnothing, \end{cases} \qquad s_m(I) = \{j \in [m] : T_j \subseteq I\}.$$

Theorem 5.6 (Coordinate-level chain–cube comparison)**.** *For $I \in B_m$ and $T \in C_m$,*

$$r_m(I) \subseteq T \iff I \subseteq i_m(T), \qquad i_m(T) \subseteq I \iff T \subseteq s_m(I).$$

Thus $r_m \dashv i_m \dashv s_m$ in finite posets. The map r_m preserves joins and $r_m i_m = \mathrm{id}_{C_m}$, so C_m is a split join-subsemilattice of B_m. The fibres are exactly

$$r_m^{-1}(T_r) = \begin{cases} \{I \subseteq T_r : r \in I\}, & 1 \leq r \leq m, \\ \{\varnothing\}, & r = m+1, \end{cases}$$

and

$$s_m^{-1}(T_r) = \begin{cases} \{[m]\}, & r = 1, \\ \{I : T_r \subseteq I,\ r-1 \notin I\}, & 2 \leq r \leq m+1. \end{cases}$$

In the last line $T_{m+1} = \varnothing$, so the condition is $m \notin I$. Consequently the two adjoints forget, respectively, every coordinate above the first occupied coordinate and every coordinate below the terminal occupied run.

Proof. The set $r_m(I)$ is the least terminal segment containing I, while $s_m(I)$ is the greatest terminal segment contained in I. These two universal properties are exactly the displayed adjunctions. Also

$$r_m(I \cup J) = r_m(I) \cup r_m(J) :$$

if both sets are nonempty, both sides are $T_{\min(\min I, \min J)}$, and the empty cases are immediate. A terminal segment is fixed by terminal closure, proving $r_m i_m = \mathrm{id}$.

For $1 \leq r \leq m$, the equality $r_m(I) = T_r$ says precisely that r is the least member of I, equivalently $r \in I \subseteq T_r$. The empty fibre is obvious. For $2 \leq r \leq m$, the equality $s_m(I) = T_r$ says that the whole tail T_r lies in I but the next longer tail does not; this is equivalent to $T_r \subseteq I$ and $r - 1 \notin I$. The endpoint cases follow from the same description. These formulas also exhibit the information loss rather than merely counting it. ☐

For reference, the fibre cardinalities obtained from the proof are

$$|r_m^{-1}(T_r)| = 2^{m-r}\ (r \le m),$$

and

$$|s_m^{-1}(T_1)| = 1, \qquad |s_m^{-1}(T_r)| = 2^{r-2}\ (2 \le r \le m+1).$$

When $m \ge 2$, C_m has width 1 whereas B_m contains the incomparable vertices $\{1\}$ and $\{2\}$. This is an explicit poset-isomorphism obstruction, not an assertion that the objects have no relation: their positive relation is the adjoint triple just proved.

Define product sheaves

$$\mathcal{I}_D(U) := \prod_{x \in U \cap Z} \mathrm{Idl}(Q_{x,m_x}), \qquad \mathcal{B}_D(U) := \prod_{x \in U \cap Z} B_{m_x}.$$

The ideal-chain identification and Θ_{m_x} give $\mathcal{I}_D \cong \prod C_{m_x}$ and identify $\mathcal{B}_D$ with $\mathcal{V}_D$ after forgetting from A-semimodules to join-semilattices. Applying i_{m_x} and r_{m_x} coordinatewise gives morphisms of sheaves of join-semilattices

$$\mathcal{I}_D \xrightarrow{i_D} \mathcal{B}_D \xrightarrow{r_D} \mathcal{I}_D, \qquad r_D i_D = \mathrm{id}_{\mathcal{I}_D},$$

Applying s_{m_x} coordinatewise gives a morphism of sheaves of posets $s_D : \mathcal{B}_D \to \mathcal{I}_D$; it is not asserted to preserve joins. Stalkwise $r_D \dashv i_D \dashv s_D$. Every map commutes with restriction because restriction is coordinate projection. This is the exact global morphism between the intrinsic Artin model and the Boolean multiplicity model.

5.5 Enhanced jets and the exact support morphism

Now take $X = \mathbb{C}$ with its literal global coordinate s. For $x \in Z$, put $t_x = s - x$ and define

$$J_x := \mathbb{C}[t_x]/(t_x^{m_x}), \qquad E_x := t_x\mathbb{C}[t_x]/(t_x^{m_x+1}), \qquad L_x := \mathbb{C}t_x^{m_x} \subset E_x.$$

Thus J_x has ordered basis $1, t_x, \ldots, t_x^{m_x-1}$ and E_x has ordered basis $t_x, \ldots, t_x^{m_x}$. The coordinate t_x gives an algebra isomorphism $Q_{x,m_x} \cong J_x$ and a linear degree-shift isomorphism

$$d_x : E_x \longrightarrow J_x, \qquad d_x\left(\sum_{r=1}^{m_x} c_r t_x^r\right) = \sum_{r=1}^{m_x} c_r t_x^{r-1}.$$

Its inverse is multiplication by t_x followed by reduction modulo $t_x^{m_x+1}$. The map d_x is a vector-space map, not a claim that E_x has the quotient-ring multiplication of J_x.

Define the coefficient-support map

$$\kappa_x : E_x \longrightarrow A^{m_x}, \qquad \kappa_x\left(\sum_{r=1}^{m_x} c_r t_x^r\right)_r = \begin{cases} e, & c_r \neq 0, \\ \tau, & c_r = 0. \end{cases}$$

and the basepoint-preserving section

$$\iota_x : A^{m_x} \longrightarrow E_x, \qquad \iota_x(a) = \sum_{\{r : a_r = e\}} t_x^r.$$

Proposition 5.7 (Support fibres and the additive obstruction)**.** *The maps κ_x and ι_x are morphisms of pointed sets and*

$$\kappa_x\iota_x = \mathrm{id}_{A^{m_x}} .$$

For the vertex corresponding to $I \subseteq [m_x]$, the fibre is

$$\kappa_x^{-1}(I) = \left\{ \sum_{r\in I} c_r t_x^r : c_r \in \mathbb{C}^\times \right\} \cong (\mathbb{C}^\times)^{|I|},$$

with the empty fibre interpreted as the singleton $\{0\}$. The map κ_x is not additive. Indeed, for every $m_x \geq 1$,

$$\kappa_x(t_x) = \kappa_x(-t_x) = \{1\}, \qquad \kappa_x(t_x - t_x) = \varnothing \neq \{1\} \cup \{1\}.$$

Finally, the additive monoids E_x and A^{m_x} are not isomorphic: the latter is idempotent, whereas $t_x + t_x = 2t_x \neq t_x$ in characteristic zero.

Proof. The coefficient 1 in $\iota_x(a)$ is nonzero in exactly the positions in which $a_r = e$, proving the retraction identity. Prescribing a support I forces the coefficients in I to be nonzero and all other coefficients to vanish, which gives the fibre formula. The displayed cancellation proves nonadditivity. An additive monoid isomorphism preserves the sentence $v+v = v$ for every v; the final two monoids satisfy different such sentences, yielding the stated obstruction. □

The product assignments

$$\mathcal{J}_D(U) = \prod_{x\in U\cap Z} J_x, \quad \mathcal{E}_D(U) = \prod_{x\in U\cap Z} E_x, \quad \mathcal{L}_D(U) = \prod_{x\in U\cap Z} L_x$$

are sheaves by theorem 5.2. The maps d_x, κ_x, ι_x assemble coordinatewise into sheaf maps, including the pointed-set retraction

$$\mathcal{E}_D \xrightarrow{\kappa_D} \mathcal{V}_D \xrightarrow{\iota_D} \mathcal{E}_D, \qquad \kappa_D\iota_D = \mathrm{id}_{\mathcal{V}_D} .$$

Thus the additive obstruction and the positive pointed-set morphism are retained simultaneously.

Proposition 5.8 (Terminal jet and the Artin socle)**.** *Under $d_x : E_x \to J_x$, the line L_x maps isomorphically to*

$$\mathrm{Soc}(J_x) := \{q \in J_x : t_x q = 0\} = \mathbb{C}t_x^{m_x-1}.$$

Its nonzero elements have Boolean support $T_{m_x} = \{m_x\}$, which corresponds under theorem 5.6 to the smallest nonzero Artin ideal I_{m_x-1}.

Proof. If $q = \sum_{r=0}^{m_x-1} c_r t_x^r$, then $t_x q = 0$ modulo $t_x^{m_x}$ exactly when every c_r with $0 \leq r \leq m_x - 2$ vanishes; this condition is vacuous when $m_x = 1$. Hence the socle is the displayed line. The degree shift sends $t_x^{m_x}$ to $t_x^{m_x-1}$, while coefficient support sends every nonzero scalar multiple of $t_x^{m_x}$ to $\{m_x\} = T_{m_x}$. The ideal-chain coordinate sends I_{m_x-1} to the same terminal segment. □

Theorem 5.9 (Local leading class with the required hypothesis)**.** *Let $f \not\equiv 0$ be meromorphic on $\mathbb{C}$, let Z_f be its zero set, and assume only that f is holomorphic in a neighbourhood of every $x \in Z_f$. If $m_x = \mathrm{ord}_x(f)$, then its truncated class in E_x is*

$$[f]_x = \frac{f^{(m_x)}(x)}{m_x!}\, t_x^{m_x} \in L_x \setminus \{0\}.$$

Its coefficient support is T_{m_x}, and its degree shift is a nonzero generator of the Artin socle.

Proof. Holomorphy near x and the definition of vanishing order give the Taylor expansion

$$f(s) = \sum_{r \geq m_x} \frac{f^{(r)}(x)}{r!} t_x^r, \qquad f^{(m_x)}(x) \neq 0.$$

Reduction modulo $t_x^{m_x+1}$ leaves precisely the displayed term. The support and socle claims are theorem 5.8. No global entire-function hypothesis is used. □

5.6 Compactification and the tail stalk

Let $j : \mathbb{C} \hookrightarrow \mathbb{P}^1(\mathbb{C})$ be the standard inclusion and let $i_\infty : \{\infty\} \hookrightarrow \mathbb{P}^1(\mathbb{C})$. For a family of pointed sets $(M_x)_{x \in Z}$, define

$$\mathrm{Tail}(M) := \left(\prod_{x \in Z} M_x\right) / \sim, \qquad a \sim b \iff a_x = b_x \text{ for all but finitely many } x.$$

For vector spaces V_x, this is the vector quotient

$$\mathrm{Tail}_{\mathbb{C}}(V) = \left(\prod_{x \in Z} V_x\right) \Big/ \left(\bigoplus_{x \in Z} V_x\right).$$

Let $j_* \mathcal{F}_M(U) = \mathcal{F}_M(U \cap \mathbb{C})$. Define explicitly $j_!^{\mathrm{fin}} \mathcal{F}_M$ by the same formula when $\infty \notin U$ and, when $\infty \in U$, by retaining exactly the families with finite support.

Lemma 5.10 (The finite extension is a sheaf)**.** *The assignment $j_!^{\mathrm{fin}} \mathcal{F}_M$ is a sheaf, and its inclusion into $j_* \mathcal{F}_M$ is an isomorphism away from ∞.*

Proof. Only gluing over an open set U containing ∞ requires proof. Let $U = \bigcup_i U_i$ and let finite-support sections on the U_i be compatible. Choose i_0 with $\infty \in U_{i_0}$. The complement $K = \mathbb{P}^1(\mathbb{C}) \setminus U_{i_0}$ is a compact subset of $\mathbb{C}$. Because Z is closed discrete, $K \cap Z$ is finite. The section on U_{i_0} has finite support, and compatibility forces the glued product section to vanish at every point of $(U \cap Z) \cap U_{i_0}$ outside that finite support. The only remaining possible support lies in $K \cap Z$, also finite. Hence the glued section has finite support. Locality and the statement away from ∞ follow directly from the definition. □

Theorem 5.11 (Exact tail stalk)**.** *The stalks of $j_!^{\mathrm{fin}} \mathcal{F}_M \to j_* \mathcal{F}_M$ agree at every finite point. At infinity,*

$$(j_!^{\mathrm{fin}} \mathcal{F}_M)_\infty \cong *, \qquad (j_* \mathcal{F}_M)_\infty \cong \mathrm{Tail}(M).$$

For vector spaces there is a short exact sequence

$$0 \longrightarrow j_!^{\mathrm{fin}} \mathcal{F}_V \longrightarrow j_* \mathcal{F}_V \longrightarrow i_{\infty *} \mathrm{Tail}_{\mathbb{C}}(V) \longrightarrow 0.$$

Both the middle and right sheaves are flasque. Therefore

$$H^q\Big(\mathbb{P}^1(\mathbb{C}), j_!^{\mathrm{fin}} \mathcal{F}_V\Big) = 0 \quad (q \geq 1),$$

and

$$\mathrm{Tail}_{\mathbb{C}}(V) \cong \Gamma(j_* \mathcal{F}_V) / \Gamma(j_!^{\mathrm{fin}} \mathcal{F}_V).$$

Proof. The finite stalks are computed as in theorem 5.2. A finite-support germ near infinity becomes the basepoint after shrinking past its finite support, so the first infinite stalk is terminal. A neighbourhood of infinity has compact complement in $\mathbb{C}$, and therefore meets Z in a cofinite subset. Extending a family by basepoints across the finite complement identifies a germ of $j_*\mathcal{F}_M$ with a global family modulo eventual equality. This proves the stalk formula.

For vector spaces, the quotient map sends a family on a neighbourhood of ∞ to its class after zero extension over the finite complement. It is surjective, and its kernel is exactly finite support, proving the sheaf sequence. Every restriction map of $j_*\mathcal{F}_V$ is a coordinate projection and is surjective by zero extension. The same is immediate for the skyscraper at ∞; hence both are flasque. The flasque-acyclicity theorem [81, pp. 184–188, Thm. 6.34] and the long exact sequence give vanishing in degrees at least two. In degree one, the relevant segment is

$$\Gamma(j_*\mathcal{F}_V) \longrightarrow \mathrm{Tail}_{\mathbb{C}}(V) \longrightarrow H^1(j_!^{\mathrm{fin}}\mathcal{F}_V) \longrightarrow 0.$$

The first arrow is the defining product-to-product/direct-sum quotient and is surjective. Thus H^1 vanishes as well, and its kernel gives the global-section quotient formula. $\square$

5.7 The zeta divisor, its jet section, and its nonzero tails

The Riemann zeta function is meromorphic on $\mathbb{C}$, with its only pole a simple pole at 1 [51, §25.2(i)]. Define

$$Z_\zeta = \{\rho \in \mathbb{C} : \zeta(\rho) = 0\}, \qquad D_\zeta = \sum_{\rho \in Z_\zeta} m_\rho[\rho], \quad m_\rho = \mathrm{ord}_\rho(\zeta).$$

The zero set is closed and discrete: every zero has a neighbourhood on which ζ is holomorphic and not identically zero, so local factorization makes the zero isolated; the pole at 1 is not a finite accumulation point of zeros. Thus all preceding sheaves apply to D_ζ.

Theorem 5.12 (Zeta divisor shadow and global leading jet)**.** *At every $\rho \in Z_\zeta$,*

$$(\mathcal{V}_{D_\zeta})_\rho \cong A^{m_\rho}, \qquad (\mathcal{A}_{D_\zeta})_\rho \cong A,$$

and the second stalk is the literal fixed set from theorem 5.1. The family

$$[\zeta]_\rho = \lambda_\rho (s-\rho)^{m_\rho}, \qquad \lambda_\rho = \frac{\zeta^{(m_\rho)}(\rho)}{m_\rho!} \neq 0,$$

is a global section of $\mathcal{E}_{D_\zeta}$. Its support at ρ is T_{m_ρ} and its degree shift generates the socle of $\mathbb{C}[s-\rho]/(s-\rho)^{m_\rho}$.

Proof. The stalk formulas are theorem 5.2; the fixed-set statement is theorem 5.1. Although ζ is not entire, it is holomorphic near each of its zeros. Hence theorem 5.9 applies at every ρ. Arbitrary families are sections of the product sheaf, which proves globality. The support and socle conclusions were proved in theorem 5.8. $\square$

The reflection formula

$$\zeta(s) = 2(2\pi)^{s-1}\sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)\zeta(1-s)$$

is DLMF equation 25.4.2 [51, §25.4, Eq. 25.4.2]; it also implies the trivial zeros [51, §25.10(i)].

Proposition 5.13 (Trivial-zero multiplicity and leading coordinate)**.** *For every $n \ge 1$, the zero $-2n$ is simple and*

$$\zeta'(-2n) = (-1)^n \frac{(2n)!}{2(2\pi)^{2n}}\,\zeta(2n+1).$$

Proof. At $s = -2n$, every factor in the reflection formula except the sine is holomorphic and nonzero: in particular $\Gamma(1+2n) = (2n)!$ and $\zeta(1+2n) > 0$ by its positive Dirichlet series. The sine has derivative $(\pi/2)\cos(-n\pi) = (\pi/2)(-1)^n$, so the zero is simple. Differentiating the product at that simple sine zero leaves

$$2(2\pi)^{-2n-1}\Gamma(1+2n)\zeta(1+2n)\cdot\frac{\pi}{2}(-1)^n,$$

which is the displayed expression after collecting powers of 2 and π. ☐

Corollary 5.14 (Nontriviality of both tail objects)**.** *The Boolean tail*

$$\mathrm{Tail}\big((A^{m_\rho})_{\rho\in Z_\zeta}\big)$$

has more than one element, and the vector tail

$$\left(\prod_{\rho\in Z_\zeta} E_\rho\right)\Big/\left(\bigoplus_{\rho\in Z_\zeta} E_\rho\right)$$

contains the nonzero class of $([\zeta]_\rho)_\rho$.

Proof. The trivial zeros $-2,-4,\ldots$ form an infinite subset. A family equal to a fixed nonbasepoint at every trivial-zero coordinate is not eventually the basepoint, so it gives a nonbasepoint Boolean tail class. By theorem 5.13, every coordinate $[\zeta]_{-2n}$ is nonzero. Hence the jet family is not finitely supported and its class in the vector quotient is nonzero. ☐

5.8 Equivariance over nonidentity base maps

Before specializing to the completed function, we state the base-map convention explicitly. Let $h : X \to X$ be a homeomorphism satisfying $h(Z) = Z$ and $m_{h(x)} = m_x$ for every $x \in Z$. For every open $U \subset X$, define

$$T_h(U) : \mathcal{V}_D(U) \longrightarrow \mathcal{V}_D(hU), \qquad \big(T_h(U)a\big)_{h(x),r} = a_{x,r} \quad (1 \le r \le m_x).$$

The same coordinate transport defines T_h on $\mathcal{I}_D$.

Proposition 5.15 (Exact transport and cocycle)**.** *Every $T_h(U)$ is an isomorphism, with inverse $T_{h^{-1}}(hU)$, and*

$$\mathrm{res}_{hU,hV}\,T_h(U) = T_h(V)\,\mathrm{res}_{U,V} \quad (V \subset U).$$

For two divisor-preserving homeomorphisms g, h,

$$T_g(hU)T_h(U) = T_{g\circ h}(U).$$

Thus these maps are equivariant sheaf data over the displayed base maps. They are not being silently recast as endomorphisms over id_X. Moreover the Artin–Boolean comparison is equivariant:

$$T_h i_D = i_D T_h, \qquad T_h r_D = r_D T_h, \qquad T_h s_D = s_D T_h.$$

Proof. Each formula merely relabels the coordinate (x, r) as $(h(x), r)$, so the inverse and restriction identities follow by substitution. Relabelling first by h and then by g relabels by $g \circ h$, proving the cocycle. The three chain–cube maps act on the r-coordinates at one support point and do not change the support point; relabelling and applying them therefore commute coordinate by coordinate. ☐

5.9 Completed-zeta packet descent and phase coordinates

Use Riemann's completed function in the literal convention

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big)\,\zeta(s).$$

DLMF gives this normalization and the symmetry $\xi(1-s) = \xi(s)$ [51, §25.4, Eqs. 25.4.3–25.4.4]. Lagarias and Montague record explicitly that ξ is entire and that its zeros are exactly the zeta zeros in $0 < \Re(s) < 1$ [83, §1, Eq. (1.1) and the following paragraph]. On this open strip the remaining factor $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$ is holomorphic and nonzero, so the vanishing orders of ξ and ζ agree. Let D_ξ be this divisor and set

$$c(s) = \overline{s}, \qquad w(s) = 1-s, \qquad W = \{1, c, w, cw\} \cong C_2 \times C_2.$$

The identity $\overline{\xi(\overline{s})} = \xi(s)$ follows first in a right half-plane from the real Dirichlet coefficients and the conjugation law for Γ, and then everywhere by analytic continuation. Hence both c and w preserve D_ξ, including multiplicities.

Let $Y_\xi = |D_\xi|/W$ with quotient map $q : |D_\xi| \to Y_\xi$. Since the source is discrete and W is finite, Y_ξ is discrete. For an orbit O, write $r_O = |O|$ and m_O for its common multiplicity.

Theorem 5.16 (Packet descent with explicit diagonal inverse)**.** *On the restriction of the divisor sheaves to $|D_\xi|$,*

$$(q_*\mathcal{V}_{D_\xi})_O \cong (A^{m_O})^{r_O}, \qquad (q_*\mathcal{I}_{D_\xi})_O \cong (C_{m_O})^{r_O}.$$

The fixed parts under the transport action of W are the diagonal subobjects. Projection to any chosen factor and diagonal repetition are mutually inverse on the fixed part. Consequently

$$(q_*\mathcal{V}_{D_\xi})^W(U) \cong \prod_{O\in U} A^{m_O}, \qquad (q_*\mathcal{I}_{D_\xi})^W(U) \cong \prod_{O\in U} C_{m_O}$$

for every open $U \subseteq Y_\xi$. The descended chain–cube maps still satisfy $r \dashv i \dashv s$ stalkwise and $ri = \mathrm{id}$.

Proof. Because the orbit space is discrete, the stalk of a pushforward at O is the product of the stalks over $q^{-1}(O) = O$. Multiplicity preservation gives the two displayed products. The W-action permutes the factors transitively. A tuple is fixed exactly when every factor equals every other factor, which is the diagonal. Projection of a diagonal tuple to one component is independent of the chosen component, and diagonal repetition is its two-sided inverse. These maps commute with restrictions, so the stalkwise descriptions assemble into the displayed product sheaves. The equivariance equations in theorem 5.15 descend the three comparison maps and their equations. □

For a zero ρ of ξ, define the nonzero leading coefficient and its phase

$$\mu_\rho := \frac{\xi^{(m_\rho)}(\rho)}{m_\rho!} \in \mathbb{C}^\times, \qquad \phi_\rho := \frac{\mu_\rho}{|\mu_\rho|} \in S^1.$$

Proposition 5.17 (Exact phase transport)**.** *For every such ρ,*

$$\mu_{\overline{\rho}} = \overline{\mu_\rho}, \qquad \mu_{1-\rho} = (-1)^{m_\rho}\mu_\rho, \qquad \mu_{1-\overline{\rho}} = (-1)^{m_\rho}\overline{\mu_\rho},$$

and the identical formulas hold for ϕ.

Proof. Expand $\overline{\xi(\overline{s})} = \xi(s)$ in local power series at ρ; coefficient conjugation gives the first formula. Differentiating $\xi(1-s) = \xi(s)$ exactly m_ρ times gives

$$(-1)^{m_\rho} \xi^{(m_\rho)}(1-s) = \xi^{(m_\rho)}(s).$$

Evaluation at $s = \rho$ and division by $m_\rho!$ gives the second formula; the third is its conjugate. Absolute values are preserved by conjugation and by the sign, so division by absolute value gives the phase formulas. □

For each orbit O, let Γ_{m_O} be the subgroup of $\mathrm{Homeo}(S^1)$ generated by

$$z \longmapsto \overline{z}, \qquad z \longmapsto (-1)^{m_O} z,$$

and put $\Phi_O = S^1/\Gamma_{m_O}$. By theorem 5.17, the image of ϕ_ρ in Φ_O is independent of $\rho \in O$. These images form a section of the product sheaf

$$\mathcal{P}\langle_\xi(U) = \prod_{O \in U} \Phi_O$$

on Y_ξ. This is a quotient-valued phase decoration of the packet sheaf; no claim about the location, multiplicity, or spectral origin of any individual zero is inferred from it.

5.10 Certificate boundary and retained interfaces

The accompanying exact finite certificate exhaustively checks, for $1 \le m \le 8$, the two adjunctions, the retraction, join preservation, every fibre formula and fibre cardinality, the reduced-shadow retraction, coefficient support fibres over a finite coefficient alphabet, and the Klein-four phase-action relations. Those checks are a replay of finite algebra, not a replacement for the all-m proofs above. In particular, they do not certify analytic continuation, the functional equation, divisor discreteness, sheaf cohomology, or any statement about individual zeros.

Three programme boundaries remain explicit. First, A^{m_x} is a constructed multiplicity shadow, while Q_{x,m_x} is the intrinsic analytic thickening; their proved interface is $r_m \dashv i_m \dashv s_m$ and not an untyped identification. Second, the coefficient-support map from jets is a pointed-set retraction with explicit torus fibres and a proved additive obstruction. Third, the symmetry maps are transport over c and w, followed by fixed-point descent along q; they are not endomorphisms over an unchanged base. Later blueprint, analytic packet-product, operator-algebraic, or physical interpretations require their own typed arrows and are not supplied by the present construction.

6 Phase sheaves, affine charts, and Taylor atoms

The preceding section constructed the divisor, jet, tail, and packet objects. We now develop the complementary phase layer and its exact maps to those objects. Two classical inputs are used at their literal scope. Lee records that a nonvanishing holomorphic function on a simply connected domain has a holomorphic logarithm [81, p. 145, Problem 5-11]; below we give the normalized local-disc construction directly from the logarithm power series. Connes and Consani construct their circle cycles and prove the exact positive-integer family of lengths attached to a critical-line ordinate [21, §6]; that result is stated separately in theorem 6.17. No map from a local derivative phase to their Hilbert-space construction is presumed.

6.1 The general fixed locus and Boolean affine points

We first retain two algebraic refinements which are independent of the analytic phase construction. For a commutative ring R, write

$$G(R) = R \sqcup \{\tau\},$$

with the same operations as in section 4: τ is the additive identity and the multiplicative absorber, while the operations between supported elements are the ring operations.

Proposition 6.1 (Fixed locus over a general coefficient ring)**.** *Let $u \in R^{\times}$. Multiplication by u on $G(R)$ has fixed set*

$$\mathrm{Fix}_{G(R)}(u \cdot -) = \{\tau\} \sqcup \mathrm{Ann}_R(u-1),$$

where

$$\mathrm{Ann}_R(u-1) = \{r \in R : (u-1)r = 0\}.$$

If R is an integral domain and $u \neq 1$, this reduces to

$$\mathrm{Fix}_{G(R)}(u \cdot -) = \{\tau, 0_R\}.$$

Proof. The absorber is fixed because $u\tau = \tau$. A supported element $r \in R \subset G(R)$ is fixed precisely when

$$ur = r \quad \Longleftrightarrow \quad (u-1)r = 0.$$

This proves the first formula, including every possible zero-divisor contribution. If R is a domain and $u - 1 \neq 0$, its annihilator is $\{0_R\}$, which gives the second formula. $\square$

Retain the Boolean coordinate isomorphism $\beta : A \xrightarrow{\sim} \mathbb{B}$ from theorem 5.1. The multiplicity cube has a precise functor-of-points description; this is an additional map, not a replacement for the Artin-chain comparison of theorem 5.6.

Proposition 6.2 (Boolean affine points and the two polynomial maps)**.** *For every $m \geq 1$, evaluation at the variables gives a bijection*

$$\mathrm{ev}_m : \mathbb{B}^m \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{B}\text{-}alg}\big(\mathbb{B}[T_1, \dots, T_m], \mathbb{B}\big),$$
$$(b_1, \dots, b_m) \longmapsto \big(P \longmapsto P(b_1, \dots, b_m)\big).$$

Under this bijection, the Boolean diagonal

$$\delta_m : \mathbb{B} \longrightarrow \mathbb{B}^m, \qquad b \longmapsto (b, \dots, b),$$

is induced contravariantly by

$$\delta_m^* : \mathbb{B}[T_1, \dots, T_m] \longrightarrow \mathbb{B}[T], \qquad T_j \longmapsto T,$$

and the nonempty-support map

$$\sigma_m : \mathbb{B}^m \longrightarrow \mathbb{B}, \qquad (b_1, \dots, b_m) \longmapsto b_1 + \cdots + b_m,$$

is induced contravariantly by

$$\sigma_m^* : \mathbb{B}[T] \longrightarrow \mathbb{B}[T_1, \dots, T_m], \qquad T \longmapsto T_1 + \cdots + T_m.$$

Moreover,

$$\sigma_m \delta_m = \mathrm{id}_{\mathbb{B}}.$$

Transport through β recovers exactly the maps $\Delta_m : A \to A^m$ and $\Sigma_m : A^m \to A$ of theorem 5.4.

Proof. A $\mathbb{B}$-algebra homomorphism out of the polynomial semiring is determined by the m values assigned to $T_1, \dots, T_m$. Conversely, any $(b_1, \dots, b_m) \in \mathbb{B}^m$ defines evaluation on a finite Boolean polynomial:

$$\sum_{\alpha \in \mathbb{N}^m} c_\alpha T^\alpha \longmapsto \sum_{\alpha \in \mathbb{N}^m} c_\alpha \prod_{j=1}^{m} b_j^{\alpha_j},$$

where only finitely many c_α are nonzero. Evaluation preserves $0, 1$, addition, multiplication, and the coefficient copy of $\mathbb{B}$. These two constructions are inverse, proving the bijection.

If $b \in \mathbb{B}$ is a point of the affine line, composition of its evaluation map with δ_m^* sends every T_j to b; this is $\delta_m(b)$. If $(b_1, \dots, b_m)$ is an m-dimensional point, composition of its evaluation map with σ_m^* sends T to $b_1 + \cdots + b_m$; this is σ_m. Finally, Boolean addition is idempotent, so

$$\sigma_m \delta_m(b) = \underbrace{b + \cdots + b}_{m \text{ terms}} = b.$$

The last statement follows by applying β in every coordinate. $\square$

6.2 A phase-decorated sheaf over the simple-zero locus

Let $\Omega \subset \mathbb{C}$ be open and let f be a holomorphic function on Ω which is not identically zero on any connected component. Its simple-zero set

$$Z_f^{\text{simp}} := \{x \in \Omega : f(x) = 0,\ f'(x) \neq 0\}$$

is closed and discrete in Ω. The derivative below is explicitly the derivative in the displayed global coordinate on $\Omega \subset \mathbb{C}$. Write

$$D_f^{\text{simp}} := \sum_{x \in Z_f^{\text{simp}}} [x].$$

Define the pointed set

$$M := \{\star\} \sqcup S^1$$

with basepoint $\star$, and define

$$\pi : M \longrightarrow A, \qquad \pi(\star) = \tau, \qquad \pi(u) = e \quad (u \in S^1).$$

There is a pointed section

$$\iota : A \longrightarrow M, \qquad \iota(\tau) = \star, \qquad \iota(e) = 1,$$

so $\pi\iota = \mathrm{id}_A$.

Theorem 6.3 (Phase-decorated simple-zero sheaf)**.** *For an open $U \subset \Omega$, set*

$$\mathcal{M}_f^{\text{ph}}(U) := \prod_{x \in U \cap Z_f^{\text{simp}}} M.$$

Then $\mathcal{M}_f^{\text{ph}}$ is a sheaf of pointed sets. Coordinatewise application of π and ι gives a split morphism

$$\mathcal{M}_f^{\text{ph}} \underset{\iota_f}{\overset{\Pi_f}{\rightleftarrows}} \mathcal{A}_{D_f^{\text{simp}}}, \qquad \Pi_f \iota_f = \mathrm{id}\,.$$

At a simple zero x, the two fibres of the stalk map are

$$\pi^{-1}(\tau) = \{\star\}, \qquad \pi^{-1}(e) = S^1.$$

The family

$$\varphi_f(U) := \left(\frac{f'(x)}{|f'(x)|}\right)_{x \in U \cap Z_f^{\mathrm{simp}}}$$

is a section compatible with restriction, and $\Pi_f(\varphi_f)$ is the everywhere-e section of the reduced Boolean shadow on the simple locus.

Proof. The sheaf proof is coordinatewise. For a cover $U = \bigcup_i U_i$, compatible families on the U_i assign a unique value in M to every $x \in U \cap Z_f^{\mathrm{simp}}$, and those values form the unique glued family. Coordinatewise application of a pointed map commutes with coordinate projections, hence defines a sheaf morphism. The equation $\Pi_f \iota_f = \mathrm{id}$ follows coordinatewise from $\pi\iota = \mathrm{id}_A$. The displayed fibres follow directly from the definition of π: discreteness gives a neighbourhood W of a simple zero x with $W \cap Z_f^{\mathrm{simp}} = \{x\}$, so the two stalks identify with M and A, respectively, and the induced stalk map is literally π.

At a simple zero x, $f'(x) \neq 0$, so $f'(x)/|f'(x)| \in S^1$. The resulting families restrict by forgetting coordinates and therefore define a compatible section. Every one of its coordinates is sent by π to e, which proves the last assertion. $\square$

Thus Π_f forgets the entire circle-valued derivative phase while retaining whether the stalk is occupied. The section ι_f selects the phase 1; it is not an inverse on $\mathcal{M}_f^{\mathrm{ph}}$. In particular, the fibre calculation states the exact information loss rather than identifying the two sheaves.

Corollary 6.4 (The negative-even derivative phase)**.** *On the negative-even zero sector of ζ,*

$$\varphi_\zeta(-2n) = (-1)^n \qquad (n \geq 1).$$

Proof. Apply theorem 6.3 on $\Omega = \mathbb{C} \setminus \{1\}$, where ζ is holomorphic. By theorem 5.13,

$$\zeta'(-2n) = (-1)^n \frac{(2n)!}{2(2\pi)^{2n}} \zeta(2n+1).$$

The factor after $(-1)^n$ is positive, so division by the absolute value leaves exactly $(-1)^n$. $\square$

6.3 Normalized real affine factors and their complete fibres

Write

$$S^1_+ := \{z \in S^1 : \Re z > 0\}.$$

For $a \in \mathbb{R}$, define the normalized real affine factor

$$\mathsf{u}(a) := \frac{1 + \mathrm{i}a}{\sqrt{1 + a^2}}.$$

This notation is reserved for the phase-factor map; unit directions will be denoted by ν.

Proposition 6.5 (One-factor coordinate and its differential)**.** *The map*

$$\mathsf{u} : \mathbb{R} \longrightarrow S^1_+$$

is a real-analytic bijection with inverse

$$\mathsf{u}^{-1}(z) = \frac{\Im z}{\Re z}.$$

It satisfies

$$\mathsf{u}(a) = \exp\bigl(\mathrm{i}\arctan a\bigr), \qquad \mathsf{u}'(a) = \frac{\mathrm{i}}{1+a^2}\mathsf{u}(a).$$

Proof. Its modulus is one and its real part is $(1+a^2)^{-1/2} > 0$, so the image lies in S^1_+. If $z = x + \mathrm{i}y \in S^1_+$, put $a = y/x$. Since $x > 0$ and $x^2 + y^2 = 1$,

$$\frac{1}{\sqrt{1+a^2}} = \frac{1}{\sqrt{1+y^2/x^2}} = x,$$

and consequently $\mathsf{u}(a) = x + \mathrm{i}y = z$. This proves surjectivity and the inverse formula, hence also injectivity.

Because $\arctan a \in (-\pi/2, \pi/2)$,

$$\cos(\arctan a) = \frac{1}{\sqrt{1+a^2}}, \qquad \sin(\arctan a) = \frac{a}{\sqrt{1+a^2}},$$

which proves the exponential formula. Differentiating it gives

$$\mathsf{u}'(a) = \frac{\mathrm{i}}{1+a^2}\exp(\mathrm{i}\arctan a) = \frac{\mathrm{i}}{1+a^2}\mathsf{u}(a).$$

□

For $N \geq 1$, define

$$P_N : \mathbb{R}^N \longrightarrow S^1, \qquad P_N(a_1, \ldots, a_N) := \prod_{k=1}^{N} \mathsf{u}(a_k).$$

Theorem 6.6 (Finite phase products, fibres, and a three-factor section)**.** *For every* $(a_1, \ldots, a_N) \in \mathbb{R}^N$*,*

$$P_N(a_1, \ldots, a_N) = \exp\left(\mathrm{i}\sum_{k=1}^{N} \arctan a_k\right).$$

If $z = e^{\mathrm{i}\theta}$ *with* $\theta \in (-\pi, \pi]$*, then*

$$P_N^{-1}(z) = \bigcup_{\substack{\ell \in \mathbb{Z} \\ \theta + 2\pi\ell \in (-N\pi/2, N\pi/2)}} \left\{ a \in \mathbb{R}^N : \sum_{k=1}^{N} \arctan a_k = \theta + 2\pi\ell \right\}.$$

In particular,

$$\operatorname{im} P_1 = S^1_+, \qquad \operatorname{im} P_2 = S^1 \setminus \{-1\}, \qquad \operatorname{im} P_N = S^1 \quad (N \geq 3).$$

An explicit set-theoretic section of P_3 *is*

$$s_3(e^{\mathrm{i}\theta}) = \left(\tan\frac{\theta}{3}, \tan\frac{\theta}{3}, \tan\frac{\theta}{3}\right), \qquad \theta \in (-\pi, \pi].$$

Proof. Multiplying the exponential identity in theorem 6.5 proves the first formula. Equality with $e^{\mathrm{i}\theta}$ holds exactly when

$$\sum_{k=1}^{N} \arctan a_k = \theta + 2\pi\ell$$

for some integer ℓ. Every arctangent lies in $(-\pi/2, \pi/2)$, so only the integers satisfying the displayed interval condition can occur. Conversely, every point on any displayed level set maps to z. This proves the complete fibre formula.

For $N = 1$, the angle range is $(-\pi/2, \pi/2)$, which is S^1_+. For $N = 2$, the sum ranges over $(-\pi, \pi)$, whose exponential image omits only -1. If $N = 3$, then for every principal argument $\theta \in (-\pi, \pi]$, one has $\theta/3 \in (-\pi/2, \pi/2)$, and therefore

$$P_3\left(\tan\frac{\theta}{3}, \tan\frac{\theta}{3}, \tan\frac{\theta}{3}\right) = e^{\mathrm{i}\theta}.$$

For $N > 3$, append zero coordinates, since $\mathsf{u}(0) = 1$. The displayed principal-argument section is not claimed to be continuous at -1. $\square$

Proposition 6.7 (Orthogonal-walk expansion with signed steps). *Let $e_q = e_q(a_1, \ldots, a_N)$ be the elementary symmetric polynomials and put*

$$W_r := \sum_{q=0}^{r} \mathrm{i}^q e_q \qquad (0 \le r \le N).$$

Then

$$\prod_{k=1}^{N}(1 + \mathrm{i}a_k) = \sum_{q=0}^{N} \mathrm{i}^q e_q = A_N + \mathrm{i}B_N,$$

where

$$A_N = \sum_{0 \le 2j \le N} (-1)^j e_{2j}, \qquad B_N = \sum_{0 \le 2j+1 \le N} (-1)^j e_{2j+1}.$$

For $1 \le q \le N$, the q-th increment is

$$W_q - W_{q-1} = \mathrm{i}^q e_q;$$

it is horizontal for even q, vertical for odd q, has signed scalar e_q, and has Euclidean length $|e_q|$. Moreover,

$$A_N^2 + B_N^2 = \prod_{k=1}^{N}(1 + a_k^2),$$

and the normalized endpoint is

$$\frac{A_N + \mathrm{i}B_N}{\sqrt{A_N^2 + B_N^2}} = P_N(a_1, \ldots, a_N).$$

Proof. In expanding the product, selecting $\mathrm{i}a_k$ from exactly the indices in a subset $I \subseteq \{1, \ldots, N\}$ contributes $\mathrm{i}^{|I|} \prod_{k \in I} a_k$. Grouping by $|I| = q$ gives

$$\prod_{k=1}^{N}(1 + \mathrm{i}a_k) = \sum_{q=0}^{N} \mathrm{i}^q \sum_{|I|=q} \prod_{k \in I} a_k = \sum_{q=0}^{N} \mathrm{i}^q e_q.$$

Separating even and odd q gives $A_N + \mathrm{i}B_N$. The increment formula is immediate, including the sign and absolute Euclidean length. Taking squared absolute values in the product identity gives

$$A_N^2 + B_N^2 = \prod_{k=1}^{N} |1 + \mathrm{i}a_k|^2 = \prod_{k=1}^{N} (1 + a_k^2).$$

Division by the positive square root factorizes coordinatewise and gives $\prod_k \mathsf{u}(a_k) = P_N(a)$. □

Corollary 6.8 (Equal-step limit)**.** *For every $\theta \in \mathbb{R}$,*

$$\lim_{N\to\infty} \mathsf{u}(\theta/N)^N = e^{\mathrm{i}\theta}.$$

Proof. By theorem 6.6,

$$\mathsf{u}(\theta/N)^N = \exp\bigl(\mathrm{i}N \arctan(\theta/N)\bigr).$$

Since $\arctan x/x \to 1$ as $x \to 0$, $N \arctan(\theta/N) \to \theta$, and continuity of the exponential proves the claim. □

6.4 Branch-correct affine cyclotomic charts

The next theorem uses the continuous argument on the entire upper-half-plane path, so its formula remains valid even when the real part changes sign.

Theorem 6.9 (Affine phase chart in a fixed upper-half-plane direction)**.** *Fix $0 < \phi < \pi$, put $\omega = e^{\mathrm{i}\phi}$, and define*

$$L_\phi(t) := 1 + t\omega \qquad (t > 0).$$

There is a unique continuous argument

$$\Theta_\phi : (0, \infty) \longrightarrow (0, \phi), \qquad \frac{L_\phi(t)}{|L_\phi(t)|} = e^{\mathrm{i}\Theta_\phi(t)}.$$

It is real analytic and satisfies

$$\Theta_\phi'(t) = \frac{\sin\phi}{1 + 2t\cos\phi + t^2} > 0.$$

It is a bijection, with exact inverse and radius

$$t_\phi(\theta) = \frac{\sin\theta}{\sin(\phi - \theta)}, \qquad |L_\phi(t_\phi(\theta))| = \frac{\sin\phi}{\sin(\phi - \theta)} \quad (0 < \theta < \phi).$$

Equivalently,

$$1 + t_\phi(\theta)e^{\mathrm{i}\phi} = \frac{\sin\phi}{\sin(\phi - \theta)} e^{\mathrm{i}\theta}.$$

Proof. The imaginary part of $L_\phi(t)$ is $t\sin\phi > 0$. Thus the path lies entirely in the upper half-plane and has a unique continuous argument in $(0, \pi)$. Since $L_\phi(t)$ is a positive linear combination of the rays with arguments 0 and ϕ, and the open sector between those rays is convex for $0 < \phi < \pi$, this argument lies in $(0, \phi)$.

For a nonzero differentiable complex path $z(t)$, differentiation of a local logarithm gives

$$\frac{d}{dt} \arg z(t) = \Im \frac{z'(t)}{z(t)}.$$

With $z = L_\phi$,

$$\Theta'_\phi(t) = \Im \frac{e^{\mathrm{i}\phi}}{1 + te^{\mathrm{i}\phi}} = \Im \frac{e^{\mathrm{i}\phi} + t}{|1 + te^{\mathrm{i}\phi}|^2} = \frac{\sin\phi}{1 + 2t\cos\phi + t^2} > 0.$$

Hence Θ_ϕ is real analytic and strictly increasing. Directly,

$$\lim_{t\to 0^+} \Theta_\phi(t) = 0, \qquad \lim_{t\to\infty} \Theta_\phi(t) = \phi,$$

so it is a bijection onto $(0, \phi)$.

For $0 < \theta < \phi$, write

$$1 + te^{\mathrm{i}\phi} = re^{\mathrm{i}\theta}$$

with $r, t > 0$. Equating imaginary and real parts gives

$$t \sin\phi = r \sin\theta, \qquad 1 + t\cos\phi = r\cos\theta.$$

Eliminating r yields

$$t \sin(\phi - \theta) = \sin\theta,$$

and therefore $t = \sin\theta/\sin(\phi - \theta)$. Substitution in the first equation gives $r = \sin\phi/\sin(\phi - \theta)$. These positive values satisfy the original complex equation, proving both inverse formulas. □

Corollary 6.10 (Cyclotomic-direction charts)**.** *Let p be an odd prime, let $u \in \{1, \dots, (p-1)/2\}$, and put*

$$\omega_{p,u} = e^{2\pi\mathrm{i}u/p},$$

Then $t \mapsto (1 + t\omega_{p,u})/|1 + t\omega_{p,u}|$ is a bijection from $(0, \infty)$ to the open unit-circle arc with arguments $(0, 2\pi u/p)$, with the inverse in theorem 6.9. In particular, for $p = 5, u = 1$,

$$t(\theta) = \frac{\sin\theta}{\sin(2\pi/5 - \theta)} \qquad (0 < \theta < 2\pi/5).$$

Proof. Apply theorem 6.9 with $\phi = 2\pi u/p$, which lies in $(0, \pi)$ by the bound on u. □

The term *cyclotomic* here describes the fixed direction $\omega_{p,u}$. For an arbitrary real t, the affine factor $1 + t\omega_{p,u}$ need not belong to the number field $\mathbb{Q}(\omega_{p,u})$: subtracting 1 and dividing by the nonzero $\omega_{p,u}$ shows that membership would imply $t \in \mathbb{Q}(\omega_{p,u})$, whereas every transcendental real t lies outside this algebraic number field. No arithmetic condition on a zeta derivative is implied by placing its phase in one of these real-analytic charts.

6.5 Uniform Taylor atomization of a real-analytic phase

Theorem 6.11 (Exact Taylor atomization)**.** *Let*

$$\Theta(t) = \sum_{n=1}^{\infty} \beta_n t^n$$

be real analytic for $|t| < R$, with $\beta_n \in \mathbb{R}$ and $\Theta(0) = 0$. There is $r \in (0, R)$ such that

$$\sum_{n=1}^{\infty} |\beta_n| r^n < \frac{\pi}{4}.$$

For $|t| \le r$, define

$$\alpha_n(t) := \tan(\beta_n t^n).$$

Then:

1. $\sum_{n\geq 1} |\alpha_n(t)|$ *converges uniformly on* $[-r,r]$*;*
2. *for every* $N \geq 1$*,*
$$\prod_{n=1}^{N} \mathsf{u}(\alpha_n(t)) = \exp\left(\mathrm{i}\sum_{n=1}^{N} \beta_n t^n\right);$$
3. *the infinite product, defined as the uniform limit of its partial products, satisfies*
$$\prod_{n=1}^{\infty} \mathsf{u}(\alpha_n(t)) = e^{\mathrm{i}\Theta(t)}$$
uniformly on $[-r,r]$*;*
4. *the unnormalized* N*-th product is the exact orthogonal-walk endpoint*
$$\prod_{n=1}^{N}(1+\mathrm{i}\alpha_n(t)) = \sum_{q=0}^{N} \mathrm{i}^q e_q(\alpha_1(t),\ldots,\alpha_N(t));$$
5. *for* $|t| < r$*,*
$$\frac{d}{dt}e^{\mathrm{i}\Theta(t)} = \mathrm{i}\Theta'(t)e^{\mathrm{i}\Theta(t)}, \qquad \frac{\alpha_n'(t)}{1+\alpha_n(t)^2} = n\beta_n t^{n-1}.$$

Proof. Absolute convergence of the power series at a sufficiently small positive radius permits a choice of r with the displayed strict bound. For $|t| \leq r$, every term satisfies
$$|\beta_n t^n| \leq |\beta_n| r^n < \frac{\pi}{4}.$$
On $[-\pi/4, \pi/4]$, the mean-value theorem and $\sec^2 x \leq 2$ give
$$|\tan x| \leq 2|x|.$$
Hence
$$|\alpha_n(t)| \leq 2|\beta_n| r^n,$$
and the Weierstrass test proves uniform absolute convergence in (1).

Because $\beta_n t^n \in (-\pi/2, \pi/2)$,
$$\arctan \alpha_n(t) = \arctan(\tan(\beta_n t^n)) = \beta_n t^n.$$
Applying theorem 6.6 gives (2). The Taylor partial sums $\Theta_N(t) = \sum_{n\leq N} \beta_n t^n$ converge uniformly to Θ on $[-r,r]$. Since $z \mapsto e^{\mathrm{i}z}$ is uniformly continuous on the compact real interval containing their ranges, (2) converges uniformly to $e^{\mathrm{i}\Theta(t)}$, proving (3). Part (4) is theorem 6.7 applied to the finite slope vector.

The first identity in (5) is differentiation of the analytic composite. For the second,
$$\alpha_n'(t) = \sec^2(\beta_n t^n) n\beta_n t^{n-1} = (1+\alpha_n(t)^2) n\beta_n t^{n-1}.$$
□

The atomization is unique after the Taylor coefficients and the small tangent branch have been fixed. It is not a uniqueness statement among all possible finite or infinite factorizations of the same unit phase; the complete finite fibres in theorem 6.6 show why such a stronger assertion would fail.

6.6 Constructive logarithms and directional phase germs

For $w \in \mathbb{C}^\times$, write

$$\operatorname{ph}(w) := \frac{w}{|w|} \in S^1.$$

Theorem 6.12 (Normalized logarithm and all Taylor coordinates)**.** *Let g be holomorphic near 0 and suppose $g(0) \neq 0$. After shrinking to a disc $|z| < \varepsilon$, there is a unique holomorphic function*

$$h(z) = \sum_{n=1}^{\infty} c_n z^n, \qquad h(0) = 0,$$

such that

$$g(z) = g(0)e^{h(z)}.$$

If

$$\frac{g(z)}{g(0)} = 1 + \sum_{n=1}^{\infty} a_n z^n,$$

then every logarithmic coefficient is explicitly

$$c_n = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} \sum_{\substack{n_1+\cdots+n_k=n \\ n_j \geq 1}} a_{n_1} \cdots a_{n_k}.$$

For a unit direction $\nu \in S^1$, put

$$\Theta_{g,\nu}(t) := \Im h(t\nu) = \sum_{n=1}^{\infty} \beta_{g,\nu,n} t^n, \qquad \beta_{g,\nu,n} := \Im(c_n \nu^n).$$

For real $|t| < \varepsilon$,

$$|g(t\nu)| = |g(0)| \exp\bigl(\Re h(t\nu)\bigr),$$
$$\operatorname{ph}(g(t\nu)) = \operatorname{ph}(g(0))e^{\mathrm{i}\Theta_{g,\nu}(t)}.$$

After decreasing ε, the final factor has the uniform atomization of theorem 6.11.

Proof. Continuity and $g(0) \neq 0$ allow a disc on which

$$\left|\frac{g(z)}{g(0)} - 1\right| < \frac{1}{2}.$$

Put $q(z) = g(z)/g(0) - 1$. The series

$$h(z) := \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} q(z)^k$$

converges normally on every smaller closed disc, hence is holomorphic. It is the usual logarithm series, so $e^{h(z)} = 1 + q(z) = g(z)/g(0)$, and $h(0) = 0$. If h_1, h_2 are two such logarithms normalized by zero at the origin, then $e^{h_1 - h_2} = 1$. The continuous function $h_1 - h_2$ takes values in the discrete set $2\pi\mathrm{i}\mathbb{Z}$; on the connected disc it is constant, and its value at zero is zero. Thus $h_1 = h_2$.

Expanding the k-th power,

$$q(z)^k = \sum_{n \geq k} \left(\sum_{\substack{n_1+\cdots+n_k=n \\ n_j \geq 1}} a_{n_1} \cdots a_{n_k} \right) z^n.$$

For fixed n, only $1 \leq k \leq n$ contributes. Taking the coefficient of z^n in the logarithm series proves the displayed formula for c_n.

Substitute $z = t\nu$:

$$g(t\nu) = g(0) \exp \left(\sum_{n \geq 1} c_n \nu^n t^n \right).$$

Taking the absolute value keeps the real part of the exponent; dividing by the absolute value keeps the imaginary part. This proves the modulus and phase formulas. The real power series for $\Theta_{g,\nu}$ has zero constant term, so theorem 6.11 applies on a smaller interval. □

Corollary 6.13 (First two directional atoms)**.** *In the notation of theorem 6.12,*

$$c_1 = a_1, \qquad c_2 = a_2 - \frac{1}{2} a_1^2,$$

and therefore

$$\beta_{g,\nu,1} = \Im(\nu a_1), \qquad \beta_{g,\nu,2} = \Im \left(\nu^2 \left(a_2 - \frac{1}{2} a_1^2 \right) \right).$$

Proof. The coefficient formula in theorem 6.12 gives

$$c_1 = a_1, \qquad c_2 = a_2 - \frac{1}{2} a_1 a_1.$$

Taking $\Im(c_n \nu^n)$ gives the two directional formulas. □

6.7 Local completed-function phase packets

Let ξ be the completed function used in theorems 5.16 and 5.17. Its entirety and exact zero set are sourced in section 5; the calculations below use only local holomorphy, exact order of vanishing, and the two functional symmetries.

Fix a zero ρ of multiplicity $m = m_\rho$, and set

$$\mu_\rho := \frac{\xi^{(m)}(\rho)}{m!} \in \mathbb{C}^\times, \qquad \psi_\rho(z) := \frac{\xi(\rho + z)}{\mu_\rho z^m} \quad (z \neq 0), \qquad \psi_\rho(0) := 1.$$

Theorem 6.14 (Local phase packet with explicit derivative coordinates)**.** *There is a disc on which a unique holomorphic*

$$h_\rho(z) = \sum_{n=1}^{\infty} c_{\rho,n} z^n, \qquad h_\rho(0) = 0,$$

satisfies

$$\xi(\rho + z) = \mu_\rho z^m e^{h_\rho(z)}.$$

Writing

$$\psi_\rho(z) = 1 + \sum_{j=1}^{\infty} a_{\rho,j} z^j,$$

one has the exact derivative coordinates

$$a_{\rho,j} = \frac{m!\,\xi^{(m+j)}(\rho)}{(m+j)!\,\xi^{(m)}(\rho)} \qquad (j \geq 1),$$

and

$$c_{\rho,n} = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} \sum_{\substack{n_1+\cdots+n_k=n \\ n_j \geq 1}} a_{\rho,n_1} \cdots a_{\rho,n_k}.$$

For a unit direction $\nu \in S^1$, define

$$\Theta_{\rho,\nu}(t) = \Im h_\rho(t\nu) = \sum_{n=1}^{\infty} \beta_{\rho,\nu,n} t^n, \qquad \beta_{\rho,\nu,n} := \Im(c_{\rho,n}\nu^n).$$

For all sufficiently small $t > 0$,

$$|\xi(\rho + t\nu)| = t^m |\mu_\rho| \exp\bigl(\Re h_\rho(t\nu)\bigr),$$

$$\mathrm{ph}(\xi(\rho + t\nu)) = \nu^m \,\mathrm{ph}(\mu_\rho) e^{\mathrm{i}\Theta_{\rho,\nu}(t)}.$$

After decreasing the radius, put

$$\alpha_{\rho,\nu,n}(t) := \tan(\beta_{\rho,\nu,n} t^n).$$

Then

$$e^{\mathrm{i}\Theta_{\rho,\nu}(t)} = \prod_{n=1}^{\infty} \mathsf{u}(\alpha_{\rho,\nu,n}(t))$$

uniformly on the resulting closed interval, and every finite truncation has the orthogonal-walk endpoint of theorem 6.7.

Proof. Exact order m means that the Taylor series begins

$$\xi(\rho + z) = \sum_{j=0}^{\infty} \frac{\xi^{(m+j)}(\rho)}{(m+j)!} z^{m+j}, \qquad \xi^{(m)}(\rho) \neq 0.$$

Dividing by $\mu_\rho z^m$ and using $\mu_\rho = \xi^{(m)}(\rho)/m!$ gives a holomorphic extension

$$\psi_\rho(z) = 1 + \sum_{j=1}^{\infty} \frac{m!\,\xi^{(m+j)}(\rho)}{(m+j)!\,\xi^{(m)}(\rho)} z^j.$$

This proves the formula for $a_{\rho,j}$. Apply theorem 6.12 to $g = \psi_\rho$, whose value at zero is one. It produces the unique normalized h_ρ, the composition formula for $c_{\rho,n}$, and

$$\psi_\rho(z) = e^{h_\rho(z)}.$$

Multiplication by $\mu_\rho z^m$ gives the factorization of ξ.

At $z = t\nu$, with $t > 0$ and $|\nu| = 1$,

$$\xi(\rho + t\nu) = \mu_\rho t^m \nu^m e^{h_\rho(t\nu)}.$$

Its absolute value and phase are exactly the two displayed formulas. The final product and finite-walk assertions are theorems 6.7 and 6.11 applied to the real-analytic phase $\Theta_{\rho,\nu}$. $\square$

Corollary 6.15 (First two local packet atoms)**.** *For every unit direction ν,*

$$\beta_{\rho,\nu,1} = \Im\left(\nu \frac{\xi^{(m+1)}(\rho)}{(m+1)\xi^{(m)}(\rho)}\right),$$

and

$$\beta_{\rho,\nu,2} = \Im\left[\nu^2\left(\frac{\xi^{(m+2)}(\rho)}{(m+2)(m+1)\xi^{(m)}(\rho)} - \frac{1}{2}\left(\frac{\xi^{(m+1)}(\rho)}{(m+1)\xi^{(m)}(\rho)}\right)^2\right)\right].$$

Proof. The derivative-coordinate formula gives

$$a_{\rho,1} = \frac{\xi^{(m+1)}(\rho)}{(m+1)\xi^{(m)}(\rho)}, \qquad a_{\rho,2} = \frac{\xi^{(m+2)}(\rho)}{(m+2)(m+1)\xi^{(m)}(\rho)}.$$

Substitute these into theorem 6.13. □

Proposition 6.16 (Exact packet transport of every local atom)**.** *After choosing a common sufficiently small radius,*

$$\mu_{1-\rho} = (-1)^m \mu_\rho, \qquad \mu_{\overline{\rho}} = \overline{\mu_\rho},$$
$$\psi_{1-\rho}(z) = \psi_\rho(-z), \qquad \psi_{\overline{\rho}}(z) = \overline{\psi_\rho(\overline{z})},$$
$$h_{1-\rho}(z) = h_\rho(-z), \qquad h_{\overline{\rho}}(z) = \overline{h_\rho(\overline{z})}.$$

Consequently,

$$c_{1-\rho,n} = (-1)^n c_{\rho,n}, \qquad c_{\overline{\rho},n} = \overline{c_{\rho,n}},$$

and for every unit direction ν,

$$\beta_{1-\rho,-\nu,n} = \beta_{\rho,\nu,n}, \qquad \beta_{\overline{\rho},\overline{\nu},n} = -\beta_{\rho,\nu,n},$$

$$\alpha_{1-\rho,-\nu,n}(t) = \alpha_{\rho,\nu,n}(t), \qquad \alpha_{\overline{\rho},\overline{\nu},n}(t) = -\alpha_{\rho,\nu,n}(t).$$

Proof. The leading-coefficient formulas are theorem 5.17. From $\xi(1-s) = \xi(s)$,

$$\xi(1-\rho+z) = \xi(\rho-z) = \mu_\rho(-z)^m\psi_\rho(-z) = \mu_{1-\rho}z^m\psi_\rho(-z),$$

so division by the nonzero leading factor gives $\psi_{1-\rho}(z) = \psi_\rho(-z)$. Likewise,

$$\xi(\overline{\rho}+z) = \overline{\xi(\rho+\overline{z})} = \overline{\mu_\rho \overline{z}^m\, \psi_\rho(\overline{z})} = \mu_{\overline{\rho}} z^m \overline{\psi_\rho(\overline{z})},$$

which gives the conjugation formula for ψ.

Both sides of each proposed h-identity exponentiate to the same ψ-germ and vanish at zero. Uniqueness in theorem 6.12 proves the two h-identities. Comparing Taylor coefficients gives

$$c_{1-\rho,n} = (-1)^n c_{\rho,n}, \qquad c_{\overline{\rho},n} = \overline{c_{\rho,n}}.$$

Therefore

$$\Im\big(c_{1-\rho,n}(-\nu)^n\big) = \Im(c_{\rho,n}\nu^n),$$

while

$$\Im\big(c_{\overline{\rho},n}\overline{\nu}^n\big) = \Im\overline{c_{\rho,n}\nu^n} = -\Im(c_{\rho,n}\nu^n).$$

These are the two β-formulas. The α-formulas follow because tan is odd. □

For a packet O, a representative $\rho \in O$, and $m = m_O$, the leading phase $\mathrm{ph}(\mu_\rho)$ descends to the quotient S^1/Γ_{m_O} constructed after theorem 5.17. The local Taylor factor is instead a germ of a curve

$$t \longmapsto e^{\mathrm{i}\Theta_{\rho,\nu}(t)}$$

based at 1. Let

$$\mathfrak{G}_1 := \{\gamma : (\mathbb{R}, 0) \to (S^1, 1) : \gamma \text{ is a real-analytic germ}\},$$

and, with $b_{\rho,\nu} := \nu^m \, \mathrm{ph}(\mu_\rho)$, let

$$\mathfrak{G}_{\rho,\nu} := \{\eta : (\mathbb{R}, 0) \to (S^1, b_{\rho,\nu}) : \eta \text{ is a real-analytic germ}\}.$$

Multiplication gives the exact bijection

$$T_{\rho,\nu} : \mathfrak{G}_1 \xrightarrow{\sim} \mathfrak{G}_{\rho,\nu}, \qquad T_{\rho,\nu}(\gamma)(t) = b_{\rho,\nu}\gamma(t),$$

with inverse $T_{\rho,\nu}^{-1}(\eta)(t) = b_{\rho,\nu}^{-1}\eta(t)$. The Taylor phase germ is the image of $\gamma_{\rho,\nu}(t) = e^{\mathrm{i}\Theta_{\rho,\nu}(t)}$. Evaluation at zero on $\mathfrak{G}_{\rho,\nu}$ has the constant value $b_{\rho,\nu}$. Thus it forgets the complete sequence $(\beta_{\rho,\nu,n})_{n\geq 1}$, while the full phase germ retains exactly that sequence for the chosen direction ν. It need not determine the real parts of the complex coefficients $c_{\rho,n}$. This is coordinate-exact: if two normalized real phase germs have the same exponential, their difference is a continuous $2\pi\mathbb{Z}$-valued germ vanishing at zero, so it is zero and their Taylor coefficients agree. For fixed (ν, n), the fibre of $c \mapsto \Im(c\nu^n)$ through c_0 is $c_0 + \nu^{-n}\mathbb{R}$. This is the precise interface between the leading phase decoration and the Taylor-atomic refinement.

6.8 The independent circle-length coordinate from the literature

The following is a typed literature interface, not a consequence of the phase atomization.

Proposition 6.17 (Critical ordinate to the circle-length family)**.** *Let*

$$\mathcal{C}_{\mathrm{cyc}} := \left\{(s, n) : s > 0,\ \zeta\left(\frac{1}{2} + \mathrm{i}s\right) = 0,\ n \in \mathbb{Z}_{>0}\right\}.$$

The coordinate map

$$\ell : \mathcal{C}_{\mathrm{cyc}} \longrightarrow \mathbb{R}_{>0}, \qquad \ell(s, n) = \frac{2\pi n}{s},$$

has fibre

$$\ell^{-1}(L) = \left\{\left(\frac{2\pi n}{L}, n\right) : n \geq 1,\ \zeta\left(\frac{1}{2} + \frac{2\pi \mathrm{i} n}{L}\right) = 0\right\}.$$

For every $s > 0$ with $\zeta(\frac{1}{2} + \mathrm{i}s) = 0$, the restriction $n \mapsto \ell(s, n)$ has image the positive-integer family

$$\left\{\frac{2\pi n}{s} : n \geq 1\right\},$$

whose least member is $2\pi/s$. Connes and Consani prove that every circle with one of these lengths is a zeta-cycle and that its orthogonal complement carries the scaling character $u \mapsto u^{\mathrm{i}s}$ [21, §6].

Proof. The fibre formula follows by solving

$$L = \frac{2\pi n}{s}$$

for s, while retaining the defining zero condition of $\mathcal{C}_{\mathrm{cyc}}$. For each such fixed s, strict increase in n gives the displayed image and its least member. The final sentence is the cited theorem: with $L = 2\pi n/s$ and $\mu = e^L$,

$$\mu^{\mathrm{i}s} = e^{\mathrm{i}sL} = e^{2\pi \mathrm{i}n} = 1,$$

so the character descends to the circle $\mathbb{R}_+^*/\mu^{\mathbb{Z}}$; the source proves its orthogonality to the relevant Riemann-sum image. □

Different pairs (s, n) can in principle lie in one fibre of ℓ, as the displayed fibre formula makes explicit. Hence $2\pi/s$ is only the least member of the family attached to that fixed s; it is not a global shortest-cycle assertion. No arrow from $(\beta_{\rho,\nu,n})_{n\geq 1}$ to the Connes–Consani Hilbert space has been constructed here.

6.9 Certificate and formalization boundary

The finite certificate accompanying this chapter checks the Boolean polynomial point maps, the split retraction, complete finite-product fibres on a bounded grid, the elementary-symmetric endpoint identity, the affine-chart inverse, the first logarithmic coefficients, and the packet sign/conjugation laws. Its floating-point checks use explicit error bounds and do not replace the standard proofs above. The uniform-convergence and holomorphic-logarithm arguments are proved in the text and remain separate Lean targets.

The retained object classes are now explicit. The Boolean affine cube is a functor-of-points object; the phase decoration is a sheaf of pointed sets; the coefficient support map forgets an S^1-fibre at a simple zero; finite affine factors map to phases through P_N with the complete displayed fibres; the Taylor atomization maps one fixed real power series to a uniformly convergent phase product; and the circle-length coordinate is a separately sourced map $(s, n) \mapsto 2\pi n/s$. These positive morphisms are the asserted connections. No categorical nonrelation, arithmetic nature of an individual derivative, zero-location conclusion, or spectral realization by the local phase atoms is asserted.

7 Centered packet products and the global zeta factorization

This section reconstructs the analytic packet-product branch at the level of functions, divisors, coordinates, products, and local germs. The published inputs are kept separate from the deductions made here. DLMF supplies the normalization and reflection symmetry of Riemann's completed function [51, §25.4, Eqs. 25.4.3–25.4.4], the genus-one product over the nontrivial zeros [51, §25.2(iv), Eq. 25.2.12], and the reciprocal-gamma product [53, §5.8, Eq. 5.8.2]. McMullen states and proves the canonical-product and Hadamard factorization theorems used to pass from an order-one entire function to its zero product [54, Theorems 3.14–3.15, pp. 79–82]. Lagarias and Montague record that the completed function is entire and that its zeros are exactly the nontrivial zeta zeros [83, §1, Eq. (1.1) and the following paragraph].

The local source under audit used a multiset of arbitrarily chosen representatives of sign pairs and then said that conjugation permuted that multiset. That sentence need not hold for an arbitrary choice. We avoid the choice entirely: sign orbits are the primary objects, conjugation acts on the orbit set, and the quotient is exhibited explicitly as the packet quotient of theorem 5.16. The second repair is analytic rather than set-theoretic: every termwise logarithmic differentiation below is justified by a compact-uniform $O(n^{-2})$ or $O(|\alpha|^{-2})$ estimate.

7.1 Centered coordinates and their two involutions

Retain the literal convention

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \tag{592}$$

and introduce the centered coordinate

$$z = s - \frac{1}{2}, \qquad \Xi(z) = \xi\left(\frac{1}{2} + z\right). \tag{593}$$

No rescaling of z, ξ, or Ξ is made.

Proposition 7.1 (Evenness, real structure, and the nonzero center)**.** *The function Ξ is entire and satisfies*

$$\Xi(-z) = \Xi(z), \qquad \overline{\Xi(\overline{z})} = \Xi(z). \tag{594}$$

Moreover,

$$\Xi(0) = \xi\left(\frac{1}{2}\right) > 0. \tag{595}$$

Consequently 0 *is not a zero of* Ξ*, every odd Taylor coefficient of* Ξ *at* 0 *vanishes, and every even Taylor coefficient is real.*

Proof. Entirety is the cited completed-function input. The functional equation gives

$$\Xi(-z) = \xi\left(\frac{1}{2} - z\right) = \xi\left(1 - \left(\frac{1}{2} + z\right)\right) = \Xi(z).$$

The Dirichlet coefficients of ζ are real in its half-plane of absolute convergence, and the gamma and exponential factors in (592) respect complex conjugation there. Analytic continuation therefore gives the second identity in (594) on all of $\mathbb{C}$.

For $0 < x < 1$, DLMF's alternating-series identity [51, §25.2, Eq. 25.2.3] reads

$$\eta(x) := \sum_{n\geq 1} \frac{(-1)^{n-1}}{n^x} = \left(1 - 2^{1-x}\right)\zeta(x).$$

Pairing consecutive summands gives

$$\eta(x) = \sum_{k\geq 1}\left((2k-1)^{-x} - (2k)^{-x}\right) > 0.$$

Since $1 - 2^{1-x} < 0$, it follows that $\zeta(x) < 0$. At $x = 1/2$,

$$\xi\left(\frac{1}{2}\right) = -\frac{1}{8}\pi^{-1/4}\Gamma\left(\frac{1}{4}\right)\zeta\left(\frac{1}{2}\right) > 0.$$

This proves (595). The Taylor assertions now follow from evenness and the real structure. $\square$

Let Z_Ξ denote the set of distinct zeros of Ξ. For $\alpha \in Z_\Xi$, let $m_\alpha = \mathrm{ord}_\alpha(\Xi)$. Define

$$\iota_-(\alpha) = -\alpha, \qquad \iota_c(\alpha) = \overline{\alpha}.$$

By theorem 7.1, these commuting involutions preserve Z_Ξ and its multiplicity function. Since $0 \notin Z_\Xi$, ι_- has no fixed point. Put

$$S_\Xi := Z_\Xi/\langle \iota_- \rangle, \qquad p_- : Z_\Xi \longrightarrow S_\Xi. \tag{596}$$

For $u = p_-(\alpha)$, set $m_u = m_\alpha$ and

$$Q_u(z) := 1 - \frac{z^2}{\alpha^2}. \tag{597}$$

Both definitions are independent of the representative because replacing α by $-\alpha$ changes neither m_α nor α^2. Conjugation induces

$$\kappa : S_\Xi \longrightarrow S_\Xi, \qquad \kappa\big(p_-(\alpha)\big) = p_-(\overline{\alpha}). \tag{598}$$

Theorem 7.2 (Centered-zero packet quotient)**.** *Let D_ξ, $W = \{1, c, w, cw\}$, and $Y_\xi = |D_\xi|/W$ be as in theorem 5.16. Translation by $1/2$,*

$$t : Z_\Xi \longrightarrow |D_\xi|, \qquad t(\alpha) = \frac{1}{2} + \alpha,$$

is a multiplicity-preserving bijection satisfying

$$t\iota_- = wt, \qquad t\iota_c = ct. \tag{599}$$

It induces a bijection

$$\Psi : S_\Xi/\langle \kappa \rangle \xrightarrow{\sim} Y_\xi, \qquad \Psi(\langle \kappa \rangle p_-(\alpha)) = W\left(\frac{1}{2} + \alpha\right). \tag{600}$$

The composite

$$Z_\Xi \longrightarrow S_\Xi \longrightarrow S_\Xi/\langle \kappa \rangle \xrightarrow{\Psi} Y_\xi$$

has fibre over $\Psi(\langle \kappa \rangle p_-(\alpha))$ equal to

$$\{\alpha, -\alpha, \overline{\alpha}, -\overline{\alpha}\}, \tag{601}$$

with repetitions removed as a set. It has two elements precisely when $\overline{\alpha} = -\alpha$, and otherwise has four elements.

Proof. Equation (593) gives a bijection between zeros, including equality of vanishing orders. Direct coordinate calculation gives

$$t(-\alpha) = \frac{1}{2} - \alpha = 1 - \left(\frac{1}{2} + \alpha\right) = w(t(\alpha))$$

and

$$t(\overline{\alpha}) = \frac{1}{2} + \overline{\alpha} = \overline{\frac{1}{2} + \alpha} = c(t(\alpha)),$$

which proves (599). Thus t maps every orbit of $\langle \iota_-, \iota_c \rangle$ onto one W-orbit and induces (600). Conversely, if $t(\beta) = g(t(\alpha))$ for $g \in W$, then β is respectively $\alpha, \overline{\alpha}, -\alpha$, or $-\overline{\alpha}$. This proves injectivity and the fibre formula. Since $0 \notin Z_\Xi$, the points α and $-\alpha$ are distinct. A fixed sign orbit under conjugation would have $\overline{\alpha} = \alpha$ or $\overline{\alpha} = -\alpha$. The first case would make $1/2 + \alpha$ a real zero in $0 < s < 1$, which was ruled out in the proof of theorem 7.1. Hence the only two-element fibre has $\overline{\alpha} = -\alpha$, exactly the purely imaginary case. $\square$

7.2 The centered sign-orbit product

The order-one property and the genus-one zero product are classical analytic inputs. DLMF records the genus-one product explicitly for ζ, citing Titchmarsh §2.12 [51, §25.2(iv), Eq. 25.2.12]. Hadamard factorization in the form used below is proved in [54, Theorems 3.14–3.15, pp. 79–82]. We retain the exponential factor until evenness removes it; it is not silently normalized away.

Theorem 7.3 (Choice-free centered product)**.** *The series*

$$\sum_{u\in S_\Xi} \frac{m_u}{|\alpha_u|^2} \tag{602}$$

converges, where α_u is either representative of u. The product

$$\Xi(z) = \Xi(0) \prod_{u\in S_\Xi} Q_u(z)^{m_u} = \Xi(0) \prod_{u\in S_\Xi} \left(1 - \frac{z^2}{\alpha_u^2}\right)^{m_u} \tag{603}$$

converges absolutely and locally uniformly on $\mathbb{C}$. It is independent of the ordering of S_Ξ and of every representative α_u.

Proof. Hadamard factorization of the order-one entire function Ξ, using $\Xi(0) \neq 0$, has the form

$$\Xi(z) = e^{a+bz} \prod_{\alpha\in Z_\Xi} E_1\Big(\frac{z}{\alpha}\Big)^{m_\alpha}, \qquad E_1(w) = (1-w)e^w. \tag{604}$$

The exponent of convergence of the zeros is at most one; in particular

$$\sum_{\alpha\in Z_\Xi} \frac{m_\alpha}{|\alpha|^2} < \infty.$$

Taking one term from each two-element sign orbit gives (602).

For either representative α_u of u,

$$E_1\left(\frac{z}{\alpha_u}\right) E_1\left(-\frac{z}{\alpha_u}\right) = (1 - z/\alpha_u)e^{z/\alpha_u}(1 + z/\alpha_u)e^{-z/\alpha_u} = Q_u(z). \tag{605}$$

The canonical product in (604) is locally normally convergent, so the two factors belonging to each sign orbit may be grouped. This gives

$$\Xi(z) = e^{a+bz} \prod_{u\in S_\Xi} Q_u(z)^{m_u}.$$

The product on the right is even. Dividing the identity at z by the identity at $-z$, initially away from the discrete zero set and then by analytic continuation, gives $e^{2bz} = 1$ for every z. Differentiation at $z = 0$ gives $b = 0$. Evaluation at $z = 0$ then gives $e^a = \Xi(0)$, proving (603).

It remains to check the stronger convergence assertion for the grouped product. Fix $R > 0$. For $|z| \leq R$,

$$\sum_{u\in S_\Xi} m_u \left|\frac{z^2}{\alpha_u^2}\right| \leq R^2 \sum_{u\in S_\Xi} \frac{m_u}{|\alpha_u|^2} < \infty.$$

The standard infinite-product criterion therefore gives absolute and uniform convergence on $|z| \leq R$. Since R is arbitrary, convergence is local uniform on $\mathbb{C}$. Absolute convergence makes the product invariant under reordering, and $\alpha_u^2 = (-\alpha_u)^2$ removes representative choice. $\square$

7.3 Packet polynomials and their coordinate inverse

For $O \in Y_\xi$, define the normalized packet polynomial

$$P_O(s) := \prod_{\eta \in O} \frac{s-\eta}{\frac{1}{2}-\eta}. \tag{606}$$

The denominator is nonzero because $\xi(1/2) = \Xi(0) > 0$. This normalization is displayed because it is mathematical data: it fixes the otherwise undetermined nonzero scalar of the polynomial.

Proposition 7.4 (All packet coordinates, without a representative convention)**.** *Every P_O lies in $\mathbb{R}[s]$, satisfies $P_O(1/2) = 1$, and has simple root set exactly O.*

If O has two points, then for either nonzero real coordinate γ with

$$O = \left\{ \frac{1}{2} + \mathrm{i}\gamma, \frac{1}{2} - \mathrm{i}\gamma \right\},$$

one has

$$P_O\left(\frac{1}{2} + z\right) = 1 + \frac{z^2}{\gamma^2}. \tag{607}$$

If O has four points, then for any coordinates $a, \gamma \in \mathbb{R} \setminus \{0\}$ with

$$O = \left\{ \frac{1}{2} + a + \mathrm{i}\gamma,\ \frac{1}{2} + a - \mathrm{i}\gamma,\ \frac{1}{2} - a + \mathrm{i}\gamma,\ \frac{1}{2} - a - \mathrm{i}\gamma \right\},$$

and $\alpha = a + \mathrm{i}\gamma$, one has

$$P_O\left(\frac{1}{2} + z\right) = \frac{\left((z-a)^2 + \gamma^2\right)\left((z+a)^2 + \gamma^2\right)}{(a^2+\gamma^2)^2} \tag{608}$$

$$= \left(1 - \frac{z^2}{\alpha^2}\right)\left(1 - \frac{z^2}{\overline{\alpha}^2}\right) \tag{609}$$

$$= 1 - 2\Re(\alpha^{-2})z^2 + |\alpha|^{-4}z^4. \tag{610}$$

All displayed expressions are invariant under the sign changes of a and γ generated by the packet action.

Proof. The orbit O is closed under conjugation, so conjugate roots occur with the same multiplicity in the defining product. Its coefficients are therefore real. Substitution of $s = 1/2$ makes every factor equal to one, and the numerator shows that the root set is O.

For the two-point packet,

$$\frac{(z - \mathrm{i}\gamma)(z + \mathrm{i}\gamma)}{(-\mathrm{i}\gamma)(\mathrm{i}\gamma)} = \frac{z^2 + \gamma^2}{\gamma^2},$$

which proves (607). For the four-point packet, pairing conjugate roots first gives

$$\prod_{\eta \in O} \left(\frac{1}{2} + z - \eta\right) = \left((z-a)^2 + \gamma^2\right)\left((z+a)^2 + \gamma^2\right).$$

At $z = 0$, the same expression equals $(a^2 + \gamma^2)^2$. This proves (608). Pairing instead by the sign involution gives (609). Multiplication of those two factors gives (610). The final invariance follows either from the root set or by direct substitution. ☐

Theorem 7.5 (Coordinate inverse and exact fibres)**.** *The map*

$$\mathcal{P} : Y_\xi \longrightarrow \mathbb{R}[s], \qquad O \longmapsto P_O, \tag{611}$$

is injective. More explicitly:

1. *If*
$$P_O\left(\frac{1}{2} + z\right) = 1 + c_2 z^2$$
is quadratic, then $c_2 > 0$*,* $\gamma^2 = c_2^{-1}$*, and its root set recovers* $O = \{1/2 \pm \mathrm{i}\gamma\}$*.*

2. *If*
$$P_O\left(\frac{1}{2} + z\right) = 1 + c_2 z^2 + c_4 z^4$$
is quartic, then $c_4 > 0$*, and the squared coordinates are recovered by*
$$R^2 = c_4^{-1/2}, \qquad \Delta = \frac{c_2}{2c_4}, \qquad a^2 = \frac{R^2 - \Delta}{2}, \qquad \gamma^2 = \frac{R^2 + \Delta}{2}. \tag{612}$$

The composite

$$|D_\xi| \xrightarrow{q} Y_\xi \xrightarrow{\mathcal{P}} \mathbb{R}[s]$$

has fibre over P_O *exactly* O*. The augmented coordinate* (P_O, m_O) *recovers the effective packet divisor*

$$D_O = m_O \sum_{\eta \in O} [\eta]. \tag{613}$$

Proof. For a quadratic packet, (607) gives $c_2 = \gamma^{-2}$, proving the first inverse. For a quartic packet, put $R^2 = a^2 + \gamma^2 = |\alpha|^2$. From (610),

$$c_4 = R^{-4}$$

and

$$c_2 = -2\frac{a^2 - \gamma^2}{R^4} = 2c_4(\gamma^2 - a^2).$$

Thus $\Delta = \gamma^2 - a^2$, and solving the two linear equations

$$a^2 + \gamma^2 = R^2, \qquad \gamma^2 - a^2 = \Delta$$

gives (612). The signs of a and γ are not discarded accidentally: all four sign choices are exactly the points already identified inside the packet O. Hence the coefficients recover the packet, proving injectivity.

The root set of P_O is O, so two points of $|D_\xi|$ have the same polynomial precisely when they lie in the same packet. This proves the fibre formula. The polynomial has each packet point once; adjoining the common multiplicity and multiplying the corresponding point divisors proves (613). ∎

7.4 The product on the packet quotient

Theorem 7.6 (Packet-grouped product). *For every $O \in Y_\xi$, let m_O be the common vanishing order on that packet. Then*

$$\xi(s) = \xi\left(\frac{1}{2}\right) \prod_{O \in Y_\xi} P_O(s)^{m_O}, \tag{614}$$

and the product converges absolutely and locally uniformly on $\mathbb{C}$.

Proof. Put $z = s - 1/2$. By theorem 7.2, a κ-orbit in S_Ξ corresponds to exactly one packet. If $u = p_-(\alpha)$ is fixed by κ, then $\overline{\alpha} = -\alpha$, and (607) gives

$$Q_u(z) = P_{\Psi(\langle\kappa\rangle u)}(s).$$

If $u \neq \kappa(u)$, then (609) gives

$$Q_u(z)Q_{\kappa(u)}(z) = P_{\Psi(\langle\kappa\rangle u)}(s).$$

Multiplicities are preserved by both involutions. Grouping the absolutely convergent product of theorem 7.3 over these one- or two-element κ-orbits therefore gives (614). Grouping finitely many factors at a time preserves absolute and local uniform convergence. □

Remark 7.7 (Degree records orbit size, and nothing further is inferred). At this exact coordinate level, a two-point packet contributes one quadratic factor and a four-point packet contributes one quartic factor. This is a description of the quotient and its product. No statement about the location of all packets, the simplicity of their zeros, or a spectral realization is deduced from the factor degrees.

7.5 Centered Taylor atoms as symmetric coordinates

To retain multiplicity without choosing repeated representatives, define the countable index set

$$I_\Xi := \{(u, \ell) : u \in S_\Xi,\ 1 \le \ell \le m_u\}, \qquad a_{(u,\ell)} := \alpha_u^{-2}. \tag{615}$$

The number $a_{(u,\ell)}$ is independent of the representative of u, and equation (602) says

$$\sum_{i \in I_\Xi} |a_i| < \infty.$$

For $n \ge 1$, set

$$\sigma_n := \sum_{\substack{J \subset I_\Xi \\ |J| = n}} \prod_{i \in J} a_i, \qquad \sigma_0 := 1. \tag{616}$$

Lemma 7.8 (Absolute symmetric expansion). *Every sum in* (616) *is absolutely convergent, and*

$$\prod_{i \in I_\Xi} (1 - a_i z^2) = \sum_{n=0}^{\infty} (-1)^n \sigma_n z^{2n} \tag{617}$$

with absolute and locally uniform convergence on $\mathbb{C}$.

Proof. For fixed n,

$$\sum_{\substack{J\subset I_\Xi\\|J|=n}}\prod_{i\in J}|a_i| \le \frac{1}{n!}\left(\sum_{i\in I_\Xi}|a_i|\right)^n < \infty.$$

For a finite subset $F \subset I_\Xi$, ordinary polynomial multiplication gives

$$\prod_{i\in F}(1-a_iz^2) = \sum_{n=0}^{|F|}(-1)^n\sigma_{n,F}z^{2n}.$$

As F increases through finite subsets, absolute convergence gives $\sigma_{n,F}\to\sigma_n$. On $|z|\le R$,

$$\sum_{n\ge 0}|\sigma_n|R^{2n} \le \prod_{i\in I_\Xi}(1+|a_i|R^2) < \infty.$$

The Weierstrass M-test now permits passage to the limit uniformly on the disc and proves (617). $\square$

Theorem 7.9 (Exact centered Taylor coordinates)**.** *The Taylor expansion of the completed function at its symmetry center is*

$$\Xi(z) = \Xi(0)\sum_{n=0}^{\infty}(-1)^n\sigma_n z^{2n}. \tag{618}$$

Equivalently, for every $n \ge 0$,

$$\frac{\Xi^{(2n)}(0)}{(2n)!\Xi(0)} = (-1)^n\sigma_n, \qquad \Xi^{(2n+1)}(0) = 0, \tag{619}$$

$$\frac{\xi^{(2n)}(1/2)}{(2n)!\xi(1/2)} = (-1)^n\sigma_n, \qquad \xi^{(2n+1)}(1/2) = 0. \tag{620}$$

In particular,

$$\frac{\xi''(1/2)}{2\xi(1/2)} = -\sum_{i\in I_\Xi}a_i, \tag{621}$$

$$\frac{\xi^{(4)}(1/2)}{24\xi(1/2)} = \sum_{\substack{J\subset I_\Xi\\|J|=2}}\prod_{i\in J}a_i. \tag{622}$$

Proof. Write the product in theorem 7.3 with one factor for each index $i = (u,\ell) \in I_\Xi$. Apply theorem 7.8, multiply by $\Xi(0)$, and compare the result with the Taylor series of Ξ. Translation by $s = 1/2 + z$ gives the formulas for ξ. The last two identities are the cases $n = 1$ and $n = 2$. $\square$

Proposition 7.10 (Power-sum form on the zero-free centered disc)**.** *Let*

$$r_0 := \inf_{\alpha\in Z_\Xi}|\alpha| > 0,$$

and for $k \ge 1$ define the absolutely convergent power sum

$$p_k := \sum_{i\in I_\Xi}a_i^k.$$

Indeed, only finitely many i have $|a_i| > 1$, while $|a_i|^k \le |a_i|$ for every remaining index. Thus $\sum_i |a_i|^k < \infty$ follows directly from $\sum_i |a_i| < \infty$, with no rearrangement convention. For $|z| < r_0$,

$$\log \frac{\Xi(z)}{\Xi(0)} = -\sum_{k\ge 1} \frac{p_k}{k} z^{2k}, \tag{623}$$

where the logarithm is the branch taking value 0 at $z = 0$.

Proof. The zero set of a nonzero entire function is discrete, and $0 \notin Z_\Xi$, so $r_0 > 0$. Fix $r < r_0$. Then $|a_i z^2| \le r^2/|\alpha_i|^2 < 1$ for $|z| \le r$, and

$$\sum_i \sum_{k\ge 1} \frac{|a_i z^2|^k}{k} \le \sum_i \frac{|a_i| r^2}{1 - |a_i| r^2}.$$

Only finitely many denominators can be close to zero; the remaining tail is bounded by a fixed multiple of $\sum_i |a_i|$. Thus the double series is absolutely and uniformly convergent on $|z| \le r$. Applying $\log(1-w) = -\sum_{k\ge 1} w^k/k$ term by term to equation (603) proves (623). □

7.6 The pole, trivial sector, and packet sector

Proposition 7.11 (Reciprocal-gamma product in the s-coordinate)**.** *For every $s \in \mathbb{C}$,*

$$\frac{1}{\Gamma(s/2)} = \frac{s}{2} e^{\gamma s/2} \prod_{n\ge 1} \left(1 + \frac{s}{2n}\right) e^{-s/(2n)}, \tag{624}$$

with local uniform convergence. Consequently,

$$\zeta(s) = \frac{\pi^{s/2} e^{\gamma s/2}}{s-1} \left[\prod_{n\ge 1} \left(1 + \frac{s}{2n}\right) e^{-s/(2n)} \right] \xi(s) \tag{625}$$

as an identity of meromorphic functions.

Proof. DLMF's reciprocal-gamma product [53, §5.8, Eq. 5.8.2] is

$$\frac{1}{\Gamma(w)} = w e^{\gamma w} \prod_{n\ge 1} \left(1 + \frac{w}{n}\right) e^{-w/n}.$$

Substitution of $w = s/2$ proves (624). Solving (592) for ζ gives

$$\zeta(s) = \frac{2\pi^{s/2}}{s(s-1)} \frac{\xi(s)}{\Gamma(s/2)}.$$

Substitution of (624) cancels the displayed factor $2/s$ and proves (625). Both sides are meromorphic, so the identity extends across removable intermediate singularities. □

Theorem 7.12 (Global pole-trivial-packet factorization)**.** *The exact factorization*

$$\zeta(s) = \frac{\xi(1/2) e^{(\gamma + \log \pi)s/2}}{s-1} \prod_{n\ge 1} \left(1 + \frac{s}{2n}\right) e^{-s/(2n)} \prod_{O \in Y_\xi} P_O(s)^{m_O} \tag{626}$$

holds on $\mathbb{C}$ as an identity of meromorphic functions. The displayed coordinates retain separately:

$$\frac{1}{s-1}, \qquad e^{(\gamma+\log\pi)s/2}\prod_{n\geq 1}\left(1+\frac{s}{2n}\right)e^{-s/(2n)}, \qquad \prod_{O\in Y_\xi} P_O(s)^{m_O}.$$

They have, respectively, the pole at 1, the simple zeros at the negative even integers, and the nontrivial zero divisor. No factor is absorbed into another.

Proof. Insert (614) into (625) and use

$$\pi^{s/2}e^{\gamma s/2} = e^{(\gamma+\log\pi)s/2}.$$

The first factor has divisor $-[1]$. In the second sector, every exponential is nonzero, while $1+s/(2n)$ has one simple zero at $-2n$; the reciprocal-gamma identity proves that there are no additional zeros. By theorem 7.5, the final sector has effective divisor $\sum_{O\in Y_\xi} m_O \sum_{\eta\in O}[\eta] = D_\xi$. This proves both the function identity and the stated divisor accounting. □

7.7 Normally convergent logarithmic derivatives

Lemma 7.13 (Compact-uniform derivative bounds)**.** *Let $K\subset\mathbb{C}$ be compact and put*

$$R_s = \max_{s\in K}|s|, \qquad R_z = \max_{s\in K}\left|s-\frac{1}{2}\right|.$$

Then:

1. *For every integer $n > R_s$ and $s\in K$,*

$$\left|\frac{1}{s+2n}-\frac{1}{2n}\right| = \left|\frac{-s}{2n(s+2n)}\right| \leq \frac{R_s}{2n^2}. \tag{627}$$

2. *For $u = p_-(\alpha)\in S_\Xi$ with $|\alpha| > 2R_z$, $z = s-1/2$, and $s\in K$,*

$$\left|\frac{d}{dz}\log Q_u(z)\right| = \left|\frac{-2z}{\alpha^2-z^2}\right| \leq \frac{8R_z}{3|\alpha|^2}. \tag{628}$$

Consequently the two derivative series obtained from the trivial factors and the sign-orbit factors converge normally on compact subsets avoiding their respective divisors. The same is true after grouping sign orbits into packets.

Proof. If $n > R_s$, then

$$|s+2n| \geq 2n-|s| \geq 2n-R_s > n,$$

which proves (627). If $|\alpha| > 2R_z$, then

$$|\alpha^2-z^2| \geq |\alpha|^2-|z|^2 > \frac{3}{4}|\alpha|^2,$$

which gives (628). The first majorant is a constant multiple of $\sum n^{-2}$; the second is a constant multiple of (602). Only finitely many excluded small indices remain on a fixed compact set. The Weierstrass M-test proves normal convergence. A packet contains one or two sign orbits, so finite grouping preserves it. □

Proposition 7.14 (One packet's rational logarithmic derivative)**.** *For $s \notin O$,*

$$\frac{P_O'(s)}{P_O(s)} = \sum_{\eta \in O} \frac{1}{s-\eta}. \tag{629}$$

The two sides are the same rational function in $\mathbb{R}(s)$.

Proof. Apply the finite product rule to (606). The constant denominators have derivative zero, so

$$\frac{d}{ds} \log P_O(s) = \sum_{\eta \in O} \frac{d}{ds} \log(s-\eta) = \sum_{\eta \in O} \frac{1}{s-\eta}.$$

The left side belongs to $\mathbb{R}(s)$ because $P_O \in \mathbb{R}[s]$. □

Theorem 7.15 (Packet-grouped partial fractions)**.** *On*

$$\mathbb{C} \setminus (\{1\} \cup \{-2,-4,-6,\ldots\} \cup |D_\xi|),$$

the normally convergent identity

$$\frac{\zeta'(s)}{\zeta(s)} = \frac{\gamma + \log \pi}{2} - \frac{1}{s-1} + \sum_{n \geq 1} \left(\frac{1}{s+2n} - \frac{1}{2n} \right) + \sum_{O \in Y_\xi} m_O \frac{P_O'(s)}{P_O(s)} \tag{630}$$

$$= \frac{\gamma + \log \pi}{2} - \frac{1}{s-1} + \sum_{n \geq 1} \left(\frac{1}{s+2n} - \frac{1}{2n} \right) + \sum_{O \in Y_\xi} m_O \sum_{\eta \in O} \frac{1}{s-\eta} \tag{631}$$

holds. On $\mathbb{C} \setminus |D_\xi|$,

$$\frac{\xi'(s)}{\xi(s)} = \sum_{O \in Y_\xi} m_O \frac{P_O'(s)}{P_O(s)} = \sum_{O \in Y_\xi} m_O \sum_{\eta \in O} \frac{1}{s-\eta}. \tag{632}$$

Proof. Fix a point outside the displayed divisor and choose a closed disc K around it which still avoids the divisor. All factors in (626) are nonzero on K. Choose local holomorphic logarithms there. Finite partial products satisfy the ordinary finite logarithmic derivative formula. By theorem 7.13, both infinite derivative tails converge uniformly on K. The products themselves converge locally uniformly by theorems 7.6 and 7.11. Passing to the limit in the finite identities is therefore valid and gives (630). Substitution of equation (629) gives (631).

Applying the same argument directly to (614) removes the pole and gamma sectors and proves (632). Since the point was arbitrary, the identities hold on the stated domains. □

Remark 7.16 (Residues retain multiplicity). At $s = 1$, the term $-1/(s-1)$ has residue -1, as required for a simple pole of ζ. At $s = -2n$, the n-th trivial term has residue 1. At $\eta \in O$, the packet term has residue m_O. Thus equation (631) retains the signed divisor coordinate at every pole and zero.

7.8 Exact global-to-local morphisms

For $O \in Y_\xi$, remove its factor from the global packet product and set

$$H_O(s) := \xi\left(\frac{1}{2}\right) \prod_{\substack{O' \in Y_\xi \\ O' \neq O}} P_{O'}(s)^{m_{O'}}. \tag{633}$$

Absolute local uniform convergence makes H_O entire, and its zero divisor is the union of all packets other than O. Hence

$$\xi(s) = H_O(s) P_O(s)^{m_O}, \qquad H_O(\rho) \neq 0 \quad (\rho \in O). \tag{634}$$

Theorem 7.17 (Leading jet and normalized germ from the packet system)**.** *Let $\rho \in O$, let $m = m_O$, and retain the leading coefficient*

$$\mu_\rho = \frac{\xi^{(m)}(\rho)}{m!}$$

from theorem 5.17. Then

$$\mu_\rho = H_O(\rho)\big(P_O'(\rho)\big)^m. \tag{635}$$

On a sufficiently small disc about ρ, the normalized local germ is

$$\frac{\xi(s)}{\mu_\rho (s-\rho)^m} = \frac{H_O(s)}{H_O(\rho)} \left(\frac{P_O(s)}{P_O'(\rho)(s-\rho)} \right)^m. \tag{636}$$

Both factors on the right extend holomorphically across $s = \rho$ and take the value 1 there. Thus the complete global packet system

$$(\xi(1/2), \{(P_O, m_O) : O \in Y_\xi\})$$

determines every leading jet, its phase, and every normalized Taylor atom of theorems 6.11 and 6.12.

Proof. The roots of P_O are simple because the elements of O are distinct. Taylor expansion at ρ gives

$$P_O(s) = P_O'(\rho)(s-\rho) + O((s-\rho)^2),$$

with $P_O'(\rho) \neq 0$. Equation (634) and $H_O(s) = H_O(\rho) + O(s-\rho)$ therefore give

$$\xi(s) = H_O(\rho)\big(P_O'(\rho)\big)^m (s-\rho)^m + O((s-\rho)^{m+1}).$$

Comparison with the definition of μ_ρ proves (635). Dividing the exact decomposition by that coefficient and by $(s-\rho)^m$ gives (636). The removable factor

$$\frac{P_O(s)}{P_O'(\rho)(s-\rho)}$$

has value 1 at ρ, as does $H_O(s)/H_O(\rho)$. Their product therefore supplies the normalized germ, whose Taylor coefficients and normalized logarithm are the atoms constructed in Chapter 16. □

Proposition 7.18 (Exact information loss of the one-packet projection)**.** *The packet coordinate (P_O, m_O) recovers D_O by (613), but it does not by itself determine μ_ρ. The missing coordinate is exactly the nonzero complementary value $H_O(\rho)$ in (635).*

More concretely, fix O, $m \geq 1$, and $\rho \in O$. Choose $\lambda \in \mathbb{R}_{>0}$ such that

$$Q_\lambda(s) := 1 + \lambda(s - 1/2)^2$$

has no zero in O and $Q_\lambda(\rho) \neq 1$. The two finite functions

$$F_1(s) = P_O(s)^m, \qquad F_2(s) = Q_\lambda(s) P_O(s)^m$$

have the same packet polynomial and multiplicity at O, but their leading coefficients at ρ are

$$\big(P_O'(\rho)\big)^m \quad \text{and} \quad Q_\lambda(\rho)\big(P_O'(\rho)\big)^m.$$

Both functions remain even in $s-1/2$ and respect complex conjugation; the additional roots of Q_λ are the two-point packet $\{1/2 \pm \mathrm{i}/\sqrt{\lambda}\}$, chosen disjoint from O. Thus the fibre of the forgetful map from a full factor system to one packet contains distinct leading-jet values whenever the complementary coordinate is allowed to vary.

Proof. The first assertion is exactly equation (635). Only finitely many values of $\lambda > 0$ make Q_λ vanish at a point of O. Moreover $\rho \neq 1/2$, so $Q_\lambda(\rho) = 1$ would force $\lambda = 0$; every admissible positive choice therefore avoids it. Choose any positive λ outside the finite excluded set. The displayed leading coefficients follow by the same first-order expansion used in theorem 7.17. This supplies two explicit points in one fibre and identifies the omitted coordinate rather than inferring nonidentity from different presentations. □

Theorem 7.19 (Reconstruction map on admissible packet data). *Let $\mathfrak{P}_1$ be the set of data*

$$\mathcal{D} = (C, \{(P_O, m_O)\}_{O\in Y})$$

such that:

1. *$C \in \mathbb{R}^\times$;*
2. *each $P_O \in \mathbb{R}[s]$ has one of the explicit forms in theorem 7.4 and $P_O(1/2) = 1$;*
3. *m_O is a positive integer; and*
4. *after resolving every P_O into its one or two sign-orbit factors, the resulting reciprocal-square sum converges and the reconstructed entire function has order at most one.*

For data whose union of root packets is discrete, define

$$\mathcal{R}(\mathcal{D})(s) := C \prod_{O\in Y} P_O(s)^{m_O}. \tag{637}$$

Then $\mathcal{R}(\mathcal{D})$ is an even function of $s - 1/2$, respects complex conjugation, takes the value C at $1/2$, and has effective zero divisor

$$\sum_{O\in Y} m_O \sum_{\eta\in O} [\eta].$$

Conversely, within the class of order-at-most-one entire functions satisfying those two symmetries and nonvanishing at $1/2$, the center value and packet divisor recover the function uniquely. Hence extraction of packet data and $\mathcal{R}$ are mutually inverse on that class.

Proof. The convergence condition and the proof of theorem 7.3 give an entire product. Each packet factor is even in $s - 1/2$, has real coefficients, and is one at the center, proving the three stated properties. Its roots and multiplicity give the displayed divisor.

For uniqueness, let F_1, F_2 be two functions in the stated class with the same packet divisor and center value. Apply Hadamard factorization separately to $F_1(1/2 + z)$ and $F_2(1/2 + z)$. Their canonical zero products are the same, including multiplicities, so division gives

$$\frac{F_1(1/2 + z)}{F_2(1/2 + z)} = e^{a+bz}.$$

Both functions are even in z, so the quotient is even and $b = 0$, by the same derivative-at-zero argument used in theorem 7.3. Equality of the center values gives $e^a = 1$. Thus $F_1 = F_2$. Extraction followed by reconstruction and reconstruction followed by extraction are therefore both identities. □

7.9 Certificate and theorem boundaries

The finite algebra in this chapter admits a deterministic certificate: the two- and four-point packet formulas, the coefficient inverse (612), the one-packet logarithmic derivative, the first symmetric coefficients, and the leading-jet formula can all be checked as polynomial or rational identities over exact symbolic coordinates. The companion script `verify_centered_packet_products.py` passes seventeen named exact checks under Python/SymPy 3.13.9/1.13.1 and 3.13.1/1.13.3; receipt `CERT-GNTE-PACKET-20260828-0001` binds both runs. Such a certificate checks those identities only. It does not certify Hadamard factorization, the DLMF products, convergence of the actual zero product, analytic continuation, or any assertion about an individual zero. Those analytic statements are proved above from their cited inputs and the explicit convergence estimates.

The resulting interface is exact. The sign-orbit quotient maps bijectively to the packet quotient; the packet polynomial has an explicit coordinate inverse; the augmented packet polynomial recovers the packet divisor; the complete packet system reconstructs the completed function in the stated order-one class; and (635) maps the global product to the local phase and Taylor-atom layer. No endpoint, simplicity, or spectral claim is added.

8 Spectral packet coordinates, the Connes–Consani Laplacian, and the Scaling Site

The packet divisor of Chapters 15–17 has two further published spectral realizations. Connes and Consani first factor the multiplicative Fourier transform of a scale-invariant Riemann sum by $\zeta(1/2 - \mathrm{i}z)$, then realize the values $(\rho - 1/2)^2 - 1/4$ as the spectrum of $\Delta = H(1 + H)$, and finally identify a quotient sheaf on the Scaling Site with a Schwartz quotient that detects the critical zeros and their possible multiplicities [22, Eq. (2.2), Prop. 4.2, and the theorem on pp. 96–98]. This chapter constructs the exact maps from the packet coordinates already proved here to those three realizations. In particular, the affine map is restricted to its correct open strip, the Laplacian's two-to-one fibre is displayed, and an entire four-point packet is represented by a real polynomial whose roots are a conjugate pair of Laplacian eigenvalues.

8.1 The integral and complex split-zero models

The local source for this programme uses $G(\mathbb{Z})$, whereas Chapters 14–17 use $G(\mathbb{C})$. These are related by a literal embedding; they are not silently identified.

For a nonzero commutative ring R, recall

$$G(R) = R \sqcup \{\tau\},$$

with

$$r \oplus r' = r + r', \qquad r \oplus \tau = \tau \oplus r = r,$$

and

$$r \odot r' = rr', \qquad r \odot \tau = \tau \odot r = \tau, \qquad \tau \odot \tau = \tau.$$

The additive zero of $G(R)$ is τ, while the supported element over the ring zero is $e_R := 0_R$. Put $A_R = \{\tau, e_R\}$.

Proposition 8.1 (Exact change of coefficient ring)**.** *The map*

$$j : G(\mathbb{Z}) \longrightarrow G(\mathbb{C}), \qquad j(n) = n + 0\mathrm{i}, \qquad j(\tau) = \tau, \tag{638}$$

is an injective unital semiring homomorphism. Its image is $\mathbb{Z} \sqcup \{\tau\}$, and every fibre over a point of its image is a singleton. It intertwines multiplication by -1, and its restriction

$$j|_{A_{\mathbb{Z}}} : A_{\mathbb{Z}} \xrightarrow{\sim} A_{\mathbb{C}}$$

is the Boolean-semiring isomorphism $\tau \mapsto \tau$, $e_{\mathbb{Z}} \mapsto e_{\mathbb{C}}$. Equivalently,

$$j\big(\mathrm{Fix}_{G(\mathbb{Z})}(-1)\big) = \mathrm{Fix}_{G(\mathbb{C})}(-1) = A_{\mathbb{C}}.$$

The map j is not surjective. The inclusions $A_R \hookrightarrow G(R)$ are not unit-preserving, because the internal unit of A_R is e_R, whereas the ambient unit is 1_R; this does not alter the fact that j itself and $j|_{A_{\mathbb{Z}}}$ are unital in their stated categories.

Proof. For supported integers m, n, preservation of the two operations is the ordinary inclusion calculation

$$j(m \oplus n) = j(m+n) = j(m) + j(n), \qquad j(m \odot n) = j(mn) = j(m)j(n).$$

If either entry is τ, the defining tables on both sides give the same answer: τ is the additive identity and multiplicative absorber. Also $j(\tau) = \tau$ and $j(1_{\mathbb{Z}}) = 1_{\mathbb{C}}$, so the map preserves zero and one in the ambient semirings. Injectivity and the image formula follow from the injectivity of $\mathbb{Z} \hookrightarrow \mathbb{C}$ together with the fact that the adjoined symbol τ is disjoint from both supported copies.

For every $x \in G(\mathbb{Z})$, $j((-1) \odot x) = (-1) \odot j(x)$. The fixed-set computation of theorem 5.1 is valid over either torsion-free ring: $-z = z$ implies $2z = 0$, hence $z = 0$, while τ is fixed because it is absorbing. Thus the fixed sets are precisely the two displayed Boolean pairs, and the restriction of j is a bijection respecting their complete operation tables. A complex number outside $\mathbb{Z}$ is not in the image, so j is not surjective. Finally, $e_R \neq 1_R$ for the nonzero rings in question, which proves the assertion about the two inclusions. $\square$

8.2 The affine spectral strip and all four symmetries

Let

$$\mathfrak{S}_s = \{s \in \mathbb{C} : 0 < \Re s < 1\}, \qquad \mathfrak{S}_z = \{z \in \mathbb{C} : |\Im z| < 1/2\},$$

and retain the unscaled affine coordinate

$$\Phi(s) = \mathrm{i}\left(s - \frac{1}{2}\right). \tag{639}$$

Proposition 8.2 (Biholomorphic strip coordinate and equivariance)**.** *The map $\Phi : \mathfrak{S}_s \to \mathfrak{S}_z$ is a biholomorphism with*

$$\Phi^{-1}(z) = \frac{1}{2} - \mathrm{i}z. \tag{640}$$

Writing $s = \beta + \mathrm{i}\gamma$, one has the coordinate formula

$$\Phi(s) = -\gamma + \mathrm{i}\left(\beta - \frac{1}{2}\right). \tag{641}$$

For the functional involution $w(s) = 1 - s$ and conjugation $c(s) = \overline{s}$, put

$$\sigma(z) = -z, \qquad \kappa(z) = -\overline{z}.$$

Then

$$\Phi \circ w = \sigma \circ \Phi, \qquad \Phi \circ c = \kappa \circ \Phi. \tag{642}$$

Consequently the packet $\{s, 1-s, \overline{s}, 1-\overline{s}\}$ *maps, with repetitions removed, to*

$$\{z, -z, -\overline{z}, \overline{z}\}, \qquad z = \Phi(s). \tag{643}$$

Moreover, $\Re s = 1/2$ *if and only if* $\Phi(s) \in \mathbb{R}$.

Proof. The affine map has nonzero derivative i, and direct substitution gives (640). Formula (641) follows from

$$\mathrm{i}\left(\beta - \frac{1}{2} + \mathrm{i}\gamma\right) = -\gamma + \mathrm{i}\left(\beta - \frac{1}{2}\right).$$

It also shows

$$0 < \beta < 1 \quad \Longleftrightarrow \quad -\frac{1}{2} < \Im\Phi(s) < \frac{1}{2},$$

which proves the strip bijection. The two equivariance identities are the literal calculations

$$\Phi(1-s) = \mathrm{i}(1/2 - s) = -\Phi(s)$$

and

$$\Phi(\overline{s}) = \mathrm{i}(\overline{s} - 1/2) = -\overline{\mathrm{i}(s - 1/2)}.$$

Applying these generators and their composite gives the four displayed spectral points. Finally, $\Im\Phi(s) = \beta - 1/2$, proving the last equivalence. □

8.3 The common-zero divisor, including multiplicities

For an even Schwartz function f, set

$$(\mathcal{E}f)(u) = u^{1/2} \sum_{n \geq 1} f(nu), \qquad \psi_f(z) = \int_{\mathbb{R}_+^*} f(u) u^{1/2 - \mathrm{i}z} \, d^*u, \tag{644}$$

and let

$$\mathcal{S}(\mathbb{R})_1^{\mathrm{ev}} = \left\{ f \in \mathcal{S}(\mathbb{R})^{\mathrm{ev}} : \int_{\mathbb{R}} f(x) \, dx = 0 \right\}.$$

Connes and Consani prove that, for every such f, the multiplicative Fourier transform

$$T_f(z) := \mathcal{F}(\mathcal{E}f)(z)$$

satisfies

$$T_f(z) = g(z)\psi_f(z), \qquad g(z) := \zeta\left(\frac{1}{2} - \mathrm{i}z\right), \tag{645}$$

throughout $\Im z > -1/2$ [22, Eq. (2.2) and the continuation immediately following it].

For a family of holomorphic functions $\mathcal{T}$ on a domain, define its common-zero divisor by

$$D_{\mathrm{com}}(\mathcal{T}) = \sum_z \left(\min_{T \in \mathcal{T}} \mathrm{ord}_z \, T \right) [z], \tag{646}$$

where a point with minimum zero contributes nothing. The identically zero member is assigned the convention $\mathrm{ord}_z(0) = +\infty$, and the minimum is finite because of the explicit nonzero member used in the proof.

Theorem 8.3 (Exact common divisor in the open spectral strip)**.** *Let D_ξ be the completed-zeta zero divisor, including multiplicities, and let*

$$\mathcal{T}_E = \{T_f : f \in \mathcal{S}(\mathbb{R})_1^{\mathrm{ev}}\}.$$

Then, as effective divisors on $\mathfrak{S}_z$,

$$\Phi_* D_\xi = \mathrm{div}_0(g)|_{\mathfrak{S}_z} = D_{\mathrm{com}}(\mathcal{T}_E)|_{\mathfrak{S}_z}. \tag{647}$$

Thus both the support and the vanishing order at every point agree. The inverse from the common divisor to the completed divisor is the affine map $z \mapsto 1/2 - \mathrm{i}z$, and every point fibre of Φ is a singleton.

Proof. All zeros of ξ lie in $\mathfrak{S}_s$, and at such a zero ρ the explicit factor

$$\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$$

is holomorphic and nonzero. Hence ξ and ζ have the same vanishing order at ρ. Since

$$g(z) = \zeta(\Phi^{-1}(z))$$

and $(\Phi^{-1})'(z) = -\mathrm{i} \neq 0$, composition with Φ^{-1} preserves that order. This proves the first equality in (647), point by point and with multiplicity.

Equation (645) implies

$$\mathrm{ord}_{z_0} T_f \geq \mathrm{ord}_{z_0} g$$

for every f and every $z_0 \in \mathfrak{S}_z$. It remains to prove that the inequality is attained simultaneously at every point. Use the explicit even Schwartz function from Connes–Consani,

$$f_0(x) = e^{-\pi x^2}(2\pi x^2 - 1). \tag{648}$$

The Gaussian identities

$$\int_{\mathbb{R}} e^{-\pi x^2}\,dx = 1, \qquad \int_{\mathbb{R}} x^2 e^{-\pi x^2}\,dx = \frac{1}{2\pi}$$

give $\int f_0 = 0$, so $f_0 \in \mathcal{S}(\mathbb{R})_1^{\mathrm{ev}}$. The cited computation is

$$\psi_{f_0}(z) = \frac{1}{4}\pi^{-1/4+\mathrm{i}z/2}(-1-2\mathrm{i}z)\Gamma\left(\frac{1}{4} - \frac{\mathrm{i}z}{2}\right). \tag{649}$$

Each factor is finite and nonzero in $\mathfrak{S}_z$. The exponential power of π never vanishes. The linear factor vanishes only at $z = \mathrm{i}/2$, which is on the boundary. If $|\Im z| < 1/2$, then

$$0 < \Re\left(\frac{1}{4} - \frac{\mathrm{i}z}{2}\right) < \frac{1}{2};$$

hence its argument is not a pole of Γ, and Γ has no zeros. Therefore ψ_{f_0} is holomorphic and zero-free throughout the open strip. It follows that

$$\mathrm{ord}_{z_0} T_{f_0} = \mathrm{ord}_{z_0} g$$

for every $z_0 \in \mathfrak{S}_z$, proving the second divisor equality. The inverse and fibre assertions follow from (640). □

Remark 8.4 (Why the strip cannot be deleted). The first equality in (647) is not a global equality of zero sets on $\mathbb{C}$. For every $n \geq 1$, the trivial zero $s = -2n$ gives

$$z = \Phi(-2n) = -\mathrm{i}\left(2n + \frac{1}{2}\right), \qquad g(z) = 0,$$

which lies below $\Im z = -1/2$ and is not the image of a zero of ξ. The pole $s = 1$ of ζ maps to the other boundary point $z = \mathrm{i}/2$. The correct statement is therefore exactly the open-strip divisor identity proved above.

8.4 The Laplacian map, its quotient, and its exceptional point

Connes and Consani let $H = x\partial_x$ be the scaling generator and define $\Delta = H(1+H)$. On the quotient of the strong Schwartz space $\mathcal{S}_\infty(\mathbb{R}_+^*)$ by the closure of the range of the trace map composed with Δ, they obtain the spectral values

$$\left(\rho - \frac{1}{2}\right)^2 - \frac{1}{4} \quad (\zeta(\rho) = 0,\ \rho \notin \mathbb{R}), \tag{650}$$

counted with possible multiplicities [22, Prop. 4.2].

Define on the spectral plane

$$\lambda_\Delta(z) = -z^2 - \frac{1}{4}. \tag{651}$$

Proposition 8.5 (Exact Laplacian fibres and symmetry transport)**.** *For every $s \in \mathbb{C}$,*

$$\lambda_\Delta(\Phi(s)) = -s(1-s) = \left(s - \frac{1}{2}\right)^2 - \frac{1}{4}. \tag{652}$$

For any $z_1, z_2 \in \mathbb{C}$,

$$\lambda_\Delta(z_1) = \lambda_\Delta(z_2) \quad \Longleftrightarrow \quad z_2 = z_1 \text{ or } z_2 = -z_1. \tag{653}$$

Thus the fibre is $\{z, -z\}$ for $z \neq 0$ and is the singleton $\{0\}$ at the branch point. The branch point is absent from $\Phi(|D_\xi|)$, since $\xi(1/2) > 0$. Consequently, if $\mathrm{Spec}_{\mathrm{val}}(\Delta)$ denotes the underlying set of distinct spectral values in (650),

$$\Phi(|D_\xi|)/\langle z \mapsto -z \rangle \xrightarrow{\sim} \mathrm{Spec}_{\mathrm{val}}(\Delta), \qquad [z] \longmapsto \lambda_\Delta(z), \tag{654}$$

is a bijection. Its inverse sends a value λ to the unordered root set of $z^2 = -(\lambda + 1/4)$.

The two packet involutions act by

$$\lambda_\Delta(-z) = \lambda_\Delta(z), \qquad \lambda_\Delta(-\overline{z}) = \overline{\lambda_\Delta(z)}. \tag{655}$$

Hence, on $\Phi(|D_\xi|)$, a full packet maps to the conjugation orbit $\{\lambda, \overline{\lambda}\}$, with repetitions removed. This orbit is a singleton precisely when $z \in \mathbb{R}$, equivalently when the original zero packet is critical and has two points.

Proof. Using $\Phi(s) = \mathrm{i}(s - 1/2)$,

$$-\Phi(s)^2 - \frac{1}{4} = \left(s - \frac{1}{2}\right)^2 - \frac{1}{4} = s^2 - s = -s(1-s),$$

which proves (652). Equality of two values of λ_Δ is equality of their squares:

$$z_1^2 = z_2^2 \quad \Longleftrightarrow \quad (z_1 - z_2)(z_1 + z_2) = 0,$$

proving the complete fibre formula. The only collapsed fibre is at zero; its inverse affine point is $1/2$, and theorem 7.1 proves $\xi(1/2) > 0$. Therefore no exceptional fibre occurs on the completed-zero support, and the quotient map is bijective with the stated root-set inverse.

The identities in (655) are direct substitutions. By theorem 8.2, the functional involution is the sign map and conjugation is $z \mapsto -\overline{z}$; therefore the image of the four-point packet has the two displayed conjugate values. If $z = x + \mathrm{i}y$, then

$$\Im\lambda_\Delta(z) = -2xy.$$

On $\Phi(|D_\xi|)$, the alternative $x = 0$ would make $\Phi^{-1}(z)$ a real point of $(0, 1)$, where ζ has no zero by the eta-series argument in theorem 7.1. Hence a real spectral value on this support forces $y = 0$, which is exactly a real spectral packet. The converse is immediate. $\square$

8.5 A real polynomial coordinate for a whole packet

For $O \in Y_\xi$, let

$$\mathrm{Sp}_\Delta(O) = \{\lambda_\Delta(\Phi(\rho)) : \rho \in O\}$$

with repetitions removed, and define the monic spectral packet polynomial

$$R_O(t) = \prod_{\lambda \in \mathrm{Sp}_\Delta(O)} (t - \lambda). \tag{656}$$

Theorem 8.6 (Spectral packet polynomial and coordinate inverse)**.** *The polynomial R_O lies in $\mathbb{R}[t]$, and the map*

$$\mathcal{R} : Y_\xi \longrightarrow \mathbb{R}[t], \qquad O \longmapsto R_O, \tag{657}$$

is injective.

If

$$O = \left\{\frac{1}{2} + \mathrm{i}\gamma, \frac{1}{2} - \mathrm{i}\gamma\right\}, \qquad \gamma \in \mathbb{R} \setminus \{0\},$$

then

$$R_O(t) = t + \gamma^2 + \frac{1}{4}. \tag{658}$$

If O has four points and

$$\alpha = a + \mathrm{i}\gamma, \qquad O = \left\{\frac{1}{2} \pm \alpha, \frac{1}{2} \pm \overline{\alpha}\right\}, \qquad a\gamma \neq 0,$$

then

$$R_O(t) = \left(t - \alpha^2 + \frac{1}{4}\right)\left(t - \overline{\alpha}^2 + \frac{1}{4}\right). \tag{659}$$

The inverse takes the roots λ of R_O, forms $A = \lambda + 1/4$, takes the complete unordered square-root fibre $\{\alpha, -\alpha\}$ of A, adjoins its conjugate fibre, and finally translates every point by $1/2$. No square-root branch is selected.

Proof. The sign pair $\{\rho, 1-\rho\}$ has centered coordinates $\{\alpha, -\alpha\}$. By (652), both points give the same spectral value $\alpha^2 - 1/4$. Conjugation supplies $\overline{\alpha}^2 - 1/4$. This proves (659); its two factors are conjugate, so their product has real coefficients. In the critical case $\alpha = \mathrm{i}\gamma$, the two conjugate values coincide and equal $-\gamma^2 - 1/4$, giving (658).

The roots of R_O recover the unordered spectral-value set. Adding $1/4$ recovers the unordered set $\{\alpha^2, \overline{\alpha}^2\}$. The complete square-root fibres are $\{\pm\alpha\}$ and $\{\pm\overline{\alpha}\}$; their union is exactly the centered packet, with repetitions removed in the critical case. Translation by $1/2$ therefore recovers O. This is an explicit inverse to $\mathcal{R}$, proving injectivity. □

The packet polynomial P_O of Chapter 17 and the spectral polynomial R_O contain the same packet coordinate in different target spaces. The next result gives both directions at coefficient level.

Proposition 8.7 (Exact conversion from packet coefficients to spectral coefficients)**.** *Write the centered packet polynomial as*

$$P_O\left(\frac{1}{2} + u\right) = 1 + c_2 u^2$$

in the two-point case, or

$$P_O\left(\frac{1}{2} + u\right) = 1 + c_2 u^2 + c_4 u^4$$

in the four-point case.

In the two-point case,

$$R_O(t) = t - \lambda_O, \qquad \lambda_O = -\frac{1}{c_2} - \frac{1}{4}. \tag{660}$$

In the four-point case,

$$R_O(t) = t^2 + \left(\frac{c_2}{c_4} + \frac{1}{2}\right) t + \left(\frac{1}{c_4} + \frac{c_2}{4c_4} + \frac{1}{16}\right). \tag{661}$$

Conversely, if λ is either root of the latter polynomial and $A = \lambda + 1/4$, then the squared real coordinates of the packet are

$$a^2 = \frac{|A| + \Re A}{2}, \qquad \gamma^2 = \frac{|A| - \Re A}{2}. \tag{662}$$

The alternative root conjugates A and leaves these two numbers unchanged. Their sign choices are exactly the four packet points, not discarded data. Thus $P_O \leftrightarrow R_O$ is a bijective change of packet coordinate.

Proof. In the critical case, theorem 7.4 gives $c_2 = \gamma^{-2}$, while (658) has root $\lambda_O = -\gamma^2 - 1/4$. Substitution proves (660).

For a four-point packet, put $A = \alpha^2$ and $B = \overline{\alpha}^2$. The factorization from Chapter 17 is

$$P_O\left(\frac{1}{2} + u\right) = \left(1 - \frac{u^2}{A}\right)\left(1 - \frac{u^2}{B}\right).$$

Coefficient comparison gives

$$AB = \frac{1}{c_4}, \qquad A + B = -\frac{c_2}{c_4}. \tag{663}$$

The two spectral roots are $\lambda = A - 1/4$ and $\overline{\lambda} = B - 1/4$. Therefore

$$\lambda + \overline{\lambda} = -\frac{c_2}{c_4} - \frac{1}{2}$$

and

$$\lambda\overline{\lambda} = AB - \frac{A+B}{4} + \frac{1}{16} = \frac{1}{c_4} + \frac{c_2}{4c_4} + \frac{1}{16}.$$

Expanding $(t-\lambda)(t-\overline{\lambda})$ proves (661).

For the inverse, write $\alpha = a + \mathrm{i}\gamma$. Then

$$A = \alpha^2 = (a^2 - \gamma^2) + 2\mathrm{i}a\gamma, \qquad |A| = |\alpha|^2 = a^2 + \gamma^2.$$

Adding and subtracting these last two real equations gives (662). Replacing A by $\overline{A}$ changes the sign of $a\gamma$ but not a^2 or γ^2. The four choices $(a,\gamma),(-a,-\gamma),(a,-\gamma),(-a,\gamma)$ are exactly $\alpha,-\alpha,\overline{\alpha},-\overline{\alpha}$, so the complete packet fibre is retained. Since theorem 7.5 recovers P_O from that same packet, the two displayed constructions are mutual coordinate conversions. □

8.6 Divisor multiplicity under the quadratic spectral map

Let $D_\Phi = \Phi_* D_\xi$. The polynomial $\lambda_\Delta(z) = -z^2 - 1/4$ is proper. Explicitly, if a compact set $K \subset \mathbb{C}$ contains $\lambda_\Delta(z)$, then

$$|z|^2 = |z^2| = |\lambda_\Delta(z) + 1/4| \leq \max_{\lambda \in K} |\lambda + 1/4|,$$

so $\lambda_\Delta^{-1}(K)$ is closed and bounded. Since ξ is a nonzero entire function, $|D_\xi|$ is a closed discrete zero set; hence $\Phi(|D_\xi|)$ is closed and discrete as well. Its intersection with the compact set $\lambda_\Delta^{-1}(K)$ is therefore finite. Thus the following divisor pushforward is locally finite, not merely finite on each point fibre:

$$(\lambda_\Delta)_* D_\Phi = \sum_\lambda \left(\sum_{z \in \Phi(|D_\xi|),\ \lambda_\Delta(z) = \lambda} m_z \right) [\lambda].$$

Corollary 8.8 (Exact aggregation of zero multiplicities)**.** *Let $O \in Y_\xi$ have common multiplicity m_O. If O is critical, its contribution to $(\lambda_\Delta)_* D_\Phi$ is*

$$2m_O[\lambda_O]. \tag{664}$$

If O has four points, its contribution is

$$2m_O[\lambda_O] + 2m_O[\overline{\lambda}_O]. \tag{665}$$

No two distinct sign orbits contribute to the same scalar value.

This aggregation is compatible with the multiplicity mechanism proved by Connes–Consani: a zero ρ of order m_O supplies an m_O-dimensional jet space, and multiplication by $-s(1-s)$ has a triangular matrix with that spectral value repeated m_O times on the diagonal [22, proof of Prop. 4.2]. Applying their per-zero calculation to the two distinct zeros ρ and $1-\rho$, which have the same spectral value, gives two such jet blocks. Their algebraic multiplicities therefore aggregate to the coefficient $2m_O$ in this chapter's divisor pushforward.

Proof. At every point of D_Φ, the affine map preserves the multiplicity m_O. By (653), a scalar spectral value has exactly the sign fibre $\{z, -z\}$, and both points occur in the same packet with multiplicity m_O. Their sum is $2m_O$. A critical packet consists of one sign fibre and hence gives (664). A four-point packet is the disjoint union of the sign fibres $\{z, -z\}$ and $\{\overline{z}, -\overline{z}\}$, giving the two conjugate terms in (665). Finally, equality of two spectral values forces equality up to sign by (653); hence distinct sign orbits cannot collide. The final paragraph is the direct coordinate comparison with the cited jet calculation. $\square$

8.7 Critical packets: spectral value to every cycle length

Let Y_ξ^{crit} be the set of two-point packets. For $O \in Y_\xi^{\mathrm{crit}}$, let λ_O be the unique root of R_O. Then $\lambda_O < -1/4$, and define

$$\nu(O) = \sqrt{-\lambda_O - \frac{1}{4}} > 0. \tag{666}$$

Proposition 8.9 (The cycle family factors through the Laplacian value)**.** *For every $O \in Y_\xi^{\mathrm{crit}}$ and every $n \in \mathbb{Z}_{\geq 1}$, set*

$$\ell_n(O) = \frac{2\pi n}{\nu(O)} = \frac{2\pi n}{\sqrt{-\lambda_O - 1/4}}. \tag{667}$$

Then $\ell_n(O)$ is exactly the Connes–Consani circle-length family attached to the positive critical ordinate represented by O. The map factors as

$$O \longmapsto R_O \longmapsto \lambda_O \longmapsto \nu(O) \longmapsto (\ell_n(O))_{n\geq 1},$$

and every arrow has the explicit inverse or fibre described above. For this fixed packet, $\ell_1(O)$ is the least member of the displayed positive-integer family. This is not a comparison with lengths arising from other ordinates.

Proof. Write $O = \{1/2 + \mathrm{i}\gamma, 1/2 - \mathrm{i}\gamma\}$ with $\gamma \neq 0$. Equation (658) gives

$$\lambda_O = -\gamma^2 - \frac{1}{4}, \qquad \nu(O) = |\gamma|.$$

Thus (667) is $2\pi n/|\gamma|$. Connes and Consani prove that if $s = |\gamma| > 0$ and $\zeta(1/2 + \mathrm{i}s) = 0$, every circle whose length is an integral multiple of $2\pi/s$ is a zeta-cycle [21, Thm. 6.4(ii)]. This is precisely the displayed family. Since n ranges over positive integers, its least member for this fixed s occurs at $n = 1$; no comparison across different values of s is involved. $\square$

8.8 What the spectral polynomial remembers about a leading jet

For $\rho \in O$, let

$$\mu_\rho = \frac{\xi^{(m_O)}(\rho)}{m_O!}, \qquad \omega(O) = |\mu_\rho|.$$

The transport calculation of theorem 5.17 proves that $\omega(O)$ is independent of the representative.

Proposition 8.10 (Exact jet-weight dependence and one-packet information loss)**.** *The spectral coordinate R_O determines P_O, the packet O, and the common multiplicity once m_O is adjoined. It consequently determines $|P_O'(\rho)|^{m_O}$, independently of the representative $\rho \in O$. The actual packet weight satisfies*

$$\omega(O) = |H_O(\rho)|\,|P_O'(\rho)|^{m_O}, \tag{668}$$

where

$$H_O(s) = \xi(1/2) \prod_{O' \neq O} P_{O'}(s)^{m_{O'}}.$$

Thus a single (R_O, m_O) does not determine $\omega(O)$ in the class of entire functions with the same centered sign and conjugation symmetries. The missing coordinate is exactly the nonzero complementary value $|H_O(\rho)|$. The complete global packet system

$$\big(\xi(1/2), \{(R_{O'}, m_{O'}) : O' \in Y_\xi\}\big)$$

does determine $\omega(O)$, every leading jet, and the normalized local germ.

Proof. The mutual conversion of theorem 8.7 recovers P_O from R_O. Because $P_O(1/2+u)$ is even in u and has real coefficients,

$$|P_O'(\rho)| = |P_O'(1-\rho)| = |P_O'(\overline{\rho})| = |P_O'(1-\overline{\rho})|.$$

Hence its absolute derivative is packet-valued. The global-to-local theorem theorem 7.17 proves the exact, unnormalized identity

$$\mu_\rho = H_O(\rho)(P_O'(\rho))^{m_O};$$

taking absolute values gives (668).

To see that the complementary factor cannot be inferred from one packet, retain the explicit fibre witness of theorem 7.18. For a positive real λ chosen so that $Q_\lambda(s) = 1 + \lambda(s-1/2)^2$ has no root in O and $Q_\lambda(\rho) \neq 1$, the functions

$$P_O(s)^{m_O} \quad \text{and} \quad Q_\lambda(s)P_O(s)^{m_O}$$

have the same local packet polynomial and multiplicity at O, but their leading coefficients differ by the nonzero, nonidentity scalar factor $Q_\lambda(\rho) \neq 0, 1$. Both retain evenness about 1/2 and conjugation symmetry. This proves the nontrivial fibre of the one-packet projection at the exact claimed scope. If all packet factors and $\xi(1/2)$ are supplied, the displayed product gives H_O and hence every quantity in the statement. □

8.9 The Scaling Site quotient and the shared coordinate diagram

The second published realization uses a different, fully specified sheaf. For $L > 0$, put $\mu = e^L$ and $C_\mu = \mathbb{R}_+^*/\mu^{\mathbb{Z}}$. Connes and Consani construct a sheaf $\mathcal{L}^2$ whose fibre is $L^2(C_\mu)$, a closed subsheaf $\overline{\Sigma\mathcal{E}}$, and prove the equivariant isomorphism

$$H^0\left(\mathfrak{S}_{\mathrm{sc}}, \mathcal{L}^2/\overline{\Sigma\mathcal{E}}\right) \;\cong\; \frac{\mathcal{S}(\mathbb{R})}{\overline{\zeta(1/2+\mathrm{i}\,\cdot)\mathcal{S}(\mathbb{R})}}. \tag{669}$$

Here the closure is in the Schwartz topology, and $r \in \mathbb{R}_+^*$ acts at the Schwartz coordinate s by multiplication with $r^{\mathrm{i}s}$ [22, the theorem and proof on pp. 96–98]. The pointwise Fourier transform on C_{e^L} uses the kernel $u^{-2\pi\mathrm{i}k/L}$, and the published scaling action on its component of index k is multiplication by $r^{-2\pi\mathrm{i}k/L}$. Therefore the exact index-to-Schwartz-coordinate map is

$$s_k(L) = -\frac{2\pi k}{L}, \qquad \zeta\left(\frac{1}{2} - \frac{2\pi\mathrm{i}k}{L}\right) = \zeta\left(\frac{1}{2} + \mathrm{i}s_k(L)\right), \qquad \chi_k(r) = r^{\mathrm{i}s_k(L)} = r^{-2\pi\mathrm{i}k/L}. \tag{670}$$

Thus the sign of the integer Fourier index is opposite to the sign of the Schwartz spectral coordinate; neither sign is normalized away. Their proof identifies the ±1 Fourier components, proves surjectivity of the global-section quotient map, and compares the two closed ideals through this coordinate substitution. Their following remark records the nilpotent Jordan term that can occur at a multiple critical zero.

Theorem 8.11 (Positive interface between a critical packet and the Scaling Site)**.** *Let $O \in Y_\xi^{\mathrm{crit}}$, let $\nu = \sqrt{-\lambda_O - 1/4}$, and let $n \geq 1$. Put*

$$L = \frac{2\pi n}{\nu}, \qquad \mu = e^L. \tag{671}$$

Then the absolute frequency of the Fourier components with indices $+n$ and $-n$ in $L^2(C_\mu)$ is

$$\frac{2\pi n}{L} = \nu, \tag{672}$$

and

$$\zeta\left(\frac{1}{2} + \mathrm{i}\nu\right) = \zeta\left(\frac{1}{2} - \mathrm{i}\nu\right) = 0. \tag{673}$$

The resulting circle is one of the zeta-cycles of theorem 8.9, and the Scaling Site spectral characters of the two components are

$$\chi_{+n}(r) = r^{-\mathrm{i}\nu}, \qquad \chi_{-n}(r) = r^{\mathrm{i}\nu}. \tag{674}$$

Equivalently,

$$s_{+n}(L) = -\nu, \qquad s_{-n}(L) = +\nu. \tag{675}$$

Accordingly, the following coordinate chain commutes by direct substitution:

$$\begin{array}{ccccc} O & \longmapsto & \lambda_O & \longmapsto & \nu = \sqrt{-\lambda_O - 1/4} \\ & & & & \downarrow n \\ & & & & L = 2\pi n/\nu \longmapsto C_{e^L}, \end{array}$$

while the Fourier coordinate returns $2\pi n/L = \nu$.

More precisely, the displayed formulas define the map

$$\Theta(O, n) = \big(\lambda_O, \nu, L, C_{e^L}; (+n, -\nu, r \mapsto r^{-\mathrm{i}\nu}), (-n, +\nu, r \mapsto r^{\mathrm{i}\nu})\big) \tag{676}$$

from $Y_\xi^{\mathrm{crit}} \times \mathbb{Z}_{\geq 1}$ to the set of labeled packet-value, positive-frequency, circle, Fourier-index, signed-frequency, and unitary-character tuples. This is the exact shared-coordinate morphism between the packet, Laplacian, cycle, and Scaling Site presentations proved here. It does not select a nonzero vector or global section of the Scaling Site quotient, and it does not identify the Boolean packet sheaf of theorem 5.16 with $\mathcal{L}^2/\overline{\Sigma\mathcal{E}}$: a stalkwise map, its topology, and its compatibility with the $\mathbb{N}^\times$-equivariant structure have not been constructed in this chapter.

Proof. Choose the positive ordinate ν represented by the critical packet. By construction of Y_ξ^{crit}, the conjugate pair $1/2 \pm \mathrm{i}\nu$ belongs to O, and both points are zeros of ζ. This proves (673). Substitution of (671) gives

$$\frac{2\pi n}{L} = \frac{2\pi n}{2\pi n/\nu} = \nu,$$

which is (672). In the Fourier formula for the Scaling Site subsheaf, the factor for index $+n$ is therefore

$$\zeta\left(\frac{1}{2} - \frac{2\pi \mathrm{i} n}{L}\right) = \zeta\left(\frac{1}{2} - \mathrm{i}\nu\right) = 0,$$

one member of (673); index $-n$ gives the other member. The published cycle theorem then supplies the zeta-cycle. The Scaling Site action on the component of index k is $r^{-2\pi \mathrm{i} k/L}$; substituting $k = +n$ and $k = -n$ gives exactly the two characters in (674). Every edge of the diagram is one of the displayed formulas, so commutativity is literal.

The Boolean packet stalk A^{m_O} is a finite semimodule, whereas the published Scaling Site object is the stated quotient of a sheaf of complex topological vector spaces. The positive coordinate chain just proved gives a common base parameter, but no formula above defines a map between those stalks. The final sentence records precisely that remaining construction obligation; it is not a claim that such a morphism cannot exist. □

8.10 The resulting packet and the exact scope of the reconstruction

Finite certification boundary. The companion Python certificate replays twenty named exact checks in two Python/SymPy installations. They cover the split-zero operation tables, both affine inverses, the quadratic fibre, both spectral-polynomial coordinate conversions, the squared-coordinate inverse, the two divisor aggregations, and the cycle-frequency substitution. The companion Lean module proves thirteen named finite algebraic theorems, including the complete sign-fibre equivalence, under Lean 4.32.0 and the pinned Mathlib revision. These certificates deliberately do not replace the cited analytic inputs: analytic continuation, the common-zero theorem, the Laplacian quotient spectrum, and the Scaling Site isomorphism remain the human-authored results cited above.

Combining the preceding maps, a packet O with multiplicity m_O has the following explicit data and no hidden representative convention:

$$\left(R_O(t), \quad m_O, \quad A_{\mathbb{C}}^{m_O}, \quad \omega(O)\right). \tag{677}$$

The polynomial R_O recovers the complete point packet; m_O recovers its effective divisor; $A_{\mathbb{C}}^{m_O}$ is the descended Boolean multiplicity stalk; and $\omega(O)$ is the representative-independent absolute leading jet. On the critical subspace, the unique root of R_O also recovers every length in (667) and the Scaling Site frequency in (672).

Three source formulations have thereby been sharpened at their exact proof sites. First, the common-zero identification is the divisor equality on $|\Im z| < 1/2$, not a global equality after the trivial zeros are restored. Second, $2\pi/\nu$ is the least length in the positive-integer family for a fixed ordinate ν, not a globally shortest cycle claim. Third, $G(\mathbb{Z})$ and $G(\mathbb{C})$ are connected by the proved embedding (638); their different coefficient rings do not erase the common Boolean fixed pair. These are corrections and positive morphisms relative to the audited local source. No general claim about what has or has not been proved elsewhere is inferred from a gap or wording defect in that source.

9 Primitive-defect source audit: a proved coordinate core

This chapter records the part of the modern local source “Primitive Defect Augmentations, Split-Zero Support, and A-Series Realizations” which survives a line-by-line audit. The local source was read consecutively at its hash-pinned TeX edition (lines 1–1474). It is an input artifact, not a human provenance record. The present chapter therefore cites human sources for imported facts and labels every local construction by its actual proof scope. In particular, the source’s final “consolidated theorem” is not retained as one theorem: its summands live in different typed categories, and several of the proposed arrows were not defined.

9.1 Scope, source identity, and audit boundary

The audited source has 47,978 bytes and SHA-256

```
2f3289f6a04105d3f15fe5f7c36dde286515e61f465caf1
582f9808e19f3d2f7.
```

An identical TeX copy occurs in the adjacent version directory. The source contains the following distinct layers:

(a) literal split-zero semiring and support-linearization constructions;

(b) augmentation, Fourier, root-lattice, and Eisenstein calculations;

(c) a proposed semilocal crossed-product boundary model;

(d) principal parts of a shifted completed Hurwitz function;

(e) orbifold, Weyl-operator, CUE, spectral, F-theory, and moonshine interfaces; and

(f) a conjectural CUE/Epstein transform bridge.

Only (a), the finite-dimensional portions of (b), the principal-part algebra, and the explicitly typed finite CUE/Weyl calculations are proved below. The other layers are retained in the machine-readable audit as candidate interfaces with their missing definitions and dependencies. "Not proved in this chapter" is a source-relative statement; it is not a claim about the whole literature.

9.2 The split-zero semiring and its two zero fibres

Let R be a nonzero commutative ring and put

$$G_R = R \sqcup \{\tau\}.$$

For $r, s \in R$ define

$$r \oplus s = r + s, \qquad r \oplus \tau = \tau \oplus r = r, \qquad \tau \oplus \tau = \tau,$$

and

$$r \odot s = rs, \qquad r \odot \tau = \tau \odot r = \tau, \qquad \tau \odot \tau = \tau.$$

Proposition 9.1 (Semiring and coordinate model)**.** *The operations above make G_R a commutative semiring with additive zero τ and multiplicative unit 1_R. The map*

$$\iota : G_R \longrightarrow D_R := \{(0,0)\} \cup (R \times \{1\}), \qquad \iota(\tau) = (0,0), \quad \iota(r) = (r,1),$$

is an injective semiring homomorphism when D_R has the coordinatewise operations inherited from $R \times \{0,1\}$ with the same split-zero tables. The projections

$$p_R(\tau) = 0_R, \quad p_R(r) = r, \qquad \chi_R(\tau) = 0, \quad \chi_R(r) = 1$$

have fibres

$$p_R^{-1}(0_R) = \{\tau, 0_R\}, \qquad \chi_R^{-1}(0) = \{\tau\}, \qquad \chi_R^{-1}(1) = R.$$

Proof. If all entries are in R, associativity, commutativity, and distributivity are those of R. If an entry is τ, the addition table says that it is an identity, so every associativity case reduces to the case with the τ removed. For multiplication, τ is absorbing, so associativity and commutativity reduce in the same way. For distributivity, if the multiplier is τ both sides are τ; if the multiplier and both summands are in R, distributivity is inherited from R; and, for example,

$$r \odot (s \oplus \tau) = r \odot s = rs \oplus \tau = (r \odot s) \oplus (r \odot \tau).$$

The remaining cases are symmetric. Thus the semiring axioms hold. The displayed coordinate map is injective because $(0,0)$ is not in $R \times \{1\}$. Checking a pair of supported entries uses the ring laws; checking a pair containing τ uses the two defining tables, so it preserves both operations and the unit. Reading the first and second coordinates gives the three fibre identities. □

Theorem 9.2 (Support and arithmetic prime ideals)**.** *The subset* $\mathfrak{s}_R = \{\tau\}$ *is a prime ideal of* G_R. *If* R *is an integral domain, then* $\mathfrak{z}_R = \{\tau, 0_R\}$ *is also a prime ideal, and*

$$\mathfrak{s}_R \subset \mathfrak{z}_R.$$

Proof. The set $\mathfrak{s}_R$ is closed under addition and absorbs multiplication. If $x \odot y = \tau$ and neither factor is τ, then $x, y \in R$ and $x \odot y = xy \in R$, a contradiction. Hence one factor belongs to $\mathfrak{s}_R$, which is primality.

Assume now that R is a domain. The set $\mathfrak{z}_R$ is closed under addition and absorbs multiplication. If $x \odot y \in \mathfrak{z}_R$ and neither factor is τ, then $x, y \in R$ and $xy = 0_R$; the domain property gives $x = 0_R$ or $y = 0_R$. Thus one factor is in $\mathfrak{z}_R$. Both ideals are proper because $R \neq \{0_R\}$, and the inclusion is immediate. $\square$

The two ideals retain different information: the first separates support from absence, while the second also identifies the supported arithmetic zero with the semiring zero. No identification of these ideals with objects in a different category is made here.

9.3 Support linearization and the A-series kernel

For a finite set I, let $\mathcal{P}(I)$ be its Boolean support cube and put

$$\mathcal{L}_I(J) = \sum_{j \in J} e_j \in \mathbb{Z}^I, \qquad \delta_I\left(\sum_{i \in I} a_i e_i\right) = -\sum_{i \in I} a_i.$$

Theorem 9.3 (Exact kernel coordinates)**.** *If* $|I| = r$, *then*

$$\ker \delta_I = \left\{(a_1, \dots, a_r) \in \mathbb{Z}^r : \sum_i a_i = 0\right\}$$

has basis $\alpha_i = e_i - e_{i+1}$ $(1 \leq i < r)$. *Its Gram matrix for the standard inner product is the Cartan matrix of type* A_{r-1}; *for* $r = 1$ *the kernel is the zero lattice* A_0.

Proof. Choose an ordering of I. Every α_i has coordinate sum zero. If $a = (a_1, \dots, a_r)$ has sum zero, set

$$b_i = a_1 + \cdots + a_i \qquad (1 \leq i < r).$$

The jth coordinate of $\sum_{i<r} b_i(e_i - e_{i+1})$ is $b_1 = a_1$ for $j = 1$, is $b_j - b_{j-1} = a_j$ for $1 < j < r$, and is $-b_{r-1} = a_r$ for $j = r$. Thus the α_i span the kernel. If a linear combination of them is zero, its first coordinate is the coefficient of α_1, then its second new coordinate determines the coefficient of α_2, and so on; hence they are independent. Finally

$$\langle \alpha_i, \alpha_j \rangle = \begin{cases} 2 & i = j, \\ -1 & |i-j| = 1, \\ 0 & |i-j| > 1, \end{cases}$$

which is the A_{r-1} Cartan matrix. When $r = 1$ there are no roots and the kernel is 0. $\square$

The defect of a support J is exactly $\delta_I(\mathcal{L}_I(J)) = -|J|$. This is a statement about the displayed free abelian group. A Grothendieck group, representation ring, or fusion ring may have relations and primitive objects of dimension other than one; the slogan "universal negative dimension" is therefore valid only after the explicit free, dimension-one hypotheses have been imposed.

9.4 Cyclic Fourier projection

Let $C_r = \langle g \mid g^r = 1\rangle$ and $\omega = e^{2\pi i/r}$. In $\mathbb{C}[C_r]$ define

$$S_{ab} = r^{-1/2}\omega^{ab}, \qquad a, b \in \mathbb{Z}/r\mathbb{Z}.$$

Proposition 9.4 (Augmentation is the trivial Fourier coordinate)**.** *For* $x = \sum_{a=0}^{r-1} c_a g^a$,

$$\epsilon(x) = \sum_a c_a = \sqrt{r}\sum_a c_a S_{a0}.$$

The nontrivial Fourier modes span $\ker(\epsilon) \otimes_{\mathbb{Z}} \mathbb{C}$*.*

Proof. Since $S_{a0} = r^{-1/2}$, the first equality is immediate. For $b \in \mathbb{Z}/r\mathbb{Z}$ put $f_b = \sum_a \omega^{-ab} g^a$. The finite geometric sum gives

$$\sum_{a=0}^{r-1} \omega^{a(b-c)} = r\,\mathbf{1}_{b=c},$$

so the f_b form a basis. The coefficient sum of f_b is r for $b = 0$ and 0 otherwise. Therefore the kernel is exactly the span of $f_1, \ldots, f_{r-1}$. □

9.5 The C_3 augmentation ideal and the Eisenstein Epstein sum

For $C_3 = \langle g\rangle$, the augmentation ideal has basis $\alpha_1 = 1 - g$, $\alpha_2 = g - g^2$, with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The evaluation map $\pi_\omega : \mathbb{Z}[C_3] \to \mathbb{Z}[\omega]$, $g \mapsto \omega$, has kernel generated by $1 + g + g^2$, and

$$\pi_\omega(I_{C_3}) = (1 - \omega)\mathbb{Z}[\omega].$$

Indeed, the two basis images are $1 - \omega$ and $\omega - \omega^2 = \omega(1 - \omega)$, and ω is a unit.

Writing $z = m + n\omega$ gives

$$N(z) = z\overline{z} = m^2 - mn + n^2, \qquad Q_{A_2}(m, n) = 2N(z).$$

Theorem 9.5 (A_2 Epstein zeta in its initial half-plane)**.** *For* $\Re(s) > 1$,

$$Z_{A_2}(s) := \sum_{(m,n)\neq(0,0)} Q_{A_2}(m, n)^{-s} = 2^{-s}6\,\zeta(s)L(s, \chi_{-3}).$$

The identity then extends wherever the two cited meromorphic continuations are defined.

Proof. The norm is multiplicative because it is the complex norm in $K = \mathbb{Q}(\sqrt{-3})$. The Eisenstein integers are Euclidean for this norm: the triangular lattice $\mathbb{Z}[\omega] \subset \mathbb{C}$ has Voronoi cells of circumradius $1/\sqrt{3} < 1$, so for $z \in K$ choose $q \in \mathbb{Z}[\omega]$ with $|z - q| < 1$ and, after clearing a denominator, obtain Euclidean division $a = bq + r$ with $N(r) < N(b)$. Hence every nonzero ideal is principal. The six elements $\{\pm 1, \pm\omega, \pm\omega^2\}$ are exactly the units (their norm is one), so each nonzero principal ideal has six generators.

For $\Re(s) > 1$, unique ideal factorization and multiplicativity of the ideal norm give the Dedekind Euler product (the standard theorem is recorded in Milne and Neukirch [58, 59]). Consequently

$$\sum_{0\neq z\in\mathbb{Z}[\omega]} N(z)^{-s} = 6\,\zeta_K(s).$$

The quadratic character of K is χ_{-3}; comparing the Euler factor at each rational prime (split for $p \equiv 1 \pmod 3$, inert for $p \equiv 2 \pmod 3$, and ramified at 3) gives

$$\zeta_K(s) = \zeta(s)L(s, \chi_{-3}).$$

Multiplying by 2^{-s} proves the displayed formula. The series argument was made only in $\Re(s) > 1$; continuation is imported from the cited Dedekind-zeta theorem rather than silently inferred from the series. $\square$

9.6 Completed Hurwitz principal parts

The local source writes $F_t(s) = \zeta(s, 1+t)$ and

$$\Lambda_t(s) = \pi^{-s/2}\Gamma(s/2)F_t(s).$$

The value formula used here is DLMF 25.11.13. The simple pole and specialization are recorded in DLMF 25.11.1–2 [51]. The following statement uses $a = 1 + t$ in the DLMF range $\Re(a) > 0$ (or, more generally, away from nonpositive-integer parameter singularities).

Lemma 9.6 (Invariant and anti-invariant principal-part basis)**.** *Put $u = s(s-1)$ and $v = 2s - 1$. Then*

$$u(1-s) = u(s), \qquad v(1-s) = -v(s), \qquad v^2 = 1 + 4u,$$

and

$$\frac{1}{u} = \frac{1}{s-1} - \frac{1}{s}, \qquad \frac{v}{u} = \frac{1}{s-1} + \frac{1}{s}.$$

If $P(s) = a/s + b/(s-1)$, then

$$P(s) = A\frac{1}{u} + B\frac{v}{u}, \qquad A = \frac{b-a}{2}, \quad B = \frac{a+b}{2}.$$

Proof. The first three identities follow by direct substitution. Since $u = s(s-1)$ and $v = 2s-1$, partial fractions give the next two identities. Expanding $A/u + Bv/u$ gives coefficient $-A + B$ at $1/s$ and $A + B$ at $1/(s-1)$. Solving $a = -A + B$, $b = A + B$ gives the displayed A, B. $\square$

Theorem 9.7 (Shifted completed-Hurwitz endpoint calculation)**.** *Under the parameter restriction above,*

$$\operatorname*{Res}_{s=0} \Lambda_t = -1 - 2t, \qquad \operatorname*{Res}_{s=1} \Lambda_t = 1,$$

and hence

$$\mathrm{PP}_{\{0,1\}}(\Lambda_t) = (1+t)\frac{1}{u} - t\frac{v}{u}.$$

For $t = 1$ the anti-invariant term is $-v/u$, with residue -1 at both endpoints.

Proof. At $s = 1$, DLMF gives residue one for $\zeta(s, a)$, while $\pi^{-1/2}\Gamma(1/2) = 1$, proving the second residue. At $s = 0$,

$$\Gamma(s/2) = \frac{2}{s} + O(1), \qquad \pi^{-s/2} = 1 + O(s),$$

and DLMF 25.11.13 gives $\zeta(0, 1+t) = \frac{1}{2} - (1+t) = -\frac{1}{2} - t$. The first residue is therefore $2(-\frac{1}{2} - t) = -1 - 2t$. Apply Lemma 9.6 with $a = -1 - 2t$ and $b = 1$; it gives $A = 1 + t$ and $B = -t$. At $t = 1$, $-v/u = -1/s - 1/(s-1)$, so both endpoint residues are -1. $\square$

Remark 9.8 (What the critical-line calculation actually says). On $s = \frac{1}{2} + i\gamma$,

$$u = -\left(\gamma^2 + \frac{1}{4}\right) \in \mathbb{R}, \qquad v = 2i\gamma.$$

Thus $1/u$ is real and v/u is purely imaginary. These are orthogonal only after choosing the standard real inner product on $\mathbb{C} \cong \mathbb{R}^2$; there is no canonical assertion of orthogonality of complex subspaces in the source's wording. This observation has no implication about the location of zeros.

9.7 The finite Weyl algebra and the corrected $r = 3$ matrices

Let $V = \mathbb{C}^r$ with basis $|m\rangle$, $m \in \mathbb{Z}/r\mathbb{Z}$, and define

$$X|m\rangle = |m+1\rangle, \qquad Z|m\rangle = \omega^m |m\rangle, \qquad W_{a,b} = X^a Z^b.$$

Then $ZX = \omega XZ$ and

$$W_{a,b}W_{c,d} = \omega^{bc} W_{a+c,b+d}, \qquad [W_{a,b}, W_{c,d}] = (\omega^{bc} - \omega^{ad}) W_{a+c,b+d}.$$

Theorem 9.9 (Finite Weyl basis). *The r^2 operators $W_{a,b}$ form a basis of $M_r(\mathbb{C})$, and the $r^2 - 1$ nonidentity operators form a basis of $\mathfrak{sl}_r(\mathbb{C})$.*

Proof. The relation $ZX = \omega XZ$ gives the multiplication formula, and reversing the two factors gives the commutator. If $a \neq 0$, $X^a Z^b$ has zero diagonal; if $a = 0$ and $b \neq 0$, its trace is $\sum_{m=0}^{r-1} \omega^{bm} = 0$. Hence every nonidentity operator is traceless. To prove independence, suppose $\sum_{a,b} c_{a,b} X^a Z^b = 0$. The $(m + a, m)$ entry isolates $c_{a,b}\omega^{bm}$ as a finite Fourier polynomial in m; Fourier orthogonality forces every $c_{a,b} = 0$. There are r^2 independent operators in the r^2-dimensional matrix space, and the trace map has the one-dimensional span of the identity as a complementary direction. The nonidentity span is therefore exactly the traceless subspace. $\square$

For $r = 3$, preserving the declared action on kets requires

$$X = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad X^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad Z = \operatorname{diag}(1, \omega, \omega^2).$$

Direct multiplication gives $ZX = \omega XZ$. The matrix printed as X in the audited local source is the second displayed matrix, hence is actually X^2 under the source's ket convention; with that printed matrix the relation is $ZX = \omega^2 XZ$. The correction is a relabelling of the two shifts and of the four mixed products, not a change of the abstract Weyl algebra. The finite matrix calculation has no map here to a split-zero element τ: the zero matrix and unsupported absence remain distinct typed objects.

9.8 The CUE exponent: a numerical equality and an independent block map

Grover, Mezzadri, and Simm define, for Haar $U \in U(N)$,

$$\Lambda_N(z) = \det(1 - zU^\dagger), \qquad M_{\mu,\nu}(z, N) = \mathbb{E}\left[\prod_i \Lambda_N^{(\mu_i)}(z) \prod_j \overline{\Lambda_N^{(\nu_j)}(z)}\right]$$

and prove their microscopic equal-length asymptotic with exponent $|\mu|+|\nu|+s^2$ under the hypotheses stated in their equal-length microscopic theorem [28]. For $\mu = \nu = (1^s)$ this exponent is

$$s + s + s^2 = s^2 + 2s = (s+1)^2 - 1.$$

The last equality is a dimension count, not a statement made by the CUE paper that its moment has a Lie-algebra-valued functor.

Proposition 9.10 (Explicit block-vector-space isomorphism)**.** *For $W = \mathbb{C}^s$, the map*

$$\Theta : \mathrm{End}(W) \oplus W \oplus W^\vee \longrightarrow \mathfrak{sl}(W \oplus \mathbb{C}), \qquad (A, u, \varphi) \longmapsto \begin{pmatrix} A & u \\ \varphi & -\operatorname{tr} A \end{pmatrix}$$

is a vector-space isomorphism.

Proof. An element of $\mathfrak{sl}(W \oplus \mathbb{C})$ has a unique block form $\left(\begin{smallmatrix} B & u \\ \varphi & \lambda \end{smallmatrix}\right)$ with $\operatorname{tr} B + \lambda = 0$. It is therefore uniquely $\Theta(B, u, \varphi)$. Conversely every displayed block has trace zero. Linearity and the two inverse descriptions prove bijectivity. Transporting the matrix commutator through this bijection gives a Lie bracket on the source vector space, but supplies no map from the CUE moment space to that bracket. $\square$

9.9 The finite-prime boundary: local model, semilocal fibre, and exact quotient

This subsection closes the finite-prime local calculation and then compares it with the two-place semilocal geometry without discarding the p-adic unit coordinate. Fix a rational prime p and put $\ell = \log p$.

Definition 9.11 (The one-ended valuation space)**.** Let

$$\overline{\mathbb{Z}}_+ = \mathbb{Z} \sqcup \{\infty\}$$

have the topology in which every integer is isolated and a neighbourhood basis at ∞ is

$$N_M = \{\infty\} \cup \{m \in \mathbb{Z} : m \geq M\}, \qquad M \in \mathbb{Z}.$$

Thus only the positive end is compactified. Define

$$\bar{v}_p : \mathbb{Q}_p \longrightarrow \overline{\mathbb{Z}}_+, \qquad \bar{v}_p(x) = \begin{cases} v_p(x), & x \neq 0, \\ \infty, & x = 0. \end{cases}$$

Lemma 9.12 (Exact valuation quotient)**.** *The map $\bar{v}_p$ is the topological quotient by multiplication by $\mathbb{Z}_p^\times$. In particular,*

$$\mathbb{Q}_p/\mathbb{Z}_p^\times \xrightarrow{\ \sim\ } \overline{\mathbb{Z}}_+$$

is a homeomorphism, not merely a bijection of strata.

Proof. For $m \in \mathbb{Z}$ the fibre is $\bar{v}_p^{-1}(m) = p^m\mathbb{Z}_p^\times$, one $\mathbb{Z}_p^\times$-orbit, while the fibre over ∞ is $\{0\}$. The inverse images

$$\bar{v}_p^{-1}(\{m\}) = p^m\mathbb{Z}_p^\times, \qquad \bar{v}_p^{-1}(N_M) = p^M\mathbb{Z}_p$$

are open, so $\bar{v}_p$ is continuous. It is also open. An open set containing a nonzero x contains a sufficiently small ball on which the valuation is constant, hence its image contains the isolated point $v_p(x)$. An open set containing 0 contains $p^M\mathbb{Z}_p$ for some M, hence its image contains N_M. A continuous open surjection is a quotient map, and its fibres are exactly the stated unit orbits. $\square$

Set

$$Y_\ell = \overline{\mathbb{Z}}_+ \times \mathbb{R}, \qquad T^n(m, y) = (m + n, y + n\ell), \qquad T^n(\infty, y) = (\infty, y + n\ell).$$

Translation by an integer preserves the positive-end neighbourhoods N_M, so every T^n is a homeomorphism. The space Y_ℓ is locally compact and Hausdorff: an integer has a compact singleton neighbourhood, and $N_M \times [a, b]$ is a compact neighbourhood of each (∞, y) after shrinking the interval.

Proposition 9.13 (Orbit coordinates and the spiral compactification)**.** *The $\mathbb{Z}$-action on Y_ℓ is free and proper. Its orbit space is*

$$X_\ell = \mathbb{R} \sqcup S^1_\ell, \qquad S^1_\ell = \mathbb{R}/\ell\mathbb{Z},$$

under the orbit map

$$q_\ell(m, y) = y - m\ell, \qquad q_\ell(\infty, y) = [y]_\ell.$$

The topology is determined exactly as follows: the two displayed pieces have their usual topologies, and a generic net $t_j \in \mathbb{R}$ converges to $[a]_\ell \in S^1_\ell$ precisely when

$$t_j \longrightarrow -\infty, \qquad [t_j]_\ell \longrightarrow [a]_\ell.$$

Equivalently, X_ℓ is homeomorphic to the closed subset of $\mathbb{C}$ formed by the unit circle and the disjoint spiral

$$t \longmapsto (1 + e^t) \exp(2\pi i t/\ell),$$

with $[a]_\ell$ sent to $\exp(2\pi i a/\ell)$.

Proof. If $T^n(m, y) = (m, y)$, then $m + n = m$; at the boundary, $y + n\ell = y$. In either case $n = 0$, so the action is free. If compact sets $K, L \subset Y_\ell$ satisfy $T^nK \cap L \neq \varnothing$, the projections of K and L to the real coordinate are bounded, while that coordinate changes by $n\ell$. Only finitely many n can occur, which is properness for this discrete action.

On the generic part, $t = y - m\ell$ is invariant and every orbit meets $\{0\} \times \mathbb{R}$ once. On the boundary the quotient is $\mathbb{R}/\ell\mathbb{Z}$. For the cross-stratum topology, representatives (m_j, y_j) converge to (∞, a) exactly when $m_j \to +\infty$ and $y_j \to a$. Their invariant coordinates then satisfy $t_j = y_j - m_j\ell \to -\infty$ and $[t_j]_\ell = [y_j]_\ell \to [a]_\ell$. Conversely, from those two conditions choose integers m_j such that $y_j = t_j + m_j\ell \to a$; necessarily $m_j \to +\infty$, giving representatives converging to (∞, a). The spiral has radius $1 + e^t$, so it is injective; the same convergence criterion identifies its complete finite accumulation set with the unit circle. This proves the last description and the asserted homeomorphism. ☐

Let $U = \mathbb{Z} \times \mathbb{R} \subset Y_\ell$ and $F = \{\infty\} \times \mathbb{R}$. Since $\mathbb{Z}$ is amenable, full and reduced crossed products agree and the invariant open–closed decomposition gives

$$0 \longrightarrow I_p \longrightarrow E_p \longrightarrow Q_p \longrightarrow 0,$$

where

$$I_p = C_0(U) \rtimes \mathbb{Z}, \qquad E_p = C_0(Y_\ell) \rtimes \mathbb{Z}, \qquad Q_p = C_0(F) \rtimes \mathbb{Z}.$$

For a free proper action, the Green–Rieffel module identifies the crossed product with the compact operators on the quotient module [87, Theorem 1.3.22]. Applied to Y_ℓ, U, and F, it gives an extension-level Morita equivalence

$$\begin{array}{ccccccccc} 0 & \to & I_p & \to & E_p & \to & Q_p & \to & 0 \\ & & \sim_M\downarrow & & \sim_M\downarrow & & \sim_M\downarrow & & \\ 0 & \to & C_0(\mathbb{R}) & \to & C_0(X_\ell) & \to & C(S^1_\ell) & \to & 0. \end{array}$$

Here extension-level compatibility is not an extra assumption. In the standard module obtained by completing $C_c(Y_\ell)$, the closed submodule generated by $C_0(\mathbb{R}) \subset C_0(X_\ell)$ is the completion of $C_c(U)$, and its quotient is the completion of $C_c(F)$. The corresponding linking-algebra ideal and quotient therefore give the displayed commuting diagram. Functoriality of the six-term sequence transports its connecting maps.

Theorem 9.14 (The local finite-prime boundary coefficient)**.** *Choose the generator of $K_1(C_0(\mathbb{R}))$ represented by*

$$v(t) = \frac{t-i}{t+i};$$

as t increases, this loop winds counterclockwise. Under the explicit Green modules above, the connecting map satisfies

$$\partial_p : K_0(Q_p) \longrightarrow K_1(I_p), \qquad \partial_p([1]) = -[v].$$

Proof. Define a function on the entire quotient X_ℓ by

$$h(t) = \phi(t) = \frac{1}{1+e^t} \quad (t \in \mathbb{R}), \qquad h([a]_\ell) = 1 \quad ([a]_\ell \in S^1_\ell).$$

The convergence criterion in Proposition 9.13 gives continuity at every boundary phase because $\phi(t) \to 1$ as $t \to -\infty$. The only escaping end of X_ℓ is $t \to +\infty$, where $\phi(t) \to 0$, so $h \in C_0(X_\ell)$. Its quotient is the constant projection $1 \in C(S^1_\ell)$.

Blackadar's explicit exponential formula for the six-term connecting map [85, Section 9.3.2] therefore gives

$$\partial[1] = [u], \qquad u(t) = \exp(2\pi i \phi(t)) \in 1 + C_0(\mathbb{R}).$$

Both endpoint limits of u are 1. Along the oriented compactification of $\mathbb{R}$, its continuous argument $2\pi\phi(t)$ decreases from 2π to 0, hence its winding number is -1. Thus $[u] = -[v]$. Naturality for the extension-level Green modules gives the identical coefficient in the crossed product extension. □

We now retain the unit coordinate which the local model suppresses. The two-place model of Connes, Consani, and Marcolli is $(\mathbb{Q}_p \times \mathbb{R})/(\{\pm 1\} \times p^{\mathbb{Z}})$; its p-boundary is the circle $\mathbb{R}_+^\times/p^{\mathbb{Z}}$, and its logarithmic scale is $\mathbb{R}/\ell\mathbb{Z}$ [84, Section 8.2]. Restrict to the generic stratum and this p-boundary, namely

$$W_p = \mathbb{Q}_p \times \mathbb{R}^\times, \qquad G_p = \{\pm p^n : n \in \mathbb{Z}\},$$

with diagonal multiplication. This action is free because the real coordinate is nonzero. It is proper because boundedness of the absolute value of that coordinate in two compact sets bounds n, leaving only the finite sign.

Proposition 9.15 (Complete semilocal orbit coordinates)**.** *The orbit space $\mathcal{S}_p = W_p/G_p$ is*

$$\mathcal{S}_p = (\mathbb{Z}_p^\times \times \mathbb{R}) \sqcup S^1_\ell.$$

After using the sign to choose $y > 0$, its coordinates are

$$(x, y) \longmapsto \left(p^{-m}x, \log y - m\ell\right), \qquad m = v_p(x), \quad x \neq 0,$$

and

$$(0, y) \longmapsto [\log y]_\ell.$$

A generic net (a_j, t_j) converges to $[b]_\ell$ exactly when

$$t_j \to -\infty, \qquad [t_j]_\ell \to [b]_\ell;$$

there is no condition on $a_j \in \mathbb{Z}_p^\times$.

Proof. The sign choice is unique because $y \neq 0$. For $x \neq 0$, write uniquely $x = p^m a$ with $m = v_p(x)$ and $a \in \mathbb{Z}_p^\times$. Multiplication by p^n changes $(m, \log y)$ to $(m+n, \log y + n\ell)$ and leaves both a and $t = \log y - m\ell$ invariant. At $x = 0$, the remaining orbit coordinate is $[\log y]_\ell$. These formulas give a bijection.

For convergence to $(0, y)$ choose orbit representatives $(p^{m_j} a_j, y_j)$ with $y_j \to y$. They approach the boundary precisely when $m_j \to +\infty$; this convergence is uniform in the unit a_j because $|p^{m_j} a_j|_p = p^{-m_j}$. Consequently $t_j = \log y_j - m_j \ell \to -\infty$ and $[t_j]_\ell = [\log y_j]_\ell \to [\log y]_\ell$. The converse follows by choosing integers $m_j \to +\infty$ for which $t_j + m_j\ell \to b$ and taking representatives $(p^{m_j} a_j, \exp(t_j + m_j\ell))$. This also proves the topological statement. $\square$

The compact group $K_p = \mathbb{Z}_p^\times$ acts on $\mathcal{S}_p$ by multiplication in the first generic coordinate and fixes the boundary circle. Its orbit space is exactly X_ℓ. Write the quotient map as

$$\Pi_p : \mathcal{S}_p \longrightarrow X_\ell, \qquad \Pi_p(a, t) = t, \qquad \Pi_p([b]_\ell) = [b]_\ell.$$

Thus $\Pi_p^{-1}(t) = \mathbb{Z}_p^\times$ on the generic stratum and $\Pi_p^{-1}([b]_\ell) = \{[b]_\ell\}$ on the boundary. Pullback gives the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & C_0(\mathbb{R}) & \to & C_0(X_\ell) & \to & C(S^1_\ell) & \to & 0 \\ & & \downarrow{\scriptstyle \pi^*} & & \downarrow{\scriptstyle \Pi_p^*} & & \| & & \\ 0 & \to & C_0(\mathbb{Z}_p^\times \times \mathbb{R}) & \to & C_0(\mathcal{S}_p) & \to & C(S^1_\ell) & \to & 0, \end{array}$$

where $\pi(a, t) = t$. The lower extension is Morita equivalent, by the same restricted Green-module argument, to the natural semilocal crossed-product extension

$$\begin{aligned} 0 \to C_0(\mathbb{Q}_p^\times \times \mathbb{R}^\times) \rtimes G_p &\longrightarrow C_0(W_p) \rtimes G_p \\ &\longrightarrow C_0(\{0\} \times \mathbb{R}^\times) \rtimes G_p \to 0. \end{aligned}$$

Corollary 9.16 (The semilocal boundary class and its exact relation to the local coefficient)**.** *Under the suspension/Bott identification*

$$K_1\big(C_0(\mathbb{Z}_p^\times \times \mathbb{R})\big) \cong K_0(C(\mathbb{Z}_p^\times)) \cong C(\mathbb{Z}_p^\times, \mathbb{Z}),$$

the semilocal connecting map sends

$$[1] \longmapsto \big(a \mapsto -1\big).$$

It is the pullback of the local class -1 along π.

Proof. Naturality of the displayed extension diagram gives $\partial_{\mathrm{sl}} = \pi^*_* \partial_p$. Equivalently, the same lift is

$$h(a, t) = \frac{1}{1 + e^t}, \qquad h([b]_\ell) = 1,$$

and its exponential is the clockwise unitary from Theorem 9.14, constant in a. Since $\mathbb{Z}_p^\times$ is compact and totally disconnected, stable projection classes in $C(\mathbb{Z}_p^\times)$ are their locally constant integer-valued rank functions. Pointwise winding is -1 at every a, proving the formula. $\square$

Proposition 9.17 (Why the unit fibre cannot be removed by Morita equivalence)**.** *The map Π_p is a proper quotient with the fibres displayed above, but it is not a homeomorphism. Moreover, the natural semilocal crossed-product algebra $C_0(W_p) \rtimes G_p$ is not strongly Morita equivalent to E_p.*

Proof. The compact group K_p acts continuously on $\mathcal{S}_p$, so its orbit map Π_p is closed and proper. On the generic open stratum it forgets the entire K_p coordinate; for example, a continuous function supported in a proper clopen subset of K_p does not lie in the image of Π_p^*.

For a topological obstruction independent of this particular formula, note that K_p is totally disconnected and has no isolated points. Hence no point of $K_p \times \mathbb{R}$ is locally connected: the projection of a connected set to K_p is a singleton, but $\{a\} \times \mathbb{R}$ has empty interior. The generic stratum is dense in $\mathcal{S}_p$, so $\mathcal{S}_p$ has no nonempty open set consisting of locally connected points. By contrast, the generic stratum $\mathbb{R} \subset X_\ell$ is a nonempty open set of locally connected points. Thus the two orbit spaces are not homeomorphic.

Both group actions are free and proper, so their crossed products are strongly Morita equivalent to $C_0(\mathcal{S}_p)$ and $C_0(X_\ell)$, respectively [87, Theorem 1.3.22]. Strong Morita equivalence of commutative C^*-algebras induces a homeomorphism of their primitive spectra [88, p. 208]. The proved topological obstruction and transitivity of Morita equivalence rule out a Morita equivalence between the two crossed products. □

Consequently the local coefficient -1 is correct, and its natural semilocal image is the constant class $a \mapsto -1$. What fails is the audited source's claim that forgetting the $\mathbb{Z}_p^\times$ coordinate is itself a Morita reduction. A different spherical corner or unit-character projection could still define another map, but it must be stated and proved; it is not supplied by the source or by the two-place construction. At this stage no larger multi-place algebra follows from the two-place theorem alone; Subsection 9.12 constructs the natural finite-set candidate independently and records exactly what it retains and forgets.

Executable boundary check. The bounded replay is

`certificates/globalization_nte_finite_prime_boundary/verify_finite_prime_boundary.py`.

It checks the two invariant coordinates, the spiral radius limits, the derivative and endpoint values of ϕ, the clockwise winding -1, the counterclockwise Cayley generator $+1$, and the pointwise-constant semilocal winding. It does not certify the quotient topologies, the Green–Rieffel theorem, extension-level Morita compatibility, or the primitive-spectrum obstruction; those are the standard proofs printed above.

9.10 The archimedean reflection extension: exact matrices, boundary map, and domain-sensitive index comparison

The next source locus, lines 573–624, defines one genuine local object: the reflection crossed product $C_0(\mathbb{R}) \rtimes (\mathbb{Z}/2)$. It does not define the larger multi-place algebra mentioned above. Lines 626–648 then write a scalar "global component formula", but do not provide a global algebra, ideal, exact sequence, codiagonal map, or component-to-global functor. We therefore first compute the local crossed product exactly, then separate three inequivalent domains for the displayed differential expression, and only after that type the additional map required by the global formula.

Let $\sigma(x) = -x$, let u denote the unitary implementing σ, and put

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad D_2 = \left\{ \begin{pmatrix} z_+ & 0 \\ 0 & z_- \end{pmatrix} : z_\pm \in \mathbb{C} \right\}.$$

The matrix H is retained: it is the explicit change from the orbit basis $(r, -r)$ to the two character coordinates of $\mathbb{Z}/2$.

Theorem 9.18 (Reflection crossed-product boundary)**.** *There is a $*$-isomorphism*

$$\Phi : C_0(\mathbb{R}) \rtimes_\sigma (\mathbb{Z}/2) \longrightarrow \mathcal{A}_\infty := \{G \in C_0([0,\infty), M_2(\mathbb{C})) : G(0) \in D_2\}$$

whose value on $f_0 + f_1 u$ is

$$\Phi(f_0 + f_1 u)(r) = H \begin{pmatrix} f_0(r) & f_1(r) \\ f_1(-r) & f_0(-r) \end{pmatrix} H, \qquad r \geq 0. \tag{678}$$

Evaluation at the fixed point gives the exact sequence

$$0 \longrightarrow C_0((0,\infty), M_2) \longrightarrow \mathcal{A}_\infty \xrightarrow{\mathrm{ev}_0} D_2 \longrightarrow 0. \tag{679}$$

Under $D_2 \cong \mathbb{C} \oplus \mathbb{C}$, the projections

$$p_+ = \frac{1+u}{2}, \qquad p_- = \frac{1-u}{2}$$

map to e_{11} and e_{22}, respectively.

Identify

$$K_1(C_0((0,\infty), M_2)) \cong \mathbb{Z}$$

by declaring the loop

$$\beta(r) = \exp\left(2\pi i \frac{r}{1+r}\right), \qquad r \in [0,\infty],$$

to have winding $+1$. Then the exponential connecting map of (679) *is*

$$\partial_\infty : \mathbb{Z}[p_+] \oplus \mathbb{Z}[p_-] \longrightarrow \mathbb{Z}, \qquad \partial_\infty(a,b) = -(a+b). \tag{680}$$

Consequently

$$\partial_\infty[p_+] = \partial_\infty[p_-] = -1, \qquad \partial_\infty([p_+] + [p_-]) = -2, \qquad \ker \partial_\infty = \mathbb{Z}(1,-1).$$

Proof. In the orbit basis of $\{r, -r\}$, the covariant matrix of the algebraic crossed-product element $f_0 + f_1 u$ is

$$M_f(r) = \begin{pmatrix} f_0(r) & f_1(r) \\ f_1(-r) & f_0(-r) \end{pmatrix}.$$

At $r = 0$ it has the form $\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$, and direct multiplication gives

$$H \begin{pmatrix} a & b \\ b & a \end{pmatrix} H = \begin{pmatrix} a+b & 0 \\ 0 & a-b \end{pmatrix}.$$

Thus (678) lands in $\mathcal{A}_\infty$.

Conversely, for $G \in \mathcal{A}_\infty$ put $M(r) = HG(r)H$. For $r > 0$ define, without identifying any of the four coordinates,

$$\begin{aligned} f_0(r) &= M_{11}(r), \quad f_0(-r) = M_{22}(r), \\ f_1(r) &= M_{12}(r), \quad f_1(-r) = M_{21}(r). \end{aligned} \tag{681}$$

Because $G(0)$ is diagonal, $M(0)$ has the form $\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$. Hence the two definitions of $f_0(0)$ agree, the two definitions of $f_1(0)$ agree, and all four half-line coordinates glue continuously. Since G vanishes at infinity, $f_0, f_1 \in C_0(\mathbb{R})$. Formula (681) is therefore an explicit inverse to Φ.

For completeness at the norm level, Mesland–Şengün define the full and reduced crossed-product norms and the regular Hilbert-module representation in their Sections 1.3.16–1.3.20 [87, pp. 12–13]. The group $\mathbb{Z}/2$ is amenable by the explicit invariant mean $m(h) = \frac{1}{2}(h(0) + h(1))$. Applied to the semidirect-product Fell bundle of this action, Exel's approximation theorem therefore identifies the full and reduced cross-sectional algebras [90, Theorems 20.6–20.7, pp. 148–149]. The regular norm here is precisely $\sup_{r\geq 0} \|M_f(r)\|$. Thus the mutually inverse coordinate maps above extend isometrically to the stated C^*-algebras.

Evaluation at zero is onto D_2: for any (z_+, z_-), choose $h \in C_0([0,\infty))$ with $h(0) = 1$ and use $G(r) = h(r)\,\mathrm{diag}(z_+, z_-)$. Its kernel consists exactly of the matrix functions vanishing at both 0 and infinity, namely $C_0((0,\infty), M_2)$. Moreover

$$HSH = \mathrm{diag}(1, -1),$$

so p_+ and p_- become e_{11} and e_{22}.

Take the two self-adjoint lifts

$$x_+(r) = \begin{pmatrix} (1+r)^{-1} & 0 \\ 0 & 0 \end{pmatrix}, \qquad x_-(r) = \begin{pmatrix} 0 & 0 \\ 0 & (1+r)^{-1} \end{pmatrix}.$$

The exponential boundary formula gives

$$\partial_\infty[p_\pm] = [\exp(2\pi i x_\pm)]$$

[85, Theorem 9.3.1 and Section 9.3.2]. In either case the determinant of the nontrivial block is

$$\exp\left(\frac{2\pi i}{1+r}\right) = \exp\left(-2\pi i \frac{r}{1+r}\right) = \beta(r)^{-1}.$$

It starts and ends at 1 and winds once clockwise, hence represents -1 in the declared generator. Additivity gives (680), including its displayed kernel. No sign has been normalized away. □

The boundary computation is therefore correct. The source's subsequent index comparison requires a separate domain audit.

Proposition 9.19 (The displayed differential expression does not determine an APS index)**.** *Let*

$$F_t = (1-2t)I_2, \qquad 0 \leq t \leq 1, \qquad \mathcal{D}_F = \frac{d}{dt} + F_t.$$

Then the following three exact realizations must not be conflated.

(a) Extend F continuously by $+I_2$ for $t \leq 0$ and by $-I_2$ for $t \geq 1$, and let

$$D_{\mathbb{R}} : W^{1,2}(\mathbb{R}, \mathbb{C}^2) \longrightarrow L^2(\mathbb{R}, \mathbb{C}^2), \qquad D_{\mathbb{R}} = \partial_t + F_t.$$

Then $\mathrm{Ind}(D_{\mathbb{R}}) = -2$.

(b) On the compact interval, with domain $H^1([0,1], \mathbb{C}^2)$ and no endpoint condition, $\mathrm{Ind}(\mathcal{D}_F) = +2$.

(c) On the compact interval, with domain $\{v \in H^1 : v(0) = v(1) = 0\}$, $\mathrm{Ind}(\mathcal{D}_F) = -2$.

Hence the local source's notation $\mathrm{Ind}_{\mathrm{APS}}(\mathcal{D}_F) = 2$ *is underdetermined until its endpoint projectors, operator domain, adjoint domain, and orientation convention are supplied. For realization (a), the exact relation is*

$$\partial_\infty([p_+] + [p_-]) = \mathrm{Ind}(D_{\mathbb{R}}) = -2,$$

whereas realization (b) gives $\partial_\infty([p_+] + [p_-]) = -\mathrm{Ind}(\mathcal{D}_F)$.

Proof. Doll, Schulz-Baldes, and Waterstraat count negative-to-positive crossings positively and prove, for

$$D_H = \partial_t - H_t : W^{1,2}(\mathbb{R}, \mathbb{C}^N) \to L^2(\mathbb{R}, \mathbb{C}^N),$$

that

$$\mathrm{Ind}(D_H) = -\mathrm{Sf}(H) \tag{682}$$

when the asymptotic limits are invertible [89, Definition 1.1.3 and Theorem 3.5.1, pp. 3 and 78–82]. Their complete proof identifies the kernel and cokernel through the stable and unstable spaces and obtains

$$\mathrm{Ind}(D_H) = \dim E^u(H_{-\infty}) - \dim E^u(H_{+\infty}).$$

Here $D_{\mathbb{R}} = D_H$ for $H = -F$. The path F goes from $+I_2$ to $-I_2$, so the retained signature formula gives

$$\mathrm{Sf}(F) = \frac{1}{2}((-2) - 2) = -2, \qquad \mathrm{Sf}(-F) = \frac{1}{2}(2 - (-2)) = 2.$$

Equation (682) therefore yields $\mathrm{Ind}(D_{\mathbb{R}}) = -2$.

On $[0, 1]$, every homogeneous solution is

$$v(t) = e^{-t+t^2} c, \qquad c \in \mathbb{C}^2.$$

With no endpoint condition this gives a two-dimensional kernel. Variation of constants with initial value $v(0) = 0$ solves $\mathcal{D}_F v = g$ for every $g \in L^2$, so the operator is onto and its index is 2.

With both endpoint values zero, the nonvanishing scalar e^{-t+t^2} forces $c = 0$, hence the kernel is zero. Integration by parts shows that the adjoint is $-\partial_t + F_t$ with no endpoint condition. Its homogeneous solutions are

$$w(t) = e^{t-t^2} d, \qquad d \in \mathbb{C}^2,$$

so the cokernel has dimension two and the index is -2. These two compact interval domains and the whole-line Fredholm domain are distinct data. Since the source supplies none of the boundary projections or domains needed to select an APS realization, its bare differential expression cannot select one of the resulting indices. □

Proposition 9.20 (The scalar three-component formula needs an additional codiagonal)**.** *Let the already proved finite-prime boundary be* $\partial_p(k) = -k$, *and let* ∂_∞ *be* (680). *The direct sum of the two defined extensions has connecting map*

$$\partial_p \oplus \partial_\infty : \mathbb{Z}^3 \longrightarrow \mathbb{Z}^2, \qquad (k, n_+, n_-) \longmapsto (-k, -n_+ - n_-).$$

The source's scalar formula is not this direct-sum boundary. It is the composite with the additional codiagonal

$$\Sigma : \mathbb{Z}^2 \to \mathbb{Z}, \qquad \Sigma(a, b) = a + b,$$

namely

$$(\Sigma \circ (\partial_p \oplus \partial_\infty))(k, n_+, n_-) = -(k + n_+ + n_-).$$

Its kernel has the explicit A_2 basis

$$b_1 = (1, -1, 0), \qquad b_2 = (0, 1, -1), \qquad (\langle b_i, b_j \rangle)_{i,j} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

By contrast, the direct-sum boundary has the rank-one kernel $\mathbb{Z}(0, 1, -1)$. Thus the A_2 calculation is a correct calculation for the displayed scalar composite. At this stage of the source audit it is not yet the boundary map of a defined multi-place C^*-extension: that admission requires an algebraic construction inducing Σ, and the audited source supplies no such map. The independent stable Busby construction in Theorem 9.23 below closes this construction obligation after stabilization; it does not retroactively attribute the map to the local source or identify the source's undefined unstabilized global algebra.

Proof. The direct-sum formula follows component by component from the two proved connecting maps. Applying Σ gives the displayed row vector $-(1, 1, 1)$. Both b_1 and b_2 lie in its kernel, they are independent, and their dot products give the displayed Cartan matrix. The direct-sum map has two independent rows, so its kernel is exactly $\mathbb{Z}(0, 1, -1)$. This proves both kernel statements and isolates the additional information carried by Σ. $\square$

Executable archimedean check. The bounded replay is

`certificates/globalization_nte_archimedean_boundary/verify_archimedean_boundary.py`.

It checks the Hadamard coordinate change, fixed-point boundary form, character projections, explicit lifts and winding, both spectral-flow signs, the three domain-sensitive index counts, the direct-sum and codiagonal matrices, and the A_2 Gram matrix. It does not certify the crossed-product completion, the six-term theorem, Fredholmness, or an APS boundary theorem; the exact standard proofs and source dependencies are printed above.

9.11 Global place balancing: an exact lattice and a stable extension theorem

The preceding local calculations determine all coefficients that occur in the source's scalar component formula, but they do not yet make that formula a connecting map of one global extension. This distinction produces a concrete construction problem rather than a slogan.

Fix an ordered list of distinct primes

$$P = (p_1, \ldots, p_n), \qquad n \geq 0,$$

and retain the two archimedean character coordinates p_+ and p_-. In the ordered coordinate basis

$$([1]_{p_1}, \ldots, [1]_{p_n}, [p_+], [p_-])$$

define the already determined direct-sum homomorphism

$$\Delta_P : \mathbb{Z}^{n+2} \longrightarrow \mathbb{Z}^{n+1}, \qquad \Delta_P(k_1, \ldots, k_n, n_+, n_-) = (-k_1, \ldots, -k_n, -n_+ - n_-). \tag{683}$$

Its first n coordinates are the finite-prime coefficients proved in Theorem 9.14; its last coordinate is the archimedean map proved in Theorem 9.18. No place, sign, or character coordinate has been identified in writing (683).

Let

$$\Sigma_P : \mathbb{Z}^{n+1} \longrightarrow \mathbb{Z}, \qquad \Sigma_P(y_1, \ldots, y_n, y_\infty) = \sum_{i=1}^{n} y_i + y_\infty \tag{684}$$

and put

$$\delta_P^{\mathrm{bal}} = \Sigma_P \circ \Delta_P, \qquad \delta_P^{\mathrm{bal}}(k_1,\dots,k_n,n_+,n_-) = -\left(\sum_{i=1}^{n} k_i + n_+ + n_-\right). \tag{685}$$

At this point δ_P^{bal} is a completely defined homomorphism of integer lattices. It is not yet declared to be the boundary map of a global C^*-extension.

Theorem 9.21 (Exact balancing lattice)**.** *Write $m = n+2$ and let $e_1,\dots,e_m$ be the retained standard basis of $\mathbb{Z}^m$. Then δ_P^{bal} is onto and*

$$\ker \delta_P^{\mathrm{bal}} = \left\{ x \in \mathbb{Z}^m : \sum_{j=1}^{m} x_j = 0 \right\}.$$

This kernel has the ordered basis

$$\alpha_i = e_i - e_{i+1}, \qquad 1 \le i \le m-1,$$

and the coefficient inverse is explicit:

$$x = \sum_{i=1}^{m-1} c_i \alpha_i, \qquad c_i = \sum_{j=1}^{i} x_j. \tag{686}$$

For the standard Euclidean pairing,

$$\langle \alpha_i, \alpha_j \rangle = \begin{cases} 2, & i = j, \\ -1, & |i-j| = 1, \\ 0, & |i-j| > 1. \end{cases}$$

Thus the balancing kernel, with all coordinates retained, is the root lattice $A_{m-1} = A_{n+1}$. More generally, every integer fibre is the affine lattice

$$(\delta_P^{\mathrm{bal}})^{-1}(r) = -re_m + \ker \delta_P^{\mathrm{bal}}.$$

Proof. The formula in (685) shows that the kernel is the displayed sum-zero lattice. It is onto because $\delta_P^{\mathrm{bal}}(-re_m) = r$ for every $r \in \mathbb{Z}$.

Each α_i has coordinate sum zero. Conversely, let $x = (x_1,\dots,x_m)$ have sum zero and define the partial sums c_i by (686). The first coordinate of $\sum_i c_i\alpha_i$ is $c_1 = x_1$; for $2 \le j \le m-1$ its jth coordinate is

$$c_j - c_{j-1} = x_j;$$

and its last coordinate is

$$-c_{m-1} = -\sum_{j=1}^{m-1} x_j = x_m.$$

This also proves uniqueness of the coefficients, because the first coordinate recovers c_1 and successive coordinate differences recover every c_i. Direct multiplication of the coordinate vectors gives the displayed Cartan Gram matrix. Finally, adding re_m to an element of the fibre over r puts it in the kernel, proving the exact affine-fibre formula. □

For one finite prime, the theorem gives

$$\ker\bigl(-(1,1,1):\mathbb{Z}^3\to\mathbb{Z}\bigr)=\mathbb{Z}(1,-1,0)\oplus\mathbb{Z}(0,1,-1)\cong A_2.$$

This was initially the conjectural stopping point. The missing codiagonal can, however, be constructed at the level of stable C^*-extensions. We give the construction rather than infer it from the displayed integer row.

Put

$$\mathcal{K}=\mathcal{K}\bigl(\ell^2(\mathbb{N}_0)\bigr),\qquad J=C_0(\mathbb{R})\otimes\mathcal{K},$$

and let $e_{00}\in\mathcal{K}$ be the projection onto the first basis vector. If

$$c(t)=\frac{t-i}{t+i},$$

then the declared positive generator of $K_1(J)$ is represented in the unitization by

$$\gamma(t)=1+(c(t)-1)e_{00}. \tag{687}$$

Lemma 9.22 (A common oriented stabilization)**.** *For every finite prime p and for the archimedean place ∞, the proved local extension admits a stabilized transport*

$$0\longrightarrow J\longrightarrow\widehat{E}_v\longrightarrow\widehat{A}_v\longrightarrow 0,\qquad v\in\{p,\infty\}, \tag{688}$$

and a morphism from the original local extension to (688)*. The quotient algebras are*

$$\widehat{A}_p=Q_p\otimes\mathcal{K},\qquad \widehat{A}_\infty=D_2\otimes\mathcal{K}.$$

Under the corner maps on the quotients and the generator (687)*, their connecting maps remain*

$$[1]_p\longmapsto-[\gamma],\qquad (a[p_+]+b[p_-])\longmapsto-(a+b)[\gamma].$$

Every stabilization map, generator, and possible orientation reversal is retained in this assertion.

Proof. For a finite prime, the extension-level Green–Rieffel module in the preceding subsection gives $I_p\sim_M C_0(\mathbb{R})$. Both algebras are separable and therefore have countable approximate identities. The Brown–Green–Rieffel stabilization theorem gives

$$I_p\otimes\mathcal{K}\cong C_0(\mathbb{R})\otimes\mathcal{K}=J$$

[86, Theorems 1.1–1.2, pp. 349–352]; equivalently, it follows by taking compact operators after Kasparov stabilization [85, Theorem 13.6.2 and Exercise 13.7.1, pp. 131–132]. At the archimedean place, the ideal is $C_0((0,\infty),M_2)$; the logarithm $(0,\infty)\to\mathbb{R}$ and a fixed unitary $\mathbb{C}^2\otimes\ell^2(\mathbb{N}_0)\cong\ell^2(\mathbb{N}_0)$ give the required stable isomorphism explicitly.

Write B_v for the original local ideal and choose an initial isomorphism $\theta_v^0:B_v\otimes\mathcal{K}\to J$. Its map on the already declared K_1 coordinate is multiplication by a recorded sign $\varepsilon_v\in\{1,-1\}$. If $\varepsilon_v=-1$, compose θ_v^0 with the explicit reflection automorphism

$$\rho:J\longrightarrow J,\qquad(\rho f)(t)=f(-t),$$

which takes $[\gamma]$ to $-[\gamma]$; if $\varepsilon_v=1$, leave it unchanged. Denote the resulting orientation-preserving isomorphism by θ_v. This records the sign correction instead of silently discarding it.

Tensor the original exact sequence with the nuclear algebra $\mathcal{K}$. Let τ_v^{s} be its Busby invariant. The isomorphism θ_v extends to multiplier algebras and induces $\dot{\theta}_v : Q(B_v \otimes \mathcal{K}) \to Q(J)$. Define

$$\widehat{\tau}_v = \dot{\theta}_v \circ \tau_v^{\mathrm{s}} : \widehat{A}_v \longrightarrow Q(J)$$

and let $\widehat{E}_v$ be its Busby pullback. The pullback reconstruction and the induced extension morphism are the constructions of Sections 15.2–15.3 of Blackadar [85, pp. 144–145]. Composing with the original corner map $x \mapsto x \otimes e_{00}$ gives the asserted morphism from the unstabilized extension. Stability sends each displayed quotient generator to its corresponding rank-one corner, while the declared θ_v sends the ideal generator to $[\gamma]$; hence the two local boundary formulas, including their minus signs, are unchanged. □

Let $\mathcal{V} = \{\infty\} \cup \{\text{rational primes}\}$ and fix once and for all a bijection $\nu : \mathcal{V} \to \mathbb{N}_0$. For

$$\kappa(j,k) = \frac{(j+k)(j+k+1)}{2} + k$$

define an isometry S_j on $\ell^2(\mathbb{N}_0)$ by $S_j e_k = e_{\kappa(j,k)}$. Since κ is a bijection $\mathbb{N}_0^2 \to \mathbb{N}_0$,

$$S_j^* S_l = \delta_{jl} 1, \qquad \sum_{j=0}^{\infty} S_j S_j^* = 1 \quad \text{strongly.} \tag{689}$$

Write $s_v = 1 \otimes S_{\nu(v)} \in M(J)$.

Theorem 9.23 (Stable place-balancing extension)**.** *For every finite prime list $P = (p_1, \ldots, p_n)$ there is an explicitly constructed extension*

$$0 \longrightarrow J \longrightarrow \mathcal{E}_P \longrightarrow \mathcal{A}_P \longrightarrow 0, \qquad \mathcal{A}_P = \left(\bigoplus_{i=1}^{n} \widehat{A}_{p_i} \right) \oplus \widehat{A}_\infty, \tag{690}$$

with a morphism from the direct sum of the original local extensions after the declared corner stabilizations. It has all of the following exact properties.

1. *Its quotient map retains the ordered classes*

$$([1]_{p_1}, \ldots, [1]_{p_n}, [p_+], [p_-])$$

without merging primes or the two archimedean characters.

2. *The ideal map from the direct sum of the stabilized local ideals induces exactly the codiagonal Σ_P of* (684) *on K_1.*

3. *The connecting map of* (690) *is*

$$\partial_P(k_1, \ldots, k_n, n_+, n_-) = -\left(\sum_{i=1}^{n} k_i + n_+ + n_- \right),$$

with the displayed sign and all source coordinates retained. Hence its kernel is the A_{n+1} lattice of Theorem 9.21.

4. *Reordering P merely permutes the named quotient summands. If $P \subset P'$, zero-inclusion of the new quotient coordinates lifts to a morphism from the P extension to the P' extension which is the identity on J. Thus adjoining a prime and removing it by restriction to the zero coordinate are compatible at extension level, not only on K-groups.*

Proof. Put $V_P = \{p_1, \dots, p_n, \infty\}$ and form the finite direct sums

$$B_P = \bigoplus_{v \in V_P} J, \qquad \widehat{E}_P^{\oplus} = \bigoplus_{v \in V_P} \widehat{E}_v, \qquad \mathcal{A}_P = \bigoplus_{v \in V_P} \widehat{A}_v.$$

For $(b_v)_{v \in V_P} \in B_P$, define

$$\mu_P((b_v)_v) = \sum_{v \in V_P} s_v b_v s_v^* \in J. \tag{691}$$

The orthogonality in (689) gives, coordinate by coordinate,

$$\mu_P(b)\mu_P(d) = \sum_{v,w} s_v b_v s_v^* s_w d_w s_w^* = \sum_v s_v b_v d_v s_v^* = \mu_P(bd),$$

and $s_v^* \mu_P(b) s_v = b_v$. Thus μ_P is an injective $*$-homomorphism. The same finite formula defines

$$\overline{\mu}_P : M(B_P) = \bigoplus_{v \in V_P} M(J) \longrightarrow M(J).$$

It maps B_P into J and therefore induces a corona homomorphism $\dot{\mu}_P : Q(B_P) \to Q(J)$.

Let

$$\widehat{\tau}_P^{\oplus} = \bigoplus_{v \in V_P} \widehat{\tau}_v : \mathcal{A}_P \longrightarrow Q(B_P)$$

be the Busby invariant of the direct-sum extension and set

$$\tau_P = \dot{\mu}_P \circ \widehat{\tau}_P^{\oplus}. \tag{692}$$

The required middle algebra is the explicit pullback

$$\mathcal{E}_P = \{(a, m) \in \mathcal{A}_P \oplus M(J) : \tau_P(a) = \pi_J(m)\}, \tag{693}$$

where $\pi_J : M(J) \to Q(J)$ is the quotient map. Projection onto a is onto, and its kernel is exactly $\{(0, j) : j \in J\}$, proving (690). In the Busby pullback models for the local extensions, the middle map is

$$((a_v, m_v))_{v \in V_P} \longmapsto \left((a_v)_{v \in V_P}, \sum_{v \in V_P} s_v m_v s_v^* \right). \tag{694}$$

Together with μ_P on ideals and the identity on $\mathcal{A}_P$, this is a morphism of short exact sequences. It is an extension-level construction before any K-group calculation. It is the finite orthogonal-corner form of the standard Busby sum for stable ideals [85, Sections 15.6 and 15.9, pp. 149–150, 156].

It remains to compute the ideal map without appealing to its intended answer. The vth copy of (687) is sent to

$$1 + (c(t) - 1)q_v, \qquad q_v = S_{\nu(v)} e_{00} S_{\nu(v)}^*.$$

The projection q_v is rank one. A rotation in the at most two-dimensional span of e_0 and $S_{\nu(v)} e_0$ gives an explicit norm-continuous path of rank-one projections from e_{00} to q_v. Substituting that path for e_{00} in (687) proves that the vth corner map takes $[\gamma]$ to $[\gamma]$. Since $K_1(B_P) = \bigoplus_{v \in V_P} \mathbb{Z}[\gamma]$, additivity gives

$$(\mu_P)_*((y_v)_v) = \left(\sum_{v \in V_P} y_v \right) [\gamma].$$

This is exactly Σ_P, not an unidentified isomorphism.

The top extension in the morphism diagram has boundary Δ_P from (683). The connecting maps in K-theory are natural for morphisms of short exact sequences [85, Definition 21.1.1 and Example 21.1.2(a), pp. 266, 268]. Consequently

$$\partial_P = (\mu_P)_* \circ \Delta_P = \Sigma_P \circ \Delta_P = \delta_P^{\mathrm{bal}},$$

which proves the first three assertions and invokes the already proved exact lattice theorem for the kernel.

Finally, the operators s_v and the transports θ_v were fixed by the place v, not by its position in P. Reordering therefore only permutes the direct-sum notation. If $P \subset P'$, let $\iota_{P,P'} : \mathcal{A}_P \to \mathcal{A}_{P'}$ insert zero in the new place coordinates. The literal formulas give

$$\tau_{P'} \circ \iota_{P,P'} = \tau_P.$$

Hence $(a, m) \mapsto (\iota_{P,P'}(a), m)$ maps $\mathcal{E}_P$ to $\mathcal{E}_{P'}$, is the identity on J, and makes the extension diagram commute. Restriction back to the zero-coordinate summand recovers the P extension. This proves the fourth assertion. $\square$

Passage to the natural arithmetic representative. The extension-existence, codiagonal, sign, kernel, permutation, and finite-set compatibility assertions are therefore theorems after stabilization. The audited source itself does not define its larger symbol $A_{\varnothing p}$ or a simultaneous multi-place action. Subsection 9.12 therefore constructs the natural unit-retaining semilocal crossed product independently and proves its exact K-theory comparison with this stable scalar extension. Subsection 9.13 then constructs the formerly missing extension-level $*$-homomorphism by an exact multiplier/corona Busby equation. The natural boundary still collapses the two archimedean character classes in K_0, and the natural ideal still contains nonconstant unit-fibre classes; the new theorem records these as the kernel and image data of an injective stable extension morphism. No endpoint statement about zeta zeros follows from any of these extension theorems.

Literature and novelty boundary. The proof uses the content-read stable-isomorphism theorem of Brown, Green, and Rieffel. It also uses Blackadar's Busby pullback, stable extension sum and functoriality, and the naturality of K-theory connecting maps. A bounded canonical-index query for these exact dependencies routed the cited sources. The earlier bounded query for the combined terms "codiagonal", "A_2", "boundary", and "K-theory", followed by a phrase-level arXiv check, did not locate this place-indexed diagram. That search scope supports no general novelty claim. The theorem above is the explicit stable construction. The next two subsections supply the independently defined arithmetic target and the exact stable-to-natural extension morphism. The bounded search scope supports no general novelty claim.

Formal finite-coordinate replay. The file

`formal/lean/ZetaFunctionFoundation/GlobalPlaceBalancing.lean`

certifies the direct-sum boundary, codiagonal composite, scalar-kernel equation, surjectivity, and the naturality-implied formula for an arbitrary finite number of prime coordinates. Receipt `formal/lean/receipts/FORMAL-20260830-0036.receipt.json` records Lean 4.32.0, Mathlib 4.32.0, exit code zero, and a peak aggregate working set below the authorized five-gigabyte ceiling. The Busby invariant, multiplier-algebra construction, and C^*-algebraic naturality theorem remain certified by the complete standard proof above and its human literature dependencies; this Lean file does not pretend to formalize those analytic objects.

9.12 The unstabilized semilocal one-zero algebra: explicit coordinates, boundary maps, and information loss

The audited source introduces $A_{\varnothing p}$ only as an "intermediate semilocal crossed-product algebra." It does not specify its space, action, completion, inclusion, quotient, or imprimitivity module. The two-place construction of Connes, Consani, and Marcolli does, however, supply the exact seed: $\mathbb{Q}_p \times \mathbb{R}$, acted on diagonally by $\{\pm p^n : n \in \mathbb{Z}\}$, with generic and one-zero strata [84, Section 8.2]. We now perform the finite-place construction directly. This gives a natural meaning to the missing symbol; it is not an assertion that this meaning was already present in the local source or is the unique possible enrichment of that symbol.

Let P be a nonempty finite set of rational primes and put

$$S_P = P \sqcup \{\infty\}, \qquad \mathbb{A}_{S_P} = \left(\prod_{q \in P} \mathbb{Q}_q \right) \times \mathbb{R},$$

$$\Gamma_P = \mathbb{Z}[1/\prod_{q \in P} q]^\times = \left\{ \varepsilon \prod_{q \in P} q^{n_q} : \varepsilon \in \{\pm 1\},\ n_q \in \mathbb{Z} \right\}.$$

The group $\Gamma_P \cong (\mathbb{Z}/2) \times \mathbb{Z}^P$ acts by diagonal multiplication. For $x = ((x_q)_{q \in P}, y) \in \mathbb{A}_{S_P}$ define its literal zero set

$$Z(x) = \{q \in P : x_q = 0\} \cup \begin{cases} \{\infty\}, & y = 0, \\ \varnothing, & y \neq 0. \end{cases}$$

No support coordinate is identified with another one in this definition. Set

$$W_P^{(1)} = \{x : \#Z(x) \le 1\}, \qquad W_\varnothing = \{x : Z(x) = \varnothing\}, \qquad W_v = \{x : Z(x) = \{v\}\}.$$

The complement of $W_P^{(1)}$ is a finite union of closed coordinate intersections, so $W_P^{(1)}$ is an invariant locally compact open subspace.

Definition 9.24 (Natural unstabilized one-zero algebra)**.** All crossed products in this subsection are full crossed products. Define

$$\begin{aligned} \mathfrak{A}_P^{(1)} &= C_0(W_P^{(1)}) \rtimes \Gamma_P, \\ \mathfrak{I}_P &= C_0(W_\varnothing) \rtimes \Gamma_P, \\ \mathfrak{D}_{v,P} &= C_0(W_v) \rtimes \Gamma_P, \qquad \mathfrak{D}_P = \bigoplus_{v \in S_P} \mathfrak{D}_{v,P}. \end{aligned}$$

For $p \in P$, the source symbol receives the explicit candidate

$$A_{\varnothing p}^{(P)} := C_0(W_\varnothing \cup W_p) \rtimes \Gamma_P,$$

with $A_\varnothing^{(P)} = \mathfrak{I}_P$ and $A_p^{(P)} = \mathfrak{D}_{p,P}$.

Theorem 9.25 (The literal one-zero extension)**.** *Restriction to the closed one-zero boundary gives the short exact sequence*

$$0 \longrightarrow \mathfrak{I}_P \longrightarrow \mathfrak{A}_P^{(1)} \longrightarrow \mathfrak{D}_P \longrightarrow 0. \tag{695}$$

For every $v \in S_P$, the invariant open subspace $W_\varnothing \cup W_v$ gives a morphism from the vth edge extension to (695)*, equal to the identity on $\mathfrak{I}_P$ and to the summand inclusion on the quotient. Full and reduced versions of all these algebras agree.*

Proof. Inside $W_P^{(1)}$, the generic stratum is open and its complement is the finite disjoint union $\bigsqcup_{v\in S_P} W_v$. Restriction of functions therefore gives

$$0 \to C_0(W_\varnothing) \to C_0(W_P^{(1)}) \to \bigoplus_{v\in S_P} C_0(W_v) \to 0.$$

The maps are Γ_P-equivariant, and the full crossed-product functor is exact. This proves (695). Removing all boundary strata except W_v leaves an invariant open subspace; extension by zero gives the asserted middle and quotient maps and is literally the identity on the common generic ideal. Finally, Γ_P is abelian-by-finite and hence amenable. Exel's approximation and regular-representation theorems identify the full and reduced cross-sectional algebras in this case [90, Theorems 20.6–20.7]. $\square$

We next retain every unit coordinate and calculate a finite-prime edge. Put

$$K_P = \prod_{q\in P} \mathbb{Z}_q^\times, \qquad \ell_q = \log q.$$

For $p \in P$, let $K_{P\setminus p} = \prod_{q\in P\setminus\{p\}} \mathbb{Z}_q^\times$, with the empty product understood as one point, and define

$$\beta_p((b_q)_{q\neq p}, s) = ((pb_q)_{q\neq p}, s + \ell_p).$$

The finite-prime boundary carrier is the compact mapping torus

$$B_{p,P} = (K_{P\setminus p} \times \mathbb{R})/\langle \beta_p \rangle. \tag{696}$$

Theorem 9.26 (Complete coordinates for the P-semilocal p-edge)**.** *The action of Γ_P on $W_\varnothing \cup W_p$ is free and proper. As a set, its orbit space is*

$$\mathcal{S}_{P,p} = (K_P \times \mathbb{R}) \sqcup B_{p,P}. \tag{697}$$

The symbol $\sqcup$ here records the two strata, not the topological-sum topology; the cross-stratum convergence is specified below. For a generic point, write

$$m_q = v_q(x_q), \qquad u_q = q^{-m_q} x_q \in \mathbb{Z}_q^\times, \qquad \epsilon = \operatorname{sgn}(y).$$

Its orbit coordinates are, without suppressing any cross-prime factor,

$$a_q = \epsilon \prod_{r\in P\setminus\{q\}} r^{-m_r} u_q, \qquad t = \log|y| - \sum_{r\in P} m_r \ell_r. \tag{698}$$

For a point of W_p, define for $q \neq p$

$$b_q = \epsilon \prod_{r\in P\setminus\{p,q\}} r^{-m_r} u_q, \qquad s = \log|y| - \sum_{r\neq p} m_r \ell_r; \tag{699}$$

its orbit coordinate is $[(b_q)_{q\neq p}, s] \in B_{p,P}$.

A generic net $(a^{(j)}, t_j)$ converges to $[b, s] \in B_{p,P}$ exactly when

$$t_j \longrightarrow -\infty, \qquad [(a_q^{(j)})_{q\neq p}, t_j] \longrightarrow [b, s] \quad \text{in } B_{p,P}; \tag{700}$$

there is no condition on the discarded coordinate $a_p^{(j)} \in \mathbb{Z}_p^\times$.

Proof. Let $\gamma = \delta \prod_{r \in P} r^{n_r} \in \Gamma_P$. On the generic stratum,

$$m'_q = m_q + n_q, \qquad u'_q = \delta \prod_{r \neq q} r^{n_r} u_q, \qquad \epsilon' = \delta\epsilon.$$

Substitution into (698) cancels every n_r and both factors δ; the logarithmic coordinate is invariant because $\log|\gamma| = \sum_r n_r \ell_r$. Conversely, multiplication by the unique element

$$\epsilon \prod_{r \in P} r^{-m_r}$$

makes y positive and every finite valuation zero. Hence (a,t) is a complete generic orbit coordinate, not only an invariant.

On W_p, the same normalization is available for the sign and for all valuations m_q with $q \neq p$. The unused power p^k acts on the resulting coordinates by β_p^k, which proves the boundary formula and its uniqueness modulo (696).

For the topology, let a generic representative have p-valuation m_p. Equations (698) and (699) give the literal relations

$$b_q = p^{m_p} a_q \quad (q \neq p), \qquad s = t + m_p \ell_p. \tag{701}$$

Thus $[(b_q), s] = [(a_q)_{q \neq p}, t]$ in $B_{p,P}$. Convergence of the p-adic coordinate to zero is precisely $m_p \to +\infty$; if a boundary representative remains in a compact set, the second identity forces $t \to -\infty$ and gives (700). Conversely, use a local section of the proper $\mathbb{Z}$-quotient defining $B_{p,P}$ to choose integers m_j such that

$$\beta_p^{m_j}((a_q^{(j)})_{q \neq p}, t_j) \longrightarrow (b, s).$$

Because $t_j \to -\infty$ while the lifted real coordinates converge to s, necessarily $m_j \to +\infty$. Take all other valuations zero, set $u_q^{(j)} = p^{m_j} a_q^{(j)}$ for $q \neq p$, $x_p^{(j)} = p^{m_j} a_p^{(j)}$, and $y_j = \exp(t_j + m_j \ell_p)$. These are valid unit coordinates and converge to the required boundary representative. This proves both directions and the absence of a condition on $a_p^{(j)}$.

The action is free because $y \neq 0$. For properness, let C, D be compact subsets of the edge and suppose $\gamma C \cap D \neq \varnothing$. Every $q \neq p$ coordinate is nonzero throughout the edge, so its valuations on C and D bound n_q. The real coordinate is also nonzero, so its absolute values on C, D bound $\sum_q n_q \ell_q$. The already bounded n_q for $q \neq p$ then bound n_p, and the sign has only two values. Only finitely many γ occur, proving properness. $\square$

There is now an exact morphism to the one-ended local model, rather than a Morita identification that erases the unit carrier. Define

$$\rho_{p,P} : B_{p,P} \to S^1_{\ell_p}, \qquad \rho_{p,P}([(b_q), s]) = [s]_{\ell_p},$$

and

$$\Pi_{P,p} : \mathcal{S}_{P,p} \to X_{\ell_p}, \qquad \Pi_{P,p}(a,t) = t, \quad \Pi_{P,p}([b,s]) = [s]_{\ell_p}. \tag{702}$$

The mapping-torus relation changes s by an integral multiple of ℓ_p, so $\rho_{p,P}$ is well defined. The compact group K_P acts by

$$k \cdot (a,t) = (ka, t), \qquad k \cdot [b,s] = [(k_q b_q)_{q \neq p}, s].$$

The second formula is well defined because coordinatewise multiplication by k commutes with β_p; the factor k_p acts trivially on the boundary. The convergence criterion proves that this action is continuous and that $\Pi_{P,p}$ is exactly its orbit map. Hence $\Pi_{P,p}$ is closed and proper. Its generic fibres are K_P; its boundary fibres are copies of $K_{P\setminus p}$. Pullback therefore gives

$$\begin{array}{ccccccccc} 0 & \to & C_0(\mathbb{R}) & \to & C_0(X_{\ell_p}) & \to & C(S^1_{\ell_p}) & \to & 0 \\ & & \downarrow\pi_P^* & & \downarrow\Pi^*_{P,p} & & \downarrow\rho^*_{p,P} & & \\ 0 & \to & C_0(K_P\times\mathbb{R}) & \to & C_0(\mathcal{S}_{P,p}) & \to & C(B_{p,P}) & \to & 0, \end{array} \tag{703}$$

where $\pi_P(a,t)=t$. Green imprimitivity, applied with the restricted modules exactly as in the two-place proof, identifies the lower row with the crossed-product edge extension

$$0\to\mathfrak{I}_P\to A^{(P)}_{\varnothing p}\to\mathfrak{D}_{p,P}\to 0$$

at the Morita level [87, Theorem 1.3.22].

Corollary 9.27 (Finite-place unit class with every unit fibre retained)**.** *Under*

$$K_1(\mathfrak{I}_P)\cong K_1(C_0(K_P\times\mathbb{R}))\cong K_0(C(K_P))\cong C(K_P,\mathbb{Z}),$$

the connecting map for the p-edge sends the Green image $\eta_{p,P}$ of $[1_{B_{p,P}}]$ to

$$\partial_{p,P}(\eta_{p,P})=-\mathbf{1}_{K_P}. \tag{704}$$

Proof. The right vertical arrow of (703) is unital, so it sends the local circle unit to the mapping-torus unit. Theorem 9.14 gives -1 for the top boundary map. Naturality sends it through π_P^*, which is the constant-function inclusion $\mathbb{Z}\to C(K_P,\mathbb{Z})$. This proves (704). The last K_0 identification retains all locally constant integer-valued rank functions on the profinite space K_P; it is not a normalization to one integer. □

The archimedean edge contains a different, equally explicit, information-loss mechanism. Normalize all finite valuations but do not choose a real sign. If $x_q=q^{m_q}u_q$, put

$$c_q=\prod_{r\in P\setminus\{q\}} r^{-m_r}u_q,\qquad z=y\prod_{r\in P} r^{-m_r}. \tag{705}$$

The residual sign acts by $(c,z)\mapsto(-c,-z)$.

Theorem 9.28 (Archimedean edge and collapse of the two character labels)**.** *The action of Γ_P on $W_\varnothing\cup W_\infty$ is free and proper. After Morita reduction of the free valuation directions, its extension is*

$$0\ \to\ C_0(K_P\times\mathbb{R}^\times)\rtimes C_2\ \to\ C_0(K_P\times\mathbb{R})\rtimes C_2\ \to\ C(K_P)\rtimes C_2\ \to\ 0, \tag{706}$$

where C_2 acts diagonally by $(c,z)\mapsto(-c,-z)$ and by $c\mapsto -c$ on the boundary. Constant functions in K_P give a morphism from the local reflection extension (679) *to* (706)*. If*

$$e_\pm=\frac{1\pm U}{2}\in C(K_P)\rtimes C_2,$$

then

$$\partial_{\infty,P}[e_+]=\partial_{\infty,P}[e_-]=-\mathbf{1}_{K_P}\in C(K_P,\mathbb{Z}). \tag{707}$$

Moreover $[e_+]=[e_-]$ in $K_0(C(K_P)\rtimes C_2)$.

Proof. Formula (705) is invariant under every prime-power direction, and every orbit has a unique representative with all finite valuations zero, up to the residual sign. This gives (706). Freeness follows because all finite coordinates are nonzero on this edge. Properness follows because their valuations on two compact sets bound every prime exponent; the residual sign is finite.

The projection $K_P \times \mathbb{R} \to \mathbb{R}$ is equivariant for the diagonal and reflection actions. Pullback, followed by crossed product, is therefore the promised extension morphism. It sends the local projections $p_\pm$ to $e_\pm$ and sends the local ideal generator to the constant rank function. Equation (707) follows from Theorem 9.18 by naturality.

It remains to prove, rather than assume, the equality of the two boundary classes. Choose one $q_0 \in P$. If q_0 is odd, choose one representative from every pair $\{r, -r\}$ in $(\mathbb{Z}/q_0\mathbb{Z})^\times$. If $q_0 = 2$, choose the residue class 1 (mod 4); its negative is the class 3 (mod 4). The inverse image of the chosen classes defines a clopen set $H \subset K_P$ with

$$K_P = H \sqcup (-H).$$

Thus the principal double cover $K_P \to K_P/\{\pm 1\}$ has the explicit continuous section whose image is H. Consequently

$$C(K_P) \rtimes C_2 \cong M_2(C(H)),$$

and e_+ and e_- are rank-one projections in the same 2×2 matrix algebra. In the standard two-sheet matrix coordinates the constant unitary $\operatorname{diag}(1, -1)$ conjugates e_+ to e_-, proving $[e_+] = [e_-]$. This is the exact point at which the natural semilocal carrier forgets the local sign-character difference. $\square$

We can now compute the natural multi-boundary map without pretending that the whole action is proper. Let

$$\Lambda_P = \bigoplus_{p \in P} \mathbb{Z}\epsilon_p \oplus \mathbb{Z}\epsilon_+ \oplus \mathbb{Z}\epsilon_-.$$

Define the boundary-label morphism

$$R_P : \Lambda_P \longrightarrow K_0(\mathfrak{D}_P), \qquad R_P(\epsilon_p) = \eta_{p,P}, \quad R_P(\epsilon_\pm) = [e_\pm], \tag{708}$$

where the last two classes are transported through the valuation Morita module. Also define the exact constant-coordinate morphism

$$c_P : \mathbb{Z} \longrightarrow C(K_P, \mathbb{Z}), \qquad c_P(n) = n\mathbf{1}_{K_P}.$$

Theorem 9.29 (Unstabilized arithmetic realization of the codiagonal shadow)**.** *For the connecting map of* (695),

$$\partial_P^{(1)} R_P((k_p)_{p\in P}, n_+, n_-) = -\left(\sum_{p \in P} k_p + n_+ + n_- \right) \mathbf{1}_{K_P}. \tag{709}$$

Equivalently, the square

$$\begin{array}{ccc} \Lambda_P & \xrightarrow{\delta_P^{\mathrm{bal}}} & \mathbb{Z} \\ {\scriptstyle R_P}\big\downarrow & & \big\downarrow{\scriptstyle c_P} \\ K_0(\mathfrak{D}_P) & \xrightarrow{\partial_P^{(1)}} & K_1(\mathfrak{I}_P) \end{array} \tag{710}$$

commutes under the displayed Morita and suspension coordinates.

The morphism R_P does not preserve all formal labels injectively:

$$\ker R_P = \mathbb{Z}(\epsilon_+ - \epsilon_-). \tag{711}$$

Hence the formal scalar kernel $A_{|P|+1}$ maps onto the actual kernel on the distinguished boundary-unit span, with exact kernel $\mathbb{Z}(\epsilon_+ - \epsilon_-)$. The latter actual kernel has rank $|P|$ and an explicit $A_{|P|}$ basis after replacing the two archimedean labels by their common image.

Proof. For each v, the edge algebra is an invariant open ideal in $\mathfrak{A}_P^{(1)}$, and Theorem 9.25 gives a morphism of its extension into the global one-zero extension. The connecting maps are natural [85, Definition 21.1.1 and Example 21.1.2(a)]. Therefore Corollary 9.27 supplies the term $-k_p\mathbf{1}_{K_P}$ for every finite p, while Theorem 9.28 supplies $-n_+\mathbf{1}_{K_P}$ and $-n_-\mathbf{1}_{K_P}$. Additivity proves (709) and the square.

The quotient $\mathfrak{D}_P$ is a direct sum over the distinct boundary strata. Each finite unit class has nonzero rank in its own summand and is therefore independent of all other finite and archimedean summands. In the archimedean summand, the preceding theorem proves that the two classes agree; their common rank is one, so this is their only relation. This proves (711). The formal kernel of δ_P^{bal} is the already proved root lattice $A_{|P|+1}$. Quotienting by the displayed simple difference identifies the two archimedean coordinates and leaves

$$\left\{((k_p)_{p\in P}, n) \in \mathbb{Z}^P \oplus \mathbb{Z} : \sum_p k_p + n = 0\right\},$$

whose consecutive-difference basis is $A_{|P|}$. No lattice coordinate has been silently deleted: the lost coordinate and the map that loses it are both displayed. □

There is a final analytic distinction. Although every individual edge action is free and proper, the simultaneous action on $W_P^{(1)}$ is not proper.

Proposition 9.30 (Explicit obstruction to a global Green reduction)**.** *For every nonempty P, the action of Γ_P on $W_P^{(1)}$ is not proper. Consequently one may not replace* (695) *by the commutative orbit-space extension using Green imprimitivity. This does not affect its crossed-product exactness or the edgewise boundary calculation above.*

Proof. Fix $p \in P$. Choose integers $n_j \to \infty$ along a subsequence for which the tuple $(p^{n_j})_{q\in P\setminus\{p\}}$ converges in the compact group $K_{P\setminus p}$. Put

$$z_j = ((x_{q,j})_{q\in P}, y_j), \qquad x_{p,j} = p^{n_j}, \quad x_{q,j} = p^{n_j}\ (q \neq p), \quad y_j = 1.$$

Then z_j converges to a point of the p-boundary. For $\gamma_j = p^{-n_j}$,

$$\gamma_j z_j = ((1)_{q\in P}, p^{-n_j}),$$

which converges to a point of the archimedean boundary. The union of both sequences and their limits is compact in $W_P^{(1)}$, while the distinct γ_j escape every finite subset of the discrete group Γ_P. Thus the inverse image of a compact subset of $W_P^{(1)} \times W_P^{(1)}$ under $(\gamma, z) \mapsto (\gamma z, z)$ is not compact. This is exactly failure of properness. □

Exact comparison boundary before the extension lift. The natural algebra (695) is now an independently defined unstabilized arithmetic representative, and (710) is an exact morphism from the stable scalar boundary calculation to its unit-retaining K-theory shadow. It is not an isomorphism: the explicit morphism R_P has kernel (711), while the ideal map c_P retains only the constant subgroup of the much larger group $C(K_P, \mathbb{Z})$. This proves the precise information loss; it does not assert that the two constructions are categorically unrelated.

Subsection 9.13 resolves that narrower problem after stabilization. It writes the ideal, quotient, middle, multiplier, and corona maps explicitly, proves the Busby equation, and gives a minimal symmetric split enrichment when the two sign classes must remain distinct in K_0. The nonconstant unit-fibre classes remain outside the stable scalar image and are retained as such. No statement about an individual zeta zero follows from the construction.

Bounded coordinate replay. The script

`certificates/globalization_nte_multiplace_unstabilized/`
`verify_multiplace_unstabilized.py`

replays 362 named checks for one through eight finite primes. They cover exponents and signs, screw coordinates, the lattice square, the clopen section, and the nonproperness sequence. The accompanying JSON receipt has stable ID

`CERT-GNTE-MULTIPLACE-`
`UNSTABILIZED-20260830-0001`.

It records two Python 3.13/SymPy runs with exit code zero. It does not replace the printed topology, crossed-product, Green-imprimitivity, or six-term naturality proofs, and it launches no Lean process.

9.13 Lifting the balancing square: an exact stable embedding into the natural one-zero extension

The preceding subsection left an extension-level comparison as an open problem. The comparison does exist after stabilizing the natural one-zero extension. The compact unit fibre and the archimedean sign collapse do not obstruct the morphism. Instead, they become its exact image and kernel data. We construct the ideal, quotient, middle, multiplier, and corona maps before using their induced maps on K-theory.

Retain a nonempty finite prime set P. Abbreviate

$$\mathfrak{N}_P = \mathfrak{A}_P^{(1)}, \qquad \mathfrak{I}_P^{\mathrm{s}} = \mathfrak{I}_P \otimes \mathcal{K}, \qquad \mathfrak{N}_P^{\mathrm{s}} = \mathfrak{N}_P \otimes \mathcal{K}, \qquad \mathfrak{D}_P^{\mathrm{s}} = \mathfrak{D}_P \otimes \mathcal{K}.$$

Tensoring (695) with the nuclear algebra $\mathcal{K}$ gives

$$0 \longrightarrow \mathfrak{I}_P^{\mathrm{s}} \longrightarrow \mathfrak{N}_P^{\mathrm{s}} \xrightarrow{q_P^{\mathrm{s}}} \mathfrak{D}_P^{\mathrm{s}} \longrightarrow 0. \tag{712}$$

Lemma 9.31 (Essential generic ideal and its literal Busby pullback)**.** *The ideal $\mathfrak{I}_P$ is essential in $\mathfrak{N}_P$. Consequently $\mathfrak{I}_P^{\mathrm{s}}$ is essential in $\mathfrak{N}_P^{\mathrm{s}}$, and the latter is canonically isomorphic to*

$$\left\{(d,n) \in \mathfrak{D}_P^{\mathrm{s}} \oplus M(\mathfrak{I}_P^{\mathrm{s}}) : \tau_P^{\mathrm{nat}}(d) = \pi_{\mathfrak{I}}(n)\right\}, \tag{713}$$

where $\tau_P^{\mathrm{nat}} : \mathfrak{D}_P^{\mathrm{s}} \to Q(\mathfrak{I}_P^{\mathrm{s}})$ is the Busby invariant and $\pi_{\mathfrak{I}}$ is the multiplier quotient.

Proof. Every one-zero boundary point is a limit of points for which the vanishing coordinate has been replaced by a nonzero coordinate tending to zero. Thus $W_\varnothing$ is dense in $W_P^{(1)}$, and $C_0(W_\varnothing)$ is an essential ideal of $C_0(W_P^{(1)})$.

It remains to check that crossing with Γ_P does not lose essentiality. Use the reduced crossed product, which equals the full crossed product because Γ_P is amenable. Let E be its faithful conditional expectation and, for $g \in \Gamma_P$, write

$$x_g = E(xu_g^*) \in C_0(W_P^{(1)})$$

for the gth Fourier coefficient. Suppose that $x\mathfrak{I}_P = 0$. For every $f \in C_0(W_\varnothing)$,

$$0 = E(xfu_g^*) = x_g\, \alpha_g(f).$$

The invariant ideal $C_0(W_\varnothing)$ is essential, so $x_g = 0$ for every g. Faithfulness of the regular Fourier representation gives $x = 0$. Applying the same argument to x^* removes the right annihilator as well. Hence $\mathfrak{I}_P$ is essential.

Stabilization preserves essentiality. The canonical homomorphism $\mathfrak{N}_P^{\mathrm{s}} \to M(\mathfrak{I}_P^{\mathrm{s}})$ is therefore injective. Busby reconstruction identifies an essential extension with the pullback of its Busby invariant and the multiplier quotient [85, Sections 15.2–15.3, pp. 144–145], proving (713). □

The generic action is free and proper, and the complete orbit coordinates of Theorem 9.26 identify its orbit space with $K_P \times \mathbb{R}$. The Green module therefore gives

$$\mathfrak{I}_P \sim_M C_0(K_P \times \mathbb{R})$$

[87, Theorem 1.3.22]. Fix one stabilization isomorphism, constructed from that module and its linking algebra,

$$\chi_P : \mathfrak{I}_P^{\mathrm{s}} \xrightarrow{\cong} C(K_P) \otimes C_0(\mathbb{R}) \otimes \mathcal{K} = C(K_P) \otimes J. \tag{714}$$

This is a declared coordinate choice. It does not identify K_P with a point.

Definition 9.32 (Constant-unit-fibre ideal embedding)**.** Define the nondegenerate injective $*$-homomorphism

$$\jmath_P : J \longrightarrow \mathfrak{I}_P^{\mathrm{s}}, \qquad \chi_P(\jmath_P(b)) = 1_{C(K_P)} \otimes b. \tag{715}$$

Write $\bar{\jmath}_P : M(J) \to M(\mathfrak{I}_P^{\mathrm{s}})$ and $\dot{\jmath}_P : Q(J) \to Q(\mathfrak{I}_P^{\mathrm{s}})$ for its multiplier and corona extensions.

For every $k \in K_P$, evaluation at k after χ_P gives a $*$-homomorphism

$$\epsilon_{P,k} : \mathfrak{I}_P^{\mathrm{s}} \longrightarrow J, \qquad \epsilon_{P,k}\jmath_P = \mathrm{id}_J\,.$$

It is nondegenerate and extends to multipliers and coronas. Hence $\jmath_P$, $\bar{\jmath}_P$, and $\dot{\jmath}_P$ are all split injective. On the declared K_1 coordinates,

$$(\jmath_P)_*(n[\gamma]) = n\mathbf{1}_{K_P}, \tag{716}$$

which is exactly the map c_P in (710).

We next make the placewise edge maps compatible before adding them.

Lemma 9.33 (Compatible stabilized edge morphisms)**.** *For each $v \in V_P = P \sqcup \{\infty\}$, the stable transports in Lemma 9.22 may be chosen together with injective $*$-homomorphisms*

$$\lambda_{v,P} : \widehat{E}_v \longrightarrow \mathfrak{N}^{\mathrm{s}}_P, \qquad \alpha_{v,P} : \widehat{A}_v \longrightarrow \mathfrak{D}_{v,P} \otimes \mathcal{K} \subset \mathfrak{D}^{\mathrm{s}}_P,$$

so that

$$\begin{tikzcd} 0 \arrow[r] & J \arrow[r] \arrow[d, "j_P"'] & \widehat{E}_v \arrow[r] \arrow[d, "\lambda_{v,P}"'] & \widehat{A}_v \arrow[r] \arrow[d, "\alpha_{v,P}"'] & 0 \\ 0 \arrow[r] & \mathfrak{I}^{\mathrm{s}}_P \arrow[r] & \mathfrak{N}^{\mathrm{s}}_P \arrow[r, "q^{\mathrm{s}}_P"] & \mathfrak{D}^{\mathrm{s}}_P \arrow[r] & 0 \end{tikzcd} \tag{717}$$

commutes. Equivalently, its Busby equation is

$$\tau^{\mathrm{nat}}_P \alpha_{v,P} = j_P \widehat{\tau}_v. \tag{718}$$

At a finite place, the quotient map is the stabilized form of $\rho^*_{p,P}$ in (703)*. At infinity it sends $\mathrm{p}_\pm$ to $\mathrm{e}_\pm$ as in Theorem 9.28.*

Proof. We first expose the coordinate which forces the ideal arrow. The generic normalization in (698) is a global section of the principal Γ_P-bundle: the representative of (a,t) is

$$((a_q)_{q \in P}, e^t) \in \left(\prod_{q \in P} \mathbb{Q}_q^\times \right) \times \mathbb{R}^\times.$$

Consequently the generic orbit is literally $K_P \times \mathbb{R}$. Every finite local-to-semilocal orbit map restricts on that stratum to

$$\pi_P : K_P \times \mathbb{R} \longrightarrow \mathbb{R}, \qquad (a,t) \longmapsto t,$$

and its function pullback is exactly $f(t) \mapsto \mathbf{1}_{K_P}(a) f(t)$. At infinity, the quotient of $K_P \times \mathbb{R}^\times$ by $(c,z) \mapsto (-c,-z)$ has the equally explicit section $z > 0$ and coordinates $(a,t) = (\mathrm{sgn}(z)c, \log|z|)$. The quotient of the equivariant projection to $\mathbb{R}^\times$ is again $(a,t) \mapsto t$. Thus all placewise ideal arrows are the same constant-unit-fibre map before stabilization; this is not an appeal to equality of their induced K-classes.

For finite p, the two commutative rows in (703) form a literal injective morphism of extensions. The local crossed-product row and the upper commutative row are Morita equivalent at extension level by the restricted Green modules used in the proof of Theorem 9.14; the semilocal edge row and the lower commutative row are Morita equivalent by the same construction. At infinity, the equivariant projection $K_P \times \mathbb{R} \to \mathbb{R}$ gives the literal crossed-product morphism, and its generic free-action Green module has the orbit map just calculated.

We now stabilize these *extension triples*, rather than three unrelated algebras. The linking algebra of each extension-level Green module contains the linking algebra of the restricted ideal as a closed ideal and has the quotient-module linking algebra as its quotient. After tensoring with $\mathcal{K}$, Brown–Green–Rieffel equivalence of the two complementary full linking projections supplies one multiplier partial isometry. Conjugation by that partial isometry gives stable isomorphisms of the ideal, middle, and quotient corners simultaneously. Since multipliers carry the linking ideal into itself, the same conjugation restricts to the ideal and descends to the quotient, so the three stable isomorphisms form a commuting extension diagram. This is the linking-algebra construction in [86, Theorems 1.1–1.2, pp. 349–352]; the compact operator model of the Green modules is [87, Theorem 1.3.22, p. 14].

Choose the source transports in Lemma 9.22 and the target coordinate χ_P by these simultaneous stable diagrams, using fixed Hilbert-space unitaries for their $\mathcal{K}$ factors. In those declared coordinates the ideal component of the composite is the already calculated map $b \mapsto \mathbf{1}_{K_P} \otimes b$, hence is literally $\jmath_P$. The middle and quotient components are $\lambda_{v,P}$ and $\alpha_{v,P}$. Extension by zero from the edge to the global one-zero algebra completes the bottom arrow. All components are injective: the orbit maps are onto and hence their function pullbacks are injective; the archimedean quotient map sends the two minimal projections to the nonzero orthogonal projections e_+ and e_-; and every other component is a corner embedding or a stable isomorphism.

This proves (717). Busby functoriality for that proved morphism gives (718). The choices are part of the displayed coordinate model; no choice-independence or canonicality statement is being made. $\square$

Let S_j, $s_v = 1 \otimes S_{\nu(v)} \in M(J)$, and μ_P be the fixed Cantor-pairing isometries and ideal codiagonal of (689)–(691). Put

$$T_v = 1_{M(\mathfrak{N}_P)} \otimes S_{\nu(v)} \in M(\mathfrak{N}^{\mathrm{s}}_P), \qquad R_v = \overline{q}^{\mathrm{s}}_P(T_v) \in M(\mathfrak{D}^{\mathrm{s}}_P),$$

where $\overline{q}^{\mathrm{s}}_P$ is the strict multiplier extension of the quotient map. Their restrictions to $M(\mathfrak{I}^{\mathrm{s}}_P)$ satisfy

$$T_v|_{M(\mathfrak{I}^{\mathrm{s}}_P)} = \bar{\jmath}_P(s_v) \tag{719}$$

in the coordinate (714). Define

$$\alpha_P((a_v)_v) = \sum_{v \in V_P} R_v \alpha_{v,P}(a_v) R_v^*, \tag{720}$$

$$\Lambda_P((x_v)_v) = \sum_{v \in V_P} T_v \lambda_{v,P}(x_v) T_v^*. \tag{721}$$

The sums are finite. Orthogonality gives

$$R_v^* \alpha_P((a_w)_w) R_v = \alpha_{v,P}(a_v), \qquad T_v^* \Lambda_P((x_w)_w) T_v = \lambda_{v,P}(x_v),$$

so both maps are injective. On the direct-sum ideal,

$$\begin{aligned} \Lambda_P((b_v)_v) &= \sum_v T_v \jmath_P(b_v) T_v^* \\ &= \jmath_P\left(\sum_v s_v b_v s_v^*\right) = \jmath_P \mu_P((b_v)_v). \end{aligned} \tag{722}$$

Theorem 9.34 (Stable balancing extension embeds in the natural one-zero extension)**.** *For every nonempty finite prime set P, there is an injective morphism of short exact sequences*

$$\begin{tikzcd} 0 \arrow[r] & J \arrow[r] \arrow[d, "\jmath_P"'] & \mathcal{E}_P \arrow[r] \arrow[d, "\Phi_P"'] & \mathcal{A}_P \arrow[r] \arrow[d, "\alpha_P"'] & 0 \\ 0 \arrow[r] & \mathfrak{I}^{\mathrm{s}}_P \arrow[r] & \mathfrak{N}^{\mathrm{s}}_P \arrow[r, "q^{\mathrm{s}}_P"] & \mathfrak{D}^{\mathrm{s}}_P \arrow[r] & 0. \end{tikzcd} \tag{723}$$

In the two Busby pullback coordinates (693) *and* (713), *its middle map is*

$$\Phi_P(a, m) = \big(\alpha_P(a), \bar{\jmath}_P(m)\big). \tag{724}$$

The required multiplier/corona identity is the literal equation

$$\tau_P^{\mathrm{nat}}\alpha_P = \dot{\jmath}_P\tau_P. \tag{725}$$

Moreover the direct-sum local middle map factors exactly:

$$\Lambda_P = \Phi_P \circ \left[((a_v, m_v))_v \longmapsto \left((a_v)_v, \sum_v s_v m_v s_v^*\right)\right].$$

Proof. The orthogonal corner relations and (719) show before passing to coronas that Λ_P has quotient component α_P and ideal component $\jmath_P\mu_P$; the latter is the literal identity (722). Hence $(\jmath_P\mu_P, \Lambda_P, \alpha_P)$ is a morphism from the direct-sum local extension to the natural extension. Busby functoriality gives, for every $a = (a_v)_v \in \mathcal{A}_P$,

$$\begin{aligned}\tau_P^{\mathrm{nat}}\alpha_P(a) &= (\jmath_P\dot{\mu}_P)\,\widehat{\tau}_P^{\oplus}(a)\\ &= \dot{\jmath}_P\dot{\mu}_P\widehat{\tau}_P^{\oplus}(a) = \dot{\jmath}_P\tau_P(a).\end{aligned}$$

Thus (725) is forced by the displayed middle, ideal, and quotient maps, not inferred from the desired K-theory square.

If $(a, m) \in \mathcal{E}_P$, then $\tau_P(a) = \pi_J(m)$. The displayed Busby equation gives

$$\tau_P^{\mathrm{nat}}(\alpha_P(a)) = \dot{\jmath}_P(\pi_J(m)) = \pi_{\mathfrak{J}}(\bar{\jmath}_P(m)),$$

so (724) lies in the natural pullback. Coordinatewise multiplication and involution show that Φ_P is a $*$-homomorphism. It restricts to $\jmath_P$ on the ideal and induces α_P on the quotient, proving commutativity of (723).

Both vertical outside maps are injective. If $\Phi_P(e) = 0$, then $\alpha_P(q(e)) = 0$, hence $q(e) = 0$ and e lies in J. Its image there is $\jmath_P(e) = 0$, so $e = 0$. Thus the middle map is injective. Finally, substituting the direct-sum pullback coordinates and using (722) proves the stated factorization of Λ_P. ☐

The embedding has an exact inverse on its image. In the natural pullback, take a pair (d, n) with

$$d \in \alpha_P(\mathcal{A}_P), \qquad n \in \bar{\jmath}_P(M(J)).$$

Recover each a_v by the corner compression $R_v^* d R_v$ followed by $\alpha_{v,P}^{-1}$, and recover m by applying the multiplier extension of any evaluation map $\epsilon_{P,k}$. Since $\dot{\jmath}_P$ has the corresponding corona left inverse, the natural pullback equation implies $\tau_P(a) = \pi_J(m)$. Hence $(a, m) \in \mathcal{E}_P$ and maps back to (d, n). Therefore

$$\operatorname{im}\Phi_P = \{(d, n) \text{ in } (713) : d \in \operatorname{im}\alpha_P, \quad n \in \operatorname{im}\bar{\jmath}_P\}. \tag{726}$$

This records the fibres and inverse data, not merely existence.

Corollary 9.35 (The earlier K-theory square is induced by the extension morphism)**.** *On the displayed boundary labels and ideal generator,*

$$(\alpha_P)_* = R_P, \qquad (\jmath_P)_* = c_P.$$

Consequently naturality of (723) *is exactly* (710)*.*

Proof. Each external corner $R_v(-)R_v^*$ induces the standard rank-one corner map on K-theory. The finite quotient maps send $[1]_p$ to $\eta_{p,P}$, and the archimedean map sends $[p_\pm]$ to $[e_\pm]$. This is R_P. Equation (716) is c_P. Naturality of connecting maps for the proved extension morphism [85, Definition 21.1.1 and Example 21.1.2(a)] gives the commuting square. ☐

There is no contradiction between injectivity of α_P and the fact that $(\alpha_P)_*$ kills $[p_+]-[p_-]$. The two projections e_+ and e_- remain distinct nonzero orthogonal elements of the archimedean boundary algebra, so the algebra map from $D_2 \otimes \mathcal{K}$ is injective. They are nevertheless unitarily equivalent in $C(K_P) \rtimes C_2 \cong M_2(C(H))$, so their K_0 classes agree. Likewise $\jmath_P$ is split injective, while its K_1 image is exactly

$$\mathbb{Z}\mathbf{1}_{K_P} \subset C(K_P, \mathbb{Z}).$$

All nonconstant locally constant rank functions remain in the target and outside this image. Thus the morphism is an embedding, not an isomorphism or a stable Morita equivalence.

For applications that require the two sign labels to remain distinct even in K_0, the smallest finite-dimensional commutative symmetric split enrichment is the one below.

Proposition 9.36 (Minimal symmetric split enrichment retaining the sign labels)**.** *Let* $\mathrm{pr}_\infty : \mathcal{A}_P \to \widehat{A}_\infty$ *be the archimedean quotient projection, and write* $q_{\mathcal{E}} : \mathcal{E}_P \to \mathcal{A}_P$ *for the top quotient map. Define the split-enriched natural extension*

$$0 \to \mathfrak{I}_P^{\mathrm{s}} \longrightarrow \mathfrak{N}_P^{\mathrm{s}} \oplus \widehat{A}_\infty \longrightarrow \mathfrak{D}_P^{\mathrm{s}} \oplus \widehat{A}_\infty \to 0, \tag{727}$$

with ideal map $i \mapsto (i,0)$ *and quotient* $(x,b) \mapsto (q_P^{\mathrm{s}}(x), b)$*. Then*

$$\begin{aligned} \alpha_P^{\mathrm{sgn}}(a) &= (\alpha_P(a), \mathrm{pr}_\infty(a)), \\ \Phi_P^{\mathrm{sgn}}(e) &= (\Phi_P(e), \mathrm{pr}_\infty(q_{\mathcal{E}}(e))), \\ \jmath_P^{\mathrm{sgn}}(j) &= (\jmath_P(j), 0) \end{aligned}$$

form an injective morphism from the stable balancing extension to (727)*. Its* K_0 *map sends*

$$[p_+] \longmapsto ([e_+],[p_+]), \qquad [p_-] \longmapsto ([e_-],[p_-]),$$

so the sign difference survives.

Among finite-dimensional commutative split boundary carriers which retain two nonzero orthogonal sign projections and the involution interchanging them, $D_2 = \mathbb{C} \oplus \mathbb{C}$ *is minimal: every such carrier has at least two minimal central summands, and* D_2 *has exactly two.*

Proof. The second summand in (727) is split, so its Busby invariant is zero. The formulas are $*$-homomorphisms and the previous Busby equation remains unchanged in the first component. The second component commutes because it is the same quotient projection on the top middle and quotient algebras. Injectivity follows already from Φ_P. The displayed K_0 images are distinct in the second direct summand.

A finite-dimensional commutative algebra is $\mathbb{C}^r$. Two nonzero orthogonal projections require disjoint nonempty supports among its r minimal central projections, hence $r \geq 2$. An automorphism interchanging the two projections permutes those supports. The algebra $D_2 = \mathbb{C} \oplus \mathbb{C}$ with the coordinate swap realizes $r = 2$, proving minimality within the stated symmetric split class. □

Bounded coordinate replay. The script

```
certificates/globalization_nte_stable_to_natural/
verify_stable_to_natural.py
```

replays 2,361 named finite checks for one through eight finite primes. It passes in two independent local runtimes. Both use Python 3.13, with distinct SymPy versions, and verify the Cantor-corner compression identities, the constant-unit ideal factorization, the finite matrix form of the Busby corner equation, the boundary naturality matrix, the exact sign-difference kernel, injectivity of the split-enriched label map, evaluation left inverses, displayed nonconstant unit-fibre classes, and the minimal two-coordinate sign carrier. The stable receipt ID is

This finite replay does not replace the essential-ideal, Green-imprimitivity, linking-algebra, multiplier, corona, Busby-pullback, or crossed-product arguments printed above. It launched no Lean or Lake process.

Exact boundary of the result. The formerly missing extension-level arrow is now constructed after stabilizing the natural one-zero extension. No global Green reduction was used: only the generic ideal and the individual boundary edges enter the compatible Green modules. Proposition 9.30 remains valid. The map Φ_P is not claimed to be onto, and (726) shows exactly what its image omits. The split sign enrichment retains the local character difference but does not manufacture the nonconstant unit-fibre classes in the stable source. No endpoint, individual-zero, critical-line, or global zeta conclusion follows.

9.14 The full nonconstant unit-fibre calculation: profinite orbit coordinates, connecting maps, and the natural middle algebra

The stable extension embedding of the preceding subsection sees the constant subgroup $\mathbb{Z}\mathbf{1}_{K_P} \subset C(K_P, \mathbb{Z})$. We now calculate the remaining classes rather than referring to them collectively as "nonconstant." The answer has three distinct pieces: invariant functions on finite-place profinite quotients, coinvariants with nontrivial finite-level transition multiplicities, and an archimedean sign-even subgroup. All three are needed for the K-theory of the natural one-zero algebra.

Fix a nonempty finite prime set P. For $p \in P$, put

$$Y_{p,P} = K_{P\setminus p} = \prod_{q\in P\setminus\{p\}} \mathbb{Z}_q^\times, \qquad g_{p,P} = (p)_{q\neq p} \in Y_{p,P}. \tag{728}$$

The empty product is one point. Let

$$a_{p,P}(y) = g_{p,P}y, \qquad (\sigma_{p,P}f)(y) = f(g_{p,P}^{-1}y), \tag{729}$$

and let

$$H_{p,P} = \overline{\langle g_{p,P}\rangle} \subset Y_{p,P}, \qquad q_{p,P} : Y_{p,P} \longrightarrow Y_{p,P}/H_{p,P} \tag{730}$$

be the closed procyclic subgroup and quotient map. If $Y_{p,P}$ is not a point, the $\mathbb{Z}$-action generated by $a_{p,P}$ is free: an equality $p^n y = y$ in one factor $\mathbb{Z}_q^\times$ would imply $p^n = 1$ in $\mathbb{Q}_q$, and hence as a rational number, so $n = 0$. For $P = \{p\}$, the action on the one-point $Y_{p,P}$ is instead trivial; this exceptional stabilizer is retained below and gives the ordinary circle.

Lemma 9.37 (Mapping-torus model and complete finite-place K-groups)**.** *There is a literal isomorphism*

$$C(B_{p,P}) \cong M_{\sigma_{p,P}} = \{F \in C([0,1], C(Y_{p,P})) : F(1) = \sigma_{p,P}(F(0))\}. \tag{731}$$

Under this isomorphism and the Green-module coordinate for $\mathfrak{D}_{p,P}$,

$$K_0(\mathfrak{D}_{p,P}) \cong \ker(1 - \sigma_{p,P,*}) \cong C(Y_{p,P}/H_{p,P}, \mathbb{Z}), \tag{732}$$

$$K_1(\mathfrak{D}_{p,P}) \cong \operatorname{coker}(1 - \sigma_{p,P,*}) \cong \frac{C(Y_{p,P}, \mathbb{Z})}{(1 - \sigma_{p,P})C(Y_{p,P}, \mathbb{Z})}. \tag{733}$$

No constant-function reduction occurs in either group.

Proof. Write $\ell_p = \log p$. For $G \in C(B_{p,P})$, define

$$(\Theta G)(u)(y) = G([y, u\ell_p]), \qquad 0 \le u \le 1.$$

Since $[y, \ell_p] = [g_{p,P}^{-1} y, 0]$,

$$(\Theta G)(1)(y) = (\Theta G)(0)(g_{p,P}^{-1} y) = \sigma_{p,P}((\Theta G)(0))(y).$$

Conversely, the endpoint equation makes a path in the displayed mapping torus descend uniquely to the quotient $(Y_{p,P} \times \mathbb{R})/\langle \beta_p \rangle$. These two maps preserve multiplication, involution, and the supremum norm, proving (731).

We also record the coefficient groups rather than hiding them behind a topological K-theory identification. For every compact profinite space Y, a projection over Y is, after refining a finite clopen partition, unitarily equivalent to a constant matrix projection on each cell. Stable equivalence is therefore determined exactly by its locally constant rank function, so

$$K_0(C(Y)) = C(Y, \mathbb{Z}).$$

Moreover every continuous unitary matrix over Y is uniformly approximated within distance less than two by one that is constant on a finite clopen partition. The two unitaries are homotopic, and each constant value is homotopic to the identity inside a unitary group. Thus $K_1(C(Y)) = 0$.

Blackadar's mapping-torus exact sequence and its explicit connecting maps are $1 - \sigma_*$ under the Bott coordinate [85, Theorem 10.2.1, Definition 10.3.1, and Proposition 10.4.1, pp. 83–87]. Substitution of the two coefficient groups just proved gives the kernel and cokernel in (732)– (733). A locally constant function is fixed by $\sigma_{p,P}$ exactly when it is constant on every orbit of $g_{p,P}$, hence on every coset of the closure $H_{p,P}$. Pullback along $q_{p,P}$ therefore identifies the kernel with $C(Y_{p,P}/H_{p,P}, \mathbb{Z})$. Finally, the restricted Green module from Theorem 9.26 transports these groups to $\mathfrak{D}_{p,P}$ without changing the extension boundary coordinate. $\square$

The coinvariant group in (733) generally does not reduce to a free group on a final orbit set. Its exact finite-level coordinates are as follows. Let $\mathcal{M}_{p,P}$ be the directed set of tuples $\mathbf{m} = (m_q)_{q \ne p}$, with $m_q \ge 1$, ordered coordinatewise. For an empty tuple set all objects below equal to the one-point object and put $L_\varnothing = 1$. Otherwise define

$$G_{\mathbf{m}} = \prod_{q \ne p} (\mathbb{Z}/q^{m_q}\mathbb{Z})^\times, \quad g_{\mathbf{m}} = (p \bmod q^{m_q})_{q \ne p}, \quad L_{\mathbf{m}} = \operatorname{ord}_{G_{\mathbf{m}}}(g_{\mathbf{m}}), \quad \Omega_{\mathbf{m}} = G_{\mathbf{m}}/\langle g_{\mathbf{m}} \rangle. \tag{734}$$

Every orbit has exactly $L_{\mathbf{m}}$ points. Thus the stabilizer of a point for the finite-level $\mathbb{Z}$-action is precisely $L_{\mathbf{m}}\mathbb{Z}$; these finite-level stabilizers disappear in the nontrivial inverse limit but are not deleted from the transition maps.

For an orbit $O \in \Omega_{\mathbf{m}}$, let

$$u_O = \mathbf{1}_O \quad \text{in the invariant group,} \qquad v_O = [\delta_x] \quad (x \in O) \quad \text{in the coinvariant group.}$$

The second class is independent of $x \in O$. If $\mathbf{m}' \ge \mathbf{m}$, let $\pi_{\mathbf{m}',\mathbf{m}} : G_{\mathbf{m}'} \to G_{\mathbf{m}}$ be reduction. Then $L_{\mathbf{m}} \mid L_{\mathbf{m}'}$, and the two transition maps are

$$\iota^0_{\mathbf{m}',\mathbf{m}}(u_O) = \sum_{\substack{O' \in \Omega_{\mathbf{m}'} \\ O' \subset \pi^{-1}_{\mathbf{m}',\mathbf{m}}(O)}} u_{O'}, \tag{735}$$

$$\iota^1_{\mathbf{m}',\mathbf{m}}(v_O) = \frac{L_{\mathbf{m}'}}{L_{\mathbf{m}}} \sum_{\substack{O' \in \Omega_{\mathbf{m}'} \\ O' \subset \pi^{-1}_{\mathbf{m}',\mathbf{m}}(O)}} v_{O'}. \tag{736}$$

Proposition 9.38 (Finite-level coordinate form, including multiplicities)**.** *The two groups in* (732)–(733) *are respectively*

$$C(Y_{p,P}/H_{p,P},\mathbb{Z}) \cong \varinjlim_{\mathbf{m}} (\mathbb{Z}[\Omega_{\mathbf{m}}],\iota^0), \tag{737}$$

$$C(Y_{p,P},\mathbb{Z})_{\sigma_{p,P}} \cong \varinjlim_{\mathbf{m}} (\mathbb{Z}[\Omega_{\mathbf{m}}],\iota^1). \tag{738}$$

Both displayed transition maps are injective. In particular the coinvariant group is torsion-free, but no free decomposition or canonical splitting is asserted.

Proof. Every locally constant integer-valued function on $Y_{p,P}$ factors through some $G_{\mathbf{m}}$, and pullback along the reduction maps gives

$$C(Y_{p,P},\mathbb{Z}) = \varinjlim_{\mathbf{m}} \mathbb{Z}^{G_{\mathbf{m}}}.$$

At level $\mathbf{m}$, the invariant functions have basis $(u_O)_{O\in\Omega_{\mathbf{m}}}$, and pullback of $\mathbf{1}_O$ is the sum of the characteristic functions of all fine orbits over O. This is (735).

The coinvariants of the permutation module of one orbit are $\mathbb{Z}$, with every point mass representing the same generator v_O. Fix $x \in O$. Pullback of δ_x is the sum of the point masses over $\pi^{-1}(x)$. Every fine orbit O' above O maps onto O with degree $L_{\mathbf{m}'}/L_{\mathbf{m}}$, so exactly that many of its points lie above x. Hence its contribution is $(L_{\mathbf{m}'}/L_{\mathbf{m}})v_{O'}$, proving (736). Equivalently, the number of fine orbits above O is

$$|\ker \pi_{\mathbf{m}',\mathbf{m}}|\frac{L_{\mathbf{m}}}{L_{\mathbf{m}'}},$$

which verifies that all points in the pullback have been counted exactly once.

Filtered colimits commute with the cokernel of $1-\sigma_{p,P}$, giving (738). For invariants, a function that factors through $G_{\mathbf{m}}$ is invariant in the inverse limit exactly when its finite-level function is constant on the $\langle g_{\mathbf{m}}\rangle$-orbits, which gives (737). Distinct coarse orbits have disjoint inverse images. Both transition formulas therefore send a nonzero integer combination to a nonzero combination on disjoint supports. They are injective. An injective filtered union of torsion-free groups is torsion-free, proving the final statement. ☐

We next compute the connecting map on every invariant rank function. Let

$$\mathrm{pr}_{p,P} : K_P \longrightarrow Y_{p,P} \tag{739}$$

drop the p-coordinate, and define

$$\iota_{p,P} : C(Y_{p,P}/H_{p,P},\mathbb{Z}) \longrightarrow C(K_P,\mathbb{Z}), \qquad \iota_{p,P}(h) = h \circ q_{p,P} \circ \mathrm{pr}_{p,P}\,. \tag{740}$$

Theorem 9.39 (Full finite-edge connecting map and middle K-groups)**.** *In the coordinates above, the connecting map of the p-edge is*

$$\partial_{p,P}(h) = -\iota_{p,P}(h) \qquad (h \in C(Y_{p,P}/H_{p,P},\mathbb{Z})). \tag{741}$$

It is injective. Consequently

$$K_0(A^{(P)}_{\varnothing p}) = 0 \tag{742}$$

and there is an exact sequence

$$0 \longrightarrow \frac{C(K_P,\mathbb{Z})}{\iota_{p,P}C(Y_{p,P}/H_{p,P},\mathbb{Z})} \longrightarrow K_1(A^{(P)}_{\varnothing p}) \longrightarrow C(Y_{p,P},\mathbb{Z})_{\sigma_{p,P}} \longrightarrow 0. \tag{743}$$

No splitting of (743) *is selected.*

Proof. It is enough by additivity to take $h = \mathbf{1}_E$ for a clopen subset $E \subset Y_{p,P}/H_{p,P}$. Put $\widetilde{E} = q_{p,P}^{-1}(E)$. It is clopen and invariant under $g_{p,P}$. The corresponding clopen part of the mapping-torus boundary is

$$B_E = (\widetilde{E} \times \mathbb{R})/\langle \beta_p \rangle \subset B_{p,P}.$$

In the orbit coordinates of Theorem 9.26, the scalar function

$$x_E(a,t) = \frac{\mathbf{1}_{\widetilde{E}}(\mathrm{pr}_{p,P}(a))}{1+e^t}, \qquad x_E([b,s]) = \mathbf{1}_{\widetilde{E}}(b) \tag{744}$$

is continuous. Invariance of $\widetilde{E}$ makes the boundary formula independent of the representative (b,s), while the exact cross-convergence criterion makes the generic value tend to it as $t \to -\infty$. The only escaping end is $t \to +\infty$, where the function vanishes. Thus x_E is a self-adjoint lift of the boundary projection $\mathbf{1}_{B_E}$.

Its exponential is the identity off $\mathrm{pr}_{p,P}^{-1}(\widetilde{E})$. On every retained generic unit fibre above $\widetilde{E}$, its argument decreases from 2π to zero as t increases, so it has winding -1 in the previously declared orientation. Hence

$$\partial_{p,P}[\mathbf{1}_{B_E}] = -\mathbf{1}_{\mathrm{pr}_{p,P}^{-1}(\widetilde{E})}.$$

Every locally constant integer function is an integer combination of such characteristic functions, proving (741). The map $q_{p,P} \circ \mathrm{pr}_{p,P}$ is onto, so its pullback is injective.

Finally, $\mathfrak{I}_P$ is Morita equivalent to $C_0(K_P \times \mathbb{R})$, whose K-groups in the retained suspension coordinate are

$$K_0(\mathfrak{I}_P) = 0, \qquad K_1(\mathfrak{I}_P) = C(K_P, \mathbb{Z}).$$

Insert these, Lemma 9.37, and the injective connecting map into the edge six-term sequence. Exactness gives (742) and (743). □

The archimedean summand contributes all sign-even functions, not only its constant unit. Retain the clopen half $H_P^+ \subset K_P$ constructed in the proof of Theorem 9.28, so

$$K_P = H_P^+ \sqcup (-H_P^+). \tag{745}$$

Define the continuous representative map

$$r_P : K_P \longrightarrow H_P^+, \qquad r_P(c) = \begin{cases} c, & c \in H_P^+, \\ -c, & c \in -H_P^+. \end{cases} \tag{746}$$

Theorem 9.40 (Full archimedean boundary map)**.** *The explicit matrix coordinate gives*

$$K_0(\mathfrak{D}_{\infty,P}) \cong C(H_P^+, \mathbb{Z}), \qquad K_1(\mathfrak{D}_{\infty,P}) = 0. \tag{747}$$

Under the generic coordinate $K_1(\mathfrak{I}_P) = C(K_P, \mathbb{Z})$, the complete archimedean connecting map is

$$\partial_{\infty,P}(h) = -r_P^* h. \tag{748}$$

Its image is exactly the sign-even subgroup

$$C(K_P, \mathbb{Z})^{\mathrm{ev}} = \{f : f(c) = f(-c) \text{ for all } c \in K_P\}. \tag{749}$$

Proof. The clopen decomposition (745) trivializes the free double cover, and the already displayed orbit matrices give
$$C(K_P) \rtimes C_2 \cong M_2(C(H_P^+)).$$
The profinite K-theory calculation in the proof of Lemma 9.37 yields (747).

For the sign and orientation, write every $c \in K_P$ uniquely as $c = \varepsilon h$, with $h \in H_P^+$ and $\varepsilon \in \{1, -1\}$. The orbit map for the diagonal action on $K_P \times \mathbb{R}$ is explicitly

$$[(\varepsilon h, z)] \longmapsto (h, w = \varepsilon z) \quad \text{in } H_P^+ \times \mathbb{R}. \tag{750}$$

On the generic part $w \neq 0$, its relation with the retained ideal coordinate $(a, t) \in K_P \times \mathbb{R}$ is

$$a = \operatorname{sgn}(w)h, \qquad t = \log|w|. \tag{751}$$

For a clopen $E \subset H_P^+$, the function

$$x_E(h, w) = \frac{\mathbf{1}_E(h)}{1 + |w|}$$

on the commutative orbit extension is a self-adjoint lift of $\mathbf{1}_E$ at $w = 0$. On both components $w > 0$ and $w < 0$, the argument of its exponential decreases from 2π to zero as $t = \log|w|$ increases. Formula (751) therefore gives pointwise winding -1 on $E \sqcup (-E) = r_P^{-1}(E)$ and zero elsewhere. The extension-level Green module transports this calculation to the crossed-product extension, so

$$\partial_{\infty,P}[\mathbf{1}_E] = -\mathbf{1}_{r_P^{-1}(E)}.$$

Additivity proves (748). Pullback by r_P is injective and its image consists exactly of the functions constant on each pair $\{c, -c\}$, proving (749). □

We can now assemble all places. Abbreviate

$$\mathcal{F}_{p,P} = C(Y_{p,P}/H_{p,P}, \mathbb{Z}), \qquad \mathcal{C}_{p,P} = C(Y_{p,P}, \mathbb{Z})_{\sigma_{p,P}}. \tag{752}$$

Theorem 9.41 (Full K-theory and connecting map of the natural one-zero extension)**.** *The quotient groups are*

$$K_0(\mathfrak{D}_P) \cong \left(\bigoplus_{p \in P} \mathcal{F}_{p,P} \right) \oplus C(H_P^+, \mathbb{Z}), \tag{753}$$

$$K_1(\mathfrak{D}_P) \cong \bigoplus_{p \in P} \mathcal{C}_{p,P}. \tag{754}$$

In these exact coordinates the connecting map is

$$\partial_P^{(1)}((h_p)_{p \in P}, h_\infty) = -\sum_{p \in P} \iota_{p,P}(h_p) - r_P^* h_\infty \quad \textit{in } C(K_P, \mathbb{Z}). \tag{755}$$

For a finite tuple $h = (h_p)_{p \in P}$, put

$$S_h(c) = \sum_{p \in P} \iota_{p,P}(h_p)(c). \tag{756}$$

Then projection to the finite-place coordinates gives an isomorphism

$$K_0(\mathfrak{N}_P) \xrightarrow{\cong} \left\{ h \in \bigoplus_{p \in P} \mathcal{F}_{p,P} : S_h(c) = S_h(-c) \text{ for every } c \in K_P \right\}, \tag{757}$$

whose inverse is

$$h \longmapsto ((h_p)_p, -S_h|_{H_P^+}). \tag{758}$$

Define

$$d_{p,P} : \mathcal{F}_{p,P} \longrightarrow C(H_P^+, \mathbb{Z}), \qquad d_{p,P}(h)(c) = \iota_{p,P}(h)(c) - \iota_{p,P}(h)(-c), \tag{759}$$

and

$$\mathcal{Q}_P = \frac{C(H_P^+, \mathbb{Z})}{\sum_{p \in P} d_{p,P}(\mathcal{F}_{p,P})}. \tag{760}$$

There is a short exact sequence

$$0 \longrightarrow \mathcal{Q}_P \longrightarrow K_1(\mathfrak{N}_P) \longrightarrow \bigoplus_{p \in P} \mathcal{C}_{p,P} \longrightarrow 0. \tag{761}$$

No canonical splitting of (761) *is asserted.*

Proof. The quotient $\mathfrak{D}_P$ is the finite direct sum of its boundary strata. Lemma 9.37 and Theorem 9.40 therefore give (753) and (754). Extension by zero from every edge to the global one-zero algebra is a morphism of extensions. Naturality and additivity of the connecting maps, now applied to the complete edge groups rather than only their units, combine (741) and (748) into (755).

Since $K_0(\mathfrak{I}_P) = 0$ and $K_1(\mathfrak{I}_P) = C(K_P, \mathbb{Z})$, the six-term sequence gives

$$K_0(\mathfrak{N}_P) = \ker \partial_P^{(1)}$$

and

$$0 \to \operatorname{coker} \partial_P^{(1)} \to K_1(\mathfrak{N}_P) \to K_1(\mathfrak{D}_P) \to 0. \tag{762}$$

Equation (755) vanishes exactly when $S_h = -r_P^* h_\infty$. The right side is sign-even. Thus S_h must be sign-even; when it is, the unique solution is $h_\infty = -S_h|_{H_P^+}$. This proves (757) and its explicit inverse.

It remains to identify the cokernel without discarding the archimedean image. Define

$$\mathrm{Odd}_{H_P^+} : C(K_P, \mathbb{Z}) \longrightarrow C(H_P^+, \mathbb{Z}), \qquad \mathrm{Odd}_{H_P^+}(f)(c) = f(c) - f(-c). \tag{763}$$

This map is onto: prescribe a function on H_P^+ and extend it by zero on $-H_P^+$. Its kernel is exactly the sign-even subgroup $r_P^* C(H_P^+, \mathbb{Z})$. Hence it induces the exact isomorphism

$$\frac{C(K_P, \mathbb{Z})}{r_P^* C(H_P^+, \mathbb{Z})} \xrightarrow{\cong} C(H_P^+, \mathbb{Z}).$$

Under this map, the finite-place summand $\iota_{p,P}(h)$ becomes exactly $d_{p,P}(h)$. Therefore $\operatorname{coker} \partial_P^{(1)} \cong \mathcal{Q}_P$. Substitution into (762) proves (761). ☐

The one-prime coordinate check. If $P = \{p\}$, then $Y_{p,P}$, $H_{p,P}$, and every finite quotient are one point. Thus $\mathcal{F}_{p,P} = \mathcal{C}_{p,P} = \mathbb{Z}$. The finite edge has

$$K_0(A_{\varnothing p}^{(P)}) = 0, \qquad 0 \to C(K_P, \mathbb{Z})/\mathbb{Z}\mathbf{1}_{K_P} \to K_1(A_{\varnothing p}^{(P)}) \to \mathbb{Z} \to 0.$$

For the global natural algebra, the K-zero kernel consists exactly of $(n, -n\mathbf{1}_{H_P^+})$, so $K_0(\mathfrak{N}_P) \cong \mathbb{Z}$. The finite odd-difference map is zero, and therefore

$$0 \to C(H_P^+, \mathbb{Z}) \to K_1(\mathfrak{N}_P) \to \mathbb{Z} \to 0.$$

This last sequence splits abstractly because its quotient is $\mathbb{Z}$, but the calculation selects no splitting. The term $C(H_P^+, \mathbb{Z})$ is the exact residual sign-asymmetric unit-fibre coordinate after the archimedean boundary has filled every sign-even function.

Exact comparison with the stable balancing extension. For constant $h_p = k_p$ and constant $h_\infty = n$, (755) reduces to

$$-\left(\sum_{p \in P} k_p + n\right) \mathbf{1}_{K_P}.$$

The two stable archimedean character projections both map to the same constant $n = 1$ class, exactly as proved in Corollary 9.35. Thus the earlier codiagonal square is the constant-coordinate restriction of the full map, not a substitute for it. The new calculation also shows precisely what lies outside that image: finite invariant functions, finite coinvariants with the multiplicities (736), and the odd-difference quotient $\mathcal{Q}_P$. It proves no surjectivity, Morita equivalence, choice-independence across prime sets, endpoint statement, or global zeta consequence.

Bounded finite-coordinate replay. The executable certificate and its receipt are stored under

`certificates/globalization_nte_nonconstant_unit_k/`.

The receipt has stable ID `CERT-GNTE-NONCONSTANT-UNIT-K-20260830-0001`. They verify the finite quotient orbit lengths, finite-level stabilizers, invariant and coinvariant bases, both transition formulas including their $L_{\mathbf{m}'}/L_{\mathbf{m}}$ multiplicities, injectivity of the transition matrices, the finite and archimedean pullback matrices, and the global kernel/cokernel coordinate identities on bounded examples. The certificate does not replace the mapping-torus theorem, Green imprimitivity, six-term exactness, or the proofs above, and it launches no Lean or Lake process.

9.15 Finite-set coherence: exact tail coordinates, transition morphisms, and the multiplier obstruction

The preceding stable-to-natural theorem was proved for one finite prime set P at a time. Its stabilization coordinate χ_P was explicitly declared to be a choice. Equality of the formulas for two different sets does not prove that the chosen isomorphisms commute when a prime is adjoined. We now construct one simultaneous coordinate system and prove every transition square in that system. The construction also exposes a genuine obstruction: when $P \subsetneq P'$, the natural middle-algebra map occupies the sector in which every new finite coordinate is nonzero. It is therefore degenerate as a map into the full P' algebra and has no unital strict multiplier extension there. The generic-ideal map is nevertheless nondegenerate, so the multiplier and corona maps required for the Busby square do exist and are explicit.

Let $\mathbb{P}$ denote the set of rational primes. For a finite prime set P define the positive complementary prime group

$$\mathcal{L}_P = \left\{ \prod_{r\in\mathbb{P}\setminus P} r^{n_r} : n_r \in \mathbb{Z},\ n_r = 0 \text{ for all but finitely many } r \right\}. \tag{764}$$

If $P \subset P'$, put

$$R = P' \setminus P, \qquad \Gamma_R^+ = \left\{ \prod_{r\in R} r^{n_r} : n_r \in \mathbb{Z} \right\}, \qquad K_R = \prod_{r\in R} \mathbb{Z}_r^\times. \tag{765}$$

Unique prime factorization gives literal direct products

$$\Gamma_{P'} = \Gamma_P \times \Gamma_R^+, \qquad \mathcal{L}_P = \Gamma_R^+ \times \mathcal{L}_{P'}, \qquad \Gamma_P \times \mathcal{L}_P = \mathbb{Q}^\times. \tag{766}$$

The sign belongs to Γ_P; the other two groups in the first two factorizations contain only positive rationals. Hence no sign coordinate is hidden in these equations.

9.15.1 The old-boundary sector and its complete coordinates

Inside $W_{P'}^{(1)}$ define the invariant open subset

$$U_{P,P'} = \{x \in W_{P'}^{(1)} : x_r \neq 0 \text{ for every } r \in R\}. \tag{767}$$

It contains the generic stratum and exactly the one-zero strata labelled by $P \sqcup \{\infty\}$; the new-prime boundary strata are not identified with those old strata.

Theorem 9.42 (Coordinate homeomorphism retaining every new unit)**.** *For $x = ((x_q)_{q\in P'}, y) \in U_{P,P'}$, write*

$$m_r = v_r(x_r), \qquad u_r = r^{-m_r} x_r \in \mathbb{Z}_r^\times \quad (r \in R), \qquad h_R(x) = \prod_{r\in R} r^{m_r} \in \Gamma_R^+.$$

Define

$$z_q = h_R(x)^{-1} x_q \qquad (q \in P), \qquad z_\infty = h_R(x)^{-1} y, \tag{768}$$

$$b_r = h_R(x)^{-1} x_r = \left(\prod_{s\in R\setminus\{r\}} s^{-m_s} \right) u_r \in \mathbb{Z}_r^\times \qquad (r \in R). \tag{769}$$

Then

$$\Theta_{P,P'} : U_{P,P'} \xrightarrow{\cong} \Gamma_R^+ \times \big(W_P^{(1)} \times K_R\big), \qquad x \longmapsto \big(h_R(x), z, (b_r)_{r\in R}\big) \tag{770}$$

is a homeomorphism. Its inverse is the coordinatewise formula

$$x_q = hz_q \ (q \in P), \qquad x_r = hb_r \ (r \in R), \qquad y = hz_\infty. \tag{771}$$

For $\gamma \in \Gamma_P$ and $\rho \in \Gamma_R^+$, the action is

$$\Theta_{P,P'}((\gamma\rho)x) = \big(\rho h_R(x), \gamma z, (\gamma b_r)_{r\in R}\big). \tag{772}$$

Thus Γ_R^+ translates only the first discrete factor, whereas Γ_P acts diagonally on $W_P^{(1)} \times K_R$.

Proof. Since $v_r(h_R(x)) = m_r$, equation (769) has r-adic valuation zero. Every factor s^{-m_s} with $s \neq r$ is an r-adic unit, so the expanded formula retains all cross-prime factors and proves $b_r \in \mathbb{Z}_r^\times$. Multiplication by the nonzero rational $h_R(x)^{-1}$ does not change which of the old coordinates is zero. Hence $z \in W_P^{(1)}$.

Given (h, z, b) on the right, write $h = \prod_{r\in R} r^{m_r}$ and use (771). Its new r coordinate has valuation m_r, and applying (769) returns b_r. Applying (768) returns z. These calculations prove both inverse identities.

On $\mathbb{Q}_r^\times$, the valuation is locally constant and the unit map $x_r \mapsto r^{-v_r(x_r)}x_r$ is continuous. The group Γ_R^+ is discrete. Therefore $\Theta_{P,P'}$ is continuous on each clopen valuation sector, and the inverse formula is continuous. It is a homeomorphism.

Finally, γ has zero r-adic valuation for every $r \in R$, whereas $v_r(\rho x_r) = v_r(\rho) + m_r$. Thus $h_R((\gamma\rho)x) = \rho h_R(x)$. Substitution cancels ρ in every old and new unit coordinate and leaves multiplication by γ, proving (772). $\square$

The cross-prime factors in (769) are essential. Replacing b_r by u_r would make Γ_R^+ alter the alleged base coordinate and would destroy both the product action and the triple-inclusion calculation below.

Put

$$\mathfrak{B}_{P,R} = C_0(W_P^{(1)} \times K_R) \rtimes \Gamma_P.$$

Projection to $W_P^{(1)}$ is proper and equivariant, because K_R is compact. It therefore gives a nondegenerate injective map

$$\mathrm{pr}^{\rtimes}_{P,R} : \mathfrak{N}_P \longrightarrow \mathfrak{B}_{P,R}, \qquad fu_\gamma \longmapsto (f \circ \mathrm{pr}_{P,R})u_\gamma. \tag{773}$$

Injectivity follows directly from the faithful conditional expectations: if the image of a is zero, then the image of $E(a^*a)$ is zero; coefficient pullback is injective, so $E(a^*a) = 0$ and $a = 0$. We use reduced crossed products in this argument and then full equals reduced because all groups are amenable.

Let $e_{\rho,\sigma}$ be the matrix units on $\ell^2(\Gamma_R^+)$. The homeomorphism above gives a crossed-product isomorphism

$$\Xi_{P,P'} : \mathcal{K}(\ell^2\Gamma_R^+) \otimes \mathfrak{B}_{P,R} \xrightarrow{\cong} C_0(U_{P,P'}) \rtimes \Gamma_{P'}. \tag{774}$$

On the dense algebraic span, with the action convention of (772), it is exactly

$$e_{\rho,\sigma} \otimes (fu_\gamma) \longmapsto (\mathbf{1}_{\{\rho\}} \otimes f)u_{(\rho\sigma^{-1},\gamma)}. \tag{775}$$

Multiplying two displayed generators gives $\delta_{\sigma,\rho'}e_{\rho,\sigma'}$ on both sides, and involution interchanges the two matrix indices. The regular covariant representation identifies $C_0(\Gamma_R^+) \rtimes \Gamma_R^+$ with $\mathcal{K}(\ell^2\Gamma_R^+)$, so the dense formula extends to the asserted isomorphism.

9.15.2 Tail stabilization and the natural transition extension

Fix $H_0 = \ell^2(\mathbb{N}_0)$ and put

$$\begin{aligned}
\widetilde{\mathfrak{N}}_P &= \mathfrak{N}_P \otimes \mathcal{K}(\ell^2\mathcal{L}_P) \otimes \mathcal{K}(H_0),\\
\widetilde{\mathfrak{I}}_P &= \mathfrak{I}_P \otimes \mathcal{K}(\ell^2\mathcal{L}_P) \otimes \mathcal{K}(H_0),\\
\widetilde{\mathfrak{D}}_P &= \mathfrak{D}_P \otimes \mathcal{K}(\ell^2\mathcal{L}_P) \otimes \mathcal{K}(H_0).
\end{aligned} \tag{776}$$

These form the exact tail-stabilized natural extension. For $P \subset P'$, the unique factorization $\ell = \rho\ell'$ in $\mathcal{L}_P = \Gamma_R^+ \times \mathcal{L}_{P'}$ gives the unitary

$$V_{P,P'} : \ell^2(\mathcal{L}_P) \longrightarrow \ell^2(\Gamma_R^+) \otimes \ell^2(\mathcal{L}_{P'}), \qquad \delta_{\rho\ell'} \longmapsto \delta_\rho \otimes \delta_{\ell'}. \tag{777}$$

Extension by zero from the invariant open set $U_{P,P'}$, together with (773)– (777), defines an injective homomorphism

$$\Psi_{P,P'} : \widetilde{\mathfrak{N}}_P \longrightarrow \widetilde{\mathfrak{N}}_{P'}. \tag{778}$$

Written as a composition with every carrier visible, it is

$$\begin{aligned}
\mathfrak{N}_P \otimes \mathcal{K}(\ell^2\mathcal{L}_P) \otimes \mathcal{K}(H_0) &\xrightarrow{\operatorname{Ad} V_{P,P'}} \mathcal{K}(\ell^2\Gamma_R^+) \otimes \mathfrak{N}_P \otimes \mathcal{K}(\ell^2\mathcal{L}_{P'}) \otimes \mathcal{K}(H_0) \\
&\xrightarrow{1\otimes \mathrm{pr}^{\rtimes}_{P,R} \otimes 1} \mathcal{K}(\ell^2\Gamma_R^+) \otimes \mathfrak{B}_{P,R} \otimes \mathcal{K}(\ell^2\mathcal{L}_{P'}) \otimes \mathcal{K}(H_0) \\
&\xrightarrow{\Xi_{P,P'}\otimes 1} (C_0(U_{P,P'}) \rtimes \Gamma_{P'}) \otimes \mathcal{K}(\ell^2\mathcal{L}_{P'}) \otimes \mathcal{K}(H_0) \\
&\xrightarrow{\mathrm{ext}_U \otimes 1} \widetilde{\mathfrak{N}}_{P'}.
\end{aligned} \tag{779}$$

It restricts to an ideal map $\Psi^I_{P,P'}$ and induces a quotient map $\Psi^D_{P,P'}$, giving a morphism

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \widetilde{\mathfrak{I}}_P & \longrightarrow & \widetilde{\mathfrak{N}}_P & \longrightarrow & \widetilde{\mathfrak{D}}_P & \longrightarrow & 0 \\
 & & {\scriptstyle \Psi^I_{P,P'}}\big\downarrow & & {\scriptstyle \Psi_{P,P'}}\big\downarrow & & {\scriptstyle \Psi^D_{P,P'}}\big\downarrow & & \\
0 & \longrightarrow & \widetilde{\mathfrak{I}}_{P'} & \longrightarrow & \widetilde{\mathfrak{N}}_{P'} & \longrightarrow & \widetilde{\mathfrak{D}}_{P'} & \longrightarrow & 0.
\end{array} \tag{780}$$

On the quotient, every old boundary summand maps to the identically labelled summand and every new-prime summand has zero preimage. Thus "old" and "new" are labels in an explicit direct-sum map, not a claim of categorical separation.

We next remove the last arbitrary generic stabilization. For a generic $x = ((x_p), y)$ put

$$\begin{aligned}
g_P(x) &= \operatorname{sgn}(y) \prod_{p\in P} p^{v_p(x_p)} \in \Gamma_P, \\
a_p(x) &= \operatorname{sgn}(y) \prod_{q\in P\setminus\{p\}} q^{-v_q(x_q)} p^{-v_p(x_p)} x_p, \\
t_P(x) &= \log|y| - \sum_{p\in P} v_p(x_p) \log p.
\end{aligned} \tag{781}$$

This is the same retained coordinate as (698), with the unit $u_p = p^{-v_p(x_p)} x_p$ written inside the formula. The map

$$\Omega_P : W_{\varnothing,P} \xrightarrow{\cong} \Gamma_P \times K_P \times \mathbb{R}, \qquad x \longmapsto (g_P(x), (a_p(x))_p, t_P(x)) \tag{782}$$

has inverse

$$(g, a, t) \longmapsto g \cdot ((a_p)_{p\in P}, e^t), \tag{783}$$

and $\delta \in \Gamma_P$ acts by $(g, a, t) \mapsto (\delta g, a, t)$. The same valuation-sector argument as in Theorem 9.42 proves the homeomorphism. Consequently

$$\mathfrak{I}_P \cong \mathcal{K}(\ell^2\Gamma_P) \otimes C(K_P) \otimes C_0(\mathbb{R}) \tag{784}$$

by the exact matrix formula

$$(\mathbf{1}_{\{g\}}f)u_{gh^{-1}} \longmapsto e_{g,h} \otimes f.$$

Let

$$\mathcal{H}_{\mathbb{Q}} = \ell^2(\mathbb{Q}^{\times}) \otimes H_0, \qquad J_{\mathbb{Q}} = C_0(\mathbb{R}) \otimes \mathcal{K}(\mathcal{H}_{\mathbb{Q}}). \tag{785}$$

The basis unitary

$$\ell^2(\Gamma_P) \otimes \ell^2(\mathcal{L}_P) \xrightarrow{\cong} \ell^2(\mathbb{Q}^{\times}), \qquad \delta_\gamma \otimes \delta_\ell \longmapsto \delta_{\gamma\ell} \tag{786}$$

turns (784) into the simultaneous isomorphism

$$\widetilde{\chi}_P : \widetilde{\mathfrak{I}}_P \xrightarrow{\cong} C(K_P) \otimes J_{\mathbb{Q}}. \tag{787}$$

No unspecified bijection between separable Hilbert spaces occurs here: every basis vector is labelled by the displayed rational and the retained H_0 index.

Let

$$r_{P,P'} : K_{P'} = K_P \times K_R \longrightarrow K_P \tag{788}$$

be projection. It is onto, its fibre over every point is K_R, and its kernel as a compact-group homomorphism is $\{1\}_{K_P} \times K_R$. Direct substitution of (768)– (769) into (781) proves the exact ideal transition formula

$$\widetilde{\chi}_{P'} \Psi^I_{P,P'} \widetilde{\chi}_P^{-1} = r^*_{P,P'} \otimes \mathrm{id}_{J_{\mathbb{Q}}}. \tag{789}$$

Indeed, the P unit coordinates become the P coordinates of $K_{P'}$; after the residual Γ_P representative is fixed, the b_r become the R coordinates. On Hilbert-space matrix units, both sides use the same identity

$$\Gamma_P \times \Gamma_R^+ \times \mathcal{L}_{P'} = \Gamma_{P'} \times \mathcal{L}_{P'} = \mathbb{Q}^{\times}.$$

Equation (789) is nondegenerate and has the unique strict multiplier extension. In the standard multiplier-section coordinate, for a bounded strictly continuous $M : K_P \to M(J_{\mathbb{Q}})$, it is

$$(\overline{\Psi}^I_{P,P'} M)(c_P, c_R) = M(c_P), \tag{790}$$

$$(\dot{\Psi}^I_{P,P'}[M])(c_P, c_R) = [M(c_P)] \quad \text{in } Q(C(K_{P'}) \otimes J_{\mathbb{Q}}). \tag{791}$$

The second equation is a formula on multiplier representatives; it does not assert that every corona class has a globally chosen continuous multiplier representative.

Define the constant-unit embedding in these simultaneous coordinates by

$$\widetilde{\jmath}_P : J_{\mathbb{Q}} \longrightarrow \widetilde{\mathfrak{I}}_P, \qquad \widetilde{\chi}_P(\widetilde{\jmath}_P(b)) = 1_{K_P} \otimes b. \tag{792}$$

Then, before and after passing to multipliers or coronas,

$$\Psi^I_{P,P'} \widetilde{\jmath}_P = \widetilde{\jmath}_{P'}, \qquad \overline{\Psi}^I_{P,P'} \bar{\widetilde{\jmath}}_P = \bar{\widetilde{\jmath}}_{P'}, \qquad \dot{\Psi}^I_{P,P'} \dot{\widetilde{\jmath}}_P = \dot{\widetilde{\jmath}}_{P'}. \tag{793}$$

The image of the first map in (789) consists exactly of the sections constant on each K_R fibre. All nonconstant new-unit sections remain in the target; none has been declared equal to an old section.

Proposition 9.43 (Triple inclusions compose on the nose)**.** *If $P \subset P' \subset P''$, then, with the tensor factors ordered by the increasing numerical order of their prime labels,*

$$\Psi_{P',P''} \Psi_{P,P'} = \Psi_{P,P''} \tag{794}$$

and the same identities hold for the ideal and quotient maps and their ideal multiplier and corona extensions.

Proof. Put $R = P' \setminus P$ and $T = P'' \setminus P'$. For $x \in U_{P,P''}$, direct extraction gives

$$h_{R \sqcup T}(x) = h_R(x) h_T(x).$$

First extracting T divides every remaining coordinate, including every R coordinate, by $h_T(x)$. Extracting R next therefore gives

$$(h_R h_T)^{-1} x_q \quad (q \in P), \qquad (h_R h_T)^{-1} x_s \quad (s \in R \sqcup T),$$

which are exactly the direct old and unit coordinates. Thus the two homeomorphisms agree after the displayed reassociation of the labelled factors.

On tail bases, unique factorization gives the same equality

$$\mathcal{L}_P = \Gamma_R^+ \times \Gamma_T^+ \times \mathcal{L}_{P''}.$$

Hence both compositions take $e_{\rho\tau\ell,\,\sigma\upsilon m}$ to $e_{\rho,\sigma} \otimes e_{\tau,\upsilon} \otimes e_{\ell,m}$. Coefficient pullbacks and extension by zero compose literally. Formula (775) therefore proves (794) on a dense algebraic span, and continuity proves it everywhere. Restriction, quotient, and the unique strict extension of the nondegenerate ideal map preserve the equality.

This is the concrete instance of the standard induction-in-stages principle: composition of groupoid correspondences becomes the internal tensor product, and staged induction equals direct induction [91, Section 8.4.3, p. 35]. Li's notes place that principle in the groupoid framework developed from Williams's text [92]; the equality used here has nevertheless been proved above on the actual coordinates and matrix units. □

9.15.3 Boundary carriers and the connecting-map square

For $p \in P$, recall $Y_{p,P} = \prod_{q \in P \setminus \{p\}} \mathbb{Z}_q^\times$. There are literal factorizations and projections

$$Y_{p,P'} = Y_{p,P} \times K_R, \qquad \pi_{p;P,P'} : Y_{p,P'} \longrightarrow Y_{p,P}. \tag{795}$$

They intertwine translation by $g_{p,P'} = (g_{p,P}, (p)_{r \in R})$ and by $g_{p,P}$. Therefore

$$\pi_{p;P,P'}(H_{p,P'}) = H_{p,P} \tag{796}$$

and there is a surjection

$$\overline{\pi}_{p;P,P'} : Y_{p,P'}/H_{p,P'} \longrightarrow Y_{p,P}/H_{p,P}. \tag{797}$$

At the mapping-torus level the map is

$$B_{p,P'} \longrightarrow B_{p,P}, \qquad [(y, k_R), s] \longmapsto [y, s]. \tag{798}$$

It is well defined because both mapping-torus actions multiply every retained unit by p and translate s by $\log p$.

Under the complete K-groups of Lemma 9.37, the old finite summand of $(\Psi^D_{P,P'})_*$ is

$$h \longmapsto h \circ \overline{\pi}_{p;P,P'} \qquad \text{on } K_0, \tag{799}$$

$$[f] \longmapsto [f \circ \pi_{p;P,P'}] \qquad \text{on } K_1. \tag{800}$$

The second formula is well defined because pullback commutes with $1 - \sigma_{p,P}$. No injectivity assertion about the coinvariant map is being inserted without proof. The K_0 map is injective because (797) is onto.

For the archimedean summand use the choice-free orbit carrier

$$\overline{K}_P = K_P/\{c \sim -c\}, \qquad q_P^{\pm} : K_P \longrightarrow \overline{K}_P. \tag{801}$$

The projection $r_{P,P'}$ is sign-equivariant and induces

$$\overline{r}_{P,P'} : \overline{K}_{P'} \longrightarrow \overline{K}_P, \qquad [(c_P, c_R)] \longmapsto [c_P]. \tag{802}$$

Under the canonical Morita coordinate $K_0(\mathfrak{D}_{\infty,P}) = C(\overline{K}_P, \mathbb{Z})$, its transition is $\overline{r}^*_{P,P'}$. Choosing a clopen half H_P^+ recovers the earlier $C(H_P^+, \mathbb{Z})$ display, but no compatibility of independently chosen halves is required here.

Define the complete old-boundary K_0 transition by

$$T^0_{P,P'}((h_p)_{p\in P}, h_\infty) = \big((h_p \circ \overline{\pi}_{p;P,P'})_{p\in P}, (0)_{r\in R}, h_\infty \circ \overline{r}_{P,P'}\big). \tag{803}$$

The complete natural connecting map, in the same choice-free archimedean coordinate, is

$$\partial_P^{(1)}((h_p)_p, h_\infty) = -\sum_{p\in P} \iota_{p,P}(h_p) - h_\infty \circ q_P^{\pm}. \tag{804}$$

This is equation (755) transported through the homeomorphism $H_P^+ \cong \overline{K}_P$.

Theorem 9.44 (Exact finite-set connecting square)**.** *For every $P \subset P'$,*

$$r^*_{P,P'} \partial_P^{(1)} = \partial_{P'}^{(1)} T^0_{P,P'}. \tag{805}$$

Equivalently, this is the K-theory naturality square induced by (780)*. It retains all finite invariant functions, the complete archimedean sign orbit, and zero-labelled new boundary summands.*

Proof. The definitions give the two literal identities

$$r^*_{P,P'} \iota_{p,P}(h) = \iota_{p,P'}(h \circ \overline{\pi}_{p;P,P'}), \tag{806}$$

$$r^*_{P,P'}(h_\infty \circ q_P^{\pm}) = (h_\infty \circ \overline{r}_{P,P'}) \circ q_{P'}^{\pm}. \tag{807}$$

For the first, both sides evaluate h after dropping the p coordinate and then forgetting every new-prime unit coordinate. For the second, both sides send (c_P, c_R) to the sign orbit $[c_P]$. Substitute (806)– (807) into (804); every new-prime boundary coordinate in (803) is zero. The displayed equality follows with its minus sign unchanged. It also follows from the published naturality theorem for the already constructed extension morphism [85, Definition 21.1.1 and Example 21.1.2(a)], but the coordinate calculation proves exactly which functions that abstract square contains. $\square$

On $K_1(\mathfrak{D}_P)$, the transition is the direct sum of (800) in the old prime coordinates and zero in the new coordinates. The six-term sequence then supplies the induced middle-algebra maps. No splitting of the middle K_1 extension and no freeness claim for the profinite coinvariant groups is selected.

9.15.4 A coherent stable-to-natural family

We now apply the natural transition maps to the stable balancing source. Tensor the stable source extension (690) by $\mathcal{K}(\ell^2\mathbb{Q}^\times)$, reorder the two explicitly labelled Hilbert factors, and denote it by

$$0 \longrightarrow J_{\mathbb{Q}} \longrightarrow \widetilde{\mathcal{E}}_P \longrightarrow \widetilde{\mathcal{A}}_P \longrightarrow 0. \tag{808}$$

The source inclusion for $P \subset P'$ is

$$\widetilde{\iota}^{\mathcal{E}}_{P,P'}(a,m) = \big(\iota_{P,P'}(a), m\big), \tag{809}$$

where $\iota_{P,P'}$ inserts zero in every new-prime quotient coordinate. It is the identity on $J_{\mathbb{Q}}$ and uses the same fixed place-indexed corner isometries at every finite stage.

There is a particularly economical way to make the target edge maps coherent. For a finite place p, fix once the base edge morphism at $P_p = \{p\}$ supplied by Lemma 9.33, after the displayed tail stabilization. For the archimedean place use $P_\infty = \varnothing$. Only for this base object, extend the preceding definitions by

$$K_\varnothing = \{1\}, \qquad \Gamma_\varnothing = \{\pm 1\}, \qquad \mathbb{A}_{S_\varnothing} = \mathbb{R},$$

and take the literal reflection extension $C_0(\mathbb{R}) \rtimes \{\pm 1\}$ with generic open set $\mathbb{R}^\times$ and boundary $\{0\}$. Equations (764)– (779) then apply with $R = P$: the old-boundary sector is precisely the archimedean edge. Its generic, middle, and boundary base maps are the identity before the same tail stabilization. For every $P \supset P_v$ define

$$(\widetilde{\jmath}_P, \widetilde{\lambda}_{v,P}, \widetilde{\alpha}_{v,P}) = (\Psi^I_{P_v,P}, \Psi_{P_v,P}, \Psi^D_{P_v,P}) \circ (\widetilde{\jmath}_{P_v}, \widetilde{\lambda}_{v,P_v}, \widetilde{\alpha}_{v,P_v}). \tag{810}$$

The ideal component agrees with (792) by (793). Proposition 9.43 makes this definition independent of an intermediate finite set and gives

$$\Psi_{P,P'}\widetilde{\lambda}_{v,P} = \widetilde{\lambda}_{v,P'}, \tag{811}$$

$$\Psi^D_{P,P'}\widetilde{\alpha}_{v,P} = \widetilde{\alpha}_{v,P'} \qquad (v \in P \sqcup \{\infty\}). \tag{812}$$

Let $S_{\nu(v)}$ be the fixed isometries of (689), now acting on the final H_0 factor. Because every $\Psi_{P,P'}$ is the identity on that factor, its corner operation commutes exactly with transition. Form the quotient sum

$$\widetilde{\alpha}_P((a_v)_v) = \sum_{v \in P \sqcup \{\infty\}} (1 \otimes S_{\nu(v)})\widetilde{\alpha}_{v,P}(a_v)(1 \otimes S_{\nu(v)})^*. \tag{813}$$

Its Busby equation is

$$\widetilde{\tau}^{\mathrm{nat}}_P \widetilde{\alpha}_P = \dot{\widetilde{\jmath}}_P \widetilde{\tau}_P. \tag{814}$$

It follows place by place from the base Busby equations, then from (793), and finally from the orthogonal corner sum. Define in the two Busby pullbacks

$$\widetilde{\Phi}_P(a,m) = \big(\widetilde{\alpha}_P(a), \bar{\widetilde{\jmath}}_P(m)\big). \tag{815}$$

Theorem 9.45 (Simultaneous finite-set coherence of the stable embedding)**.** *The maps* (815) *are injective morphisms of the tail-stabilized source extensions into the tail-stabilized natural extensions. For every inclusion $P \subset P'$ of nonempty finite prime sets,*

$$\Psi_{P,P'}\widetilde{\Phi}_P = \widetilde{\Phi}_{P'}\widetilde{\iota}^{\mathcal{E}}_{P,P'}. \tag{816}$$

The same square commutes on ideals, quotients, ideal multipliers, ideal coronas, and connecting maps. For three nested finite sets, the square is strictly associative by (794)*.*

Proof. Equation (814) puts (815) in the natural Busby pullback. The constant ideal map and every orthogonal quotient corner are injective. The same two-outside-arrows argument as in Theorem 9.34 proves injectivity of the middle map.

For $a = (a_v)_{v \in P \sqcup \{\infty\}}$, equations (812) and (813), together with identity on the H_0 corner factor, give

$$\Psi^D_{P,P'} \widetilde{\alpha}_P(a) = \widetilde{\alpha}_{P'}(\iota_{P,P'} a).$$

The new quotient coordinates on the right are zero. On multiplier coordinates, equation (793) gives

$$\overline{\Psi}^I_{P,P'} \bar{\bar{\jmath}}_P(m) = \bar{\bar{\jmath}}_{P'}(m).$$

The two coordinates in the natural Busby pullback are therefore equal, which proves (816). Restriction and quotient give the ideal and boundary squares; strict extension gives the ideal multiplier and corona squares; and Theorem 9.44 gives the complete connecting map. Triple associativity is already proved on the middle algebras and hence persists on every induced component. □

This theorem constructs a coherent replacement for the independently chosen coordinates in Subsection 9.13. It does not assert that an arbitrary prior family $(\chi_P)_P$ was secretly coherent. For any such family, the exact unresolved datum for an inclusion is the pair of homomorphisms

$$\chi_{P'} \Psi^I_{P,P'} \quad \text{and} \quad (r^*_{P,P'} \otimes \mathrm{id})\chi_P. \tag{817}$$

Coherence means equality of this pair after placing both maps in the same declared stabilization. Formula (787) makes the pair equal by construction. No quotient by a vague "unitary equivalence" is taken.

9.15.5 The exact middle-multiplier obstruction

For $P \subsetneq P'$, let

$$\widetilde{\mathfrak{O}}_{P,P'} = (C_0(U_{P,P'}) \rtimes \Gamma_{P'}) \otimes \mathcal{K}(\ell^2 \mathcal{L}_{P'}) \otimes \mathcal{K}(H_0). \tag{818}$$

The map $\Psi_{P,P'}$ is nondegenerate as a map into $\widetilde{\mathfrak{O}}_{P,P'}$ and therefore has a unique strict extension

$$M(\widetilde{\mathfrak{N}}_P) longrightarrow M(\widetilde{\mathfrak{O}}_{P,P'}). \tag{819}$$

The sector is an essential ideal in $\widetilde{\mathfrak{N}}_{P'}$: it contains the dense generic coefficient ideal, and the Fourier-coefficient argument of Lemma 9.31 applies verbatim. It is nevertheless proper. For every $r \in R$, the nonzero new boundary summand $\widetilde{\mathfrak{D}}_{r,P'}$ is annihilated by the sector quotient.

Consequently $\Psi_{P,P'}$ is degenerate as a map into the full $\widetilde{\mathfrak{N}}_{P'}$. If it had a unital strict extension $M(\widetilde{\mathfrak{N}}_P) \to M(\widetilde{\mathfrak{N}}_{P'})$, the image of an approximate unit would converge strictly to one. Applying the quotient onto $\widetilde{\mathfrak{D}}_{r,P'}$ would make the identically zero image converge strictly to its nonzero identity multiplier, a contradiction. Thus the canonical middle multiplier target is $M(\widetilde{\mathfrak{O}}_{P,P'})$, not $M(\widetilde{\mathfrak{N}}_{P'})$. Essentiality embeds $M(\widetilde{\mathfrak{N}}_{P'})$ into $M(\widetilde{\mathfrak{O}}_{P,P'})$ in the opposite direction; reversing that arrow would be an unsupported categorical assertion.

This obstruction does not damage the Busby construction. The generic ideal map is nondegenerate into the complete generic ideal and has exactly the multiplier and corona maps (790)– (791). Those are the maps used in (814) and in the pullback formula (815).

Exact boundary of the result. The finite-set system now has explicit maps on compact unit carriers, generic ideals, finite and archimedean boundary quotients, tail Hilbert spaces, orthogonal corners, Busby pullbacks, ideal multipliers, ideal coronas, and connecting maps. All triple-inclusion squares commute in the declared basis. The proper old-sector ideal proves why there is no unital full-middle multiplier map when a prime is genuinely added. The theorem is an existence and construction result for a coherent coordinate family; it is not a choice-independence theorem for every possible stabilization. It does not select a splitting of any coinvariant or middle K_1 sequence, identify a new-prime unit fibre with a point, or imply an endpoint, individual-zero, critical-line, or global zeta statement.

9.16 Candidate interfaces retained without overclaiming

The following broader source interfaces remain useful research leads. The table separates the bounded theorems now proved from the additional evidence that would be required for any stronger interface claim.

Source layer	Precisely retained datum	Missing typed evidence before admission
Global finite-prime assembly	The local boundary is -1; the natural unit-retaining multi-place algebra is Definition 9.24; Theorem 9.34 lifts the stable codiagonal square to an injective extension morphism; and Theorem 9.41 computes the full nonconstant finite invariants, finite coinvariants, archimedean sign-even image, natural connecting map, K_0 kernel, and K_1 extension	Any stronger surjectivity, isomorphism, or Morita-equivalence claim must account for the explicit odd-difference quotient $\mathcal{Q}_P$, the finite-level coinvariant multiplicities, and coherence under enlarging P. Proposition 9.36 retains the sign difference in a separate split carrier; it does not identify that carrier with the natural boundary algebra
Archimedean APS/global splice	The reflection crossed product, its matrix model, quotient, two boundary generators, map $\partial_\infty(a,b) = -(a+b)$, the whole-line and two compact-interval index realizations, and the stable multi-place codiagonal and the natural arithmetic representative are proved separately	A specified APS domain, endpoint projectors, adjoint domain, and an exact morphism from that operator realization to the extension data
Orbifold labels	The conditional (C, ρ) label set and the S_3 count $(1, 1, 2, 3, 3, 2, 2, 2)$	The exact VOA/orbifold hypotheses and a content-read primary source for the claimed fusion setting
Leech/moonshine	The formal A_2 direct-sum rank count and the displayed theta identity as candidate data	A human-source chain for the $12A_2$ neighbour construction and the stated simple-current identifications
Spectral/F-theory layers	Rank-one, ADE, Mordell–Weil, and BSD expressions as separately typed analogies or sourced dictionaries	Exact operators, bases, arithmetic fibration, and maps; no common object follows from shared notation
CUE/Epstein bridge	The four listed structural coincidences	An explicit transform-level map between compact pushforward measures and discrete lattice-shell measures

Consequently, the local source's broad "primitive defect realization theorem" is replaced by a collection of theorems above plus an open-interface ledger. No endpoint statement, individual-zero map, critical-line conclusion, or Riemann-zeta zero result is inferred from the finite algebraic coincidences.

9.17 Open interface obligations from this tranche

The remaining obligations are the following.

1. Give a direct distribution-level proof of (32), rather than only the present subtraction proof, and extend the verified common interface to the sharp-Chebyshev presentation while retaining every pole, trivial-zero, summation-order, test-space, and jump term.

2. Recover both (19) and (21) as typed kernel specializations of that common statement, including the two prime ranges, logarithmic-derivative term, pole term, trivial-zero series, the $a_n(x)$ split, and every remainder.

3. Close the explicit gamma/trivial-zero commutator after (34). Guinand, Weil, and Montgomery have now been read completely; the remaining issue is a termwise proof, not source identification.

4. Read the primary Bogomolny–Keating finite-height correlation papers before transporting their formulas into any local programme. Keating–Snaith has now been read completely; its 2000 moment conjecture must next be crosswalked against the subsequent moment literature before any claim about present knowledge.

5. Formalize the exp/log, Fourier, convolution, even-extension, finite-place, zero-pairing, and Poisson-kernel maps. Also formalize the finite-height scale transition, Fourier reflection, and weighted pair measure, retaining every integrability hypothesis and exceptional locus. These are explicit paper proofs, not Lean receipts.

10 Vacuum fluids, thermal time, and arithmetic spectral tests

This investigation follows two concrete constructions: the Rindler fluid–gravity map and the Bost–Connes cooling map. A second, independent calculation evaluates the complete Weil form without imposing RH. The aim is to determine whether these constructions expose an off-critical zeta zero. No such zero or negative Weil certificate has been established here. The explicit calculations below are retained regardless of that outcome; no novelty claim is made for elementary consequences of the cited constructions.

10.1 A Ricci-flat fluid metric and its temperature coordinates

Bredberg–Keeler–Lysov–Strominger construct the following metric [2, §5.1, (14)–(20)]. Coordinates are $(\tau, r, x^1, \ldots, x^p)$, the timelike cutoff is $r = r_c > 0$, and the future Rindler horizon is at $r = 0$. Spatial indices use δ_{ij}. Retain their original velocity $v_i(\tau, x)$ and pressure $P(\tau, x)$:

$$\begin{aligned} ds^2 = &- r\,d\tau^2 + 2\,d\tau\,dr + \delta_{ij}dx^i dx^j \\ &- 2(1 - r/r_c)v_i\,dx^i d\tau - 2r_c^{-1}v_i\,dx^i dr \\ &+ (1 - r/r_c)\big[(v^2 + 2P)d\tau^2 + r_c^{-1}v_iv_j dx^i dx^j\big] + r_c^{-1}(v^2 + 2P)d\tau\,dr \\ &- r_c^{-1}(r^2 - r_c^2)\Delta v_i\,dx^i d\tau + \cdots . \end{aligned} \tag{820}$$

The grading is $v = O(\epsilon)$, $P = O(\epsilon^2)$, $\partial_i = O(\epsilon)$, $\partial_\tau = O(\epsilon^2)$. The displayed metric solves the vacuum equations through the displayed hydrodynamic orders when

$$\partial_i v^i = 0, \qquad \partial_\tau v_i + v^j\partial_j v_i + \partial_i P - r_c\Delta v_i = 0. \tag{821}$$

This is a source-level perturbative reconstruction, not a claimed convergent metric for an arbitrary singular fluid. The boundary conditions are a flat induced metric at r_c, regularity at the future horizon, and the incoming-data condition specified in the source. Changing these data changes the reconstruction.

Here is the exact coordinate calculation relating its viscosity to temperature. It uses the flat zeroth-order metric and introduces no identification of bulk quantum algebras. For $r > 0$ put

$$\eta = \tau - \log r, \qquad \xi = 2\sqrt{r}, \qquad r = \xi^2/4, \quad \tau = \eta + \log(\xi^2/4).$$

Substitution of $d\tau = d\eta + dr/r$ gives

$$-r\,d\tau^2 + 2\,d\tau\,dr = -r\,d\eta^2 + \frac{dr^2}{r} = -\frac{\xi^2}{4}d\eta^2 + d\xi^2. \tag{822}$$

The further map to Minkowski coordinates is $X^0 = \xi\sinh(\eta/2)$, $X^1 = \xi\cosh(\eta/2)$. Its image is the right wedge $X^1 > |X^0|$; the inverse is $\xi = \sqrt{(X^1)^2 - (X^0)^2}$ and $\eta = 2\,\mathrm{arctanh}(X^0/X^1)$. The singularity of these inverse coordinates on the horizon is not by itself a curvature singularity: the metric in (X^0, X^1, x) is flat.

On a fixed-ξ orbit proper time is $dt = (\xi/2)d\eta$. Twice differentiating its Minkowski coordinates with respect to t gives proper acceleration $a = 1/\xi$. The Rindler-vacuum temperature relation used by Eling–Fouxon–Oz is $T_{\mathrm{loc}} = a/(2\pi)$, in units $\hbar = c = k_B = 1$ [1, (32)–(33) and following discussion]. At the cutoff this yields

$$T_{\mathrm{loc}} = \frac{1}{4\pi\sqrt{r_c}}, \qquad t = \sqrt{r_c}\,\tau, \quad V_i(t,x) = \frac{v_i(t/\sqrt{r_c}, x)}{\sqrt{r_c}}, \quad \Pi(t,x) = \frac{P(t/\sqrt{r_c}, x)}{r_c}. \tag{823}$$

On this orbit the exact relation is $t = \sqrt{r_c}(\eta + \log r_c)$. The inverse field map is $v_i(\tau, x) = \sqrt{r_c}V_i(\sqrt{r_c}\tau, x)$ and $P(\tau, x) = r_c\Pi(\sqrt{r_c}\tau, x)$. Every term in (821) then transforms explicitly:

$$\partial_\tau v_i = r_c\partial_t V_i, \quad v^j\partial_j v_i = r_c V^j\partial_j V_i, \quad \partial_i P = r_c\partial_i\Pi, \quad r_c\Delta v_i = r_c^{3/2}\Delta V_i.$$

Consequently the proper-time equation is

$$\partial_t V_i + V^j\partial_j V_i + \partial_i\Pi - \nu_{\mathrm{loc}}\Delta V_i = 0, \qquad \nu_{\mathrm{loc}} = \sqrt{r_c} = \frac{1}{4\pi T_{\mathrm{loc}}}. \tag{824}$$

Both the source viscosity r_c and the proper-time viscosity $\sqrt{r_c}$ are therefore retained, together with the map between them.

Eling–Fouxon–Oz also derive Navier–Stokes by projecting the Einstein equations onto a horizon. The normal–normal equation supplies incompressibility, and the normal–tangential equation supplies momentum balance. Their Rindler specialization has $R_{AB} = 0$ and explicitly identifies the thermal interpretation with the vacuum viewed by accelerated observers [1, (21)–(25), (32)–(33)]. The reconstruction-to-singularity question is part of the published programme: Bredberg–Strominger discuss its convergence requirement in [3, §5]. To apply it at blowup, the particular fluid, boundary data, and reconstructed geometric invariant must be followed up to that time. The displayed formal expansion alone does not supply such control.

10.2 Thermal time in the arithmetic energy basis

In the rational BC representation on $\mathcal{H} = \ell^2(\mathbb{N}^\times)$, write

$$H\varepsilon_n = (\log n)\varepsilon_n, \quad \mu_q\varepsilon_n = \varepsilon_{qn}, \quad e(r)\varepsilon_n = e^{2\pi i n r}\varepsilon_n, \qquad q \in \mathbb{N}^\times,\ r \in \mathbb{Q}/\mathbb{Z}. \tag{825}$$

Here H is the self-adjoint diagonal operator with domain $\{(a_n) : \sum_n (\log n)^2|a_n|^2 < \infty\}$. Let $\gamma_t(a) = e^{itH}ae^{-itH}$. Direct basis evaluation gives $\gamma_t(\mu_q) = q^{it}\mu_q$ and $\gamma_t(e(r)) = e(r)$. For $\beta > 1$ the trace and the corresponding Gibbs state are

$$Z(\beta) = \sum_{n\geq 1} n^{-\beta} = \zeta(\beta), \qquad D_\beta = Z(\beta)^{-1}e^{-\beta H}, \qquad \varphi_\beta(a) = \mathrm{Tr}(D_\beta a). \tag{826}$$

These are the low-temperature formulas of Bost–Connes [5, §§5–7].

Proposition 10.1 (Exact modular-time calculation)**.** *With the Connes–Rovelli modular sign convention, $\alpha_u(a) = D_\beta^{-iu}aD_\beta^{iu}$ on $B(\mathcal{H})$, one has*

$$\alpha_u = \gamma_{\beta u}, \qquad \alpha_u(E_{mn}) = (m/n)^{i\beta u}E_{mn}, \quad E_{mn} = |\varepsilon_m\rangle\langle\varepsilon_n|. \tag{827}$$

The standard opposite modular convention gives $\gamma_{-\beta u}$.

Proof. The positive eigenvalues of D_β are $p_n = n^{-\beta}/Z(\beta)$. Functional calculus gives $D_\beta^{-iu}E_{mn}D_\beta^{iu} = (p_m/p_n)^{-iu}E_{mn}$, which is exactly (827). Equivalently, $D_\beta^{-iu} = Z(\beta)^{iu}e^{i\beta uH}$, and the scalar factors cancel in conjugation. The equality for all bounded a follows directly from these unitary conjugations, without extending the unbounded operator H to all vectors. In the Hilbert–Schmidt standard representation, $\Delta = L_{D_\beta}R_{D_\beta}^{-1}$; hence conjugation by Δ^{-iu} on left multiplication implements the same formula. This also identifies the sign with [4, §2.1, footnote 3; §4.1, (29), (36)–(42)]. □

On bounded observables this real flow is defined for every real u. The cancellation just proved concerns the automorphism, not the thermal expectations: the latter retain $1/Z(\beta)$. In particular, differentiating the absolutely and locally uniformly convergent Dirichlet series gives

$$\langle H\rangle_\beta = -\frac{\zeta'(\beta)}{\zeta(\beta)}, \qquad \mathrm{Var}_\beta(H) = \frac{d^2}{d\beta^2}\log\zeta(\beta). \tag{828}$$

Indeed the first two derivatives insert $-\log n$ and $(\log n)^2$; quotient differentiation subtracts the square of the first moment. The local Laurent expansion $\zeta(1+\epsilon) = \epsilon^{-1} + \gamma_E + O(\epsilon)$ therefore yields $\langle H\rangle_{1+\epsilon} = \epsilon^{-1} - \gamma_E + O(\epsilon)$ and $\mathrm{Var}_{1+\epsilon}(H) = \epsilon^{-2} + O(1)$. These are temperature-parameter divergences with an explicit variable. Bost–Connes also construct KMS states for $0 < \beta \leq 1$; the divergent ordinary trace at 1 is not their definition in that range [5, §7].

10.3 Explicit observables retaining the zeta denominator

Define the real Dirichlet character $\chi_3(n) = 0, 1, -1$ for $n \equiv 0, 1, 2 \pmod 3$, respectively.

Proposition 10.2 (A bounded arithmetic observable)**.** *In* (825),

$$A_3 = \frac{e(1/3) - e(-1/3)}{i\sqrt{3}}, \qquad A_3 = A_3^*, \quad \|A_3\| = 1, \qquad \varphi_\beta(A_3) = \frac{L(\beta,\chi_3)}{\zeta(\beta)} \quad (\beta > 1). \tag{829}$$

The meromorphic continuation of this expectation has a pole of order $m - k$ at a zeta zero of multiplicity m when the numerator has zero order $k < m$. When $k \geq m$ the singularity is removable.

Proof. Since $e(r)^* = e(-r)$, the operator is self-adjoint. Its nth eigenvalue is $2\sin(2\pi n/3)/\sqrt{3} = \chi_3(n)$, proving the norm and all matrix entries. The trace is the absolutely convergent sum $\zeta(\beta)^{-1}\sum_n \chi_3(n)n^{-\beta}$. For the pole statement write locally $\zeta(s) = (s-\rho)^m a(s)$ and $L(s,\chi_3) = (s-\rho)^k b(s)$ with $a(\rho)b(\rho) \neq 0$. Their ratio is $(s-\rho)^{k-m}b(s)/a(s)$. This explicitly records possible cancellation instead of asserting that a single observable must detect every zero. □

The full root-of-unity trace provides a second check. For $\Re s > 1$ put $S_j(s) = \sum_{n\equiv j\ (3)} n^{-s}$. Then $S_0 = 3^{-s}\zeta(s)$, $S_1 + S_2 = (1-3^{-s})\zeta(s)$ and $S_1 - S_2 = L(s, \chi_3)$. Solving this two-by-two linear system gives

$$\frac{\mathrm{Tr}(e(1/3)e^{-sH})}{\zeta(s)} = \frac{3^{1-s}-1}{2} + \frac{i\sqrt{3}}{2}\frac{L(s,\chi_3)}{\zeta(s)}. \tag{830}$$

In contrast, even–odd decomposition gives $\mathrm{Tr}(e(1/2)e^{-sH})/\zeta(s) = 2^{1-s} - 1$. Here the zeta factor cancels exactly. These two computations identify what information the respective observable retains.

Proposition 10.3 (Finite-prime projections and the reciprocal zeta trace)**.** *For a finite set of primes S, define $P_S = \prod_{p\in S}(1-\mu_p\mu_p^*)$ in the represented BC algebra. As S increases through all primes, P_S converges strongly to the rank-one projection P_1 onto $\mathbb{C}\varepsilon_1$, but not in operator norm. For real $\beta > 1$,*

$$\varphi_\beta(P_S) = \prod_{p\in S}(1-p^{-\beta}), \qquad \varphi_\beta(P_1) = \frac{1}{\zeta(\beta)}. \tag{831}$$

Proof. The range projection $\mu_p\mu_p^*$ has eigenvalue 1 precisely when $p \mid n$; these projections commute. Thus $P_S\varepsilon_n$ is ε_n when no prime in S divides n, and is zero otherwise. Each $n > 1$ eventually vanishes, while $n = 1$ survives. For $a \in \ell^2$, dominated convergence in $\sum_n |a_n|^2$ proves strong convergence. A prime $q \notin S$ satisfies $(P_S - P_1)\varepsilon_q = \varepsilon_q$, so $\|P_S - P_1\| = 1$. Inclusion–exclusion, or finite removal of Euler factors from the absolutely convergent Dirichlet series, gives

$$\sum_{(n,\prod_{p\in S} p)=1} n^{-\beta} = \zeta(\beta)\prod_{p\in S}(1-p^{-\beta}).$$

Finally $\mathrm{Tr}(P_1 e^{-\beta H}) = 1$. The normal Gibbs state extends to $B(\mathcal{H})$, so its expectation at this strong limit is defined. This argument asserts membership of P_1 in the represented von Neumann closure; it does not assert a norm-convergent infinite product in the BC C^*-algebra. □

Meromorphic continuation of the last scalar function has a pole at every zeta zero, with that zero's multiplicity. Finite products are entire in s; their limit above is justified on $\Re s > 1$, not automatically near those poles. This is an explicit limiting question to investigate. Likewise the energy characteristic function has the exact trace $\mathrm{Tr}(D_\beta e^{-itH}) = \zeta(\beta+it)/\zeta(\beta)$ for real t, $\beta > 1$. Both coordinates of its complex zeta argument are specified.

10.4 Cooling and the spectral-support formulation

The zeros are also present in a published spectral construction, not just in continued scalar expectations [6, §4.2–4.4]. Let $\mathcal{A}$ be the BC groupoid algebra of functions with finite support in k and finite level in $u \in \widehat{\mathbb{Z}}$ (factoring through some $\mathbb{Z}/N\mathbb{Z}$). Its groupoid consists of $(k,u) \in \mathbb{Q}_+^* \times \widehat{\mathbb{Z}}$ with $ku \in \widehat{\mathbb{Z}}$; composition is $(k_1, k_2u)(k_2, u) = (k_1k_2, u)$. For $x = \int x(t)U_t dt$ in the analytic crossed-product algebra $\widehat{\mathcal{A}}_\beta$, the source's coordinate map is

$$f(k,u,\lambda) = \int_{\mathbb{R}} x(t)(k,u)\lambda^{it}dt, \qquad \theta_a(x) = \int_{\mathbb{R}} a^{it}x(t)U_t dt \quad (a > 0). \tag{832}$$

The analytic strip is $I_\beta = \{z : 0 \le \Im z \le \beta\}$, with $\beta > 1$. In the source definition, the finitely many coefficient functions extend holomorphically to that strip and have Schwartz boundary behavior.

Cooling evaluates representations, followed by trace:

$$\pi : \widehat{\mathcal{A}}_\beta \longrightarrow C(\widetilde{\Omega}_\beta, \mathcal{L}^1), \quad \pi(x)(\varepsilon, H) = \pi_{\varepsilon,H}(x), \qquad \delta = \mathrm{Trace}\circ\pi. \tag{833}$$

Here $\widetilde{\Omega}_\beta \simeq \widehat{\mathbb{Z}}^* \times \mathbb{R}_+^*$ and $\mathcal{L}^1$ denotes trace-class operators. In degree zero,

$$\delta(x)(u,\lambda) = \sum_{n\geq 1} f(1, nu, n\lambda). \tag{834}$$

This formula can be checked directly: in the representation at (u,λ), the Hamiltonian has nth diagonal entry $\log n + \log\lambda$, and the nth diagonal entry of $x(t)$ is $x(t)(1,nu)$. Taking the trace of the integrated representation gives exactly (834).

The construction restricts to the trace-kernel cyclic submodule $\widehat{\mathcal{A}}^\natural_{\beta,0}$ before taking

$$\delta : \widehat{\mathcal{A}}^\natural_{\beta,0} \longrightarrow \mathcal{S}^\natural(\widetilde{\Omega}_\beta), \qquad D(\mathcal{A},\varphi) = \operatorname{coker}\delta.$$

The cokernel is in cyclic modules. This specifies the category and the discarded trace component; it is not an unrestricted algebra quotient. For the trivial character projector $p_{\mathbf{1}}$, put $M = HC_0(p_{\mathbf{1}}D(\mathcal{A},\varphi))$. Here $\mathrm{Hol}(I_\beta)$ denotes the source's holomorphic multiplier algebra of $\mathcal{S}(I_\beta)$: multiplication by h must preserve that Schwartz strip space. It does not mean all holomorphic functions with unrestricted growth. The one-dimensional module $\mathbb{C}_z$ uses evaluation at z. The source's spectral theorem [6, Theorem 4.16, (4.36)] gives, in the open critical strip,

$$M \otimes_{\mathrm{Hol}(I_\beta)} \mathbb{C}_z \neq 0 \quad \Longleftrightarrow \quad \zeta(-iz) = 0. \tag{835}$$

For $\rho = \sigma + i\gamma$ the coordinate and inverse are

$$z = i\rho = -\gamma + i\sigma, \qquad \rho = -iz. \tag{836}$$

Thus a nonzero fibre with $0 < \Im z < 1$ and $\Im z \neq 1/2$ is exactly an off-critical zeta zero in these coordinates. This is a genuine target for the investigation. No such fibre has been exhibited here.

The step to investigate on the fluid side is the image of its geometric or quantum observables under a concrete correspondence into this arithmetic construction. The metric map (820), modular map (827), and cooling map (833) have now been specified. They do not yet constitute a constructed map from a singular fluid to a nonzero off-critical fibre. This reports the current extent of the calculation, not a theorem excluding such a map.

10.5 A direct arithmetic test independent of the fluid route

Use $\widehat{f}(z) = \int_{\mathbb{R}} f(u)e^{-iuz}du$ and $f^\star(u) = \overline{f(-u)}$. For $f, g \in C_c^2(\mathbb{R})$ define

$$W(f,g) = \sum_\rho m_\rho \overline{\widehat{f}(\overline{\gamma_\rho})}\widehat{g}(\gamma_\rho), \qquad \gamma_\rho = \frac{\rho - 1/2}{i}. \tag{837}$$

The sum is over distinct nontrivial zeros and m_ρ is their multiplicity. This is conjugate-linear in its first slot, reversing the slots of [10, §1.2]; diagonal values agree. Twice integrating by parts gives $\widehat{f}(z) = O((1+|\Re z|)^{-2})$ uniformly for $|\Im z| \leq 1/2$. The zero count is $O(T\log T)$, so the displayed sum converges absolutely. Functional-equation pairing $\gamma \leftrightarrow \overline{\gamma}$ shows it is Hermitian.

Proposition 10.4 (Finite negative-certificate test)**.** *Let $G_{ij} = W(f_i, f_j)$ for a finite family and $a \in \mathbb{C}^d$. A rigorously established value $a^*Ga < 0$ refutes RH. If $\widetilde{G}$ approximates G with $|G_{ij} - \widetilde{G}_{ij}| \leq e_{ij}$, it suffices to establish*

$$\Re(a^*\widetilde{G}a) + \sum_{i,j} e_{ij}|a_i||a_j| < 0. \tag{838}$$

Proof. Sesquilinearity gives $a^*Ga = W(\sum a_i f_i, \sum a_i f_i)$. Under RH each γ_ρ is real and this last expression is $\sum m_\rho |\widehat{\sum a_i f_i}(\gamma_\rho)|^2 \geq 0$. The contrapositive proves the first statement. The entrywise triangle inequality bounds the error by the displayed sum, proving the second. All errors, including the analytic integration and omitted tails, must be included in e_{ij}. No positivity assumption is used to compute G. □

For explicit evaluation put $h_{ij} = f_j * f_i^\star$. Then $\widehat{h}_{ij}(z) = \widehat{f}_j(z)\overline{\widehat{f}_i(\overline{z})}$. The complete explicit formula is

$$W(f_i, f_j) = \widehat{h}_{ij}(i/2) + \widehat{h}_{ij}(-i/2) + \int_{\mathbb{R}} \widehat{h}_{ij}(t) \left[\frac{1}{2\pi}\Re\psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{\log\pi}{2\pi}\right] dt$$
$$- \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} \{h_{ij}(\log n) + h_{ij}(-\log n)\}. \tag{839}$$

The finite-prime terms, gamma factor, and both pole terms are all retained [26, 10]. Compact support makes the arithmetic sum finite, but does not remove the other terms.

10.6 A completely specified compact-support calibration

Let $b_4 = \mathbf{1}_{[-1/2,1/2]}^{*4}$ and $B = b_4 * b_4 = \mathbf{1}_{[-1/2,1/2]}^{*8}$. Thus $b_4 \in C_c^2(\mathbb{R})$, $B \in C_c^6(\mathbb{R})$, and their supports are $[-2, 2]$ and $[-4, 4]$. The repeated convolution, or successive integration of the indicator, gives the exact truncated-power formula

$$B(x) = \frac{1}{7!}\sum_{k=0}^{8}(-1)^k\binom{8}{k}(x+4-k)_+^7, \qquad B(0) = \frac{151}{315}. \tag{840}$$

For completeness, the convolution of m indicators of $[0, 1]$ is the first derivative of the volume $m!^{-1}\sum_{k=0}^{m}(-1)^k\binom{m}{k}(x-k)_+^m$ of the clipped simplex $\{u_i \in [0, 1], \sum u_i \leq x\}$; the volume formula follows by inclusion–exclusion on the inequalities $u_i > 1$. Differentiating once (to obtain the density), and translating by $m/2$, proves (840) for $m = 8$.

Set $f_T(u) = e^{-iTu}b_4(u)$ for real T. Its autocorrelation is $h_T(x) = e^{-iTx}B(x)$, and $\widehat{h}_T(z) = \operatorname{sinc}((z+T)/2)^8$, where $\operatorname{sinc} w = \sin(w)/w$ with value 1 at zero. Define $M(w) = (\sinh(w/2)/(w/2))^8$, also filled in continuously at zero. An equivalent real-space form of (839) is

$$Q(T) := W(f_T, f_T) = 2\Re M(1/2 + iT) - (\gamma_E + \log(4\pi))B(0)$$
$$- \int_0^4 \frac{e^{x/2}B(x)\cos(Tx) - B(0)}{\sinh x}\,dx + B(0)\log\coth 2$$
$$- 2\sum_{\substack{n=p^k\leq e^4\\ p \text{ prime},\ k\geq 1}} \frac{\log p}{\sqrt{n}} B(\log n)\cos(T\log n). \tag{841}$$

To verify the constants, substitute $F(y) = y^{-1/2}h_T(\log y)$ in (9)–(11) and then put $y = e^x$. The archimedean integrand becomes $(e^{x/2}B(x)\cos(Tx) - B(0))/\sinh x$. For $x > 4$, $B(x) = 0$; its exact contribution after the minus sign is $B(0)\int_4^\infty dx/\sinh x = B(0)\log\coth 2$. The two Mellin-pole terms give $2\Re M(1/2 + iT)$. This proves (841), including its nonzero tail.

The executable implementation uses the exact polynomial on each interval $[j, j+1]$, $j = 0, 1, 2, 3$. To remove the apparent singularity at 0 analytically, write $a_\pm = 1/2 \pm iT$ and $\operatorname{sinhc}(z) = \sinh(z)/z$. Then

$$\frac{e^{x/2}\cos(Tx) - 1}{x} = \frac{1}{2}\sum_{a\in\{a_+,a_-\}} a e^{ax/2}\operatorname{sinhc}(ax/2).$$

Thus the numerator divided by x is evaluated using the polynomial $(B(x)-B(0))/x$ plus this entire function; division is by $\operatorname{sinhc}(x)$, nonzero on the integration interval. This is an identity of analytic functions near $[0,1]$, not the deletion of a small interval near zero.

The local certificate directory is

```
certificates/rh_counterexample_20260908/
```

Its scripts `verify_bc_observables.py` and `verify_weil_bspline.py` retain their parameters, software versions, source-code hashes, and output classifications. The BC check includes exact symbolic residue identities and independent 60-digit evaluations at $\beta = 2, 3, 5$; their agreement is a numerical cross-check of the proof, not a formal proof of RH or of a complex zero. The Weil test evaluates the complete arithmetic expression; diagnostic sums over finitely many known zeros are never used as a full-form certificate.

The first run explored $T = 0, 1/2, \ldots, 100$ in floating point and then enclosed six selected full-form values using 192-bit Arb arithmetic. The following deliberately widened decimal intervals contain its enclosures:

T	lower bound	upper bound
0	$2.5040847844497\,10^{-8}$	$2.5040847844498\,10^{-8}$
2.5	$2.3576262787854\,10^{-8}$	$2.3576262787855\,10^{-8}$
7.5	$7.8968660947314\,10^{-9}$	$7.8968660947315\,10^{-9}$
8	$1.4007953026914\,10^{-8}$	$1.4007953026915\,10^{-8}$
14	0.9939671803371	0.9939671803372
40	0.8989486891251	0.8989486891252

These are positive tests, not an RH counterexample.

There is also a fully evaluated matrix test. Put $g_j(u) = b_4(u - j/4)$, $j = 0, \ldots, 4$. Its ij correlation is $B(x - (j-i)/4)$. The evenized correlation is

$$C_d(x) = \frac{B(x-d) + B(x+d)}{2}, \qquad d = |i-j|/4.$$

Replacing a correlation by its even part leaves the explicit formula unchanged: every term pairs its arguments with their negatives. Hence $G_{ij} = W(g_i, g_j)$ is real symmetric Toeplitz. For each d the formula is (841) with $B\cos(Tx)$ replaced by C_d, endpoint $4+d$, value at zero $C_d(0)$, pole term $2M(1/2)\cosh(d/2)$, and tail $C_d(0)\log\coth((4+d)/2)$. The prime sum extends to e^{4+d}; the largest endpoint here is 5.

Interval LDL^{T} elimination gives five strictly positive pivots, approximately

$$(2.50408478,\ 1.50545884,\ 1.48555915,\ 1.36380903,\ 0.02280375) \times 10^{-8}.$$

The last pivot is enclosed by $[2.2803752561005, 2.2803752561006] \times 10^{-10}$. The exact recurrences used are

$$D_j = G_{jj} - \sum_{k<j} L_{jk}^2 D_k, \qquad L_{ij} = \frac{G_{ij} - \sum_{k<j} L_{ik}L_{jk}D_k}{D_j}, \quad L_{jj} = 1.$$

All inputs are enclosing balls, each divisor has a positive lower bound, and interval arithmetic encloses each rational-operation recurrence. Thus the actual pivots are positive and the exact five-by-five matrix is positive definite. This rules out negative witnesses in this particular five-dimensional space, not in the complete test space. The numerical trust boundary is FLINT/Arb's enclosure arithmetic and integration implementation; this execution is not a Lean verification of that implementation.

10.7 Finite periodic observables with controlled numerator cancellation

The earlier A_3 is a character observable. A different finite-level observable gives a numerator that can be kept uniformly away from zero. For an integer $q \geq 2$ define

$$B_q = \frac{1}{q}\sum_{j=0}^{q-1}(e^{-2\pi ij/q} - 1)e(j/q), \qquad b_q(n) = \mathbf{1}_{n\equiv 1 \pmod q} - \mathbf{1}_{n\equiv 0 \pmod q}. \tag{842}$$

The finite geometric-series identity shows that $B_q\varepsilon_n = b_q(n)\varepsilon_n$. Thus $B_q = B_q^*$ and $\|B_q\| = 1$. It is an actual finite Fourier sum in the BC algebra, not a postulated unbounded observable.

Proposition 10.5 (A quantitative finite-level pole detector). *For $\Re s = \sigma > 0$ the grouped series*

$$D_q(s) = 1 + \sum_{k\geq 1}\big((qk+1)^{-s} - (qk)^{-s}\big) \tag{843}$$

converges absolutely and locally uniformly, and

$$|D_q(s) - 1| \leq |s|q^{-1-\sigma}\zeta(1+\sigma), \tag{844}$$

$$\left|D_q(s) - 1 - \sum_{k=1}^{N}((qk+1)^{-s} - (qk)^{-s})\right| \leq \frac{|s|q^{-1-\sigma}}{\sigma N^\sigma}. \tag{845}$$

The Gibbs expectation of B_q has meromorphic continuation $F_q(s) = D_q(s)/\zeta(s)$. On the explicitly specified region

$$\mathcal{R} = \{s : \Re s \geq 1/4,\ |s| \leq 101\}$$

one has

$$|D_{256}(s) - 1| \leq 505/1024, \qquad |D_{256}(s)| \geq 519/1024. \tag{846}$$

Consequently F_{256} has a pole at every zeta zero in $\mathcal{R}$, with exactly the zero's multiplicity and with no numerator cancellation.

Proof. For real $x > 0$ powers use the real logarithm. The fundamental theorem of calculus gives

$$(qk+1)^{-s} - (qk)^{-s} = -s\int_{qk}^{qk+1} x^{-s-1}\,dx.$$

Its absolute value is at most $|s|(qk)^{-1-\sigma}$. Summing proves (844) and local uniform convergence. The integral comparison $\sum_{k>N} k^{-1-\sigma} \leq N^{-\sigma}/\sigma$ proves (845). For real $\beta > 1$ the absolutely convergent trace may be grouped by residue class, proving $\varphi_\beta(B_q) = D_q(\beta)/\zeta(\beta)$. The grouped holomorphic numerator supplies the stated continuation.

For $\sigma \geq 1/4$, $\zeta(1+\sigma) \leq 1 + 1/\sigma \leq 5$ and $256^{1+\sigma} \geq 256^{5/4} = 1024$. This proves both bounds in (846). If $\zeta(s) = (s-\rho)^m a(s)$ with $a(\rho) \neq 0$, then $F_{256}(s) = (s-\rho)^{-m}D_{256}(s)/a(s)$, proving the pole order. □

The comparison with special-function software uses the exact identity

$$D_q(s) = q^{-s}\{\zeta(s, 1/q) - \zeta(s)\}.$$

It follows by residue-class grouping for $\Re s > 1$ and then continuation; the equal simple-pole residues at $s = 1$ cancel. Here $\zeta(s, a)$ is Hurwitz's zeta function, with the convention of [51, §25.11(i), (25.11.1)–(25.11.3)]. In particular $D_2(s) = (1 - 2^{1-s})\zeta(s)$ and $F_2(s) = 1 - 2^{1-s}$: numerator cancellation really does occur for a poorly chosen modulus. The bound above supplies a choice for which it cannot occur in the stated region.

As $q \to \infty$, $D_q \to 1$ locally uniformly on $\Re s > 0$. Thus the continued F_q converge to $1/\zeta$ away from its poles. These are continuations of real-temperature expectations, not assertions that complex temperature defines a positive state. The source of the operator is the BC representation (825); the quantitative estimates here are direct deductions with their proofs, not claims of a new literature priority.

10.8 A centered prime projector through the actual cooling map

For a prime p use just one of the projections already constructed:

$$J_p = \frac{p}{p-1}(1 - \mu_p\mu_p^*) - 1, \qquad j_p(n) = \begin{cases} 1/(p-1), & p \nmid n, \\ -1, & p \mid n. \end{cases} \tag{847}$$

The same formula defines a locally constant function $j_p(u)$ on $\widehat{\mathbb{Z}}$ of additive Haar mean zero. For $a > 0$ put

$$g_a(\lambda) = \frac{e^{-(\log\lambda)^2/(4a)}}{\sqrt{4\pi a}}, \quad x_a(t)(k,u) = \mathbf{1}_{k=1} j_p(u)\frac{e^{-at^2}}{2\pi}. \tag{848}$$

The coefficient functions e^{-at^2} are Schwartz on each closed finite horizontal strip: every derivative is a polynomial times the Gaussian, and $|e^{-a(v+iw)^2}| = e^{-av^2+aw^2}$. Thus this is an element of the source's $\widehat{\mathcal{A}}_\beta$ for each fixed $\beta > 1$. Its groupoid function is $f_a(k,u,\lambda) = \mathbf{1}_{k=1} j_p(u) g_a(\lambda)$ by Gaussian Fourier inversion. Its dual trace is zero since $\int j_p(u)\,du = 0$. These statements use the explicit source maps, not only an analogy with temperature [6, §4.4, BC groupoid representation and dual trace].

Since a unit $u \in \widehat{\mathbb{Z}}^*$ does not change divisibility by p, (834) gives

$$C_{p,a}(\lambda) = \delta(x_a)(u,\lambda) = \sum_{n\ge1} j_p(n) g_a(n\lambda). \tag{849}$$

In particular this cooled function is already in the trivial unit-character sector. It is independent of u.

Proposition 10.6 (The Mellin transform of the cooled prime observable)**.** *The Mellin integral of* (849) *is entire and*

$$\int_0^\infty C_{p,a}(\lambda)\lambda^s \frac{d\lambda}{\lambda} = e^{as^2}\frac{1 - p^{1-s}}{p-1}\zeta(s). \tag{850}$$

Inside $0 < \Re s < 1$, *its zeros are exactly those of* $\zeta(s)$, *with the same multiplicities.*

Proof. Substituting $v = \log\lambda$ in the Gaussian integral yields $\int g_a(\lambda)\lambda^s d^*\lambda = e^{as^2}$. For $\Re s > 1$ absolute Fubini is valid since $|j_p(n)| \le 1$ and $\sum n^{-\sigma} < \infty$. Therefore the left side equals $e^{as^2}\sum_n j_p(n)n^{-s}$. Splitting off multiples of p gives

$$\sum_n j_p(n)n^{-s} = \left(\frac{1-p^{-s}}{p-1} - p^{-s}\right)\zeta(s),$$

which is the right side of (850).

To justify the Mellin integral everywhere, extend g_a by zero on $(-\infty, 0]$. This is a Schwartz function on the real line: each derivative on $(0, \infty)$ is a finite sum of λ^{-r} times a polynomial in $\log \lambda$ times g_a, which tends to zero faster than any prescribed power at both endpoints. The sequence $j_p(n)$ is periodic of mean zero; express it by its finite nonconstant Fourier modes. Poisson summation of each mode gives only frequencies $(k + j/p)/\lambda$, $j = 1, \ldots, p-1$. Repeated integration by parts in the Fourier transform of g_a then bounds each sum by $O(\lambda^{r-1})$ for every integer $r \geq 2$. The sums over k converge absolutely, since $\sum_k |k + j/p|^{-r} < \infty$. At infinity, for any $B > 1$,

$$|C_{p,a}(\lambda)| \leq \sup_{x>0} |x^B g_a(x)| \, \zeta(B) \lambda^{-B}.$$

Both estimates are locally uniform in the Mellin parameter on compact vertical strips and remain so after insertion of powers of $\log \lambda$. They prove that the integral is entire. Analytic continuation now proves (850) for all s; the apparent pole at $s = 1$ is removable. Finally e^{as^2} never vanishes, while $p^{1-s} = 1$ implies $\Re s = 1$, proving the zero and multiplicity claim. $\square$

There is an exact distinction of maps, with no information discarded silently: $\varphi_\beta(J_p) = (1 - p^{1-\beta})/(p-1)$ has its zeta factor cancelled by Gibbs division, whereas cooling followed by Mellin transformation retains that factor in (850). Also $C_{p,a}$ is an *image* under δ, so its class in $\operatorname{coker} \delta$ is zero. The scalar zero detector just constructed is not itself a nonzero spectral fibre.

For numerical work the case $p = 2$ has explicit error bounds. Here $J_2 = -e(1/2)$ and $C_{2,a}(\lambda) = \sum_{n \geq 1} (-1)^{n+1} g_a(n\lambda)$. Use the ordinary Fourier transform $\mathcal{F}g(\xi) = \int_{\mathbb{R}} g(x) e^{-2\pi i \xi x} dx$. Poisson summation gives the exact alternative

$$C_{2,a}(\lambda) = -\lambda^{-1} \sum_{k \in \mathbb{Z}} \mathcal{F}g_a((k + 1/2)/\lambda).$$

Consequently

$$|C_{2,a}(\lambda)| \leq \frac{2(2^r - 1)\zeta(r)}{(2\pi)^r} \|g_a^{(r)}\|_1 \lambda^{r-1} \quad (r \geq 2).$$

For $r = 2$ one may use $B_2(a) = e^a(1 + a^{-1} + \sqrt{1 + (2a)^{-1}})$ as an upper bound for $\|g_a''\|_1$. Indeed, with $v = \log x$,

$$g_a''(x) = x^{-2} g_a(x) \left(\frac{v^2}{4a^2} + \frac{v}{2a} - \frac{1}{2a} \right).$$

Changing variables and completing the square gives a Gaussian of mean $-2a$, variance $2a$, and total mass e^a. The triangle inequality and Cauchy–Schwarz bound its three displayed terms by $1 + (2a)^{-1}$, $\sqrt{1 + (2a)^{-1}}$, and $(2a)^{-1}$, respectively.

For $\sigma = \Re s > 0$, $b > \max(1, \sigma)$, $0 < \epsilon < R$, and $N\epsilon \geq 1$, the lower tail, upper tail, and omitted terms in the middle integral are respectively bounded by

$$E_{\text{low}} = \frac{B_2(a)\epsilon^{\sigma+1}}{4(\sigma+1)}, \qquad E_{\text{high}} = \zeta(b) e^{ab^2} R^{\sigma-b},$$
$$E_{\text{middle}} = \frac{e^a}{2\epsilon} \operatorname{erfc}\left(\frac{\log(N\epsilon) - 2a}{2\sqrt{a}} \right) \frac{R^\sigma - \epsilon^\sigma}{\sigma}. \tag{851}$$

The first is obtained by integrating $B_2(a)\lambda/4$. For the second, bound $\lambda^{\sigma-b} \leq R^{\sigma-b}$ and apply absolute Fubini to the nonnegative majorant. For the third, g_a decreases on $[1, \infty)$, so $\sum_{n>N} g_a(n\lambda) \leq \lambda^{-1} \int_{N\lambda}^\infty g_a(x) dx$. Completing the square evaluates this last integral as

$$\frac{e^a}{2} \operatorname{erfc}\left(\frac{\log(N\lambda) - 2a}{2\sqrt{a}} \right);$$

bound it uniformly with $\lambda \geq \epsilon$ and integrate $\lambda^{\sigma-1}$. Thus a finite quadrature with these three errors can evaluate the detector without a zero table.

10.9 Larger matrices including the odd channel and cross terms

The finite compression in [10, §2.3] is $\widetilde{G} + \widetilde{E}$ on the full arithmetic side: $\widetilde{E}$ retains the zeros outside the chosen height window. The computations here evaluate the full expression directly, not a height truncation with its remainder omitted.

In addition to $b(u) = b_4(u)$, take $o(u) = ub_4(u)$, which is still C_c^2. Write $H(z) = \operatorname{sinc}(z/2)^4$ and $K(w) = \operatorname{sinhc}(w/2)^4$. The transforms and correlations are

$$\widehat{o}(z) = iH'(z), \qquad h_{00} = B, \qquad h_{01} = \tfrac{x}{2}B(x), \qquad h_{10} = -h_{01}, \qquad h_{11} = -o * o. \tag{852}$$

For example, in $h_{01}(x) = \int vb(v)b(x-v)\,dv$, exchanging v and $x-v$ shows that twice the integral is $xB(x)$. The final sign follows because $o^\star = -o$. Their exponential moments are respectively

$$M_{00} = K^2, \qquad M_{01} = KK', \qquad M_{10} = -KK', \qquad M_{11} = -(K')^2. \tag{853}$$

These are complex squares, not squared moduli.

For an exact polynomial description put $a_k = 2-k$ and $c_k = (-1)^k\binom{4}{k}/6$. Multiplication of the cubic truncated-power formula by u gives $o(u) = \sum_{k=0}^4 c_k[(u+a_k)_+^4 - a_k(u+a_k)_+^3]$. The identity

$$(u+a)_+^m * (u+b)_+^n = \frac{m!\,n!}{(m+n+1)!}(x+a+b)_+^{m+n+1}$$

follows by the substitution $u+a = (x+a+b)t$ and the beta integral on $0 \leq t \leq 1$. It proves

$$\begin{aligned} h_{11}(x) = -\sum_{k,l=0}^{4} c_k c_l \Bigg[&\frac{(4!)^2}{9!}(x+a_k+a_l)_+^9 \\ &- \frac{(a_k+a_l)4!3!}{8!}(x+a_k+a_l)_+^8 + \frac{a_k a_l (3!)^2}{7!}(x+a_k+a_l)_+^7 \Bigg]. \end{aligned} \tag{854}$$

In particular $h_{11}(0) = 491/5670$, $\int h_{11} = 0$, and $\int x^2 h_{11}(x)dx = -2/9$. These provide exact independent checks.

For $h = 1/8$, $j = 0, \ldots, 32$, and $T \in \{0, 14, 40, 100\}$ use

$$f_{0,j,T}(u) = e^{-iT(u-jh)}b(u-jh), \qquad f_{1,j,T}(u) = e^{-iT(u-jh)}o(u-jh).$$

For each T the full family has 66 vectors. Its four correlation blocks are the translates and modulations of (852). All correlations lie in $[-8, 8]$, so the arithmetic cutoff is e^8. All cross terms between the even and odd channels are included. Different values of T were tested separately, not as a single 264-dimensional space.

For clarity, if $k(x)$ is one of these real correlation kernels, of parity $\varepsilon \in \{1, -1\}$, the evenized shifted correlation is

$$C_{d,T}(x) = \frac{e^{iTd}}{2}\left[e^{-iTx}k(x-d) + \varepsilon e^{iTx}k(x+d)\right].$$

It has $C_{d,T}(0) = e^{iTd}k(-d)$ and endpoint $L = 4+d$ for $d \geq 0$. Its full-form value is

$$\begin{aligned} &e^{d/2}M_k(1/2 - iT) + \varepsilon e^{-d/2}M_k(1/2+iT) - (\gamma_E + \log 4\pi)C_{d,T}(0) \\ &- \int_0^L \frac{e^{x/2}C_{d,T}(x) - C_{d,T}(0)}{\sinh x}\,dx + C_{d,T}(0)\log\coth(L/2) \\ &- 2\sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} C_{d,T}(\log n). \end{aligned} \tag{855}$$

This follows term by term from (839). The calculation splits at every shifted integer knot and uses only exact rational polynomial coefficients. The numerator at the origin is divided by x algebraically, as in the scalar computation.

One may also form $f_j - 2\cosh(h/2) f_{j+1} + f_{j+2}$. Its transform is

$$2e^{-i(j+1)hz}[\cos(hz) - \cosh(h/2)]\widehat{f}_0(z),$$

which vanishes at both $z = \pm i/2$. The coefficient matrix C is three-banded, injective by its leading coefficient, and its compression is C^*GC. This is a precisely specified subspace, not an assertion that pole terms can be deleted from an arbitrary test.

The executable `search_weil_translations.py` uses 384-bit Arb arithmetic, 240-bit integration targets and one worker. It establishes positivity of the eight individual 33-dimensional matrices, their 31-dimensional pole-annihilating subspaces, and all four coupled 66-dimensional matrices by enclosing every Hermitian LDL^* recurrence. The final pivots of the coupled matrices are:

T	final pivot, rounded for display
0	$5.79841339049 \times 10^{-17}$
14	$2.05294634281 \times 10^{-16}$
40	$1.96917104792 \times 10^{-16}$
100	$2.49888863012 \times 10^{-15}$

Full enclosing balls, rather than these rounded entries, are in the machine-readable receipt. No zero ordinates are inputs to these matrices. They furnish no negative full-form witness.

An implementation correction is retained in the computation record: a first mixed-block trial accidentally formed one polynomial coefficient through binary floating-point division. In this nearly singular problem that produced apparent negative pivots. Exact rational construction and type assertions removed the error; the corrected complete rerun gives the positive enclosures above. Interval arithmetic bounds the expression actually supplied to it, so exact input validation is essential.

10.10 Evaluating the cooling detector without zeta input

For the prime-two kernel, the finite computation underlying (851) can be performed by antiderivatives rather than numerical quadrature. For $a > 0$, $0 < \epsilon < R$, and $s \in \mathbb{C}$, substitute $v = \log(n\lambda)$ and complete the square to obtain

$$\int_\epsilon^R g_a(n\lambda)\lambda^s \frac{d\lambda}{\lambda} = \frac{n^{-s}e^{as^2}}{2}\Bigg[\operatorname{erf}\left(\frac{\log(nR) - 2as}{2\sqrt{a}}\right) - \operatorname{erf}\left(\frac{\log(n\epsilon) - 2as}{2\sqrt{a}}\right)\Bigg]. \tag{856}$$

The Gaussian antiderivative is entire in s, so the calculation, initially valid for real s, also holds for complex s. Summing with coefficients $(-1)^{n+1}$ for $1 \le n \le N$ and adding the three proved errors gives an enclosure of the entire cooling Mellin transform. The upper-tail bound needs no zeta evaluation: replace $\zeta(b)$ by $1 + 1/(b-1)$, using the integral test for $b > 1$.

The executable `evaluate_cooled_mellin.py` uses $a = 1/400$, $\epsilon = 10^{-4}$, $R = 2$, $b = 100$, $N = 20000$, and 192-bit Arb arithmetic. The following numbers are rounded displays; the receipt contains the actual enclosing balls.

s	finite sum	total analytic error less than
$0.3 + 14i$	$-0.255791340038 - 0.203324126919i$	$5.050289\,10^{-4}$
$0.7 + 14i$	$0.207445883527 - 0.114179515120i$	$9.700868\,10^{-6}$

Both complete enclosures exclude zero. The finite sums, error bounds and single-term antiderivative check contain no zeta values or zero tables. A separate evaluation of (850) agrees. This certifies two point values, not a zero-free region or a counterexample.

10.11 Heat and Burgers equations from a fixed BC state

There is an explicit fluid equation associated with the represented BC Hamiltonian. Fix $\beta > 1$ and keep the Gibbs state φ_β fixed. On $\ell^2(\mathbb{N})$ put

$$F_\beta(\tau, x) = \varphi_\beta\Big(e^{-\tau H^2} e^{ixH}\Big) = \frac{1}{\zeta(\beta)} \sum_{n\geq 1} n^{-\beta} e^{-\tau(\log n)^2 + ix\log n}. \tag{857}$$

For real $\tau \geq 0, x$, both displayed factors are bounded functional calculus observables in $B(\ell^2(\mathbb{N}))$. This specifies the algebra: no membership assertion in the original BC C^*-algebra is required. For $\operatorname{Re}\tau > 0$ the series and every derivative converge locally uniformly for complex x, since a negative quadratic in $\log n$ dominates every linear term and every power of $\log n$. At $\tau = 0$ the same conclusion holds for x in compact subsets of $\beta + \operatorname{Im} x > 1$, including one-sided τ derivatives. Consequently

$$\partial_\tau F_\beta = \partial_x^2 F_\beta, \qquad F_\beta(0, x) = \frac{\zeta(\beta - ix)}{\zeta(\beta)}. \tag{858}$$

The sign follows from differentiating each summand: both derivatives introduce $-(\log n)^2$.

Here is the complete logarithmic change of variables (Cole–Hopf [19, 20]). Fix $\nu \neq 0$. For any nonvanishing heat solution $F_\tau = \nu F_{xx}$ on a simply connected coordinate patch, set $L = \log F$ and $u = -2\nu L_x$. Then $L_\tau = \nu(L_{xx} + L_x^2)$, and differentiating this equality gives

$$u_\tau = -2\nu^2 L_{xxx} - 4\nu^2 L_x L_{xx}, \quad u u_x = 4\nu^2 L_x L_{xx}, \quad \nu u_{xx} = -2\nu^2 L_{xxx}.$$

Thus $u_\tau + uu_x = \nu u_{xx}$; in (857) the viscosity is $\nu = 1$. Conversely integrate $L_x = -u/(2\nu)$. Burgers' equation says that $L_\tau - \nu(L_{xx} + L_x^2)$ is independent of x; subtracting its time integral makes e^L a heat solution. The inverse has a constant nonzero multiplicative freedom. The forward map loses that amplitude, but not the zeros: if $F = (x - x_0)^m a(x)$ with $a(x_0) \neq 0$, then $u = -2\nu m/(x - x_0) - 2\nu a'/a$.

This is a concrete fixed-state observable-to-complex-fluid map. The parameter τ changes the observable, not the Gibbs temperature or the modular automorphism. The relation to the completed zeta function will now be given explicitly.

10.12 The Hermite cooling seed and the completed zeta function

Use the Mellin measure $d\lambda/\lambda$ and the real Fourier convention $\widehat{f}(y) = \int_{\mathbb{R}} f(x) e^{-2\pi ixy} dx$. Following the arithmetic summation construction in [16, §6.2], define

$$h(x) = \frac{\pi}{2} x^2 (2\pi x^2 - 3) e^{-\pi x^2}, \quad S(\lambda) = \sum_{n\geq 1} h(n\lambda), \quad k(v) = e^{v/2} S(e^v). \tag{859}$$

These definitions retain all constants. In particular, $\xi(s)$ below means $\frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$.

The seed h is Fourier self-dual. Indeed the Gaussian is self-dual, and differentiating its transform gives

$$\widehat{x^2e^{-\pi x^2}} = \left(\frac{1}{2\pi} - y^2\right)e^{-\pi y^2}, \qquad \widehat{x^4e^{-\pi x^2}} = \left(y^4 - \frac{3y^2}{\pi} + \frac{3}{4\pi^2}\right)e^{-\pi y^2}.$$

Insert the coefficients π^2 and $-3\pi/2$ to get $\widehat{h} = h$. In particular $h(0) = \int_{\mathbb{R}} h(x)dx = 0$. Poisson summation applied to this Schwartz function therefore gives $S(\lambda) = \lambda^{-1}S(\lambda^{-1})$. It follows that $k(v) = k(-v)$. Gaussian decay of the series for $\lambda \geq 1$, followed by this equality, gives double-exponential decay of k at both ends, together with all its derivatives. Also $k(v) > 0$: for $v \geq 0$ every summand has $ne^v \geq 1$ and $2\pi n^2e^{2v} - 3 > 0$, and for negative v use evenness.

The precise BC input is the groupoid function $f(q, u, \lambda) = \mathbf{1}_{q=1}h(\lambda)$, constant in u. The cooling formula of [6, §4.4] is exactly $f \mapsto S$. Moreover its dual trace is $\int_0^\infty h(\lambda)d\lambda = 0$. In logarithmic coordinates $h(e^v) = O(e^{2v})$ at $-\infty$ and is super-exponentially decreasing at $+\infty$. Its Fourier transform and all weighted derivatives are Schwartz on each closed strip $0 \leq \operatorname{Im} w \leq \beta$, so this is an allowed smooth crossed-product input for every fixed $\beta > 1$ in that convention.

The elementary integral $\int_0^\infty x^{s+j-1}e^{-\pi x^2}dx = \frac{1}{2}\pi^{-(s+j)/2}\Gamma((s+j)/2)$ gives, for $\operatorname{Re} s > -2$,

$$\mathcal{M}h(s) = \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2). \tag{860}$$

The apparent singularities at $s = 0$ are removable. For $\operatorname{Re} s > 1$ absolute Fubini thus gives

$$\int_0^\infty S(\lambda)\lambda^s\frac{d\lambda}{\lambda} = \zeta(s)\mathcal{M}h(s) = \frac{\xi(s)}{4}. \tag{861}$$

Poisson summation above makes the left side entire in s, so this identity continues to all s. This proves the completion, including the factor $1/4$. With the printed seed of [16, §6.2, (15)], its preceding Fourier-transform display therefore needs a factor 4 on the integral. Equivalently one can use $4h$ as seed. This scalar correction does not move zeros.

There is also a direct fixed-Gibbs-observable version of the same map. For $\lambda > 0$ define $A_{\beta,\lambda} = e^{\beta H}h(\lambda e^H)$ by its diagonal eigenvalues $n^\beta h(n\lambda)$. Their Gaussian decay makes this a bounded trace-class operator in the represented von Neumann algebra. Thus, pointwise in λ,

$$S(\lambda) = \operatorname{Tr} h(\lambda e^H) = \zeta(\beta)\varphi_\beta(A_{\beta,\lambda}). \tag{862}$$

The trace-summation step is exactly that of [6, §4.3]. This gives an explicit observable through which the fixed BC state feeds the completed heat kernel. The β counterweight cancels the Gibbs weight. The observables commute with H and are modular-fixed; their parameter λ is not modular time. Take the expectation before the Mellin integral: the absolute termwise exchange proved above applies only on $\operatorname{Re} s > 1$.

10.13 Newman evolution, its exact conjugations, and cooling

Retain the conventions of Rodgers–Tao [17, §1]:

$$\Phi(u) = \sum_{n\geq 1}(2\pi^2n^4e^{9u} - 3\pi n^2e^{5u})e^{-\pi n^2e^{4u}},$$

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \qquad H_0(z) = \frac{1}{8}\xi(1/2 + iz/2). \tag{863}$$

Substituting $x = ne^{2u}$ in (859) proves $k(2u) = \Phi(u)/2$. Since k is even, the exact composite map is

$$h \longmapsto S(\lambda) \longmapsto k(v) = e^{v/2}S(e^v) \longmapsto H_t(z) = \frac{1}{2}\int_{\mathbb{R}} e^{tv^2/4}k(v)e^{izv/2}dv. \tag{864}$$

The inverse coordinate substitutions are $\lambda = e^v$, $v = 2u$, $s = 1/2 + iz/2$ and $z = -2i(s - 1/2)$. The scalar changes are displayed, not absorbed into an unnamed normalization. Double-exponential decay justifies all complex (t, z) derivatives under the integral; hence

$$\partial_t H_t = -\partial_z^2 H_t.$$

Putting $\tau = -t$ makes $F(\tau, z) = H_{-\tau}(z)$ a forward heat solution, with $u = -2F_z/F$ solving the viscosity-one Burgers equation. Equivalently $v = 2H_z/H$ obeys $v_t + vv_z = -v_{zz}$ in Newman time.

The completed arithmetic coordinate has a different, explicit viscosity. Put

$$X_t(s) = 8H_t(-2i(s - 1/2)). \quad X_0 = \xi, \qquad \partial_t X_t = \tfrac{1}{4}\partial_s^2 X_t. \tag{865}$$

Thus $w = -\frac{1}{2}X_s/X$ satisfies $w_t + ww_s = \frac{1}{4}w_{ss}$. The chain rule $\partial_z = (i/2)\partial_s$ proves the factor 1/4. On $0 < \operatorname{Re} s < 1$ the factor $Q(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$ is nonzero. Therefore $Z_t = X_t/Q$, with inverse $X_t = QZ_t$, obeys the exact conjugated equation

$$\partial_t Z_t = \frac{1}{4}\left[\partial_s^2 Z_t + 2A(s)\partial_s Z_t + (A'(s) + A(s)^2)Z_t\right], \quad Z_0 = \zeta, \tag{866}$$

where $A(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2)$. This follows by twice differentiating QZ_t. It records precisely the gamma-factor drift and potential that relate completed heat evolution to the Dirichlet function in (858).

Heat can also be compared before and after arithmetic cooling, not merely declared identical or different. Write $(Cg)(\lambda) = \sum_{n\geq 1} g(n\lambda)$ and $(Mg)(\lambda) = (\log\lambda)^2 g(\lambda)$. Direct subtraction yields

$$[M, C]g(\lambda) = -2\log\lambda\sum_{n\geq 1}(\log n)g(n\lambda) - \sum_{n\geq 1}(\log n)^2 g(n\lambda). \tag{867}$$

For $g = h$, Mellin transformation initially on $\operatorname{Re} s > 1$ turns this into $\zeta''\mathcal{M}h + 2\zeta'(\mathcal{M}h)'$, since M becomes ∂_s^2. The Newman deformation is output multiplication $e^{tM/4}$, followed by the coordinate changes in (864); the commutator gives its correction relative to input multiplication. The distinction has a concrete domain manifestation: for real $t > 0$, $e^{t(\log\lambda)^2/4}h(\lambda)$ grows at $\lambda \downarrow 0$ and is not Mellin integrable there, whereas $e^{t(\log\lambda)^2/4}S(\lambda)$ still decays at both ends. Even the infinitesimal input Mh changes the trace, because

$$\int_0^\infty (\log\lambda)^2 h(\lambda)d\lambda = (\mathcal{M}h)''(1) = \frac{2 - \gamma_E - 2\log 2 - \log\pi}{8} \neq 0.$$

These computations specify the map and its domain, not a claim of disconnection. They also prevent treating output heat automatically as an evolution of a trace-zero input or of the cyclic cokernel.

10.14 Zeros as Burgers poles and their collision coordinates

The zero dynamics needed here are not restricted to an assumed all-real-zero regime. Polymath [18, §3, Proposition 3.1] proves them for real t_0, allowing complex and multiple zeros. For a simple zero $z_j(t)$, differentiating $H_t(z_j(t)) = 0$ and using backward heat gives

$$z_j'(t) = \frac{H_t''(z_j(t))}{H_t'(z_j(t))} = 2\sum_{k\neq j}{}' \frac{1}{z_j(t) - z_k(t)}. \tag{868}$$

For completeness, the second equality follows as follows. H_t is even and, for real t, positive at the origin. Its order is at most one: the positive kernel satisfies $\Phi(u) \leq 6\pi^2 e^{9u-\pi e^{4u}}$ for $u \geq 0$, by the geometric-series estimate below. For fixed real t, absorb $\max(t,0)u^2+9u$ into $\pi e^{4u}/2+C_t$. For $|z| \leq R$ the remaining integral is bounded by a fixed constant times $\exp(\sup_{u\geq 0}(Ru-\pi e^{4u}/4))$. For $R > \pi$ this supremum is $\frac{R}{4}(\log(R/\pi)-1)$, proving $\log\max_{|z|\leq R}|H_t(z)| = O_t(R\log R)$. Pair its nonzero zeros as $z_{-k} = -z_k$. Hadamard factorization then gives a locally uniformly convergent paired product $H_t(z) = H_t(0)\prod_k'(1-z/z_k)$. The paired logarithmic derivative converges locally away from the zeros, since a pair contributes $2z/(z^2-z_k^2)$ and $\sum_k |z_k|^{-2} < \infty$. At a simple z_j, subtract $1/(z-z_j)$ from H_t'/H_t and take the limit. Taylor expansion shows the limit is $H_t''(z_j)/(2H_t'(z_j))$. The prime in (868) pairs k with $-k$, except that the term $k=-j$ is kept separately. It is not an unregularized sum.

The local collision law has equally explicit coordinates. Suppose $H_{t_0}(z) = a_m(z-z_0)^m + O((z-z_0)^{m+1})$, $a_m \neq 0$. Taylor expansion in time, with $\partial_t^\ell H = (-1)^\ell \partial_z^{2\ell} H$, shows that after $z - z_0 = \sqrt{2(t-t_0)}\,y$ the leading polynomial is

$$a_m[2(t-t_0)]^{m/2}\,\mathrm{He}_m(y), \qquad \mathrm{He}_m(y) = \sum_{\ell=0}^{\lfloor m/2\rfloor} \frac{m!(-1)^\ell y^{m-2\ell}}{2^\ell \ell!(m-2\ell)!}.$$

On bounded y sets the error divided by the displayed prefactor is $O(|t-t_0|^{1/2})$. The Hermite polynomial has simple real roots: its Gaussian derivative expression and Rolle's theorem give real roots, and its second-order differential equation excludes a repeated root by uniqueness. On small disjoint circles around those roots, Rouche's theorem and then the inverse function theorem give $z = z_0 + \sqrt{2(t-t_0)}\,y_j + O(|t-t_0|)$. For $m=2$ these are $z_0 \pm \sqrt{2(t-t_0)} + O(|t-t_0|)$. The exact polynomial heat solution $(z-z_0)^2 - 2(t-t_0)$ illustrates the leading law, but is not substituted for the arithmetic H_t.

The pole interpretation now has a precise counterexample criterion. De Bruijn and Newman give a finite threshold Λ such that H_t has only real zeros exactly for $t \geq \Lambda$; Rodgers–Tao prove $\Lambda \geq 0$ [17, §1]. Thus a certified nonreal zero at any specified $t \geq 0$ would prove $\Lambda > t \geq 0$ and refute classical RH. At $t = 0$ the coordinate of $\rho = \sigma + i\gamma$ is $z = 2\gamma + i(1-2\sigma)$, so off-critical is exactly off-real. In forward Burgers time $\tau = -t$, this target is $\tau \leq 0$. A pole somewhere is not the target: even the real critical-line zeros give poles of the logarithmic derivative. This identifies which singularity, coordinate and time must be located. The explicit BC-observable route of §10.11 is retained alongside the earlier vacuum-to-Navier–Stokes construction.

10.15 Certified local zero motion from the theta integral

The code `verify_newman_heat.py` evaluates (863) directly, without zeta values or zero tables. Here are full bounds for its truncations. They hold for $0 \leq t \leq 1/4$, $|\operatorname{Im} z| \leq 1$, and $0 \leq j \leq 4$ derivatives with respect to z. On $0 \leq u \leq 2$ keep $1 \leq n \leq N$, where $N = 8$. Every theta summand is positive for real $u \geq 0$ and is at most $2\pi^2 n^4 e^{9u} e^{-\pi n^2 e^{4u}}$. Since $|\partial_z^j \cos(zu)| \leq u^j e^{|\operatorname{Im} z|u}$, the discarded part of the integral over $[0,2]$ is bounded by

$$E_n = 64\pi^2 e^{21-\pi N^2}\left(\frac{N^3}{2\pi} + \frac{3N}{4\pi^2} + \frac{3}{8\pi^3 N}\right). \tag{869}$$

Indeed $u^j \leq 16$ there, the interval length is 2, and the other exponentials contribute at most e^{1+2+18}. The function $x^4 e^{-\pi x^2}$ is decreasing for $x \geq N$, so its sum for $n > N$ is bounded by its integral from N to infinity. Two integrations by parts, followed by $\int_N^\infty e^{-\pi x^2}dx \leq e^{-\pi N^2}/(2\pi N)$, give the parenthesis in (869).

For the remaining $u \geq 2$ tail, use $n^4 \leq 16^{n-1}$ (induction) and $n^2 - 1 \geq 3(n-1)$ to obtain

$$\sum_{n \geq 1} n^4 e^{-\pi n^2 e^{4u}} \leq \frac{e^{-\pi e^{4u}}}{1 - 16e^{-3\pi}} < 3e^{-\pi e^{4u}}.$$

For $u \geq 2$, $4 \log u + u^2/4 + 10u \leq 29u^2/4 \leq 8u^2 \leq e^{4u} \leq (\pi/2)e^{4u}$. The middle exponential inequality follows at $u = 2$ and persists because $4 - 2/u > 0$ is the logarithmic derivative of e^{4u}/u^2. Consequently the integrand tail, including all the stated derivatives, is at most $6\pi^2 e^{-(\pi/2)e^{4u}}$. Set $v = e^{4u}$ and bound $1/v \leq e^{-8}$ to get

$$E_u = 3\pi e^{-8} \exp\left(-\frac{\pi}{2} e^8\right). \tag{870}$$

Outward-rounded values are $E_n < 3.272678\,10^{-74}$ and $E_u < 8.425984\,10^{-2037}$.

Arb integrates the retained finite sum on eight subintervals of $[0, 2]$. It then adds $E_n + E_u$ to every required value or derivative. Floating-point quadrature proposes the following decimal-rational endpoints; their proof uses only the complete interval evaluations.

t	unique simple real root in this interval	local speed
0	[28.269450279469, 28.269450287469]	$-0.5796306 \pm 4.70\,10^{-8}$
1/20	[28.240474250649, 28.240474258649]	$-0.5794105 \pm 4.99\,10^{-8}$
1/5	[28.153612284449, 28.153612292449]	$-0.5787486 \pm 4.90\,10^{-8}$

At each pair of endpoints, the complete H_t enclosures have opposite signs. The enclosure of H_t' over the whole interval excludes zero. The intermediate value theorem gives existence, and the derivative sign gives uniqueness and simplicity. Dividing the complete interval enclosures of H_t'' and H_t' gives the displayed local speeds by (868). These are three local certificates, not an asserted connecting continuation tube, nor exclusion of nonreal zeros elsewhere. They are sufficient, however, to show that the arithmetic heat flow really moves a certified zero at time zero.

10.16 An explicit arithmetic lift and its Mellin singularities

The preceding commutator does not prevent an arithmetic lift in a larger function class. It tells us to construct that lift explicitly. For real t set

$$B_t(\lambda) = e^{t(\log \lambda)^2/4} S(\lambda), \qquad g_t(\lambda) = \sum_{n \geq 1} \mu(n) B_t(n\lambda). \tag{871}$$

For each fixed $\lambda > 0$ the latter series converges absolutely, as do all its derivatives locally in (t, λ): $B_t(x)$ decreases faster than every power as $x \to \infty$, uniformly on compact t intervals. Absolute double summation and the divisor identity $\sum_{d|m} \mu(d) = \mathbf{1}_{m=1}$ prove $Cg_t = B_t$ and $g_0 = h$. The same calculation proves uniqueness among smooth functions that decrease faster than every power at infinity and for which the indicated sums converge absolutely. This is an actual inverse, not an unsupported existence assertion.

For $\sigma > 1$, Tonelli's theorem applied to the absolute majorant $\sum_n |B_t(n\lambda)|$ gives the integrable bound $\zeta(\sigma) \int_0^\infty |B_t(\lambda)| \lambda^\sigma d\lambda/\lambda$. Mellin transformation is therefore justified and yields

$$\mathcal{M}g_t(s) = \frac{X_t(s)}{4\zeta(s)} \quad (\operatorname{Re} s > 1), \qquad \mathcal{M}\dot{g}_0(s) = \frac{\xi''(s)}{16\zeta(s)}. \tag{872}$$

The second formula follows by differentiating (865); both quotients specify meromorphic continuations, not continued positive Gibbs states.

There is a certified, explicit loss of smooth Mellin-strip membership in this lift. Let a be the unique root in the first interval of §10.15 and $\rho = 1/2 + ia/2$. Then $\xi(\rho) = 0$, $\xi'(\rho) \neq 0$ and $\xi''(\rho) \neq 0$: indeed $H_0'(a) = i\xi'(\rho)/16$ and $H_0''(a) = -\xi''(\rho)/32$, while the two derivative enclosures exclude zero. Since $Q(\rho) \neq 0$, ζ has a simple zero at ρ. Thus $\mathcal{M}\dot{g}_0$ has the nonzero residue

$$\operatorname*{Res}_{s=\rho} \mathcal{M}\dot{g}_0(s) = \frac{\xi''(\rho)}{16\zeta'(\rho)} = \frac{Q(\rho)}{8i}\, a'(0). \tag{873}$$

A Mellin transform holomorphic on a strip containing ρ cannot have this residue. More precisely, among the rapid-decay-at-infinity lifts just specified, the lift is uniquely g_t. Its pointwise derivative cannot have a holomorphic Mellin transform on a strip containing ρ and connected to $\operatorname{Re} s > 1$. For example, fix $\beta > 1$ and require differentiability in every weighted L^1 seminorm $\int_0^\infty |g(\lambda)|\lambda^\sigma d\lambda/\lambda$, $0 < \sigma < \beta$. Mellin transformation is then differentiable and holomorphic on that strip (use two adjacent weights to control logarithmic derivatives); the nonzero residue excludes such a path. Formula (871) still realizes it in the larger pointwise class, and (872) states exactly the extra singularity. For sufficiently small nonzero t, $X_t(\rho) \neq 0$ as well, by its nonzero time derivative at zero. This demonstrates a moving-zero mechanism and an exact change of function space, not a counterexample to RH: the certified ρ itself is on the critical line.

10.17 Finite certificates for Newman collisions and nonreal zeros

The general zero-motion statement of [18, §3] includes complex and multiple zeros. It gives a direct two-variable target, with no real-zero assumption in the calculation:

$$\mathcal{F}(t,x) = (H_t(x), H_t'(x)), \qquad D\mathcal{F}(t,x) = \begin{pmatrix} -H_t''(x) & H_t'(x) \\ -H_t'''(x) & H_t''(x) \end{pmatrix}. \tag{874}$$

At a real double zero with $H_t''(x) \neq 0$, the determinant is $-(H_t''(x))^2$, so the system is nonsingular. A certified solution with $t_c > 0$ gives nonreal zeros at times just below t_c, still positive, by the collision expansion proved above. Hence it gives $\Lambda > 0$ and disproves RH. The derivative bounds in §10.15 cover $\mathcal{F}$ and its Jacobian on the displayed time range; interval Newton can therefore certify this system without a new unbounded integral representation. No such positive-time solution is reported here.

A complementary scalar test follows from the classical Jensen criterion; see [94, §2 and §4.2] for its statement and for the loss of detecting power under repeated differentiation. At a real center x, use the degree-two polynomial

$$J_{t,x}^2(w) = H_t(x) + 2H_t'(x)w + H_t''(x)w^2, \qquad L_1(t,x) = H_t'(x)^2 - H_t(x)H_t''(x). \tag{875}$$

Here is the required necessity directly. If all zeros r_j of H_t are real, its paired Hadamard product gives, away from them,

$$-\left(\frac{H_t'}{H_t}\right)'(x) = \sum_j \frac{1}{(x-r_j)^2} \geq 0.$$

The sum converges locally away from the zeros since the entire function has order one; the exponential factor in a genus-one product has logarithm of degree at most one and disappears after two derivatives. Multiplication by $H_t(x)^2$ proves $L_1 \geq 0$. At a zero the same assertion follows by continuity, or directly from $L_1 = H_t'^2$. Consequently a rigorously negative value at any $t \geq 0$ refutes RH by the de Bruijn–Newman threshold theorem. This is a finite rejection certificate, not an assertion that a bounded collection of nonnegative values characterizes RH.

Write a_j for approximations to $H_t^{(j)}(x)$ with errors at most e_j, $j = 0, 1, 2$. Expanding the two products and using the triangle inequality proves

$$|L_1 - (a_1^2 - a_0 a_2)| \le E, \tag{876}$$

$$E = 2|a_1|e_1 + e_1^2 + |a_0|e_2 + |a_2|e_0 + e_0 e_2. \tag{877}$$

Thus $a_1^2 - a_0 a_2 + E < 0$ suffices. The theta integral gives $H_t = I_0$, $H_t' = -I_1$, $H_t'' = -I_2$, where I_0, I_2 use the cosine with weights $1, u^2$ and I_1 uses $u \sin(xu)$. In particular $L_1 = I_1^2 + I_0 I_2$. At $x = 0$ both cosine integrals are positive, so that center cannot supply a negative witness. For a real, nonvanishing multiplier q on an interval, differentiation also gives the exact correction

$$L_1(qH) = q^2 L_1(H) - q^2 H^2 (\log|q|)''.$$

An x-dependent rescaling must retain that correction. The polynomial $Q_t(z) = (z-a)^2 + b^2 - 2t$ satisfies the same backward heat equation and has $L_1(t, a) = -2(b^2 - 2t)$; this checks the sign and detects its nonreal pair, but is not a zeta example.

The bounded implementation `verify_centered_laguerre.py` evaluates the full three derivative integrals at exactly $t \in \{0, 1/20, 1/5\}$ and $x \in \{20, 24, 28, \ldots, 60\}$. All 33 resulting interval enclosures are strictly positive, with no inconclusive enclosure. For example, $L_1(1/5, 60) = 5.021267296249904426999643023181472271150519 \times 10^{-18}$ with absolute error at most 5.43×10^{-61}. The polynomial control gives the exact negative values $-2, -9/5, -6/5$ at its center at the same times. An independent implementation re-evaluates nine of the positive points from the Hermite seed with nine summands and sixteen panels, and reconstructs all 33 signs from the saved enclosing derivative balls. These are pointwise calibration results, not positivity on the intervals between the points.

Finally, a candidate nonreal zero admits a small contour certificate. Fix $t \ge 0$, $0 < r < |\operatorname{Im} z_0|$, and a proved bound $M_2 \ge \sup_{|z - z_0| \le r} |H_t''(z)|$. Taylor's formula along the line segment from z_0 gives a remainder at most $M_2 r^2/2$. If

$$|H_t(z_0)| + M_2 r^2/2 < r|H_t'(z_0)|, \tag{878}$$

then on the circle the remainder after subtracting $H_t'(z_0)(z - z_0)$ is strictly smaller than that linear function. Rouché's theorem gives exactly one zero in the disk, which lies strictly off the real axis. This is the local version of the boundary/winding method used in [18, §8.3].

10.18 Connes's exact arithmetic heat kernel and its Newman map

We use the Gaussian and explicit-formula conventions of [95, §2–4, equations (5), (9), (13), (25)]. For $a > 0$, put

$$F_a(e^y) = (4\pi a)^{-1/2} e^{-y^2/(4a)}, \qquad \widehat{F}_a(\gamma) = e^{-a\gamma^2}, \tag{879}$$

$$\Psi(a) = \frac{1}{\sqrt{\pi a}} \sum_{n \ge 2} \Lambda(n) n^{-1/2} e^{-(\log n)^2/(4a)}, \tag{880}$$

$$\begin{aligned} W_{\mathbb{R}}(F_a) &= (\log(4\pi) + \gamma_E) F_a(1) \\ &\quad + \int_0^\infty [2F_a(e^y) - 2e^{-y/2} F_a(1)] \frac{e^{y/2}}{e^y - e^{-y}}\, dy, \end{aligned} \tag{881}$$

$$Z(a) = 2e^{a/4} - W_{\mathbb{R}}(F_a) - \Psi(a). \tag{882}$$

Here $\Lambda(n)$ is the von Mangoldt function, not the Newman constant, and γ_E is Euler's constant. The integrand in the archimedean term has a removable limit at zero: its numerator is $F_a(1)y + O(y^2)$ and its denominator is $2y + O(y^3)$. The series and integral converge absolutely at infinity, uniformly on compact positive a intervals, as do all a derivatives.

Apply the unrestricted explicit formula to $f_a(x) = x^{-1/2}F_a(x)$. Its Mellin transform is $\widetilde{f}_a(s) = e^{a(s-1/2)^2}$, by the elementary Gaussian integral. The two pole terms are each $e^{a/4}$, and the finite-place terms combine into Ψ. Thus, without imposing RH,

$$Z(a) = \sum_{\rho} e^{a(\rho-1/2)^2}, \tag{883}$$

where the sum is over nontrivial zeros with multiplicity. Its absolute local convergence follows from $0 < \operatorname{Re}\rho < 1$ and the zero count $O(T\log T)$: the summands are bounded on positive compact a intervals by a constant times $e^{-c(\operatorname{Im}\rho)^2}$. This also proves convergence after any number of a derivatives. It distinguishes the unconditional arithmetic definition from the self-adjoint trace interpretation used under RH in the cited paper.

There is an exact transform relation to the Newman evolution, not just common terminology. For $\tau > 0$, the kernel k of §10.12 gives

$$\begin{aligned} H_{-\tau}(z) &= \frac{1}{2}\sqrt{\frac{4\pi}{\tau}} \int_{\mathbb{R}} k(v) F_{1/\tau}(e^v) e^{izv/2}\, dv \\ &= \frac{1}{\sqrt{4\pi\tau}} \int_{\mathbb{R}} e^{-(z-y)^2/(4\tau)} H_0(y)\, dy. \end{aligned} \tag{884}$$

For the first equality substitute the definition of $F_{1/\tau}$. For the second insert the Fourier integral for H_0 and evaluate the inner Gaussian integral: the multiplier is $e^{-\tau(v/2)^2}$. Fubini is justified by integrability of k and Gaussian decay; the identity extends to complex z by dominated holomorphic continuation. This gives the reciprocal width $a = 1/\tau$ in logarithmic kernel coordinates and the forward heat time $\tau = -t$ in Newman coordinates. Evaluating that Gaussian on the fixed zero spectrum gives $Z(a)$; multiplying the theta kernel and Fourier transforming gives $H_{-\tau}$. The formulas specify both operations and their shared kernel exactly.

Two finite arithmetic matrices now give possible rejection certificates. For any $a_i > 0$, $a > 0$ and nonnegative integers i, j, form

$$K_{ij} = Z(a_i + a_j), \qquad M_{ij} = (-1)^{i+j} Z^{(i+j)}(a). \tag{885}$$

If RH holds, write $\rho = 1/2 + i\gamma$. For every finite real vector c the absolutely convergent sums give

$$c^T K c = \sum_{\gamma} \left(\sum_i c_i e^{-a_i\gamma^2} \right)^2 \geq 0, \qquad c^T M c = \sum_{\gamma} e^{-a\gamma^2} \left(\sum_i c_i \gamma^{2i} \right)^2 \geq 0.$$

Therefore a certified negative quadratic form evaluated by (882), including its full prime sum, would disprove RH. This proves a finite test criterion; it is not a conditional construction of the desired example. No such negative matrix is reported here.

The prime term must not be replaced by heat asymptotics. For $0 < a \leq \log 6/8$ and $n \geq 2$, $\log n \leq \sqrt{n}$ and

$$e^{-(\log n)^2/(4a)} \leq e^{-(\log 2)^2/(4a)} (n/2)^{-2},$$

because for $n \geq 3$ the factorization $\log^2 n - \log^2 2 = \log(n/2)\log(2n)$ is at least $\log 6 \log(n/2) \geq 8a\log(n/2)$; $n = 2$ is exact. The inequality $\log n \leq \sqrt{n}$ follows by maximizing $\log x/\sqrt{x}$, whose maximum on $x \geq 1$ is $2/e < 1$. Consequently

$$0 \leq \Psi(a) \leq 4\left(\frac{\pi^2}{6} - 1\right)\frac{e^{-(\log 2)^2/(4a)}}{\sqrt{\pi a}}. \tag{886}$$

Indeed $\Lambda(n)n^{-1/2} \leq 1$ and the comparison series is $\sum_{n\geq 2} 4n^{-2}$. This reproduces the cited bound and proves $\Psi(a) = O(a^A)$ for every fixed $A > 0$. Thus power-series heat coefficients cannot distinguish the complete arithmetic expression from one with its prime term omitted. Exact finite-a tests retain it.

11 Transporting the retained counterexample mechanism

This section starts from the explicit polynomial used in the accompanying fluid reconstruction, not from a search over unrelated test functions. Tao's displayed map and geometric explanation[97], Speyer's marked-factor discussion[98], and Poplett's complete polynomial-coordinate proof[99] identify the source. The calculations below retain its original coefficients. The interfaces developed here are the full inverse-fibre cubic, its coefficient heat flow, and the conductor model obtained from its two-sheet fibre product. These calculations establish precise local and dynamical connections; they do not identify a zero of the Riemann zeta function off its critical line.

11.1 Original coordinates and the entire inverse fibre

Use spatial coordinates (x, y, w): w is the third source coordinate called t in the cited polynomial, with the explicit identity renaming $(x, y, t) \mapsto (x, y, w)$. Evolution time below is τ or s. Write the target as (a, b, c) and set

$$\begin{aligned}
F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy),\\
F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy),\\
F_3 &= 2x - 3x^2 y - x^3 w.
\end{aligned} \tag{887}$$

Here is a short exact determinant check. On $x \neq 0$, put $v = 1+xy$, $h = 2-3xy-x^2w$, $A = v^2+v-v^3h$ and $B = 4v + 2 - 3v^2h$. Then $F = (A/x^2, B/x, xh)$ and

$$\det\frac{\partial(x, v, h)}{\partial(x, y, w)} = -x^3, \qquad \det\frac{\partial F}{\partial(x, v, h)} = x^{-3}\det\begin{pmatrix} -2A & 2v + 1 - 3v^2h & -v^3 \\ -B & 4 - 6vh & -3v^2 \\ h & 0 & 1 \end{pmatrix} = 2x^{-3}.$$

The numerator in the last equality expands to $(4 - 6vh)(-2v^2 - 2v + 3v^3h) + (2v + 1 - 3v^2h)(4v + 2 - 6v^2h) = 2$. The product is -2. Since $\det DF$ is a polynomial, the identity on $x \neq 0$ proves $\det DF = -2$ everywhere, including $x = 0$.

Proposition 11.1 (An exact inverse-fibre chart)**.** *For fixed $(a, b, c) \in \mathbb{C}^3$, define*

$$P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \qquad \alpha(r) = \frac{1}{2}P'_{a,b,c}(r). \tag{888}$$

The points in $F^{-1}(a, b, c)$ with $x \neq 0$ correspond bijectively to the simple roots r of $P_{a,b,c}$, by

$$x = \alpha^{-1}, \qquad y = r - \alpha, \qquad w = 5\alpha^2 - 3r\alpha - c\alpha^3. \tag{889}$$

When $c = 0$ there is one additional point $(0, b, a - 4b^2)$. When $c \neq 0$ there is no point with $x = 0$.

Proof. At a point with $x \neq 0$ define $r = y + 1/x$ and $\alpha = 1/x$. Solving $F_3 = c$ gives the formula for w in (889). Substitution in the other two original polynomials gives, before imposing their target values,

$$F_2 = 4r + 2\alpha - 3cr^2, \qquad F_1 = r^2 + r\alpha - cr^3.$$

Thus $F_2 = b$ is precisely $2\alpha = 3cr^2 - 4r + b = P'(r)$. After this substitution, $2(F_1 - a) = P(r)$. These identities prove both directions, and recovering $r = y + 1/x$ proves injectivity of the root parametrization. The requirement $\alpha \neq 0$ is exactly the simple-root requirement; a multiple root is not a finite point of this chart. Finally $F(0, y, w) = (w + 4y^2, y, 0)$, which proves the separate boundary statement without division by x. $\square$

For $c \neq 0$ the cubic discriminant is

$$\operatorname{Disc}_r P = 4\big(b^2 - cb^3 + 18abc - 16a - 27a^2c^2\big). \tag{890}$$

This follows by substitution of $(c, -2, b, -2a)$ into the expanded cubic discriminant, or by eliminating r between P and P'. The same polynomial remains defined at $c = 0$, but the polynomial then has degree two and the extra $x = 0$ point must still be included. For example $(a, b, c) = (-1/4, 0, 0)$ gives $r = \pm 1/2$ and $\alpha = -2r$, yielding $(1, -3/2, 13/2)$ and $(-1, 3/2, 13/2)$; the additional point is $(0, 0, -1/4)$. At $(a, b, c) = (0, 0, 0)$ the quadratic has a double root, hence neither of these two chart points survives; the additional point is $(0, 0, 0)$. In particular, loss of the two inverse branches is not a zero of $\det DF$ at a finite source point.

11.2 The same branch as an incompressible trajectory

Let $J = DF$ on $\mathbb{R}^3$ and define

$$W = J^{-1}e_1 = -\tfrac{1}{2}\nabla F_2 \times \nabla F_3 = \operatorname{curl}(-\tfrac{1}{2}F_2 \nabla F_3). \tag{891}$$

The scalar triple products with ∇F_i prove $JW = e_1$; the determinant computation shows that its coefficients are polynomial. Commuting mixed derivatives gives $\operatorname{div} W = 0$. For $s < 1/4$ define

$$\gamma_\pm(s) = \left(\pm(1-4s)^{-1/2}, \mp\tfrac{3}{2}(1-4s)^{1/2}, \tfrac{13}{2}(1-4s)\right).$$

Both satisfy $F(\gamma_\pm(s)) = (s - 1/4, 0, 0)$ by the inverse chart. Differentiation gives $J\gamma_\pm' = e_1$, hence $\gamma_\pm' = W(\gamma_\pm)$. Their first coordinate escapes at $s = 1/4$. The inverse-root coordinate on the positive branch is $r = -\sqrt{1-4s}/2$, with $r^2 = 1/4 - s$ and $r' = -1/(2r)$.

The metric making this transport an isometry is explicitly $g = J^\mathsf{T} J$, with $\det g = 4$. Locally $X = F(x, y, w)$ is a coordinate system and $g = \sum dX_i^2$. Thus W is the pullback of the constant field e_1, is parallel, and has g-norm one. It solves the intrinsic incompressible viscous equation for every positive viscosity with constant pressure and the Hodge Laplacian on one-forms. Its displayed trajectories have finite g-length before escaping, proving metric incompleteness: a sequence of their times tending to $1/4$ is metric Cauchy but has no finite coordinate limit. A smooth positive metric gives the ordinary local topology, as follows by bounding its eigenvalues above and below on a small closed coordinate ball.

This transport retains its metric, rather than silently replacing it by the original Euclidean one. For a scalar target function f, set $W_i = J^{-1}e_i$. The chain rule proves

$$W_i(f \circ F) = (\partial_i f) \circ F, \qquad \sum_i W_i^2 (f \circ F) = (\Delta_X f) \circ F. \tag{892}$$

The fields commute: their bracket annihilates each F_j, and invertibility of J then makes that bracket zero. The ordinary source Laplacian is not the operator in (892): $\Delta_{x,y,w}F_1(0) = 8$ whereas $\Delta_X X_1 = 0$. These are explicit operator coordinates for the fluid transfer, not a conclusion that the constructions are unrelated.

11.3 The released fluid amplifier in material coordinates

The released Boussinesq construction[101, Section 3.1, Lemma 3.1] provides the following local wave. We retain its profile rather than replace it by a different trial wave. Write $J_2(a,b) = (-b,a)$, $\kappa = |\zeta|^2$, $\zeta(t_0) \neq 0$, $\lambda > 0$, and $s = \lambda\zeta \cdot x$. The earlier fields are $u_{\text{old}} = Dx$, $\theta_{\text{old}} = G \cdot x$, with $\operatorname{tr} D = 0$. A smooth odd periodic profile f has an even periodic primitive p, $p' = f$. The new fields are

$$\vartheta = \Theta f(s), \quad \psi = \frac{\Omega}{\lambda^2\kappa}p(s), \quad v = \frac{\Omega}{\lambda\kappa}J_2\zeta f(s), \quad \varpi = \Omega f'(s).$$

Here $v = J_2\nabla\psi$, $\varpi = \operatorname{curl} v = \Delta\psi$. The equations for the inviscid scalar and vorticity have right-hand sides 0 and $\partial_1\theta$, respectively, before the specified forcing residuals. Their exact increments vanish when

$$\dot\zeta = -D^{\mathsf T}\zeta, \qquad \dot\Theta = -\frac{J_2\zeta \cdot G}{\lambda\kappa}\Omega, \qquad \dot\Omega = \lambda\zeta_1\Theta. \tag{893}$$

For completeness, $\dot\zeta = -D^{\mathsf T}\zeta$ makes $(\partial_t + Dx\cdot\nabla)s = 0$. Since $J_2\zeta\cdot\zeta = 0$, $v\cdot\nabla\vartheta = v\cdot\nabla\varpi = 0$. The old vorticity is spatially constant. The scalar residual is $[\dot\Theta + \Omega(J_2\zeta\cdot G)/(\lambda\kappa)]f(s)$; the vorticity residual is $[\dot\Omega - \lambda\zeta_1\Theta]f'(s)$. These expressions prove every cancellation in (893). They use no sinusoidal identity.

The covector map has an exact inverse-flow origin. If $\dot M = DM$, $M(t_0) = I$, then $\zeta = M^{-\mathsf T}\zeta(t_0)$, as differentiation proves. Jacobi's determinant identity gives $\partial_t \det M = (\operatorname{tr} D)\det M = 0$, so $\det M = 1$. More generally a smooth incompressible material flow $\dot X(a,t) = u(X(a,t),t)$ with $X(a,t_0) = a$ has

$$\partial_t(D_aX) = (D_xu)\circ X\, D_aX, \quad \det D_aX = 1, \quad \nabla_x(\phi_0\circ X^{-1})\circ X = (D_aX)^{-\mathsf T}\nabla_a\phi_0.$$

The first identity follows by differentiating the flow in its label; the other two follow by the determinant identity and chain rule. These formulas exhibit exactly how gradient information is transported while the determinant stays fixed. The particular amplification and reset sequence, compact localization, and all-derivative forcing control belong to the source construction, not to the determinant identity alone.

One can retain viscosity directly in this calculation. Scalar diffusivity $\eta \geq 0$ and vorticity viscosity $\nu \geq 0$ add $\eta\lambda^2\kappa\Theta f''$ and $\nu\lambda^2\kappa\Omega f'''$ to the two right-hand sides. For the special profile $f(s) = \sin s$ and stationary background $D = 0$, $G = -Ae_2$, $A > 0$, put $\zeta = r(\sin\phi, \cos\phi)$ with $r > 0$, and $k = \lambda r$. Then the amplitude matrix and its two eigenvalues are exactly

$$B_{\eta,\nu} = \begin{pmatrix} -\eta k^2 & A\sin\phi/k \\ k\sin\phi & -\nu k^2 \end{pmatrix},$$
$$\mu_\pm = -\frac{(\eta+\nu)k^2}{2} \pm \sqrt{\frac{(\eta-\nu)^2k^4}{4} + A\sin^2\phi}.$$

Expanding $\det(\mu I - B)$ proves the formula; comparing the nonnegative square root with $(\eta+\nu)k^2/2$ gives $\mu_+ > 0$ exactly when $A\sin^2\phi > \eta\nu k^4$. For the actual localized profile, the displayed f'' and f''' terms remain; there is no replacement of those functions by $-f, -f'$.

There is likewise an exact, reversible force map in the original three-dimensional Cartesian coordinates. From an Euler triple $\partial_t u + (u \cdot \nabla)u + \nabla p_E = f_E$, $\operatorname{div} u = 0$, all Navier–Stokes triples with the same velocity and viscosity ν are

$$p_{NS} = p_E + q, \qquad f_{NS} = f_E - \nu\Delta u + \nabla q, \qquad \operatorname{curl} f_{NS} = \operatorname{curl} f_E - \nu\Delta \operatorname{curl} u.$$

Substitution proves sufficiency; subtraction proves necessity. The curl identity follows from commuting distributional derivatives, so it retains every pressure adjustment. The companion Euler manuscript[102, Theorem 1.1 and Sections 3–4] is the source of the three-dimensional material-layer construction. This section reconstructs the amplifier and its exact transfer terms; it does not certify that manuscript's full multiscale proof or replace the viscous transfer by dropping $\nu\Delta u$.

11.4 Exact heat transport of the fibre cubic

Keep a_0, b_0, c fixed and move the target by

$$a(\tau) = a_0 + 2\tau, \qquad b(\tau) = b_0 + 6c\tau. \tag{894}$$

For $P(\tau, r) = P_{a(\tau),b(\tau),c}(r)$ direct differentiation gives

$$P_\tau = 6cr - 4 = P_{rr}. \tag{895}$$

Thus the original inverse-fibre family is closed under the forward heat operator, with every coefficient and the time sign specified. On a simple root, $r_\tau = -P_{rr}/P_r$. The velocity of its lifted point in the original source coordinates is $J^{-1}(2, 6c, 0)^\mathsf{T}$, by differentiating its target. No such finite lift is assigned when $P_r = 0$.

With Newman time $t = -\tau$, write $\mathcal{P}(t, r) = P_{a_0-2t,b_0-6ct,c}(r)$. Then $\mathcal{P}_t = -\mathcal{P}_{rr}$ and $r_t = \mathcal{P}_{rr}/\mathcal{P}_r$, the same local zero equation as in Section 10.14. For $b_0 = c = 0$ this becomes exactly

$$\mathcal{P}(t, r) = -2\big(r^2 - 2(t - t_c)\big), \qquad t_c = a_0/2.$$

For real a_0 and real t, the double root occurs at $(t_c, 0)$; the two roots are real for $t > t_c$ and nonreal for $t < t_c$. The point $a_0 = -1/4$ of the original three-point collision corresponds to $t_c = -1/8$ in this particular heat orbit. Neither shifting this model time origin nor choosing another a_0 changes the distinguished initial function $H_0(z) = \xi(1/2 + iz/2)/8$ of the actual Newman family.

There is also an exact nonlinear velocity equation. Wherever $P \neq 0$, set $u = -2P_r/P$. Writing locally $L = \log P$ gives $L_\tau = L_{rr} + L_r^2$. Differentiating this equality proves $u_\tau + uu_r = u_{rr}$. A zero of multiplicity m gives $u = -2m/(r - r_*) + O(1)$ by factoring $P = (r - r_*)^m A$, $A(r_*) \neq 0$. This is the singularity of a specified logarithmic-derivative map.

The arithmetic transport can already be written without guessing a new zero. Define

$$\widehat{H}(t, x, y, w) = H_t(F_1(x, y, w)).$$

The established Newman equation and (892) give

$$\partial_t \widehat{H} = -W_1^2 \widehat{H}. \tag{896}$$

Its values are constant on every fibre of F. In particular the nonzero functional $D_s = \delta_{\gamma_+(s)} - \delta_{\gamma_-(s)}$ annihilates every scalar $f \circ F$, including $\widehat{H}(t, \cdot)$, while $D_s(x) = 2(1 - 4s)^{-1/2}$ diverges. This proves exactly which branch information scalar descent loses and supplies the retained branch-sensitive observable. The cancellation is not a negative Weil value or an off-critical zero.

Even a local replacement of H_t by a polynomial factor must carry its analytic unit. The exact product identity is

$$(\partial_t + \partial_z^2)(AP) = A(\partial_t + \partial_z^2)P + (A_t + A_{zz})P + 2A_z P_z. \tag{897}$$

At a simple zero of P, the ratio $(AP)_{zz}/(AP)_z$ is $P_{zz}/P_z + 2A_z/A$. Thus discarding the unit would change the zero dynamics by an explicitly known term. The present reconstruction transfers the mechanism and its equations; it does not replace the arithmetic initial datum by a chosen cubic.

11.5 Two inverse sheets produce a conductor crossing

On $b = c = 0$ the inverse-root cover is the map $\pi : \mathbb{C}_r \to \mathbb{C}_a$, $a = -r^2$. Its two-fold self-fibre product has coordinate ring

$$B = \mathbb{C}[r_1, r_2]/(r_1^2 - r_2^2).$$

The linear changes

$$u = r_1 - r_2, \quad v = r_1 + r_2, \qquad r_1 = (u + v)/2, \quad r_2 = (v - u)/2 \tag{898}$$

give the exact ring isomorphism $B \simeq R = \mathbb{C}[u, v]/(uv)$. The component $u = 0$ records equal roots, and $v = 0$ records opposite roots. For the physical $x \neq 0$ inverse chart one must remove $r_1 r_2 = 0$; the crossing at the origin belongs to its algebraic completion. Adjoining a free coordinate z gives the corresponding local surface $uv = 0$ in three-space. This is an actual map from the retained polynomial to the double-crossing model used in the analysis of Campana–Demailly–Peternell[100, Section 7, p. 692], through a stated fibre-product construction.

We first compute in the curve ring R; the free surface coordinate will be restored below. The normalization ring is $\widetilde{R} = \mathbb{C}[u] \oplus \mathbb{C}[v]$. The injection $R \to \widetilde{R}$ sends a polynomial to its restrictions $(f(u, 0), f(0, v))$. Its image consists exactly of pairs having equal constant terms: the inverse reconstruction is $p(u) + q(v) - p(0)$ for a pair with $p(0) = q(0)$. The conductor is

$$\mathfrak{c} = (u, v) \subset R, \qquad \mathfrak{c}\widetilde{R} = u\mathbb{C}[u] \oplus v\mathbb{C}[v].$$

Indeed a pair with zero constant terms can be multiplied by every pair in $\widetilde{R}$ and remains in the image. Conversely multiplication by $(1, 0)$ forces the common constant term of any conductor element to be zero.

The intrinsic differential module is explicitly

$$\Omega^1_R = (R\,du \oplus R\,dv)/R(v\,du + u\,dv). \tag{899}$$

Its class $\theta = v\,du = -u\,dv$ is nonzero. If it were zero, $(v, 0) = h(v, u)$ in R^2, so $hv = v$ and $hu = 0$. The latter forces $h \in (v)$ by the just-proved normalization description. Reducing the former in $\mathbb{C}[v]$ then forces $h(0, v) = 1$, contradicting $h \in (v)$. Also $u\theta = v\theta = 0$. Since $u + v$ is a non-zero-divisor (its two normalized restrictions are nonzero polynomials), this is a genuine torsion class. Its image in $\Omega^1_{\widetilde{R}}$ is zero on both branches.

In fact the kernel of $\Omega^1_R \to \Omega^1_{\widetilde{R}}$ is exactly $\mathbb{C}\theta$. For a class $A\,du + B\,dv$ to map to zero, $A(u, 0) = B(0, v) = 0$, so $A = va(v)$ and $B = ub(u)$. The relations $v^2 du = u^2 dv = 0$ reduce this class to $(a(0) - b(0))\theta$. This proves the kernel statement, not merely the existence of a forgotten class. In the original root coordinates, $\theta = r_2 dr_1 - r_1 dr_2$: expand vdu using (898) and use $r_1 dr_1 = r_2 dr_2$. For the surface ring $R[z]$, differentials split as $(\Omega^1_R \otimes_R R[z]) \oplus R[z]dz$. The last summand injects into the normalized ring's dz summand, and the preceding calculation works coefficient by coefficient in z. Thus its kernel is exactly $\mathbb{C}[z]\theta$.

The ambient restricted module is another explicit part of the same diagram: for $S = (uv = 0) \subset \mathbb{C}^2$ it is $\Omega^1_{\mathbb{C}^2}|_S = R\,du \oplus R\,dv$. Its nonzero conormal $d(uv) = vdu + udv$ maps to zero in (899); vdu instead maps to the nonzero class θ and subsequently to zero on normalization. These two calculated kernels are the precise objects to retain when comparing an ambient restricted differential with its torsion-free quotient. They are not interchangeable vanishings. No global line bundle, compact threefold, or CDP20 counterexample is inferred just from this local ring: the existing global construction must transport these classes through its own gluing maps.

Here is the global sheaf diagram in which the local calculation belongs. Let S be a reduced effective Cartier divisor in a smooth complex manifold X. Put $E = \Omega^1_X|_S$, $N = \mathcal{O}_S(-S)$, $T = \ker(\Omega^1_S \to j_*\Omega^1_{S_{\mathrm{reg}}})$, and $Q = \Omega^1_S/T$, where $j : S_{\mathrm{reg}} \hookrightarrow S$. Let K be the inverse image of T under $E \to \Omega^1_S$. The precise sequences are

$$0 \longrightarrow N \longrightarrow K \longrightarrow T \longrightarrow 0, \qquad 0 \longrightarrow K \longrightarrow E \longrightarrow Q \longrightarrow 0. \tag{900}$$

Indeed the Cartier conormal map sends a local frame to dg for a reduced defining equation g. At the generic smooth points of each component, dg is nonzero. A coefficient annihilating it therefore vanishes on every component and is zero since S is reduced. This proves left injectivity in the conormal sequence. Taking the inverse image of T proves the first sequence, and its quotient proves the second. On $uv = 0$, N is generated by $vdu + udv$ and T by the class calculated above.

For any line bundle L on S, twisting these sequences is exact because a line bundle is locally free. In particular the relevant segments in cohomology are

$$\begin{aligned}
&0 \longrightarrow H^0(N \otimes L) \longrightarrow H^0(K \otimes L) \longrightarrow H^0(T \otimes L) \xrightarrow{\delta_L} H^1(N \otimes L),\\
&0 \longrightarrow H^0(K \otimes L) \longrightarrow H^0(E \otimes L) \longrightarrow H^0(Q \otimes L).
\end{aligned}$$

The connecting map is explicit: lift a torsion section to sections k_i of $K \otimes L$ on a sufficiently fine trivializing cover where these local lifts exist; the differences $k_j - k_i$ lie in $N \otimes L$ and form its representing cocycle. Changing lifts adds a coboundary. In the vanishing reduction used in the cited Section 7, setting the two endpoint groups $H^0(N \otimes L)$ and $H^0(Q \otimes L)$ to zero leaves precisely $\ker \delta_L$ as the ambient group $H^0(E \otimes L)$. It does not erase T. This calculation identifies the exact additional term and the actual gluing map, without assigning any global conclusion to the local model alone.

11.6 Retaining the two sheets in the arithmetic spectral quotient

Connes–Consani–Moscovici[103, Section 3.6, Proposition 3.6] use the Hadamard topological ring $\mathcal{H}_{\leq 1}$ of entire functions of order at most one. Their quotient is

$$Q_{\mathrm{ar}} = \mathcal{H}_{\leq 1}/\overline{\mathrm{span}\{\mathcal{F}_\mu \mathcal{E}(\psi^\pm_\ell)\}}, \qquad (\mathcal{F}_\mu \mathcal{E}\psi^\pm_\ell)(s) = R^\pm_\ell(s)\Xi(s).$$

Multiplication by s induces an operator S on this quotient; the cited proposition identifies its spectrum with the parameters s for which $1/2 + is$ is a nontrivial zeta zero. We use this specified quotient, not an orthogonal complement and not an unjustified replacement of the closed subspace by an algebraic ideal. The exact relation to our heat coordinate is

$$\Xi(s) = \xi(1/2 + is) = 8H_0(2s), \qquad a = 2s.$$

Consequently the retained inverse-root cover $a = -r^2$ gives $s = -r^2/2$, including the factor two.

Form the finite free scalar extension

$$\widehat{Q}_{\mathrm{ar}} = \mathbb{C}[r] \otimes_{\mathbb{C}[s]} Q_{\mathrm{ar}}, \qquad s \longmapsto -r^2/2. \tag{901}$$

Every polynomial in r has a unique even–odd decomposition $p_0(-r^2/2) + rp_1(-r^2/2)$. Existence follows by separating its even and odd powers, and uniqueness by comparing those powers. Thus $\mathbb{C}[r]$ is free of rank two over $\mathbb{C}[s]$, and (901) identifies exactly with $Q_{\mathrm{ar}} \oplus rQ_{\mathrm{ar}}$. Give it this finite direct-sum topology. We are not asserting that this tensor product is a space of all entire functions pulled back along $s = -r^2/2$; that would alter their growth class.

In the ordered basis $(1, r)$, multiplication by r and by the original spectral coordinate s are, respectively,

$$\mathcal{R} = \begin{pmatrix} 0 & -2S \\ I & 0 \end{pmatrix}, \qquad \widehat{S} = -\tfrac{1}{2}\mathcal{R}^2 = \begin{pmatrix} S & 0 \\ 0 & S \end{pmatrix}. \tag{902}$$

Indeed $r(q_0 + rq_1) = -2Sq_1 + rq_0$, proving both matrices. The sheet involution sends (q_0, q_1) to $(q_0, -q_1)$; its trace is $2q_0$ and its kernel is precisely rQ_{ar}. This trace is $\mathbb{C}[s]$-linear, not $\mathbb{C}[r]$-linear: $\mathcal{R}(rq) = -2Sq$ lies in the even summand, so the odd summand is not in general preserved by $\mathcal{R}$ itself. The normalized trace $\pi(q_0, q_1) = q_0$ has section $\iota(q) = (q, 0)$. Therefore the sequence

$$0 \longrightarrow rQ_{\mathrm{ar}} \longrightarrow \widehat{Q}_{\mathrm{ar}} \xrightarrow{\pi} Q_{\mathrm{ar}} \longrightarrow 0$$

is split, and $\widehat{S}$ preserves both summands. These are the explicit branch-retention, branch-forgetting, and inverse maps.

The spectrum in the original arithmetic coordinate is unchanged: $\operatorname{Spec} \widehat{S} = \operatorname{Spec} S$. Here spectrum means failure of a continuous two-sided inverse in the source quotient topology. If $S - \lambda$ has inverse T, $\operatorname{diag}(T, T)$ inverts $\widehat{S} - \lambda$. Conversely, for an inverse B of $\widehat{S} - \lambda$, $\pi B \iota$ is an inverse of $S - \lambda$, since $\pi(\widehat{S} - \lambda) = (S - \lambda)\pi$ and $(\widehat{S} - \lambda)\iota = \iota(S - \lambda)$. Continuity follows from the finite direct-sum topology in both directions. This proof carries the actual operator and its lost summand; it does not turn a complex value of r into an off-real value of the original spectral coordinate s. For an eigenvector $Sq = \lambda q$ and a scalar ρ with $\rho^2 = -2\lambda$, $(\rho q, q)$ is directly an eigenvector of $\mathcal{R}$ with eigenvalue ρ. Thus even that change of spectral coordinate is completely explicit.

11.7 The covariant branch derivative and its exact obstruction

The finite extension retains two coordinates, so differentiating a section requires retaining the derivative of the branch itself. On the punctured base $s \neq 0$, write a section as

$$q(s, r) = q_0(s) + rq_1(s), \qquad r^2 = -2s.$$

Since $2r\,dr/ds = -2$, one has

$$\frac{dr}{ds} = -\frac{1}{r} = \frac{r}{2s}.$$

In the ordered basis (q_0, q_1) define

$$D = \frac{d}{ds} + A(s), \qquad A(s) = \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2s} \end{pmatrix}, \qquad R(s) = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}. \tag{903}$$

Here R is multiplication by r, not a new identification of the two sheets. Direct differentiation gives the exact commutator

$$[D, R] = R' + [A, R] = -R^{-1}, \qquad R^{-1} = \begin{pmatrix} 0 & 1 \\ -\frac{1}{2s} & 0 \end{pmatrix}. \tag{904}$$

For completeness, squaring the connection gives

$$D^2 = \frac{d^2}{ds^2} + 2A\frac{d}{ds} + A' + A^2 = \begin{pmatrix} \frac{d^2}{ds^2} & 0 \\ 0 & \frac{d^2}{ds^2} + \frac{1}{s}\frac{d}{ds} - \frac{1}{4s^2} \end{pmatrix}. \tag{905}$$

The inverse matrix exists precisely on the punctured base because $R^2 = -2sI$. Thus the inverse-square term is forced by the ramified map $s = -r^2/2$, rather than being a coordinate convention.

Proposition 11.2 (Exact heat transport on the two-sheet module)**.** *Let $H_t(a)$ be the scalar heat-coordinate function already used above and put $Z_t(s) = 8H_t(2s)$. If $\partial_t H_t(a) = -\partial_a^2 H_t(a)$, then*

$$\partial_t Z_t(s) = -\frac{1}{4}\partial_s^2 Z_t(s). \tag{906}$$

On a two-component section, the corresponding branch-covariant second derivative is D^2, not the componentwise operator ∂_s^2. Consequently the odd component acquires the exact inverse-square term in (905)*.*

Proof. The change of variable $a = 2s$ gives $\partial_a^2 H_t(2s) = \frac{1}{4}\partial_s^2 H_t(2s)$, proving (906) after multiplication by 8. For a section $q_0 + rq_1$, the displayed formula for dr/ds gives $dq/ds = (q_0', r(q_1' + q_1/(2s)))$, exactly $D(q_0, q_1)$. Applying this once more gives D^2 and its lower-right entry in (905). No differentiation has been asserted to descend to the closed quotient Q_{ar} : the calculation is on the explicit punctured finite extension, with its domain stated above. □

The commutator records the precise information lost by treating the branch variable as a scalar coordinate. In particular, D does not commute with multiplication by r; the failure is the explicit map $-R^{-1}$. This is the typed obstruction for transporting the material generator through the arithmetic module. It is not a negative Weil pairing, an off-critical zero, or a classical Navier–Stokes disproof.

For a scalar analytic unit $A(s,t)$ and a two-component section v, the same connection gives the exact product rule

$$\widehat{L}(Av) = A\widehat{L}v + \left(A_t + \frac{1}{4}A_{ss}\right)v + \frac{1}{2}A_s Dv, \qquad \widehat{L} = \partial_t + \frac{1}{4}D^2. \tag{907}$$

This is a coordinate identity on the punctured extension. In particular, if v solves the homogeneous branch heat equation, the two displayed derivative terms are the complete analytic-unit defect; they cannot be discarded by passing to the scalar quotient. On a simple scalar zero $Z_t = AP$ the corresponding scalar specialization is

$$\lambda'(t) = \frac{1}{4}\frac{P_{ss}}{P_s}(t, \lambda(t)) + \frac{1}{2}\frac{A_s}{A}(t, \lambda(t)), \tag{908}$$

on the open locus $P_s \neq 0$. At a multiple zero the quotient is undefined, and (907), rather than a scalarized formula, is the retained statement.

The scalar transported arithmetic generator from (934) has a direct two-sheet lift on this punctured extension:

$$\widehat{B}_t = \widehat{S} - \frac{t}{2}D, \qquad \widehat{S} = -\frac{1}{2}R^2, \qquad [\widehat{B}_t, R] = \frac{t}{2}R^{-1}. \tag{909}$$

Indeed, $\widehat{S}$ is a polynomial in R^2 and hence commutes with R, while (904) gives the last identity. This is the exact branch-coupling term that must be retained when the material generator is transported through the cover. It is defined on the punctured two-sheet extension; it is not asserted to induce an operator on the frozen quotient Q_{ar}.

There is, however, an exact typed interface with the scalar arithmetic generator before taking that quotient. Let

$$\mathcal{J} : \mathcal{O}(U) \longrightarrow \mathcal{O}(U)^2, \qquad \mathcal{J}f = (f, 0),$$

on the punctured s-domain U. Since $R^2 = -2sI$ and the displayed connection is diagonal in the (q_0, q_1) frame,

$$D\mathcal{J}f = \mathcal{J}(f'), \qquad \widehat{S}\,\mathcal{J}f = \mathcal{J}(sf), \qquad \widehat{B}_t\mathcal{J}f = \mathcal{J}\left(sf - \frac{t}{2}f'\right).$$

Consequently

$$\widehat{B}_t\mathcal{J} = \mathcal{J}B_t, \qquad B_t = M - \frac{t}{2}\partial_s, \tag{910}$$

as an equality of maps on the stated analytic-germ domain. This is an intertwiner for the even section only; it neither identifies the odd sheet with a scalar function nor supplies descent of D or $\widehat{B}_t$ to the frozen quotient. In particular, the branch commutator above remains visible on the full two-component module.

Verification and continuation. The accompanying exact checker replays the determinant, inverse chart, entire collision fibre, discriminant, coefficient heat flow, escaping trajectory, and the fibre-product change of variables. It uses rational polynomial arithmetic, not numerical evidence or a Lean certification. The conductor and differential-module arguments above are full written proofs. The next arithmetic step carries the actual heat and material generators through the now-retained two-component arithmetic module, including the branch coupling and analytic-unit derivative. No independent generic zero scan is prescribed by this section.

12 The retained cover as an exact Weil test algebra

The inverse-fibre construction supplies the relation $r^2 = -2s$. We now realize that relation on the actual test space of the Weil form, calculate every finite polynomial channel and its inverse domain, and identify precisely which pairing evaluates a zeta-zero witness. This is a transport of the retained mechanism, not a replacement by a generic zero search. We use the Fourier and Weil conventions of (837); compare Weil[26] and Alpöge–Furman[10, Sections 1.1–1.2 and 2]. The elementary module calculations below are proved here in full; no novelty is claimed for polynomial division, compactly supported ordinary differential equations, or the resulting linear algebra.

12.1 The arithmetic-coordinate operator and its full form

Put $\mathcal{E} = C_c^\infty(\mathbb{R};\mathbb{C})$ and retain the variable u in the test domain and s in the spectral domain. Define

$$(Tf)(u) = -if'(u), \qquad \widehat{f}(s) = \int_{\mathbb{R}} f(u)e^{-ius}\,du.$$

Integration by parts, with no endpoint term because f has compact support, gives $\widehat{Tf}(s) = s\widehat{f}(s)$ for every complex s. If $\operatorname{supp} f \subset [-L, L]$, repeated integration by parts gives arbitrarily rapid decay in $|\Re s|$ in each fixed horizontal strip. Consequently every polynomial in T acts on $\mathcal{E}$ without enlarging support, and all polynomially weighted zero sums below converge absolutely. This uses the usual zero-count bound $N(T) = O(T\log T)$, not RH.

Proposition 12.1 (The coordinate is symmetric for the actual Weil form)**.** *For $f, g \in \mathcal{E}$,*

$$W(Tf, g) = W(f, Tg). \tag{911}$$

For a polynomial p, let $p^\#(s) = \overline{p(\bar{s})}$. Then $W(p(T)f, g) = W(f, p^\#(T)g)$.

Proof. Write $s_\rho = (\rho - 1/2)/i$, exactly as in (837). The contribution to the first expression is

$$m_\rho \overline{\bar{s}_\rho \widehat{f}(\bar{s}_\rho)} \widehat{g}(s_\rho) = m_\rho \overline{\widehat{f}(\bar{s}_\rho)} s_\rho \widehat{g}(s_\rho),$$

which is the contribution to the second expression. Absolute convergence permits the equality of sums. Iteration and conjugate-linearity in the first slot prove the polynomial statement. No s_ρ was assumed real. □

Here is also the exact arithmetic expression for these weighted entries. For $h = g * f^\star$, where $f^\star(u) = \overline{f(-u)}$, let $\mathcal{L}(h)$ denote the right side of (839), with correlation h. For integers $j, k \geq 0$,

$$W(T^j f, T^k g) = \mathcal{L}\big((-i)^{j+k} h^{(j+k)}\big). \tag{912}$$

Indeed $(Tf)^\star = T(f^\star)$ by differentiating the reflection and conjugating $-i$. Derivatives commute with convolution and $(T^k g) * (T^j f)^\star = T^{j+k}(g * f^\star)$. Applying the explicit formula proves the identity, including both pole terms, the gamma term and every prime-power term. Thus a finite branch calculation gives explicit correlations for the full arithmetic form; it does not authorize dropping its archimedean or prime contributions.

12.2 Branch multiplication and the two different pairings

On $\mathcal{E}^2$ in the ordered basis $(1, r)$ set

$$\mathcal{R}(f_0, f_1) = (-2Tf_1, f_0), \qquad \mathcal{R}^2 = -2\,\mathrm{diag}(T, T), \qquad \pi(f_0, f_1) = f_0. \tag{913}$$

These are the same coefficients as (902), now on a concrete test-function module with a specified domain. Fourier transformation on each component intertwines this operator with multiplication by the matrix $\left(\begin{smallmatrix} 0 & -2s \\ 1 & 0 \end{smallmatrix}\right)$.

The componentwise arithmetic form is

$$\mathcal{W}(U, V) = W(f_0, g_0) + W(f_1, g_1), \qquad U = (f_0, f_1), \quad V = (g_0, g_1).$$

The algebra trace with the real structure $s^* = s$, $r^* = r$ instead gives the form

$$\mathcal{B}(U, V) = 2W(f_0, g_0) - 4W(f_1, Tg_1). \tag{914}$$

To derive the coefficients, multiply $(q_0^\# + rq_1^\#)(p_0 + rp_1)$, replace r^2 by $-2s$, and take the algebra trace, which is twice the even coefficient. The result is $2q_0^\# p_0 - 4sq_1^\# p_1$; evaluating by the Weil distribution gives (914). This derivation retains the involution: choosing $r^* = -r$ would change the odd sign and is a different specified real structure, not a silent convention change.

Proposition 12.2 (Exact adjoint and trace comparison)**.** *Both forms are Hermitian. For $\mathcal{W}$ the adjoint identity is*

$$\mathcal{W}(\mathcal{R}U, V) = \mathcal{W}\left(U, \begin{pmatrix} 0 & I \\ -2T & 0 \end{pmatrix} V\right).$$

For $\mathcal{B}$, the operator $\mathcal{R}$ is symmetric: $\mathcal{B}(\mathcal{R}U, V) = \mathcal{B}(U, \mathcal{R}V)$.

Proof. Hermiticity follows from that of W and (911). Direct substitution gives

$$\mathcal{W}(\mathcal{R}U, V) = -2W(f_1, Tg_0) + W(f_0, g_1).$$

Similarly each of $\mathcal{B}(\mathcal{R}U, V)$ and $\mathcal{B}(U, \mathcal{R}V)$ equals $-4W(f_1, Tg_0) - 4W(f_0, Tg_1)$. □

The distinction has an explicit fibre calculation. At a real positive spectral parameter $s = \lambda$, the algebra trace matrix is $\mathrm{diag}(2, -4\lambda)$, whereas the componentwise evaluation form has matrix I_2. Thus the former already has a negative odd direction over a real positive parameter. For example, at $\lambda = 1$ the vector $(0, 1)$ has trace value -4 and componentwise value 1. This is a calculation in the fibre algebra, not a claim of a zeta zero at 1. It proves why a negative branch-trace value alone is not a negative Weil value. The exact route to an arithmetic witness is the scalar channel calculated next; no relation between the structures is discarded by this comparison.

12.3 Every finite polynomial channel: range, kernel and lifts

Every $P(r) \in \mathbb{C}[r]$ has a unique expression

$$P(r) = A(s) + rB(s), \qquad s = -r^2/2.$$

If $P(r) = \sum_j p_j r^j$, the coefficients are exactly $A(s) = \sum_n p_{2n}(-2s)^n$ and $B(s) = \sum_n p_{2n+1}(-2s)^n$. The induced scalar channel is

$$L_P = \pi P(\mathcal{R}), \qquad L_P(f_0, f_1) = A(T)f_0 + C(T)f_1, \qquad C(s) = -2sB(s). \tag{915}$$

This follows by multiplying $\mathcal{R}$ into the two components, not by identifying the two sheets.

Proposition 12.3 (Constructive channel classification)**.** *Let $P \neq 0$. Choose polynomials d, a, c, u, v satisfying*

$$A = da, \qquad C = dc, \qquad ua + vc = 1.$$

They can be obtained by the Euclidean algorithm; retain the chosen nonzero constant factor in d and its corresponding factors in a, c, u, v. Then

$$\operatorname{im} L_P = d(T)\mathcal{E}, \qquad\qquad \ker L_P = \{(-c(T)h, a(T)h) : h \in \mathcal{E}\}. \tag{916}$$

The unique preimage $h \in \mathcal{E}$ of an element $d(T)h$ gives the explicit section $(u(T)h, v(T)h)$ of L_P onto its image. All these maps preserve any common compact support interval. For $P = 0$, the channel is zero, its image is zero and its kernel is all of $\mathcal{E}^2$.

Proof. First $p(T)$ is injective on $\mathcal{E}$ for every nonzero polynomial p: $p(s)\widehat{f}(s) = 0$ implies $\widehat{f} = 0$ away from finitely many points, hence identically by analyticity, and Fourier uniqueness gives $f = 0$. Now the displayed section maps to $d(T)(ua+vc)(T)h = d(T)h$, proving the image claim. A kernel vector satisfies $a(T)f_0 + c(T)f_1 = 0$ by injectivity of $d(T)$. Put $h = -v(T)f_0 + u(T)f_1$. Commutativity of polynomial differential operators gives

$$c(T)h = -(vc + ua)(T)f_0 = -f_0, \qquad a(T)h = (vc + ua)(T)f_1 = f_1.$$

Conversely $(-c(T)h, a(T)h)$ maps to zero. If both components of this parameterized vector vanish, $(ua + vc)(T)h = h = 0$, proving uniqueness of the kernel parameter. Derivatives preserve support. $\square$

The image condition is also fully explicit; it is not an unspecified solvability assumption. Write $d(s) = d_0 \prod_j (s - \lambda_j)^{k_j}$ with $d_0 \neq 0$ and distinct λ_j. Then

$$d(T)\mathcal{E} = \left\{ g \in \mathcal{E} : \int_{\mathbb{R}} u^\ell e^{-i\lambda_j u} g(u)\, du = 0 \quad (0 \leq \ell < k_j, \text{ every } j) \right\}. \tag{917}$$

These conditions are equivalently $\widehat{g}^{(\ell)}(\lambda_j) = 0$ because differentiating the Fourier integral inserts $(-iu)^\ell$.

Here is the inverse proving sufficiency, including the endpoint behavior. For one repeated factor, set

$$[(T-\lambda)^{-k}g](u) = \frac{i^k e^{i\lambda u}}{(k-1)!}\int_{-\infty}^{u}(u-v)^{k-1}e^{-i\lambda v}g(v)\,dv. \tag{918}$$

To verify the equation, conjugate $T-\lambda$ by $e^{i\lambda u}$: it becomes $-i\partial_u$. Differentiating the k-fold primitive k times gives g, with $(-i)^k i^k = 1$. If g is supported in $[a,b]$, the expression vanishes for $u < a$. For $u > b$, expand $(u-v)^{k-1}$: every coefficient is a constant multiple of one of the k weighted moments in (917), so it vanishes there too. Smoothness follows from the integral and from smooth compact support of g; the differential equation and the same derivative formulas show matching of all derivatives at a, b. Necessity follows by Fourier transformation of $g = (T-\lambda)^k f$. For distinct factors, apply (918) successively and divide by d_0. After division by one factor the Fourier transform still has the required vanishing orders at the other roots, since the removed factor is nonzero there. This proves sufficiency for the full product. Injectivity already proved makes the result independent of the order of successive divisions.

In particular $T^{-1}g = i\int_{-\infty}^{u} g(v)\,dv$ exists in $\mathcal{E}$ exactly when $\int g = 0$. Therefore

$$\mathcal{R}^{-1}(g_0, g_1) = (g_1, -\tfrac{1}{2}T^{-1}g_0), \qquad \operatorname{dom}\mathcal{R}^{-1} = \{(g_0, g_1) : \textstyle\int g_0 = 0\}. \tag{919}$$

Both compositions give the identity on their exact domains by (913). The factor $-1/2$ and the lost zero-moment condition have not been absorbed into a convention.

For a concrete polynomial family tied to the same root operator, take $P(r) = r(r^2 + 2\lambda)$. Then $A = 0$, $C(s) = 4s(s-\lambda)$, and

$$L_P(f_0, f_1) = 4T(T-\lambda)f_1.$$

The kernel is exactly $\mathcal{E} \oplus 0$. For $\lambda \neq 0$, the image consists of tests whose Fourier transforms vanish at 0 and λ. At $\lambda = 0$ it instead requires a double zero at 0, equivalently $\int g = \int ug = 0$. The inverse of the nonzero component is $\frac{1}{4}(T-\lambda)^{-1}T^{-1}g$, with the repeated-factor formula when $\lambda = 0$. This exhibits the collision of two image conditions without losing their multiplicity.

12.4 Explicit finite jet interpolation with retained support

Let $\lambda_1, \ldots, \lambda_q$ be distinct complex numbers, let $k_j \geq 1$, and put $D = \sum_j k_j$ and $d(s) = d_0 \prod_j (s - \lambda_j)^{k_j}$ with $d_0 \neq 0$. There is an exact $\mathbb{C}[s]$-module isomorphism

$$\mathcal{E}/d(T)\mathcal{E} \longrightarrow \bigoplus_{j=1}^{q} \mathbb{C}[s]/(s-\lambda_j)^{k_j}, \qquad [f] \longmapsto \left[\sum_{\ell < k_j} \frac{\widehat{f}^{(\ell)}(\lambda_j)}{\ell!}(s-\lambda_j)^\ell\right]_j. \tag{920}$$

The module action on the left is through T. Its kernel statement is exactly (917). We prove surjectivity by an explicit construction inside any prescribed nonempty open interval I.

Choose $u_0 \in I$ and $\epsilon > 0$ so $[u_0 - \epsilon, u_0 + \epsilon] \subset I$ and $\epsilon \max_j |\lambda_j| < \log 2$ (the latter is automatic if all nodes are zero). Put

$$b(v) = \begin{cases} \exp(-1/(1-v^2)), & |v| < 1, \\ 0, & |v| \geq 1, \end{cases} \qquad B = \int_{-1}^{1} b(v)\,dv > 0, \quad \phi_\epsilon(u) = \frac{b((u-u_0)/\epsilon)}{\epsilon B}.$$

Every derivative of b on $(-1,1)$ is its exponential times a rational function with a finite-order pole at the endpoints, so all derivatives tend to zero there; hence b is smooth on $\mathbb{R}$. The constants ϵ and B are retained in the construction. Writing $\Phi(s) = \widehat{\phi_\epsilon}(s)$, we have

$$\Phi(s) = e^{-iu_0 s} B^{-1} \int_{-1}^{1} b(v) e^{-i\epsilon v s}\, dv, \qquad \left| B^{-1} \int_{-1}^{1} b(v) e^{-i\epsilon v \lambda_j}\, dv - 1 \right| \le e^{\epsilon|\lambda_j|} - 1 < 1.$$

Thus $\Phi(\lambda_j) \neq 0$ at every node, with a direct lower bound.

For prescribed jets let $H_j(s) = \sum_{\ell<k_j} h_{j,\ell}(s-\lambda_j)^\ell$ and $M_j(s) = \prod_{a\neq j}(s-\lambda_a)^{k_a}$. Let $A_j(s)$ be the Taylor polynomial of degree $k_j - 1$ at λ_j of $H_j(s)/(M_j(s)\Phi(s))$. Its denominator is nonzero near that point, so its coefficients are obtained by ordinary finite power-series division. Define

$$Q(s) = \sum_{j=1}^{q} M_j(s) A_j(s), \qquad f = Q(T)\phi_\epsilon. \tag{921}$$

At λ_j, every summand other than the jth vanishes to order at least k_j, and the jth satisfies $M_j A_j \Phi = H_j + O((s-\lambda_j)^{k_j})$. Therefore $\widehat{f} = Q\Phi$ has exactly the prescribed truncated jets. Its support is inside I, and $\deg Q \le D - 1$. This proves surjectivity in (920), not merely an existence claim for an unspecified interpolant. Injectivity on the quotient and the polynomial module action follow from the already proved kernel and $\widehat{Tf} = s\widehat{f}$. For a nonzero constant d, both sides are the zero module, with an empty direct sum on the right.

12.5 Arithmetic evaluation jets and the original spectral coordinate

The following map records every zeta-zero multiplicity without imposing RH:

$$J : \mathcal{E} \longrightarrow \prod_\rho \mathbb{C}^{m_\rho}, \qquad (Jf)_{\rho,j} = \frac{\widehat{f}^{(j)}(s_\rho)}{j!}, \quad 0 \le j < m_\rho. \tag{922}$$

Here one factor is used for each distinct nontrivial zero. It is an injective linear map. To prove this, suppose $Jf = 0$ and $\widehat{f} \not\equiv 0$. Choose s_0 where $\widehat{f}(s_0) \neq 0$. Compact support in $[-L, L]$ gives $|\widehat{f}(s)| \le \|f\|_1 e^{L|\Im s|}$. Jensen's formula on a disc of radius $2R$ about s_0 therefore bounds the number of zeros in the radius-R disc, counted with multiplicity, by $2LR/\log 2 + O(1)$: each such zero contributes at least $\log 2$ to the Jensen sum, and the boundary logarithm is at most $2LR + L|\Im s_0| + \log \|f\|_1$. Radii with boundary zeros follow by taking decreasing slightly larger admissible radii. But $Jf = 0$ requires zeros of order at least m_ρ at every s_ρ. Since $|\Im s_\rho| < 1/2$, the Riemann–von Mangoldt count $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$ places $\gg R\log R$ of these required zeros in that disc for large R, a contradiction. The cited source recalls this count in Section 1.1[10]. Fourier uniqueness now gives $f = 0$. This proof uses multiplicities, not an assumption that all zeros are simple, and claims no surjectivity onto arbitrary jet data. The finite interpolation construction proves that the image is dense for the product topology on the displayed product: every basic neighborhood specifies only finitely many jet coordinates, and those can be matched exactly. This topology statement does not control an infinite weighted Weil sum or identify the topology of a different arithmetic quotient.

In the factorial coordinates of (922), the exact intertwining formula is

$$(JTf)_{\rho,j} = s_\rho (Jf)_{\rho,j} + (Jf)_{\rho,j-1}, \qquad (Jf)_{\rho,-1} := 0. \tag{923}$$

It follows by applying the product rule to $s\widehat{f}(s)$ and dividing by $j!$. Thus the m_ρ-jet carries precisely $S_\rho = s_\rho I + N_\rho$, where N_ρ has ones immediately below the diagonal and zeros elsewhere. On two-component jets the branch operator is $\left(\begin{smallmatrix} 0 & -2S_\rho \\ I & 0 \end{smallmatrix}\right)$, whose square is $-2\,\mathrm{diag}(S_\rho, S_\rho)$. This retains both the arithmetic coordinate and the nilpotent jet information. It makes no unproved identification with the topology of the closed Connes quotient; that quotient has its own specified source construction in the preceding section.

The Weil form evaluates only the zeroth coordinates of these jets, with the original multiplicity weights and conjugate-zero pairing:

$$W(f,g)=\sum_{\rho} m_\rho \overline{\widehat{f}(\bar{s}_\rho)}\widehat{g}(s_\rho).$$

Higher jets and the branch trace are not silently substituted into this expression. For any finite family $U_j\in\mathcal{E}^2$ and specified polynomial channels P_j, put $h_j=L_{P_j}U_j$. Then, exactly,

$$c^*\big[W(h_i,h_j)\big]c=W\left(\sum_j c_jh_j,\sum_j c_jh_j\right). \tag{924}$$

Each entry can be evaluated by (912) and the complete explicit formula. Thus a certified negative value in this transported family would be an actual negative Weil value, while the fibre trace example above is not such a certificate. The present results supply the exact map, every finite-channel kernel and the inverse-domain conditions; they do not exhibit a negative arithmetic value.

13 The arithmetic trace quotient and exact heat transport

13.1 The primary arithmetic object and the topology actually specified

The starting point is Connes–Consani–Moscovici, *Zeta zeros and prolate wave operators: Semilocal adelic operators*, Section 3.6, proof-PDF pages 17–19, especially Lemma 3.3, equations (3.16)–(3.22), and Proposition 3.6 [103]. The preceding arithmetic map is

$$\mathcal{E}f(x)=x^{1/2}\sum_{n\geq 1}f(nx),\qquad \mathcal{F}_\mu g(s)=\int_0^\infty g(x)x^{-is}\,\frac{dx}{x}.$$

The input consists of even Schwartz functions with $f(0)=\widehat{f}(0)=0$. The paper's two Hermite families are

$$\psi_\ell^+=h_{4\ell}-\frac{h_{4\ell}(0)}{h_0(0)}h_0,\qquad \psi_\ell^-=-h_{4\ell+2}+\frac{h_{4\ell+2}(0)}{h_2(0)}h_2\quad(\ell\geq 1).$$

These signs and the minus sign in the Mellin exponent are retained. The completed function is

$$\Xi(s)=\xi(\tfrac12+is),\qquad \xi(z)=\tfrac12 z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z),\qquad \Xi(s)=8H_0(2s). \tag{925}$$

The paper proves

$$\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm=R_\ell^\pm\Xi,\qquad P_\ell^\pm=-\tfrac12(\tfrac14+s^2)R_\ell^\pm. \tag{926}$$

It also states immediately before Proposition 3.6 that these give all polynomial multiples of Ξ. Here is the precise algebra behind the last assertion. The top summands of (3.19) and (3.20) are nonzero; their degrees are 2ℓ and $2\ell+1$. Division by $-\frac12(\frac14+s^2)$ gives degrees $2\ell-2$ and $2\ell-1$, respectively. The even and odd parities remain. Thus the ordered family $R_1^+,R_1^-,R_2^+,R_2^-,\dots$ has one polynomial of each degree, with a nonzero leading coefficient. Subtracting the appropriate multiple of the polynomial of largest degree, and iterating down to degree zero, expresses every polynomial as a finite linear combination. Linear independence follows by the largest-degree coefficient of a finite relation. Consequently

$$N_{\mathrm{ar}} = \overline{\mathrm{span}\{\mathcal{F}_\mu \mathcal{E}\psi_\ell^\pm\}}^{\tau} = \overline{\mathbb{C}[s]\Xi}^{\tau}, \qquad Q_{\mathrm{ar}} = (\mathcal{H}_{\le 1}, \tau)/N_{\mathrm{ar}}. \tag{927}$$

Here τ denotes precisely the source's topology. No equality of this closed space with a principal algebraic ideal is being asserted. Multiplication $Mf(s) = sf(s)$ is continuous in the topological ring; it preserves the dense generating subspace, and hence N_{ar}. Its induced operator S is the operator of Proposition 3.6. We use the source's spectral assertion as an explicitly imported theorem:

$$\mathrm{Spec}(S) = \{\lambda \in \mathbb{C} : \zeta(\tfrac{1}{2} + i\lambda) = 0 \text{ at a nontrivial zero}\}. \tag{928}$$

Spectrum here means failure of a continuous two-sided inverse. No proof of the entire source spectral theorem is claimed as a new result of this manuscript.

There is a material source limitation. Page 19 calls $\mathcal{H}_{\le 1}$ the Hadamard topological ring of entire functions of order at most one; it does not state seminorms or a convergence criterion. The cited Connes–Marcolli Proposition 2.24, printed pages 379–381, proves the polynomial-image assertion and explicitly discusses different function-space realizations [104]. It does not supply that missing specification. We therefore retain τ in every source quotient. Below, all assertions depending on another topology name and define that topology. The comparison with the source is an exact quotient diagram, rather than an identification by terminology.

13.2 A globally defined heat map on the original underlying functions

For an entire function f, let $M_f(r) = \max_{|s|\le r} |f(s)|$ and define its order by

$$\limsup_{r\to\infty} \frac{\log\log \max(e, M_f(r))}{\log r}.$$

The zero function is included. For $1 < p < 2$, $a > 0$ set

$$\|f\|_{p,a} = \sup_{s\in\mathbb{C}} |f(s)| e^{-a|s|^p}.$$

The underlying set $\mathcal{H}_{\le 1}$ is exactly the set where all these seminorms are finite. Indeed order at most one gives, for $1 < q < p$, $M_f(r) \le \exp(r^q)$ for sufficiently large r; the expression $r^q - ar^p$ is bounded above. Conversely finiteness gives order at most p for every $p > 1$. Write γ for this explicitly defined growth topology. It is enough to take $p = 1 + 1/j$, $j \ge 2$, and $a = 1/k$, $k \ge 1$: given $p > 1, a > 0$, select one of these $p_j < p$ and use the boundedness of $|s|^{p_j}/k - a|s|^p$. This proves equivalence of the countable family with all displayed seminorms. A sequence Cauchy in these seminorms converges uniformly on compact sets to an entire function. Taking pointwise limits in each weighted Cauchy inequality gives convergence in that seminorm and finiteness of its norm. Thus $(\mathcal{H}_{\le 1}, \gamma)$ is a complete metrizable locally convex space.

Lemma 13.1 (Derivative bounds and the heat group)**.** *Put $D = \partial_s$. Differentiation and multiplication by s map the original underlying set $\mathcal{H}_{\le 1}$ into itself. For $m \ge 1$,*

$$\|D^m f\|_{p,a} \le m! \left(\frac{epa}{m}\right)^{m/p} \|f\|_{p,a/2^{p-1}}. \tag{929}$$

For every $t \in \mathbb{C}$, the series

$$U_t f = \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} D^{2k} f \tag{930}$$

converges in γ, locally uniformly in t; U_t is a continuous automorphism of $(\mathcal{H}_{\le 1},\gamma)$ and $U_tU_u = U_{t+u}$, $U_t^{-1} = U_{-t}$.

Proof. Cauchy's circle formula on a circle of radius R about s gives

$$|D^m f(s)| \le m! R^{-m} \max_{|z-s|=R} |f(z)|.$$

For $b = a/2^{p-1}$ the inequality $|s+z|^p \le 2^{p-1}(|s|^p + |z|^p)$ bounds the last expression, after multiplication by $e^{-a|s|^p}$, by $m!R^{-m}e^{aR^p}\|f\|_{p,b}$. Choosing $R = (m/(ap))^{1/p}$ proves (929). For multiplication use

$$\|Mf\|_{p,a} \le \sup_{r\ge 0} re^{-ar^p/2}\|f\|_{p,a/2}.$$

The kth heat summand is at most

$$\frac{(|t|/4)^k}{k!}(2k)!\left(\frac{epa}{2k}\right)^{2k/p}\|f\|_{p,b}.$$

Since $(2k)!/k! \le (2k)^k$, its scalar factor is at most $[C_{p,a}|t|k^{1-2/p}]^k$, whose kth root tends to zero. This proves summability, continuity, and locally uniform convergence of every time-differentiated series by the same estimate with the additional polynomial factors in k. Commutation with D follows from continuity of D. In the double composition series group terms with $n = k+\ell$. The sum of the absolute scalar coefficients at $D^{2n}f$ is $((|t|+|u|)/4)^n/n!$; the preceding estimate justifies regrouping. The binomial theorem gives U_{t+u}. The value at zero is the identity, proving the inverse. $\square$

Lemma 13.2 (Polynomial approximation)**.** *The Taylor polynomials of any $f \in \mathcal{H}_{\le 1}$ converge to f in γ. Multiplication of two elements is continuous in γ.*

Proof. If $f = \sum c_n s^n$, Cauchy's coefficient bound with parameter $b < a$ gives, for $n \ge 1$,

$$|c_n| \le \|f\|_{p,b}(epb/n)^{n/p}.$$

Also $\|s^n\|_{p,a} = (n/(epa))^{n/p}$. Therefore $\|c_ns^n\|_{p,a} \le \|f\|_{p,b}(b/a)^{n/p}$. The tail is bounded by a convergent geometric series. Finally $\|fg\|_{p,a} \le \|f\|_{p,a/2}\|g\|_{p,a/2}$. $\square$

Retain the actual real heat family, rather than substitute a polynomial:

$$\Phi(u) = \sum_{n\ge 1}(2\pi^2n^4e^{9u} - 3\pi n^2e^{5u})e^{-\pi n^2 e^{4u}},$$

$$Z_t(s) = 8H_t(2s) = 8\int_0^\infty e^{tu^2}\Phi(u)\cos(2su)\,du, \qquad \partial_t Z_t = -\tfrac{1}{4}D^2Z_t, Z_0 = \Xi. \tag{931}$$

These are the Rodgers–Tao conventions, Section 1 [17]. For clarity the analytic exchanges in this transport can be proved directly. Since $n^2e^{4u} \ge (n^2+e^{4u})/2$, the absolutely convergent sums $\sum n^j e^{-\pi n^2/2}$ give constants $C, c > 0$ such that $|\Phi(u)| \le Ce^{9u}e^{-ce^{4u}}$ for $u \ge 0$. For $|t| \le T$ and $|s| \le R$, every derivative of the integrand is bounded by a polynomial in u times $C\exp(Tu^2 + (2R+9)u - ce^{4u})$, which is integrable. The exponential series in t can therefore be integrated absolutely. Because $D^{2k}\cos(2su) = (-1)^k(2u)^{2k}\cos(2su)$, it gives exactly $Z_t = U_t\Xi$ with the coefficient $(-t/4)^k/k!$. In particular Z_t lies in the original order-at-most-one set for every complex t, by Lemma 13.1; it is never identically zero because U_t is invertible and $\Xi(0) \ne 0$.

13.3 Transport of the actual closed arithmetic image

Define τ_t on the same underlying vector space $\mathcal{H}_{\le 1}$ by declaring $U_t : (\mathcal{H}_{\le 1}, \tau) \longrightarrow (\mathcal{H}_{\le 1}, \tau_t)$ a homeomorphism. This definition specifies every open set: V is τ_t-open exactly when $U_{-t}V$ is τ-open. It uses the globally defined bijection just proved and does not require unproved continuity in τ. Set

$$N_t = U_t N_{\mathrm{ar}} = \overline{\operatorname{span}\{U_t(R_\ell^{\pm}\Xi)\}}^{\tau_t}, \qquad Q_t = (\mathcal{H}_{\le 1}, \tau_t)/N_t. \tag{932}$$

The closure equality follows because homeomorphisms carry closure to closure. In particular N_t is closed. The quotient map

$$\mathcal{U}_t : Q_{\mathrm{ar}} \longrightarrow Q_t, \qquad [f] \longmapsto [U_t f] \tag{933}$$

is well defined, bijective, and continuous, with continuous inverse $[g] \mapsto [U_{-t}g]$. Indeed changing f by $n \in N_{\mathrm{ar}}$ changes $U_t f$ by $U_t n \in N_t$, and the inverse gives necessity; the quotient universal property proves both continuity statements.

Theorem 13.3 (Exact arithmetic heat intertwiner)**.** *On the entire-function set,*

$$B_t = U_t M U_{-t} = M - \tfrac{t}{2} D. \tag{934}$$

This operator preserves N_t, is continuous in τ_t, and induces S_t on Q_t. It satisfies $S_t\mathcal{U}_t = \mathcal{U}_t S$ and

$$\operatorname{Spec}(S_t) = \operatorname{Spec}(S) = \{\lambda : \zeta(\tfrac{1}{2} + i\lambda) = 0 \text{ nontrivially}\}. \tag{935}$$

The time-dependent trace functions are exactly $U_t(R_\ell^{\pm}\Xi) = R_\ell^{\pm}(B_t)Z_t$.

Proof. The product rule gives $D^{2k}Mf = MD^{2k}f + 2kD^{2k-1}f$. Insert this identity in the convergent heat series to obtain $U_t M = (M - (t/2)D)U_t$, proving (934). If $n \in N_{\mathrm{ar}}$, then $B_t U_t n = U_t M n \in N_t$. Continuity is conjugation of the continuous M by the defining homeomorphism. The induced identity follows on representatives. If T is a continuous inverse of $S - \lambda$, then $\mathcal{U}_t T \mathcal{U}_t^{-1}$ is a continuous inverse of $S_t - \lambda$. Conjugating in the opposite direction proves the converse, and hence the spectrum equality using (928). Apply the polynomial identity $U_t p(M) = p(B_t)U_t$ to Ξ for the last assertion. $\square$

The heat map also transports the full multiplication, with every cross term retained. For $f, g \in \mathcal{H}_{\le 1}$ define

$$f \star_t g = U_t\big((U_{-t}f)(U_{-t}g)\big) = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!}(D^n f)(D^n g). \tag{936}$$

We prove the series identity, including convergence. For $F_t = U_t f$, $G_t = U_t g$, the series

$$J_t = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!}(D^n F_t)(D^n G_t)$$

converges in γ, uniformly on compact time sets: applying (929) with output norm $(p, a/2)$ to both factors bounds its nth term by

$$K\left[\frac{T}{2}\left(\frac{epa}{2}\right)^{2/p} n^{1-2/p}\right]^n \qquad (|t| \le T),$$

where K bounds the product of the two input norms $(p, a/2^p)$, and is finite by the heat estimate. The root test applies. Cauchy's formula in time permits termwise time derivatives; continuity of D permits spatial derivatives. Differentiating the coefficient contributes

$$-\tfrac{1}{2}\sum_{n\ge 0}(-t/2)^n(D^{n+1}F_t)(D^{n+1}G_t)/n!.$$

Differentiating the factors contributes the two pure second-derivative terms with coefficient $-1/4$. The complete product rule therefore gives $\partial_t J_t = -D^2 J_t/4$ and $J_0 = fg$. Its entire time Taylor coefficients are uniquely $(-1/4)^k D^{2k}(fg)/k!$, so $J_t = U_t(fg)$. Replacing the inputs by $U_{-t}f, U_{-t}g$ proves (936). Conjugation of ordinary multiplication proves associativity, commutativity, unit 1, and continuity for τ_t, even when continuity of the series in the source topology is unknown. The identity $s\star_t f = sf - tf'/2$ recovers exactly B_t. No ideal property of N_{ar} beyond its stated polynomial invariance is required for this identity or for the quotient in Theorem 13.3.

The transported actual trace image has no common point-evaluation zero at any nonzero time. Suppose λ were such a common zero. It would annihilate each $U_t(s^n\Xi)$. For every $a \in \mathbb{C}$, the Taylor series of e^{as} converges in γ by Lemma 13.2. Multiplication, U_t, and evaluation are continuous in γ, so

$$0 = \sum_{n=0}^{\infty} \frac{a^n}{n!} U_t(s^n\Xi)(\lambda) = e^{a\lambda - ta^2/4} Z_t(\lambda - ta/2).$$

The last equality follows from the just-proved product formula and the entire Taylor formula for Z_t. At $t \neq 0$ varying a forces Z_t to vanish everywhere, a contradiction. This argument uses only the polynomial generators contained in N_t; it assumes no source-topology convergence of the exponential series. Thus the source arithmetic spectrum in (935) is carried by the explicitly transported operator and relation, even though the transported relation has no common ordinary evaluation zero.

The first two nonconstant polynomial generators make the extra terms visible without changing any polynomial coefficient:

$$\begin{aligned} U_t(s\Xi) &= sZ_t - \tfrac{t}{2}Z_t', \\ U_t(s^2\Xi) &= s^2 Z_t - tsZ_t' - \tfrac{t}{2}Z_t + \tfrac{t^2}{4}Z_t''. \end{aligned} \tag{937}$$

For $L = \partial_t + \frac{1}{4}D^2$, the complete product rule is

$$L(AP) = A\big(P_t + \tfrac{1}{4}P_{ss}\big) + \big(A_t + \tfrac{1}{4}A_{ss}\big)P + \tfrac{1}{2}A_s P_s. \tag{938}$$

Thus $L(R(s)Z_t) = \frac{1}{4}R''Z_t + \frac{1}{2}R'Z_t'$. The family $R(B_t)Z_t$ solves the homogeneous heat equation, whereas the displayed expression is the full defect of replacing it by $R(s)Z_t$. For a local factorization $Z_t = AP$ with $A \neq 0$, the same formula retains the analytic unit A and its derivatives. At a simple zero $\lambda(t)$, differentiating $Z_t(\lambda(t)) = 0$ gives

$$\lambda'(t) = \frac{1}{4}\frac{Z_{ss}}{Z_s}(t, \lambda(t)) = \frac{1}{4}\frac{P_{ss}}{P_s}(t, \lambda(t)) + \frac{1}{2}\frac{A_s}{A}(t, \lambda(t)). \tag{939}$$

This statement is on the open locus $Z_s \neq 0$, on which the analytic implicit-function theorem supplies the local curve; it claims no chosen zero is simple. At a multiple zero the division is undefined, and the full equation (938) remains the equation to use.

13.4 Derivative domains, multivalued operators, and jet extensions

Every derivative has the full underlying domain $\mathcal{H}_{\leq 1}$ by Lemma 13.1. Continuity in the unnamed source topology is not assumed. The following gives exact source-quotient operators and their domains without imposing it. For $k \geq 1$ put

$$K_k = \{n \in N_{\mathrm{ar}} : D^k n \in N_{\mathrm{ar}}\}, \qquad W_k = (D^k N_{\mathrm{ar}} + N_{\mathrm{ar}})/N_{\mathrm{ar}} \subseteq Q_{\mathrm{ar}}.$$

The derivative correspondence is the actual linear relation

$$\mathfrak{D}_k = \{([f], [D^k f]) : f \in \mathcal{H}_{\leq 1}\} \subseteq Q_{\mathrm{ar}} \times Q_{\mathrm{ar}}. \tag{940}$$

For fixed $[f]$, its output fibre is precisely $[D^k f]+W_k$: all representatives are $f+n$ with $n\in N_{\rm ar}$, and linearity gives all and only the stated outputs. The kernel of its presenting map $f\mapsto([f],[D^k f])$ is exactly K_k. The relation has domain all of $Q_{\rm ar}$; its multivalued part at zero is exactly W_k. Thus it is the graph of an operator exactly when $D^k N_{\rm ar}\subseteq N_{\rm ar}$. This is a calculated equivalence, with an explicit obstruction space, rather than an assumption of invariance.

There is no nonempty domain of quotient classes on which the same representative formula is unambiguous if $W_k\neq 0$ and every representative of a class is admitted: addition of the same offending n gives two different outputs for every class. A chosen linear lift is a different map. More precisely any vector subspace $A\subseteq\mathcal{H}_{\le 1}$ supplies a single-valued map on its quotient image $q(A)$ exactly when $D^k(A\cap N_{\rm ar})\subseteq N_{\rm ar}$; the identical difference-of-representatives argument proves both directions. The first heat generator is $-D^2/4$, so its ambiguity space is $-W_2/4$, with the coefficient and sign unchanged.

Even when W_k is not closed, the smallest source-closed enlargement of the target relation subspace is explicit:

$$[f]\longmapsto[D^k f]:Q_{\rm ar}\longrightarrow\mathcal{H}_{\le 1}/\overline{N_{\rm ar}+D^k N_{\rm ar}}^{\tau}. \tag{941}$$

It is well defined algebraically. Equip its source with the quotient of the graph topology $f\mapsto(f,D^k f)$ in $(\mathcal{H}_{\le 1},\tau)^2$ to obtain a continuous map. This specifies a topology change explicitly, without claiming it is the original one. The lost part of the original target $Q_{\rm ar}$ is exactly $\overline{N_{\rm ar}+D^k N_{\rm ar}}^{\tau}/N_{\rm ar}$.

Here is an extension which retains correlated derivatives rather than quotienting them away. For $j\ge 1$ define the bijection

$$T_j(f_0,\dots,f_j)=(f_0,f_1-Df_0,\dots,f_j-D^j f_0).$$

Its inverse is $(f_0,g_1,\dots,g_j)\mapsto(f_0,g_1+Df_0,\dots,g_j+D^j f_0)$. Give $V_j=\mathcal{H}_{\le 1}^{j+1}$ the topology transported by T_j from $(\mathcal{H}_{\le 1},\tau)\times(\mathcal{H}_{\le 1},\gamma)^j$. The residual factors are explicitly given the proved Hausdorff growth topology; the source factor keeps its original topology. For $J_j f=(f,Df,\dots,D^j f)$, $T_j(J_j N_{\rm ar})=N_{\rm ar}\oplus 0^j$, so $J_j N_{\rm ar}$ is closed. Define

$$X_j=V_j/J_j N_{\rm ar}. \tag{942}$$

Then the map

$$[f_0,\dots,f_j]\longmapsto([f_0],f_1-Df_0,\dots,f_j-D^j f_0)$$

is a homeomorphism $X_j\simeq Q_{\rm ar}\oplus(\mathcal{H}_{\le 1},\gamma)^j$; independence, the inverse, and continuity follow directly from T_j and the quotient definition. In particular there is the split exact sequence

$$0\longrightarrow\mathcal{H}_{\le 1}^j\longrightarrow X_j\xrightarrow{[f_0,\dots,f_j]\mapsto[f_0]}Q_{\rm ar}\longrightarrow 0,\qquad [f]\longmapsto[J_j f]. \tag{943}$$

Its relation on a source generator retains every product term:

$$D^k(R\Xi)=\sum_{h=0}^{k}\binom{k}{h}R^{(h)}\Xi^{(k-h)}\quad(0\le k\le j). \tag{944}$$

The binomial formula follows by induction from the ordinary product rule, including the two edge terms at each step.

The original spectral coordinate acts on V_j by

$$(\mathcal{M}_j f)_k=sf_k+kf_{k-1}\quad(0\le k\le j),\qquad f_{-1}=0.$$

The identity $D^k(sf) = sD^kf + kD^{k-1}f$ proves $\mathcal{M}_j J_j f = J_j(sf)$; it therefore preserves the exact relation subspace. In the T_j coordinates it acts by S on Q_{ar} and by $g_k \mapsto sg_k + kg_{k-1}$ on the residual terms ($g_0 = 0$). This also proves continuity in the stated topology. Notice that (943) adds a free residual module: the spectrum of the entire extension is not claimed to equal the source spectrum. The arithmetic quotient and its split inclusion are explicitly preserved.

13.5 A fully computed analytic realization and its exact comparison

We now compute the closures and the failure of descent in specified topologies. This does not identify them with τ. Let κ be the topology of uniform convergence on compact sets on the same set $\mathcal{H}_{\le 1}$, and define

$$N_\gamma = \overline{\mathbb{C}[s]\Xi}^{\gamma}, \qquad N_\kappa = \overline{\mathbb{C}[s]\Xi}^{\kappa}.$$

Evaluations of all derivatives are continuous in both topologies by Cauchy's formula. The growth topology is finer than κ, so $N_\gamma \subseteq N_\kappa$. If λ is a zero of Ξ with its actual multiplicity m_λ, every element of either closed subspace has vanishing derivatives of orders $0, \dots, m_\lambda - 1$ at λ.

Theorem 13.4 (Compact closure, multiplicities, and coordinate spectrum)**.** *The compact closure is exactly*

$$\begin{aligned} N_\kappa &= \{f \in \mathcal{H}_{\le 1} : f/\Xi \text{ extends to an entire function}\} \\ &= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0 \ (h < m_\lambda),\ \Xi(\lambda) = 0\}. \end{aligned} \tag{945}$$

On $Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/N_\kappa$, multiplication by s has spectrum exactly the zeros of Ξ in the original s coordinate. For a zero λ of multiplicity m, there is a continuous surjection

$$Q_\kappa \longrightarrow \mathbb{C}[\epsilon]/(\epsilon^m), \qquad [f] \longmapsto \sum_{h=0}^{m-1} \frac{D^h f(\lambda)}{h!} \epsilon^h,$$

intertwining multiplication by s with $\lambda + \epsilon$.

Proof. The vanishing conditions are closed and hold on every $p\Xi$, so the closure is contained in the last displayed set. Local Taylor division at each isolated zero identifies that set with entire divisibility by Ξ. For such an f, put $g = f/\Xi \in \mathcal{O}(\mathbb{C})$. Its Taylor polynomials p_n converge uniformly on every compact; $p_n\Xi \to f$ in κ. Every $p_n\Xi$ belongs to the original order class even though no growth assertion about g is needed. This proves (945) with the closure retained. The local-jet map is independent of representatives; the polynomials $1, (s-\lambda), \dots, (s-\lambda)^{m-1}$ prove surjectivity. Leibniz's rule gives the stated action, and the finite jet evaluations are continuous.

If $\Xi(\lambda) \ne 0$, define

$$T_\lambda f(s) = \frac{f(s) - f(\lambda)\Xi(s)/\Xi(\lambda)}{s - \lambda},$$

filling in the removable value. Division by a linear factor preserves the order bound: outside a disk about λ the denominator is bounded below, and inside a disk the filled entire function is bounded. It is continuous in κ by Cauchy's formula on a larger disk applied to the numerator. For $n \in N_\kappa$ the quotient $T_\lambda n/\Xi = (n/\Xi - (n/\Xi)(\lambda))/(s-\lambda)$ is entire; thus it descends. Direct multiplication proves $(S_\kappa - \lambda)[T_\lambda f] = [f]$. Applying T_λ to $(s-\lambda)f$ gives f exactly, proving the other inverse identity. If $\Xi(\lambda) = 0$, evaluation at λ descends to a nonzero continuous functional which annihilates the image of $S_\kappa - \lambda$. Hence that operator is not surjective. This proves the spectrum claim. $\square$

Theorem 13.5 (Direct differential and heat obstructions)**.** *Neither N_γ nor N_κ is invariant under D or D^2. For either frozen quotient, neither representative formula $[f']$ nor $[-f''/4]$ defines an* *operator on any nonempty class domain admitting all representatives. For every $t \neq 0$, neither $U_tN_\gamma \subseteq N_\gamma$ nor $U_tN_\kappa \subseteq N_\kappa$ holds. Ordinary multiplication by s preserves neither U_tN_γ nor U_tN_κ.*

Proof. Choose any zero λ of the actual Ξ and let $m \geq 1$ be its multiplicity. The existence of its zeros is part of the classical arithmetic datum in the primary source; no realness or simplicity is used. Write $\Xi = (s-\lambda)^m A$ with $A(\lambda) \neq 0$. Then $\Xi' = (s-\lambda)^{m-1}(mA + (s-\lambda)A')$ does not vanish to order m. Since $\Xi \in N_\gamma \subseteq N_\kappa$, both derivative invariance assertions fail. If $m \geq 2$, Ξ'' has leading coefficient $m(m-1)A(\lambda)$ at order $m-2$, proving the second-derivative failure. If $m = 1$, use the actual polynomial multiple $n = (s-\lambda)\Xi$ instead: $n''(\lambda) = 2\Xi'(\lambda) \neq 0$. This retains the product-rule cross term and covers all multiplicities. The nonempty-domain assertion follows from the explicit fibre computation in (940).

For $a \in \mathbb{C}$, $e^{as}\Xi \in N_\gamma$ by polynomial approximation of e^{as} and continuous multiplication, and hence belongs to N_κ. The conjugation $D(e^{as}f) = e^{as}(D+a)f$ and the absolutely convergent heat and translation series give

$$U_t(e^{as}\Xi)(s) = e^{as-ta^2/4}Z_t(s-ta/2). \tag{946}$$

For completeness, the translation series for f is its Taylor series about s and sums to $f(s-ta/2)$; its local uniform convergence and the derivative estimates justify commutation of $e^{-(ta/2)D}$ with U_t. If $U_tN_\bullet \subseteq N_\bullet$ for either choice, evaluate (946) at λ. The exponential never vanishes, so $Z_t(\lambda - ta/2) = 0$ for every a. When $t \neq 0$ these arguments fill $\mathbb{C}$; this contradicts $Z_t \neq 0$. Finally if M preserved $U_tN_\bullet$, conjugation would imply $(M + (t/2)D)N_\bullet \subseteq N_\bullet$. Since M already preserves $N_\bullet$, this forces $DN_\bullet \subseteq N_\bullet$, contrary to the first part. $\square$

The analytic moving-divisor quotient, which does carry the zeros of the actual Z_t under ordinary s, is

$$I_t = \overline{\mathbb{C}[s]Z_t}^{\kappa} = \{f \in \mathcal{H}_{\leq 1} : f/Z_t \text{ is entire}\}, \qquad A_t = (\mathcal{H}_{\leq 1}, \kappa)/I_t. \tag{947}$$

The proof of Theorem 13.4 applies verbatim with the nonzero entire order-at-most-one function Z_t: it uses only local Taylor division and compact polynomial approximation. Consequently $\mathrm{Spec}(M \text{ on } A_t) = \{\lambda : Z_t(\lambda) = 0\}$, with the full local multiplicity jets, and with the explicit inverse $[f] \mapsto [(f - f(\lambda)Z_t/Z_t(\lambda))/(s-\lambda)]$ off that set. Equation (939) is the evolution of this original-coordinate spectrum on its simple-zero locus. This spectrum can move; (935) is the spectrum of the different, exactly specified transported operator B_t.

We now give the promised comparison morphisms with every kernel retained. Put $J = N_{\mathrm{ar}} \cap N_\kappa$ and $C = (\mathcal{H}_{\leq 1}, \tau \vee \kappa)/J$, where $\tau \vee \kappa$ is the topology generated by both topologies. Since each summand is closed in its own topology, J is closed in their join. The identity maps give continuous surjections

$$Q_{\mathrm{ar}} \longleftarrow C \longrightarrow Q_\kappa, \qquad \ker(C \to Q_{\mathrm{ar}}) = N_{\mathrm{ar}}/J, \quad \ker(C \to Q_\kappa) = N_\kappa/J. \tag{948}$$

Each assertion follows by taking the same representative f and testing membership in the respective relation subspace; the quotient universal property proves continuity. An algebraic common quotient is $\mathcal{H}_{\leq 1}/(N_{\mathrm{ar}} + N_\kappa)$ with the quotient topology from the join; its Hausdorff version uses $\overline{N_{\mathrm{ar}} + N_\kappa}^{\tau\vee\kappa}$. The map from C to this further quotient is continuous. Continuity of maps to it from each of the two separately topologized quotients is not asserted. No closedness of the sum is presumed. This describes the precise information lost in that further quotient.

At time t replace the two subspaces by N_t and I_t and the join by $\tau_t \vee \kappa$. The same construction gives continuous surjections from their intersection quotient to the genuine transported arithmetic quotient Q_t and to A_t, with kernels $N_t/(N_t \cap I_t)$ and $I_t/(N_t \cap I_t)$, respectively. The two generator systems are linked by the explicit formula $R(B_t)Z_t$ versus $R(s)Z_t$ and the defects (937)–(938).

These are exact morphisms and closure formulas for the actual source space. Determining that either kernel vanishes for τ is not established by the two inspected primary texts. Accordingly the source-frozen statements $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ and $D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ have not been decided solely from that unnamed topology. The unqualified noninvariance results above have been proved in the two explicitly specified analytic realizations.

13.6 The inverse-root cover, the branch locus, and arithmetic fibres

The retained polynomial inverse-fibre slice is $P_{a,0,0}(r) = -2r^2 - 2a$. Its root equation is $a = -r^2$. The arithmetic coordinate is $a = 2s$, so

$$s = -r^2/2. \tag{949}$$

The finite free extension of the actual quotient is

$$\widehat{Q} = \mathbb{C}[r] \otimes_{\mathbb{C}[s]} Q_{\mathrm{ar}} = Q_{\mathrm{ar}} \oplus rQ_{\mathrm{ar}}, \qquad \mathcal{R} = \begin{pmatrix} 0 & -2S \\ 1 & 0 \end{pmatrix}, \qquad -\tfrac{1}{2}\mathcal{R}^2 = \begin{pmatrix} S & 0 \\ 0 & S \end{pmatrix}.$$

The even–odd decomposition of a polynomial in r proves freeness with basis $(1, r)$ and uniqueness coefficient by coefficient. Multiplying $q_0 + rq_1$ by r proves the displayed matrices; the finite direct-sum topology makes every induced map continuous. The involution is $(q_0, q_1) \mapsto (q_0, -q_1)$; the trace is $\mathrm{Tr}(q_0 + rq_1) = 2q_0$, with kernel rQ_{ar}. It is $\mathbb{C}[s]$-linear. It does not provide a $\mathbb{C}[r]$-module quotient with this kernel unless $S = 0$: multiplying rq by r gives $-2Sq$ in the even summand. This proves the exact limitation, rather than suppressing the odd component. The same construction at time t uses S_t and is conjugated by $\mathrm{diag}(\mathcal{U}_t, \mathcal{U}_t)$.

The original s-spectrum is unchanged by the extension: an inverse of $S - \lambda$ duplicates to a diagonal inverse, and conversely projection of an inverse of the doubled matrix to its first component, following inclusion of that component, gives an inverse of $S - \lambda$. Both directions are continuous. Moreover

$$\mathrm{Spec}(\mathcal{R}) = \{\rho \in \mathbb{C} : -\rho^2/2 \in \mathrm{Spec}(S)\}. \tag{950}$$

To prove this without an unsupported spectral-mapping citation, solve $(\mathcal{R} - \rho)(u, v) = (a, b)$. The second row gives $u = b + \rho v$ and the first becomes $-2(S + \rho^2/2)v = a + \rho b$. A continuous inverse of $S + \rho^2/2$ therefore gives a continuous inverse matrix. Conversely the right-hand side $(a, 0)$ and the second projection of an inverse matrix supply an inverse of $-2(S + \rho^2/2)$, with both identities checked by the same two rows.

Before any quotient by Ξ, let $\mathcal{O}_s$ be the sheaf of holomorphic functions in s and put $B = \mathcal{O}_s[r]/(r^2 + 2s)$, identified with $\mathcal{O}_s \oplus r\mathcal{O}_s$. On the punctured base $s \neq 0$, implicit differentiation gives $dr/ds = -1/r = r/(2s)$. Thus the actual lifted derivative is

$$\nabla_s \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} = \begin{pmatrix} f_0' \\ f_1' + f_1/(2s) \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \quad [\nabla_s, R] = -R^{-1}. \tag{951}$$

Indeed with $A = \mathrm{diag}(0, 1/(2s))$, $R' + AR - RA = \left(\begin{smallmatrix} 0 & -1 \\ 1/(2s) & 0 \end{smallmatrix}\right) = -R^{-1}$. A second application gives

$$\nabla_s^2 = \mathrm{diag}\left(D^2, D^2 + s^{-1}D - \frac{1}{4s^2}\right). \tag{952}$$

Equivalently, on a function of r the derivative is $-r^{-1}\partial_r$ and the heat generator is $-\frac{1}{4}r^{-2}\partial_r^2 + \frac{1}{4}r^{-3}\partial_r$. Squaring $-r^{-1}\partial_r$ by the product rule proves this with both signs preserved.

The branch extension is exact. The meromorphic connection sends B into $s^{-1}B$ and the connection one-form has a simple pole with residue $\mathrm{diag}(0, 1/2)$. Its maximal domain for holomorphic output at $s = 0$ is

$$\mathrm{Dom}_{\mathrm{hol}}\, \nabla_s = \mathcal{O}_s \oplus rs\mathcal{O}_s. \tag{953}$$

In fact the odd coefficient is $f_1' + f_1/(2s)$; its only possible pole has coefficient $f_1(0)/2$, proving necessity and sufficiency. For the square, write $f_1 = \sum_{n\geq 0} a_n s^n$. Its image has coefficient $(n^2 - 1/4)a_n s^{n-2}$. The negative powers vanish exactly when $a_0 = a_1 = 0$, so

$$\mathrm{Dom}_{\mathrm{hol}}\, \nabla_s^2 = \mathcal{O}_s \oplus rs^2\mathcal{O}_s.$$

For k iterates the odd coefficient is $\prod_{h=0}^{k-1}(n + 1/2 - h)a_n s^{n-k}$, whose prefactor is never zero for integral n. Thus its domain is $\mathcal{O}_s \oplus rs^k\mathcal{O}_s$. No derivation $B \to B$ extending ∂_s exists at the branch: derivation of $r^2 + 2s = 0$ would give $2r\delta(r) + 2 = 0$, forcing r to be a unit although it vanishes at the branch point. The logarithmic derivative $s\nabla_s$ does extend to all of B, with matrix $sD + \mathrm{diag}(0, 1/2)$.

These are sheaf and representative calculations; they do not assert that ∇_s descends through the frozen arithmetic relation. For a holomorphic local multiplier $R_0(s)$ the relation vector $(R_0\Xi, 0)$ acquires $(R_0'\Xi + R_0\Xi', 0)$, and the odd relation $(0, R_0\Xi)$ acquires $(0, R_0'\Xi + R_0\Xi' + R_0\Xi/(2s))$. The second iterate includes the full $2R_0'\Xi'$ term, as well as the odd $s^{-1}D - 1/(4s^2)$ terms of (952). Jet extension (942), or the moving closed quotient (932), retains these relations exactly.

Finally, the branch point is not an arithmetic spectral point at $t = 0$. The source gives $\Xi(0) \neq 0$, so its Proposition 3.6 makes S continuously invertible; then $\mathcal{R}$ is invertible by (950). In the analytic divisor realization the stronger local statement follows from Ξ being a holomorphic unit on a disk about zero: the quotient stalk is zero there. For real t, $Z_t(0) > 0$ directly from the retained kernel:

$$2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u} = \pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0 \quad (u \geq 0),$$

and the positive integral converges by the bound already proved. Thus the actual real heat divisor never meets the cover branch point $s = 0$. On a simple zero curve its retained lift satisfies $r^2 = -2\lambda$, $r' = -\lambda'/r$; together with (939) these are the original-coordinate and cover equations, including the unit contribution. A nonreal r alone does not specify whether $\lambda = -r^2/2$ is nonreal.

13.7 Exact ingress from compact Weil tests

The accompanying retained Weil-test calculation supplies the concrete space $\mathcal{E} = C_c^\infty(\mathbb{R}; \mathbb{C})$ with coordinate operator $T = -i\partial_u$ and Fourier transform

$$(\mathscr{F}f)(s) = \int_{\mathbb{R}} f(u)e^{-ius}\, du.$$

Here we give the complete heat ingress and its inverse domains; none of the following claims requires identifying the source quotient topology with a test-space topology. Integration by parts gives $\mathscr{F}(Tf) = s\mathscr{F}f$ with no endpoint term. If $\mathrm{supp}\, f \subset [-L, L]$, then

$$|\mathscr{F}f(s)| \leq \|f\|_1 e^{L|\operatorname{Im} s|}, \qquad \|\mathscr{F}f\|_{p,a} \leq \|f\|_1 \sup_{r\geq 0} e^{Lr - ar^p} < \infty.$$

Thus the Fourier transform maps the actual test space into the original entire-function set, continuously in γ on every fixed-support smooth-function space and hence in its usual inductive-limit topology.

Define, for all complex t,

$$(G_t f)(u) = e^{tu^2/4} f(u), \qquad U_t \mathscr{F} f = \mathscr{F} G_t f, \qquad T_t = G_t T G_{-t} = T + \frac{it}{2} u. \tag{954}$$

The multiplication G_t is a continuous automorphism on each fixed-support smooth-function space, with inverse G_{-t}. Indeed each derivative of the smooth multiplier is bounded on the fixed compact support, and Leibniz's finite sum bounds every output derivative seminorm by finitely many input ones. The same argument for the inverse proves the assertion. The exponential series and every u-derivative converge uniformly on a fixed compact set, locally uniformly in t; consequently the family is entire in time in that test topology. In the Fourier integral, $D^{2k}\mathscr{F} f = \mathscr{F}((-1)^k u^{2k} f)$. The compact support permits absolute interchange with the exponential series, proving the middle identity with the exact factor $1/4$. Finally differentiation of $e^{-tu^2/4} f$ gives the displayed T_t, and

$$\mathscr{F}(T_t f) = B_t \mathscr{F} f, \qquad D\mathscr{F} f = \mathscr{F}(-iuf).$$

Both equations follow directly from the integral, including their signs. The test-space version of the retained cover is therefore

$$\mathcal{R}_t^{\text{test}} = \begin{pmatrix} 0 & -2T_t \\ 1 & 0 \end{pmatrix}, \qquad (\mathcal{R}_t^{\text{test}})^2 = -2\operatorname{diag}(T_t, T_t).$$

Componentwise Fourier transformation intertwines this matrix with the representative matrix using B_t, and then with the actual quotient matrix using S_t.

To retain every possible arithmetic relation in this ingress, put

$$K = \{f \in \mathcal{E} : \mathscr{F} f \in N_{\text{ar}}\}, \qquad K_t = \{f \in \mathcal{E} : \mathscr{F} f \in N_t\}. \tag{955}$$

Then $K_t = G_t K$: by (954), $\mathscr{F} f \in U_t N_{\text{ar}}$ exactly when $\mathscr{F}(G_{-t} f) \in N_{\text{ar}}$. Thus there are injective linear maps onto their explicitly specified images

$$\mathcal{E}/K \longrightarrow Q_{\text{ar}}, \quad [f] \longmapsto [\mathscr{F} f], \qquad \mathcal{E}/K_t \longrightarrow Q_t, \quad [f] \longmapsto [\mathscr{F} f],$$

and the square formed by G_t and $\mathcal{U}_t$ commutes. The kernel statements follow exactly from (955); no claim that $K = 0$ in the unnamed source topology is inserted. To make the ingress continuous without that identification, give $\mathcal{E}$ the join η of its usual test topology with the inverse-image topology of τ under $\mathscr{F}$. Then K is closed and the first quotient map is continuous. Give the time-t test space the transported topology $(G_t)_*\eta$; the second map and the commutative square are continuous by (954). This specifies the exact topologies rather than claiming an unproved topological embedding of a quotient image equipped with its subspace topology.

There is a fully explicit inverse-domain calculation in the same test coordinates. For $\lambda \in \mathbb{C}$, $k \geq 1$, the image is

$$(T_t - \lambda)^k \mathcal{E} = \left\{ g \in \mathcal{E} : \int_{\mathbb{R}} v^j e^{-tv^2/4 - i\lambda v} g(v)\, dv = 0 \quad (0 \leq j < k) \right\}. \tag{956}$$

On that precise image the inverse is

$$[(T_t - \lambda)^{-k} g](u) = \frac{i^k e^{tu^2/4 + i\lambda u}}{(k-1)!} \int_{-\infty}^{u} (u - v)^{k-1} e^{-tv^2/4 - i\lambda v} g(v)\, dv. \tag{957}$$

To prove both directions, write $f = e^{tu^2/4 + i\lambda u} h$. Direct differentiation gives

$$(T_t - \lambda) f = e^{tu^2/4 + i\lambda u}(-ih').$$

Iteration reduces the inverse equation to a kth primitive, giving (957) because $(-i)^k i^k = 1$. For g supported in $[a, b]$, that formula vanishes below a. Above b, expansion of $(u-v)^{k-1}$ shows it vanishes identically exactly when all k moments in (956) vanish; the binomial coefficients and the nonzero exponential do not change that equivalence. Its smoothness follows by repeated differentiation of the integral; smoothness of g with compact support gives matching derivatives at both endpoints. Necessity for any compact solution also follows by integrating $h^{(k)}$ against $1, v, \ldots, v^{k-1}$ and integrating by parts k times. Uniqueness follows since a compactly supported solution of $h^{(k)} = 0$ is a polynomial of degree at most $k - 1$ which vanishes on a ray, and hence is zero. This proves the exact image, inverse, kernel zero, and preservation of the original support interval.

In particular the inverse of $\mathcal{R}_t^{\text{test}}$ is

$$(g_0, g_1) \longmapsto (g_1, -\tfrac{1}{2}T_t^{-1}g_0), \qquad \operatorname{Dom}(\mathcal{R}_t^{\text{test}})^{-1} = \left\{(g_0, g_1) : \int e^{-tv^2/4} g_0(v)\, dv = 0\right\}.$$

Substitution into its two rows proves both inverse identities. Thus the branch-locus obstruction in the concrete test realization is a retained Gaussian-weighted zero-moment condition. The exact arithmetic quotient instead has invertible S_t and root operator, as proved above. Their ingress map and its kernels explain precisely how these statements coexist; one operator's inverse domain is not silently assigned to the other realization.

13.8 Proof scope and reproducibility

The new source-topology assertions proved here are the complete transported closure, homeomorphic quotient, conjugated spectral operator, source-domain differential correspondence, correlated jet extension, finite free cover, compact-test heat ingress with its weighted-moment inverse domains, and exact comparison diagrams. The explicit compact and growth realizations have proved frozen differential and heat noninvariance; the compact realization also has a complete divisor and spectrum calculation with multiplicities. The missing identification of the paper's unnamed topology is recorded explicitly and is not replaced by a conditional arithmetic theorem. Nothing in these results establishes an off-critical zero, a negative Weil value, or an arithmetic endpoint by analogy with the inverse-fibre collision. The exact transport through that collision cover and the derivative information it requires have been calculated above.

The companion checker uses rational polynomial arithmetic to verify the heat conjugation on finite polynomial inputs, the retained product terms, the matrix commutator, the iterated branch-domain coefficients, the original-coordinate relation, and sample finite jet multiplication. These are reproducible checks of the displayed identities, not a substitute for their written proofs. Source hashes and exact reading locators are in the companion provenance receipt; no numerical zero search or Lean run is part of this work.

14 The entire inverse fibre and its incompressible coefficient heat flow

We continue the original polynomial construction of Section 11, with the same coordinates and human-source provenance[97, 98, 99]. The complete inverse map is retained below to verify its dynamical lift; no finite inverse point is discarded.

14.1 Every finite inverse point, including the excluded roots

Retain the original map, with the identity renaming of its third spatial coordinate from t to w:

$$\begin{aligned} F_1 &= (1+xy)^3 w + y^2(1+xy)(4+3xy), \\ F_2 &= y + 3x(1+xy)^2 w + 3xy^2(4+3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned} \tag{958}$$

The target coordinates are (a, b, c). All algebraic statements first hold over $\mathbb{C}$. Real parameters and real roots give precisely the real points. For the determinant, on $x \neq 0$ set

$$v = 1 + xy, \qquad h = 2 - 3xy - x^2 w, \qquad A = v^2 + v - v^3 h, \qquad B = 4v + 2 - 3v^2 h.$$

Direct substitution gives $F = (A/x^2, B/x, xh)$ and

$$\det \frac{\partial(x, v, h)}{\partial(x, y, w)} = -x^3.$$

The other determinant is x^{-3} times

$$\det \begin{pmatrix} -2A & 2v + 1 - 3v^2 h & -v^3 \\ -B & 4 - 6vh & -3v^2 \\ h & 0 & 1 \end{pmatrix} = 2.$$

Indeed expansion gives

$$(4 - 6vh)(-2v^2 - 2v + 3v^3 h) + (2v + 1 - 3v^2 h)(4v + 2 - 6v^2 h) = 2.$$

Their product is -2. The original Jacobian determinant is a polynomial; agreement on $x \neq 0$ proves

$$\det DF = -2 \quad \text{everywhere.} \tag{959}$$

For fixed (a, b, c) define

$$P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \qquad \alpha = \tfrac{1}{2} P_r(r).$$

For each simple root, the entire inverse point is

$$H(r, \alpha, c) = \left(\alpha^{-1},\ r - \alpha,\ 5\alpha^2 - 3r\alpha - c\alpha^3\right). \tag{960}$$

Here and below $\alpha \neq 0$ is part of the domain. To prove both directions, start with $x \neq 0$, set $\alpha = 1/x$, $r = y + 1/x$, and solve $F_3 = c$. This gives exactly (960). Before either other target equation is imposed, direct substitution gives

$$F \circ H = \left(r^2 + r\alpha - cr^3,\ 4r + 2\alpha - 3cr^2,\ c\right). \tag{961}$$

The second equation equals b precisely when $2\alpha = 3cr^2 - 4r + b = P_r(r)$. After imposing that identity, $2(F_1 - a) = P(r)$. Thus the formulas give every point with $x \neq 0$, and recovering $r = y + 1/x$ proves injectivity of the parametrization. A multiple root has $\alpha = 0$ and gives no finite point in this chart.

The excluded hyperplane is handled without division:

$$F(0, y, w) = (w + 4y^2, y, 0).$$

It contributes exactly $(0, b, a - 4b^2)$ when $c = 0$, and contributes no point when $c \neq 0$. These statements exhaust the whole fibre.

For completeness the multiplicities are explicit. If $c \neq 0$, put

$$\mathscr{D} = 4(b^2 - cb^3 + 18abc - 16a - 27a^2c^2).$$

Substitution of $(c, -2, b, -2a)$ in the cubic discriminant gives this formula. If $\mathscr{D} \neq 0$, there are three simple complex roots and three finite complex inverse points. If $\mathscr{D} = 0$ and there is a double root and a simple root, the entire fibre is the single point corresponding to the simple root. The remaining case is a triple root. Comparing coefficients of $c(r - r_*)^3$ gives exactly

$$r_* = \frac{2}{3c}, \qquad b = \frac{4}{3c}, \qquad a = \frac{4}{27c^2}. \tag{962}$$

All three root multiplicities then belong to the excluded $\alpha = 0$ locus, and the original finite fibre is empty. These are conclusions about this displayed polynomial and its inverse formulas, not about any other coordinate model.

When $c = 0$, the degree is exactly two, since its leading coefficient remains -2. Its discriminant is $\delta = b^2 - 16a$. If $\delta \neq 0$, its two simple roots and the separate $x = 0$ point give three complex points. If $\delta = 0$, its double root supplies no chart point and the separate point is the whole fibre. For real parameters, a real cubic with $\mathscr{D} > 0$ has three real simple roots, while $\mathscr{D} < 0$ has one real simple root and a nonreal conjugate pair. This follows, for example, from $c^4 \prod_{i<j}(r_i - r_j)^2$: it is positive for three distinct real roots and negative for one real root and a conjugate pair. In the double-root cubic case both the double and simple root are real, because a nonreal multiple root would require its conjugate with the same multiplicity. The real quadratic case has two real chart points when $\delta > 0$ and none when $\delta < 0$, always retaining the additional real boundary point.

14.2 The coefficient heat flow and every time constant

Let τ denote forward heat time. The complete target flow is

$$\Phi_\tau(a, b, c) = (a + 2\tau,\ b + 6c\tau,\ c). \tag{963}$$

It is a group of affine maps with determinant one:

$$D\Phi_\tau = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 6\tau \\ 0 & 0 & 1 \end{pmatrix}, \qquad \Phi_{\tau+s} = \Phi_\tau \circ \Phi_s.$$

For $P(\tau, r) = P_{\Phi_\tau(a,b,c)}(r)$,

$$P_\tau = 6cr - 4 = P_{rr}. \tag{964}$$

On a simple root the implicit function theorem applies because $P_r \neq 0$; differentiating $P(\tau, r(\tau)) = 0$ gives $r' = -P_{rr}/P_r$. No derivative formula is assigned to a multiple root.

Two further identities determine this coefficient motion exactly:

$$\frac{d}{d\tau}(b - 3ca) = 0,$$
$$\frac{d}{d\tau}\mathscr{D} = (2\partial_a + 6c\partial_b)\mathscr{D} = -8(3bc - 4)^2. \tag{965}$$

Indeed differentiating the displayed polynomial $\mathscr{D}$ gives $4(48bc-18c^2b^2-32)$, which equals the last expression. Writing $A_0 = 3bc-4$ at $\tau = 0$ and integrating the exact square gives

$$\mathscr{D}(\tau) = \mathscr{D}(0) - 8A_0^2\tau - 144A_0c^2\tau^2 - 864c^4\tau^3.$$

For real $c \neq 0$ this is strictly decreasing: the integral of the nonnegative square over every interval of positive length is positive, since $A_0 + 18c^2\tau$ cannot vanish throughout such an interval. Its cubic leading term gives opposite infinite limits at the two ends of the real line. Thus each such real heat orbit meets the discriminant locus exactly once. For $c = 0$, the derivative is instead the constant -128, with the degree-two interpretation above still required.

For $c \neq 0$, the exact coordinate substitution $r = z + 2/(3c)$ gives

$$P = cz^3 + \left(b - \frac{4}{3c}\right)z + K, \qquad K = \frac{2b}{3c} - \frac{16}{27c^2} - 2a.$$

The original coefficient c is retained, and K is constant under (963). If the orbit passes through (962) at τ_c, its polynomial is exactly

$$P = cz\big(z^2 + 6(\tau-\tau_c)\big).$$

For $\tau \neq \tau_c$, the central root has $\alpha = 3c(\tau-\tau_c)$; either outer root has $\alpha = -6c(\tau-\tau_c)$. These explicit denominators record the escape of every chart inverse point at the triple-root time. At that time the finite fibre is empty, as proved above.

Newman time is the specific reversal $t = -\tau$. Consequently

$$\mathcal{P}(t,r) = P_{a_0-2t,b_0-6ct,c}(r), \qquad \mathcal{P}_t = -\mathcal{P}_{rr}.$$

For $b_0 = c = 0$,

$$\mathcal{P}(t,r) = -2\big(r^2 - 2(t-a_0/2)\big).$$

Thus $a_0 = -1/4$ gives the model collision time $-1/8$ in this Newman convention, while its forward heat time is $+1/8$. Neither number specifies the Newman constant of the zeta family: the polynomial here has its own explicitly fixed initial datum.

14.3 A global polynomial incompressible lift

Write $J = DF$ and define all three inverse columns, with cyclic indices,

$$W_i = J^{-1}e_i = -\tfrac{1}{2}\nabla F_{i+1} \times \nabla F_{i+2}.$$

The scalar triple products and (959) prove $JW_i = e_i$. Commutation of mixed derivatives proves $\operatorname{div} W_i = 0$. Each column is polynomial. Moreover $W_iF_j = \delta_{ij}$, and $[W_i, W_j]F_k = 0$ for every k; invertibility of J proves $[W_i, W_j] = 0$.

The exact heat lift is

$$U = 2W_1 + 6F_3W_2, \qquad JU = (2, 6F_3, 0)^{\mathsf{T}}. \tag{966}$$

It is a polynomial vector field on the whole original $\mathbb{R}^3$. Its ordinary Euclidean divergence is zero, since

$$\operatorname{div} U = 2\operatorname{div} W_1 + 6F_3 \operatorname{div} W_2 + 6W_2F_3 = 0.$$

A global vector potential is also explicit:

$$U = \operatorname{curl}\left(-F_2\nabla F_3 - \tfrac{3}{2}F_3^2\nabla F_1\right). \tag{967}$$

Expanding the curl gives $-\nabla F_2 \times \nabla F_3 - 3F_3 \nabla F_3 \times \nabla F_1$, which is exactly (966).

The complete component formula can be kept polynomial without an inverse denominator. With the same original $h = 2 - 3xy - x^2 w$, put $Q = 3h(1 + xy) - 2$. Then

$$\begin{aligned} U_x &= x^3(Q^2 - 3h), \\ U_y &= x(Q^2 - Q - 3h), \\ U_w &= 3Q + (3h + 3xy - 7)Q^2 + (21 - 9xy)h - 9h^2. \end{aligned} \tag{968}$$

These identities follow either by expanding (966), or by substituting the inverse chart into the differential calculation below and using $F_3 = xh$. They hold on $x \neq 0$ and then everywhere by polynomial identity. At $x = 0$ they give $U = (0, 0, 2)$, so the separate inverse point has exactly the required boundary evolution.

Let φ_τ be the maximal local flow of U. Smooth polynomial coefficients give local existence and uniqueness. Differentiating $F(\varphi_\tau(q))$ using (966), and using uniqueness for the affine target ODE, proves

$$F \circ \varphi_\tau = \Phi_\tau \circ F \tag{969}$$

on every common interval of definition. This identity includes the time variable, not only a matching of fibres. Differentiating it in the original initial point gives

$$D\varphi_\tau(q) = J(\varphi_\tau(q))^{-1} D\Phi_\tau(F(q)) J(q), \qquad \det D\varphi_\tau = 1. \tag{970}$$

Both determinants of J are -2, so their ratio and $\det D\Phi_\tau = 1$ give the last equality without a limit argument. For a retained initial covector ξ_0, its transported covector is

$$(D\varphi_\tau(q))^{-\mathsf{T}} \xi_0 = J(\varphi_\tau(q))^{\mathsf{T}} (D\Phi_\tau)^{-\mathsf{T}} J(q)^{-\mathsf{T}} \xi_0.$$

This is the full inverse-transpose material dictionary in these coordinates.

There is an exact scalar operator identity as well. Define

$$\Pi(q, r) = F_3(q) r^3 - 2r^2 + F_2(q) r - 2F_1(q).$$

Using $UF_1 = 2$, $UF_2 = 6F_3$, $UF_3 = 0$ gives, on the entire product of the original space with the root coordinate,

$$U\Pi = 6F_3 r - 4 = \partial_r^2 \Pi. \tag{971}$$

Thus $\Pi(\varphi_\tau(q), r)$ obeys the exact forward heat equation in the displayed auxiliary variable r. This variable remains the inverse-root coordinate; it is not relabelled as a Euclidean spatial diffusion coordinate of the three-dimensional fluid.

14.4 The full inverse-chart dynamics and its preserved area form

In (960), (r, α, c) are actual coordinates on $x \neq 0$, with inverse $(y+1/x, 1/x, F_3)$. Direct determinants of (960) and (961) give

$$\det DH = -\alpha, \qquad \det D(F \circ H) = 2\alpha.$$

They multiply consistently with $\det J = -2$. Put $q_r = 3cr - 2$, where the subscript is a name for this polynomial, not a derivative. The complete lifted ODE in this chart is

$$r' = -\frac{q_r}{\alpha}, \qquad \alpha' = 3c - \frac{q_r^2}{\alpha}, \qquad c' = 0. \tag{972}$$

Indeed $P_{rr} = 2q_r$, $P_r = 2\alpha$ give the first equation. Differentiating $\alpha = (3cr^2 - 4r + b)/2$ along $b' = 6c$ gives $\alpha' = q_r r' + 3c$, which is the second equation. Differentiating each component of H then yields exactly $U \circ H$. For example $x' = -\alpha'/\alpha^2 = -3cx^2 + q_r^2 x^3$ and $y' = r' - \alpha' = (q_r^2 - q_r)x - 3c$. For the third component the unaltered expression is

$$w' = -3\alpha r' + (10\alpha - 3r - 3c\alpha^2)\alpha',$$

which gives (968) after $c = xh$, $q_r = 3h(1 + xy) - 2$ are substituted.

The heat invariant $I = b - 3ca$ becomes the specific polynomial

$$I(r, \alpha, c) = 4r + 2\alpha - 6cr^2 - 3cr\alpha + 3c^2 r^3.$$

Its two derivatives are

$$I_\alpha = -q_r, \qquad I_r = q_r^2 - 3c\alpha.$$

Consequently, for each fixed real c, the nondegenerate two-form

$$\omega_c = \alpha\, dr \wedge d\alpha \quad (\alpha \neq 0)$$

satisfies $\iota_{(r',\alpha')}\omega_c = dI$. Indeed its contraction is $\alpha r'\, d\alpha - \alpha\alpha'\, dr = I_\alpha\, d\alpha + I_r\, dr$. This states the exact Hamiltonian convention and proves the claimed Hamiltonian correspondence.

The preserved signed volume can also be checked directly:

$$\partial_r(\alpha r') + \partial_\alpha(\alpha\alpha') = \partial_r(-q_r) + \partial_\alpha(3c\alpha - q_r^2) = 0.$$

The Euclidean volume of the original source is $|\alpha|\, dr\, d\alpha\, dc$ in this chart. On either component of $\alpha \neq 0$, the absolute value has a fixed sign relative to α, so this same calculation proves its preservation. The density tending to zero at $\alpha = 0$ describes a boundary of this inverse chart; $x = 1/\alpha$ shows that boundary is not a finite original coordinate point.

14.5 The exact positive-viscosity fluid equation and escape

On the target $\mathbb{R}^3_{a,b,c}$ with Euclidean metric, standard orientation, and any retained viscosity $\nu > 0$, set

$$v(a, b, c) = (2, 6c, 0), \qquad p = 0, \qquad f = 0.$$

Every term of its stationary Navier–Stokes equation can be computed: $\operatorname{div} v = 0$, $(v \cdot \nabla)v = (2\partial_a + 6c\partial_b)v = 0$, and $\Delta v = 0$. Its oriented curl is $(-6, 0, 0)$. Its material flow is exactly (963).

On the original source set $g = J^{\mathsf{T}} J$, using the standard source orientation. This positive metric has $\det g = 4$ and volume $2\, dx\, dy\, dw$. In a local F chart it is precisely the Euclidean target metric. The Levi–Civita connection pulls back under this local isometry: this follows from its metric compatibility and zero torsion, which characterize that connection uniquely. The nonpositive vector diffusion operator is

$$\mathcal{L}_g Y = \left[-(d\delta_g + \delta_g d)Y_g^\flat\right]_g^\sharp.$$

The Hodge Laplacian commutes with local metric pullback, including an orientation-reversing pullback, because the two orientation signs in the codifferential cancel. In Euclidean coordinates this operator is the componentwise Laplacian. Therefore

$$\operatorname{div}_g U = 0, \qquad \nabla^g_U U = 0, \qquad \mathcal{L}_g U = 0.$$

These prove the complete intrinsic stationary unforced positive-viscosity Navier–Stokes equation with constant pressure. Its vorticity retains the orientation sign:

$$\operatorname{curl}_g U = 6W_1.$$

In fact $\det J = -2$ makes $*_g F^*\eta = -F^*(*_{\text{target}}\eta)$. Applying this to $d(v^\flat)$ negates the pullback of $\operatorname{curl} v = -6e_1$, giving the displayed result.

There are explicit real finite-time escaping material trajectories. On the initial fibre $(-1/4, 0, 0)$, for $\tau < 1/8$ take

$$\gamma_\pm(\tau) = \left(\pm(1-8\tau)^{-1/2}, \mp\tfrac{3}{2}(1-8\tau)^{1/2}, \tfrac{13}{2}(1-8\tau)\right).$$

Substitution in the full map gives $F(\gamma_\pm(\tau)) = (-1/4 + 2\tau, 0, 0)$. Differentiation and J^{-1} prove $\gamma_\pm' = U(\gamma_\pm)$. Their first coordinate escapes to infinity at $\tau = 1/8$; their g-speed is $|v| = 2$ throughout. Hence their length from $\tau = 0$ to the endpoint is $1/4$. Their times approaching $1/8$ form a metric Cauchy sequence because the remaining curve length tends to zero. It has no point limit: a smooth positive Riemannian metric induces the ordinary coordinate topology locally, while x diverges. Thus the metric is incomplete. The separate inverse point $(0, 0, -1/4 + 2\tau)$ remains a finite trajectory throughout. The coefficient heat collision and its escaping branches have all been retained by these formulas.

14.6 Its exact force in the original Euclidean metric

The same polynomial U, now using the original Euclidean metric and its component Laplacian, has stationary Navier–Stokes triples precisely of the form

$$p = q, \qquad f = (U \cdot \nabla)U - \nu\Delta U + \nabla q.$$

Substitution proves sufficiency; rearrangement of the momentum equation proves necessity. Thus every pressure adjustment has been retained.

The force has the following exact curl on the original coordinate axis:

$$\operatorname{curl} f(0, 0, w) = (0, 0, 108\nu w). \tag{973}$$

Here is a direct derivative certificate from the full component formula. Its transverse polynomial terms of degree at most three are

$$U_x = 10x^3 + R_x, \quad U_y = 6x - 12x^2y - 18wx^3 + R_y, \quad U_w = 2 - 54xy - 18wx^2 + R_w,$$

where the remainders are defined by these exact differences and

$$R_x, R_y \in (x, y)^5\mathbb{R}[x, y, w], \qquad R_w \in (x, y)^4\mathbb{R}[x, y, w].$$

These ideal memberships follow by expanding (968); thus no unspecified coefficient enters a derivative of the required order. On the axis,

$$U = 2e_3, \qquad DU = \begin{pmatrix} 0 & 0 & 0 \\ 6 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} =: N.$$

The derivative of the convective acceleration there is $N^2 + 2\partial_w DU = 0$, so its curl vanishes. For the diffusive part,

$$\Delta \operatorname{curl} U(0, 0, w) = (0, 0, -108w):$$

the third component is $\partial_x \Delta U_y - \partial_y \Delta U_x = -108w$, from $-18wx^3$, and the first two components vanish from the displayed terms and remainder ideals. Commuting derivatives and using $\operatorname{curl} \nabla q = 0$ proves (973), also distributionally wherever a pressure is interpreted as a distribution.

Thus this exact same Euclidean velocity cannot have a rapidly decaying smooth force for any pressure choice: its force curl is unbounded on the axis for $\nu > 0$. The source field is also a nonzero polynomial, hence has infinite Euclidean L^2 norm. To see the latter fact directly, a nonzero leading homogeneous component of a polynomial is bounded away from zero on some open patch of the unit sphere. The full polynomial then has a positive multiple of its leading radial power on the corresponding cone for all sufficiently large radii. Radial integration diverges. A nonzero constant component already gives the same conclusion at degree zero. Here the field is nonconstant, as $U_y = 6x$ to first order at the origin proves, and a polynomial periodic in every unit coordinate direction must be constant by the one-variable polynomial difference identity. It therefore does not define the required periodic velocity either.

These calculations identify both fluid realizations completely: the target shear and its intrinsic lift are exact unforced viscous solutions, while the original Euclidean lift requires the precisely computed nondecaying force. The finite-time material escape belongs to the incomplete metric realization. No classical Euclidean Navier–Stokes counterexample or off-critical zeta zero is asserted.

15 The original material generator in the arithmetic coordinate

We retain the polynomial F and the incompressible vector field U of Section 14, originating in the polynomial construction cited there[97, 98, 99]. The arithmetic functions and their heat parameter are precisely those of Section 13, with $\Xi(s) = \xi(1/2 + is)$ and $Z_t(s) = 8H_t(2s)$ [103, 17]. The following calculations give their actual pullback by F and the original Euclidean diffusion, rather than replacing that diffusion by a chosen scalar heat operator.

15.1 An injective arithmetic pullback and the full material drift

Write $q = (x, y, w)$ and define

$$s(q) = F_1(q)/2, \qquad b(q) = F_2(q), \qquad c(q) = F_3(q).$$

These names denote specific functions on the original space, with no claim that F has a global inverse. For $W_j = (DF)^{-1} e_j$ and $U = 2W_1 + 6F_3 W_2$, the identity $W_j F_k = \delta_{jk}$ gives

$$Us = 1, \qquad Ub = 6c, \qquad Uc = 0, \qquad U\big(\phi(s, b, c)\big) = (\partial_s \phi + 6c\partial_b \phi)(s, b, c). \tag{974}$$

The chain rule proves the last identity for every differentiable ϕ, over $\mathbb{R}$, and every holomorphic ϕ over $\mathbb{C}$. In particular $b - 6cs = F_2 - 3F_3F_1$ is a first integral. This retains the b drift when $c \neq 0$.

For entire functions let

$$\mathcal{I} : \mathcal{O}(\mathbb{C}) \longrightarrow \mathcal{O}(\mathbb{C}^3), \qquad (\mathcal{I}f)(q) = f(s(q)).$$

It is an injective algebra map. Indeed, the polynomial section $\iota(s) = (0, 0, 2s)$ satisfies $s(\iota(s)) = s$, so $\iota^* \mathcal{I} f = f$. Both maps are continuous for uniform convergence on compact sets, since continuous images of compact sets are compact. The same section identifies the image with the original function space when that image is given the transported topology. Repeated application of (974) proves

$$U^k \mathcal{I} f = \mathcal{I} f^{(k)} \quad (k \geq 0). \tag{975}$$

Thus this is an actual intertwiner, including its explicit left inverse.

On the order-at-most-one function space of Section 13.2, the already proved heat group gives $\mathscr{H}_t = \mathcal{I} U_t \iota^*$ on its image. Its series is

$$\mathscr{H}_t \mathcal{I} f = \sum_{k \geq 0} \frac{(-t/4)^k}{k!} U^{2k} \mathcal{I} f = \mathcal{I}(U_t f).$$

Convergence follows from the proved derivative estimates there, transported by $\mathcal{I}$; no convergence assertion on every entire function of three variables is needed or made. In particular the original arithmetic input, not a polynomial proxy, obeys

$$\widetilde{Z}_t(q) = Z_t(s(q)), \qquad \partial_t \widetilde{Z}_t = -\tfrac{1}{4} U^2 \widetilde{Z}_t, \qquad \widetilde{Z}_0 = \mathcal{I}\Xi. \tag{976}$$

If $X_\tau(q)$ is the maximal material flow where it exists, integration of (974) gives

$$(s, b, c)(X_\tau q) = (s(q) + \tau, b(q) + 6c(q)\tau, c(q)).$$

Hence $(\mathcal{I} f)(X_\tau q) = f(s(q) + \tau)$ on that domain. The material parameter τ translates s; the arithmetic heat parameter t generates $-U^2/4$. Their different actions are related by these equations, not identified by a change of name.

15.2 The branch derivative with every coefficient retained

In the inverse-root space use the full relation

$$P(s, b, c; r) = cr^3 - 2r^2 + br - 4s = 0.$$

At $P_r \neq 0$, differentiation with b, c fixed gives $\partial_s r = 4/P_r$. The lifted material vector field, including its coefficient drift, is instead

$$U_{\mathrm{root}} = \frac{4 - 6cr}{P_r} \partial_r + 6c \partial_b. \tag{977}$$

Indeed $U_{\mathrm{root}} P = (4 - 6cr) + 6cr - 4 = 0$ when the independent s component is included as in (974). More explicitly, on the root hypersurface with local coordinates (r, b, c) one has $s = (cr^3 - 2r^2 + br)/4$, and the displayed operator gives $U_{\mathrm{root}} s = 1$, $U_{\mathrm{root}} b = 6c$, $U_{\mathrm{root}} c = 0$. This proves the stated lift without deleting the transverse drift.

At $c = 0$ the coefficient b is constant along the flow. Retain it:

$$s = \frac{br}{4} - \frac{r^2}{2}, \quad \delta_b = b^2 - 32s = (b - 4r)^2, \quad \alpha = \frac{b - 4r}{2}, \quad D_b = \frac{4}{b - 4r} \partial_r.$$

On a base open set on which $\delta_b \neq 0$, the quadratic algebra $\mathcal{O}[r]/(r^2 - br/2 + 2s)$ is free in the ordered basis $(1, r)$. Its multiplication and connection matrices are

$$R_b = \begin{pmatrix} 0 & -2s \\ 1 & b/2 \end{pmatrix}, \qquad D_b = \partial_s I + A_b, \qquad A_b = \begin{pmatrix} 0 & 4b/\delta_b \\ 0 & -16/\delta_b \end{pmatrix}. \tag{978}$$

To check the second formula, use $D_b r = 4/(b - 4r) = (4b - 16r)/\delta_b$ and differentiate $f_0(s) + r f_1(s)$ by the product rule. In particular

$$\begin{aligned} R_b^2 - \tfrac{b}{2} R_b + 2sI &= 0, \\ [D_b, R_b] &= 4(bI - 4R_b)/\delta_b, \\ D_b^2 &= \partial_s^2 I + 2A_b \partial_s + (16/\delta_b) A_b. \end{aligned} \tag{979}$$

For the commutator calculate $R_b' + A_bR_b - R_bA_b$. For the square use $A_b' = (32/\delta_b)A_b$ and $A_b^2 = -(16/\delta_b)A_b$. These computations prove every entry, including the zeroth-order term. At $b=0$ the connection is $\mathrm{diag}(\partial_s, \partial_s + 1/(2s))$, the earlier two-sheet connection with exactly the same constants.

The sheet involution $r \mapsto b/2 - r$ has matrix $J_b = \left(\begin{smallmatrix} 1 & b/2 \\ 0 & -1 \end{smallmatrix}\right)$, with $J_b^2 = I$ and $J_bR_bJ_b = bI/2 - R_b$. The algebra trace is

$$\mathrm{tr}(f_0 + rf_1) = 2f_0 + \tfrac{b}{2}f_1, \qquad \ker \mathrm{tr} = \{(-bh/4, h) : h \in \mathcal{O}\}.$$

This proves its kernel as well as the retained sheet symmetry.

The excluded locus in this inverse chart is $r = b/4$, $s = s_* := b^2/32$, $\alpha = 0$. Writing *explicitly* $\eta = r - b/4$ gives

$$s = s_* - \eta^2/2, \qquad H = \left(-\frac{1}{2\eta}, \frac{b}{4} + 3\eta, 26\eta^2 + \frac{3b\eta}{2}\right). \tag{980}$$

This is a variable substitution with the inverse $r = \eta + b/4$, not deletion of b. The separate point with $x = 0$ remains $(0, b, 2s - 4b^2)$, including at $s = s_*$. For complex b every complex value of s_* occurs; for real b the values are exactly $[0, \infty)$. Thus zero-freeness at the single $b = 0$ branch origin does not exclude arithmetic zeros at the other branch values.

For precision, if the actual function at a fixed time has $Z_t(s) = (s - s_*)^m A(s)$ near s_*, where $A(s_*) \neq 0$, then substitution proves the complete local identity

$$Z_t(s_* - \eta^2/2) = (-1/2)^m \eta^{2m} A(s_* - \eta^2/2).$$

The ramification doubles that existing multiplicity; it does not supply a new zero. For $Z_t(s_*) \neq 0$ the same pullback is nonzero at $\eta = 0$. Both cases follow directly by evaluating the analytic factor. These are local formulas for the actual function, not an assumption about the location or multiplicity of its zeros.

15.3 The original Euclidean diffusion and its fibre dependence

For real q, put $v = 1 + xy$. Direct differentiation of the retained polynomial gives

$$\begin{aligned} s_x &= \tfrac{1}{2}\{3yv^2w + y^3(1+6v)\}, \\ s_y &= \tfrac{1}{2}\{3xv^2w + 2y(v + 3v^2) + xy^2(1+6v)\}, \\ s_w &= \tfrac{1}{2}v^3, \\ \Delta_E s &= 3wv(x^2+y^2) + 18x^2y^2 + 21xy + 3y^4 + 4. \end{aligned}$$

For completeness, use $F_1 = v^3w + y^2(v + 3v^2)$, with $v_x = y$, $v_y = x$, $v_w = 0$, to obtain the three first derivatives. The second x derivative is $3y^2vw + 3y^4$. The second y derivative is $3x^2vw + v + 3v^2 + 2xy(1+6v) + 3x^2y^2$. Their sum, with $s_{ww} = 0$ and $v = 1 + xy$, gives the displayed Laplacian. The scalar chain rule is therefore exactly

$$\Delta_E(\mathcal{I}f) = |\nabla_E s|^2 \mathcal{I}f'' + (\Delta_E s)\mathcal{I}f'. \tag{981}$$

The squared gradient here is the Euclidean real squared norm. For holomorphic algebraic extension its expression is $\sum_j s_{q_j}^2$, not a silently inserted Hermitian norm.

Along the actual escaping trajectory

$$q(z) = (z^{-1}, -3z/2, 13z^2/2), \qquad z = \sqrt{1 - 8\tau} > 0,$$

substitution gives $v=-1/2$, $(s,b,c)=(-z^2/8,0,0)$ and

$$\nabla_E s=\left(-\frac{9z^3}{32},-\frac{3z}{16},-\frac{1}{16}\right),\quad |\nabla_E s|^2=\frac{81z^6+36z^2+4}{1024},\quad \Delta_E s=\frac{13-27z^4}{4}. \tag{982}$$

For example the x component is $\{-117z^3/16+27z^3/4\}/2=-9z^3/32$, by direct substitution into the preceding derivative formula. The squared norm is the sum of the three displayed squares; the Laplacian follows from its polynomial formula. Consequently

$$(\Delta_E\mathcal{I}f)(q(z))=\frac{81z^6+36z^2+4}{1024}f''(-z^2/8)+\frac{13-27z^4}{4}f'(-z^2/8).$$

Both coefficients have finite limits, $1/256$ and $13/4$, as $z\downarrow 0$. Meanwhile $q(z)$ leaves every bounded set. This is a statement about the specified arithmetic observable along that trajectory, not a boundedness statement for every transported covector or for the full material metric.

There is an exact, finite-point reason that the full diffusion cannot be encoded by a scalar operator depending only on s. The points

$$q_1=(1,-3/2,13/2),\qquad q_0=(0,0,-1/4)$$

both have $F(q_j)=(-1/4,0,0)$ and hence $s=-1/8$. At q_1, $\Delta_E s=-7/2$; at q_0, $\Delta_E s=4$. Thus $\Delta_E s$ is not constant even on this full F fibre. If an operator L on scalar functions of s satisfied $\Delta_E\mathcal{I}=\mathcal{I}L$, applying it to $f(s)=s$ would give the same value at these two points, contradicting $-7/2\neq 4$. The actual relation is instead the completely specified two-coefficient formula (981). On the escaping trajectory one can write its coefficients as $-81s^3/2-9s/32+1/256$ and $13/4-432s^2$, respectively, but that trajectory formula is not an autonomous reduction valid on every fibre.

15.4 Transport of the complete arithmetic form

To carry the same heat map into arithmetic tests, retain $\mathcal{E}=C_c^\infty(\mathbb{R};\mathbb{C})$ and set, for real t,

$$G_tf(u)=e^{tu^2/4}f(u),\qquad W_t(f,g)=W(G_{-t}f,G_{-t}g).$$

Multiplication by the smooth nonvanishing function and its inverse preserves each compact support and its smooth topology. Thus G_t is an automorphism of the test space. Section 13.7 proved $\mathcal{F}G_t=U_t\mathcal{F}$ and $T_t=G_tTG_{-t}=-i\partial_u+itu/2$ for the retained Fourier sign. The symmetry of T for W now gives, directly,

$$W_t(T_tf,g)=W(TG_{-t}f,G_{-t}g)=W(G_{-t}f,TG_{-t}g)=W_t(f,T_tg).$$

Moreover, for any finite family of tests,

$$[W_t(G_tf_i,G_tf_j)]_{ij}=[W(f_i,f_j)]_{ij}. \tag{983}$$

This is equality of every matrix entry, including its entire quadratic form and inertia. In particular transport of both tests and pairing does not manufacture a negative value.

The exact arithmetic expression for fixed, untransported tests uses

$$h_t(x)=\int_{\mathbb{R}}g(x+v)\overline{f(v)}e^{-t((x+v)^2+v^2)/4}\,dv=(G_{-t}g)*(G_{-t}f)^\star(x).$$

The convolution identity follows by substituting the definition of reflection-conjugation. It keeps support inside the fixed difference of the two supports. Applying the full explicit formula [26, 10] yields

$$\begin{aligned} W_t(f,g) &= \widehat{h}_t(i/2) + \widehat{h}_t(-i/2) \\ &\quad + \int_{\mathbb{R}} \widehat{h}_t(\lambda)\left\{\frac{1}{2\pi}\Re\psi\left(\frac{1}{4}+\frac{i\lambda}{2}\right) - \frac{\log\pi}{2\pi}\right\} d\lambda \\ &\quad - \sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\{h_t(\log n) + h_t(-\log n)\}. \end{aligned} \tag{984}$$

The prime-power sum is finite by compact support. Uniform bounds on all derivatives on a fixed compact t interval give rapid Fourier decay in horizontal strips by integration by parts; this justifies both the integral and its parameter derivatives. Differentiating G_{-t}, or equivalently the displayed compact integral, therefore gives

$$\frac{d}{dt}W_t(f,g) = -\tfrac{1}{4}\{W_t(u^2 f, g) + W_t(f, u^2 g)\}.$$

Every pole, gamma and prime term is still present in (984). This is the congruent transport of the original Weil form; a form defined instead by the moving zero divisor of Z_t has not been substituted for it. The exact transported quotient and the moving divisor comparison were proved in Section 13. The present calculation supplies the actual material generator, the full Euclidean diffusion and the retained arithmetic test map. It does not exhibit a negative value of the original Weil form.

16 Nonstandard GCT: the coefficient, its arithmetic spectrum, and heat connection

16.1 The original source and the direction of its programme

Mulmuley's discussion of nonstandard Riemann hypotheses concerns a proposed extension of the finite-field geometry behind canonical-basis positivity. The actual maps in his account start with the plethysm representation $H = GL_n(\mathbb{C}) \longrightarrow GL(V_\mu(H))$, its nonstandard quantum deformation, and then the class-variety triple $\bar{H}_{\hat{h}} \longrightarrow \bar{H} \longrightarrow GL(\bar{W})$. The intended destination is explicit representation-theoretic obstructions to characteristic-zero complexity inclusions [106, section "How to prove PH1 and why should it hold?"]. In that account, positivity supplies proposed polyhedral counting descriptions; emptiness and eventual nonemptiness of the two different polytopes supply the obstruction comparison. The source separately states the positivity and obstruction hypotheses in its refined GCT6 theorem. These are recorded as the author's research programme, not promoted here to proved premises or reversed into an implication about classical RH. The maps calculated below concern one explicit coefficient in the later two-row construction, and the existing arithmetic spectral module.

Blasiak, Mulmuley and Sohoni's GCT IV prints a mixed-sign expression [107, §17.2, example labelled "ex not positive"]. Their parameter macro denotes q. Retain exactly their scalar expression

$$\begin{aligned} C(q) &= \frac{[6]_q([7]_q - [3]_q) - [3]_q[8]_q}{[2]_q} \\ &= q^{10} - q^4 - q^{-4} + q^{-10} = (q-q^{-1})^2[7]_q[3]_q, \\ [j]_q &= q^{j-1} + q^{j-3} + \cdots + q^{1-j}. \end{aligned} \tag{985}$$

The first line is the source's printed computation with $n_3 = 3, n_2 = 2$. Its literal target tableau has a weight mismatch, proved below; the earlier identification of this expression as the coefficient for that literal target is withdrawn. None of the polynomial, spectrum or heat calculations below uses that identification. Comparing its numerator with $[2]_q$ times the claimed Laurent polynomial and expanding the finite sums proves the equality without division at specializations. In particular the last two lines define the continued coefficient even at $q = \pm 1, \pm i$. At $q = \pm i$ the first line itself has a zero denominator and is not an evaluation prescription. The source's preceding theorem gives sign-definite Laurent coefficients for the individual Chevalley generators; this divided-power example is explicitly outside that stronger sign assertion. Its §1.7 retains the tableau rescaling $(-1/[2]_q)^{\deg T}$ and the integral form used at $q = 1$. Thus continuation of this single Laurent coefficient does not by itself specialize the entire source basis at $[2]_q = 0$.

The literal target and its exact weight projection. Write a column from top to bottom in parentheses, and put $p = [2]_q$. Over $\mathbb{Q}(q)$ the source's displayed tableaux specialize to

$$\begin{aligned} T &= (123)(123)(124)(12)(12)(1)(1), \\ P &= -p^{-1}(123)(123)(124)(23)(12)(4)(1), \\ T_1 &= (123)(123)(124)(23)(12)(1)(1), \\ T_2 &= -p^{-1}(123)(123)(123)(12)(12)(4)(1). \end{aligned}$$

The source's alphabet is $1 \leftrightarrow (1,1)$, $2 \leftrightarrow (1,2)$, $3 \leftrightarrow (2,1)$, $4 \leftrightarrow (2,2)$. Its nonstandard-column dictionary preserves the resulting counts: for example $(23) = c^{VW}_{21,12} + p^{-1} c^{VW}_{21,21}$ has V-weight $(1,1)$ and W-weight $(1,1)$ in both summands, and the two terms of (124) both have weights $(2,1), (1,2)$. Tensor products add weights; the source's based products and quotient maps are $U_q(V) \otimes U_q(W)$-module maps and preserve them [107, nonstandard-column dictionary and Proposition labelled "p NSC facts"]. Counting gives

	wt_V	wt_W
T	$(12,3)$	$(9,6)$
T_1, T_2	$(11,4)$	$(9,6)$
P	$(10,5)$	$(8,7)$
$Q = -p^{-1}(123)(123)(124)(13)(12)(4)(1)$	$(10,5)$	$(9,6)$.

Here Q records the explicit alternative endpoint, not a renaming of P. On the weight decomposition $V_\nu = \bigoplus_{\alpha,\beta} (V_\nu)_{\alpha,\beta}$ let $\pi_{\alpha,\beta}$ be the projection which is the identity on the named summand and zero on the others. Its kernel is the direct sum of all the other summands and its image is exactly the named summand. Since F_V has weight $((-1,1),(0,0))$, the map

$$F_V^{(2)} : (V_\nu)_{(12,3),(9,6)} \longrightarrow (V_\nu)_{(10,5),(9,6)}$$

is followed by zero under $\pi_{(10,5),(8,7)}$. Thus the coefficient of the literal printed P is zero. This conclusion uses the generic source field, not a root-of-unity specialization, and is unaffected by the nonzero rescaling $-p^{-1}$. Replacing the indicated column (23) by (13) restores the target weight and gives Q. Weight compatibility alone does not compute its coefficient; its contextual straightening must be retained in a two-step generator calculation. The subsequent scalar base change applies to the explicit $C(q)$ in (985) independently of that calculation.

16.2 A finite-free map retaining the original arithmetic coordinate

Work first over a characteristic-zero field k. The original slice of the polynomial inverse construction has $b = c = 0$ and $r^2 = -a$. Adjoin p by $p^2 = 4 - a$. This gives an exact base change

$$k[p] \otimes_{k[a]} k[a, r]/(r^2 + a) = k[p, r]/(p^2 - r^2 - 4) \simeq k[q, q^{-1}], \quad a \longmapsto 4 - p^2. \tag{986}$$

The isomorphism and its inverse are

$$p = q + q^{-1}, \quad r = q - q^{-1}, \qquad q = (p + r)/2, \quad q^{-1} = (p - r)/2.$$

Indeed the Laurent expressions have $p^2 - r^2 = 4$, and in the conic ring $(p + r)(p - r) = 4$. The compositions fix all four generators, proving both maps and their inverse identities. Retain $a = 2s$, exactly as in the preceding arithmetic construction. Then

$$s = s(q) = -\frac{(q - q^{-1})^2}{2} = 1 - \frac{q^2 + q^{-2}}{2}. \tag{987}$$

No original source coordinate is renamed: p is the new cover coordinate, whereas the original triple (x, y, w) on the nonboundary chart is

$$x = -\frac{1}{2r}, \quad y = 3r, \quad w = 26r^2, \qquad F(x, y, w) = (-r^2, 0, 0).$$

These expressions are defined for $r \neq 0$, equivalently $q \neq 0, \pm 1$. The other original fibre point $(0, 0, -r^2)$ remains present separately.

The Laurent ring is free of rank four over $k[s]$:

$$k[q, q^{-1}] \simeq k[s, q]/(q^4 + (2s - 2)q^2 + 1). \tag{988}$$

To check this, the quartic yields $q^{-1} = -q^3 + (2 - 2s)q$; substitution of (987) gives the inverse map. Division by the monic quartic gives a unique representative of degree at most three. The second free basis is $(1, r, p, rp)$, because $r^2 = -2s$ and $p^2 = 4 - 2s$ are successive monic quadratic relations. Explicit conversion is

$$\begin{gathered} r = q^3 + (2s - 1)q, \qquad p = -q^3 + (3 - 2s)q, \qquad rp = 2q^2 + 2s - 2, \\ q = (p + r)/2, \qquad q^2 = (rp + 2 - 2s)/2, \qquad q^3 = r - (2s - 1)(p + r)/2. \end{gathered}$$

These identities also prove independence of the second basis directly.

For the actual topological module (Q, S) of the preceding arithmetic construction, take $k = \mathbb{C}$ and give $Q^{(4)} = \mathbb{C}[q, q^{-1}] \otimes_{\mathbb{C}[s]} Q$ the fourfold direct-sum topology in $(1, q, q^2, q^3)$. Here s acts by the already defined continuous operator S, not by a freely chosen replacement spectrum. Set

$$K = \begin{pmatrix} 0 & 0 & 0 & -I \\ I & 0 & 0 & 0 \\ 0 & I & 0 & 2I - 2S \\ 0 & 0 & I & 0 \end{pmatrix}, \qquad \widehat{S} = \operatorname{diag}(S, S, S, S). \tag{989}$$

The quartic proves

$$\begin{gathered} K^{-1} = -K^3 + (2I - 2\widehat{S})K, \qquad \widehat{S} = -\tfrac{1}{2}(K - K^{-1})^2, \\ (K + K^{-1})^2 = 4I - 2\widehat{S}. \end{gathered}$$

All expressions are continuous. The constant-coefficient inclusion and projection retain S itself. This constructs the action of the Laurent coefficient algebra; a representation of all the nonstandard quantum group generators is not asserted by this tensor product.

16.3 The complete resolvent and all spectral fibres

Spectrum here means failure of a continuous two-sided inverse on the specified topological space. No Banach-space spectral mapping theorem is needed.

Theorem 16.1. *For $\lambda \neq 0$, put $s_\lambda = s(\lambda)$ and $A_\lambda = S - s_\lambda I$. Define*

$$\begin{aligned} E_\lambda v &= (-\lambda^{-1}v, -\lambda^{-2}v, \lambda v, v), \\ L_\lambda y &= \lambda^{-2}y_0 + \lambda^{-1}y_1 + y_2 + \lambda y_3, \\ H_\lambda y &= (-\lambda^{-1}y_0, -\lambda^{-2}y_0 - \lambda^{-1}y_1, y_3, 0), \\ J_0 v &= (0,0,v,0), \qquad P_3 x = x_3. \end{aligned}$$

Then $K - \lambda I$ is continuously invertible exactly when A_λ is continuously invertible. In that case

$$(K-\lambda I)^{-1} = H_\lambda - \tfrac{1}{2}E_\lambda A_\lambda^{-1} L_\lambda, \qquad A_\lambda^{-1} = P_3(K-\lambda I)^{-1}(-2J_0). \tag{990}$$

Moreover

$$\operatorname{Spec} K = s^{-1}(\operatorname{Spec} S) \subset \mathbb{C}^\times, \qquad \ker(K-\lambda I) = E_\lambda \ker(S - s_\lambda I), \tag{991}$$

with inverse P_3 on the displayed kernels.

Proof. Writing $B = K - \lambda I$, its four equations $Bx = y$ give, successively,

$$\begin{aligned} x_0 &= -\lambda^{-1}(x_3 + y_0), \\ x_1 &= -\lambda^{-2}x_3 - \lambda^{-2}y_0 - \lambda^{-1}y_1, \\ x_2 &= \lambda x_3 + y_3, \\ -2(S - s_\lambda I)x_3 &= L_\lambda y. \end{aligned}$$

Conversely these equations imply all four rows of $Bx = y$ by substitution. Thus the last equation determines x_3 uniquely when A_λ is invertible, and the other three determine x by (990). This inverse is continuous, since it is a finite sum of continuous maps.

In the reverse direction, direct multiplication gives $BE_\lambda = -2J_0A_\lambda$ and $P_3E_\lambda = I$. If B is invertible, $A_\lambda v = 0$ forces $v = 0$. For any v, solve $Bx = -2J_0v$; the last eliminated equation gives $A_\lambda P_3 x = v$. This proves surjectivity, the second inverse formula, and its continuity. Taking $y = 0$ in the same equations proves the kernel isomorphism. Finally K itself has the polynomial inverse already displayed, so zero adds no point to its spectrum. □

Proposition 16.2. *The complete inverse image of the real arithmetic axis is*

$$\Gamma = \mathbb{R}^\times \ \cup\ \{q : |q| = 1\}\ \cup\ i\mathbb{R}^\times.$$

The respective images are $(-\infty, 0]$, $[0, 2]$, and $[2, \infty)$. Consequently

$$\operatorname{Spec} S \subset \mathbb{R} \quad\Longleftrightarrow\quad \operatorname{Spec} K \subset \Gamma. \tag{992}$$

Proof. Write $q = \rho e^{i\theta}$ with $\rho > 0$. Equation (987) gives $\operatorname{Im} s(q) = -(\rho^2 - \rho^{-2})\sin(2\theta)/2$. This is zero exactly when $\rho = 1$ or q lies on one of the two axes. Substitution on the axes and circle gives the stated ranges. For completeness, the full fibres, with independent signs $\varepsilon, \delta \in \{1, -1\}$, are

$$\begin{aligned} s < 0 : \quad & q = \tfrac{1}{2}\big(\varepsilon\sqrt{4-2s} + \delta\sqrt{-2s}\big), \\ 0 < s < 2 : \quad & q = \tfrac{1}{2}\big(\varepsilon\sqrt{4-2s} + i\delta\sqrt{2s}\big), \\ s > 2 : \quad & q = \tfrac{i}{2}\big(\varepsilon\sqrt{2s-4} + \delta\sqrt{2s}\big). \end{aligned}$$

They follow by solving for p, r in (986). Their products $qq^{-1} = 1$ ensure none is zero; away from $s = 0, 2$ they are distinct. At $s = 0$ the quartic is $(q^2 - 1)^2$; at $s = 2$ it is $(q^2 + 1)^2$. Every complex s has a nonzero quartic root, so (991) proves both directions of (992), including its converse. $\square$

The two symmetries of each fibre are $q \mapsto -q$ and $q \mapsto q^{-1}$. In the displayed (p, r) signs, inversion changes δ and negation changes both signs. The branch points are exactly $q = \pm 1, \pm i$:

$$s'(q) = -q + q^{-3}, \qquad s''(q) = -1 - 3q^{-4} = -4 \quad (q^4 = 1).$$

Thus their ramification index is two, not a removable multiplicity. Indeed let $Sv = s_0 v$ with $v \neq 0$, $s_\lambda = s_0$ and $\lambda^4 = 1$. The value $s_0 = 0$ is included. Then

$$(K - \lambda I)E_\lambda v = 0, \qquad (K - \lambda I)(\lambda^{-2}v, 2\lambda^{-3}v, v, 0) = E_\lambda v. \tag{993}$$

To prove the second identity, differentiate $(K - \lambda I)E_\lambda v = -2J_0(S - s_\lambda I)v$ in λ; $s'_\lambda = 0$ at the branch point. The two vectors are independent: applying $K - \lambda I$ to a proposed dependence proves that its second coefficient is zero. A semisimple source eigenvalue can therefore acquire a genuine two-step Jordan chain on this ramified extension.

16.4 Exact signs and three real structures

The recurrence $U_0 = 1, U_1 = p, U_j = pU_{j-1} - U_{j-2}$ gives $U_j = [j+1]_q$ by cancellation of the interior Laurent terms. Hence

$$C(q) = -a(7 - 14a + 7a^2 - a^3)(3 - a) = c(s), \qquad c(s) = -2s(7 - 28s + 28s^2 - 8s^3)(3 - 2s). \tag{994}$$

In particular $C(K) = c(\widehat{S})$. At $q = 1$ the original vanishing factor has $a = -(q-1)^2(q+1)^2/q^2$, so $C(q)/(q-1)^2 \to 84$. In the original a coordinate the zero is simple with coefficient -21. Both orders are retained.

Let $\alpha_j = 2\sin^2(j\pi/7)$, $j = 1, 2, 3$. On the unit circle $[7]_q = \sin(7\theta)/\sin\theta$, and therefore

$$c(s) = -32s(s - \alpha_1)(s - \alpha_2)(s - \tfrac{3}{2})(s - \alpha_3).$$

To check the factorization, the cubic factor in (994) vanishes at the three distinct α_j and has leading coefficient -8; multiplication by $-2s(3 - 2s)$ proves the formula. Since sine is strictly increasing on $[0, \pi/2]$ and $2\pi/7 < \pi/3 < 3\pi/7 < \pi/2$, the root order is $0 < \alpha_1 < \alpha_2 < 3/2 < \alpha_3 < 2$. The signs of c on the six successive complementary real intervals are $+, -, +, -, +, -$. This proves $C(q) > 0$ on real nonzero $q \neq \pm 1$, and $C(q) < 0$ on every nonzero purely imaginary q; in particular $C(\pm i) = -4$. The exact circle example

$$q_0 = (\sqrt{14} + i\sqrt{2})/4, \qquad s(q_0) = 1/4, \qquad C(q_0) = -65/32$$

follows by multiplication, with $q_0^{-1} = \overline{q_0}$.

Three conjugate-linear, multiplicative involutions on the Laurent algebra extend the same involution $s^* = s$:

$$q^{*_\mathrm{r}} = q, \qquad q^{*_\mathrm{u}} = q^{-1}, \qquad q^{*_\mathrm{i}} = -q.$$

Each squares to the identity on q and scalars, hence on every Laurent polynomial. Their characters are respectively the real axis, unit circle and imaginary axis, since the character condition is $\overline{q_0} = q_0$, q_0^{-1}, or $-q_0$. They fix s by (987). This is an explicit relation between the three real forms, not an identification of their positivity cones. For example, in a positive Hilbert-space representation with bounded invertible K, the first involution gives $\widehat{S} = -(K - K^{-1})^2/2 \le 0$. The second makes K unitary and gives $0 \le \widehat{S} = I - \operatorname{Re}(K^2) \le 2I$. For the third write $K = iA$ with bounded self-adjoint invertible A. Then $\widehat{S} = (A + A^{-1})^2/2 \ge 2I$, since $(A + A^{-1})^2 - 4I = (A - A^{-1})^2 \ge 0$. These are statements about those specified representations; they do not assume a positive Hilbert realization of the arithmetic quotient.

The algebraic fibre trace of multiplication by C is $4c(s)$ even at ramification, because the module is free of rank four and $C = c(s)$. It is not the quadratic Weil value of a scalar test. The precise test map explains the distinction. If h is a scalar transform in the retained test space, multiplication sends it to $c(s)h(s)$; the reflected sesquilinear spectral summand is consequently multiplied by $c(s)\overline{c(\bar{s})}$. At real s this is $|c(s)|^2$, not $c(s)$. Equivalently, for an S-symmetric sesquilinear form B on an invariant polynomial domain,

$$B(c(S)u, c(S)u) = B(u, c(S)^2 u),$$

because c has real coefficients and the adjoint can be moved one factor at a time. An inserted value $B(u, c(S)u)$ and the actual value $B(c(S)u, c(S)u)$ are thus related by different, explicit operators. The negative coefficient above supplies no negative scalar Weil value by this multiplication map. A spectral point of K outside Γ, on the other hand, would transfer exactly by Theorem 16.1 to an off-real arithmetic spectral point.

16.5 All four material and arithmetic heat sectors

Put $B_0 = \mathbb{C}[s, s^{-1}, (s-2)^{-1}]$ and

$$\mathcal{A} = B_0[r, p]/(r^2 + 2s,\ p^2 + 2s - 4).$$

This is the same cover localized away from its two branch values. There is a unique derivation D extending ∂_s on B_0:

$$Dr = -1/r = r/(2s), \qquad Dp = -1/p = p/(2(s-2)), \qquad Dq = -q/(rp). \tag{995}$$

Differentiating the two defining relations forces the first two identities, because r, p are invertible. Conversely these assignments annihilate both relations, so they define a derivation. The conic map then gives the last identity. Thus D is the original material coefficient derivative with $a' = 2, b = c = 0$, and $Ds = 1$.

Proposition 16.3. *For a section $f = f_0 + rf_1 + pf_2 + rpf_3$ of this module, the coefficient vector of Df is $(\partial_s + A)\mathbf{f}$, where*

$$A = \operatorname{diag}\left(0, \frac{1}{2s}, \frac{1}{2(s-2)}, \frac{1}{2s} + \frac{1}{2(s-2)}\right). \tag{996}$$

Its full square, including its zeroth-order terms, is

$$D^2 = \partial_s^2 + 2A\partial_s + \operatorname{diag}\left(0, -\frac{1}{4s^2}, -\frac{1}{4(s-2)^2}, -\frac{1}{s^2(s-2)^2}\right). \tag{997}$$

The lifted arithmetic heat generator is $-D^2/4$, retaining the original sign and factor. The field trace is $\operatorname{Tr} f = 4f_0$, with kernel $rB_0 \oplus pB_0 \oplus rpB_0$, and $\operatorname{Tr} Df = \partial_s \operatorname{Tr} f$.

Proof. Apply the product rule and (995) to each term of f. Squaring ∂_s+A gives $\partial_s^2+2A\partial_s+A'+A^2$. The middle entries of $A'+A^2$ are $-1/(4s^2)$ and $-1/(4(s-2)^2)$. The last one is

$$-\frac{1}{4s^2}-\frac{1}{4(s-2)^2}+\frac{1}{2s(s-2)}=-\frac{1}{s^2(s-2)^2},$$

proving every lower-order term. The four sign automorphisms $(r,p)\mapsto(\delta r,\varepsilon p)$ have sum $4f_0$; alternatively the multiplication matrices of r,p,rp have zero diagonal, giving the trace even before localization. Linear independence of the free basis identifies the kernel. The first row of A is zero, proving the trace-derivative identity without discarding the other three sectors. □

On a simply connected analytic neighbourhood avoiding $0,2$, choose nonvanishing branches $r(s),p(s)$. The four evaluations are

$$h_{\delta,\varepsilon}(s)=f_0(s)+\delta r(s)f_1(s)+\varepsilon p(s)f_2(s)+\delta\varepsilon r(s)p(s)f_3(s).$$

Their inverse recovers f_0 by averaging, f_1 by averaging $\delta h/r$, f_2 by averaging $\varepsilon h/p$, and f_3 by averaging $\delta\varepsilon h/(rp)$, each with factor $1/4$. These formulas prove an invertible local map and show directly that evaluation intertwines D with ∂_s and D^2 with ∂_s^2. Equivalently, with $E=\operatorname{diag}(1,r,p,rp)$, $\partial_s+A=E^{-1}\partial_sE$ on coefficient vectors. This does not erase the factors: the inverse ceases to exist at a branch value. Around $s=0$, r and rp change sign; around $s=2$, p and rp change sign. The residues of A are respectively $\operatorname{diag}(0,1/2,0,1/2)$ and $\operatorname{diag}(0,0,1/2,1/2)$, retaining exactly this monodromy.

For a finite material increment τ, local continuation gives

$$s_\tau=s+\tau,\quad r_\tau^2=r^2-2\tau,\quad p_\tau^2=p^2-2\tau,\quad q_\tau=(p_\tau+r_\tau)/2.$$

The initial branches fix $r_0=r,p_0=p$ and their continuation until a branch value is met. Since $p_\tau^2-r_\tau^2=4$, this preserves the conic and $q_\tau^{-1}=(p_\tau-r_\tau)/2$. Differentiation recovers (995). The original source triple and its velocity are therefore

$$(-1/(2r_\tau),3r_\tau,26r_\tau^2),\qquad(-1/(2r_\tau^3),-3/r_\tau,-52).$$

At $r_\tau=0$ its first coordinate escapes; at $p_\tau=0$ that source triple remains finite and only the additional conic coordinate is ramified. The finite boundary section $(0,0,2(s+\tau))$ has velocity $(0,0,2)$.

These derivations are on the displayed localized coefficient module and its analytic sections. On the actual arithmetic quotient, multiplication has already been constructed by (989), while differentiation uses the transported image and topology established in the preceding heat section. For scalar initial data the present formulas give exactly the pullback of $\Xi(s)=\xi(1/2+is)=8H_0(2s)$ and of its established heat family, through (987); no new initial function, discarded coefficient factor, or substituted spectral coordinate occurs.

All ring, resolvent, ramification, coefficient and connection identities have exact reproducible symbolic checks accompanying these full proofs. The spectral criterion is an exact reduction, not a constructed point outside Γ or a complexity separation theorem.

17 Material endpoint with the original parameter retained

The polynomial map and its provenance are those of Section 14 [97, 98, 99]. We calculate the original Euclidean operator, not a diffusion chosen on the target. The coordinate-gradient and divergence conventions agree with Lychev–Koifman, Sections 2.3.3–2.3.4, especially (2.26) [96]; all uses here are proved by the chain rule below. The preceding cancellation at $b=0$ is not a cancellation at every b.

17.1 The full scalar diffusion on every retained real branch

Keep $s = F_1/2$, $b = F_2$, $c = F_3$ and the original coordinates (x, y, w). Fix a real b and introduce only the abbreviations

$$\beta = b/4, \qquad s_* = b^2/32 = \beta^2/2, \qquad \eta = r - b/4.$$

Their inverses are $b = 4\beta$, $r = \eta + b/4$; the original parameter has not been set to zero. On $c = 0$, the inverse chart of (980) is

$$H_b(\eta) = \left(-\frac{1}{2\eta}, \beta + 3\eta, 26\eta^2 + 6\beta\eta\right), \qquad s(H_b(\eta)) = s_* - \eta^2/2, \quad \eta \neq 0. \tag{998}$$

It satisfies $F(H_b(\eta)) = (2s, b, 0)$. The separate finite point $(0, b, 2s - 4b^2)$ remains on that same fibre, including at $s = s_*$. On either real branch, the original material time satisfies

$$\frac{d\eta}{d\tau} = -\eta^{-1}, \qquad \eta(\tau) = \operatorname{sgn}(\eta_0)\sqrt{\eta_0^2 - 2\tau}, \qquad \tau < \eta_0^2/2.$$

Indeed $Us = 1$, $Ub = Uc = 0$ on this slice, and differentiation of $s = s_* - \eta^2/2$ proves the equation. Thus H_b is an actual escaping material orbit, not merely a curve on a fixed fibre.

Here are all three gradient entries and the full Laplacian, evaluated only after differentiation in the original three spatial variables:

$$\begin{aligned}
g_x(\eta) &= \frac{(\beta + 3\eta)(3\beta^3 + 17\beta^2\eta + 9\beta\eta^2 + 3\eta^3)}{4\eta},\\
g_y(\eta) &= \frac{3\beta^3 + 9\beta^2\eta + 17\beta\eta^2 + 3\eta^3}{8\eta^2},\\
g_w(\eta) &= -\frac{(\beta + \eta)^3}{16\eta^3},\\
L_b(\eta) &= \frac{9\beta^2}{4\eta^2} + \frac{9\beta}{2\eta} + \frac{13}{4} - 6\beta^4 - 66\beta^3\eta - 246\beta^2\eta^2 - 342\beta\eta^3 - 108\eta^4,\\
N_b(\eta) &= g_x(\eta)^2 + g_y(\eta)^2 + g_w(\eta)^2.
\end{aligned} \tag{999}$$

Here $g_j = (\partial_j s) \circ H_b$, $L_b = (\Delta_E s) \circ H_b$, and $N_b = |\nabla_E s|^2 \circ H_b$ for real variables. These three displayed squares specify every coefficient of N_b, including all regular terms. For holomorphic continuation the same algebraic sum of squares is used; it is not called a Hermitian norm there.

To verify the formulas, set $v = 1 + xy$ in $s = (v^3 w + y^2(v + 3v^2))/2$. Differentiate using $v_x = y$, $v_y = x$, $v_w = 0$ as in (981). Substitute $v(H_b) = -(\beta + \eta)/(2\eta)$, together with (998), into those three first derivatives and the displayed polynomial formula for $\Delta_E s$ in that section. Collecting powers gives precisely (999). The scalar chain rule, before restriction, now proves for every C^2 function f near s_*

$$(\Delta_E(f \circ s))(H_b(\eta)) = N_b(\eta) f''(s_* - \eta^2/2) + L_b(\eta) f'(s_* - \eta^2/2). \tag{1000}$$

This is a map from the scalar function and its two jets to a function on the retained chart; no fibre-independent scalar generator is inferred.

For $b \neq 0$, the exact leading terms are

$$\eta^3(g_x, g_y, g_w) \longrightarrow (0, 0, -\beta^3/16), \qquad \eta^6 N_b \longrightarrow \beta^6/256, \qquad \eta^2 L_b \longrightarrow 9\beta^2/4. \tag{1001}$$

They follow by multiplying the rational expressions by the stated powers and evaluating at zero. For instance, along either branch $\nu N_b \sim \nu\beta^6/(2048(T-\tau)^3)$, with the original viscosity $\nu > 0$ and $T = \eta_0^2/2$. For $b = 0$, instead,

$$(g_x, g_y, g_w) = (9\eta^3/4, 3\eta/8, -1/16), \qquad N_0 = 81\eta^6/16 + 9\eta^2/64 + 1/256, \quad L_0 = 13/4 - 108\eta^4.$$

Taking $\eta = -z/2$ recovers every coefficient of (982). This explicitly locates the earlier finite limit at the special value $b = 0$.

17.2 Exact analytic-jet classification of bounded diffusion

Theorem 17.1. *Let $b \in \mathbb{R} \setminus \{0\}$, and let f be complex analytic near $s_* = b^2/32$, with its actual convergent expansion $f(s_* + h) = \sum_{j\geq 0} a_j h^j$. Along either real branch of* (998)*:*

1. *The physical gradient $\nabla_E(f \circ s)$ is bounded exactly when $a_1 = a_2 = 0$. In that case it tends to zero.*

2. *The full Euclidean Laplacian in* (1000) *is bounded exactly when $a_1 = a_2 = a_3 = a_4 = 0$. Its limit is then $-5\beta^6 a_5/512$.*

No condition is imposed on a_0 in either statement.

Proof. The gradient is exactly $f'(s_* - \eta^2/2)(g_x, g_y, g_w)$. Its last component has successive terms $-\beta^3 a_1/(16\eta^3)$ and, once $a_1 = 0$, $\beta^3 a_2/(16\eta)$. Their coefficients are nonzero when the indicated a_j is nonzero, so boundedness forces both to vanish. Conversely $f' = O(\eta^4)$ if they vanish; the gradient entries are $O(\eta^{-3})$, proving convergence to zero.

For the second statement it suffices to retain the first six potentially negative Laurent powers. A convergent Taylor remainder after $a_5 h^5$ has second derivative $O(h^4) = O(\eta^8)$ and first derivative $O(h^5)$; multiplying by $N_b = O(\eta^{-6})$ and $L_b = O(\eta^{-2})$ leaves no pole and gives a vanishing remainder. The coefficients of η^{-6} and η^{-5} are

$$\frac{\beta^6 a_2}{128}, \qquad \frac{3\beta^5 a_2}{64}.$$

Boundedness therefore forces $a_2 = 0$. After that substitution, the coefficients of η^{-4} and η^{-3} are

$$-\frac{3\beta^6 a_3}{256}, \qquad -\frac{9\beta^5 a_3}{128},$$

so $a_3 = 0$. After both substitutions, the remaining pole coefficients are exactly

$$[\eta^{-2}]\Delta_E(f \circ s) = \frac{3\beta^2}{256}(192a_1 + \beta^4 a_4), \qquad [\eta^{-1}]\Delta_E(f \circ s) = \frac{9\beta}{128}(64a_1 + \beta^4 a_4). \tag{1002}$$

All these coefficients follow by expanding the three squares in (999) and $f' = \sum j a_j(-\eta^2/2)^{j-1}$, $f'' = \sum j(j-1)a_j(-\eta^2/2)^{j-2}$. Subtracting the two zero equations in parentheses gives $128a_1 = 0$; then $a_4 = 0$, since $\beta \neq 0$. A finite Laurent principal part cannot be bounded on a real punctured half-interval unless every negative-power coefficient vanishes: multiply by its highest pole power and take the limit. This proves necessity on either branch.

If a_1 through a_4 vanish, then $f'' = -(5/2)a_5\eta^6 + O(\eta^8)$ and $f' = O(\eta^8)$. Equation (1001) proves both sufficiency and the limit $-(5/2)a_5\beta^6/256$. This also checks all possible cancellations between the two terms of the actual Euclidean operator. □

In particular this theorem applies to the actual $Z_t(s) = 8H_t(2s)$ at each fixed real heat time, with $a_j = Z_t^{(j)}(s_*)/j!$. It does not replace Z_t by its Taylor polynomial: the proof explicitly controls the analytic remainder. At an existing zero of order $m \geq 2$, the first nonzero diffusion term is

$$\frac{\beta^6}{256} m(m-1)a_m(-1/2)^{m-2}\eta^{2m-10}.$$

For $m \geq 5$ it is bounded; for $m = 2, 3, 4$ it diverges. At a simple zero it also diverges, by the complete pole classification, regardless of possible cancellations involving a_4. These statements concern the Euclidean derivative of the pulled-back function. A zero is not created by this calculation, and its location has not been presumed.

17.3 All fixed covectors at the original $b = 0$ trajectory

The scalar cancellation and the full material tensor can now be compared without discarding transverse covectors. Put $q_0 = (1, -3/2, 13/2)$, $z = \sqrt{1-8\tau} > 0$, and $\gamma = (z^{-1}, -3z/2, 13z^2/2)$ as before. Direct differentiation of the original F gives

$$J_0 = \begin{pmatrix} -9/16 & -3/8 & -1/8 \\ 3/8 & 25/4 & 3/4 \\ -17/2 & -3 & -1 \end{pmatrix}, \quad J_\gamma = \begin{pmatrix} -9z^3/16 & -3z/8 & -1/8 \\ 3z^2/8 & 25/4 & 3/(4z) \\ -17/2 & -3/z^2 & -1/z^3 \end{pmatrix}.$$

The full material derivative and inverse are $A = J_\gamma^{-1} B J_0$, $C = J_0^{-1} B^{-1} J_\gamma$, where $B = I + 3(1 - z^2)E_{23}/4$. Differentiating $F \circ X_\tau = (F_1 + 2\tau, F_2 + 6\tau F_3, F_3)$ proves these identities for all initial directions, not just the orbit tangent. Both determinants of J are -2 and $\det B = 1$, so $C = A^{-1}$ and $\det C = 1$.

For an arbitrary fixed real initial covector $\ell = (\ell_1, \ell_2, \ell_3)^\mathsf{T}$, define

$$\lambda = \frac{2\ell_1 + 3\ell_2 - 18\ell_3}{8}, \qquad \mu = \ell_2 - 3\ell_3, \qquad \kappa = \ell_3 - \lambda.$$

These are coordinates on the entire covector space, with inverse

$$\ell_1 = (17\lambda - 3\mu + 9\kappa)/2, \quad \ell_2 = 3\lambda + \mu + 3\kappa, \quad \ell_3 = \lambda + \kappa.$$

Matrix multiplication in the displayed factorizations gives

$$C^\mathsf{T}\ell = \left(\frac{17\lambda}{2} - \frac{3\mu z^2}{2} + \frac{9\kappa z^3}{2}, \quad \frac{3\lambda}{z^2} + \mu + 3\kappa z, \quad \frac{\lambda}{z^3} + \kappa \right)^\mathsf{T}. \tag{1003}$$

One can check every entry without inverting a symbolic flow: substitute the inverse coordinates for ℓ into $J_0^{-\mathsf{T}}\ell$, apply $B^{-\mathsf{T}}$, and then apply J_γ^T. This proves the formula for arbitrary ℓ.

It follows that $C^\mathsf{T}\ell$ is bounded exactly on the plane $2\ell_1 + 3\ell_2 - 18\ell_3 = 0$. For $\lambda \neq 0$, $z^3 C^\mathsf{T}\ell \to \lambda e_3$ and $z^6 \ell^\mathsf{T} C C^\mathsf{T} \ell \to \lambda^2$. For $\lambda = 0$, its complete diffusion cost is

$$\nu \ell^\mathsf{T} C C^\mathsf{T} \ell = \nu \left\{ \frac{9}{4} z^4(-\mu + 3\kappa z)^2 + (\mu + 3\kappa z)^2 + \kappa^2 \right\} \longrightarrow \nu(\mu^2 + \kappa^2). \tag{1004}$$

The limit is positive for every nonzero covector on that plane, by the explicit inverse coordinate map. There is no additional intermediate pole order for fixed covectors: setting $\lambda = 0$ cancels both negative powers in (1003). Time-dependent covectors are not covered by this fixed-input assertion.

The actual initial arithmetic covector is $ds(q_0) = (-9/32, -3/16, -1/16)$, giving $(\lambda, \mu, \kappa) = (0, 0, -1/16)$. Formula (1003) recovers exactly $ds(\gamma) = (-9z^3/32, -3z/16, -1/16)$ and the finite cost $\nu/256$. The generic example $\ell = e_3$ has $\lambda = -9/4$ and cost asymptotic $81\nu z^{-6}/16$. Thus the two behaviors are obtained from one full inverse transpose.

17.4 Exact compact Weil-test channels at a shifted branch

Retain $\mathcal{E} = C_c^\infty(\mathbb{R};\mathbb{C})$, $\widehat{g}(s) = \int g(u)e^{-isu}\,du$ and $T = -i\partial_u$. Set $\sigma = s_*$ for this subsection. Define the actual endpoint test subspace

$$\mathcal{B}_\sigma = \{g \in \mathcal{E} : \widehat{g}^{(j)}(\sigma) = 0,\ 1 \le j \le 4\}.$$

By Theorem 17.1 this is exactly the compact test space whose Fourier transform has bounded pulled-back Euclidean Laplacian at this nonzero-b branch. Notice that its zeroth Fourier value is unrestricted. Equivalently its four moments $\int u^j e^{-i\sigma u} g(u)\,du$, $1 \le j \le 4$, vanish.

Here is a complete constructive parametrization, including that unrestricted value. Choose any real smooth bump χ supported in an interval $[a,d]$, $a<d$, with $\int \chi \ne 0$, and put $\chi_\sigma(u) = e^{i\sigma u}\chi(u)$. For example, on (a,d) one may take $\chi(u) = \exp(-1/((u-a)(d-u)))$, extended by zero; every derivative vanishes at each endpoint since an exponential dominates every inverse power, so this is a positive smooth compact bump. Then $\widehat{\chi}_\sigma(\sigma) = \int \chi \ne 0$. Let $P_4(s)$ be the degree-four Taylor polynomial at σ of $1/\widehat{\chi}_\sigma(s)$, which exists in a neighborhood of σ. Set $k_\sigma = P_4(T)\chi_\sigma$. Integration by parts gives

$$\widehat{k}_\sigma(s) = P_4(s)\widehat{\chi}_\sigma(s) = 1 + O((s-\sigma)^5).$$

Every operation preserves $[a,d]$. We have the direct decomposition

$$\mathcal{B}_\sigma = \mathbb{C}k_\sigma \ \oplus\ (T-\sigma)^5\mathcal{E}. \tag{1005}$$

To prove both directions, for $g \in \mathcal{B}_\sigma$ subtract $\widehat{g}(\sigma)k_\sigma$. The remainder g_0 has all five Fourier jets of orders zero through four equal to zero at σ. The explicit compact inverse is

$$f(u) = \frac{i^5 e^{i\sigma u}}{4!}\int_{-\infty}^{u}(u-v)^4 e^{-i\sigma v} g_0(v)\,dv, \qquad (T-\sigma)^5 f = g_0.$$

Differentiating five times after conjugation by $e^{i\sigma u}$ proves the equation. Above the support interval, expansion of $(u-v)^4$ gives a polynomial all of whose coefficients are the five vanishing moments; below it the integral is zero. Smoothness and all endpoint derivatives follow from the integral. This proves $f \in \mathcal{E}$ with no enlarged support. Conversely the Fourier transform of $(T-\sigma)^5 f$ has all five zero jets, proving membership. Evaluation at σ proves that the sum is direct; injectivity of $(T-\sigma)^5$ follows because its Fourier product can vanish identically only if the entire Fourier transform is zero. This also proves uniqueness of f and of the scalar coefficient.

The exact Weil form on this space is still the original one. If $g = ck_\sigma + (T-\sigma)^5 f$, it is

$$\begin{aligned} W(g,g) = {} & |c|^2 W(k_\sigma,k_\sigma) + \bar{c}\,W(k_\sigma,(T-\sigma)^5 f) \\ & + c\,W((T-\sigma)^5 f, k_\sigma) + W((T-\sigma)^5 f,(T-\sigma)^5 f). \end{aligned} \tag{1006}$$

This is sesquilinearity, including both cross terms. It can be evaluated using the complete pole, gamma and prime formula (984) at $t=0$ [26, 10]. On a spectral value s_ρ, the last term has the retained factor $(s_\rho-\sigma)^5\overline{(\overline{s_\rho}-\sigma)^5}$ in the ordering of Section 12; no assumption that s_ρ is real is used. The smooth bump generator and the polynomial channel give actual compact tests, not formal rational functions with unverified inverse domains.

These calculations establish singular diffusion of generic arithmetic observables at nonzero retained b, its exact exceptional jet space, and the full scalar Weil channels on that space. Neither diffusion singularity nor the loss of a uniform material bound is itself a negative value of (1006). No such value, or RH counterexample, is claimed here. The next calculation must use these actual channels and all cross terms, not replace their pairing by the positive Euclidean diffusion norm or a signed coefficient.

18 The actual Weil form on the endpoint channels

We now evaluate the arithmetic form, not the positive Euclidean diffusion cost. The parameter remains $\sigma = b^2/32$. The finite calculation below uses $b = 8$, hence $\sigma = 2$; this choice is not a claim that 2 is a zero ordinate. Both the exceptional bounded-diffusion space of Theorem 17.1 and all four complementary derivative channels are retained. The form and Fourier conventions are those of (837)–(839); compare Alpöge–Furman, Section 2.1 [10], and Weil's explicit formula [26].

18.1 Exact finite coordinates for every endpoint jet

Keep $b_4 = \mathbf{1}_{[-1/2,1/2]}^{*4}$, and set

$$d_j = (j-8)/4, \quad v_j(u) = e^{i\sigma u} b_4(u - d_j), \qquad 0 \le j \le 16.$$

These are globally C^2 compact tests, not smooth tests. Their common support lies in $[-4, 4]$. The passage to smooth compact tests is proved below with its exact moment and support maps. For a coefficient vector $c \in \mathbb{C}^{17}$ put $g = \sum_j c_j v_j$. Its Fourier transform is

$$\widehat{g}(s) = \operatorname{sinc}((s-\sigma)/2)^4 \sum_{j=0}^{16} c_j e^{-i(s-\sigma)d_j}. \tag{1007}$$

The centered spline moments of orders zero through four are $(1, 0, 1/3, 0, 3/10)$. They follow either by integrating its four truncated-power pieces, or by taking derivatives at zero of $(\sinh(z/2)/(z/2))^4$. Thus the full five-moment matrix is

$$M_{rj} = \int u^r b_4(u - d_j)\, du, \qquad (M_{0j}, \dots, M_{4j}) = (1, d_j, d_j^2 + 1/3, d_j^3 + d_j, d_j^4 + 2d_j^2 + 3/10). \tag{1008}$$

Precisely, $\widehat{g}^{(r)}(\sigma) = (-i)^r (Mc)_r$. The bounded-Laplacian constraints are rows 1 through 4, not row 0.

Here is a rational basis and its inverse moment coordinates. The following five columns are supported on the five central translates, with every unlisted coefficient zero:

d_j	k	q_1	q_2	q_3	q_4
$-1/2$	$62/15$	$17/3$	-22	$-16/3$	$32/3$
$-1/4$	$-96/5$	$-40/3$	96	$32/3$	$-128/3$
0	$467/15$	0	-148	0	64
$1/4$	$-96/5$	$40/3$	96	$-32/3$	$-128/3$
$1/2$	$62/15$	$-17/3$	-22	$16/3$	$32/3$

The same symbols denote the functions with those coefficients in the v_j basis. Direct multiplication by (1008) gives $Mk = e_0$, $Mq_r = e_r$ for $1 \le r \le 4$. There are twelve further columns

$$w_\ell = \sum_{r=0}^{5} (-1)^r \binom{5}{r} v_{\ell+r}, \qquad 0 \le \ell \le 11. \tag{1009}$$

Every row of M is a polynomial of degree at most four in the translate index. The fifth difference annihilates it, so $Mw_\ell = 0$. These twelve columns are independent: their first nonzero entries occur successively in rows $0, \dots, 11$, each with coefficient 1. The five rows of M are independent because they have distinct degrees and leading coefficient one; a polynomial of degree at most four which vanishes on the seventeen distinct d_j is zero. The same argument gives rank four for rows 1 through 4. It follows that, inside this entire seventeen-dimensional test family,

$$\begin{aligned} \ker M &= \operatorname{span}\{w_0, \ldots, w_{11}\}, \\ \ker M_{1:4} &= \mathbb{C}k \oplus \ker M, \\ \operatorname{span}\{v_0, \ldots, v_{16}\} &= \mathbb{C}k \oplus \ker M \oplus \operatorname{span}\{q_1, q_2, q_3, q_4\}. \end{aligned} \tag{1010}$$

The last sum is direct by applying M and then independence of the w_ℓ. Independence of the original translates follows directly from (1007): after division near $s = \sigma$ by the nonzero sinc factor, the remaining Laurent polynomial in $e^{-i(s-\sigma)/4}$ is identically zero only when all its coefficients vanish. Thus these are coordinates on functions, not just a redundant coefficient list.

For any g in this family, its k coefficient is $\widehat{g}(\sigma)$; its q_r coefficient is $i^r \widehat{g}^{(r)}(\sigma)$. Subtracting those five specified columns leaves the unique fifth-difference part. This gives the exact inverse map in (1010). The four q_r channels have unbounded original Euclidean Laplacian at this branch, by Theorem 17.1; they are nevertheless valid Weil tests and have not been removed from the calculation.

18.2 Complete arithmetic matrix and all cross terms

Let $B = b_4 * b_4$ be the piecewise polynomial in (840), and put

$$d = d_j - d_i, \quad H_d(x) = e^{i\sigma x} B(x - d), \quad E_d(x) = \tfrac{1}{2}\{e^{i\sigma x} B(x - d) + e^{-i\sigma x} B(x + d)\}.$$

Convolution proves $H_d = v_j * v_i^\star$ and E_d is its even part. In particular $H_d(0) = B(d)$. Write $\mathcal{M}(a) = (\sinh(a/2)/(a/2))^8$, filled in at zero, and $a_+ = 1/2 + i\sigma$, $a_- = -1/2 + i\sigma$. With $L_d = 4 + |d|$, the exact matrix entry $G_{ij} = W(v_i, v_j)$ is

$$\begin{aligned} G_{ij} = {} & e^{a_+ d} \mathcal{M}(a_+) + e^{a_- d} \mathcal{M}(a_-) - (\gamma_E + \log(4\pi)) B(d) \\ & - \int_0^{L_d} \frac{e^{x/2} E_d(x) - B(d)}{\sinh x}\, dx + B(d) \log\coth(L_d/2) \\ & - 2 \sum_{\substack{n = p^r < e^{L_d} \\ p \text{ prime},\ r \geq 1}} \frac{\log p}{\sqrt{n}} E_d(\log n). \end{aligned} \tag{1011}$$

Indeed the two pole terms are the bilateral exponential moments of H_d, which give the first two displayed terms after $x \mapsto x + d$. The change of variables underlying (841) applies to this complex even part as well. Above L_d it is zero, and $\int_{L_d}^\infty dx/\sinh x = \log\coth(L_d/2)$ retains the exact archimedean tail. Every prime power in the finite support is included. This proves (1011); it uses neither a truncated zero sum nor an assumption on the zeros' real parts. Conjugation gives $G_{ji} = \overline{G_{ij}}$.

For clarity about the implementation's phase, its existing translate is $\widetilde{v}_j = e^{i\sigma(u - d_j)} b_4(u - d_j)$. Our $v_j = e^{i\sigma d_j} \widetilde{v}_j$, so its arithmetic entry must be multiplied by $e^{i\sigma(d_j - d_i)}$. The code performs this exact diagonal congruence. The origin singularity is removed analytically using

$$\frac{P(x)e^{ax} - P(0)}{x} = \frac{P(x) - P(0)}{x} e^{ax} + P(0) a e^{ax/2} \operatorname{sinhc}(ax/2).$$

The first quotient is a polynomial; division afterward is by $\operatorname{sinhc} x$. No interval near the origin is omitted. The integration cells are exactly the positive knots of $B(x - d)$ and $B(x + d)$ together with $0, L_d$. All polynomial coefficients, shifts and knot comparisons are exact rational data; prime-power cell membership uses strict Arb inequalities.

Let A have columns $(k, w_0, \ldots, w_{11})$, Q columns $(q_1, \ldots, q_4)$, and $S = (A, Q)$ in the original translate coordinates. The complete matrices are

$$H = A^*GA, \quad K = S^*GS = \begin{pmatrix} H & R \\ R^* & J \end{pmatrix}.$$

In particular no k–w_ℓ, regular–singular or singular–singular cross entry is dropped. Write $H = \left(\begin{smallmatrix} \alpha & r \\ r^* & D \end{smallmatrix}\right)$. For a test with zeroth value 1 and all four derivative jets zero, its cost in these coordinates is exactly

$$W(k + \sum y_\ell w_\ell, k + \sum y_\ell w_\ell) = \alpha - rD^{-1}r^* + (y + D^{-1}r^*)^* D(y + D^{-1}r^*). \tag{1012}$$

Here D is proved positive in the certificate below, so this identity gives the actual minimum and minimizer, not a discarded cross term. Likewise $J - R^*H^{-1}R$ retains the optimal regular contribution to each prescribed four-component derivative-jet vector.

At $b = 8$, the complete interval calculation gives

$$0.00013342655218 < W(k, k) < 0.00013342655219,$$

whereas the exact unit-zeroth-value minimum in (1012) lies in

$$8.5095890260\,10^{-10} < \ \alpha - rD^{-1}r^* \ < \ 8.5095890261\,10^{-10}. \tag{1013}$$

Both bounds are outward weakenings of the saved Arb enclosures. The cross terms therefore matter substantially, but do not yield a negative value here. All thirteen pivots of H, all seventeen pivots of K, and all four pivots of $J - R^*H^{-1}R$ are strictly positive. To give the numerical scale without confusing a pivot with a smallest eigenvalue, the latter four pivots lie respectively in these outward intervals:

$$\begin{aligned} &(2.13817, 2.13818)10^{-10}, \qquad &&(4.45947, 4.45948)10^{-12}, \\ &(2.41563, 2.41564)10^{-14}, \qquad &&(1.06660, 1.06661)10^{-15}. \end{aligned}$$

These are basis-dependent elimination pivots, not uniform spectral gaps for an infinite test space. Positive pivots give an invertible unit lower-triangular L and $K = LDL^*$ with positive real diagonal D; hence every nonzero vector in this entire seventeen-dimensional family has positive cost. The finite identity does not extend that assertion to untested functions or other parameters.

The replayable computation is `verify_endpoint_weil.py`, with complete parameters, all matrix entries and the component integrals in `endpoint_weil_results.json`, in the current certificate directory. It uses 384-bit arithmetic, 2^{-240} integration tolerances, one worker and a 5,000,000,000-byte process-tree cap. As separate checks, the autocorrelations of k and $k + w_0 + iw_1$ are assembled first as single piecewise polynomials, then integrated directly with their own complete prime cutoffs and tails. They overlap the corresponding Gram values; their correlation endpoints are 5 and 13/2, with 47 and 143 prime powers respectively. The read-only `replay_endpoint_weil.py` independently derives the rational moments from the generating function, checks all seventeen inverse coordinates, and proves all seventeen leading principal determinants positive using FLINT matrix determinants, without calling the production LDL routine. It independently obtains (1013) from $1/(H^{-1})_{00}$. The trust boundary is exact rational algebra plus FLINT Arb integration and ball arithmetic, not a Lean verification.

18.3 Smooth compact tests preserving the exact jets

Ordinary positive mollification need not preserve this test space: when its zeroth moment is m_0, it adds $\varepsilon^2\mu_2 m_0$ to the second moment. Here is an explicit repair which retains all five moments, not a projection which drops the zeroth channel. Choose the even nonnegative bump

$$\phi(u) = Z^{-1}e^{-1/(1-u^2)} \quad (|u|<1), \qquad \phi(u)=0 \quad (|u|\ge 1), \qquad Z=\int_{-1}^{1} e^{-1/(1-u^2)}du.$$

Let $\mu_j = \int u^j\phi(u)du$ and define the signed smooth kernel

$$K_0 = \phi - \frac{\mu_2}{2}\phi'' + \left(\frac{\mu_2^2}{4} - \frac{\mu_4}{24}\right)\phi''''. \tag{1014}$$

Integrating by parts r times gives $\int u^n\phi^{(r)} = (-1)^r n!\mu_{n-r}/(n-r)!$ for $n\ge r$, and zero otherwise. Hence $m_0(K_0)=1$, odd moments vanish by parity, $m_2(K_0)=\mu_2-\mu_2=0$, and $m_4(K_0) = \mu_4 - 6\mu_2^2 + 6\mu_2^2 - \mu_4 = 0$.

For every compact C^2 test $g = e^{i\sigma u}h$ put

$$K_{0,\varepsilon}(u) = \varepsilon^{-1}K_0(u/\varepsilon), \qquad g_\varepsilon(u) = e^{i\sigma u}(K_{0,\varepsilon} * h)(u).$$

Then $g_\varepsilon \in C_c^\infty$, and Fubini and the binomial theorem give, for $0\le n\le 4$,

$$\int u^n e^{-i\sigma u} g_\varepsilon(u)du = \sum_{r=0}^{n}\binom{n}{r}\varepsilon^r m_r(K_0)m_{n-r}(h) = m_n(h).$$

This preserves every channel of (1010), not just the constrained ones. Equivalently $\widehat{g}_\varepsilon(s) = \widehat{K}_0(\varepsilon(s-\sigma))\widehat{g}(s)$. If $\operatorname{supp} h\subset[a,d]$, its new support is in $[a-\varepsilon, d+\varepsilon]$. For a fixed $\varepsilon_0>0$ and $0<\varepsilon\le\varepsilon_0$, all supports lie in the one interval $I=[a-\varepsilon_0, d+\varepsilon_0]$. For $p_2(f) = \max_{0\le j\le 2}\|f^{(j)}\|_\infty$, uniform continuity of h, h', h'' and

$$\|(K_{0,\varepsilon}*h)^{(j)} - h^{(j)}\|_\infty \le \|K_0\|_1 \sup_{|t|\le\varepsilon}\|h^{(j)}(\cdot - t) - h^{(j)}\|_\infty$$

prove $p_2(g_\varepsilon - g)\to 0$. This is convergence in the fixed support C^2 space, not smooth-topology convergence to a nonsmooth spline.

The complete Weil form is continuous in precisely this topology. For f, g supported in an interval of length ℓ, the two poles in (839) are bounded together by $2\ell^2 e^{\ell/2}p_2(f)p_2(g)$: express each as the double integral with weight $e^{\pm(u-v)/2}$ and use $|u-v|\le\ell$. Twice integrating each Fourier transform by parts gives

$$|\widehat{g * f^\star}(t)| \le \ell^2 p_2(f)p_2(g)\min(1,|t|^{-4}).$$

An entirely explicit gamma-integral constant is

$$C_\Gamma = \frac{\frac{8}{3}(\gamma_E + 8 + \log\pi) + 4}{2\pi}.$$

Indeed, for $a=1/4$, the digamma series [53, (5.7.6)] yields

$$\Re\psi(a+it/2) - \psi(a) = \sum_{n\ge 0}\frac{t^2/4}{(n+a)((n+a)^2+t^2/4)} \le 4 + \tfrac{1}{2}\log(1+4t^2) \le 4 + 2|t|.$$

For the first inequality bound the $n=0$ term by 4 and the decreasing positive remaining summands by their integral from a to infinity. The series or recurrence and monotonicity give $-\gamma_E - 4 \le \psi(1/4) \le -\gamma_E$. Thus the gamma weight's absolute value is at most $(\gamma_E + 8 + \log\pi + 2|t|)/(2\pi)$. Integrating against $\min(1, |t|^{-4})$ uses the exact integrals $8/3$ and 2 for its constant and $|t|$ moments, proving the bound. Finally $|g * f^\star(x)| \le \ell p_2(f) p_2(g)$ and the correlation support is $[-\ell, \ell]$. All prime powers, including newly admitted ones after smoothing, are bounded using $P_\ell = \sum_{2\le n\le e^\ell} \Lambda(n)/\sqrt{n}$. We have proved

$$|W(f,g)| \le C_\ell p_2(f) p_2(g), \qquad C_\ell = \ell^2(2e^{\ell/2} + C_\Gamma) + 2\ell P_\ell. \tag{1015}$$

There is no discarded prime or archimedean tail in this estimate.

For an explicit spline error, let $C = \sum |c_j|$ and $A_r = \int |u|^r |K_0(u)| du$. The four cubic pieces show that b_4'' is globally Lipschitz with constant 3. Taylor's theorem through orders two, one and zero, respectively, and the vanished moments give

$$\|h_\varepsilon - h\|_\infty \le CA_3\varepsilon^3/2, \quad \|h_\varepsilon' - h'\|_\infty \le 3CA_2\varepsilon^2/2, \quad \|h_\varepsilon'' - h''\|_\infty \le 3CA_1\varepsilon.$$

Since $A_3, A_2 \le A_1$, for $\varepsilon \le 1$ the product rule proves

$$p_2(g_\varepsilon - g) \le B_\sigma\varepsilon, \quad B_\sigma = 3CA_1(1 + |\sigma| + \sigma^2/6), \quad p_2(g) \le G_\sigma = C(2 + 4|\sigma|/3 + 2\sigma^2/3).$$

The last bound follows from the exact spline suprema $2/3, 2/3, 2$. Applying (1015) to the two cross terms and the quadratic error gives the complete certificate-transport estimate

$$|W(g_\varepsilon, g_\varepsilon) - W(g,g)| \le C_\ell\{2G_\sigma B_\sigma\varepsilon + B_\sigma^2\varepsilon^2\}, \qquad \ell = d - a + 2\varepsilon_0. \tag{1016}$$

Thus an arithmetic strict-sign margin can be transferred with an explicit smoothing error, preserving all jets and the zeroth value. This result does not assign a negative margin to the positive calculation above. The fixed-support continuity also proves entrywise convergence of the whole smoothed seventeen-channel matrix, including every cross term. In particular its strict finite positivity persists for sufficiently small smoothing; no positivity of the full infinite-dimensional endpoint space is asserted.

The finite search has therefore tested both the bounded-diffusion and singular-derivative channels at their complete arithmetic scope. It supplies no negative Weil witness and no RH counterexample. The retained parameter, the finite support and the finite-dimensional span in this statement are essential parts of what was calculated.

19 The separated-link coefficient on actual compact Weil tests

The full supporting calculation in Section 20 uses the original finite-box Hamiltonian convention of Kogut–Susskind [108]. Its fourth coefficient is derived there, not attributed to that historical paper. Here we carry that coefficient into the *classical* Weil form with its explicit inverse, and then evaluate its actual test functions. The arithmetic form is still (839).

19.1 The retained physical parameters and arithmetic observable

For integer $L \ge 2$ retain the $N = 3(2L)(2L+1)^2$ physical links and $M = 3(2L)^2(2L+1)$ elementary faces of the original open box. Write

$$H = \kappa(H_0 - \xi S) + 2b_{\text{mag}}MI, \qquad H_0 = \sum_e E_e, \quad S = \sum_p W_p,$$
$$\kappa = 2g^2/a_{\text{lat}}, \qquad b_{\text{mag}} = 1/(2g^2 a_{\text{lat}}), \quad \xi = 1/(4g^4).$$

The subscripts retain the lattice spacing and magnetic coupling, whose source names are a, b; they do not identify them with the polynomial target coordinates. More precisely, on the product of the positive parameter plane with the original polynomial coordinate space, the map is $(a_{\mathrm{lat}}, g, q) \mapsto (a_{\mathrm{lat}}, g, F(q))$. It has derivative $\mathrm{diag}(I_2, DF)$ and arithmetic projection $F_1(q)/2$. Both physical parameters are unchanged. The positive parameter chart has the explicit inverse

$$(a_{\mathrm{lat}}, g) \mapsto (\kappa, \xi), \qquad (\kappa, \xi) \mapsto (1/(\kappa\sqrt{\xi}), (4\xi)^{-1/4}), \qquad b_{\mathrm{mag}} = \kappa\xi.$$

Substitution gives both identities. The original generators $T_a = -i\sigma_a/2$ satisfy $-\operatorname{tr}(T_aT_b)/2 = \delta_{ab}/4$. Thus the original metric's orthonormal frame is $2T_a$ and $-\Delta^c = 4E_e$; its electric coefficient is $\kappa/4$.

Let ψ_ξ be the actual positive unit vacuum, with energy $2b_{\mathrm{mag}}M + \kappa e(\xi)$, and set $A_\xi = H_0 - \xi S - e(\xi)I$ and $v_e = (E_e - \langle\psi_\xi, E_e\psi_\xi\rangle)\psi_\xi$. For distinct links sharing no face, the supporting proof gives

$$\lim_{\xi\to 0} \xi^{-4}\langle v_e, h(A_\xi)v_f\rangle = a_{ef}\mathcal{D}(h), \qquad \mathcal{D} = c_0\delta_3 + \delta_3'/336 + c_1\delta_{9/2} + c_2\delta_{13/2}, \tag{1017}$$

for bounded continuous h differentiable at 3, where

$$c_0 = -59/275184, \qquad c_1 = 1/1296, \qquad c_2 = 3/132496.$$

The integer a_{ef} is the exact ordered adjacent-face count there. In particular $\delta_3'(h) = -h'(3)$, $\mathcal{D}(1) = 127/219024$, and $\mathcal{D}(\lambda) = 0$. The zero first moment retains the band derivative and both higher spin channels; it does not remove the positive zeroth moment. For this general class of h, (1017) gives a little-$o(\xi^4)$ remainder. An even $O(\xi^6)$ remainder requires the further analytic dependence proved for the specified source observables, not just bounded continuity of h.

Here is the direct connection to the unchanged arithmetic function. Put $k(v) = \Phi(|v|/2)/2$, with Φ from (931). Changing variables in the cosine integral gives

$$Z_t(s) = 4\int_{\mathbb{R}} k(v)e^{tv^2/4+isv}\,dv.$$

The superexponential estimate proved below (931) absorbs every fixed quadratic, linear and polynomial weight. Hence for every complex t, s the function $h(q) = Z_t(s+q)$ and all its real q derivatives are bounded: the factor e^{iqv} has modulus one. For real coupling the spectral theorem and Fubini, applied to each pair of Hilbert-space vectors, give the actual bounded operator

$$Z_t(s + A_\xi) = 4\int_{\mathbb{R}} k(v)e^{tv^2/4+isv}e^{ivA_\xi}\,dv.$$

The integral is strong on each vector and has operator norm at most $4\int |k(v)|e^{\mathrm{Re}(t)v^2/4+|\operatorname{Im} s||v|}dv$. This dominating integral also proves norm-entire dependence on t, s by integrating their absolutely convergent Taylor series. Applying (1017) therefore yields the actual coefficient $a_{ef}\mathcal{T}_{\mathcal{D}}Z_t$, where

$$(\mathcal{T}_{\mathcal{D}}f)(s) = c_0f(s+3) - f'(s+3)/336 + c_1f(s+9/2) + c_2f(s+13/2). \tag{1018}$$

In particular its initial value contains the actual Ξ and its derivative at these translated arguments. No freely chosen entire function has replaced the Riemann datum.

19.2 Exact compact-support map, inverse and all jets

Use the original minus-sign Fourier transform $\mathscr{F}g(s) = \int g(u)e^{-isu}du$. Define

$$
\begin{aligned}
m(v) &= e^{3iv}(c_0 - iv/336) + c_1 e^{9iv/2} + c_2 e^{13iv/2}, \\
w(u) &= m(-u) = e^{-3iu}(c_0 + iu/336) + c_1 e^{-9iu/2} + c_2 e^{-13iu/2}.
\end{aligned} \tag{1019}
$$

Translation and differentiation under the integral prove the exact typed correspondence

$$\mathcal{T}_{\mathcal{D}}\mathscr{F} = \mathscr{F}M_w, \qquad M_w g(u) = w(u)g(u). \tag{1020}$$

The sign reversal in $w = m(-u)$ is essential.

For all real v there is a uniform rational lower bound $|m(v)| \geq \delta = 575/1752192$. Here is its full elementary proof. For $|v| \leq 1/2$, factoring e^{3iv} and using $\cos x \geq 1 - x^2/2$ gives

$$\Re(e^{-3iv}m(v)) \geq c_0 + (23/32)c_1 - (17/32)c_2 = 575/1752192.$$

The cosine inequality follows by twice integrating $1 - \cos x \geq 0$. For $|v| \geq 1/2$, the reverse triangle inequality gives

$$|m(v)| \geq |v|/336 - (|c_0| + c_1 + c_2) \geq 10291/21464352 > 575/1752192.$$

These ranges cover the whole line, proving the bound without removing a Fourier point or choosing a branch.

For every compact interval I and $r = 2$ or ∞, M_w is an automorphism of the functions in $C_c^r(\mathbb{R})$ supported in I, with inverse $M_{1/w}$ and *exactly* unchanged support. Indeed multiplication by a nowhere-zero smooth function preserves the nonzero set. Leibniz's rule bounds each derivative seminorm of wg by finitely many of those of g. The inverse has the same property: differentiate $w(1/w) = 1$ recursively and use $|w| \geq \delta$ on I. Both pointwise compositions are the identity. Thus the map also gives a continuous automorphism of C_c^∞ with its usual fixed-support inductive-limit topology, and of its Fourier image with the transported topology.

The full inverse jet coordinates, rather than an identification of pointwise zero sets, are particularly useful. For an output $G = \mathscr{F}g$ define

$$\mathcal{L}_{z,n}(G)_r = \int_I (-iu)^r \frac{g(u)}{w(u)} e^{-izu}du, \quad 0 \leq r < n. \tag{1021}$$

Then $\mathcal{L}_{z,n}\mathcal{T}_{\mathcal{D}}\mathscr{F}f = ((\mathscr{F}f)^{(r)}(z))_{r<n}$. This follows by substituting $g = wf$ and differentiating an integral over a fixed compact set. Its absolute value is bounded by $\delta^{-1}\int_I |u|^r|g(u)|e^{|\operatorname{Im} z||u|}du$, which proves continuity. Conversely the forward coordinates are, exactly,

$$(\mathcal{T}_{\mathcal{D}}\mathscr{F}f)^{(r)}(z) = c_0\widehat{f}^{(r)}(z+3) - \widehat{f}^{(r+1)}(z+3)/336 + c_1\widehat{f}^{(r)}(z+9/2) + c_2\widehat{f}^{(r)}(z+13/2).$$

Thus a zero and all its multiplicity data are transported by (1021); no equality of untransported evaluations is imposed.

For completeness, each finite ordinary jet map onto $\mathbb{C}^n$ is surjective on the compact smooth Fourier image. Choose a nonnegative nonzero smooth bump χ inside an interval with nonempty interior. The n polynomials times $e^{izu}\chi(u)$ give a moment matrix $\left(\int u^{j+k}\chi(u)du\right)_{j,k<n}$, times invertible diagonal powers of $-i$. The moment matrix is positive definite: its quadratic form is $\int |p(u)|^2\chi(u)du$, which vanishes only when the polynomial vanishes on an interval and hence identically. Matrix inversion gives every prescribed jet supported in $\operatorname{supp}\chi$. To retain exactly this support

even for the zero jet, use the same moment matrix to construct the monic degree-n polynomial p_n orthogonal to degrees below n for weight χ. Adding p_n to the degree-less-than-n interpolant changes none of its prescribed moments. The resulting polynomial is nonzero and has only finitely many real roots, so its product with $e^{izu}\chi$ has exactly $\operatorname{supp}\chi$. Composition with M_w proves surjectivity of (1021). If $\mathcal{J}$ is the ordinary jet map, then $\mathcal{T}_{\mathcal{D}}\ker\mathcal{J} = \ker\mathcal{L}$; the induced quotient isomorphism and its inverse are the displayed maps on representatives. It is the identity in their $\mathbb{C}^n$ jet coordinates. The entire recovered function also retains every analytic-unit coefficient: at a zero of order n they are $\mathcal{L}_{z,n+j+1}(G)_{n+j}/(n+j)!$, for $j \geq 0$.

The endpoint constraints in Section 18 therefore transport to coordinates 1 through 4 of $\mathcal{L}_{\sigma,5}$, leaving coordinate 0 unrestricted. This describes their actual image under M_w, not another four constraints chosen on the output. The heat multiplier $e^{tu^2/4}$ commutes with w and $1/w$ on every compact-support space. Conjugating by $\mathscr{F}$ gives commutation with the full complex-time heat group of (954), in both directions.

19.3 The full arithmetic pullback and its evaluated channels

For arbitrary compact tests f, g the transformed correlation is

$$H^w_{f,g}(x) = \int_{\mathbb{R}} w(v)g(v)\,\overline{w(v-x)f(v-x)}\,dv. \tag{1022}$$

Substitute this expression, its Fourier transform and its values at every $\pm\log n$ into (839). This is the complete pulled-back form $W_w(f,g) = W(M_w f, M_w g)$. It retains both poles, the whole gamma integral and all prime powers. The map on ordered test pairs is the bijection $(f,g) \mapsto (wf, wg)$ with inverse $(f/w, g/w)$; hence it gives the exact correspondence between strict negative witnesses of W_w and those of W. It is not a claim that $W_w(f,g) = W(f,g)$. In particular the distributional coefficient $\mathcal{D}$ itself is not substituted for this arithmetic pairing.

At the previously retained endpoint $b_{\mathrm{pol}} = 8$, $\sigma = 2$, start with $f_0 = e^{2iu}b_4(u)$. The source image is exactly

$$wf_0 = c_0 f_1 + f_2/336 + c_1 f_3 + c_2 f_4, \quad (f_1, f_2, f_3, f_4) = (e^{-iu}b_4,\ iue^{-iu}b_4,\ e^{-5iu/2}b_4,\ e^{-9iu/2}b_4).$$

Every test is supported in $[-2, 2]$. We evaluate the entire five-by-five Gram matrix, not just the displayed linear combination. The functions are independent: on an open interval where $b_4 > 0$, a vanishing combination is an exponential polynomial with distinct frequencies $2, -1, -5/2, -9/2$ and a degree-one coefficient at -1. Successive application of the differential factors $\partial_u - i\sigma_j$ annihilating the other terms isolates each highest polynomial coefficient; descending its degree proves all coefficients zero.

Here the gamma integral was independently evaluated in frequency coordinates. Set $\varphi(z) = \operatorname{sinc}(z/2)^4$. The five Fourier functions are

$$(\Psi_0, \ldots, \Psi_4)(z) = (\varphi(z-2), \varphi(z+1), -\varphi'(z+1), \varphi(z+5/2), \varphi(z+9/2)).$$

They are real on the real axis. Thus the two pole terms are $2\Re(\Psi_i(i/2)\Psi_j(i/2))$, the gamma integrand is $\Psi_i(t)\Psi_j(t)$ times the exact weight in (839), and the prime terms use the real parts of $f_j * f_i(\log n)$. This last convolution identity follows because each $f_i^\star = f_i$. Its four cubic spline pieces and their shifted knots are integrated without approximation of the phase. All prime powers $n < e^4$ are included.

Here is the explicit bound for the discarded *integration range*, which is added back as an interval error. Take $T = 96$, $A = 9/2$, $\ell = T - A$. For $|t| \geq T$ one has $|\Psi_i(t)| \leq C_i/(|t| - A)^4$, where $C_i = 16$

except $C_2 = 32 + 64/\ell$. Indeed $\varphi(q) = 16\sin^4(q/2)/q^4$ and $|\varphi'(q)| \le 32/|q|^4 + 64/|q|^5$ on the real line. The digamma bound already proved in (1015), keeping its logarithmic bound, gives

$$|\text{gamma weight}(t)| \le \{\gamma_E + 8 + \log\pi + \log(1+2|t|)\}/(2\pi).$$

Concavity bounds the last logarithm, for $t \ge T$, by $\log(1+2T)+(t-T)/(T+1/2)$. Direct integration on the two tails therefore bounds the absolute omitted matrix entry by

$$\frac{C_iC_j}{\pi}\left\{\frac{\gamma_E + 8 + \log\pi + \log(1+2T)}{7\ell^7} + \frac{1}{42(T+1/2)\ell^6}\right\}. \tag{1023}$$

This is an error enclosure, not deletion of a tail or truncation of a zero sum.

The saved 192-bit Arb calculation, with 2^{-100} quadrature tolerances, proves all five leading principal determinants positive. All cross entries are retained. Its source vector and optimal subtraction of the seed obey the deliberately outward bounds

$$\begin{aligned}&6.63\,10^{-14} < W(wf_0, wf_0) < 6.77\,10^{-14},\\ &4.70\,10^{-14} < W(wf_0, wf_0) - |W(f_0, wf_0)|^2/W(f_0, f_0) < 4.90\,10^{-14}.\end{aligned}$$

The second expression is the exact minimum over multiples of f_0, by completing the square, with minimizer $wf_0 - f_0W(f_0, wf_0)/W(f_0, f_0)$. The certificate also verifies overlap of its $(0,0)$ entry with the earlier independently evaluated real-space gamma formula. The full entries, quadrature pieces, prime correlations and error bounds are retained in `xi4_weil_results.json`; `verify_xi4_weil.py` reproduces them under a one-worker 5,000,000,000-byte process-tree ceiling. No zero ordinates enter. The read-only `replay_xi4_weil.py` independently computes all 360 prime correlations by exponential-polynomial antiderivatives instead of quadrature. On each exact spline cell, for a polynomial P of degree d and $a \ne 0$, it uses

$$\int P(v)e^{av}dv = e^{av}\sum_{r=0}^{d}(-1)^r P^{(r)}(v)/a^{r+1}.$$

Differentiation telescopes to the integrand; at $a = 0$ it integrates each polynomial monomial directly. All 15 independently assembled matrix entries overlap the original enclosures. Independent interval LDL elimination proves all five pivots positive both for the saved matrix and for this reconstructed matrix. The replay also recomputes the pole terms, the sum of the saved 192 gamma integration cells, the explicit tail bound, and both displayed quadratic values. The frequency-side gamma integrations remain trusted Arb integrals: this replay does not mislabel their saved values as a second integration. Serialization enlarges the replayed Schur enclosure to $[4.8\,10^{-14} \ \pm 4.02\,10^{-16}]$; the displayed bounds contain it as well as the original enclosure.

These tests are C_c^2 splines. Write S_ε for the moment-preserving smoothing at $\sigma = 2$ in Section 18. For each computed output test h_i, the exact smooth approximation retaining its inverse jet coordinates is

$$h_{i,\varepsilon} = M_w S_\varepsilon M_{1/w} h_i.$$

The reciprocal map first gives a compact C^2 input. S_ε preserves its moments of orders zero through four at σ, and multiplication back by w gives the displayed transported coordinates unchanged. On one common enlarged compact interval both multiplier maps are bounded in C^2 by the product-rule estimates above. Thus $h_{i,\varepsilon} \to h_i$ in C^2. Applying the complete Weil continuity bound to every pair proves convergence of the entire Gram matrix, so each strictly positive leading determinant stays positive for all sufficiently small ε. Directly smoothing h_i and then applying M_w would instead converge to M_wh_i; that is not the family calculated here. The positive five-dimensional result is not positivity of the infinite compact-test space. It supplies no RH counterexample, but evaluates the actual source multiplier and its full arithmetic cross terms.

20 Complete source proof of the separated-link fourth coefficient

In the spectral-band construction below, the contour projection is that of $K_\xi = H_0 - \xi S$. The excitation operator is $A_\xi = K_\xi - e(\xi)I$ and has the same eigenspaces. This identifies the contour operator and the K_ξ used in the semigroup argument; neither energy nor scalar shift is changed.

8 September 2026. Two links which share no elementary face nevertheless develop a nonzero connected electric covariance through two adjacent faces. This note calculates that coefficient in the actual full-box vacuum. The calculation keeps both intermediate spin channels and the disconnected expectation subtraction. It does not assume a volume-uniform remainder.

20.1 Original Hilbert space and the coefficient to be calculated

Fix $L \geq 2$, vertices $\{-L, \dots, L\}^3$, all positive contained links e, and all contained elementary faces p. Write their complete counts as $N = 3(2L)(2L+1)^2$ and $M = 3(2L)^2(2L+1)$. Link variables are $U_e \in SU(2)$, with the inverse on reverse traversal. The face trace is

$$W_{(n;i,j)} = \operatorname{tr}\{U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}\}, \qquad i<j.$$

On product Haar probability space use $T_a = -i\sigma_a/2$, left derivatives $X_{e,a}$, $E_e = -\sum_a X_{e,a}^2$, and

$$H = \kappa \sum_e E_e + b\sum_p (2 - W_p), \quad \kappa = 2g^2/a, \quad b = 1/(2g^2 a), \quad \xi = b/\kappa = 1/(4g^4). \tag{S1}$$

The fundamental Casimir is $3/4$. In the original metric $-\operatorname{tr}(AB)/2$, the Laplacian is four times the sum-of-squares Laplacian in these generators, so the electric coefficient in that metric is $\kappa/4$. Neither this factor nor the Wilson scalar $2bM$ is changed.

The compact connected configuration manifold and bounded smooth potential give a self-adjoint compact-resolvent operator of domain H^2, form domain H^1, and unique smooth positive unit ground vector ψ_ξ. For precision, the modulus inequality supplies a nonnegative form minimizer, elliptic regularity and the maximum principle make it positive, and the identity $q_{H-\mathcal{E}}(\psi f) = \kappa\sum_{e,a}\int \psi^2|X_{e,a}f|^2$ implies uniqueness by division by ψ. Gauge transformations preserve this vector and every link Casimir. All following vectors are thus physical.

Set $r_e = \#\{p : e \in \partial p\}$, $\gamma_e = \langle\psi, E_e\psi\rangle$, $v_e = (E_e - \gamma_e)\psi$, and

$$C_{ef} = \langle v_e, v_f\rangle = \langle\psi, E_eE_f\psi\rangle - \gamma_e\gamma_f. \tag{S2}$$

The operators commute for different links. Smoothness makes (S2) legitimate as an operator identity on the vacuum, not only a formal product of forms.

For distinct e,f that share no face, define the integer

$$a_{ef} = \#\{(p,q) : e \in \partial p,\ f \in \partial q, |\partial p \cap \partial q| = 1\}. \tag{S3}$$

The pair in (S3) is ordered by which face contains e and which contains f; there is no double counting because no face contains both links. All faces are those of the original open box. The result is

$$\boxed{C_{ef} = \frac{127}{219024}\, a_{ef}\xi^4 + O_L(\xi^6) \quad (e,f \text{ share no face}).} \tag{S4}$$

In particular the coefficient is positive whenever a two-face channel exists, even though the energy matrix between these links is exactly zero. The latter assertion is proved in Section 20.4.

20.2 The full second vacuum coefficient, including both shared-link channels

Put $H_0 = \sum E_e$ and $S = \sum W_p$, so $H = \kappa(H_0 - \xi S) + 2bM$. On $|z| = 3/8$ in the full Hilbert space the resolvent of H_0 has norm at most $8/3$, since its first nonzero eigenvalue is $3/4$. Also $\|S\| = 2M$. Therefore the bounded perturbation Neumann series converges for $|\xi| < 3/(16M)$, gives the analytic rank-one bottom projection, and gives an analytic positive unit eigenvector for real ξ near zero. Elliptic regularity applied repeatedly to the eigen-equation upgrades this to analyticity in each fixed Sobolev space. Its radius and estimates may depend on L. This suffices for the finite derivative calculations below.

Product Haar integration gives $\int W_p = 0$, $\langle W_p, W_q\rangle = \delta_{pq}$, and $H_0 W_p = 3W_p$. Indeed a link in only one of two distinct face boundaries has odd central parity; for the same face integrate its Haar holonomy and use the fundamental character squared norm one. Comparing coefficients in the actual equation $(H_0 - \xi S)\psi = e(\xi)\psi$, including its unit-vector condition, gives

$$\psi = 1 + \xi A_1 + \xi^2 A_2 + O_{H^k,L}(\xi^3), \quad A_1 = S/3, \quad e(\xi) = -M\xi^2/3 + O_L(\xi^3),$$
$$H_0 A_2 = (S^2 - M)/3, \qquad \int A_2 = -M/18. \tag{S5}$$

The inverse in (S5) can be resolved entirely. For adjacent distinct faces p,q sharing g put

$$Z_{pq,0} = \int W_p W_q \, dU_g, \qquad Z_{pq,1} = W_p W_q - Z_{pq,0}.$$

After reversing one real trace if needed, their words are $\mathrm{tr}(U_g B)$, $\mathrm{tr}(U_g^{-1} D)$, with six distinct outer links. The fundamental Haar identity gives $Z_{pq,0} = \mathrm{tr}(BD)/2$. The shared-edge spin tensor $1/2 \otimes 1/2$ consists of spins zero and one, while all outer links have spin 1/2. Thus

$$H_0 Z_{pq,0} = \tfrac{9}{2} Z_{pq,0}, \quad H_0 Z_{pq,1} = \tfrac{13}{2} Z_{pq,1}, \quad \|Z_{pq,0}\|^2 = \tfrac{1}{4}, \quad \|Z_{pq,1}\|^2 = \tfrac{3}{4}. \tag{S6}$$

The norms use orthogonal Haar projection on g and total product norm one. A repeated face has $W_p^2 - 1 = \chi_1(U_p)$ with free energy 8 and norm one. An edge-disjoint pair has energy 6 and norm one, even when the faces meet at a vertex. Consequently the full coefficient is

$$\boxed{\begin{aligned} A_2 = &-\frac{M}{18} + \sum_p \frac{W_p^2 - 1}{24} \\ &+ \sum_{\{p,q\}\ \text{edge-disjoint}} \frac{W_p W_q}{9} \\ &+ \sum_{\{p,q\}\ \text{adjacent}} \left(\frac{4}{27} Z_{pq,0} + \frac{4}{39} Z_{pq,1} \right). \end{aligned}} \tag{S7}$$

All distinct-pair sums in (S7) are unordered; the factor two in S^2 has already been included. No finite-face subsystem is substituted for H.

Different unordered pairs in (S7) have different odd-link parity supports. If their symmetric boundaries agreed, their symmetric face difference would be a nonempty closed mod-two collection of at most four squares. A face in that collection needs another face on each of its four boundary edges; two distinct elementary squares share at most one edge, so at least four other squares would be needed, a contradiction. Haar integration on a link where parity differs makes the cross term zero. Repeated-face terms are even everywhere, and different repeated-face terms have different joint link spins. Thus all the displayed components, with their two spin channels, are mutually orthogonal. This remains true after application of any link Casimir, which preserves those parity and spin subspaces.

20.3 Exact connected fourth coefficient

Suppose e,f share no face. Then $E_eE_f1 = E_eE_fA_1 = 0$: each term in A_1 is a single face and depends on at most one of these links. Self-adjointness moves E_eE_f to either of these vectors, so every term of degree less than four in $\langle\psi, E_eE_f\psi\rangle$ vanishes, and its degree-four term is exactly

$$\langle A_2, E_eE_fA_2\rangle. \tag{S8}$$

In particular no third or fourth unknown coefficient of the vacuum is being omitted from (S8): its possible pairing with 1 or A_1 is identically zero for the stated reason. Similarly

$$\gamma_e = \xi^2\langle A_1, E_eA_1\rangle + O_L(\xi^3) = \frac{r_e}{12}\xi^2 + O_L(\xi^3). \tag{S9}$$

Only a pair p,q with $e \in p$, $f \in q$ contributes to (S8). Repeated faces contribute zero. There are exactly r_er_f such distinct unordered pairs when they are labeled as in (S3). For an edge-disjoint pair, both e and f carry fundamental spin, so its contribution is $(3/4)^2/9^2 = 1/144$. For an adjacent pair, neither e nor f can be its shared edge: otherwise the face containing the other link would contain both e and f. They are therefore both outer fundamental edges in each of the two channels (S6), and the contribution is

$$\left(\frac{3}{4}\right)^2\left[\left(\frac{4}{27}\right)^2\frac{1}{4} + \left(\frac{4}{39}\right)^2\frac{3}{4}\right] = \frac{1}{324} + \frac{3}{676}. \tag{S10}$$

Equation (S9) subtracts $r_er_f/144$ at degree four in (S2). Hence every edge-disjoint pair cancels its disconnected contribution exactly; every adjacent pair leaves

$$\frac{1}{324} + \frac{3}{676} - \frac{1}{144} = \frac{127}{219024} > 0. \tag{S11}$$

This proves the coefficient in (S4).

For its stated even error, the center transformation $U_{(n,i)} \mapsto (-1)^{\sum_{j<i} n_j}U_{(n,i)}$ changes every face trace sign, commutes with every E_e and preserves Haar measure. It intertwines $H_0 - \xi S$ with $H_0 + \xi S$ and hence the positive unit vacua at ξ and$-\xi$. Both terms in (S2) are consequently even analytic functions of ξ near zero. Their next possible order after four is six. This proves the error in (S4); it does not supply a volume-uniform analytic radius.

20.4 Exact support map and relation to the energy matrix

Define the edge adjacency matrix B by $B_{eg} = 1$ when e,g are distinct physical links belonging to a common elementary face, and zero otherwise. Two distinct links belong to at most one elementary square: perpendicular incident links determine its two plane directions and base, while parallel links in a square must be opposite sides and also determine that square. Thus, for the e,f considered above,

$$\boxed{a_{ef} = (B^2)_{ef}.} \tag{S12}$$

Indeed a pair p,q counted in (S3) has a unique shared edge g, which is a common neighbor of e,f. Conversely a common neighbor g determines the unique p containing e,g and q containing g,f; these are distinct because e,f share no face, and their unique shared edge is g. These two maps are inverse. The covariance coefficient thus has support at graph distance two, including boundary incidences, and no support beyond distance two at this order. It makes no all-orders finite-range claim about C.

There are actual such channels in every $L \geq 2$. Take $e = ((0,0,0),1)$ and $f = ((0,2,0),1)$. They share no face, but the faces in directions $1,2$ based at $(0,0,0)$ and $(0,1,0)$ share the edge $((0,1,0),1)$, giving $a_{ef} \geq 1$. All of their vertices are in the original box, including at L=2.

Put $\mathcal{N}_{ef} = \langle v_e, (H-\mathcal{E})v_f\rangle$. Commuting the link operators on the smooth real vacuum gives

$$\mathcal{N}_{ef} = \tfrac{1}{2}\langle \psi, [E_e, [H, E_f]]\psi\rangle. \tag{S13}$$

To check (S13), expand the double commutator, use $H\psi = \mathcal{E}\psi$, commute E_e,E_f, and use the real symmetry of their two matrix elements. Only potential faces containing f occur in $[H, E_f]$. If e shares no face with f, E_e commutes with every one of their multiplication and derivative terms. Thus

$$\boxed{\mathcal{N}_{ef} = 0 \text{ exactly, at every positive coupling, when e,f share no face.}} \tag{S14}$$

There is no contradiction between (S4) and (S14): a matrix-valued spectral measure can have nonzero zeroth moment and zero first moment off diagonal. Explicitly, for the spectral resolution of A=H−E on the physical space,

$$d\Sigma_{ef}(\omega) = \langle v_e, \mathbf{1}_{d\omega}(A)v_f\rangle, \quad C_{ef} = \int d\Sigma_{ef}, \quad \mathcal{N}_{ef} = \int \omega\, d\Sigma_{ef}. \tag{S15}$$

The measure matrix is positive semidefinite on every Borel set because its quadratic form is the squared norm of a projected linear combination of the v_e. An individual off-diagonal entry, however, need not be a nonnegative measure; (S15) retains its signs. Smoothness gives the finite moments used here. These identities are the exact relation between the nonlocal covariance contribution and the local energy matrix, not an assumption that one determines the other's support.

Finally $\xi^4 = 1/(256g^{16})$. The physical coefficient in (S4) is therefore $127a_{ef}/(56070144g^{16})$; C has no extra energy factor, whereas $\mathcal{N}$ has one. The scalar energy shift remains in H and cancels only in H−E. The fixed-box coefficient is a concrete next input to the extensive denominator calculation; control of all higher orders and summability at fixed coupling is not inferred from its finite range at order four.

20.5 The complete fourth-order spectral functional and time correlation

The same calculation can resolve the energy dependence, rather than only its zeroth moment. Put $A_\xi = (H-\mathcal{E})/\kappa$, and write

$$F_{ef,\xi}(h) = \langle v_e, h(A_\xi)v_f\rangle.$$

For each fixed box, for distinct links sharing no face, and every bounded continuous real or complex function h which is differentiable at 3,

$$\boxed{\begin{aligned}\lim_{\xi\to 0} \xi^{-4} F_{ef,\xi}(h) = a_{ef}\Big[&-\frac{59}{275184}h(3) - \frac{1}{336}h'(3) \\ &+ \frac{1}{1296}h(9/2) + \frac{3}{132496}h(13/2)\Big].\end{aligned}} \tag{S16}$$

Here the limit is a functional on the specified test functions. In particular the derivative term must not be omitted or represented as an ordinary positive spectral atom. Each original matrix-valued spectral measure is still the exact positive-semidefinite measure (S15).

We prove (S16), including the relation to the actual first interacting spectral subspace. Let P project onto the free physical energy-three space $\mathcal{P} = \mathrm{span}\{W_p\}$, and let

$$R_3 = (I-P)((H_0 - 3)|_{\mathcal{P}^\perp})^{-1}(I-P).$$

The inverse is on the physical space: its constant channel is -1/3. To verify isolation, a nonzero physical spin network has no degree-one support vertex. Its support contains a cycle of length at least four in the bipartite cubic graph, costing at least 3. With fewer than six nontrivial edges it must be one elementary square: a fifth edge would have a new degree-one endpoint, or be a forbidden same-bipartition chord of the four-cycle. The degree-two invariant contractions require equal spins on that square. Only spin 1/2 has energy below 9/2. A six-edge rectangular loop realizes energy 9/2. Thus the first physical free band is exactly P, with the next energy 9/2.

For sufficiently small real ξ, the contour $|z-3| = 3/4$ defines the actual spectral projection P_ξ, and the exact isometry

$$V_\xi = P_\xi P (P P_\xi P|_{\mathcal{P}})^{-1/2} : \mathcal{P} \longrightarrow \operatorname{ran} P_\xi \tag{S17}$$

uses its full Gram factor. Indeed the free contour resolvent has norm at most 4/3. For $|\xi| \le 3/(64M)$, its Neumann factor is at most 1/8, and integrating the resolvent difference gives $\|P_\xi - P\| \le 1/7$. The displayed inverse square root therefore exists by its binomial series; idempotence and orthogonality of the real projection show $V_\xi^* V_\xi = I$. Its range is smooth, and the finite-rank resolvent formula and elliptic regularity justify all operator products and fixed Sobolev derivatives below. It is orthogonal to the actual vacuum for a smaller interval, for example $|\xi| \le 3/(128M)$, by the free gaps and the bounded perturbation estimate $2M|\xi|$.

Let Q be the unchanged open-face signless adjacency matrix, with diagonal the actual face degrees and off-diagonal 1 for shared-edge faces. Direct virtual-channel calculation gives

$$V_\xi^* A_\xi V_\xi = 3I + \xi^2 F + O_L(\xi^4), \qquad F = \frac{7}{15} I - \frac{1}{21} Q. \tag{S18}$$

Here is the coefficient calculation rather than an assumption of (S18). Differentiating (S17) gives $V_0' = R_3 S P$. The second matrix coefficient before vacuum subtraction is $-PSR_3SP$. The vacuum channel gives $+1/3$ to every entry; a repeated-face channel gives -1/5 to its diagonal; an edge-disjoint pair gives -1/3 to its two diagonals and mutual entries; an adjacent pair gives $-(1/4)/(9/2-3) - (3/4)/(13/2-3) = -8/21$ to that same four-entry pattern. Thus the diagonal is $7/15 - M/3 - d_p/21$, and the off-diagonal is -1/21 exactly for adjacent faces. The actual vacuum coefficient -M/3 from (S5) gives (S18). The central cochain used in Section 20.3 acts as -I on P; it makes the matrix in (S18) even and accounts for the fourth-order remainder.

The electric vector can be treated with arbitrary fixed real link weights w, retaining its linear dependence on all of them. Define $\Gamma_w = \sum w_e E_e$, $a_p(w) = \sum_{e \in p} w_e/4$, and $v(w) = (\Gamma_w - \langle \Gamma_w \rangle)\psi$. Put

$$y_\xi(w) = V_\xi^* v(w), \qquad \rho_\xi(w) = (I - P_\xi) v(w).$$

These are exact orthogonal components, both orthogonal to the actual vacuum. Coefficient comparison using (S7) and $V_0' = R_3 S P$ gives

$$y_\xi(w) = \xi x(w) + \xi^3 z(w) + O_L(\xi^5), \qquad x(w) = \sum_p a_p(w) W_p,$$

$$\rho_\xi(w) = \xi^2 r(w) + O_{H^k,L}(\xi^3), \tag{S19}$$

where, for $s = a_p(w) + a_q(w)$ and g the shared edge,

$$r(w) = \sum_p \frac{2a_p(w)}{15}(W_p^2 - 1) + \sum_{\{p,q\} \text{ adjacent}} \left[-\frac{2(s + w_g)}{9} Z_{pq,0} + \frac{2(3s + 7w_g)}{273} Z_{pq,1} \right]. \tag{S20}$$

For completeness, Γ_w has eigenvalues $8a_p$ on the repeated face, $3s$ on an edge-disjoint pair, and $3s - 3w_g/2, 3s + w_g/2$ on the adjacent zero and one channels. The shared edge is counted twice as fundamental in 3s; replacing that by its actual spin accounts for precisely the last two corrections. In $\Gamma_w A_2$, the respective coefficients are $a_p/3, s/3, (4s-2w_g)/9, (12s+2w_g)/39$. Subtract the full dressing $R_3 Sx(w)$, whose coefficients are $a_p/5, s/3, 2s/3, 2s/7$. The two constant coefficients are both $-\sum_p a_p/3$ and also cancel. This proves (S20) with no omitted channel. The centered projection identity $V_\xi^* \psi_\xi = 0$ is exact. The same central cochain makes $y_\xi(w)$ odd, proving the first expansion (S19).

For a single link e use the original weight $w_g^{(e)} = \delta_{eg}$, and denote the resulting x,z,r by $x_e, z_e, r_e^{\text{vec}}$. This superscript distinguishes the vector from the incidence count r_e. Then $x_e = \sum_{p \ni e} W_p/4$. If e,f share no face, $\langle x_e, x_f\rangle = 0$, and (S18) gives exactly

$$\langle x_e, F x_f\rangle = -a_{ef}/336. \tag{S21}$$

Each ordered adjacent pair contributes $-1/(4 \cdot 21 \cdot 4)$. In the cross spectral measure of $r_e^{\text{vec}}, r_f^{\text{vec}}$, only the pairs counted by (S3) occur. Repeated-face cross terms vanish. For every contributing pair, e and f are outer links of different faces, $s = 1/4$ for each weight, and $w_g = 0$. Thus their zero-channel coefficients in (S20) are both -1/18, and their one-channel coefficients are both 1/182. Orthogonality and (S6) give the complete free cross measure

$$\langle r_e^{\text{vec}}, h(H_0) r_f^{\text{vec}}\rangle = a_{ef}\left[\frac{1}{1296}h(9/2) + \frac{3}{132496}h(13/2)\right]. \tag{S22}$$

No edge-disjoint channel survives in (S20).

As $A_\xi - H_0 = -\xi S - e(\xi)I$ tends to zero in operator norm, its resolvents converge in norm. For bounded continuous h, this implies convergence of the cross expectations on ρ_ξ/ξ^2 to (S22): first use continuous functions vanishing at infinity and norm resolvent convergence, then approximate h on a compact interval and control the discarded tail by the uniformly bounded second spectral moments. Those moments follow from the H^2 convergence in (S19). Polarization gives the cross statement from the four corresponding diagonal statements. Hence (S22) is the entire fourth-order out-of-band contribution.

The fourth coefficient of the band mass is obtained without knowing the third vacuum coefficient: subtract (S22) at h=1 from (S4). It is

$$\langle x_e, z_f\rangle + \langle z_e, x_f\rangle = a_{ef}\left(\frac{127}{219024} - \frac{1}{1296} - \frac{3}{132496}\right) = -\frac{59}{275184}a_{ef}. \tag{S23}$$

Differentiability of h at 3 and spectral calculus of the finite matrix (S18) give $h(V_\xi^* A_\xi V_\xi) = h(3)I + \xi^2 h'(3)F + o_L(\xi^2)$. Combine this with (S19), (S21), (S23), and the exact orthogonal spectral decomposition into the band and its complement. This proves (S16).

In particular the complete dimensionful ground-state time correlation, with physical $\kappa > 0$ and physical time $t \geq 0$ fixed, is

$$\boxed{\begin{aligned}\langle v_e, e^{-t(H-\mathcal{E})} v_f\rangle = a_{ef}\xi^4 \Bigg[&-\frac{59}{275184}e^{-3\kappa t} + \frac{\kappa t}{336}e^{-3\kappa t} \\ &+ \frac{1}{1296}e^{-9\kappa t/2} + \frac{3}{132496}e^{-13\kappa t/2}\Bigg] + O_{L,t,\kappa}(\xi^6).\end{aligned}} \tag{S24}$$

Here $h(q) = e^{-\kappa t q}$ is used on the nonnegative excitation spectrum, with any bounded continuous extension below zero, so it belongs to the test class of (S16). The bounded potential perturbation gives $e^{-\kappa t K_\xi}$ its norm-convergent Duhamel series: each order is bounded by $(2M|\xi|\kappa t)^n/n!$. The excitation semigroup retains the additional analytic scalar $e^{\kappa te(\xi)}$. The vacuum and vectors are analytic in fixed Sobolev norms, and the central cochain makes the whole expression even. These facts justify the even remainder in (S24), not only a pointwise little-o limit. For varying physical κ, the same proof uses the explicit parameter $\tau = \kappa t$ and keeps its dependence; no estimate uniform in τ or L is implied.

At t=0 the bracket is 127/219024. Its derivative at t=0 is exactly zero, because

$$3\left(-\frac{59}{275184}\right) - \frac{1}{336} + \frac{9}{2\cdot 1296} + \frac{39}{2\cdot 132496} = 0. \tag{S25}$$

This agrees with the exact identity (S14). It exhibits which separated-link spectral contributions cancel in the first moment: the actual band-mass correction, the actual band-energy derivative, and both higher spin channels. The term proportional to t is essential to that cancellation. Equations (S16)–(S25) describe the fixed-box fourth coefficient of the original interacting correlations; they do not exchange a volume limit with the coupling expansion or assert a continuum spectrum.

20.6 Primary context

The original Hamiltonian framework is J. Kogut and L. Susskind, *Hamiltonian formulation of Wilson's lattice gauge theories*, Physical Review D **11** (1975), 395–408, `https://doi.org/10.1103/PhysRevD.11.395`. Full conventions appear in (S1). For historical linked-cluster calculations in the same spatial dimension, see A. C. Irving, T. E. Preece and C. J. Hamer, *Cluster expansion approach to non-abelian lattice gauge theory in (3+1)D (I). SU(2)*, Nuclear Physics B **270** (1986), 536–552, `https://doi.org/10.1016/0550-3213(86)90567-5`. This note derives (S4) directly, with no historical novelty assertion or imported coefficient.

21 The actual separated-link operator at a Riemann spectral fibre

Keep the original $\Xi(s) = \xi(1/2 + is) = 8H_0(2s)$ and the operator $\mathcal{T}_{\mathcal{D}}$ of (1018). Put $K = \mathcal{T}_{\mathcal{D}}\Xi$. The following calculation concerns this actual function, not an arbitrary entire replacement for Ξ.

21.1 A nonzero coefficient at a certified simple zero

There is a unique simple real zero γ of Ξ in

$$\frac{28269450279469}{2000000000000} < \gamma < \frac{28269450287469}{2000000000000}. \tag{1024}$$

The certificate evaluates the original infinite theta integral, with the kernel and tails in Section 19; its normalization is the one of [17, 18]. Here are the explicit inequalities used to certify this particular zero. For the two doubled rational endpoints l, r of (1024), the outward-rounded integrals satisfy

$$H_0(l) > 3.4571\,10^{-13}, \qquad H_0(r) < -3.4564\,10^{-13},$$
$$-0.001382720 < 16H_0'([l,r]) < -0.001382718.$$

Continuity gives a root; the strictly negative derivative proves uniqueness and simplicity throughout that interval. In every integral the sum over $n \geq 9$ and the range $u \geq 2$ are bounded separately and added as symmetric errors. Their respective absolute bounds are less than $3.273\,10^{-74}$ and $8.427\,10^{-2037}$ for this real-argument calculation. Large real translations do not change the sine/cosine bounds used in those tails.

Fresh integrations over the enclosing interval, with all original coefficients retained, give

$$\begin{aligned}\Xi(\gamma+3) &= [-0.000508120 \ \pm 5.76\,10^{-10}],\\ \Xi(\gamma+9/2) &= [-0.000180084 \ \pm 4.62\,10^{-10}],\\ \Xi(\gamma+13/2) &= [-9.117\,10^{-6} \ \pm 2.49\,10^{-10}],\\ \Xi'(\gamma+3) &= [0.0002611693 \ \pm 4.80\,10^{-11}].\end{aligned}$$

Using $\Xi' = 16H_0'(2s)$ and the negative derivative-delta term gives

$$\begin{aligned}&K(\gamma) = c_0\Xi(\gamma+3) - \Xi'(\gamma+3)/336 + c_1\Xi(\gamma+9/2) + c_2\Xi(\gamma+13/2),\\ &-8.07510\,10^{-7} < K(\gamma) < -8.07506\,10^{-7},\\ &0.000583998 < r_\gamma := K(\gamma)/\Xi'(\gamma) < 0.000584002.\end{aligned} \tag{1025}$$

These deliberately widened bounds also contain an independent recombination of the serialized scalar intervals. They use neither an assumed zero ordinate nor RH. The reproducible one-worker 5,000,000,000-byte calculation and its complete receipt are

```
verify_xi4_trace_pole.py
xi4_trace_pole_results.json.
```

Taylor expansion at the certified simple root now proves

$$\frac{K(s)}{\Xi(s)} = \frac{r_\gamma}{s-\gamma} + \text{a holomorphic function near } \gamma. \tag{1026}$$

Indeed $\Xi(s) = (s-\gamma)A_\gamma(s)$ with $A_\gamma(\gamma) = \Xi'(\gamma) \neq 0$; dividing the Taylor series of K/A_γ gives exactly the displayed principal part. Thus the original source coefficient does not preserve frozen Ξ-divisibility. This is a pole at an on-critical-line zero, not an off-critical zero or an RH counterexample.

The full Weil calculation has a precise, different connection to this same operator. For a test g, $\mathcal{T}_{\mathcal{D}}\mathscr{F}g = \mathscr{F}(wg)$ and its pulled-back quadratic value is $W_w(g,g) = W(wg,wg)$, using the whole weighted autocorrelation (1022). Negating g negates its linear Fourier value and leaves this quadratic value unchanged. For the compact test $g = e^{2iu}b_4(u)$ of Section 19, exact interval recombination of every cross term yields

$$\begin{aligned}\text{explicit-formula pole terms:}\quad & [1.9093530770084217927\,10^{-6} \ \pm 10^{-25}],\\ \text{gamma contribution:}\quad & [-1.155956081\,10^{-6} \ \pm 2.64\,10^{-16}],\\ \text{prime term to subtract:}\quad & [7.5339692863834583529\,10^{-7} \ \pm 10^{-26}],\\ \text{complete value:}\quad & [6.7\,10^{-14} \ \pm 6.34\,10^{-16}] > 0.\end{aligned}$$

The displayed component intervals have been widened outwards; the complete value is calculated from the full-precision components, not their printed decimal centers. This compact test is not the noncompact theta input whose transform is Ξ. Its arithmetic balance does not cancel the pole (1026). The read-only replay `report_xi4_components.py` retains all fifteen matrix entries and the factor two on cross terms.

21.2 Its exact original Schwartz–Mellin meaning

Retain the arithmetic map of Connes–Consani–Moscovici [103, §3.1 and §3.6]:

$$(\mathcal{E}\psi)(x) = x^{1/2}\sum_{n\geq 1}\psi(nx), \qquad F(s) = \mathcal{F}_\mu\mathcal{E}\psi(s) = \int_0^\infty \mathcal{E}\psi(x)x^{-is}\frac{dx}{x}.$$

Here ψ is even Schwartz and $\psi(0) = \widehat{\psi}(0) = 0$, with the source Fourier convention $\widehat{\psi}(y) = \int \psi(x)e^{-2\pi ixy}dx$. Its Mellin transform $M_\psi(z) = \int_0^\infty \psi(x)x^{z-1}dx$ is holomorphic for $\Re z > -2$. At the origin this follows from evenness and $\psi(0) = 0$, which imply $\psi(x) = O(x^2)$; at infinity it follows from the Schwartz bounds. Derivatives in z add powers of $\log x$, still dominated on each compact substrip. For $\Re z > 1$, absolute convergence and $y = nx$ give

$$F(i(z-1/2)) = \zeta(z)M_\psi(z). \tag{1027}$$

The trace is entire: Poisson summation and the two zero conditions give

$$\sum_{n\geq 1}\psi(nx) = x^{-1}\sum_{n\geq 1}\widehat{\psi}(n/x).$$

Thus $\mathcal{E}\psi$ decays faster than every power at both ends, as do its logarithmic derivatives. Its defining Mellin integral converges locally uniformly for every complex s. Analytic continuation therefore extends (1027) to $\Re z > -2$, with the removable value at $z = 1$ since $M_\psi(1) = \frac{1}{2}\widehat{\psi}(0) = 0$.

The seed Ξ itself is genuinely in this source image. Take

$$\psi_0(x) = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}, \qquad A(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2). \tag{1028}$$

Direct substitution $y = \pi x^2$ and the gamma recurrence yield $M_{\psi_0}(z) = A(z)$. Both source zero conditions hold, and $\zeta(z)A(z) = \xi(z)$; the functional equation $\xi(1-z) = \xi(z)$ gives $\mathcal{F}_\mu\mathcal{E}\psi_0 = \Xi$ with the original minus Mellin sign. This proves the seed's provenance by its explicit input.

Set $\rho = 1/2 - i\gamma$. It is a simple nontrivial zero of ζ. For any original Schwartz input, (1027) vanishes at ρ. But its putative value for $F = K$ is $K(\gamma) \neq 0$. Therefore no original even Schwartz input with the two source zero conditions has trace K. The exact meromorphic expression, not merely this exclusion, is

$$\frac{K(i(z-1/2))}{\zeta(z)} = \frac{c_0\xi(z-3i) + (i/336)\xi'(z-3i) + c_1\xi(z-9i/2) + c_2\xi(z-13i/2)}{\zeta(z)}. \tag{1029}$$

The sign $+i/336$ follows by differentiating $\Xi(i(z-1/2)+a) = \xi(z-ia)$ with respect to a. Using $\Xi'(\gamma) = -i\xi'(\rho)$ and $\xi'(\rho) = A(\rho)\zeta'(\rho)$, its nonzero residue is

$$\operatorname*{Res}_{z=\rho}\frac{K(i(z-1/2))}{\zeta(z)} = -iA(\rho)r_\gamma. \tag{1030}$$

All factors of $A(\rho)$ are finite and nonzero. This retains the gamma factor and the orientation of the coordinate change.

21.3 The comparison retains the lost coordinate

Let E be the entire functions of order at most one and define

$$e(F) = F(\gamma), \qquad \ell(F) = (\mathcal{T}_\mathcal{D}F)(\gamma), \qquad d_0 = \mathcal{D}(1) = 127/219024.$$

Finite translations and differentiation preserve E. The actual relation between the original and transformed evaluations is the map $J : E \to \mathbb{C}^2$, $J(F) = (e(F), \ell(F))$, whose explicit right inverse is

$$R(a,b) = a + \frac{b - d_0a}{K(\gamma)}\Xi. \tag{1031}$$

Indeed $e(1) = 1$, $\ell(1) = d_0$, $e(\Xi) = 0$ and $\ell(\Xi) = K(\gamma)$. These identities prove $JR = I_2$ and the full splitting

$$E = \ker J \oplus \operatorname{span}\{1, \Xi\}, \qquad F = (F - RJF) + RJF.$$

Thus every fibre is $R(a, b) + \ker J$. Projection onto the first coordinate discards the second, and vice versa; each projection has its usual one-dimensional coordinate-axis kernel. In particular ℓ cannot factor through e, since $\Xi \in \ker e$ but $\ell(\Xi) \neq 0$. This is the precise local representative defect, with both measurements and the source vector producing it retained.

There is also a full algebraic moving-relation comparison, without assuming invertibility of $\mathcal{T}_{\mathcal{D}}$ on E. For any specified source relation $I \subset E$, put $T = \mathcal{T}_{\mathcal{D}}$ and $\Gamma_T = \{(F, TF) : F \in E\}$. Its subspace $\Gamma_T(I) = \{(n, Tn) : n \in I\}$ gives

$$\Gamma_T/\Gamma_T(I) \simeq E/I, \quad [(F, TF)] \mapsto [F], \qquad [F] \mapsto [(F, TF)].$$

The second projection is a surjection onto $T(E)/T(I)$ with kernel $(I + \ker T)/I$. To prove the kernel formula, $TF \in T(I)$ means $TF = Tn$ for some $n \in I$, exactly $F - n \in \ker T$. Every output class has a preimage by the definition of $T(E)$. These formulas apply both to the actual Schwartz trace image intersected with E and to the analytic relation ΞE. They neither assert invariance of those frozen relations nor erase the possible kernel of T. The compact-test automorphism in Section 19 retains its separately proved inverse $1/w$; it was never an assertion of global invertibility on the entire-function ring or on an unspecified source topology.

22 The original arithmetic input: explicit inverse and endpoint growth

The nonzero pole in Section 21 has an explicit preimage on the positive half-line. We construct it without changing the four-term source coefficient, prove its endpoint estimates, and give the inverse coordinate maps on a space containing the actual theta input. The arithmetic trace and Hermite construction are those of Connes–Consani–Moscovici [103, §3.6]; the heat normalization remains $Z_t(s) = 8H_t(2s)$ of Rodgers–Tao [17, §1]. The estimates and applications below are derived here, not attributed to those sources.

22.1 The two-ended theta input and its exact inverse

Retain ψ_0 and A from (1028), and put

$$q(x) = \sum_{m \geq 1} \psi_0(mx), \qquad g_0(u) = e^{u/2} q(e^u), \qquad K_t = \mathcal{T}_{\mathcal{D}} Z_t.$$

In particular $\mathscr{F} g_0 = \Xi$ for $\mathscr{F} g(s) = \int_{\mathbb{R}} g(u) e^{-isu} du$. Here is a direct check of the two-ended bounds used below. The Gaussian Fourier formulas are

$$\widehat{x^2 e^{-\pi x^2}}(y) = (1/(2\pi) - y^2) e^{-\pi y^2}, \quad \widehat{x^4 e^{-\pi x^2}}(y) = (y^4 - 3y^2/\pi + 3/(4\pi^2)) e^{-\pi y^2}.$$

They follow by differentiating the Gaussian transform twice and four times, with its original 2π convention. Substitution proves $\widehat{\psi}_0 = \psi_0$, including cancellation of the constant term. Since $\psi_0(0) = \widehat{\psi}_0(0) = 0$, Poisson summation gives $q(x) = x^{-1} q(1/x)$ and hence $g_0(-u) = g_0(u)$.

For $u \geq 0$, every derivative of the defining series of g_0 is a finite sum of terms bounded by $Cn^B e^{Au} \exp(-\pi n^2 e^{2u})$, with fixed nonnegative A, B. The inequality $n^2 e^{2u} \geq (n^2 + e^{2u})/2$ separates a summable Gaussian in n from $\exp(-\pi e^{2u}/2)$. This justifies all termwise derivatives locally and bounds each full derivative. Evenness gives the same estimate at the other endpoint. Consequently, for every $j \geq 0$,

$$|g_0^{(j)}(u)| \leq C_j \exp(A_j|u| - \kappa_j e^{2|u|}) \quad (u \in \mathbb{R}), \qquad C_j, A_j, \kappa_j > 0. \tag{1032}$$

For $t \in \mathbb{C}$ and $x > 0$ define

$$q_{\mathcal{D},t}(x) = e^{t(\log x)^2/4} \left[x^{-3i}(c_0 + i\log x/336) + c_1 x^{-9i/2} + c_2 x^{-13i/2} \right] q(x). \tag{1033}$$

All powers use the real logarithm of the positive argument. This is exactly $e^{t(\log x)^2/4} w(\log x) q(x)$, with the c_j and w of Section 19. The sign of the logarithmic term is fixed by the original derivative distribution. The required inverse is the explicit double series

$$\begin{aligned} \psi_{\mathcal{D},t}(x) &= \sum_{n\geq 1} \mu(n) q_{\mathcal{D},t}(nx) \\ &= \sum_{n\geq 1} \mu(n) e^{t(\log(nx))^2/4} \Big[(nx)^{-3i}(c_0 + i\log(nx)/336) \\ &\qquad\qquad + c_1(nx)^{-9i/2} + c_2(nx)^{-13i/2} \Big] \\ &\qquad \cdot \sum_{m\geq 1} (4\pi^2 m^4 n^4 x^4 - 6\pi m^2 n^2 x^2) e^{-\pi m^2 n^2 x^2}. \end{aligned} \tag{1034}$$

The function $q_{\mathcal{D},t}$ is smooth, flat at zero, and rapidly decreasing with every derivative at infinity. Indeed $x = e^u$ turns each x derivative into a finite combination of logarithmic derivatives times powers of e^{-u}. Estimate (1032) absorbs these powers, every derivative of w, and the fixed factor $e^{tu^2/4}$ at both ends. For a compact interval $x \geq \varepsilon > 0$, the jth derivative of the nth inverse summand is bounded by $C\varepsilon^{-B} n^{j-B}$, for arbitrarily large B. Thus (1034) converges locally with every derivative. The same estimate for $x \geq 1$ proves rapid decrease of the inverse and all its derivatives at infinity.

For each fixed $x > 0$ the series $\sum_{m,n\geq 1} |q_{\mathcal{D},t}(mnx)| = \sum_{k\geq 1} d(k)|q_{\mathcal{D},t}(kx)|$ converges, since $d(k) \leq k$ and arbitrary decay powers are available. Regrouping and using $\sum_{n|k} \mu(n) = \mathbf{1}_{k=1}$ proves

$$\sum_{m\geq 1} \psi_{\mathcal{D},t}(mx) = q_{\mathcal{D},t}(x), \qquad \mathscr{F}_\mu \mathcal{E} \psi_{\mathcal{D},t} = K_t. \tag{1035}$$

In the second identity the trace is formed first: its logarithmic representative is $e^{tu^2/4} w(u) g_0(u)$ and its Fourier integral is absolutely convergent. This does not presume that the inverse itself is Schwartz at zero. Uniqueness holds among functions rapidly decreasing at infinity whose trace sum is the specified one: apply the same absolutely convergent divisor rearrangement to their trace.

For $\Re z > 1$ absolute integration gives

$$\sum_{n\geq 1} \int_0^\infty |q_{\mathcal{D},t}(nx)| x^{\Re z - 1} dx = \zeta(\Re z) \int_0^\infty |q_{\mathcal{D},t}(y)| y^{\Re z - 1} dy < \infty.$$

Hence the original Mellin map, with its exact denominator, is

$$M_{\psi_{\mathcal{D},t}}(z) = \frac{H_{\mathcal{D},t}(z)}{\zeta(z)}, \qquad H_{\mathcal{D},t}(z) = K_t(i(z - 1/2)), \quad \Re z > 1. \tag{1036}$$

The numerator is entire and

$$H_{\mathcal{D},0}(z) = c_0 \xi(z - 3i) + (i/336)\xi'(z - 3i) + c_1 \xi(z - 9i/2) + c_2 \xi(z - 13i/2).$$

The factor $+i/336$ and all shifted completion factors are retained.

22.2 A proved endpoint singularity and its exponent threshold

For every fixed complex t and every $j \geq 0$,

$$\psi_{\mathcal{D},t}^{(j)}(x) = O(x^{-j-1}) \quad (x \downarrow 0). \tag{1037}$$

To prove this with specified finite constants, set $r_j(y) = y^j q_{\mathcal{D},t}^{(j)}(y)$, $C_{j,0} = \sup_{y>0} |r_j(y)|$ and $C_{j,2} = \sup_{y>0} y^2 |r_j(y)|$. Both are finite by the two-ended estimates. For $0 < x \leq 1$ put $N = \lfloor 1/x \rfloor \geq 1/(2x)$. Differentiating the convergent inverse series and using $|\mu(n)| \leq 1$ gives

$$\begin{aligned} x^j |\psi_{\mathcal{D},t}^{(j)}(x)| &\leq \sum_{n \geq 1} |r_j(nx)| \\ &\leq N C_{j,0} + C_{j,2} x^{-2} \sum_{n>N} n^{-2} \leq (C_{j,0} + 2C_{j,2}) x^{-1}. \end{aligned}$$

This proves (1037), not a claimed sharp asymptotic.

For the actual time-zero coefficient there are also unconditional lower restrictions:

$$\limsup_{x \downarrow 0} x^{\beta} |\psi_{\mathcal{D},0}(x)| = \infty \quad \text{for every real } \beta < \tfrac{1}{2}, \tag{1038}$$

and

$$\int_0^1 |\psi_{\mathcal{D},0}(x)| x^{\sigma-1} dx = \infty \quad \text{for every real } \sigma \leq \tfrac{1}{2}. \tag{1039}$$

These follow from the certified nonzero pole at $\rho = 1/2 - i\gamma$, with residue $-iA(\rho) r_\gamma \neq 0$ in (1030). Here is the full analytic argument. If the inverse were $O(x^{-\beta})$ for some $\beta < 1/2$, its Mellin integral would be holomorphic on $\Re z > \beta$: on each compact substrip, the bound at zero and rapid decrease at infinity dominate every z derivative, including powers of $\log x$. Equation (1036) would then imply $(z-1)\zeta(z) M_\psi(z) = (z-1) H_{\mathcal{D},0}(z)$ throughout that half-plane by the identity theorem. Evaluation at ρ gives zero on the left and a nonzero value on the right. This contradiction proves that no such big-O bound exists, which is equivalent to (1038) because the inverse is continuous on every compact subinterval of $(0, \infty)$.

For (1039), it suffices to exclude integrability at $\sigma = 1/2$; smaller σ gives a larger weight on $(0, 1)$. Such integrability would make the Mellin integral holomorphic for $\Re z > 1/2$ and bounded along $z = \rho + \delta$, $0 < \delta \leq 1$. The bound follows directly from $|x^\delta| \leq 1$ on $(0, 1)$ and rapid decrease on $(1, \infty)$. But the same identity theorem and the nonzero simple residue give $M_\psi(\rho+\delta) = -iA(\rho) r_\gamma/\delta + O(1)$. This contradicts boundedness. No assumption on zeros away from this certified point has entered either proof.

Thus this particular arithmetic inverse really is unbounded, while its trace (1035) still has superexponential decay in logarithmic coordinates. The divisor cancellations in that trace do not remove the pole of its Mellin inverse. The threshold $1/2$ here comes from the real part of a certified on-line zero. Neither a larger lower exponent nor a full leading $x^{-\rho}$ expansion is asserted; the latter would require justified contour and remainder estimates. The full physical coefficient is $a_{ef} \psi_{\mathcal{D},t}$ with the original incidence factor: nonzero a_{ef} retains these conclusions; at $a_{ef} = 0$ the coefficient and its unique inverse are identically zero.

22.3 The logarithmic divisor term is part of the inverse

At time zero, grouping (1034) by $k = mn$ gives an alternative exact expression. Define

$$J_a(k) = \sum_{d|k} \mu(d) d^{-ia} = \prod_{p|k} (1 - p^{-ia}), \qquad L_a(k) = \sum_{d|k} \mu(d) d^{-ia} \log d.$$

The finite product follows by expanding over squarefree divisors; the coefficient of the divisor d is $\mu(d)$. For the double series at fixed x, the heat factor is absent and the remaining weights are bounded by a constant times $1+|\log x|+\log k$. The Gaussian $\psi_0(kx)$ therefore permits absolute regrouping. It yields

$$\begin{aligned}\psi_{\mathcal{D},0}(x)=\sum_{k\geq 1}\psi_0(kx)\bigl\{x^{-3i}&[(c_0+i\log x/336)J_3(k)+(i/336)L_3(k)]\\&+c_1x^{-9i/2}J_{9/2}(k)+c_2x^{-13i/2}J_{13/2}(k)\bigr\}.\end{aligned}\tag{1040}$$

In particular L_a is not discarded and no factor $1-p^{-ia}$ is divided out at a possible zero.

22.4 An inverse domain containing the original noncompact input

Let $\mathcal{B}$ be the vector space of all complex smooth functions whose every derivative satisfies a bound of the form (1032), with constants depending on the derivative and function, and let $V=\mathscr{F}\mathcal{B}$. The space contains g_0 before transformation and Ξ afterwards. Each Fourier transform is entire: differentiation on a compact set of complex arguments is dominated by the superexponential bound. Moreover V consists of entire functions of order at most one. Indeed split off half of $\kappa_0e^{2|u|}$ in the exponent of the integral bound, and maximize $(R+A_0)|u|-(\kappa_0/2)e^{2|u|}$; the maximum is $O((R+1)\log(R+2))$. The remaining integrable factor is independent of R, proving that bound for $\log M_F(R)$. Fourier injectivity on $\mathcal{B}\subset L^1$ can be checked without an inverse-growth assumption: convolve a function with zero Fourier transform with a Gaussian approximate identity. Fourier inversion for the Gaussian and Fubini show each convolution vanishes, and convergence in L^1 gives the original function zero.

Every derivative of w grows at most linearly. The established lower bound $|w|\geq 575/1752192$ implies polynomial growth for every derivative of $1/w$, by induction from $w(1/w)=1$. Leibniz's rule and (1032) now show that w, $1/w$, $e^{tu^2/4}$ and $e^{-tu^2/4}$ are multipliers of $\mathcal{B}$ for every fixed $t\in\mathbb{C}$. The same is true for differentiation and multiplication by u. Consequently the following are proved mutual inverses on V:

$$A_t=\mathcal{T}_{\mathcal{D}}U_t=\mathscr{F}M_{we^{tu^2/4}}\mathscr{F}^{-1},\qquad A_t^{-1}=\mathscr{F}M_{e^{-tu^2/4}/w}\mathscr{F}^{-1}.\tag{1041}$$

This includes the actual identity $A_t^{-1}K_t=\Xi$; it is not a claim of an inverse on every entire function.

The conjugated coordinate operator, including its lower terms, is

$$B_{\mathcal{D},t}=A_tM_sA_t^{-1}=M_s-\frac{t}{2}D_s+\mathscr{F}M_{iw'/w}\mathscr{F}^{-1}.\tag{1042}$$

Integration by parts gives $M_s\mathscr{F}=\mathscr{F}(-iD_u)$, with zero boundary terms by (1032). Applying this to $e^{-tu^2/4}h/w$ and multiplying back gives $-ih'+i(tu/2+w'/w)h$. Since $D_s\mathscr{F}h=\mathscr{F}(-iuh)$, this is precisely (1042). Explicitly,

$$iw'=e^{-3iu}(3c_0-1/336+iu/112)+(9/2)c_1e^{-9iu/2}+(13/2)c_2e^{-13iu/2}.$$

Thus at $t=0$, $B_{\mathcal{D},0}=M_s+C_{\mathcal{D}}\mathcal{T}_{\mathcal{D}}^{-1}$, where

$$C_{\mathcal{D}}f=(3c_0-1/336)f(s+3)-f'(s+3)/112+(9/2)c_1f(s+9/2)+(13/2)c_2f(s+13/2).$$

No logarithmic branch for w occurs.

For $G = \mathscr{F}h \in V$, all inverse jet coordinates are

$$\mathcal{L}^{\mathcal{D},t}_{\lambda,n}(G)_j = \int_{\mathbb{R}} (-iu)^j e^{-i\lambda u} e^{-tu^2/4} \frac{h(u)}{w(u)} du, \qquad 0 \le j < n. \tag{1043}$$

The bounds prove absolute convergence for every complex λ. Substitution gives $\mathcal{L}^{\mathcal{D},t}_{\lambda,n}(A_t f)_j = f^{(j)}(\lambda)$. In particular these coordinates of K_t are the full original Ξ jets. Their order-zero coordinate vanishes at γ, although ordinary output evaluation $K_0(\gamma)$ does not.

For a precise algebraic comparison put $E = \mathcal{H}_{\le 1}$, $I_V = V \cap \Xi E$, $I_t = A_t I_V$ and $J_t = I_V \cap I_t$. The maps $[f] \mapsto [A_t f]$ and $[G] \mapsto [A_t^{-1} G]$ are mutual inverse maps $V/I_V \simeq V/I_t$, with well-definedness following from $A_t I_V = I_t$. Both quotient projections from V/J_t are surjective, with kernels I_V/J_t and I_t/J_t, respectively. At time zero $K_0 \in I_0$ but $K_0 \notin I_V$, by its certified nonzero value. Hence K_0 is a specified nonzero class in I_0/J_0: it is lost by projection to the moving quotient and detected in the frozen one. The same inverse construction works for the actual unclosed Schwartz trace image intersected with V, because the seed Ξ has its explicit Schwartz preimage. These are algebraic maps; no identification with a closure in an unspecified source topology is being assumed.

22.5 All pole orders, heat jets and inverse-root coordinates

At any actual nontrivial zero ρ of multiplicity m, let $\lambda = i(\rho - 1/2)$ and write

$$\zeta(\rho + v) = v^m a_\rho(v), \quad a_\rho(0) \ne 0, \qquad a_\rho(v)^{-1} = \sum_{j \ge 0} \alpha_j v^j.$$

Multiplying the convergent Taylor series gives every principal coefficient, not only the residue:

$$\begin{aligned}[v^{-m+q}] \frac{H_{\mathcal{D},t}(\rho+v)}{\zeta(\rho+v)} = \sum_{n=0}^{q} \frac{i^n \alpha_{q-n}}{n!} \big[& c_0 Z_t^{(n)}(\lambda+3) - Z_t^{(n+1)}(\lambda+3)/336 \\ & + c_1 Z_t^{(n)}(\lambda + 9/2) + c_2 Z_t^{(n)}(\lambda + 13/2) \big], \quad 0 \le q < m. \end{aligned} \tag{1044}$$

For $b_n(t) = H^{(n)}_{\mathcal{D},t}(\rho)/n!$, differentiation under the superexponentially convergent integral yields $\partial_t b_n = (n+2)(n+1) b_{n+2}/4$. The positive sign follows from $D_z = iD_s$ and $\partial_t K_t = -D_s^2 K_t/4$. Thus the evolution of the principal part retains higher numerator jets; it is not a closed finite system on its negative powers alone. At a trivial zero the residue is $K_t(-i(2k+1/2))/\zeta'(-2k)$, $k \ge 1$. For the unbounded time-zero inverse, these meromorphic residues are not Taylor derivatives of a smooth extension at the origin.

There is an exact restriction on return times for this particular input. The entire function $t \mapsto K_t(\gamma)$ is not identically zero, since its time-zero value is certified nonzero. Its zeros form a closed discrete set. Every time at which K_t is in ΞE, or has an original even Schwartz preimage with the two zero conditions, belongs to that set. Each subset of a closed discrete subset of $\mathbb{C}$ is itself closed discrete: every compact set meets the parent set finitely, so no omitted finite accumulation point can occur. Both return sets therefore are closed discrete and exclude zero; neither is claimed empty.

Finally keep the original cover $s = -r^2/2$ and the two points $r_\pm = \pm i\sqrt{2\gamma}$. Set

$$d_\gamma = \frac{K_0'(\gamma)}{\Xi'(\gamma)} - \frac{K_0(\gamma)\Xi''(\gamma)}{2\Xi'(\gamma)^2}.$$

Since $s-\gamma = -r_\pm h - h^2/2$ for $h = r - r_\pm$, division gives

$$\frac{K_0(-r^2/2)}{\Xi(-r^2/2)} = -\frac{r_\gamma}{r_\pm h} + d_\gamma + \frac{r_\gamma}{2r_\pm^2} + O(h). \tag{1045}$$

The constant term follows by expanding $1/(s-\gamma) = -1/(r_\pm h) + 1/(2r_\pm^2) + O(h)$; it retains the second derivative of Ξ. On the two-component moving module the root-coordinate matrix is $\left(\begin{smallmatrix} 0 & -2B_{\mathcal{D},t} \\ 1 & 0 \end{smallmatrix}\right)$, whose square is $-2\,\mathrm{diag}(B_{\mathcal{D},t}, B_{\mathcal{D},t})$. Both components and the complete conjugated operator are required. The two imaginary inverse-root coordinates still lie over a certified on-critical-line zero; changing coordinates does not change that fact.

The replay

`verify_xi4_mobius_transport.py`

supplies 455 exact regressions: the original coefficient moments, all operator and Mellin signs on polynomial degrees zero through eight, the Gaussian self-duality, 128 divisor/product/logarithmic tests, the complete cover constant term and all principal coefficients through multiplicity seven. Its receipt is

`xi4_mobius_transport_results.json`

These finite regressions supplement the full convergence and endpoint proofs above; they are not represented as a formalization of those proofs.

23 The full Weil form on the actual theta input

The compact spline calculation above does not test the noncompact input whose Fourier transform is Ξ. We now evaluate the complete Weil form on that input after the actual multiplier w. This section retains both pole terms, the entire gamma integral, every prime power, all cross terms, and explicit infinite-tail bounds. It uses the classical explicit formula (839), with the conventions of Weil [26] and Alpöge–Furman [10, Sections 1.1–1.2 and 2].

23.1 The input, its involution, and the exact scale

Put

$$\psi_0(x) = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}, \qquad g_0(u) = e^{u/2}\sum_{n\ge 1}\psi_0(ne^u).$$

The Gaussian differentiation in (1028) gives $\widehat{\psi_0} = \psi_0$ and $\psi_0(0) = 0$. Poisson summation therefore gives $g_0(-u) = g_0(u)$. Writing the source theta kernel as

$$\Phi(v) = \sum_{n\ge 1}(2\pi^2n^4e^{9v} - 3\pi n^2e^{5v})e^{-\pi n^2 e^{4v}},$$

one has, by direct substitution,

$$g_0(u) = \sum_{n\ge 1}(4\pi^2n^4e^{9u/2} - 6\pi n^2e^{5u/2})e^{-\pi n^2e^{2u}} = 2\Phi(u/2). \tag{1046}$$

Since $H_0(z) = \int_0^\infty \Phi(v)\cos(zv)\,dv$ and $\Xi(s) = 8H_0(2s)$, the change of variable $u = 2v$ proves

$$\mathscr{F}g_0(s) = \int_{\mathbb{R}} g_0(u)e^{-isu}\,du = \Xi(s).$$

No rescaling of the source coefficient or of w is made.

Set $q(u) = w(u)g_0(u)$ and $Q = \mathscr{F}q = \mathcal{T}_{\mathcal{D}}\Xi$. The relation $g_0(-u) = g_0(u)$ and $w(-u) = \overline{w(u)}$ gives $q^\star(u) := \overline{q(-u)} = q(u)$. Hence

$$h(x) := (q * q^\star)(x) = \int_{\mathbb{R}} q(v)q(x-v)\,dv, \qquad \widehat{h}(z) = Q(z)^2.$$

On the real axis $Q(t)$ is real, so the Weil value is the real number

$$W(q,q) = 2\Re Q(i/2)^2 + \int_{\mathbb{R}} Q(t)^2 G(t)\,dt - \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}}\{h(\log n) + h(-\log n)\}, \tag{1047}$$

where

$$G(t) = \frac{1}{2\pi}\left(\Re\psi(1/4 + it/2) - \log\pi\right).$$

The correlation in (1047) is not a pointwise value of Q; it is the full convolution and therefore retains every cross term.

23.2 Uniform theta and correlation bounds

For $u \ge 0$ write $\Theta_N(u)$ for the first N terms of (1046). Every summand is nonnegative, because $2\pi n^2 e^{2u} - 3 \ge 2\pi - 3 > 0$. For $N = 8$, the omitted tail is bounded by

$$0 \le g_0(u) - \Theta_8(u) \le E_8 := \frac{4\pi^2 9^4 e^{-\pi 9^2}}{1 - (10/9)^4 e^{-\pi(19)}}. \tag{1048}$$

Indeed $e^{9u/2}e^{-\pi n^2 e^{2u}} \le e^{-\pi n^2}$ for $u \ge 0$, and the ratio of consecutive n^4 majorants is at most the denominator ratio displayed. The negative quadratic term can only reduce a summand, so it need not be added to this upper bound. The full positive kernel has the bound

$$|g_0(u)| \le C_\Phi e^{-\pi}, \quad C_\Phi = \frac{4\pi^2}{1 - 16e^{-3\pi}}. \tag{1049}$$

For $x = \log n \ge 0$, split the convolution integral into $[-3, 0]$, $[0, x]$, and $[x, x+3]$. On the first and last cells use the evenness of g_0 to replace negative arguments by positive ones; the remaining two rays have one argument of size at least 3. Replacing both factors by (1049) and integrating the elementary majorants gives the exterior error

$$E_{\text{ext}}(x) = \frac{2C_\Phi^2 e^{-\pi}\exp(\frac{9}{2}3 - \pi e^6)(B + (x+3)/336)^2}{2\pi e^6 - \frac{9}{2} - [168(B + (x+3)/336)]^{-1}}, \qquad B = |c_0| + c_1 + c_2. \tag{1050}$$

Replacing g_0 by Θ_8 on the three cells changes the product by at most

$$E_{\text{trunc}}(x) = (x+6)(B + (x+3)/336)^2(2C_\Phi e^{-\pi}E_8 + E_8^2). \tag{1051}$$

The bounds use only $|w(u)| \le B + |u|/336$ and the product identity $(a+\delta)(b+\delta) - ab = \delta(a+b) + \delta^2$. The three finite cell integrals are evaluated by Arb interval quadrature, split at the actual reflection locations 0 and x. On each cell the two reflected theta arguments have fixed signs. Thus the reported ball plus $E_{\text{ext}} + E_{\text{trunc}}$ encloses the exact $h(x)$.

For the prime tail use a bound on the whole convolution line. Positivity and the ratio of successive theta terms give, for every real u,

$$|g_0(u)| \le Ce^{9|u|/2 - \pi e^{2|u|}}, \qquad C = C_\Phi = \frac{4\pi^2}{1 - 16e^{-3\pi}} < 40.$$

For the last bound, the finite exponential series gives $e^3 > 20$; using $3 < \pi < 22/7$ yields $C < 4(22/7)^2/(1 - 16/8000) = 968000/24451 < 40$. Put $x = \log n$, $v = x/2 + y$, and $A_n = B + x/672$. The identities and inequalities

$$|v| + |x - v| = \max(x, 2|y|) \le x + 2|y|,$$
$$e^{2|v|} + e^{2|x-v|} \ge e^{2v} + e^{2(x-v)} = 2n\cosh(2y) \ge 2n + 4ny^2,$$
$$|w(v)w(x - v)| \le (A_n + |y|/336)^2, \qquad 9|y| \le 2\pi ny^2 + \frac{81}{8\pi n}$$

bound $|h(x)|$ by

$$C^2 n^{9/2} e^{-2\pi n + 81/(8\pi n)} \int_{\mathbb{R}} (A_n + |y|/336)^2 e^{-2\pi ny^2}\, dy.$$

Use $(a + b)^2 \le 2a^2 + 2b^2$ and the exact Gaussian moments

$$\int_{\mathbb{R}} e^{-2\pi ny^2}\, dy = (2n)^{-1/2}, \qquad \int_{\mathbb{R}} y^2 e^{-2\pi ny^2}\, dy = \frac{1}{4\pi n\sqrt{2n}}.$$

The resulting coefficient multiplying $A_n^2 n^4 e^{-2\pi n}$ is

$$C^2\sqrt{2}\, e^{81/(8\pi n)} \left(1 + \frac{1}{4\pi n\, 336^2 A_n^2}\right).$$

For $n \ge 17$, use $A_n \ge \log n/672$, $\log 17 > 2$, $\pi > 3$, and $\sqrt{2} < 10/7$. Also $81/(8\pi n) < 1/5$ and $e^{1/5} < 5/4$, the latter obtained by bounding its Taylor series by the geometric series. Thus this coefficient is less than

$$1600 \frac{10}{7} \frac{5}{4} \frac{205}{204} = \frac{4100000}{1428} < 3000.$$

This proves the conservative envelope

$$|h(\log n)| \le 3000 \left(B + \frac{\log n}{672}\right)^2 n^4 e^{-2\pi n}. \tag{1052}$$

Use $\Lambda(n) \le \log n$. The resulting continuous majorant is $F(t) = t^{7/2} \log t\, (B + \log t/672)^2 e^{-2\pi t}$. Its logarithmic derivative

$$\frac{7}{2t} + \frac{1}{t \log t} + \frac{2}{t(672B + \log t)} - 2\pi$$

is decreasing for $t > 1$. Hence $\log F$ is concave and its consecutive ratios for integers $n \ge 17$ are at most

$$r = e^{-2\pi} (18/17)^{7/2} \frac{\log 18}{\log 17} \left(\frac{B + \log 18/672}{B + \log 17/672}\right)^2 < 1.$$

For example the three ratios are bounded by $2, 2, 4$, respectively, so $r < 16e^{-6} < 1/25$. This is a ratio bound on the majorant, not on the sparse von Mangoldt sequence. Therefore the omitted prime-power sum after integer cutoff 16 is at most

$$E_{\text{prime}} = 6000\, 17^{7/2} \log 17 \left(B + \frac{\log 17}{672}\right)^2 e^{-34\pi}/(1 - r). \tag{1053}$$

For the gamma integral, shift the two theta Fourier contours by $\pm i/2$. The resulting exponential factor is $a = \pi \cos 1 > 0$; summing the geometric theta majorants gives

$$C_\theta = \frac{4\pi^2}{1 - 16e^{-3a}} + \frac{6\pi}{1 - 4e^{-3a}}.$$

The polynomial factor and one derivative term in w give the explicit constant

$$C_K = 2C_\theta e^{13/4}\left(\frac{9}{2ae}\right)^{9/4} e^{-a/2}\left(\frac{B+1/672}{a}+\frac{1}{336a^2}\right).$$

The standard digamma bound

$$|G(t)| \le \{\gamma_E + 8 + \log\pi + \log(1+2|t|)\}/(2\pi)$$

then yields, for $T = 64$,

$$E_\Gamma(T) = \frac{C_K^2}{\pi} e^{-T}\left(\gamma_E + 8 + \log\pi + \log(1+2T) + \frac{1}{T+1/2}\right). \tag{1054}$$

Here are the contour details. For $u \ge 0$, the theta series on $u \pm i/2$ has absolute value at most $C_\theta e^{9u/2-ae^{2u}}$; evenness supplies the other half-line. The multiplier on that strip is at most $e^{13/4}(B + 1/672 + |u|/336)$. Split the theta exponential into two equal factors and use

$$e^{9u/2-(a/2)e^{2u}} \le \left(\frac{9}{2ae}\right)^{9/4}, \qquad e^{-(a/2)e^{2u}} \le e^{-a/2-au}.$$

Integrating e^{-au} and ue^{-au} gives exactly C_K. Horizontal contour displacement, with the vertical ends tending to zero by these estimates, proves $|Q(t)| \le C_K e^{-|t|/2}$. For the digamma bound, use its convergent series at $z = 1/4 + ib$:

$$\frac{1}{n} - \frac{n+1/4}{(n+1/4)^2+b^2} = \frac{1/4}{n(n+1/4)} + \frac{b^2}{(n+1/4)((n+1/4)^2+b^2)}.$$

Splitting the second sum at $\max(1, \lceil |b| \rceil)$ bounds its tail by $1/2$ and its initial part by $1+\log(1+|b|)$. The $n = 0$ real term has absolute value at most 4. Together these imply the stated, larger bound for $|G(t)|$. Finally $\log(1+2t) \le \log(1+2T) + (t-T)/(T+1/2)$ for $t \ge T$. Integrating e^{-t} and $(t-T)e^{-t}$ proves the displayed tail. No zero ordinates or RH input occurs.

23.3 Passage from compact tests to the actual input

Here is the noncompact-test justification for (1047). Choose an even real $\chi \in C_c^\infty(\mathbb{R})$, with $0 \le \chi \le 1$ and $\chi = 1$ near zero, and put $q_R(u) = \chi(u/R)q(u)$, $R \ge 1$. Then $q_R^\star = q_R$. Every fixed derivative of q is superexponentially decaying: differentiating its theta series adds only fixed polynomial factors in the summation index, $e^{2|u|}$, and $|u|$, still dominated by the negative exponential. The differentiated series converge uniformly on compact intervals; evenness supplies the negative half-line. It follows by Leibniz's rule that $\|q_R\|_1 + \|q_R''\|_1$ is uniformly bounded. Twice integrating by parts gives

$$|\widehat{q}_R(t)| \le \min\{\|q_R\|_1, \|q_R''\|_1/|t|^2\} \le C_*(1+|t|)^{-2}$$

with one constant independent of R. Thus the gamma integrands are dominated by an integrable multiple of $(1+|t|)^{-4}\log(2+|t|)$, and converge by dominated convergence. The pole evaluations converge because $|q(u)|e^{|u|/2}$ is integrable.

For the arithmetic terms,

$$|(q_R * q_R)(x)| \le \int_{\mathbb{R}} |q(v)q(x-v)|\, dv.$$

The proof of (1052) bounds this absolute-product integral, not merely a cancellation in h. It consequently supplies one summable, R-independent majorant after von Mangoldt weighting for every $n \geq 17$. Each fixed convolution converges by dominated convergence; the finitely many smaller integers cause no difficulty. The compact-test explicit formula therefore passes to (1047) with all its terms retained. For completeness, applying the same integration-by-parts argument to the weighted derivatives gives uniform Fourier decay on $|\operatorname{Im} z| \leq 1/2$. Together with the zero-count estimate in (837), this also proves absolute convergence and passage to the spectral sum. No assumption about the location of the zeros is used.

23.4 The certified value

The one-worker Arb calculation uses 192-bit balls, $N = 8$, $T = 64$, prime integer cutoff 16, margin 3, and absolute/relative tolerance 2^{-90}. It integrates all 128 gamma cells and the three correlation cells for each prime power $n \leq 16$, then adds (1053), (1054), (1050) and (1051). The exact receipt is

```
agents/connes_actual_input_review/actual_theta_weil/actual_theta_weil_results.json
```

with checker hash

```
f937f8a919e9a45cf00db975064723c8c28af93a26e5376d306005ab98340a5b
```

The component enclosures are

$$\text{pole terms} \in [2.1389982672320679,\ 2.1389982672320681]\,10^{-7},$$
$$\text{prime terms through 16} \in [1.4681655033429545,\ 1.4681655033429548]\,10^{-8},$$
$$E_{\text{prime}} < 3.850\,10^{-43}, \qquad E_{\Gamma}(64) < 8.103\,10^{-27},$$

and the complete outward-rounded value is

$$W(wg_0, wg_0) = [2.0984855607004\,10^{-11} \pm 6.08\,10^{-25}] > 0. \tag{1055}$$

Thus the actual theta input is not a negative Weil witness. This positive value is separate from the already proved negative linear coefficient $K(\gamma)$ and from the uncanceled pole K/Ξ: one is a quadratic full-form value and the other is a linear shifted evaluation. The calculation establishes positivity for this one actual noncompact input, not global Weil positivity and not RH.

24 The source radial heat profile and its exact arithmetic transform

This section starts from the integral in Lemma A.6, equations (A.32)–(A.38), of the preserved 165-page edition of *Finite Time Blowup for Navier–Stokes* [7]. It concerns that explicit heat-exterior profile, not an independent verification of the manuscript's complete nonlinear construction. The integral, all dilation coefficients, and the construction's parameter range $0 < h < 1/100$ are retained.

24.1 The integral, its differential equation, and the physical field

Fix $0 < h < 1/100$, and put $A = 1/2 + h$. Define, on the real nonnegative half-line,

$$H_h(x) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1 + xv)^{-h}\, dv. \tag{1056}$$

Every power in this formula uses its positive real base. For $j \geq 0$, with $(h)_0 = 1$ and $(h)_j = h(h+1)\cdots(h+j-1)$, differentiation under the integral gives

$$H_h^{(j)}(x) = \frac{(-1)^j (h)_j}{\Gamma(1+h)} \int_0^\infty e^{-v} v^{h+j} (1+xv)^{-h-j}\, dv. \tag{1057}$$

Indeed, the absolute value of each differentiated integrand on $x \geq 0$ is bounded by a constant times $e^{-v}v^{h+j}$, which is integrable. Dominated convergence also gives the right endpoint derivatives and their continuity. Thus H_h is smooth on $[0,\infty)$, with

$$H_h(0) = 1, \qquad H_h^{(j)}(0) = (-1)^j (h)_j (1+h)_j.$$

This statement requires no convergent Taylor series at zero.

The exact source equation is

$$x^2 H_h''(x) + [1 + 2(1+h)x] H_h'(x) + h(1+h) H_h(x) = 0. \tag{1058}$$

Here is a direct proof retaining all its coefficients. Set

$$F(v) = e^{-v} v^h (1+xv)^{-h}, \qquad B(v) = e^{-v} v^{1+h} (1+xv)^{-1-h}.$$

Differentiation gives

$$B'(v) = F(v) \left\{ \frac{1+h}{(1+xv)^2} - \frac{v}{1+xv} \right\}.$$

Substitution from (1057) into the left side of (1058) gives $h\Gamma(1+h)^{-1} \int_0^\infty B'(v)\, dv$: the remaining factor reduces exactly by

$$(1+h) \left[1 - \frac{2xv}{1+xv} + \frac{x^2 v^2}{(1+xv)^2} \right] - \frac{v}{1+xv} = \frac{1+h}{(1+xv)^2} - \frac{v}{1+xv}.$$

Both boundary values of B vanish, by $h > 0$ at zero and exponential decay at infinity. This proves the equation, including $x = 0$ by right continuity.

Use a distinct symbol $\sigma = r^2/2$ for the physical radial variable, so that it is not confused with the Mellin argument below. The manuscript's heat exterior is

$$K(r,t) = c_\infty \sigma^{-A} H_h(2\tau/\sigma), \qquad \sigma = r^2/2, \quad \tau = 1 - t, \quad c_\infty > 0, \tag{1059}$$

on $r > 0,\ t < 1$. Its swirl term must be retained. To derive its physical equation, write $Z = 2\tau/\sigma$ and $W(\sigma,\tau) = \sigma^{-A} H_h(Z)$. Direct differentiation yields

$$\begin{aligned} W_\sigma &= \sigma^{-A-1}[-AH_h - ZH_h'], \\ W_{\sigma\sigma} &= \sigma^{-A-2}[A(A+1)H_h + 2(A+1)ZH_h' + Z^2 H_h''], \\ \partial_t W &= -2\sigma^{-A-1} H_h'. \end{aligned}$$

Since $\partial_r = r\partial_\sigma$,

$$(\partial_{rr} + r^{-1}\partial_r - r^{-2})W = 2\sigma W_{\sigma\sigma} + 2W_\sigma - \frac{W}{2\sigma} = 2\sigma^{-A-1}[Z^2 H_h'' + (2A+1)ZH_h' + (A^2 - 1/4)H_h].$$

The identities $2A + 1 = 2(1+h)$ and $A^2 - 1/4 = h(1+h)$, together with (1058), prove

$$\partial_t K = (\partial_{rr} + r^{-1}\partial_r - r^{-2})K. \tag{1060}$$

The equation concerns this explicitly defined exterior field; it does not replace the nonlinear convection, pressure, or stress corrections of the full construction.

24.2 Two exact dilation differences and their derivative bounds

Define

$$\phi_h(x) = H_h(x) - 2^h H_h(2x), \tag{1061}$$
$$\begin{aligned}\psi_h(x) &= \phi_h(x) - \tfrac{1}{2}\phi_h(x/2)\\ &= (1+2^{h-1})H_h(x) - 2^h H_h(2x) - \tfrac{1}{2}H_h(x/2).\end{aligned} \tag{1062}$$

All three source evaluations and coefficients in the last line are retained. We first prove the all-order bounds needed for every subsequent integral and inverse series.

For $a > 0$ put

$$I_a(\varepsilon) = \int_0^\infty e^{-v}\left(\frac{v}{v+\varepsilon}\right)^a dv.$$

For $0 < \varepsilon \le 1$,

$$0 \le 1 - I_a(\varepsilon) \le \varepsilon[1 + a + a\log(1/\varepsilon)]. \tag{1063}$$

To prove it, integrate

$$0 \le 1 - (1+\varepsilon/v)^{-a} \le \min(1, a\varepsilon/v).$$

The second inequality follows by integrating $a(1+w)^{-a-1} \le a$. On $0 < v < \varepsilon$ use the bound 1; on $\varepsilon < v < 1$ use $a\varepsilon/v$; on $v > 1$ use the latter bound with $e^{-v}/v \le e^{-v}$. Their integrals give (1063).

Let $c_j = (-1)^j (h)_j/\Gamma(1+h)$. Equation (1057) has the exact form

$$H_h^{(j)}(x) = c_j x^{-h-j} I_{h+j}(1/x),$$

and differentiation of (1061) therefore gives the cancellation

$$\phi_h^{(j)}(x) = c_j x^{-h-j}\{I_{h+j}(1/x) - I_{h+j}(1/(2x))\}.$$

Using (1063) in both terms, and then differentiating (1062), proves, for every $j \ge 0$,

$$|\phi_h^{(j)}(x)| + |\psi_h^{(j)}(x)| \le C_{h,j} x^{-1-h-j}(1+\log x) \qquad (x \ge 1). \tag{1064}$$

At zero their derivatives are bounded by (1057). In particular ϕ_h, ψ_h, and every derivative of each, are integrable. An ordinary substitution, now justified by that integrability, gives

$$\int_0^\infty \psi_h(x)\,dx = \int_0^\infty \phi_h(x)\,dx - \tfrac{1}{2}\,2\int_0^\infty \phi_h(y)\,dy = 0. \tag{1065}$$

Their endpoint values are

$$\phi_h(0) = 1 - 2^h, \qquad \psi_h(0) = \tfrac{1}{2}(1-2^h).$$

We will also need a lower-order asymptotic with a signed leading term. For fixed $a > 0$,

$$I_a(\varepsilon) = 1 - a\varepsilon\log(1/\varepsilon) + O_a(\varepsilon). \tag{1066}$$

The intervals $v < \varepsilon$ and $v > 1$ contribute $O_a(\varepsilon)$ to $1 - I_a$. On the middle interval Taylor's formula on $0 \le w \le 1$ gives $1 - (1+w)^{-a} = aw + O_a(w^2)$. Hence its contribution is

$$a\varepsilon\int_\varepsilon^1 \frac{e^{-v}}{v}\,dv + O_a\left(\varepsilon^2\int_\varepsilon^1 v^{-2}\,dv\right) = a\varepsilon\log(1/\varepsilon) + O_a(\varepsilon).$$

This proves (1066), rather than assuming an asymptotic expansion under a nonuniform integral. Substitution first into H_h, then into the two differences, yields

$$\begin{aligned}
H_h(x) &= \frac{x^{-h}}{\Gamma(1+h)} - \frac{h\,x^{-1-h}\log x}{\Gamma(1+h)} + O_h(x^{-1-h}),\\
\phi_h(x) &= -\frac{h\,x^{-1-h}\log x}{2\Gamma(1+h)} + O_h(x^{-1-h}),\\
\psi_h(x) &= a_h x^{-1-h}\log x + O_h(x^{-1-h}), \qquad a_h = \frac{h(2^h-1)}{2\Gamma(1+h)} > 0.
\end{aligned} \tag{1067}$$

24.3 The arithmetic trace, including every endpoint derivative

For $x > 0$, define the genuine arithmetic dilation sum

$$S_h(x) = \sum_{n\geq 1} \psi_h(nx). \tag{1068}$$

After j differentiations its n-th term is $n^j \psi_h^{(j)}(nx)$. On every compact subinterval of $(0,\infty)$, (1064) bounds it, apart from finitely many terms, by $Cn^{-1-h}(1+\log n)$. This is summable, so all differentiated series converge locally uniformly and S_h is smooth there. For $x \geq 1$, the same argument gives

$$S_h^{(j)}(x) = O_{h,j}(x^{-1-h-j}(1+\log x)). \tag{1069}$$

Using the uniform remainder in (1067) and summing absolutely further proves

$$S_h(x) = a_h\zeta(1+h)x^{-1-h}\log x + O_h(x^{-1-h}). \tag{1070}$$

Indeed $\sum n^{-1-h}\log n < \infty$, and the sum of the individual remainders is bounded by $C_h x^{-1-h}\zeta(1+h)$.

We prove the needed endpoint summation formula explicitly. Let $B_m(v)$ be the Bernoulli polynomials defined by $B_0 = 1$, $B_m' = mB_{m-1}$, and $\int_0^1 B_m(v)\,dv = 0$ for $m \geq 1$; write $B_m = B_m(0)$. In particular $B_1(v) = v - 1/2$. For $m \geq 2$ integration of B_m' shows $B_m(1) = B_m(0)$. Uniqueness of this defining recursion proves $B_m(1-v) = (-1)^m B_m(v)$, so $B_m(0) = 0$ for odd $m \geq 3$. Let $\widetilde{B}_m(v) = B_m(\{v\})$; the values at the integers are immaterial inside integrals.

Suppose f and its derivatives through order $2M$ are integrable, have the indicated one-sided values at zero, and tend to zero at infinity through order $2M-1$. Summing integration by parts over the intervals $[nx,(n+1)x]$ gives

$$\sum_{n\geq 1} f(nx) = x^{-1}\int_0^\infty f(v)\,dv - \frac{f(0)}{2} + \int_0^\infty \widetilde{B}_1(v/x)f'(v)\,dv.$$

For example, the integral on one interval equals the half-sum of its endpoint values minus x^{-1} times its integral. For every subsequent integration the identity $\frac{d}{dv}\widetilde{B}_m(v/x) = mx^{-1}\widetilde{B}_{m-1}(v/x)$ holds between endpoints. The endpoint values match for $m \geq 2$, and the odd endpoint constants vanish as just proved. Repeating these integrations gives

$$\begin{aligned}
\sum_{n\geq 1} f(nx) &= x^{-1}\int_0^\infty f(v)\,dv - \frac{f(0)}{2} - \sum_{k=1}^{M} \frac{B_{2k}}{(2k)!} f^{(2k-1)}(0)x^{2k-1} + R_M(f,x), \\
|R_M(f,x)| &\leq \frac{\|B_{2M}\|_{L^\infty[0,1]}}{(2M)!} x^{2M-1}\int_0^\infty |f^{(2M)}(v)|\,dv.
\end{aligned} \tag{1071}$$

The remainder is the negative of $x^{2M-1}(2M)!^{-1}\int \widetilde{B}_{2M}(v/x)f^{(2M)}(v)\,dv$; its displayed bound justifies the infinite endpoint limit as well.

For every fixed $j \geq 0$, apply this formula to

$$f_j(v) = v^j \psi_h^{(j)}(v), \qquad x^j S_h^{(j)}(x) = \sum_{n\geq 1} f_j(nx).$$

Each derivative $f_j^{(k)}$ is a finite sum of constant multiples of $v^{j-\ell}\psi_h^{(j+k-\ell)}(v)$, with $0 \leq \ell \leq \min(j,k)$. It is bounded near zero and $O(v^{-1-h-k}(1+\log v))$ at infinity, so every integrability and boundary assumption of (1071) holds. Integrating by parts j times, with every boundary term zero by these same estimates, gives

$$\int_0^\infty f_j(v)\,dv = (-1)^j j! \int_0^\infty \psi_h(v)\,dv = 0.$$

At the origin

$$f_j^{(r)}(0) = \begin{cases} 0, & r < j, \\ \dfrac{r!}{(r-j)!}\psi_h^{(r)}(0), & r \geq j. \end{cases} \tag{1072}$$

Thus, after division by x^j, every nonzero term of (1071) has a nonnegative power of x. Choose $2M-1 > j$; its remainder tends to zero after this division. This proves that $S_h^{(j)}$ is bounded at zero for every j and, more precisely, has a finite limit. With $C_{h,j}$ denoting that limit,

$$C_{h,0} = -\frac{\psi_h(0)}{2} = \frac{2^h-1}{4}, \qquad C_{h,j} = -\frac{B_{j+1}}{j+1}\psi_h^{(j)}(0) \quad (j \geq 1). \tag{1073}$$

For even $j \geq 2$ the last expression is zero. The formula follows from (1072): the possible constant term has $2k-1 = j$, and its coefficient is $-B_{j+1}j!/(j+1)!$.

These are actual smooth endpoint jets, not merely unrelated limits. For $0 < a < x$, the fundamental theorem of calculus gives $S_h^{(j)}(x) - S_h^{(j)}(a) = \int_a^x S_h^{(j+1)}(v)\,dv$. Boundedness permits $a \downarrow 0$, and continuity of the limiting derivative proves that extending each $S_h^{(j)}$ by $C_{h,j}$ preserves the derivative relation. Induction proves $S_h \in C^\infty([0,\infty))$. In particular

$$S_h(x) = \frac{2^h-1}{4} + O_h(x) \qquad (x \downarrow 0). \tag{1074}$$

The endpoint constant is positive. It is not a scalar Weil value.

24.4 The exact Mellin detector and its removable points

Use the Mellin convention

$$\mathcal{M}f(s) = \int_0^\infty f(x)x^{s-1}\,dx, \qquad x^{s-1} = \exp((s-1)\log x),$$

where $s \in \mathbb{C}$ and $\log x$ is real. For $0 < \Re s < h$, absolute interchange in (1056) and substitution $w = xv$ give

$$\mathcal{M}H_h(s) = G_h(s) := \frac{\Gamma(s)\Gamma(h-s)\Gamma(1+h-s)}{\Gamma(h)\Gamma(1+h)}. \tag{1075}$$

For completeness, the inner integral is $\int_0^\infty w^{s-1}(1+w)^{-h}dw$. In the absolutely convergent product integral $\Gamma(s)\Gamma(h-s)$, substitute $(a,b) = (ry, r(1-y))$, $r > 0$, $0 < y < 1$; its Jacobian is r. The result is $\Gamma(h)\int_0^1 y^{s-1}(1-y)^{h-s-1}dy$. Substitution $w = y/(1-y)$ proves that the inner integral is $\Gamma(s)\Gamma(h-s)/\Gamma(h)$. The remaining v-integral is $\Gamma(1+h-s)$, proving (1075).

For $a > 0$, substitution gives exactly $\mathcal{M}[f(a\,\cdot)](s) = a^{-s}\mathcal{M}f(s)$. Consequently, initially on $0 < \Re s < h$,

$$\mathcal{M}\phi_h(s) = (1-2^{h-s})G_h(s), \tag{1076}$$

$$\mathcal{M}\psi_h(s) = (1-2^{s-1})(1-2^{h-s})G_h(s). \tag{1077}$$

The bounds already proved make the left sides holomorphic on $0 < \Re s < 1+h$, with locally uniform logarithmic moments justifying all complex derivatives. On the right, the only gamma pole in that open strip is at $s = h$. The factor $1 - 2^{h-s}$ removes it, since

$$\lim_{s\to h}(1-2^{h-s})\Gamma(h-s) = -\log 2.$$

The identity theorem therefore proves (1076) and (1077) throughout the enlarged strip, without interchanging a divergent source integral. In particular their removable values are

$$\mathcal{M}\phi_h(h) = -\frac{\log 2}{\Gamma(1+h)}, \qquad \mathcal{M}\psi_h(h) = -\frac{(1-2^{h-1})\log 2}{\Gamma(1+h)}, \tag{1078}$$

$$\mathcal{M}\phi_h(1) = \frac{1-2^{h-1}}{h(h-1)}, \qquad \mathcal{M}\psi_h(1) = 0. \tag{1079}$$

The last equality agrees with the direct mean-zero calculation.

On the nonempty strip $1 < \Re s < 1+h$, the sum defining S_h can be integrated termwise absolutely: the sum of the absolute Mellin integrals is $\zeta(\Re s)\int_0^\infty |\psi_h(x)|x^{\Re s-1}dx < \infty$. It follows that

$$\mathcal{M}S_h(s) = \zeta(s)(1-2^{s-1})(1-2^{h-s})\frac{\Gamma(s)\Gamma(h-s)\Gamma(1+h-s)}{\Gamma(h)\Gamma(1+h)}. \tag{1080}$$

Equations (1074) and (1069) make its left side holomorphic on $0 < \Re s < 1+h$. The right side's point at h is removable as above, and its point at 1 is removable because $\zeta(s)$ has residue 1 there and $1-2^{s-1} = -(s-1)\log 2 + O((s-1)^2)$. The identity therefore holds on the full open strip. Its values are

$$\mathcal{M}S_h(h) = -\frac{\zeta(h)(1-2^{h-1})\log 2}{\Gamma(1+h)},$$

$$\mathcal{M}S_h(1) = \frac{(1-2^{h-1})\log 2}{h(1-h)} > 0. \tag{1081}$$

Here $\Gamma(h-1)/\Gamma(1+h) = 1/[h(h-1)]$ supplies the second value; no sign or scale is suppressed.

For fixed h, throughout

$$h < \Re s < 1 \tag{1082}$$

the multiplier of $\zeta(s)$ in (1080) is holomorphic and nowhere zero. The dilation factors can vanish only on the lines $\Re s = h$ and $\Re s = 1$, respectively. The gamma functions have no zeros, and their poles do not lie in (1082). If a point s_0 in that strip is a zero of order m of ζ, write $\zeta(s) = (s-s_0)^m a(s)$, $a(s_0) \neq 0$; multiplication by the nonvanishing holomorphic factor preserves this order. Division proves the converse. Thus the detector preserves every zero and its exact multiplicity on (1082). The formula itself does not supply a new zero.

The absolute Mellin strip is exactly $0 < \Re s < 1+h$. Sufficiency was proved above. Necessity follows from the positive nonzero endpoint constant (1074) at zero and the positive leading term (1070) at infinity: the absolute integral diverges for $\Re s \le 0$ at zero and for $\Re s \ge 1+h$ at infinity. This is the domain of the defining integral, not a denial of the meromorphic continuation supplied by (1080).

24.5 All three inverse maps and their convergence

The source kernel is recovered by the following explicit series:

$$\phi_h(x) = \sum_{j\geq 0} 2^{-j}\psi_h(x/2^j), \tag{1083}$$

$$H_h(x) = -\sum_{k\geq 1} 2^{-kh}\phi_h(x/2^k), \tag{1084}$$

$$\psi_h(x) = \sum_{n\geq 1} \mu(n)S_h(nx). \tag{1085}$$

The first finite sum, through $j = J$, telescopes exactly to

$$\phi_h(x) - 2^{-J-1}\phi_h(x/2^{J+1}).$$

The term subtracted tends to zero because ϕ_h is bounded near zero. The negative second finite sum, through $k = K$, equals

$$H_h(x) - 2^{-Kh}H_h(x/2^K);$$

its last term tends to zero because $h > 0$ and $H_h(0) = 1$. For derivative order ℓ, the terms of these two series have the additional factors $2^{-j\ell}$ and $2^{-k\ell}$, respectively. All source derivatives are bounded on bounded intervals, so the series and every differentiated series converge uniformly on such intervals. Their remainders vanish there as well. Thus the two inverse identities hold smoothly, including at zero. Boundedness at zero also makes the inverses unique: iterating either corresponding homogeneous dilation equation gives its value times 2^{-J}, respectively 2^{-Jh}, at a shrinking argument, and hence zero.

For the arithmetic inverse, at fixed $x > 0$ the double series is absolutely bounded by

$$\sum_{a,b\geq 1} |\psi_h(abx)| = \sum_{k\geq 1} d(k)|\psi_h(kx)| \leq C_x \sum_{k\geq 1} d(k)k^{-1-h}(1 + \log k) < \infty.$$

To verify the last convergence directly, set $k = ab$, use $\log(ab) = \log a + \log b$, and factor into products of $\sum a^{-1-h}$ and $\sum a^{-1-h}\log a$, both convergent by the integral test. Since $|\mu(a)| \leq 1$, exchanging sums is legitimate and gives

$$\sum_{a\geq 1} \mu(a)S_h(ax) = \sum_{k\geq 1} \psi_h(kx) \sum_{a|k} \mu(a) = \psi_h(x).$$

The divisor coefficient is 1 for $k = 1$, while for $k > 1$ it is $\prod_{p|k}(1-1) = 0$, by the defining squarefree expansion of μ. For derivative order ℓ, the double-series terms have the factor $(ab)^\ell\psi_h^{(\ell)}(abx)$; the same majorant works by (1064), locally uniformly in $x > 0$. Thus (1085) is also an equality of smooth functions with all derivatives, and is an actual inverse on this decay class. The three steps preserve the original source through the arithmetic sum; they are not merely identities between formal series.

24.6 The logarithmic Fourier map and its heat domain

For $0 < c < 1 + h$ define the logarithmic input

$$g_{h,c}(u) = e^{cu}S_h(e^u), \qquad u \in \mathbb{R}, \tag{1086}$$

and use the same minus-Fourier convention as the earlier sections. The change of variable $x = e^u$ gives

$$\mathscr{F} g_{h,c}(s) = \mathcal{M}S_h(c - is). \tag{1087}$$

This is an exact coordinate map, including its scale and vertical line. The domain of absolute convergence of the defining Fourier integral is exactly

$$-c < \Im s < 1 + h - c. \tag{1088}$$

Indeed $\Re(c - is) = c + \Im s$; the exact Mellin strip just proved therefore supplies both necessity and sufficiency.

The real input $g_{h,c}$ is a Schwartz function, despite the finite absolute complex strip. To check every seminorm, differentiate $e^{cu}S_h(e^u)$ repeatedly. Each derivative is a finite linear combination of $e^{cu}x^jS_h^{(j)}(x)$, $x = e^u$, with $0 \le j \le k$ for derivative order k. At $u \to -\infty$, every $S_h^{(j)}$ is bounded by the complete endpoint proof, so these derivatives are $O_{h,c,k}(e^{cu})$. At $u \to +\infty$, (1069) bounds them by

$$O_{h,c,k}((1 + u)e^{-(1+h-c)u}).$$

Multiplication by any power of $|u|$ preserves boundedness and decay at both ends. This proves all Schwartz seminorm estimates. Fourier inversion recovers $g_{h,c}$. To check the inversion directly on this domain, two integrations by parts give $\mathscr{F} g_{h,c}(s) = O((1+|s|)^{-2})$ on the real axis: the boundary terms vanish and $g''_{h,c}$ is integrable. Insert the factor $e^{-\varepsilon s^2}$, $\varepsilon > 0$, in the inverse integral. Absolute Fubini and the Gaussian integral identify it with

$$\int_{\mathbb{R}} \frac{e^{-(u-v)^2/(4\varepsilon)}}{\sqrt{4\pi\varepsilon}}\, g_{h,c}(v)\, dv.$$

This tends to $g_{h,c}(u)$: split at $|u - v| = \delta$, use continuity on the inner interval, and use boundedness of $g_{h,c}$ and the vanishing Gaussian tail outside it. On the Fourier side the integrable bound just proved permits $\varepsilon \downarrow 0$ by dominated convergence. Thus, explicitly on the real axis,

$$g_{h,c}(u) = \frac{1}{2\pi}\int_{\mathbb{R}} \mathcal{M}S_h(c - is)e^{isu}\, ds, \qquad S_h(x) = x^{-c}g_{h,c}(\log x).$$

Together with (1083)–(1085), these are inverse coordinates all the way back to H_h.

The earlier arithmetic heat convention multiplies a logarithmic input by $e^{tu^2/4}$, corresponding to generator $-\partial_s^2/4$ on its Fourier transform. On this actual input the domain must be checked, not inferred from its Schwartz property. Equation (1074) gives

$$g_{h,c}(u) \sim \frac{2^h - 1}{4} e^{cu} \qquad (u \to -\infty). \tag{1089}$$

For any complex t with $\Re t > 0$ and any complex s, the absolute integrand $|e^{tu^2/4}g_{h,c}(u)e^{-isu}|$ is therefore bounded below on a sufficiently negative half-line by a positive constant times

$$\exp\big((\Re t)u^2/4 + (c + \Im s)u\big).$$

Its integral diverges. In particular the source input does not belong to the superexponentially decreasing two-sided heat domain used for the theta input. This explicit endpoint obstruction concerns that defining integral, not every possible analytic continuation.

For $\Re t < 0$, on the other hand, set

$$F_{h,c}(s,t) = \int_{\mathbb{R}} e^{tu^2/4} g_{h,c}(u) e^{-isu}\, du. \tag{1090}$$

On any compact set of such (s,t), there is a uniform negative quadratic coefficient in u. Every differentiated integrand is bounded by an integrable Gaussian times an exponential-linear factor and a fixed polynomial in $|u|$, by the preceding endpoint estimates. Dominated differentiation proves joint holomorphy, entirety in s, and the exact equation

$$\partial_t F_{h,c}(s,t) = -\tfrac{1}{4}\partial_s^2 F_{h,c}(s,t). \tag{1091}$$

Both sides equal the integral with multiplier $u^2/4$. As t tends to zero through $\Re t < 0$, dominated convergence on each compact substrip of (1088) gives the boundary value (1087), with every s-derivative. On $\Re t = 0$ absolute convergence is again exactly (1088), since the heat multiplier has modulus one. No entire-time heat continuation is asserted.

The constructions above give an invertible, source-specific route from the actual radial heat exterior to an arithmetic transform with exactly the zeta zero multiplicities on (1082). They establish neither a new off-critical-line zero nor a negative Weil witness. The physical exterior equation (1060) and the valid spectral equation (1091) have both been proved with their actual domains; the full nonlinear Navier–Stokes construction has not been replaced by either one.

25 Exact arithmetic profiles driven by a supplied incompressible flow

25.1 Public source, retained input class and coordinates

The public manuscript *Finite Time Blowup for Navier–Stokes* [7, Theorem 1.1] claims a smooth forced velocity on $\mathbb{R}^3 \times [0,1)$, with one fixed compact spatial support, initial velocity zero, bounded kinetic energy and unbounded terminal supremum norm, for every $\nu > 0$. Its complete proof and formal verification are not imported as independently certified results in this section. We construct and prove an operator correspondence for its stated smooth input class. Section 26 applies the complete correspondence to the source's actual cutoff-summed corrected field. The correspondence itself is established below without assuming any unproved endpoint. The first source statement is an attribution; the formulas and estimates that follow are our proofs.

For attribution, Buckmaster's primary statement [8, p. 1] identifies his collaboration with Levent Alpöge on smooth-forced Euler and credits Diego Córdoba and Luis Martínez-Zoroa for the underlying forced-blowup programme. The revised NS manuscript [7, pp. 2–3] now discusses that programme and the hypodissipative extension with Zheng explicitly. These statements identify the respective work. No dependence of that historical programme on the polynomial coordinates below is asserted.

Let $\mathcal{U}_{K,T,\nu}$ be the set of smooth real triples (u,p,f) on $[0,T) \times \mathbb{R}^3$ satisfying, component by component,

$$\partial_t u_i + \sum_{j=1}^{3} u_j \partial_j u_i = \nu \sum_{j=1}^{3} \partial_j^2 u_i - \partial_i p + f_i, \qquad \sum_i \partial_i u_i = 0, \qquad \operatorname{supp} u(t,\cdot) \subset K, \tag{1092}$$

where $K \subset \mathbb{R}^3$ is compact, $0 < T < \infty$, and $\nu > 0$ is retained. We do not require compact support of p, f for the following map. The actual compactly supported source input in Section 26 satisfies these conditions on every preterminal interval. No assertion of existence of a singular member is part of the definition of this set.

In the original real polynomial coordinates $x = (x_1, x_2, x_3) = (x, y, w)$, retain every coefficient of F and put

$$S(x) = (F_1(x)/2, F_2(x), F_3(x))^{\mathsf{T}}, \quad J(x) = DS(x) = \operatorname{diag}(1/2, 1, 1)DF(x). \tag{1093}$$

The determinant proof in (959) gives $\det J = -1$, with the exact inverse

$$J^{-1} = (DF)^{-1} \operatorname{diag}(2, 1, 1) = [2W_1\ W_2\ W_3]. \tag{1094}$$

Thus $S_1 = \sigma$ is precisely the earlier arithmetic coordinate. The other two coordinates retain the entire original target. No global injectivity of S or F is used; J^{-1} is a pointwise matrix inverse. Both J and J^{-1} have polynomial entries. Here the columns $W_i = (DF)^{-1}e_i = -\frac{1}{2}\nabla F_{i+1} \times \nabla F_{i+2}$, with cyclic indices, are the full inverse columns proved in Section 14. The first-coordinate observation alone has the exact velocity kernel $\ker(d\sigma)_x = \operatorname{span}_{\mathbb{R}}\{W_2(x), W_3(x)\}$. Indeed $d\sigma(W_i) = \delta_{1i}/2$ and the W_i form a basis, by the full inverse matrix. The retained two additional target coordinates recover precisely those two directions in (1094).

25.2 The actual arithmetic heat function

We use the convention of [17, Section 1, equations (1)–(4)]:

$$\begin{aligned}
\Phi(u) &= \sum_{n\geq 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}}, \\
H_t(z) &= \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \\
Z_t(s) &= 8H_t(2s), \qquad Z_0(s) = \Xi(s) = \xi(1/2 + is), \\
\xi(\sigma) &= \tfrac{1}{2}\sigma(\sigma-1)\pi^{-\sigma/2}\Gamma(\sigma/2)\zeta(\sigma).
\end{aligned} \tag{1095}$$

The Mellin and spectral coordinates are related by $\sigma = 1/2 + is$, with inverse $s = -i(\sigma - 1/2)$. They are not silently identified.

We rederive the integral identity and its analytic bounds. Set

$$h(x) = \frac{\pi}{2}x^2(2\pi x^2 - 3)e^{-\pi x^2}, \quad S(\lambda) = \sum_{n\geq 1} h(n\lambda), \quad k(v) = e^{v/2}S(e^v).$$

For the Fourier convention $\widehat{h}(q) = \int_{\mathbb{R}} h(x)e^{-2\pi ixq}dx$, the Gaussian transform and its second and fourth derivatives give

$$\begin{aligned}
\widehat{x^2e^{-\pi x^2}} &= (1/(2\pi) - q^2)e^{-\pi q^2}, \\
\widehat{x^4e^{-\pi x^2}} &= (q^4 - 3q^2/\pi + 3/(4\pi^2))e^{-\pi q^2}.
\end{aligned}$$

Multiplying by $-3\pi/2$ and π^2 respectively gives $\widehat{h} = h$. Since $h(0) = 0$, Poisson summation gives $S(\lambda) = \lambda^{-1}S(1/\lambda)$ and $k(v) = k(-v)$. For $\lambda \geq 1$, $|S(\lambda)| \leq C\lambda^4 e^{-\pi\lambda^2}$, because the remaining series is bounded by a constant times $\sum_{n\geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. The symmetry gives the corresponding bound as $\lambda \downarrow 0$. Each fixed number of v derivatives introduces only finitely many additional powers of ne^v. It follows, with constants depending on the derivative order, that

$$|k^{(j)}(v)| \leq C_j e^{C_j|v|} e^{-\pi e^{2|v|}}. \tag{1096}$$

Termwise differentiation on compact sets is justified by the same summable bounds. On the negative half-line differentiate the proved evenness identity. This proves (1096) on both ends.

For $\Re\sigma > 1$, absolute integrability permits summation inside the Mellin integral. The substitution $q = \pi x^2$ gives

$$\begin{aligned}\int_0^\infty h(x)x^{\sigma-1}dx &= \pi^{-\sigma/2}\left[\tfrac{1}{2}\Gamma(\sigma/2+2) - \tfrac{3}{4}\Gamma(\sigma/2+1)\right] \\ &= \frac{\sigma(\sigma-1)}{8}\pi^{-\sigma/2}\Gamma(\sigma/2), \\ \int_0^\infty S(\lambda)\lambda^{\sigma-1}d\lambda &= \xi(\sigma)/4.\end{aligned}$$

The two-ended bounds make the last integral entire in σ, so the equality extends by analytic continuation. Substituting $\lambda = e^v$ and $\sigma = 1/2 + is$ proves

$$Z_t(s) = 4\int_{\mathbb{R}} e^{tv^2/4}k(v)e^{isv}\,dv. \tag{1097}$$

Indeed direct expansion also gives $\Phi(u) = 2k(2u)$; the substitution $v = 2u$ and the evenness of k carry the stated H_t integral to (1097), retaining the factors $2, 4, 8$. The bound (1096) dominates every fixed derivative uniformly on compact complex (t, s) sets. Thus Z is jointly entire and differentiation under its integral proves exactly

$$\partial_t Z_t = -\tfrac{1}{4}\partial_s^2 Z_t. \tag{1098}$$

This supplies the actual initial datum and the actual time orientation.

25.3 An explicit domain for the heat evolution

The differential expression alone does not specify a heat evolution on every entire-function space. We now give a precise invariant domain containing the actual kernel.

Definition 25.1. Let $\mathscr{E}_{\mathrm{nfl}}$ consist of $\phi \in C^\infty(\mathbb{R}, \mathbb{C})$ for which

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\,dv < \infty$$

for every nonnegative integer A, B, m, j. Give it these seminorms. Define

$$T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\,dv, \qquad \mathscr{A}_{\mathrm{nfl}} = T\mathscr{E}_{\mathrm{nfl}},$$

and equip $\mathscr{A}_{\mathrm{nfl}}$ with the transported topology.

The map T is injective. To give the needed uniqueness argument, let $T\phi$ vanish on the real line and let $g_\epsilon(v) = (4\pi\epsilon)^{-1/2}e^{-v^2/(4\epsilon)}$. Its Gaussian integral representation is

$$g_\epsilon(v) = (2\pi)^{-1}\int_{\mathbb{R}} e^{-\epsilon s^2}e^{isv}ds.$$

Fubini is valid because $\phi \in L^1$ and the Gaussian is integrable; it expresses $\phi * g_\epsilon$ as a Gaussian-weighted integral of the vanishing Fourier transform. Thus $\phi * g_\epsilon = 0$. As $\epsilon \downarrow 0$, these convolutions converge to ϕ in L^1: translations are continuous in L^1, first for continuous compactly supported functions by uniform continuity, then for all L^1 functions by approximation; splitting the Gaussian integral into a small translation ball and its vanishing tail proves the claim. Consequently $\phi = 0$ almost everywhere and, being continuous, everywhere. This also justifies the transported topology without an unspecified choice of representative.

Proposition 25.2 (Heat evolution with its inverse)**.** *For every complex t, multiplication $E_t\phi(v) = e^{tv^2/4}\phi(v)$ is a continuous automorphism of $\mathscr{E}_{\mathrm{nfl}}$. The operators $U_t = TE_tT^{-1}$ on $\mathscr{A}_{\mathrm{nfl}}$ satisfy*

$$\begin{gathered} U_tU_u = U_{t+u}, \qquad U_t^{-1} = U_{-t}, \qquad \partial_t U_t = -\tfrac{1}{4}\partial_s^2 U_t, \\ U_t\Xi = Z_t, \qquad U_t\partial_s = \partial_s U_t, \qquad U_tM_sU_{-t} = M_s - \tfrac{t}{2}\partial_s. \end{gathered} \tag{1099}$$

Here M_s is multiplication by the unchanged spectral coordinate.

Proof. Every v derivative of $e^{tv^2/4}\phi$ is a finite sum of $e^{tv^2/4}$ times a polynomial in v times a derivative of ϕ. Each target seminorm is bounded by finitely many source seminorms with larger A, m; on compact t sets these bounds are uniform. The inverse multiplier is $e^{-tv^2/4}$. Differentiation in t inserts $v^2/4$, while two s derivatives in the transform insert $-v^2$, proving the generator and the group identities. The kernel bound proves $k \in \mathscr{E}_{\mathrm{nfl}}$, so $U_t\Xi = Z_t$ follows from (1097).

Integration by parts gives $M_sT\phi = T(i\phi')$. For completeness the boundary terms vanish: each derivative of $e^{isv}\phi$ is integrable on a horizontal real-v line and the function itself is integrable, so its limits at both ends exist and are zero. The same argument applies after exponential weights. Thus multiplication by s preserves $\mathscr{A}_{\mathrm{nfl}}$; differentiation does also, since $\partial_sT\phi = T(iv\phi)$. Finally

$$e^{tv^2/4}i\partial_v(e^{-tv^2/4}\phi) = i\phi' - \tfrac{t}{2}iv\phi.$$

Applying T proves the last identity in (1099). All displayed operators have the domains just specified. □

In particular,

$$k(0) = \sum_{n\geq 1} \frac{\pi}{2}n^2(2\pi n^2 - 3)e^{-\pi n^2} > 0,$$

because each term is positive. The same exact derivative estimates imply that $T\phi$ and every spectral derivative are Schwartz on the real line for every $\phi \in \mathscr{E}_{\mathrm{nfl}}$. Indeed $\partial_s^jT\phi = T((iv)^j\phi)$, and integration by parts n times expresses $s^n\partial_s^jT\phi$ as $T(i^n\partial_v^n((iv)^j\phi))$. The boundary terms vanish by the weighted integrability argument above, and the last transform is bounded in absolute value by four times the L^1 norm of its amplitude. All pairs n, j are allowed, which gives every Schwartz seminorm with no change of Fourier factor.

25.4 An invertible arithmetic map on the full retained Fourier domain

Retain $c_0 = -59/275184$, $c_1 = 1/1296$, $c_2 = 3/132496$ from (1017). Define the unchanged full profile

$$K_\theta(s) = \mathcal{K}_\theta(s) = c_0Z_\theta(s+3) - \frac{Z_\theta'(s+3)}{336} + c_1Z_\theta(s+9/2) + c_2Z_\theta(s+13/2). \tag{1100}$$

The two symbols K_θ and $\mathcal{K}_\theta$ denote this identical function throughout the present section. Retain the exact spaces $\mathscr{E}_{\mathrm{nfl}}, \mathscr{A}_{\mathrm{nfl}}$ and transform T already defined above, including every seminorm

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\,dv, \qquad T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\,dv.$$

Define the source multiplier and arithmetic operator by

$$m(v) = c_0e^{3iv} - \frac{iv}{336}e^{3iv} + c_1e^{9iv/2} + c_2e^{13iv/2}, \tag{1101}$$

$$(\mathcal{T}_{\mathcal{D}}f)(s) = c_0f(s+3) - \frac{f'(s+3)}{336} + c_1f(s+9/2) + c_2f(s+13/2). \tag{1102}$$

Lemma 25.3 (A uniform exact lower bound for the original multiplier)**.** *For every real v,*

$$|m(v)| \geq \delta, \qquad \delta = \frac{575}{1752192} > 0. \tag{1103}$$

In particular its real-axis reciprocal exists everywhere, with no branch choice or removed spectral point.

Proof. Factoring the nonzero phase gives the exact identity

$$e^{-3iv} m(v) = c_0 - \frac{iv}{336} + c_1 e^{3iv/2} + c_2 e^{7iv/2}.$$

For $|v| \leq 1/2$, the elementary inequality $\cos x \geq 1 - x^2/2$ yields

$$\begin{aligned} \operatorname{Re}(e^{-3iv} m(v)) &= c_0 + c_1 \cos(3v/2) + c_2 \cos(7v/2) \\ &\geq c_0 + \frac{23}{32} c_1 - \frac{17}{32} c_2 = \frac{575}{1752192}. \end{aligned}$$

For completeness, this cosine bound follows directly by integration: for $x \geq 0$, $1 - \cos x = \int_0^x \sin u \, du \leq \int_0^x u \, du$, using $\sin u \leq u$; the latter follows from $u - \sin u = \int_0^u (1 - \cos w) \, dw \geq 0$. Evenness gives the negative case. Since the real part is positive, its displayed lower bound also bounds the modulus.

For $|v| \geq 1/2$, the triangle inequality with the original linear term gives

$$\begin{aligned} |m(v)| &\geq \frac{|v|}{336} - (|c_0| + c_1 + c_2) \\ &\geq \frac{1}{672} - \frac{10825}{10732176} = \frac{10291}{21464352} > \frac{575}{1752192}. \end{aligned}$$

All coefficients in these estimates are the original rational coefficients. The two intervals cover the whole real axis and prove (1103). $\square$

Theorem 25.4 (Exact arithmetic automorphism and heat conjugacy)**.** *Multiplication by m is a continuous automorphism of $\mathscr{E}_{\mathrm{nfl}}$. Its inverse is literal pointwise multiplication by $1/m$ on the unchanged real Fourier axis. The operator $\mathcal{T}_{\mathcal{D}}$ is a continuous automorphism of $\mathscr{A}_{\mathrm{nfl}}$, with exact inverse*

$$\mathcal{T}_{\mathcal{D}}^{-1} f = T\left(\frac{T^{-1} f}{m}\right). \tag{1104}$$

For every complex t,

$$\mathcal{T}_{\mathcal{D}} = T M_m T^{-1}, \qquad \mathcal{T}_{\mathcal{D}} U_t = U_t \mathcal{T}_{\mathcal{D}}, \qquad \mathcal{T}_{\mathcal{D}}^{-1} U_t = U_t \mathcal{T}_{\mathcal{D}}^{-1}, \tag{1105}$$

$$\mathcal{K}_t = \mathcal{T}_{\mathcal{D}} Z_t, \qquad Z_t = \mathcal{T}_{\mathcal{D}}^{-1} \mathcal{K}_t. \tag{1106}$$

Thus the actual arithmetic function is exactly recoverable from the propagated source coefficient, on this fully specified domain.

Proof. Every derivative of m has at most linear growth on the real axis. More explicitly, for $j \geq 0$ it satisfies $|m^{(j)}(v)| \leq C_j(1 + |v|)$, where one may take

$$C_j = |c_0| 3^j + c_1 (9/2)^j + c_2 (13/2)^j + \frac{3^j + \mathbf{1}_{j \geq 1} j 3^{j-1}}{336}.$$

The product rule gives the seminorm estimate

$$p_{A,B,m,j}(m\phi) \le \sum_{\ell=0}^{j} \binom{j}{\ell} C_\ell p_{A,B,m+1,j-\ell}(\phi).$$

This proves continuity of M_m on the original $\mathscr{E}_{\mathrm{nfl}}$.

Let $r(v) = 1/m(v)$, which is smooth by Lemma 25.3. The exact product identity $mr = 1$ and its derivatives give

$$r^{(j)} = -\frac{1}{m}\sum_{\ell=1}^{j} \binom{j}{\ell} m^{(\ell)} r^{(j-\ell)} \quad (j \ge 1), \qquad |r| \le \delta^{-1}.$$

Set $Q_0 = \delta^{-1}$ and recursively define the finite positive constants

$$Q_j = \delta^{-1}\sum_{\ell=1}^{j} \binom{j}{\ell} C_\ell Q_{j-\ell}.$$

Induction proves $|r^{(j)}(v)| \le Q_j(1+|v|)^j$: the term with index $\ell \ge 1$ has exponent at most $1+j-\ell \le j$, and the denominator is bounded by δ^{-1}. Therefore

$$p_{A,B,m,j}(r\phi) \le \sum_{\ell=0}^{j} \binom{j}{\ell} Q_\ell p_{A,B,m+\ell,j-\ell}(\phi).$$

This proves that reciprocal multiplication is continuous on exactly the same $\mathscr{E}_{\mathrm{nfl}}$. The two pointwise compositions are the identity, proving the Fourier automorphism and its inverse.

Applying the retained transform to each term of $m\phi$ gives the three translations and the derivative in (1102); the factor iv is exactly the Fourier multiplier of ∂_s. The injectivity of T already proved above makes its inverse on $\mathscr{A}_{\mathrm{nfl}}$ unambiguous. This proves the first identity in (1105) and (1104). Multiplication by m and by $1/m$ each commutes pointwise with multiplication by $e^{tv^2/4}$. All three maps preserve $\mathscr{E}_{\mathrm{nfl}}$, so applying T proves the two full-domain heat commutation identities for every complex t. Finally (1100) is exactly $\mathcal{T}_{\mathcal{D}} Z_t$, and applying the proved inverse gives (1106). □

25.5 Flow, inverse and volume form on every preterminal interval

For an input of (1092), define $X_t(q)$ by

$$\partial_t X_t(q) = u(t, X_t(q)), \qquad X_0(q) = q.$$

Here $q \in \mathbb{R}^3$ is the retained initial label and $x = X_t(q)$ the physical coordinate. This is a global smooth diffeomorphism for each $t < T$. To prove existence with its actual domain, fix $T_0 < T$. On $[0, T_0] \times K$, u and every spatial derivative are bounded by compactness; they vanish outside K. In particular u is globally Lipschitz in space, uniformly on this time interval. On a sufficiently short time interval, the map $Y \mapsto q + \int_0^t u(s, Y(s))\,ds$ is a contraction in the uniform norm on continuous curves, because its Lipschitz constant is at most the interval length times $\sup|Du|$. Successive contractions cover $[0, T_0]$; bounded u prevents escape in each such interval. The same argument backward in time gives the inverse flow. Differentiation of the integral equation gives successive linear integral equations for each label derivative, proving their existence and continuity by the same local iteration, and proves smoothness by induction. Since T_0 is arbitrary this proves all the assertions on $[0, T)$.

Set $A = D_q X_t$. Differentiation gives

$$A' = Du(t, X_t)A, \qquad A(0) = I_3, \qquad (\det A)' = (\operatorname{div} u)(t, X_t) \det A = 0.$$

The determinant identity follows first near $t = 0$ by the product rule for the determinant and A^{-1}; it gives $\det A = 1$, which prevents loss of invertibility and continues that identity to every $t < T$. Thus $\det A = 1$ with its original positive orientation. Every point outside K has the constant trajectory, by uniqueness. If a trajectory starting in K ended outside K, the backward constant trajectory from its endpoint would contradict the inverse-flow uniqueness. Consequently

$$X_t(K) = K, \qquad X_t(q) = q \ (q \notin K), \qquad X_t(L) \subset K \cup L \quad \text{for every compact } L \subset \mathbb{R}^3. \tag{1107}$$

The inverse equality follows in the same way. These conclusions require no bound on $\sup_{t<T} |u|$.

25.6 An invertible profile correspondence with a complex spectral coordinate

Write $\mathcal{B} = C^\infty(\mathbb{R}^3; \mathcal{O}(\mathbb{C}_\zeta))$, with seminorms given by uniform suprema of finitely many real q derivatives and complex ζ derivatives on compact products. Let

$$s_{j,t}(q) = S_j(X_t(q)), \qquad \mathcal{P}_t : \mathcal{O}(\mathbb{C})^3 \longrightarrow \mathcal{B}^3, \quad (h_j)_j \longmapsto \big(h_j(s_{j,t}(q) + \zeta)\big)_j.$$

Every $s_{j,t}$ is real. The domain variables of h_j are complex; the spatial labels remain real, since a general smooth NS velocity need not have a holomorphic extension in space. The map on the full product is explicitly

$$H_{j,t} : \mathbb{R}^3 \times \mathbb{C}_\zeta \longrightarrow \mathbb{R}^3 \times \mathbb{C}_s, \quad (q, \zeta) \mapsto (q, s_{j,t}(q) + \zeta), \qquad H_{j,t}^{-1}(q, s) = (q, s - s_{j,t}(q)). \tag{1108}$$

These are smooth in the real variables and biholomorphic on every complex fibre. For any fixed label q_*, an exact left inverse of $\mathcal{P}_t$, and inverse on its image, is

$$(\mathcal{R}_t G)_j(s) = G_j(q_*, s - s_{j,t}(q_*)), \qquad \mathcal{R}_t \mathcal{P}_t = I. \tag{1109}$$

Substitution proves both directions on the image. On each compact product the image argument stays in a compact subset of $\mathbb{C}$, and all the finitely many derivatives of $s_{j,t}$ involved are bounded. The chain and product rules prove continuity of $\mathcal{P}_t$; evaluation and translation prove continuity of $\mathcal{R}_t$. Thus the image carries a topology for which the stated maps are an actual topological isomorphism, namely its subspace topology in $\mathcal{B}^3$. The map retains the full original complex profile.

For comparison with the earlier spatial observable, let $e_0(q) = (q, 0)$. The precise restriction is

$$e_0^*(\mathcal{P}_t h)_1(q) = h_1(\sigma(X_t(q))) = X_t^*(\sigma^* h_1)(q).$$

The inverse in (1109) uses the complex fibre; no holomorphic spatial flow is silently supplied to this real restriction.

25.7 Full first and second time generators, and Euclidean diffusion

Let $\vartheta : [0, T) \to \mathbb{R}$ be smooth. It is the chosen Newman-time map, independent of physical t; no value for it is inferred from NS. Take either $h_\theta = Z_\theta$ or the full four-term coefficient $h_\theta = K_\theta$ of Section 19, so that $\partial_\theta h_\theta = -\partial_s^2 h_\theta / 4$ with its original factor and sign. Set

$$\Psi_j(t, q, \zeta) = h_{\vartheta(t)}(s_{j,t}(q) + \zeta), \qquad v_j(t, q) = [\nabla S_j \cdot u](t, X_t(q)).$$

Then $\partial_t s_{j,t} = v_j$ and the chain rule proves the exact equation

$$\partial_t \Psi_j = -\frac{\vartheta'}{4}\partial_\zeta^2 \Psi_j + v_j \partial_\zeta \Psi_j. \tag{1110}$$

The exact connection on the moving image of $\mathcal{P}_t$ is $\nabla_t = \partial_t - \mathrm{diag}(v_j)\partial_\zeta$: $\nabla_t(\mathcal{P}_t h_t) = \mathcal{P}_t(\partial_t h_t)$. This follows by the same chain rule for every smooth curve of entire profiles. It is a connection between the displayed moving images; multiplication by $v_j(q)$ is not being asserted to preserve a fixed image of the profile space. There is no ζ dependence in v_j. Differentiating this complete equation once more, including the time derivatives of its coefficients, gives

$$\partial_t^2 \Psi_j = \frac{(\vartheta')^2}{16}\partial_\zeta^4 \Psi_j - \frac{\vartheta' v_j}{2}\partial_\zeta^3 \Psi_j + \left(v_j^2 - \frac{\vartheta''}{4}\right)\partial_\zeta^2 \Psi_j + a_j \partial_\zeta \Psi_j, \tag{1111}$$

where the acceleration coefficient is, with all NS terms retained,

$$a_j = \partial_t v_j = \left[\sum_i (\nu \Delta u_i - \partial_i p + f_i)\partial_i S_j + \sum_{i,l} u_i u_l \partial_i \partial_l S_j\right](t, X_t(q)). \tag{1112}$$

Indeed apply $\partial_t + u \cdot \nabla_x$ to $\sum_i u_i \partial_i S_j$ and substitute (1092); the derivative of $\partial_i S_j$ is exactly the displayed Hessian term. In (1111), the cubic derivative occurs twice, each with coefficient $-\vartheta' v_j/4$, proving its coefficient $-\vartheta' v_j/2$.

The full physical Laplacian has an exact material counterpart. Define

$$G_t = A^{-1}A^{-\mathsf{T}}, \qquad \mathcal{L}_t = \sum_{a,b} \partial_{q_a}(G_t^{ab}\partial_{q_b}).$$

For every smooth scalar $b(x)$,

$$\mathcal{L}_t(X_t^* b) = X_t^*(\Delta_x b). \tag{1113}$$

Here is a proof retaining the first-order terms in $\mathcal{L}_t$. The gradient chain rule gives $\nabla_q X_t^* b = A^{\mathsf{T}}(\nabla_x b) \circ X_t$. For a compactly supported smooth test ϕ, integration by parts and the change of variables $x = X_t(q)$, whose Jacobian is one, give

$$\int \phi\, \mathcal{L}_t X_t^* b \, dq = -\int \nabla_q \phi \cdot A^{-1}(\nabla_x b) \circ X_t \, dq = -\int \nabla_x(\phi \circ X_t^{-1}) \cdot \nabla_x b \, dx = \int \phi\, X_t^* \Delta_x b \, dq.$$

All functions are smooth, so the distributional equality is pointwise. Apply this at fixed ζ to the full profile to obtain

$$\mathcal{L}_t \Psi_j = (|\nabla S_j|^2 \circ X_t)\partial_\zeta^2 \Psi_j + (\Delta S_j \circ X_t)\partial_\zeta \Psi_j. \tag{1114}$$

Consequently its complete advective viscous residual is

$$(\partial_t - \nu \mathcal{L}_t)\Psi_j = -\left(\frac{\vartheta'}{4} + \nu |\nabla S_j|^2 \circ X_t\right)\partial_\zeta^2 \Psi_j + (v_j - \nu \Delta S_j \circ X_t)\partial_\zeta \Psi_j.$$

These formulas identify the two differential operators through an explicit coordinate map and calculate their remaining coefficients; they do not delete the viscosity or equate two time variables.

25.8 Recovering the actual velocity from the arithmetic temporal defect

For real θ the functions Z_θ, K_θ and their spectral derivatives are Schwartz on the real s line, by the weighted Fourier space proved above. With the original real kernel k and multiplier m, their Fourier representations are

$$Z_\theta(s) = 4\int_{\mathbb{R}} k(\omega)e^{\theta\omega^2/4}e^{is\omega}\,d\omega, \qquad K_\theta(s) = 4\int_{\mathbb{R}} m(\omega)k(\omega)e^{\theta\omega^2/4}e^{is\omega}\,d\omega.$$

No source energy, derivative channel, or Fourier factor is changed. Define $d_h(\theta)^2 = \int_{\mathbb{R}} |h'_\theta(s)|^2\,ds$. Then

$$\begin{aligned} d_Z(\theta)^2 &= 32\pi\int_{\mathbb{R}} \omega^2 k(\omega)^2 e^{\theta\omega^2/2}\,d\omega, \\ d_K(\theta)^2 &= 32\pi\int_{\mathbb{R}} \omega^2|m(\omega)|^2 k(\omega)^2 e^{\theta\omega^2/2}\,d\omega. \end{aligned} \tag{1115}$$

For completeness, the factor follows without a Fourier convention change. For a Schwartz amplitude b, insert $e^{-\epsilon s^2}$ in the squared norm of $4\int b(\omega)e^{is\omega}d\omega$ and use Fubini. The result is

$$16\sqrt{\pi/\epsilon}\iint b(\omega)\overline{b(\eta)}e^{-(\omega-\eta)^2/(4\epsilon)}d\omega d\eta.$$

The Gaussian convolution, with mass 2π, tends to $2\pi b$; splitting at any fixed small distance and using continuity and the Gaussian tail proves this convergence under the integral. Thus the limit is $32\pi\int|b|^2$. On the original side dominated convergence applies because the transform is Schwartz. Use respectively $b = i\omega k e^{\theta\omega^2/4}$ and $b = i\omega m k e^{\theta\omega^2/4}$ to prove (1115).

Both numbers are strictly positive. The previously proved $k(0) > 0$ and continuity give an interval of nonzero k containing nonzero ω. For the second integral the established full-real bound $|m| \geq 575/1752192$ applies. The weighted bounds also prove continuity of these integrals in θ by domination on each compact parameter interval. Hence for each compact interval $I \subset \mathbb{R}$ there are finite constants $0 < d_- \leq d_h(\theta) \leq d_+ < \infty$ on I.

Define the actual arithmetic temporal defect by

$$D_j(t,q,s) = \partial_t\Psi_j(t,q,s) + \frac{\vartheta'}{4}\partial_s^2\Psi_j(t,q,s), \qquad s \in \mathbb{R}. \tag{1116}$$

Equation (1110) gives $D_j = v_j h'_\vartheta(s + s_{j,t})$. Real translation preserves Lebesgue measure, so

$$\sum_j \|D_j(t,q,\cdot)\|^2_{L^2(\mathbb{R})} = d_h(\vartheta(t))^2|v(t,q)|^2. \tag{1117}$$

Moreover the inverse is a literal integral followed by the original matrix inverse:

$$\begin{aligned} v_j &= \frac{\int_{\mathbb{R}} \overline{h'_\vartheta(s+s_{j,t})}D_j(t,q,s)\,ds}{d_h(\vartheta)^2}, \\ u(t,X_t(q)) &= 2W_1(X_t(q))v_1 + W_2(X_t(q))v_2 + W_3(X_t(q))v_3. \end{aligned} \tag{1118}$$

Substitution proves both inverse identities. The correspondence retains $X_t(q)$ as part of both sides: $(X_t, u\circ X_t) \leftrightarrow (X_t, D)$. The data D alone are not claimed to select a globally unique inverse sheet of the polynomial map. At fixed (t,q), the formula identifies $\mathbb{R}^3$ with the three-dimensional real subspace of $L^2(\mathbb{R};\mathbb{C})^3$ spanned componentwise by the three displayed shifted profiles. It is not merely an analogy between their norms.

For a nonempty K, set $M = \max_K \|J\|_{\mathrm{op}}$ and $N = \max_K \|J^{-1}\|_{\mathrm{op}}$. They are finite and positive. If $\vartheta([0,T)) \subset I$, the exact map satisfies, for all $t < T$,

$$\frac{d_-}{N}\|u(t)\|_{L^\infty(\mathbb{R}^3)} \le \sup_q \left(\sum_j \|D_j(t,q,\cdot)\|_2^2\right)^{1/2} \le d_+ M\|u(t)\|_{L^\infty(\mathbb{R}^3)}. \tag{1119}$$

Indeed $|Ju| \le M|u|$ and $|u| \le N|Ju|$ on K, while both u and v vanish off K; (1107) preserves K and surjectivity preserves the supremum. Combine this with (1117). This proves equivalence of unbounded terminal velocity and unbounded supremum of this specific arithmetic temporal-defect norm for every input in the defined class. It supplies an exact detection map even though no singular member is established here. At $u = 0$ all these defects vanish, as the same formulas require. The full space–spectral energy has an exact identity too. Since $v = 0$ off K, (1117) is integrable in q and the volume-preserving change of variables gives

$$\sum_j \int_{\mathbb{R}^3}\int_{\mathbb{R}} |D_j(t,q,s)|^2\,ds\,dq = d_h(\vartheta(t))^2 \int_{\mathbb{R}^3} |J(x)u(t,x)|^2\,dx.$$

The same constants therefore bound its square root between $(d_-/N)\|u(t)\|_2$ and $d_+M\|u(t)\|_2$. In particular the map transfers a uniform kinetic-energy bound into a uniform joint $L^2(q,s)$ bound, while (1119) detects concentration through the separate supremum in q. This conclusion follows from the displayed integrals without a spatial truncation of the defect. In particular $\vartheta(t) = 0$ for every physical time is an explicit allowed choice. Then the entire input stays equal to the actual Ξ (or K_0), and $D_j = \partial_t\Psi_j$; the constants in (1119) use the single strictly positive value $d_h(0)$.

For an actual separated-link pair, the full coefficient is $a_{ef}K$. The defect is then $a_{ef}D^K$, and its squared norm acquires exactly a_{ef}^2; division in (1118) is available for $a_{ef} \neq 0$. If $a_{ef} = 0$ this pair supplies the zero observation, not a spurious inverse. For a pair with $a_{ef} > 0$, recovery of Z from the profiles of K additionally uses the already proved full-domain inverse $\mathcal{T}_{\mathcal{D}}^{-1}$; its translation and derivative commutation hold because multiplication by m commutes with $e^{is_{j,t}\omega}$ and $i\omega$ in the unchanged Fourier domain.

25.9 Bounded profile values and the exact transported zero divisor

For compact label set L, compact spectral set $E \subset \mathbb{C}$, and bounded $\vartheta([0,T)) \subset I$, (1107) puts every argument $s_{j,t}(q) + \zeta$ in the compact set $S_j(K \cup L) + E$. The jointly entire Z and K are bounded on that set times I. Therefore

$$\sup_{t<T,\ q\in L,\ \zeta\in E} |\Psi_j(t,q,\zeta)| < \infty.$$

The identical argument bounds every pure spectral derivative. It does not bound time or spatial-label derivatives; (1119) specifies a quantity that can grow even while these values stay bounded.

Let $h_\theta(s)$ have a zero at z_* of multiplicity m and write its actual local factorization $h_\theta(s) = (s - z_*)^m A(s)$, $A(z_*) \neq 0$. The full inverse of (1108) takes that zero to

$$\zeta_*(q) = z_* - s_{j,t}(q), \qquad \Psi_j = (\zeta - \zeta_*(q))^m A(s_{j,t}(q) + \zeta).$$

Thus it preserves multiplicity and the whole analytic unit on every complex fibre. Its complete jet formula is

$$\partial_\zeta^{m+n}\Psi_j(t,q,\zeta_*(q)) = \frac{(m+n)!}{n!}A^{(n)}(z_*), \qquad n \ge 0,$$

with all lower derivatives zero. This follows by multiplying the Taylor series of the unchanged A by $(\zeta-\zeta_*)^m$; no unit coefficient has been discarded. Since $s_{j,t}(q)$ is real, $\operatorname{Im}\zeta_*(q)=\operatorname{Im}z_*$. At Newman time $\vartheta=0$ with $h=Z$, the full map to the completed-zeta argument is

$$\rho=\tfrac{1}{2}+i(s_{j,t}(q)+\zeta),\qquad \zeta=-i(\rho-\tfrac{1}{2})-s_{j,t}(q),\qquad \operatorname{Re}\rho=\tfrac{1}{2}-\operatorname{Im}\zeta.$$

Consequently this NS-induced real translation preserves each zero's distance from the critical line exactly; the full arithmetic time evolution is still the specified Z_ϑ. For $h=K$ the same coordinate proof transports the zeros of the four-term profile; these are converted to a statement about Z only by actually applying the proved inverse operator to the whole profile. The inverse does not identify individual zeros of two different entire functions. These explicit maps give both the velocity detection and the precise zero transport of the construction, without deducing an off-critical zero from growth of the temporal defect.

26 Using the released Navier–Stokes construction as the arithmetic input

26.1 The precise imported witness and its unchanged viscosity

We now use the particular field constructed in the released manuscript [7], not merely an arbitrary member of the input class of Section 25. The source edition used in this section has 166 pages and is the later revision at the public manuscript URL in [7]; it is not identified with the 165-page profile edition used in Section 24. The complete primary file is retained in the local source collection, with its exact identity and source-task reading record in the machine-readable provenance. Its primary PDF is cited, not reproduced in this reader. The source existence and all-order residual construction are imported from that manuscript. The source task's independent reading of the witness and closing argument does not amount to independent verification of every source lemma. The formulas below specify the imported object and prove its arithmetic image in full.

Use a star to retain the source's viscosity-one fields, rather than overwriting the physical viscosity $\nu>0$. In this section fields are written with time first, $u_\nu(t,x)$, as in Section 25. The source's spatial-first notation is related by the explicit coordinate permutation $(t,x)\mapsto(x,t)$, with inverse $(x,t)\mapsto(t,x)$; thus $u_\nu(t,x)$ denotes exactly the source field evaluated at (x,t). The same permutation is used for p_ν, f_ν and starred fields. No spatial coordinate, physical parameter or time value changes. In source equation (9.21), page 115, the actual summed Cartesian vector potential, azimuthal field, and pressure are denoted by $A_*, B_*e_\theta, p_{*,\mathrm{loc}}$. These are the cutoff-summed corrected fields of that equation, including all stages and the finite initialization. Source equation (10.4), page 118, constructs

$$u_*=\operatorname{curl}(c_*A_*)+c_*B_*e_\theta,\qquad p_*=c_*p_{*,\mathrm{loc}},\qquad c_*=\chi_x\chi_t.$$

Here the source spatial cutoff is smooth in (r^2,z), supported in $r<r_0,\ |z|<z_0$, and equals one on $r\le r_0/2,\ |z|\le z_0/2$. Its time cutoff is zero for $1-t\ge\tau_0$ and one for $0\le 1-t\le\tau_0/2$, where $0<\tau_0<1/2$. Retain the actual source choices of these parameters. The source force before time one is exactly

$$f_*=\partial_t u_*+(u_*\cdot\nabla)u_*-\Delta u_*+\nabla p_*.$$

Thus every cutoff derivative belongs to the force. Its smooth extension through time one is source (10.11), page 120, with all its prescribed Taylor coefficients and cutoff sequence. The source proves that its support is contained in $K_*\times[0,2]$, where $K_*=\operatorname{supp}\chi_x$, and it vanishes near time zero. The pressure is the displayed compactly supported source pressure; no divergence-free condition is imposed on f_*.

For the physical parameter $\nu > 0$, take precisely source (10.22):

$$\begin{aligned} y &= x/\sqrt{\nu}, & x &= \sqrt{\nu}\, y, \\ u_\nu(t,x) &= \sqrt{\nu}\, u_*(t, x/\sqrt{\nu}), & p_\nu(t,x) &= \nu p_*(t, x/\sqrt{\nu}), \\ f_\nu(t,x) &= \sqrt{\nu}\, f_*(t, x/\sqrt{\nu}), & K_\nu &= \sqrt{\nu}\, K_*. \end{aligned}$$

The inverse multiplies the velocity and force by $\nu^{-1/2}$, the pressure by ν^{-1}, and substitutes $x = \sqrt{\nu}\, y$; physical time is unchanged. The chain rule gives

$$\partial_t u_\nu + (u_\nu \cdot \nabla_x) u_\nu - \nu \Delta_x u_\nu + \nabla_x p_\nu = \sqrt{\nu}\, [\partial_t u_* + (u_* \cdot \nabla_y) u_* - \Delta_y u_* + \nabla_y p_*](t,y) = f_\nu(t,x).$$

Also $\operatorname{div}_x u_\nu = \operatorname{div}_y u_* = 0$, $u_\nu(0,\cdot) = 0$, and u_ν, p_ν have support in K_ν for every $t < 1$. These are exactly the inputs of Section 25 with $T = 1$, with all coordinate and parameter maps specified.

Let

$$F_\nu(t) = \int_0^t \|f_\nu(s)\|_{L^2(\mathbb{R}^3)}\, ds, \qquad E_\nu = F_\nu(1) = \nu^{5/4} F_*(1) < \infty.$$

The last exponent follows from the factor $\sqrt{\nu}$ in f_ν and $dx = \nu^{3/2} dy$. For completeness the energy estimate used below is proved directly for this input. Compact support and incompressibility give, by integrating the equation against u_ν,

$$\frac{1}{2}\frac{d}{dt}\|u_\nu(t)\|_2^2 + \nu \|\nabla u_\nu(t)\|_2^2 = \langle f_\nu(t), u_\nu(t)\rangle.$$

The convection integral is the integral of $\operatorname{div}(u_\nu |u_\nu|^2/2)$, and the pressure integral is minus the integral of $p_\nu \operatorname{div} u_\nu$; both vanish. Divide by $(\|u_\nu\|_2^2 + \delta^2)^{1/2}$, use Cauchy–Schwarz, integrate from zero, and let $\delta \downarrow 0$. This proves $\|u_\nu(t)\|_2 \le F_\nu(t)$. Substitution back into the integrated identity proves, with its full dissipation coefficient,

$$\|u_\nu(t)\|_2^2 + 2\nu \int_0^t \|\nabla u_\nu(s)\|_2^2 ds \le 2 \int_0^t F_\nu'(s) F_\nu(s) ds = F_\nu(t)^2.$$

26.2 The source's actual growing component in two arithmetic channels

The source fixes $0 < h < 1/100$, $X_{\text{in}} > 0$, and $e_0 > 0$ through its construction. Its closing proof, (10.20)–(10.21), page 124, gives the actual corrected swirl along

$$x_{*,\tau} = (\sqrt{2X_{\text{in}}\tau}, 0, 0), \qquad t = 1 - \tau, \qquad (u_*)_2(1-\tau, x_{*,\tau}) = \tau^{-1/2-h}(e_0 + R_*(\tau)).$$

The original remainder is retained as the function $R_*(\tau) = \tau^{1/2+h}(u_*)_2(1-\tau, x_{*,\tau}) - e_0$. The source estimate supplies constants $C_* > 0, \tau_* > 0$ with $|R_*(\tau)| \le C_* \tau^{2h}$ for $0 < \tau < \tau_*$. At these points the cylindrical angle is zero, so $e_\theta = (0,1,0)$ and the displayed Cartesian component is precisely the source swirl. The annular and higher-order corrections have already been accounted for in this remainder. The parameter h is the chosen source parameter; no claim that an arbitrary prescribed h works is needed or made.

Define

$$r_{\nu,\tau} = \sqrt{2\nu X_{\text{in}}\tau}, \qquad x_{\nu,\tau} = (r_{\nu,\tau}, 0, 0).$$

The exact viscosity map proves

$$(u_\nu)_2(1-\tau, x_{\nu,\tau}) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{1120}$$

Use the actual material flow $X_{\nu,t}$ of u_ν proved to exist on each interval before one in Section 25. Put

$$X_{\nu,t}(q) = \sqrt{\nu}\, X_{*,t}(q/\sqrt{\nu}).$$

This is an equality of the actual flows: its right side equals q at zero, and its time derivative is $\sqrt{\nu}\, u_*(t, X_{*,t}(q/\sqrt{\nu})) = u_\nu(t, \sqrt{\nu}\, X_{*,t}(q/\sqrt{\nu}))$. The previously proved uniqueness gives the equality, with inverse $X_{\nu,t}^{-1}(x) = \sqrt{\nu}\, X_{*,t}^{-1}(x/\sqrt{\nu})$. Define the labels

$$q_{\nu,\tau} = X_{\nu,1-\tau}^{-1}(x_{\nu,\tau}).$$

This is the exact inverse between the Eulerian sampling points and their material labels. It is a family of labels depending on τ; the source path has not been asserted to be a single fluid trajectory.

Retain $S = (F_1/2, F_2, F_3)$ and $J = DS$ in the physical Cartesian coordinates x, not the viscosity-one coordinate y. Direct substitution into the original polynomials gives

$$S(r,0,0) = (0,0,2r), \qquad J(r,0,0) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 1 & 3r \\ 2 & -3r^2 & -r^3 \end{pmatrix}.$$

Indeed the first row comes from differentiating $F_1/2 = ((1+xy)^3 w + y^2(1+xy)(4+3xy))/2$; the second comes from $F_2 = y + 3x(1+xy)^2 w + 3xy^2(4+3xy)$; the third comes from $F_3 = 2x - 3x^2 y - x^3 w$, all evaluated at $(x,y,w) = (r,0,0)$. Every entry, including the off-diagonal powers of r, is retained. Writing $v = J(r,0,0)u$ therefore gives the exact component inverse

$$u_3 = 2v_1, \qquad u_2 = v_2 - 6rv_1,$$
$$u_1 = \tfrac{1}{2}v_3 + \tfrac{3}{2}r^2 v_2 - 8r^3 v_1.$$

Substituting these expressions into all three rows proves both directions of this matrix inverse.

Fix actual arithmetic time zero. Take $H = Z_0 = \Xi$ or $H = K_0$ with the complete four-term coefficient from the preceding proof, and write

$$d_H^2 = \int_{\mathbb{R}} |H'(s)|^2 ds > 0, \qquad \Psi_{\nu,j}(t,q,\zeta) = H(S_j(X_{\nu,t}(q)) + \zeta), \qquad D_{\nu,j} = \partial_t \Psi_{\nu,j}.$$

The exact positive constants are those of (1115) at $\theta = 0$, including 32π and, for K_0, the entire factor $|m|^2$. At the particular pair $(t,q) = (1-\tau, q_{\nu,\tau})$, the first two shifts are both zero. Consequently define the actual integral

$$I_{\nu,j}(\tau) = \frac{1}{d_H^2} \int_{\mathbb{R}} \overline{H'(s)} D_{\nu,j}(1-\tau, q_{\nu,\tau}, s) ds, \qquad j = 1,2.$$

The chain rule gives $D_{\nu,j} = v_j H'(s)$ there. It follows, without an asymptotic replacement of the profiles, that

$$I_{\nu,2}(\tau) - 6r_{\nu,\tau} I_{\nu,1}(\tau) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{1121}$$

There is also a whole-profile identity, stronger than its projection:

$$\begin{aligned} &D_{\nu,2}(1-\tau, q_{\nu,\tau}, s) - 6r_{\nu,\tau} D_{\nu,1}(1-\tau, q_{\nu,\tau}, s) \\ &\quad = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)) H'(s). \end{aligned} \tag{1122}$$

Its squared $L^2(ds)$ norm is exactly $\nu\tau^{-1-2h}(e_0 + R_*(\tau))^2 d_H^2$. Thus the source's actual component and remainder, not just an abstract possibility of growth, have been carried into the original arithmetic profile. The complex conjugation and integration above retain the complex values of K_0 when that channel is used.

Choose any $0 < \tau_{**} \le \tau_*$ such that $C_* \tau_{**}^{2h} \le e_0/2$ and the source cutoffs equal one on the path for $\tau < \tau_{**}$. Cauchy–Schwarz on the two component coefficients $(-6r_{\nu,\tau}, 1)$ yields

$$\left(\sum_{j=1}^{3} \|D_{\nu,j}(1-\tau, q_{\nu,\tau}, \cdot)\|_2^2 \right)^{1/2} \ge \frac{d_H \sqrt{\nu}\, e_0}{2\sqrt{1 + 72\nu X_{\text{in}} \tau}} \tau^{-1/2-h}. \tag{1123}$$

This lower bound proves divergence in the spatial-supremum, spectral-L^2 norm for the imported constructed field. It is not a matching upper asymptotic for its global supremum. All forcing, pressure, and material diffusion terms remain those of (1112) and (1113), with the explicit (u_ν, p_ν, f_ν) just given.

26.3 A terminal arithmetic profile with a bounded energy norm

The same constructed input gives a terminal value in a precise function space. This assertion can be proved from its energy identity without controlling its unbounded spatial supremum. Let $M_\nu = \max_{K_\nu} \|J\|_{\text{op}}$ and put

$$\mathcal{H}_\nu = L^2(K_\nu \times \mathbb{R}, dq\, ds; \mathbb{C}^3), \qquad B_{\nu,j}(t,q,s) = \Psi_{\nu,j}(t,q,s) - H(S_j(q) + s).$$

Subtracting the displayed initial value defines an affine coordinate with the explicit inverse $B \mapsto B + (H(S_j(q) + s))_j$. It does not discard the initial profile. Its domain and reference element are specified; outside K_ν the difference is identically zero. The full profile on K_ν is also square integrable because real translation gives norm $|K_\nu|^{1/2} \|H\|_2$ in each channel.

The already proved exact identity gives

$$\|D_\nu(t)\|_{\mathcal{H}_\nu}^2 = d_H^2 \int_{K_\nu} |J(x) u_\nu(t,x)|^2 dx \le d_H^2 M_\nu^2 E_\nu^2.$$

The identity follows by integrating the real spectral translation and then using $\det D_q X_{\nu,t} = 1$; no spatial coordinate is removed. On a closed time interval before one the functions are smooth and the Schwartz bounds justify their Hilbert-space derivatives. Integrating $\partial_t B_\nu = D_\nu$ there and using the triangle inequality for the integral yields, for $0 \le a < b < 1$,

$$\|B_\nu(b) - B_\nu(a)\|_{\mathcal{H}_\nu} \le d_H M_\nu E_\nu (b-a). \tag{1124}$$

Completeness constructs a unique terminal element $B_\nu(1)$ and the exact bound $\|B_\nu(1) - B_\nu(t)\| \le d_H M_\nu E_\nu (1-t)$. This constructs a Lipschitz extension on $[0,1]$ in the specified Hilbert space, while (1123) proves divergence in the stronger spatial-supremum norm of its time derivative.

We also identify the terminal element through an actual map on labels. Volume preservation gives

$$\int_{K_\nu} \int_0^1 |u_\nu(t, X_{\nu,t}(q))| dt\, dq = \int_0^1 \|u_\nu(t)\|_{L^1(K_\nu)} dt \le |K_\nu|^{1/2} E_\nu.$$

The nonnegative integral is finite, so for almost every q the full trajectory has finite total length before one. Its integral equation therefore constructs the limit $X_{\nu,1}(q) \in K_\nu$ as $t \uparrow 1$. Set its value arbitrarily in K_ν on the null exceptional set. Also the integral triangle inequality and volume preservation give

$$\|X_{\nu,b} - X_{\nu,a}\|_{L^2(K_\nu)} \le \int_a^b \|u_\nu(t)\|_2 dt \le E_\nu (b-a),$$

and the same estimate with $b = 1$ follows by the constructed limit and Fatou's lemma. For every continuous g on compact K_ν, dominated convergence gives

$$\int_{K_\nu} g(X_{\nu,1}(q))dq = \lim_{t\uparrow 1} \int_{K_\nu} g(X_{\nu,t}(q))dq = \int_{K_\nu} g(x)dx.$$

This identifies the pushforward of Lebesgue measure exactly. To see that continuous tests suffice, for each relatively closed set C use the continuous decreasing approximations $g_n(x) = \max(0, 1 - n\,\mathrm{dist}(x, C))$ to its indicator, then monotone or dominated convergence; complements give open sets and the uniqueness of finite measures on their generated Borel sigma algebra gives the stated pushforward.

For almost every q the pointwise terminal entire fibre is

$$\Psi_{\nu,j}(1, q, \zeta) = H(S_j(X_{\nu,1}(q)) + \zeta).$$

It agrees with the Hilbert-space limit: real translations are continuous in $L^2(ds)$, as follows from $\|H(\cdot+a)-H(\cdot+b)\|_2 \le d_H|a-b|$ by integrating H'. The shifts converge almost everywhere, remain bounded on K_ν, and their translated-profile squared differences are bounded by $4\|H\|_2^2$. Dominated convergence proves the identification in $\mathcal{H}_\nu$. Replacing H by $H^{(n)}$ in this proof proves the same assertions for every pure spectral derivative, with the retained constant $\|H^{(n+1)}\|_2 M_\nu E_\nu$.

The terminal label map has a further exact operator correspondence. On $L^2(K_\nu)$ define $V_t g = g \circ X_{\nu,t}$, including $t = 1$. For $t < 1$ its inverse is composition with $X_{\nu,t}^{-1}$, so it is unitary. The just-proved measure identity makes V_1 an isometry. For continuous g, dominated convergence gives $V_t g \to V_1 g$ in L^2. Continuous functions are dense for Lebesgue measure restricted to a compact set: approximate measurable-set indicators by continuous distance cutoffs between compact subsets and open supersets using finite-measure regularity, then approximate simple functions. The isometric bounds extend the convergence to every $g \in L^2(K_\nu)$. Thus $V_t \to V_1$ strongly. Its Hilbert adjoint satisfies $V_1^* V_1 = I$ by preservation of inner products, and $V_1 V_1^*$ is the orthogonal projection onto its closed range: it is self-adjoint, idempotent, acts as identity on that range, and annihilates its orthogonal complement. These are the proved terminal inverse and range statements. No pointwise inverse of $X_{\nu,1}$, or surjectivity of V_1, has been supplied by this argument.

26.4 The actual zero image, including the terminal fibres

For every time before one and almost every terminal label, the shift $a = S_j(X_{\nu,t}(q))$ is real. The full coordinate map is $\zeta \mapsto a + \zeta$ with inverse $s \mapsto s - a$. If $H(s) = (s - z_0)^m A_0(s)$ with $A_0(z_0) \neq 0$, the actual image is

$$H(a + \zeta) = (\zeta - (z_0 - a))^m A_0(a + \zeta).$$

This retains the whole analytic unit and all its derivatives. In particular its zero has imaginary part $\mathrm{Im}\, z_0$ at all these times, even for the source input whose arithmetic time-derivative combination obeys (1122). With $H = \Xi$ the precise completed-zeta argument is $\rho = 1/2 + i(a + \zeta)$, and $\mathrm{Re}\,\rho = 1/2 - \mathrm{Im}\,\zeta$. Thus the actual source witness produces the proved derivative singularity and terminal profile above; this explicit map preserves zero heights. A zero-height change is not obtained by substituting a faster-growing velocity into this same real-translation map.

26.5 The exact moving-label compensation and the derivatives it compares

The sampling path used by the source is not asserted to be a material trajectory. Its changing label has an exact derivative which accounts for the constant value of the first two sampled profiles. Fix

$0 < \tau < 1$ and write

$$t_\tau = 1-\tau, \quad r_\tau = \sqrt{2\nu X_{\rm in}\tau}, \quad x_\tau = (r_\tau, 0, 0), \quad q_\tau = X^{-1}_{\nu,t_\tau}(x_\tau), \quad A_\tau = D_q X_{\nu,t_\tau}(q_\tau).$$

These functions are smooth on every open preterminal interval by the flow and inverse construction of Section 25; no endpoint derivative of q_τ is being assigned. Since $\det A_\tau = 1$, the matrix inverse below exists. Differentiating the exact identity $X_{\nu,1-\tau}(q_\tau) = x_\tau$ gives

$$-u_\nu(1-\tau, x_\tau) + A_\tau q'_\tau = x'_\tau, \qquad q'_\tau = A_\tau^{-1}\big(x'_\tau + u_\nu(1-\tau, x_\tau)\big), \qquad r'_\tau = \frac{\nu X_{\rm in}}{r_\tau}. \tag{1125}$$

The minus sign in the first term is the derivative of $1-\tau$. Both the entire actual velocity and the Eulerian path derivative are retained, rather than equating these two vectors.

For fixed $\zeta \in \mathbb{C}$ the spatial-label differential of the full arithmetic profile is the row

$$D_q\Psi_{\nu,j}(t_\tau, q_\tau, \zeta) = H'(S_j(x_\tau) + \zeta)\,[J(x_\tau)A_\tau]_{j,:}.$$

Multiplying this row by (1125), using the original matrix value $J(r,0,0)e_1 = (0,0,2)^{\mathsf T}$, proves

$$D_q\Psi_{\nu,j}(t_\tau, q_\tau, \zeta)\, q'_\tau = D_{\nu,j}(t_\tau, q_\tau, \zeta) + 2\delta_{j3} r'_\tau H'(S_j(x_\tau) + \zeta). \tag{1126}$$

Here $D_{\nu,j} = \partial_t \Psi_{\nu,j}$ is the derivative taken *at fixed material label and fixed spectral coordinate*, then evaluated at (t_τ, q_τ, ζ). It is not the total derivative of that evaluated expression with respect to τ.

For $j = 1, 2$, the exact equations $S_j(x_\tau) = 0$ give $\Psi_{\nu,j}(t_\tau, q_\tau, \zeta) = H(\zeta)$. Their total derivatives are therefore zero, consistently with the complete chain rule

$$\frac{d}{d\tau}\Psi_{\nu,j}(1-\tau, q_\tau, \zeta) = -D_{\nu,j}(t_\tau, q_\tau, \zeta) + D_q\Psi_{\nu,j}(t_\tau, q_\tau, \zeta)\, q'_\tau = 0.$$

For the third channel no term is suppressed:

$$\Psi_{\nu,3}(t_\tau, q_\tau, \zeta) = H(2r_\tau + \zeta), \qquad \frac{d}{d\tau}\Psi_{\nu,3}(t_\tau, q_\tau, \zeta) = 2r'_\tau H'(2r_\tau + \zeta).$$

Thus the compensation is an explicitly calculated label-transport term. It does not remove the lower bound (1123) on the fixed-label partial derivative evaluated at this changing family of labels. That bound implies divergence of the supremum over labels; it does not assert divergence along a single label independent of τ, nor a limit or regularity for q_τ at zero. Its source remainder restriction remains $0 < \tau < \tau_{**}$.

There is one further coefficient derivative if the two channels are combined before taking the total derivative. The exact sampled combination of profile values is

$$\Psi_{\nu,2}(t_\tau, q_\tau, \zeta) - 6r_\tau\Psi_{\nu,1}(t_\tau, q_\tau, \zeta) = (1 - 6r_\tau)H(\zeta),$$

so its total derivative is $-6r'_\tau H(\zeta)$, not zero. Expanding its derivative before substitution gives

$$\begin{aligned} &-\big(D_{\nu,2} - 6r_\tau D_{\nu,1}\big) + \big(D_q\Psi_{\nu,2} - 6r_\tau D_q\Psi_{\nu,1}\big)q'_\tau \\ &\qquad - 6r'_\tau\Psi_{\nu,1} = -6r'_\tau H(\zeta). \end{aligned}$$

All profiles and partial derivatives on the left are evaluated at (t_τ, q_τ, ζ). The first two grouped terms cancel by (1126), while the last is exactly the derivative of the moving coefficient. Meanwhile the whole fixed-label partial-derivative combination remains the nonzero growing function of (1122), with its entire original remainder.

27 The complete source momentum equation in arithmetic profile coordinates

The preceding actual-witness calculation detects the constructed velocity. Here we transport its *complete equation*, including the mechanism that leaves a smooth force while the velocity grows. The source increment identity is the one in Section 3.3 and the correction cycle of Section 3.4 of [7]; the realized fields are (9.21), the localization is (10.4), and the force is (10.5), with its extension (10.11). Existence of those fields is imported from the released construction, not independently reproved here. Every correspondence below is proved directly, with no averaging of the constructed fields.

27.1 The retained profile bundle and its explicit inverse

Write physical Cartesian coordinates as $x = (x_1, x_2, x_3)$. Keep the original polynomial

$$\begin{aligned} &F_1 = (1 + x_1x_2)^3x_3 + x_2^2(1 + x_1x_2)(4 + 3x_1x_2), \\ &F_2 = x_2 + 3x_1(1 + x_1x_2)^2x_3 + 3x_1x_2^2(4 + 3x_1x_2), \qquad S = (F_1/2, F_2, F_3), \quad J = DS. \\ &F_3 = 2x_1 - 3x_1^2x_2 - x_1^3x_3, \end{aligned}$$

The earlier exact determinant calculation gives $\det J = -1$; its pointwise inverse is the polynomial matrix $J^{-1} = [2W_1\ W_2\ W_3]$. None of these statements supplies a globally single-valued inverse of S, and none is needed: the physical point x is retained throughout.

Fix the entire arithmetic function $H = \Xi$, or the complete $H = K_0 = T_{\mathcal{D}}\Xi$ from Section 19. Put

$$d_H^2 = \int_{\mathbb{R}} |H'(s)|^2\, ds, \qquad 0 < d_H < \infty.$$

These are the original functions, not a finite spectral truncation. Their Schwartz bounds and the positivity of this integral were proved in the preceding flow correspondence. Define the linear map

$$(\mathcal{T}_x b)_a(s) = H'(s + S_a(x)) \sum_{k=1}^{3} J_{ak}(x) b_k, \qquad b \in \mathbb{C}^3, \quad a = 1, 2, 3. \tag{1127}$$

Its codomain fibre $\mathcal{E}_x$ is its three-dimensional complex image in $L^2(\mathbb{R}; \mathbb{C}^3)$, also an image in $\mathscr{S}(\mathbb{R}; \mathbb{C}^3)$ and in entire functions of s. It has kernel zero. Explicitly, for $B \in \mathcal{E}_x$,

$$c_a = \frac{1}{d_H^2} \int_{\mathbb{R}} \overline{H'(s + S_a(x))} B_a(s)\, ds, \qquad b = J(x)^{-1}c. \tag{1128}$$

Substitution of (1127) into this formula gives $c = Jb$ by real translation in the integral; substitution of the resulting b recovers B by its membership in $\mathcal{E}_x$. In particular

$$\|\mathcal{T}_x b\|_{L_s^2}^2 = d_H^2 |J(x)b|^2. \tag{1129}$$

On every compact physical set both J, J^{-1} are bounded. The maps and inverse therefore identify smooth vector fields with smooth sections of the displayed finite-rank bundle, with uniform seminorm bounds on compact sets. Differentiating (1127) proves this assertion for each spatial derivative: only finitely many derivatives of the polynomials and translated Schwartz functions occur. The inverse follows from (1128) by the same bounds and differentiation under the absolutely convergent integral. No division by a zero of H' occurs.

27.2 The full connection, including the mixing of different shifts

For $i = 1, 2, 3$ set

$$\Gamma_i(x) = (\partial_i J(x))J(x)^{-1}, \qquad (\mathcal{R}_{ab,x}f)(s) = f(s + S_a(x) - S_b(x)).$$

Each translation has the stated inverse $\mathcal{R}_{ba,x}$ and preserves Lebesgue measure. On smooth sections of $\mathcal{E}$ define

$$(\mathfrak{D}_i B)_a = \partial_i B_a - J_{ai}\partial_s B_a - \sum_{b=1}^{3}(\Gamma_i)_{ab}\mathcal{R}_{ab,x}B_b. \tag{1130}$$

The translations in the last term are essential: the three components have different spectral shifts. This is not ordinary matrix multiplication of the three unshifted functions.

For every smooth $b(x)$, the exact identity is

$$\mathfrak{D}_i(\mathcal{T}_x b) = \mathcal{T}_x(\partial_i b). \tag{1131}$$

Indeed $\partial_i(\mathcal{T}_x b)_a$ is

$$J_{ai}H''(s + S_a)(Jb)_a + H'(s + S_a)\big((\partial_i J)b + J\partial_i b\big)_a.$$

The first subtracted term of (1130) removes the displayed H'' term. In the final term $\mathcal{R}_{ab,x}(\mathcal{T}_x b)_b = H'(s+S_a)(Jb)_b$; hence that term removes exactly $(\partial_i J)b$. The remaining expression is $\mathcal{T}_x\partial_i b$, proving the identity.

Iteration proves, with their full lower-order terms retained,

$$\mathfrak{D}_i\mathfrak{D}_j\mathcal{T}b = \mathcal{T}\partial_i\partial_j b, \qquad [\mathfrak{D}_i, \mathfrak{D}_j]B = 0, \qquad \sum_i \mathfrak{D}_i^2\mathcal{T}b = \mathcal{T}\Delta b.$$

Thus the connection is flat on this bundle. Its flatness is an exact conjugation identity, not a statement that its displayed coefficients vanish. For example the matrix coefficients themselves satisfy

$$\partial_i\Gamma_j - \partial_j\Gamma_i = [\Gamma_i, \Gamma_j].$$

To check the sign, differentiate J^{-1}: $\partial_i J^{-1} = -J^{-1}(\partial_i J)J^{-1}$; the two mixed derivatives of J cancel and leave the displayed commutator. The variable translations in (1130) are still retained separately.

For a physical stress tensor $R = (R_{ki})$, encode its i-th column by $B_i = \mathcal{T}_x(R_{\cdot i})$. With the physical convention $(\operatorname{div} R)_k = \sum_i \partial_i R_{ki}$,

$$\sum_i \mathfrak{D}_i B_i = \mathcal{T}_x(\operatorname{div} R). \tag{1132}$$

The column index has not been discarded or averaged. The inverse (1128), separately on each column, recovers the entire tensor. This convention avoids silently identifying a covariance in the cylindrical frame with its Cartesian components. If $O = (e_r, e_\theta, e_z)$, then its actual Cartesian tensor is $OR_{\mathrm{cyl}}O^{\mathsf{T}}$; derivatives in (1132) act on O as well as on R_{cyl}.

27.3 The complete nonlinear equation and the original cancellation cycle

Let u_ν, p_ν, f_ν be the actual localized fields and extended force of the preceding section, with unchanged physical $\nu > 0$. Write $V = \mathcal{T}_x u_\nu$ and $\mathcal{F} = \mathcal{T}_x f_\nu$. Velocity on the right sides below always means the full inverse $u_\nu = \mathcal{T}_x^{-1}V$, not an additional unspecified input. The equation is exactly

$$\partial_t V + \sum_i (u_\nu)_i\mathfrak{D}_i V - \nu\sum_i \mathfrak{D}_i^2 V + \mathcal{T}_x\nabla p_\nu = \mathcal{F}, \qquad \sum_i \partial_i(\mathcal{T}_x^{-1}V)_i = 0. \tag{1133}$$

Every term follows from (1131); the map is time independent, so $\partial_t \mathcal{T}_x u = \mathcal{T}_x \partial_t u$. Conversely applying $\mathcal{T}_x^{-1}$ proves the original Navier–Stokes equation, including the stated divergence equation. Thus this is an invertible equation on the specified bundle. In conservative form its nonlinear flux columns are $(u_\nu)_i V = \mathcal{T}_x(u_\nu (u_\nu)_i)$. The product rule for $\mathfrak{D}_i$, derived directly from (1130), gives the same convection because $\operatorname{div} u_\nu = 0$.

Here is the exact source correction identity in these coordinates. For a divergence-free physical increment w, pressure increment π, and $W = \mathcal{T}_x w$, define

$$\mathscr{R}_{\mathcal{E}}(V,p) = \partial_t V + \sum_i u_i \mathfrak{D}_i V - \nu \sum_i \mathfrak{D}_i^2 V + \mathcal{T}_x \nabla p.$$

Expanding every product, without taking any mean, gives

$$\begin{aligned}\mathscr{R}_{\mathcal{E}}(V+W, p+\pi) = \mathscr{R}_{\mathcal{E}}(V,p) + \partial_t W + \sum_i u_i \mathfrak{D}_i W + \sum_i w_i \mathfrak{D}_i V \\ - \nu \sum_i \mathfrak{D}_i^2 W + \mathcal{T}_x \nabla \pi + \sum_i \mathfrak{D}_i (w_i W).\end{aligned} \tag{1134}$$

The final term equals $\mathcal{T}_x \operatorname{div}(w \otimes w)$ by (1132). It includes the entire self-interaction. The two preceding transport terms are the two linearized interactions with the actual current background. The pressure increment is not omitted. Equation (1134) is therefore the exact image of the source identity at every finite stage, before the source estimates improve its remainder.

In particular, write a source residual as $-\operatorname{div} R + E$. Its image is $-\sum_i \mathfrak{D}_i \mathcal{T}_x(R_{\cdot i}) + \mathcal{T}_x E$. When the source pulse covariance supplies R, its divergence cancels this first term by (1132); all differences between the actual quadratic tensor and that selected covariance remain explicit tensor columns in (1134). The source uses an auxiliary torus and an angular average to construct the leading covariance, before physical evaluation. Here that evaluation and conversion to Cartesian components precede $\mathcal{T}_x$. We make no assertion that angular averaging commutes with $S(x)$, $J(x)$, or this connection.

The source takes curls *after* cutting off the potentials. For every actual finite-stage increment of the form $w = \operatorname{curl}(\chi A) + \chi b e_\theta$, its full image is

$$\mathcal{T}_x w = \chi \mathcal{T}_x(\operatorname{curl} A + b e_\theta) + \mathcal{T}_x(\nabla \chi \times A).$$

Consequently the last term is present both in the linear and quadratic terms of (1134). For the realized cutoff sums (9.21), the sums are locally finite before terminal time. Linearity and the proved local seminorm bounds transport their actual derivatives and sums. At the terminal point the all-order remainder estimate is imported from the source summation proof, not inferred from formal termwise summation.

27.4 A smooth arithmetic force and the actual growing field

The source force is compactly supported and smooth through time one. Therefore $\mathcal{F}(t,x,\cdot)$ is a smooth compactly space–time-supported function with values in $\mathscr{S}(\mathbb{R};\mathbb{C}^3)$. This follows at every seminorm, not only in L^2_s: differentiating in t, x, s gives finite sums of derivatives of f_ν, J, S times derivatives of H' translated by the bounded $S_a(x)$ on K_ν. For nonnegative integers m, k, bounded translations obey

$$\sup_s (1+|s|)^m |H^{(k)}(s+a)| \le (1+|a|)^m \sup_y (1+|y|)^m |H^{(k)}(y)|.$$

Every derivative of f_ν vanishes outside its fixed support; the same proof covers all support boundaries.

Near the singular point the stronger source estimate (10.9) is retained. In physical viscosity coordinates put

$$Q_\nu(x,t) = (1-t) + |x_3/\sqrt{\nu}|^{1/(1/2-h)}.$$

For every Cartesian multi-index α, time derivative order j, spectral derivative order k, weight m, and N, the source's bounds and the preceding finite product expansions give

$$\sup_s (1+|s|)^m |\partial_x^\alpha \partial_t^j \partial_s^k \mathcal{F}(t,x,s)| \le C_{\alpha,j,k,m,N,\nu}\, Q_\nu(x,t)^N \tag{1135}$$

for $0 \le t < 1$ on its stated scaled source neighborhood. The smooth continuation of the force for $t \ge 1$ was proved separately; this preterminal estimate is not asserted there. This is flatness of the actual force image, not a bound for each separate term of the nonlinear equation. Only finitely many source derivatives enter each left side, so choosing their bounds at the same N proves the estimate.

The actual velocity still grows. At

$$t = 1-\tau, \quad r = \sqrt{2\nu X_{\mathrm{in}}\tau}, \quad x_{\nu,\tau} = (r,0,0),$$

retain the source's fixed h, X_{in}, e_0 and actual remainder $R_*(\tau)$, with $|R_*(\tau)| \le C_*\tau^{2h}$ and $e_0 > 0$. Direct substitution in the original polynomials gives

$$S(r,0,0) = (0,0,2r), \qquad J(r,0,0) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 1 & 3r \\ 2 & -3r^2 & -r^3 \end{pmatrix}.$$

The first two profile shifts coincide. The two rows therefore prove the full-function identity

$$V_2(1-\tau, x_{\nu,\tau}, s) - 6rV_1(1-\tau, x_{\nu,\tau}, s) = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau))H'(s). \tag{1136}$$

No leading-profile substitution was made in the left side: it uses the complete localized constructed velocity. Taking its norm yields exactly

$$\|V_2 - 6rV_1\|_{L_s^2}^2 = \nu\tau^{-1-2h}(e_0 + R_*(\tau))^2 d_H^2.$$

For $C_*\tau^{2h} \le e_0/2$, within the unchanged source cutoff region, Cauchy–Schwarz in the two components gives

$$\|V(1-\tau, x_{\nu,\tau}, \cdot)\|_{L_s^2} \ge \frac{\sqrt{\nu}\, d_H e_0}{2\sqrt{1+72\nu X_{\mathrm{in}}\tau}}\,\tau^{-1/2-h}.$$

Meanwhile, if $M_\nu = \max_{K_\nu} \|J\|_{\mathrm{op}}$ and $E_\nu = \int_0^1 \|f_\nu(t)\|_2 dt$, the source energy estimate and (1129) give

$$\|V(t)\|_{L_{x,s}^2} \le d_H M_\nu \|u_\nu(t)\|_2 \le d_H M_\nu E_\nu.$$

Thus the source's concentration, bounded integrated energy and smooth force coexist in the exact arithmetic momentum equation (1133). No sign test or zero ordinate is substituted for the constructed field.

27.5 The original theta amplitude and the next arithmetic question

For $H = \Xi$ retain the minus-Fourier convention and the even theta input g_0 from the actual-theta calculation: $\Xi(s) = \int_{\mathbb{R}} g_0(v)e^{-isv}dv$. Fourier inversion and differentiation give

$$\widehat{H'}(\omega) = 2\pi i\omega\, g_0(\omega), \qquad \widehat{V_a}(t,x,\omega) = 2\pi i\omega\, g_0(\omega)e^{i\omega S_a(x)}(J(x)u_\nu(t,x))_a.$$

For verification of the constants, the double minus transform of g_0 is $2\pi g_0(-\omega)$, and g_0 is even; integration by parts supplies the factor $i\omega$. All functions are Schwartz, so both operations are justified. For K_0, replace g_0 by the complete reflected input $w(-\omega)g_0(\omega)$; no summand or derivative term of w is removed.

This explicit amplitude formula explains what the present map transports: the full physical equation acts on the specified three-component arithmetic subbundle. It is not an assertion that the source force is the scalar de Bruijn–Newman heat generator. The source-specific Mellin equation of the radial heat profile in Section 24 supplies a second, different exact interface with zeta. Section 28 calculates the actual corrected nonlinear radial fluxes under that Mellin map, retaining their full stress-cancellation mechanism. Section 29 then identifies an exact fixed-axis derivative of its arithmetic image.

28 Actual angular momentum, nonlinear fluxes and arithmetic dilation

28.1 Cartesian input, angular projection and its exact section

Use the complete source witness (u_ν, p_ν, f_ν) constructed in Section 26, for every physical time $0 \le t < 1$. The existence of that complete corrected field is imported from the 166-page manuscript [7]; the projection, nonlinear equation and arithmetic correspondence are proved below for that retained input. Retain the viscosity $\nu > 0$ and original Cartesian coordinates (x_1, x_2, z). For $r > 0$ put $x_1 = r\cos\theta, x_2 = r\sin\theta$, $e_r = (\cos\theta, \sin\theta, 0)$ and $e_\theta = (-\sin\theta, \cos\theta, 0)$. Write $u_r = e_r \cdot u_\nu$, $u_\theta = e_\theta \cdot u_\nu$ and $u_z = (u_\nu)_3$, all evaluated at $(t, x_1, x_2, z) = (t, r\cos\theta, r\sin\theta, z)$ in the time-first convention of Section 26. These components may depend on θ; in particular the complete corrected velocity is not assumed axisymmetric. Define $\langle a\rangle = (2\pi)^{-1}\int_0^{2\pi} a\,d\theta$ and retain the radial coordinate and inverse

$$\sigma = \frac{r^2}{2}, \qquad r = \sqrt{2\sigma}, \qquad r\,dr = d\sigma.$$

This radial σ is not the earlier polynomial coordinate $F_1/2$. They coexist as explicit functions on the original Cartesian space; no equality of their Laplacians is required below. Their exact common coordinate map, on $(0,\infty) \times (\mathbb{R}/2\pi\mathbb{Z}) \times \mathbb{R}$, is $(\sigma, \theta, z) \mapsto (\sqrt{2\sigma}\cos\theta, \sqrt{2\sigma}\sin\theta, z)$. Its inverse is $\sigma = (x_1^2 + x_2^2)/2$, the direction $(\cos\theta, \sin\theta) = (x_1, x_2)/\sqrt{x_1^2 + x_2^2}$, and the unchanged z. In these coordinates the full polynomial is

$$\frac{F_1}{2} = \frac{1}{2}\big[(1 + \sigma\sin(2\theta))^3 z + 2\sigma\sin^2\theta(1 + \sigma\sin(2\theta))(4 + 3\sigma\sin(2\theta))\big].$$

Substitution of $x_1x_2 = \sigma\sin(2\theta)$ and $x_2^2 = 2\sigma\sin^2\theta$ proves this morphism; adjoining the displayed value as an extra coordinate gives its graph map with inverse projection onto (σ, θ, z). The Cartesian definition continues over the axis where the angular chart is not defined.

The actual angular-momentum density, fluxes and force torque are

$$B(\sigma, z, t) = \langle ru_\theta\rangle, \qquad P(\sigma, z, t) = \langle r^2 u_r u_\theta\rangle,$$
$$Q(\sigma, z, t) = \langle ru_z u_\theta\rangle, \qquad F(\sigma, z, t) = \langle rf_\theta\rangle.$$

These definitions include every corrected, oscillatory and cutoff term of the actual source fields before averaging. All four functions have fixed compact support in (σ, z) on the closed half-plane $\sigma \geq 0$. They are smooth there on every closed preterminal time interval, with

$$B = O(\sigma), \quad Q = O(\sigma), \quad F = O(\sigma), \qquad P = O(\sigma^2) \quad (\sigma \downarrow 0). \tag{1137}$$

Here and below these estimates hold with all z, t derivatives on compact preterminal sets. To prove them without polar smoothness assumptions, observe that $ru_\theta = x_1u_2 - x_2u_1$ and $ru_r = x_1u_1 + x_2u_2$. Angular averaging their Cartesian expressions gives smooth rotation-invariant functions of (x_1, x_2). A smooth rotation-invariant function is smooth in σ on the closed half-line: its restriction to $(r, 0)$ is even and smooth; Taylor's formula to any even order, with its differentiated remainder, expresses it as a polynomial in r^2 plus a remainder of arbitrarily high order. The chain rule on $r > 0$ then supplies every continuous right derivative in σ at zero and the fundamental theorem of calculus identifies those derivatives successively. The same Taylor proof works with the retained z, t parameters. The constants in the averages defining B, Q, F vanish at $r = 0$; odd radial powers vanish under rotation. They therefore start at r^2. For P, the quadratic Cartesian term at zero is

$$\big\langle (x_1a + x_2b)(x_1b - x_2a) \big\rangle = 0, \qquad (a, b) = (u_1, u_2)(t, 0, 0, z),$$

because $\langle x_1^2 - x_2^2 \rangle = \langle x_1x_2 \rangle = 0$. Its constant term also vanishes, and rotation removes the odd powers, so P starts at r^4. This proves (1137).

The projection $\Pi u = B$ has the explicit linear section

$$(\mathcal{R}B)(x_1, x_2, z) = \frac{B(\sigma, z)}{2\sigma}(-x_2, x_1, 0), \qquad \Pi\mathcal{R} = I. \tag{1138}$$

The quotient at zero is the actual smooth extension: $B(\sigma, z)/\sigma = \int_0^1 \partial_\sigma B(a\sigma, z)da$. Thus the section is Cartesian smooth, compactly supported, and divergence-free. For $r > 0$ it has sole component B/r in the e_θ direction, proving the section identity. For the exact remainder $w = u_\nu - \mathcal{R}B$, one has $\Pi w = 0$ and

$$P = rB\langle w_r \rangle + r^2\langle w_r w_\theta \rangle, \qquad\qquad Q = B\langle w_z \rangle + r\langle w_z w_\theta \rangle. \tag{1139}$$

These follow by substituting $u_r = w_r$, $u_z = w_z$ and $u_\theta = B/r + w_\theta$ into the original definitions. They exhibit both mean transport and the complete two quadratic correlations. The full inverse of $u \mapsto (B, w)$ is $(B, w) \mapsto \mathcal{R}B + w$ on the stated subspace $\Pi w = 0$. In particular the projection has the precise kernel $\{w : \langle rw_\theta \rangle = 0\}$; none of that kernel's contribution to the fluxes has been discarded.

28.2 The exact equation before and after angular averaging

For the unaveraged Cartesian angular momentum $L = x_1u_2 - x_2u_1 = ru_\theta$, the product rule in the full Navier–Stokes equation gives

$$\partial_t L + \operatorname{div}(u_\nu L) = \nu\Delta L - 2\nu\omega_z - \partial_\theta p_\nu + (x_1f_2 - x_2f_1), \qquad \omega_z = \partial_1u_2 - \partial_2u_1. \tag{1140}$$

Indeed the material derivatives of x_1, x_2 contribute $u_1u_2 - u_2u_1 = 0$; incompressibility converts material transport to the displayed divergence. The Laplacian product rule gives $\Delta L = x_1\Delta u_2 - x_2\Delta u_1 + 2\omega_z$. Finally $x_1\partial_2 - x_2\partial_1 = \partial_\theta$ gives the negative pressure-torque term with its sign.

In polar coordinates $\omega_z = r^{-1}\partial_r(ru_\theta) - r^{-1}\partial_\theta u_r$. Average (1140) over a full circle. Every angular derivative has zero integral, by the fundamental theorem of calculus and periodicity, including the actual non-axisymmetric pressure. The convection term is $r^{-1}\partial_r(r\langle u_r L\rangle) + \partial_z\langle u_z L\rangle$; the averaged viscous term is $\nu(\partial_{rr} - r^{-1}\partial_r + \partial_{zz})B$. Using $\partial_r = r\partial_\sigma$ now proves

$$\partial_t B + \partial_\sigma P + \partial_z Q = \nu(2\sigma\partial_{\sigma\sigma}B + \partial_{zz}B) + F. \tag{1141}$$

The formula extends to $\sigma = 0$ by (1137) and smoothness. The pressure has disappeared here by a proved projection, with its entire original torque term exhibited in (1140). No scalar heat equation has replaced the actual nonlinear fluxes.

More generally a Cartesian momentum residual $f_i + \sum_j \partial_j T_{ij}$ contributes the torque $\langle x_1\partial_j T_{2j} - x_2\partial_j T_{1j}\rangle$. For a symmetric stress T, the product rule gives exactly

$$x_1\partial_j T_{2j} - x_2\partial_j T_{1j} = \partial_j(x_1 T_{2j} - x_2 T_{1j}) - (T_{21} - T_{12}) = \partial_j(x_1 T_{2j} - x_2 T_{1j}).$$

Writing its vector flux in polar components therefore turns this contribution into

$$\partial_\sigma\langle r^2 T_{\theta r}\rangle + \partial_z\langle r T_{\theta z}\rangle, \quad T_{\theta r} = e_\theta^\top T e_r, \quad T_{\theta z} = e_\theta^\top T e_z. \tag{1142}$$

For a nonsymmetric tensor the retained extra torque is $-\langle T_{21} - T_{12}\rangle$. Thus every symmetric stress cancellation in the source momentum residual has this exact projected image, with its sign determined by the side of the momentum equation on which the source places $\operatorname{div} T$. The map is linear in the full Cartesian tensor; its evaluation is not an identification of the source's auxiliary-torus average with the physical angular average used here.

The source cutoff sum has this exact image before the terminal limit. In its original viscosity-one coordinates let $\mathbf{b}_j$ denote the actual Cartesian vector appearing as B_j in source (9.21) [7, p. 115]. With q_* here denoting that source's concentration coordinate, retain every increment as

$$\delta u_{*,j} = \operatorname{curl}\bigl(c_*\chi(a_j q_*)A_j\bigr) + c_*\chi(a_j q_*)\mathbf{b}_j, \qquad \delta p_{*,j} = c_*\chi(a_j q_*)p_j.$$

The notation q_* in this display is the source coordinate determined by $1 - t = q_*(1 - \eta^2)$ and $z = q_*^{1/2-h}\eta$, not a material label; those two equations specify its actual dependence on (z, t). Define each viscosity-ν increment by the full scaling of the viscosity map in Section 26, and denote the correspondingly localized base by $(u_{\nu,0}, p_{\nu,0})$. The actual fields are $u_\nu = \sum_{j\geq 0} u_{\nu,j}$ and $p_\nu = \sum_{j\geq 0} p_{\nu,j}$, where the positive-index terms are the displayed increments with that scaling. Source local finiteness on every compact set before time one makes these sums, and every derivative of them, finite on a neighborhood of each sampled circle. Indeed the source coordinate is independent of θ, so its positive lower bound and its finite set of active stages can be chosen over the entire compact circle. Thus the exact nonlinear fluxes are

$$P = \sum_{a,b\geq 0}\langle r^2(u_{\nu,a})_r(u_{\nu,b})_\theta\rangle, \qquad Q = \sum_{a,b\geq 0}\langle r(u_{\nu,a})_z(u_{\nu,b})_\theta\rangle.$$

Each sum contains every diagonal term and both ordered cross terms for each pair $a \neq b$. The curl in $u_{\nu,j}$ is taken after multiplying both cutoffs, so their derivative corrections belong to these same fluxes. Projection, differentiation and the sums commute by the stated local finiteness. Hence the source's exact residual identity, including its flat remainder and final smooth force, passes to (1141) with all these contributions. This argument uses the actual summed fields for $t < 1$; it does not exchange an uncontrolled infinite series with the terminal limit.

28.3 The radial Mellin equation with all shifted fluxes

For each actual field $a = B, P, Q, F$ put

$$\widehat{a}(s,z,t) = \int_0^\infty a(\sigma,z,t)\sigma^{s-1}d\sigma, \qquad \sigma^{s-1} = e^{(s-1)\log\sigma}.$$

The original measure here is $d\sigma = r\,dr$, with real logarithm; angular averaging already contains its factor $1/(2\pi)$. The integral for B, Q, F is absolutely convergent and holomorphic on $\Re s > -1$, and that for P on $\Re s > -2$, by compact support and (1137). On compact substrips every complex derivative adds an integrable power of $|\log\sigma|$, which proves holomorphy by differentiated domination. All z, t derivatives on closed preterminal sets obey the same justification.

For $\Re s > 0$, integration by parts gives

$$\int_0^\infty \partial_\sigma P\,\sigma^{s-1}d\sigma = -(s-1)\widehat{P}(s-1),$$
$$\int_0^\infty 2\sigma\partial_{\sigma\sigma}B\,\sigma^{s-1}d\sigma = 2s(s-1)\widehat{B}(s-1).$$

All upper boundary terms vanish by compact support. The lower terms $P\sigma^{s-1}$, $B'\sigma^s$ and $B\sigma^{s-1}$ vanish by (1137) and $\Re s > 0$. Consequently the full equation is

$$\partial_t\widehat{B} - (s-1)\widehat{P}(s-1) + \partial_z\widehat{Q} = \nu\{2s(s-1)\widehat{B}(s-1) + \partial_{zz}\widehat{B}\} + \widehat{F}, \qquad \Re s > 0. \tag{1143}$$

In particular no shift $s \mapsto s-1$, polynomial factor, or force moment has been suppressed.

This is an actual injective transform with its inverse. For any real $c > 0$ and fixed (z,t) put $g(u) = e^{cu}a(e^u, z, t)$. It vanishes for sufficiently large positive u and is $O(e^{(c+1)u})$ as $u \to -\infty$, with the same bound for every derivative; for P the exponent improves to $c+2$. These facts follow from the chain rule and (1137). Thus g is Schwartz and

$$\widehat{a}(c-i\eta,z,t) = \int_{\mathbb{R}} g(u)e^{-i\eta u}du, \qquad a(\sigma,z,t) = \frac{\sigma^{-c}}{2\pi}\int_{\mathbb{R}}\widehat{a}(c-i\eta,z,t)e^{i\eta\log\sigma}d\eta.$$

To justify the inverse without a distributional shortcut, twice integrate by parts in the forward integral to obtain an integrable $O((1+|\eta|)^{-2})$ bound. Insert $e^{-\epsilon\eta^2}$ in the inverse. Fubini and the Gaussian integral give convolution by $(4\pi\epsilon)^{-1/2}e^{-u^2/(4\epsilon)}$, which converges to g by continuity and its Gaussian tail. Dominated convergence in the integrable Fourier inverse then proves the displayed identity.

28.4 The same arithmetic dilation map on the actual corrected fields

Retain the source's chosen $0 < h < 1/100$. For any smooth compactly supported half-line function a with $a(0) = 0$, define

$$\begin{aligned}
\phi_a(x) &= a(x) - 2^h a(2x),\\
\psi_a(x) &= \phi_a(x) - \tfrac{1}{2}\phi_a(x/2)\\
&= (1+2^{h-1})a(x) - 2^h a(2x) - \tfrac{1}{2}a(x/2),\\
(\mathcal{A}_h a)(x) &= \sum_{n\geq 1}\psi_a(nx), \qquad x > 0.
\end{aligned}$$

These are the exact three dilation coefficients of (1062) in Section 24; here their input is the actual B, P, Q, F. Every sum is locally finite on $x > 0$, and the trace vanishes for all sufficiently large x. Substitution gives $\int_0^\infty \psi_a(x)dx = 0$ and $\psi_a(0) = 0$. Its small-x bound has a direct proof. On each interval $[nx, (n+1)x]$, two integrations by parts give the trapezoidal identity

$$\int_{nx}^{(n+1)x} \psi_a(v)dv = \frac{x}{2}\{\psi_a(nx) + \psi_a((n+1)x)\} - \frac{1}{2}\int_{nx}^{(n+1)x} (v - nx)((n+1)x - v)\psi_a''(v)dv.$$

Summing, using compact support and the two zero quantities just proved, yields

$$|(\mathcal{A}_h a)(x)| \le \frac{x}{8}\|\psi_a''\|_1. \tag{1144}$$

Thus its Mellin integral is holomorphic on $\Re s > -1$. Parameter derivatives in z, t have the same bound because the trace is linear, preserves the zero endpoint and zero integral, and commutes with those derivatives on preterminal compact sets.

On $\Re s > 1$ absolute Fubini gives

$$\mathcal{M}(\mathcal{A}_h a)(s) = W_h(s)\widehat{a}(s), \qquad W_h(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s}). \tag{1145}$$

Indeed the sum of absolute Mellin integrals is $\zeta(\Re s)\int |\psi_a(x)|x^{\Re s - 1}dx$, and a dilation by $b > 0$ multiplies its Mellin integral by b^{-s}. The only pole of ζ is its simple pole at one, and the factor $1 - 2^{s-1}$ cancels it. Consequently W_h is entire and the identity theorem extends (1145) to $\Re s > -1$ for these actual inputs. Its multiplier is the same arithmetic factor as in the exterior calculation (1080). The gamma product $G_h(s)$ in (1075) is specifically the Mellin transform of that exterior's H_h; it is not silently substituted for the present corrected-field moment $\widehat{a}$.

The complete inverse preserves the actual field and its endpoint:

$$\begin{aligned} \psi_a(x) &= \sum_{n\ge 1} \mu(n)(\mathcal{A}_h a)(nx), \\ \phi_a(x) &= \sum_{j\ge 0} 2^{-j}\psi_a(x/2^j), \\ a(x) &= -\sum_{k\ge 1} 2^{-kh}\phi_a(x/2^k). \end{aligned}$$

The first double sum is finite at each $x > 0$, so rearranging gives $\sum_{m\ge 1}\psi_a(mx)\sum_{n|m}\mu(n) = \psi_a(x)$. The divisor sum is one for $m = 1$ and zero otherwise, since the squarefree expansion gives $\prod_{p|m}(1-1)$ for $m > 1$. The finite second sum through $j = J$ equals $\phi_a(x) - 2^{-J-1}\phi_a(x/2^{J+1})$. The negative finite third sum through $k = K$ equals $a(x) - 2^{-Kh}a(x/2^K)$. Their tails vanish by boundedness at zero and $h > 0$. After any x derivative, each summand gains the additional factor $2^{-j\ell}$ or $2^{-k\ell}$; bounded derivatives on compact half-line intervals prove uniform convergence there. The arithmetic inverse is locally finite away from zero; the subsequent recovered functions extend to zero by the displayed smooth inverse. These are actual inverse maps on this stated class, including its parameter dependence, with no division by zeta values at their zeros.

Set $\mathfrak{B} = \mathcal{M}\mathcal{A}_h B$ and define $\mathfrak{P}, \mathfrak{Q}, \mathfrak{F}$ in the identical way. Multiplication of (1143) by $W_h(s)$ gives the exact equation

$$\partial_t\mathfrak{B} + \partial_z\mathfrak{Q} - \nu\partial_{zz}\mathfrak{B} - \mathfrak{F} = (s-1)W_h(s)\{\widehat{P}(s-1) + 2\nu s\widehat{B}(s-1)\}.$$

Both hatted terms on the right are explicitly recovered by the proved inverses, so this equation retains the full dynamics. There is also the division-free shifted relation

$$\begin{aligned} &W_h(s-1)\{\partial_t\mathfrak{B}+\partial_z\mathfrak{Q}-\nu\partial_{zz}\mathfrak{B}-\mathfrak{F}\} \\ &=(s-1)W_h(s)\{\mathfrak{P}(s-1)+2\nu s\mathfrak{B}(s-1)\}, \qquad \Re s>0. \end{aligned} \tag{1146}$$

Initially this follows on $\Re s>1$; every factor is holomorphic on $\Re s>0$ by the previous convergence proofs, so it holds throughout that half-plane. The preceding inverse form retains the information even at a zero of $W_h(s-1)$; the cross-multiplied relation alone is not being asserted to reconstruct information lost by multiplying by a vanishing scalar.

28.5 The actual concentrating circle and its recoverable arithmetic image

In the source's fixed inner similarity region all annular corrections vanish. The remaining cutoff-summed slow field is axisymmetric, so the swirl estimate on the positive x_1 axis in (10.21) holds on the full circle with the same r,z,t. The source fixes this inner region in Section 5 and Proposition 9.9, proof Step 5 [7]; its axisymmetry is used only for these remaining inner slow fields, not imposed on the full velocity. The precise source reading is recorded in the independent review. Hence at $\sigma=\nu X_{\mathrm{in}}\tau, z=0, t=1-\tau$ the actual average is

$$B(\nu X_{\mathrm{in}}\tau,0,1-\tau)=\nu\sqrt{2X_{\mathrm{in}}}\,\tau^{-h}(e_0+R_*(\tau)). \tag{1147}$$

The factor follows from multiplying the full retained swirl in (1120) by $r=\sqrt{2\nu X_{\mathrm{in}}\tau}$; the physical angular average of an axisymmetric value leaves it unchanged. In particular this actual scalar angular momentum also diverges, now at the source exponent h, with its original remainder.

Applying the proved inverse series to $\mathcal{A}_hB(\cdot,0,1-\tau)$ recovers exactly the value in (1147); it is therefore a source-specific arithmetic image of the concentrating corrected field. The double dyadic inverse and the finite-at-each-argument arithmetic inverse retain their stated order of evaluation. No uniform bound for those inverses as the sampling point tends to zero has been inferred, so the recovery identity alone is not a claim that the un-inverted trace has the same pointwise blowup rate.

For completeness the resulting zeta factor also has a precise local zero map. On $h<\Re s<1$, both dilation factors are nonzero. Writing $\zeta(s)=(s-s_0)^m a_0(s)$ at a zeta zero gives

$$\mathfrak{B}(s,z,t)=(s-s_0)^m a_0(s)(1-2^{s-1})(1-2^{h-s})\widehat{B}(s,z,t).$$

This is the complete product, including the corrected-field Mellin factor. If that factor has order n at s_0, the product has order $m+n$, by multiplication of the two Taylor series; if it is nonzero, the multiplicity is exactly m. Identical statements hold for each flux and force moment. The arithmetic map and inverse are thus exact for the nonlinear witness, while no off-critical zeta zero or nonvanishing of the actual corrected-field Mellin factor is assumed.

29 An exact fixed-axis singularity of the arithmetic trace

The angular-momentum transform can see a fixed-axis derivative of the actual source field directly, without first applying its inverse. The reason is a specific feature of the construction: every positive-order azimuthal coefficient has zero axis value. We retain the source amplitude C, the parameter h, the physical viscosity ν, all cutoffs and the original time $t=1-\tau$. The calculation uses the constructed field of Section 26; it does not certify the entire existence argument of the released manuscript independently.

29.1 The axis trace of the fully corrected velocity

Here and only here let $q_{\rm src}(z,t)$ denote the scalar similarity coordinate called q in the source, not the material label in Section 25. Its source coordinates are

$$\tau = 1-t = q_{\rm src}(1-\eta^2), \qquad z = q_{\rm src}^D \eta, \qquad X = \frac{r^2}{2q_{\rm src}}, \qquad D = \frac{1}{2}-h, \quad A = \frac{1}{2}+h.$$

In particular at $z=0$ one has exactly $\eta=0$ and $q_{\rm src}=\tau$. Source (5.1), pp. 45–46, writes each azimuthal coefficient as

$$u_{\theta,n} = \frac{r}{C} q_{\rm src}^{-1-h+2nh} \varphi_n(X,\eta).$$

Its Lemma 5.1, p. 47, prescribes $\varphi_n(0,\eta)=0$ for every $n\geq 1$. The radial extensions and moment corrections are supported away from the unchanged inner interval. Source Proposition 5.5 and its proof, pp. 60–62, realize this series with cutoffs $\chi(c_n q_{\rm src})$; these depend on z,t, not r, and the realized angular component is

$$\frac{u_{\theta,B}}{r} = \frac{q_{\rm src}^{-1-h}}{C}\left(\varphi_0(X,\eta) + \sum_{n\geq 1}\chi(c_n q_{\rm src}) q_{\rm src}^{2nh} \varphi_n(X,\eta)\right). \tag{1148}$$

For every fixed preterminal z,t, this sum is finite in a neighborhood of that point: $c_n\to\infty$, while $q_{\rm src}>0$ and χ has compact support. Thus evaluation at $X=0$, and the smooth extension of the quotient $u_{\theta,B}/r$, require no exchange of a terminal limit with an infinite sum.

The leading axis datum is not an unknown positive remainder. Source (B.3), p. 145, and Proposition B.2, pp. 146–147, give

$$\varphi_0(0,\eta) = \varphi_*(\eta), \qquad \varphi_*(\eta) = \exp\left(\Lambda\int_0^\eta \zeta_*(w)\,dw\right), \qquad \varphi_*(0)=1.$$

Here ζ_* is the source's auxiliary axis function, not the Riemann zeta function. Neither its value nor C is rescaled.

All annular wave and mean potentials, including the finite initialization and the later cutoff-summed stages, vanish near the axis for each $t<1$: source Proposition 9.9, (9.21), pp. 114–117. Only finitely many stages are present locally at such a time. Their Cartesian derivatives therefore vanish on a common neighborhood of the axis. The base potential is azimuthal and independent of angle, so its curl has only radial and axial components. Consequently it contributes no additional azimuthal term to (1148). Finally the spatial and temporal cutoffs in source (10.4), p. 118, are identically one near the origin for all sufficiently small positive τ. Their derivatives vanish there as well. These are facts about the actual summed source field, not a replacement of that field by its leading approximation [7, Lemma 5.1; Proposition 5.5; (9.21); (10.4); (B.3)].

It follows that there is a source-determined $\tau_1>0$ such that

$$\left.\frac{u_{\theta,*}(r,0,1-\tau)}{r}\right|_{r=0} = \frac{1}{C}\tau^{-1-h} \qquad (0<\tau<\tau_1). \tag{1149}$$

The quotient denotes its proved smooth extension, not division by zero. For the physical viscosity map $u_\nu(x,t)=\sqrt{\nu}\,u_*(x/\sqrt{\nu},t)$, put $r_*=r/\sqrt{\nu}$. Then, for $r>0$,

$$\frac{u_{\theta,\nu}(r,0,t)}{r} = \frac{\sqrt{\nu}\,u_{\theta,*}(r_*,0,t)}{\sqrt{\nu}\,r_*} = \frac{u_{\theta,*}(r_*,0,t)}{r_*}.$$

Passing to the smooth axis extension proves the same identity (1149) for u_ν. This cancellation follows from the actual coordinate map; viscosity has not been set to one.

Use the original physical radial coordinate $\sigma = r^2/2$ and angular average $(2\pi)^{-1}\int_0^{2\pi}$. Let

$$B_\nu(\sigma, z, t) = \frac{1}{2\pi}\int_0^{2\pi} r u_{\theta,\nu}(r,\theta,z,t)\, d\theta.$$

In the axis neighborhood just established the full field is axisymmetric and $B_\nu = 2\sigma(u_{\theta,\nu}/r)$. The exact consequence is

$$B_\nu(0,0,1-\tau) = 0, \qquad \partial_\sigma B_\nu(0,0,1-\tau) = \frac{2}{C}\tau^{-1-h}. \tag{1150}$$

No $O(\tau^{2h})$ error is present in this first axis derivative: each positive-order axis value is exactly zero. This does not remove the actual remainder at the distinct nonzero X_{in} sample of Section 26.

29.2 The arithmetic operator on every endpoint derivative

We prove the endpoint map on its full smooth compactly supported domain. Let $a \in C^\infty([0,\infty))$ have compact support and $a(0) = 0$. For the same fixed h define

$$\phi_a(x) = a(x) - 2^h a(2x), \qquad \psi_a(x) = \phi_a(x) - \tfrac{1}{2}\phi_a(x/2), \qquad (\mathcal{A}_h a)(x) = \sum_{n\ge1}\psi_a(nx) \quad (x>0).$$

This is exactly the arithmetic operator used on the nonlinear fluxes. There is no radius change hidden in its argument x. Every sum and its derivatives are locally finite for $x > 0$. Substitution in the two integrals proves $\int_0^\infty \psi_a = 0$, and $\psi_a(0) = 0$. The full endpoint derivatives of the input are

$$\psi_a^{(j)}(0) = (1-2^{-j-1})(1-2^{h+j})a^{(j)}(0) \qquad (j \ge 0). \tag{1151}$$

For clarity the passage from the sum to actual endpoint derivatives is justified here, rather than differentiating a formal asymptotic series. We use the Euler–Maclaurin identity proved with its remainder in (1071); its classical provenance is [9, §2.10(i), (2.10.1)]. With $f_j(v) = v^j\psi_a^{(j)}(v)$, direct differentiation on $x > 0$ gives

$$x^j(\mathcal{A}_h a)^{(j)}(x) = \sum_{n\ge1} f_j(nx).$$

All these f_j are smooth and compactly supported. Integration by parts j times gives $\int f_j = (-1)^j j! \int \psi_a = 0$; the factors v^j make each zero-end boundary term vanish. Their axis derivatives are

$$f_j^{(r)}(0) = 0 \quad (r<j), \qquad f_j^{(r)}(0) = \frac{r!}{(r-j)!}\psi_a^{(r)}(0) \quad (r \ge j).$$

Choose the Euler–Maclaurin order M so that $2M - 1 > j$. After division by x^j, its remainder is bounded by

$$\frac{\|B_{2M}\|_\infty}{(2M)!}\, x^{2M-1-j}\|f_j^{(2M)}\|_1,$$

and hence tends to zero. Terms below degree j vanish, so every differentiated series has a finite limit at zero. The only constant term for $j \ge 1$ is the Bernoulli term with $2k - 1 = j$; for even $j \ge 2$ it is absent. The fundamental theorem of calculus on $[\epsilon, x]$, followed by $\epsilon \downarrow 0$, now identifies these derivative limits with successive derivatives of the extended function. Boundedness of the next derivative justifies the passage under each integral. Thus $\mathcal{A}_h a$ is smooth at zero, and

$$\begin{aligned}(\mathcal{A}_h a)(0) &= 0,\\ (\mathcal{A}_h a)^{(j)}(0) &= -\frac{B_{j+1}}{j+1}(1-2^{-j-1})(1-2^{h+j})a^{(j)}(0), \qquad j \geq 1.\end{aligned} \tag{1152}$$

The Bernoulli convention is $B_2 = 1/6$ and odd $B_{j+1} = 0$ for even $j \geq 2$. In particular

$$(\mathcal{A}_h a)'(0) = \frac{2^{h+1}-1}{16}\, a'(0). \tag{1153}$$

The coefficient is strictly positive for the source $h > 0$.

The preceding proofs apply also after any fixed z, t derivative of a smooth compactly supported parameter family with common support on compact preterminal parameter sets. Every bound then holds uniformly on those sets, with constants depending on the derivative order. This proves compatibility of the endpoint map with those parameter derivatives. It supplies no uniform estimate as $t \uparrow 1$.

29.3 A directly visible singularity at a fixed arithmetic coordinate

Apply this same operator to the actual compactly supported $a(\sigma) = B_\nu(\sigma, z, t)$. Write its arithmetic image as

$$\mathcal{B}_\nu(x, z, t) = \mathcal{A}_h B_\nu(x, z, t).$$

The domain and all nonlinear fluxes are those already given in the preceding angular-momentum section. Equations (1150) and (1153) now prove the exact formula

$$\boxed{\partial_x \mathcal{B}_\nu(0, 0, 1-\tau) = \frac{2^{h+1}-1}{8C}\,\tau^{-1-h}, \qquad 0 < \tau < \tau_1, \quad \nu > 0.} \tag{1154}$$

Here $x = 0, z = 0$ are fixed coordinates, not a shrinking sampling path, and $\mathcal{B}_\nu(0, z, t) = 0$ for every preterminal t. Thus the value is zero while its first x-derivative diverges positively. For any fixed $\epsilon > 0$, the $C^1([0, \epsilon])$ seminorm is at least the displayed endpoint derivative. In particular no extension continuous at $t = 1$ with values in $C^1([0, \epsilon])$ can agree with this family. This assertion follows from an exact derivative of the original arithmetic sum, not merely a lower bound obtained after its inverse.

For every integer $m \geq 0$, differentiation of the exact equality gives

$$\partial_t^m \partial_x \mathcal{B}_\nu(0, 0, t) = \frac{2^{h+1}-1}{8C}(1+h)_m(1-t)^{-1-h-m}, \qquad 1 - \tau_1 < t < 1.$$

Indeed $d_t(1-t)^{-\alpha} = \alpha(1-t)^{-\alpha-1}$. These are equalities on an open preterminal interval. They make no assignment of finite terminal derivatives.

The projection onto B_ν has not discarded the other velocity data in the full correspondence. Its exact smooth section and kernel are

$$RB_\nu = \frac{B_\nu(r^2/2, z, t)}{r^2}(-x_2, x_1, 0), \qquad w_\nu = u_\nu - RB_\nu, \qquad \langle r(w_\nu)_\theta \rangle = 0,$$

where B_ν/σ has its smooth integral extension at zero. The preceding section proves the inverse $(B_\nu, w_\nu) \mapsto RB_\nu + w_\nu$ and retains every covariance in both momentum fluxes. The signal (1154) concerns the B_ν component of that complete map. It does not replace the full Navier–Stokes equation by a closed scalar equation for B_ν.

29.4 The exact Mellin pole carrying this axis signal

For fixed preterminal parameters, put

$$\widehat{B}_\nu(s) = \int_0^\infty B_\nu(\sigma, z, t)\sigma^{s-1}\, d\sigma, \qquad \mathfrak{B}_\nu(s) = \int_0^\infty \mathcal{B}_\nu(x, z, t)x^{s-1}\, dx.$$

Both integrals are initially holomorphic on $\Re s > -1$. The full transform calculation in the preceding section gives

$$\mathfrak{B}_\nu(s) = W_h(s)\widehat{B}_\nu(s), \qquad W_h(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s}).$$

The pole of ζ at 1 is canceled by the displayed first dilation factor, making W_h entire.

Here is the continuation needed at $s = -1$. For any fixed $b > 0$, split the integral at b and subtract the linear Taylor term on $[0, b]$. Since $B_\nu(0) = 0$, the exact formula is

$$\widehat{B}_\nu(s) = \int_0^b [B_\nu(\sigma) - B_{\nu,\sigma}(0)\sigma]\sigma^{s-1} d\sigma + \int_b^\infty B_\nu(\sigma)\sigma^{s-1} d\sigma + \frac{B_{\nu,\sigma}(0)b^{s+1}}{s+1}.$$

Taylor's integral remainder is $O(\sigma^2)$; the first integral is holomorphic for $\Re s > -2$, including after any complex s derivative on a compact subset, because the additional powers of $\log \sigma$ remain integrable. The second is entire by compact support away from zero. Hence $\mathrm{Res}_{s=-1}\, \widehat{B}_\nu = B_{\nu,\sigma}(0)$. The identical argument applies to the proved smooth $\mathcal{B}_\nu$, so its residue is $\partial_x \mathcal{B}_\nu(0)$.

The classical value $\zeta(-1) = -1/12$ [51, §25.6(i)] gives

$$W_h(-1) = -\frac{1}{12}\left(1 - \frac{1}{4}\right)(1 - 2^{h+1}) = \frac{2^{h+1} - 1}{16} > 0.$$

Equality on $\Re s > -1$ therefore continues to the meromorphic functions just constructed and agrees with the direct endpoint proof. At $z = 0, t = 1 - \tau$ the pole is genuine and its residue is exactly

$$\mathrm{Res}_{s=-1}\, \mathfrak{B}_\nu(s, 0, 1 - \tau) = \frac{2^{h+1} - 1}{8C}\tau^{-1-h}. \tag{1155}$$

There is no zero-factor compensation at this pole, since $W_h(-1) > 0$. The sign of $\zeta(-1)$ alone is not the sign of the arithmetic residue: both dilation factors have been evaluated explicitly.

This identifies what the construction propagates in this channel: an exact axis derivative, equivalently a growing residue at the fixed Mellin point -1. The identity retains the full $\widehat{B}_\nu$, its flux equations, and its inverse maps. The nontrivial zeros of ζ remain the zero factors specified in the full product formula; no off-critical zero is inferred from this axis pole or from its positive residue.

30 How the actual radial diffusion recovers hidden arithmetic axis data

The fixed-axis singularity of Section 29 has a precise evolution law in the same arithmetic coordinates. An even radial Taylor coefficient disappears from the Taylor series of the arithmetic trace, but survives as a regular Mellin value at a trivial zeta zero. Radial diffusion reads that value and sends it into an odd endpoint derivative. We prove the full maps and their inverses, rather than discarding the canceled factor or replacing the Navier–Stokes system by a scalar heat equation.

30.1 The original transform and its complete input class

Fix the source parameter $0 < h < 1/100$, without changing it, and put

$$\mathcal{D}_0 = \{a \in C^\infty([0,\infty);\mathbb{C}) : \operatorname{supp} a \text{ compact},\ a(0) = 0\}.$$

The coordinate of a is the physical $\sigma = r^2/2$ when the input is a fluid field. The arithmetic argument is denoted x. Retain exactly

$$\phi_a(x) = a(x) - 2^h a(2x), \qquad \psi_a(x) = \phi_a(x) - \tfrac{1}{2}\phi_a(x/2), \qquad G_a(x) = \mathcal{A}_h a(x) = \sum_{n\geq 1} \psi_a(nx).$$

The mean of ψ_a is zero by substitution, and $\psi_a(0) = 0$. Section 29 proves by differentiated Euler–Maclaurin remainders that $G_a \in \mathcal{D}_0$, including its smooth extension at zero, and gives

$$G_a^{(j)}(0) = W_h(-j)a^{(j)}(0) \quad (j \geq 1), \qquad W_h(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s}). \tag{1156}$$

This W_h is entire: its first dilation factor cancels the simple pole of ζ at one. In particular $G_a^{(2m)}(0) = 0$ for every $m \geq 1$.

There has nevertheless been no loss of the full function. On the actual range $\mathcal{G}_h = \mathcal{A}_h\mathcal{D}_0$, the inverse is

$$\begin{aligned} \psi_a(x) &= \sum_{n\geq 1} \mu(n) G_a(nx), \\ \phi_a(x) &= \sum_{\ell\geq 0} 2^{-\ell}\psi_a(x/2^\ell), \\ a(x) &= -\sum_{k\geq 1} 2^{-kh}\phi_a(x/2^k). \end{aligned} \tag{1157}$$

For the first identity both divisor sums are finite at a fixed $x > 0$. The identity $\sum_{d|m}\mu(d) = 1$ for $m = 1$, and zero otherwise, follows by expanding $\prod_{p|m}(1-1)$; it proves that inverse. The partial second sum ends in $\phi_a(x) - 2^{-\ell-1}\phi_a(x/2^{\ell+1})$, and the partial third sum ends in $a(x) - 2^{-kh}a(x/2^k)$. The remainders tend to zero by boundedness at zero and $h > 0$. After j differentiations the extra factors are $2^{-\ell j}$ and 2^{-kj}, so the recovered series converge smoothly on compact half-line intervals. These statements concern the recovered ψ_a and ϕ_a; no uniform bound on the first inverse near $x = 0$ has been assumed. The three formulas prove both inverse compositions on the stated range, as in Section 28.

30.2 Every input Taylor coefficient in residues and regular values

Use ordinary Taylor coefficients $a_n = a^{(n)}(0)/n!$, not derivatives without their factorials. For $a \in \mathcal{D}_0$, its Mellin transform is initially holomorphic for $\Re s > -1$. For any positive b and integer $N \geq 1$, subtraction at zero gives the exact continuation

$$\begin{aligned} \widehat{a}(s) = &\int_0^b \left(a(\sigma) - \sum_{n=1}^N a_n\sigma^n\right)\sigma^{s-1}\,d\sigma + \int_b^\infty a(\sigma)\sigma^{s-1}\,d\sigma \\ &+ \sum_{n=1}^N \frac{a_n b^{s+n}}{s+n}, \qquad \Re s > -N-1. \end{aligned} \tag{1158}$$

Taylor's integral remainder is $O(\sigma^{N+1})$. Each complex derivative in s adds an integrable power of $\log\sigma$ on compact subsets of this half-plane; the second integral is entire. This proves holomorphy of the remainder integrals, not merely formal continuation. Different N, b agree on the original half-plane and then by the identity theorem. Thus $\widehat{a}$ is meromorphic on $\mathbb{C}$, with at most simple poles at $-1, -2, \ldots$, and $\operatorname{Res}_{s=-n}\widehat{a} = a_n$. The same argument applies to the proved smooth compact G_a.

The exact Mellin formula (1145), initially proved by absolute summation and then analytic continuation, therefore holds meromorphically on the entire plane:

$$\mathfrak{G}_a(s) := \mathcal{M}G_a(s) = W_h(s)\widehat{a}(s). \tag{1159}$$

The necessary local factors of W_h are computable from the classical functional equation

$$\zeta(s) = 2(2\pi)^{s-1}\sin(\pi s/2)\Gamma(1-s)\zeta(1-s),$$

with its original constants [51, §25.4, (25.4.2)]. At $s = -2m$, only the sine vanishes; differentiating it there proves

$$\zeta'(-2m) = \frac{(-1)^m(2m)!}{2(2\pi)^{2m}}\zeta(2m+1) \neq 0. \tag{1160}$$

Indeed $\cos(-m\pi) = (-1)^m$, $\Gamma(1+2m) = (2m)!$, and $\zeta(2m+1) = \sum_{n\geq 1} n^{-2m-1} > 0$. Both dilation factors at $-2m$ are nonzero. At negative odd integers the sine is itself nonzero, and the other factors of the functional equation are finite and nonzero. Hence the two exact recovery formulas are

$$\boxed{a_{2m-1} = \frac{\operatorname{Res}_{s=-(2m-1)}\mathfrak{G}_a(s)}{W_h(-(2m-1))}, \qquad a_{2m} = \frac{\mathfrak{G}_a(-2m)}{W_h'(-2m)} \quad (m \geq 1).} \tag{1161}$$

For the second formula write, in the local coordinate $w = s + 2m$,

$$\widehat{a}(s) = a_{2m}/w + H(w), \qquad W_h(s) = w\{W_h'(-2m) + wV(w)\},$$

where H, V are holomorphic. Their product is holomorphic and has value $W_h'(-2m)a_{2m}$ at $w = 0$. This also proves the formula when $a_{2m} = 0$. For odd indices the residue of a product with a holomorphic nonzero factor gives the first formula, including the zero-residue case. Every higher coefficient of the analytic unit has been retained in the displayed local product; none changes its constant term.

For later use define

$$\kappa_h = W_h(-1) = \frac{2^{h+1}-1}{16} > 0, \qquad d_h = W_h'(-2) = \frac{7(2^{h+2}-1)\zeta(3)}{32\pi^2} > 0. \tag{1162}$$

The first identity is the original $\zeta(-1) = -1/12$ calculation. The second follows from (1160) with $m = 1$, $1 - 2^{-3} = 7/8$, and the remaining negative dilation factor. In particular

$$\frac{\kappa_h}{d_h} = \frac{2\pi^2(2^{h+1}-1)}{7(2^{h+2}-1)\zeta(3)}.$$

Thus cancellation at a trivial zero removes a pole from $\mathfrak{G}_a$, but does not remove the input coefficient: it becomes the regular value in (1161). The vanishing even Taylor coefficients of G_a and its generally nonzero regular Mellin values are exact related coordinates, not contradictory assertions. These recover all Taylor coefficients of a, not its whole smooth germ from that Taylor sequence alone; smooth flat remainders are retained by the full inverse (1157).

30.3 The actual diffusion operator and a proved endpoint obstruction

The radial part of the angular-momentum diffusion in (1141) is

$$L_{\mathrm{rad}}a(\sigma) = 2\sigma a''(\sigma).$$

It maps $\mathcal{D}_0$ to itself. Its exact transported operator, with domain and codomain both specified, is

$$\mathscr{L}_h : \mathcal{G}_h \longrightarrow \mathcal{G}_h, \qquad \mathscr{L}_h G = \mathcal{A}_h L_{\mathrm{rad}} \mathcal{A}_h^{-1} G. \tag{1163}$$

The inverse in this definition is precisely the three convergent series (1157). It is not division by W_h at its zeros. For $\Re s > 0$, integration by parts twice, with the zero-end terms vanishing since $a = O(\sigma)$ and $a' = O(1)$, gives

$$\mathcal{M}(L_{\mathrm{rad}}a)(s) = 2s(s-1)\widehat{a}(s-1).$$

Upper boundary terms vanish by compact support. It follows that

$$\mathcal{M}(\mathscr{L}_h G_a)(s) = 2s(s-1)W_h(s)\widehat{a}(s-1). \tag{1164}$$

This equality then holds meromorphically everywhere by (1158). In particular

$$\boxed{(\mathscr{L}_h G_a)'(0) = \mathrm{Res}_{s=-1}\,\mathcal{M}(\mathscr{L}_h G_a)(s) = 4\kappa_h a_2 = \frac{4\kappa_h}{d_h}\mathfrak{G}_a(-2).} \tag{1165}$$

The direct derivative gives the same factor: $(2\sigma a'')'(0) = 2a''(0) = 4a_2$, followed by (1156) for its arithmetic image.

There is no operator on full Taylor sequences at $x = 0$ whose output is the full Taylor sequence of $\mathscr{L}_h G$ for every $G \in \mathcal{G}_h$. Here is an explicit witness, including its smooth cutoff. Put

$$e(v) = \begin{cases} \exp(-1/v), & v > 0, \\ 0, & v \le 0, \end{cases} \qquad \chi(\sigma) = \frac{e(2-\sigma)}{e(2-\sigma) + e(\sigma-1)}, \qquad \sigma \ge 0.$$

The denominator is strictly positive: both terms could vanish only if $\sigma \ge 2$ and $\sigma \le 1$ simultaneously. Every right derivative of e at zero is zero, because on $v > 0$ each derivative is a polynomial in $1/v$ times $e^{-1/v}$, and $y^k e^{-y} \to 0$ as $y \to +\infty$. Thus χ is smooth, equals one on $[0, 1]$, and is zero on $[2, \infty)$. Take $a(\sigma) = \sigma^2\chi(\sigma)$. Its only nonzero endpoint derivative is $a''(0) = 2$. Equation (1156) proves that G_a is flat at zero: every one of its derivatives there is zero, including its value. It is not the zero function, by injectivity of $\mathcal{A}_h$. But (1165) gives

$$(\mathscr{L}_h G_a)'(0) = 4\kappa_h \ne 0, \qquad \mathscr{L}_h 0 = 0.$$

The two inputs G_a and zero have identical full Taylor sequences and different output first derivatives. This proves the assertion without presuming any finite-order description. In particular $\mathscr{L}_h$ cannot agree on its entire stated domain with a finite-order differential operator having coefficients smooth up to $x = 0$: repeated product rules show that every such operator sends flat functions to flat functions. The obstruction concerns endpoint jets, not injectivity of the full arithmetic transform, whose exact inverse was proved above.

More generally,

$$(\mathscr{L}_h G_a)^{(j)}(0) = 2jW_h(-j)a^{(j+1)}(0) \quad (j \ge 1).$$

This follows by applying (1156) to $2\sigma a''$ and differentiating the product. For odd j it accesses precisely an even input derivative. The regular-value half of (1161) supplies that derivative with its exact factorial. No Taylor datum has been silently invented.

30.4 The complete nonlinear residue equation for the actual source

Use exactly the actual corrected-field functions of Section 28, with their physical viscosity:

$$B = \langle ru_\theta\rangle, \quad P = \langle r^2 u_r u_\theta\rangle, \quad Q = \langle ru_z u_\theta\rangle, \quad F = \langle rf_\theta\rangle, \qquad \langle\cdot\rangle = \frac{1}{2\pi}\int_0^{2\pi}(\cdot)\,d\theta.$$

Their compact supports, all original cutoff-summed cross terms, and their exact equation are already proved in (1141). Write ordinary radial Taylor coefficients

$$B = b_1\sigma + b_2\sigma^2 + O(\sigma^3), \qquad P = p_2\sigma^2 + O(\sigma^3), \qquad Q = q_1\sigma + O(\sigma^2), \quad F = f_1\sigma + O(\sigma^2).$$

These are coefficients of the complete actual angular averages; no axisymmetry has been imposed outside the source's inner region. The coefficient of σ in (1141) is exactly

$$\partial_t b_1 + 2p_2 + \partial_z q_1 = \nu(4b_2 + \partial_{zz} b_1) + f_1. \tag{1166}$$

This follows equivalently by one σ derivative at zero; the smoothness already proved justifies commutation with z, t derivatives on every compact preterminal parameter set.

Let $\mathfrak{B} = \mathcal{M}\mathcal{A}_h B$, and define $\mathfrak{P}, \mathfrak{Q}, \mathfrak{F}$ identically. Define their residues

$$J_B = \operatorname{Res}_{s=-1}\mathfrak{B}, \qquad J_Q = \operatorname{Res}_{s=-1}\mathfrak{Q}, \qquad J_F = \operatorname{Res}_{s=-1}\mathfrak{F}.$$

Equation (1161) proves, without deleting any flux, that

$$J_B = \kappa_h b_1, \quad J_Q = \kappa_h q_1, \quad J_F = \kappa_h f_1, \quad \mathfrak{B}(-2) = d_h b_2, \quad \mathfrak{P}(-2) = d_h p_2.$$

Multiplying (1166) by κ_h therefore gives the full arithmetic evolution identity

$$\boxed{\partial_t J_B + \partial_z J_Q - \nu\partial_{zz} J_B - J_F = \frac{\kappa_h}{d_h}\{4\nu\mathfrak{B}(-2) - 2\mathfrak{P}(-2)\}.} \tag{1167}$$

It holds on the entire preterminal domain where the actual source fields are smooth. The right side retains the true quadratic flux and radial diffusion, recovered by regular Mellin values. It is not a closed scalar equation for J_B.

The division-free equation (1146) yields exactly the same identity, including at its canceled factor. In the local coordinate $w = s+1$, its left side is the product of

$$W_h(s-1) = d_h w + O(w^2) \quad \text{and} \quad \frac{\partial_t J_B + \partial_z J_Q - \nu\partial_{zz} J_B - J_F}{w} + O(1).$$

Its limit is d_h times the numerator. Its right side tends to

$$-2\kappa_h\{\mathfrak{P}(-2) - 2\nu\mathfrak{B}(-2)\}.$$

Thus cancellation is a fully calculated zero-times-pole limit, not the equation $0 = 0$ obtained by substituting a zero factor before retaining the principal part.

For the particular source field at $z = 0, t = 1-\tau$, (1155) supplies the exact value

$$J_B(0, 1-\tau) = \frac{2^{h+1}-1}{8C}\tau^{-1-h}, \qquad \partial_t J_B(0, 1-\tau) = \frac{(1+h)(2^{h+1}-1)}{8C}\tau^{-2-h}.$$

These can be substituted into (1167) without discarding the transverse z derivative, the actual force, or either regular Mellin value. The residue is the first radial arithmetic derivative, as proved in Section 29. This identifies a specific part of its full dynamics: the even radial curvature is invisible to the trace's Taylor series but visible to its Mellin value at -2, and enters the growing residue through the original viscous term. The auxiliary compact cutoff above proves a domain obstruction for this exact operator; it is not a new Navier–Stokes or RH counterexample. The actual source field and the trivial zeta zero used in the evolution law have never been replaced.

30.5 The full alternating residue–value hierarchy

The same calculation retains every radial Taylor order, not just the first growing derivative. For every integer $n \geq 1$ define the nonzero constant and the corresponding Mellin coordinate

$$c_{h,n} = \begin{cases} W_h(-n), & n \text{ odd}, \\ W_h'(-n), & n \text{ even}, \end{cases} \qquad E_n(\mathfrak{G}) = \begin{cases} \operatorname{Res}_{s=-n} \mathfrak{G}(s), & n \text{ odd}, \\ \mathfrak{G}(-n), & n \text{ even}. \end{cases}$$

On the actual arithmetic image these values and residues exist by (1161), and $E_n(\mathfrak{G}_a) = c_{h,n} a_n$. This is a diagonal bijection between the input Taylor coefficients and their joint residue–value coordinates, with inverse $a_n = E_n / c_{h,n}$. It is not a claim that a smooth function is determined by its Taylor coefficients.

For $A = B, P, Q, F$, put $\widetilde{A}_n = E_n(\mathfrak{A})$ and $A_n = \partial_\sigma^n A(0, z, t)/n!$. Differentiating the exact equation (1141) n times at zero, or applying the product rule with these factorials, gives

$$\partial_t B_n + (n+1) P_{n+1} + \partial_z Q_n = \nu\{2n(n+1) B_{n+1} + \partial_{zz} B_n\} + F_n.$$

For the radial diffusion coefficient, $B_{n+1}\sigma^{n+1}$ contributes $2n(n+1)B_{n+1}\sigma^n$; every other monomial has a different degree. Taylor's differentiated remainder justifies this finite coefficient extraction without analyticity. Multiplication by $c_{h,n}$ and the proved coefficient inverse therefore yield, at every order,

$$\partial_t \widetilde{B}_n + \partial_z \widetilde{Q}_n - \nu \partial_{zz} \widetilde{B}_n - \widetilde{F}_n = (n+1) \frac{c_{h,n}}{c_{h,n+1}} \{2\nu n\, \widetilde{B}_{n+1} - \widetilde{P}_{n+1}\}, \quad n \geq 1. \tag{1168}$$

Every denominator has been proved nonzero from the functional equation, with no RH assumption. This alternates regular values at negative even integers with residues at negative odd integers. The $n = 1$ equation is precisely (1167). All original nonlinear fluxes, forcing coefficients and viscosity remain; the complete functions, including flat radial remainders, are still carried by (1159) and (1157).

31 The actual source quotient of a flow-driven arithmetic profile

31.1 Retained source data and the full division estimate

The arithmetic data are exactly those of Section 13.1 and (927): $\mathcal{H}_{\leq 1}$ is the set of entire functions of order at most one, $N_{\text{ar}} = \overline{\mathbb{C}[s]\Xi}^{\tau}$, $Q_{\text{ar}} = (\mathcal{H}_{\leq 1}, \tau)/N_{\text{ar}}$, and $S[h] = [sh]$. The source topology τ remains unnamed beyond the primary specification; it is not replaced by the explicit growth topology γ of Section 13.2. Its growth seminorms are $\|h\|_{p,a} = \sup_s |h(s)| e^{-a|s|^p}$ for every $1 < p < 2$, $a > 0$. The full-domain heat group is precisely (930). The original multiplication $Mh(s) = sh(s)$ preserves N_{ar}, as proved from its polynomial generators in Section 13.1. In the flow formulas below, S_j, J and X_t instead denote the polynomial target, its derivative and the volume-preserving preterminal material flow of Section 25; their exact formulas are (1093), (1094) and (1107). The field convention is $u(t, x)$. The particular source field and its full remainder are those of Section 26. Its cited existence theorem remains an imported primary-source result.

For the division argument the value $\Xi(0) = \xi(1/2)$ is nonzero. One direct verification retains the usual Dirichlet eta function: $\eta(1/2) = \sum_{n\geq 1}(-1)^{n-1} n^{-1/2} > 0$, since its paired terms are positive and its first pair is strictly positive. The alternating-series convergence and $\eta(s) = (1 - 2^{1-s})\zeta(s)$ give $\zeta(1/2) = \eta(1/2)/(1 - \sqrt{2}) < 0$. The other factors in $\xi(1/2) = \frac{1}{2}(1/2)(-1/2)\pi^{-1/4}\Gamma(1/4)\zeta(1/2)$ therefore give $\Xi(0) > 0$. No assertion about any noncentral zeta zero is used.

We now compute the closures and the failure of descent in specified topologies. This does not identify them with τ. Let κ be the topology of uniform convergence on compact sets on the same set $\mathcal{H}_{\le 1}$, and define

$$N_\gamma = \overline{\mathbb{C}[s]\Xi}^{\gamma}, \qquad N_\kappa = \overline{\mathbb{C}[s]\Xi}^{\kappa}.$$

Evaluations of all derivatives are continuous in both topologies by Cauchy's formula. The growth topology is finer than κ, so $N_\gamma \subseteq N_\kappa$. If λ is a zero of Ξ with its actual multiplicity m_λ, every element of either closed subspace has vanishing derivatives of orders $0, \dots, m_\lambda - 1$ at λ.

Lemma 31.1 (Entire division with its growth and topology retained)**.** *Let $g \in \mathcal{H}_{\le 1}$ with $g(0) \neq 0$. If $f \in \mathcal{H}_{\le 1}$ and $h = f/g$ extends to an entire function, then $h \in \mathcal{H}_{\le 1}$. For $1 < p < 2$, $a > 0$, put $b = a/(6 \cdot 2^p)$. There is the linear division estimate*

$$\|f/g\|_{p,a} \le \left(\frac{\max(1, \|g\|_{p,b})}{|g(0)|}\right)^3 \|f\|_{p,b}. \tag{1169}$$

Multiplication by g is a topological linear isomorphism from $(\mathcal{H}_{\le 1}, \gamma)$ onto the closed subspace of functions divisible by g, with its subspace topology. Compact-topology division is also continuous.

Proof. Choose $R > 0$ with no zero of g on $|s| = R$. Factor its finitely many zeros in the disk, retaining multiplicities. The residual is holomorphic and zero-free there, so its log modulus is harmonic. For $|z| < R$ the boundary mean of $\log|Re^{i\theta} - z|$ equals $\log R$: expand $\log(1 - (z/R)e^{-i\theta})$ in its uniformly convergent series and take real parts. The mean-value formula for the residual gives the exact Jensen identity

$$\frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta = \log|g(0)| + \sum_{|z|<R} m_z \log(R/|z|) \ge \log|g(0)|.$$

There is no zero at zero. With $\log^+ x = \max(0, \log x)$ and $\log^- x = \max(0, -\log x)$, the inequality $\log^+|h| \le \log^+|f| + \log^-|g|$ gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log^+|h(Re^{i\theta})|\, d\theta \le \log^+ M_f(R) + \log^+ M_g(R) - \log|g(0)|. \tag{1170}$$

The right side is nonnegative since $M_g(R) \ge |g(0)|$. The harmonic Poisson extension of the continuous nonnegative boundary function $\log^+|h|$ majorizes $\log|h|$ by the maximum principle and is nonnegative; hence it majorizes $\log^+|h|$ in the disk. For $|s| \le r < R$ the Poisson kernel is at most $(R + r)/(R - r)$. Take $R \downarrow 2r$ through circles avoiding zeros. Continuity of the two maximum functions gives, for every $r > 0$,

$$\log^+ M_h(r) \le 3\big(\log^+ M_f(2r) + \log^+ M_g(2r) - \log|g(0)|\big). \tag{1171}$$

No pointwise lower bound for g on a large circle has been assumed. For $f \neq 0$, put $F = \|f\|_{p,b} > 0$ and apply this to $f/F, h/F$. Then $\log^+ M_{f/F}(2r) \le b(2r)^p$, and $\log^+ M_g(2r) \le \log\max(1, \|g\|_{p,b}) + b(2r)^p$. The coefficient of r^p is $6b2^p = a$, giving (1169); continuity fills in $r = 0$. The zero numerator is immediate. All p, a are allowed, so h belongs to the original order class and division is continuous. Multiplication is continuous by Lemma 13.2. The requisite vanishing jets at every zero are closed conditions; local Taylor division identifies them with entire divisibility. The estimate proves this closed set is exactly $g\mathcal{H}_{\le 1}$. For the compact topology choose $R > r$ with no boundary zero and $\delta = \min_{|s|=R}|g(s)| > 0$. The maximum principle for the filled entire quotient gives $M_h(r) \le \delta^{-1} M_f(R)$, proving compact continuity too. $\square$

For general nonzero $g \in \mathcal{H}_{\leq 1}$, choose and retain a point c with $g(c) \neq 0$ and use disks centered at c. The functions and spectral coordinate remain unchanged. The bound $|s+c|^p \leq 2^{p-1}(|s|^p + |c|^p)$ proves continuity of translations by c and $-c$ in γ; compact containment proves it in κ. Thus all conclusions hold with the stated translated estimates. The original center $c = 0$ already applies to Ξ.

Proposition 31.2 (The computed closures are the same subspace)**.** *With their definitions unchanged,*

$$\begin{aligned} N_\gamma = N_\kappa &= \Xi \mathcal{H}_{\leq 1} \\ &= \{f \in \mathcal{H}_{\leq 1} : D^h f(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0,\ 0 \leq h < m_\lambda\}. \end{aligned} \tag{1172}$$

This is equality of subsets of the original ring, not an identification of γ, κ, or τ.

Proof. Jet continuity puts both closures inside the displayed vanishing set and gives $N_\gamma \subseteq N_\kappa$. Local Taylor division makes $h = f/\Xi$ entire for any f in that set; the division lemma then puts h in $\mathcal{H}_{\leq 1}$. Its Taylor polynomials p_n converge in γ, and multiplication gives $p_n \Xi \to f$ there. Thus the vanishing set is contained in N_γ, proving all equalities. The algebraic-ideal equality is the conclusion of this argument; it was not substituted for an unevaluated closure. □

31.2 Translation, heat, and the original spectral generator

For $a, \theta \in \mathbb{C}$ define, with the input coordinate s and output coordinate ζ retained,

$$(T_a h)(\zeta) = h(\zeta + a), \qquad V_{\theta,a} = T_a U_\theta : \mathcal{H}_{\leq 1} \longrightarrow \mathcal{H}_{\leq 1}.$$

The derivative is the complex derivative in the displayed coordinate. Translation preserves the original order class. More precisely, for $1 < p < 2$ and $c > 0$ the already defined growth seminorms satisfy

$$\|T_a h\|_{p,c} \leq e^{c|a|^p} \|h\|_{p,c\,2^{1-p}},$$

because $|\zeta + a|^p \leq 2^{p-1}(|\zeta|^p + |a|^p)$. Thus T_a and T_{-a} are continuous for γ. Ordinary differentiation gives $DT_a = T_a D$; continuity and the convergent heat series give $T_a U_\theta = U_\theta T_a$. Consequently

$$V_{\theta,a}^{-1} = U_{-\theta} T_{-a}, \qquad V_{\theta,a} D = D V_{\theta,a}.$$

These assertions hold on all of $\mathcal{H}_{\leq 1}$ and do not impose continuity of translation or differentiation for the unnamed source topology.

Retain the actual τ, N_{ar} and Q_{ar}, and define their exact transported presentations

$$\tau_{\theta,a} = (V_{\theta,a})_* \tau, \qquad N_{\theta,a} = V_{\theta,a} N_{\mathrm{ar}}, \qquad Q_{\theta,a} = (\mathcal{H}_{\leq 1}, \tau_{\theta,a})/N_{\theta,a}. \tag{1173}$$

Then $N_{\theta,a}$ is closed, and

$$\overline{V}_{\theta,a} : Q_{\mathrm{ar}} \longrightarrow Q_{\theta,a}, \qquad [h] \longmapsto [V_{\theta,a} h]$$

is a complex-linear homeomorphism. Its inverse is induced by the displayed inverse on representatives. The original arithmetic multiplication has the exact conjugate

$$B_{\theta,a} = V_{\theta,a} s V_{\theta,a}^{-1} = \zeta + a - \frac{\theta}{2} D_\zeta. \tag{1174}$$

Indeed the previous heat commutator gives $U_\theta s U_{-\theta} = s - \theta D/2$; translation sends multiplication by s to multiplication by $\zeta + a$ and commutes with D. This full combination preserves $N_{\theta,a}$ and is continuous for $\tau_{\theta,a}$ by its conjugacy. The induced quotient operator is similar through $\overline{V}_{\theta,a}$ to the source S, so its spectrum is exactly the original s-spectrum. To verify both directions, conjugate a continuous two-sided inverse of $S-\lambda$ by $\overline{V}_{\theta,a}$, or conjugate an inverse of the transported operator back. At $\theta=0$, multiplication by ζ itself descends, and its spectrum is $\mathrm{Spec}(S)-a$ because $B_{0,a}=\zeta+a$.

Along the retained real flow let

$$a_j(t,q)=S_j(X_t(q)),\qquad v_j(t,q)=\partial_t a_j(t,q),\qquad \Psi_j(t,q,\zeta)=V_{\vartheta(t),a_j(t,q)}\Xi(\zeta).$$

Here S_j denotes the original polynomial target component, whereas the unindexed S above denotes the arithmetic quotient operator. The source quotient is taken fibre by fibre at each fixed (t,q,j). If a topology on the whole family of quotient fibres is needed, give their disjoint union the topology transported from $\mathbb{C}^2\times Q_{\mathrm{ar}}$ by $((\theta,a),[h])\mapsto((\theta,a),[V_{\theta,a}h])$. This specifies joint continuity by an exact trivialization and does not assert joint parameter continuity for the untransported τ.

Because $\Xi\in N_{\mathrm{ar}}$, every profile represents zero in its actual transported quotient:

$$[\Psi_j]_{Q_{\vartheta,a_j}}=0. \tag{1175}$$

Its temporal defect is nevertheless a specific full representative,

$$\mathcal{D}_j=\partial_t\Psi_j+\frac{\vartheta'}{4}D_\zeta^2\Psi_j=v_jV_{\vartheta,a_j}\Xi'.$$

This is the complete chain rule with the original heat coefficient. Write

$$\beta=[\Xi']_{Q_{\mathrm{ar}}},\qquad \mathfrak{a}_\beta=\{c\in\mathbb{C}:c\Xi'\in N_{\mathrm{ar}}\}.$$

Then the exact quotient statement and its scalar kernel are

$$[\mathcal{D}_j]_{Q_{\vartheta,a_j}}=\overline{V}_{\vartheta,a_j}(v_j\beta),\qquad \ker(c\mapsto c\beta)=\mathfrak{a}_\beta. \tag{1176}$$

The set $\mathfrak{a}_\beta$ is a closed complex linear subspace: it is the inverse image of the closed N_{ar} under the continuous scalar map $c\mapsto c\Xi'$ into $(\mathcal{H}_{\le 1},\tau)$. Its equality with $\{0\}$ or $\mathbb{C}$ is precisely the unresolved membership question $\Xi'\notin N_{\mathrm{ar}}$ or $\Xi'\in N_{\mathrm{ar}}$. No evaluation functional on the source quotient is used to decide it.

At a fixed physical point $x=X_t(q)$, retain $v=J(x)u(t,x)$ and the full inverse $J^{-1}(x)=[2W_1(x)\ W_2(x)\ W_3(x)]$. Thus the exact kernel of the complexified velocity-to-source-class map is

$$J(x)^{-1}(\mathfrak{a}_\beta^3)\subseteq\mathbb{C}^3;$$

for real velocities take its intersection with $\mathbb{R}^3$. This follows componentwise from (1176) and the invertibility of the unchanged J.

The derivative relation explains why a zero profile class does not determine its representative derivative. At a fixed (θ,a) put

$$\mathfrak{D}_{\theta,a}=\{([h],[Dh]):h\in\mathcal{H}_{\le 1}\}\subset Q_{\theta,a}^2,\qquad \mathcal{W}_{\theta,a}=(DN_{\theta,a}+N_{\theta,a})/N_{\theta,a}.$$

It has full domain and output fibre $[Dh]+\mathcal{W}_{\theta,a}$ at $[h]$: replace h by every $h+n$ with $n\in N_{\theta,a}$. Its presenting map has kernel $\{h\in N_{\theta,a}:Dh\in N_{\theta,a}\}$. Since D commutes with $V_{\theta,a}$, conjugating both components by $\overline{V}_{\theta,a}$ gives exactly the previous source derivative relation and its ambiguity space. In particular $\overline{V}_{\theta,a}\beta$ is an actual element of its multivalued part at zero. The displayed temporal connection is an identity on the full smooth profiles; a derivative on all untransported source quotient representatives is not presumed.

31.3 An exact information-preserving refinement and its residue projection

Retain the explicit analytic ideal $I = \Xi\mathcal{H}_{\le 1}$, already proved closed for γ by the full division estimate, and put

$$H_\cap = N_{\mathrm{ar}} \cap I, \qquad C_\cap = (\mathcal{H}_{\le 1}, \tau \vee \gamma)/H_\cap, \qquad Q_I = (\mathcal{H}_{\le 1}, \gamma)/I. \tag{1177}$$

The join topology is precisely the topology induced by $h \mapsto (h,h)$ into $(\mathcal{H}_{\le 1}, \tau) \times (\mathcal{H}_{\le 1}, \gamma)$. Both subspaces in the intersection are closed in this join, so $C_\cap$ is Hausdorff. The identity on representatives induces continuous surjections

$$p_N : C_\cap \longrightarrow Q_{\mathrm{ar}}, \qquad p_I : C_\cap \longrightarrow Q_I, \qquad \ker p_N = N_{\mathrm{ar}}/H_\cap, \quad \ker p_I = I/H_\cap. \tag{1178}$$

The kernel assertions are direct membership tests; continuity follows from the quotient universal property. These kernel identifications are also homeomorphisms when N_{ar} and I have their subspace topologies from $\tau \vee \gamma$. To prove this, let $q_\cap$ be the open quotient map and let A denote either subspace. Because $H_\cap \subset A$, $q_\cap(A \cap O) = q_\cap(A) \cap q_\cap(O)$ for every open ambient O. Thus its restriction to A is an open quotient map onto the corresponding kernel with its subspace topology. The joint map (p_N, p_I) is continuous and injective since its kernel is zero. No inference of a topological embedding from those two properties is needed.

Let λ be any actual zero of Ξ, with its actual multiplicity $m_\lambda \ge 1$. Choose a disk containing no other zero and retain the full unit

$$\Xi(s) = (s-\lambda)^{m_\lambda} A_\lambda(s), \qquad A_\lambda(\lambda) \ne 0.$$

For $h \in \mathcal{H}_{\le 1}$ define

$$R_\lambda(h) = \operatorname{Res}_{s=\lambda} \frac{h(s)}{\Xi(s)} = \frac{1}{2\pi i} \int_{|s-\lambda|=r} \frac{h(s)}{\Xi(s)}\, ds, \tag{1179}$$

where the circle has positive orientation and lies in this disk. The integral is independent of the permitted r by Laurent expansion. It is continuous for γ: its absolute value is at most $r \sup_{|s-\lambda|=r} |h(s)| / \min_{|s-\lambda|=r} |\Xi(s)|$, and a compact supremum is bounded by a fixed growth seminorm times the maximum of its exponential weight on the circle. It annihilates I, since h/Ξ is entire there, and therefore annihilates $H_\cap$.

Logarithmic differentiation of the full local factor gives

$$\frac{c\Xi'(s)}{\Xi(s)} = \frac{m_\lambda c}{s-\lambda} + c\frac{A_\lambda'(s)}{A_\lambda(s)}, \qquad R_\lambda(c\Xi') = m_\lambda c. \tag{1180}$$

In particular $\Xi' \notin I$: the displayed pole is nonzero. Thus $\beta_I = [\Xi']_{Q_I} \ne 0$, and $\widetilde{\beta} = [\Xi']_{C_\cap} \ne 0$ as well, independent of the status of β in the actual source quotient. The residue functional factors continuously through $C_\cap$; write its induced functional as $\widetilde{R}_\lambda$. The formula

$$\Pi_\lambda(c) = \frac{\widetilde{R}_\lambda(c)}{m_\lambda}\, \widetilde{\beta}, \qquad c \in C_\cap \tag{1181}$$

defines a continuous projection onto the line $\mathbb{C}\widetilde{\beta}$. Indeed its square is itself by $\widetilde{R}_\lambda(\widetilde{\beta}) = m_\lambda$, and scalar multiplication on the quotient is continuous. Its inverse coordinate on that line is $\widetilde{R}_\lambda/m_\lambda$, while $c \mapsto c\widetilde{\beta}$ is its continuous inverse. Consequently this line has exactly the ordinary complex topology, and $C_\cap = \mathbb{C}\widetilde{\beta} \oplus \ker \widetilde{R}_\lambda$ is a continuous direct sum. This proves an information-preserving refinement of the actual source map, including a continuous recovery coordinate.

The full principal-part map $h \mapsto (\mathrm{PP}_\lambda(h/\Xi))_{\lambda \in Z(\Xi)}$ has kernel exactly I. Vanishing of every negative Laurent coefficient makes h/Ξ entire at every zero; it is already holomorphic elsewhere, and the proved entire division bound puts it in $\mathcal{H}_{\le 1}$. The reverse inclusion is immediate. Each finite principal-part coordinate is continuous by the same circle integral with the appropriate power of $s-\lambda$. On the velocity line its entire image at every actual zero is the single term $m_\lambda c/(s-\lambda)$ from (1180), with the unchanged analytic unit in the regular part.

For general (θ, a) let $I_{\theta,a} = V_{\theta,a} I$ and transport (1177) by $V_{\theta,a}$. Its ambient topology is $\tau_{\theta,a} \vee \gamma$, since $V_{\theta,a}$ is a γ-homeomorphism, and its relation is $N_{\theta,a} \cap I_{\theta,a} = V_{\theta,a} H_\cap$. Every map, kernel and projection above is transported exactly. In particular the transported residue coordinate is

$$R_{\lambda;\theta,a}(h) = \operatorname{Res}_{s=\lambda} \frac{(U_{-\theta} T_{-a} h)(s)}{\Xi(s)}.$$

For the actual temporal defect it gives $R_{\lambda;\vartheta,a_j}(\mathcal{D}_j) = m_\lambda v_j$. The pair of source and analytic classes therefore has the explicit injective common lift $v_j[V_{\vartheta,a_j}\Xi']$ in this transported refinement. For nonzero heat time this formula retains the inverse heat operator; it does not replace $V_{\theta,a} I$ by an unproved ordinary ideal generated by $T_a Z_\theta$.

31.4 The logarithmic connection and the public blowup residue field

Take the explicit Newman-time choice $\vartheta(t) = 0$. For the original real flow set

$$\Psi_j(t,q,\zeta) = \Xi(\zeta + a_j(t,q)), \qquad \mathcal{D}_j = \partial_t \Psi_j = v_j(t,q)\Xi'(\zeta + a_j(t,q)).$$

At every actual zero λ the translated divisor is $\zeta_{\lambda,j}(t,q) = \lambda - a_j(t,q)$, and the full factorization is

$$\Psi_j = (\zeta - \zeta_{\lambda,j})^{m_\lambda} A_\lambda(\zeta + a_j).$$

Thus all multiplicities and the actual analytic unit are retained. Since a_j is real, its imaginary coordinate is exactly $\operatorname{Im}\lambda$. Differentiating its location gives $\partial_t \zeta_{\lambda,j} = -v_j$.

On the complement of the divisor in $[0,T) \times \mathbb{R}^3_q \times \mathbb{C}_\zeta$, define the complex-valued differential form, smooth in the real variables and holomorphic in ζ,

$$\omega_j = \frac{d\Psi_j}{\Psi_j} = \frac{\Xi'(\zeta + a_j)}{\Xi(\zeta + a_j)}(d\zeta + da_j), \qquad da_j = v_j\, dt + \sum_b \partial_{q_b} a_j\, dq_b. \tag{1182}$$

All label and time coefficients are displayed. Put $s = \zeta + a_j$. Direct exterior differentiation gives $d\omega_j = (\Xi'/\Xi)'(s)\, ds \wedge ds + (\Xi'/\Xi)(s) d^2 s = 0$. Equivalently the connection $d - \omega_j$ on the trivial complex line has zero curvature and annihilates the section Ψ_j. This calculation is on the complement of its zeros and retains the singular divisor explicitly.

At fixed (t,q) a sufficiently small positively oriented circle around $\zeta_{\lambda,j}$ has

$$\frac{1}{2\pi i}\int \omega_j = m_\lambda,$$

where only its $d\zeta$ restriction is integrated. The pole term in (1180) gives this integral; the holomorphic unit term has zero integral by Cauchy's theorem. The time component and its residue are exactly

$$\begin{aligned}
\omega_j(\partial_t) &= \frac{\mathcal{D}_j}{\Psi_j} = \frac{m_\lambda v_j}{\zeta - \zeta_{\lambda,j}} + v_j \frac{A_\lambda'(\zeta + a_j)}{A_\lambda(\zeta + a_j)}, \\
r_j(t,q) &:= \operatorname{Res}_{\zeta = \zeta_{\lambda,j}} \frac{\mathcal{D}_j}{\Psi_j} = m_\lambda v_j = -m_\lambda \partial_t \zeta_{\lambda,j}.
\end{aligned} \tag{1183}$$

The coordinate change $s = \zeta + a_j$ has differential $ds = d\zeta$ on each spectral fibre, so this is precisely $R_{\lambda;0,a_j}(\mathcal{D}_j)$, including its orientation and multiplicity factor.

Write $r = (r_1, r_2, r_3)^{\mathsf{T}}$, and retain the full original polynomial target and its matrix inverse:

$$S = (F_1/2, F_2, F_3)^{\mathsf{T}}, \quad J = DS = \operatorname{diag}(1/2, 1, 1)DF, \quad J^{-1} = [2W_1 \ W_2 \ W_3].$$

The literal recovery and pointwise norm identities are

$$\begin{aligned} u(t, X_t(q)) &= \frac{2W_1(X_t(q))r_1 + W_2(X_t(q))r_2 + W_3(X_t(q))r_3}{m_\lambda}, \\ |r(t,q)|^2 &= m_\lambda^2 |J(X_t(q))u(t, X_t(q))|^2. \end{aligned} \tag{1184}$$

They follow by $r = m_\lambda v$, $v = Ju$ and the stated matrix inverse, with no selection of a global inverse sheet of S. The flow X_t is retained as part of the data.

For the nonempty compact velocity support K put

$$M_K = \max_K \|J\|_{\mathrm{op}}, \qquad N_K = \max_K \|J^{-1}\|_{\mathrm{op}}.$$

They are finite and positive by compactness and $\det J = -1$. Since $X_t(K) = K$, X_t is onto, and u, v, r vanish outside the corresponding compact labels, (1184) gives

$$\frac{m_\lambda}{N_K}\|u(t)\|_\infty \le \sup_q |r(t,q)| \le m_\lambda M_K \|u(t)\|_\infty. \tag{1185}$$

The volume-preserving change of variables $x = X_t(q)$ gives

$$\int_{\mathbb{R}^3} |r(t,q)|^2 \, dq = m_\lambda^2 \int_{\mathbb{R}^3} |J(x)u(t,x)|^2 \, dx, \tag{1186}$$

whose square root is between $(m_\lambda/N_K)\|u(t)\|_2$ and $m_\lambda M_K \|u(t)\|_2$. These are exact spatial estimates for the arithmetic residues. With the previously proved spectral normalization $d_Z(0)^2 = \int_{\mathbb{R}} |\Xi'(s)|^2 ds > 0$, the unchanged real translations also give

$$|r(t,q)| = \frac{m_\lambda}{d_Z(0)} \left(\sum_j \|\mathcal{D}_j(t,q,\cdot)\|_{L^2(\mathbb{R})}^2 \right)^{1/2}.$$

Proposition 31.3 (Application of the cited public Navier–Stokes theorem)**.** *For every $\nu > 0$, take the velocity, pressure and force supplied by the public Theorem 1.1 of [7]. The construction above, with $\vartheta = 0$, gives arithmetic logarithmic residue fields satisfying*

$$\sup_{0 \le t < 1} \int_{\mathbb{R}^3} |r(t,q)|^2 \, dq < \infty, \qquad \limsup_{t \uparrow 1} \sup_q |r(t,q)| = \infty.$$

Every residue recovers the full physical velocity by (1184)*, and each translated arithmetic divisor has the same imaginary coordinates as the original zero divisor of Ξ.*

Proof. The cited theorem supplies a smooth incompressible velocity on $[0,1) \times \mathbb{R}^3$, a fixed compact support, zero initial velocity, a uniform kinetic-energy bound, and unbounded terminal velocity supremum, with its stated smooth compactly supported force and its pressure. These are the inputs of the proved preterminal flow construction, so its volume and inverse statements apply on every $[0, T_0]$ with $T_0 < 1$. Formula (1186) and the upper bound using M_K prove the uniform residue energy bound. The lower bound in (1185) proves the unbounded terminal supremum. Recovery and imaginary-coordinate preservation were proved above for the actual same fields. The existence step in this corollary is exactly the cited Theorem 1.1; the correspondence and every displayed estimate are proved here. $\square$

31.5 The published witness's exact growing residue

Take the particular corrected field of the public construction, with the source's fixed $h \in (0, 1/100)$, $X_{\text{in}} > 0$, $e_0 > 0$, and remainder $R_*(\tau)$. Its closing formulas [7, equations (10.20)–(10.22), p. 124], as imported in (1120), are

$$|R_*(\tau)| \le C_* \tau^{2h} \quad (0 < \tau < \tau_*), \qquad (u_\nu)_2(1-\tau, x_{\nu,\tau}) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)),$$
$$r_{\nu,\tau} = \sqrt{2\nu X_{\text{in}} \tau}, \quad x_{\nu,\tau} = (r_{\nu,\tau}, 0, 0), \quad q_{\nu,\tau} = X^{-1}_{\nu,1-\tau}(x_{\nu,\tau}).$$

The remainder is the actual source remainder, including its corrected fields. The parameter h is the one chosen in that construction. The quantity $r_{\nu,\tau}$ here is the displayed physical radius; the map to the original polynomial target is retained below. The labels $q_{\nu,\tau}$ vary with τ according to the exact material inverse just displayed.

Substitution into the full original polynomials gives

$$S(r, 0, 0) = (0, 0, 2r), \qquad J(r, 0, 0) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 1 & 3r \\ 2 & -3r^2 & -r^3 \end{pmatrix}.$$

Here the same original formulas are

$$F_1 = Q^3 w + y^2 Q(4+3xy), \quad F_2 = y + 3xQ^2 w + 3xy^2(4+3xy), \quad F_3 = 2x - 3x^2 y - x^3 w, \quad Q = 1 + xy.$$

Differentiating $S = (F_1/2, F_2, F_3)$ and setting $(x, y, w) = (r, 0, 0)$ gives each matrix entry. Its componentwise inverse gives $u_3 = 2v_1$ and $u_2 = v_2 - 6rv_1$, retaining the off-diagonal $3r$. The first two arithmetic shifts at $(1-\tau, q_{\nu,\tau})$ are zero. Consequently, as an identity of full entire functions of ζ,

$$\begin{aligned} \mathcal{E}_{\nu,\tau}(\zeta) &:= \mathcal{D}_{\nu,2}(1-\tau, q_{\nu,\tau}, \zeta) - 6r_{\nu,\tau} \mathcal{D}_{\nu,1}(1-\tau, q_{\nu,\tau}, \zeta) \\ &= \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)) \Xi'(\zeta). \end{aligned} \tag{1187}$$

Indeed each defect is $v_j \Xi'$ because its shift is zero, so their combination is $(u_\nu)_2 \Xi'$, and the stated source identity applies.

For every actual zero λ of multiplicity m_λ, its complete local logarithmic image is

$$\frac{\mathcal{E}_{\nu,\tau}(\zeta)}{\Xi(\zeta)} = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)) \left(\frac{m_\lambda}{\zeta - \lambda} + \frac{A'_\lambda(\zeta)}{A_\lambda(\zeta)} \right). \tag{1188}$$

Therefore the actual unshifted arithmetic residue at the source sample is exactly

$$\operatorname{Res}_{\zeta=\lambda} \frac{\mathcal{E}_{\nu,\tau}(\zeta)}{\Xi(\zeta)} = m_\lambda \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{1189}$$

All multiplicities, the whole analytic unit, the positive viscosity, the original remainder and the exact material labels remain present. Choose $\tau_{**} > 0$ within the source interval and its cutoff region with $C_* \tau_{**}^{2h} \le e_0/2$. For $0 < \tau < \tau_{**}$ the absolute value of (1189) is at least $m_\lambda \sqrt{\nu}\, e_0 \tau^{-1/2-h}/2$, which diverges. This is an explicit pole-coefficient singularity of the published constructed field's arithmetic image.

For the three residue channels $r_{\nu,j}$ of (1183), the same evaluation and the two-component Cauchy–Schwarz inequality give the quantitative bound

$$|r_\nu(1-\tau, q_{\nu,\tau})| \ge \frac{m_\lambda \sqrt{\nu}\, e_0}{2\sqrt{1 + 72\nu X_{\text{in}} \tau}} \tau^{-1/2-h}. \tag{1190}$$

To verify the denominator, the coefficient vector $(-6r_{\nu,\tau}, 1)$ has squared norm $1 + 36r_{\nu,\tau}^2 = 1 + 72\nu X_{\mathrm{in}}\tau$. Its product with the first two residue coordinates is exactly the residue in (1189); the third coordinate can only increase the Euclidean norm on the left. If $E_\nu = \nu^{5/4}F_*(1)$ is the full source force-integral energy bound and $M_\nu = \max_{K_\nu} \|J\|_{\mathrm{op}}$, the earlier volume identity simultaneously gives

$$\int_{\mathbb{R}^3} |r_\nu(t,q)|^2 dq \le m_\lambda^2 M_\nu^2 E_\nu^2 = m_\lambda^2 M_\nu^2 \nu^{5/2} F_*(1)^2 \quad (t < 1).$$

At these particular first two channels, their zero shifts mean the source quotient is the original Q_{ar} itself. The exact classes are

$$[\mathcal{E}_{\nu,\tau}]_{Q_{\mathrm{ar}}} = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau))\beta, \qquad [\mathcal{E}_{\nu,\tau}]_{C_\cap} = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau))\widetilde{\beta}.$$

The latter line has the explicitly proved continuous inverse $\widetilde{R}_\lambda/m_\lambda$, so its scalar coordinate has exactly the displayed singularity. In particular these common classes leave every bounded subset of $C_\cap$: a continuous linear functional sends bounded sets to bounded scalar sets, whereas their $\widetilde{R}_\lambda$ values diverge. Their projection to the actual source quotient and its exact kernel remain (1178); the undecided source membership is not replaced by an identification with the analytic ideal.

The source class has an exact two-case classification as well. If $\beta = 0$, every displayed class in Q_{ar} is zero. If $\beta \ne 0$, the line $\mathbb{C}\beta$ has exactly the ordinary complex topology, and these source classes leave every bounded subset of Q_{ar}. Here is the complete topological argument, which does not require a residue functional on Q_{ar}. The closed relation makes Q_{ar} a Hausdorff topological vector space. The compact circle $\{c\beta : |c| = 1\}$ omits zero, so it has a disjoint balanced zero-neighborhood U. If $c\beta \in U$ and $|c| \ge 1$, multiplying by $1/|c|$ contradicts this disjointness. Thus $c\beta \in U$ implies $|c| < 1$; scaling U proves continuity of the inverse scalar coordinate on the line. For a bounded subset B there is $t > 0$ with $B \subset tU$, so $c\beta \in B$ implies $|c| < t$. The source lower bound makes the displayed scalar tend to infinity, proving the asserted escape. The source membership of Ξ' decides which case occurs; this argument does not decide that membership. In the same two cases the full complexified velocity map has respectively rank zero or rank three, by the exact kernel $J^{-1}(\mathfrak{a}_\beta^3)$. There is no intermediate loss of one component in this common-β construction.

There is also an exact relation between the changing sample labels and the time component of the flat logarithmic connection. Let $A_\tau = D_q X_{\nu,1-\tau}(q_{\nu,\tau})$. Differentiating $X_{\nu,1-\tau}(q_{\nu,\tau}) = x_{\nu,\tau}$ gives

$$A_\tau \frac{dq_{\nu,\tau}}{d\tau} = \frac{dx_{\nu,\tau}}{d\tau} + u_\nu(1-\tau, x_{\nu,\tau}), \qquad \frac{dx_{\nu,\tau}}{d\tau} = \frac{r_{\nu,\tau}}{2\tau}(1,0,0).$$

The sign follows from $d(1-\tau)/d\tau = -1$. Since $\nabla_q a_j$ is the jth row of $J(x_{\nu,\tau})A_\tau$, and the first two rows of $J(r,0,0)$ have zero first component,

$$\nabla_q a_j \cdot \frac{dq_{\nu,\tau}}{d\tau} = v_j, \qquad j = 1, 2.$$

Thus $a_j(1-\tau, q_{\nu,\tau}) = 0$ has total derivative $-v_j + \nabla_q a_j \cdot dq_{\nu,\tau}/d\tau = 0$, exactly as required. On the spectral complement of the zeros, the pullback of (1182) to this curve and fixed ζ is zero in the τ direction, while its original ∂_t component has the residue (1183). These statements are related by the displayed full tangent-vector identity; the sample is not silently treated as one fixed material label.

For the full four-term profile $K_\theta = U_\theta K_0$ of (1100), the corresponding actual source classes are instead

$$\kappa = [K_0]_{Q_{\mathrm{ar}}}, \qquad \beta_K = [K_0']_{Q_{\mathrm{ar}}}, \qquad [T_a K_\theta] = \overline{V}_{\theta,a}\kappa, \qquad [vT_a K_\theta'] = \overline{V}_{\theta,a}(v\beta_K).$$

These follow by the same commuting translation, heat and derivative maps and retain κ without assigning it the value zero. The residue corollary above uses the actual Ξ profile. The displayed four-term quotient formulas concern the whole K profile and its derivative; they do not identify individual zeros of K with zeros of Ξ.

31.6 The translated modular double

Define $C_E h(s) = \overline{h(\overline{s})}$, $\tau^c = (C_E)_*\tau$ and $N^c = C_E N_{\mathrm{ar}}$. This is an antilinear involution of the original order class, since conjugating the Taylor coefficients gives an entire function with exactly the same maximum modulus on each centered disk. Also $\|C_E h\|_{p,a} = \|h\|_{p,a}$, so it is a γ-homeomorphism. By definition it is a homeomorphism from $(\mathcal{H}_{\le 1}, \tau)$ to $(\mathcal{H}_{\le 1}, \tau^c)$, with inverse itself. Applying it to the convergent heat series gives $C_E U_\theta = U_{\overline{\theta}} C_E$ with the coefficient $-1/4$ retained. Direct conjugation gives $C_E T_a = T_{\overline{a}} C_E$, and the established heat identity gives $C_E U_\theta = U_{\overline{\theta}} C_E$. On the two entire sectors define

$$\widehat{V}_{\theta,a} = \mathrm{diag}(T_a U_\theta, T_a U_{-\theta}), \qquad \mathcal{J}(h,k) = (C_E k, C_E h).$$

Then the exact all-parameter identity is

$$\mathcal{J}\widehat{V}_{\theta,a} = \widehat{V}_{-\overline{\theta},\overline{a}}\mathcal{J}. \tag{1191}$$

Indeed its first component is $C_E T_a U_{-\theta} k = T_{\overline{a}} U_{-\overline{\theta}} C_E k$, and its second component is the same calculation with θ. For the actual real flow, $a \in \mathbb{R}$; for real Newman time it therefore sends (θ, a) to $(-\theta, a)$.

Give the two ambient sectors the topologies $(T_a U_\theta)_*\tau$ and $(T_a U_{-\theta})_*\tau^c$, and quotient by the closed relation

$$\widehat{N}_{\theta,a} = T_a U_\theta N_{\mathrm{ar}} \oplus T_a U_{-\theta} N^c.$$

Applying (1191) both to these relation spaces and to the definitions of their transported topologies proves an antilinear homeomorphism from this actual doubled quotient to the one indexed by $(-\overline{\theta}, \overline{a})$. Its square is the identity, and its conjugacy with the quotient heat and translation maps follows on representatives from the same displayed equality. Thus the Cartan reversal retains the full real flow translation while reversing the radial heat parameter, with both original source and conjugate-source sectors specified.

32 The exact strength of the frozen source spectral information

The primary spectrum statement in [103, Proposition 3.6(ii)] specifies the actual zero set as the spectrum of the Hadamard quotient's multiplication operator. It gives no multiplicity assertion. The cited [104, Proposition 2.24] computes the finite Hermite polynomial image; the discussion after its proof explicitly treats function-space choices. The finite multiplicity calculation in [104, Theorem 2.47 and equations (2.447)–(2.449)] belongs to the separately specified weighted Hilbert quotient. The following exact algebra records both the obstruction available from such data and the limits of transferring a spectrum set alone.

32.1 Exact algebraic spectral defects of the source relation

These calculations keep the actual N_{ar} and require only the already proved $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. For every $\lambda \in \mathbb{C}$, division by $x = s - \lambda$ gives the algebraic identities

$$\begin{aligned} \mathrm{coker}_{\mathrm{alg}}(S-\lambda) &= \mathcal{H}_{\le 1}/(N_{\mathrm{ar}} + x\mathcal{H}_{\le 1}), \\ \ker(S-\lambda) &\simeq (N_{\mathrm{ar}} \cap x\mathcal{H}_{\le 1})/(xN_{\mathrm{ar}}), \quad [n] \longmapsto [n/x]. \end{aligned} \tag{1192}$$

Division belongs to the original order class: if $f(\lambda) = 0$, the filled quotient is entire, is bounded on a fixed disk, and outside $|s| \geq 2|\lambda|+1$ its modulus is at most $2|f(s)|$. Thus $x\mathcal{H}_{\leq 1}$ is exactly the kernel of evaluation at λ. The cokernel identity follows from the image of the induced multiplication. For the second identity, changing n by xn_0 changes n/x by $n_0 \in N_{\mathrm{ar}}$; its image is in the stated kernel. Conversely any kernel representative f has $xf \in N_{\mathrm{ar}} \cap x\mathcal{H}_{\leq 1}$. Finally $[n/x] = 0$ exactly when $n \in xN_{\mathrm{ar}}$. This proves every asserted kernel and image.

The cokernel has only two possibilities. When every $n \in N_{\mathrm{ar}}$ vanishes at λ, it is one-dimensional, with its exact identification given by evaluation. Otherwise retain any $h_\lambda \in N_{\mathrm{ar}}$ with $h_\lambda(\lambda) \neq 0$. Then

$$\begin{gathered} A_\lambda f(s) = \frac{f(s) - f(\lambda)h_\lambda(s)/h_\lambda(\lambda)}{s-\lambda}, \\ f = \frac{f(\lambda)}{h_\lambda(\lambda)}h_\lambda + (s-\lambda)A_\lambda f \end{gathered} \tag{1193}$$

proves surjectivity of $S-\lambda$, and its exact remaining kernel is

$$\ker(S-\lambda) \simeq N_{\mathrm{ar}}/\big((s-\lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda\big), \qquad n \longmapsto [A_\lambda n]. \tag{1194}$$

The map is onto because a kernel class $[g]$ is the image of $n = (s-\lambda)g \in N_{\mathrm{ar}}$. Its kernel consists exactly of $n = (s-\lambda)n_0 + ch_\lambda$, $n_0 \in N_{\mathrm{ar}}$, by (1193). The sum is direct by evaluation. In this second case algebraic invertibility is therefore equivalent to the exact identity $N_{\mathrm{ar}} = (s-\lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda$. On that locus the representative formula $[f] \mapsto [A_\lambda f]$ is independent of representatives, since $A_\lambda N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; multiplication in (1193) proves the right inverse, and $A_\lambda((s-\lambda)f) = f$ proves the left inverse. These are algebraic statements; continuity of this inverse in τ has not been inserted.

The first derivative and the heat generator have exactly the same frozen invariance question for the source subspace:

$$DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \iff D^2N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \iff -\tfrac{1}{4}D^2N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}. \tag{1195}$$

Indeed the product rule is $D^2(sn) - sD^2n = 2Dn$. Since $sn \in N_{\mathrm{ar}}$, second-derivative invariance puts both terms on the left in N_{ar}, proving first-derivative invariance. Iteration proves the reverse implication; the retained nonzero scalar $-1/4$ proves the last equivalence. If these equivalent inclusions hold, all $D^j\Xi$ belong to N_{ar}. At each λ at least one is nonzero: otherwise the Taylor series would make Ξ identically zero. Thus every $S-\lambda$ is then algebraically surjective by (1193). Its prescribed topological spectral points would be represented entirely by the kernel (1194) or a noncontinuous algebraic inverse. This computes the precise alternatives without assigning either one to the source's stated spectral set.

There is also a full multiplicity consequence of algebraic derivative descent. Writing $d[f] = [Df]$ on its exact descent locus, the product rule gives $[d, S] = I$. For $0 \neq v \in \ker(S-\lambda)$,

$$(S-\lambda)d^kv = -kd^{k-1}v, \qquad (S-\lambda)^kd^kv = (-1)^kk!v \quad (k \geq 0), \tag{1196}$$

with zero right side in the first identity at $k = 0$. To prove induction, commute the next d using $(S-\lambda)d = d(S-\lambda) - I$; the coefficient $-k$ becomes $-(k+1)$. Iteration proves the second identity. A finite linear dependence among $v, dv, \ldots, d^nv$, with largest nonzero coefficient at index j, becomes that coefficient times $(-1)^jj!v$ after applying $(S-\lambda)^j$, a contradiction. Thus every such eigenvector generates an infinite independent generalized eigenvector sequence. No finite source multiplicity assertion has been assumed.

Finally even a single forward nonzero heat inclusion has a calculated consequence. Fix $t \neq 0$ and $c = t/2$, and define

$$\begin{gathered} A_0 = U_t, \qquad A_{k+1} = MA_k - A_kM, \qquad A_k = P_k(D)U_t, \\ P_0(X) = 1, \qquad P_{k+1}(X) = cXP_k(X) - P_k'(X). \end{gathered} \tag{1197}$$

The identity follows by induction from $[M, U_t] = cDU_t$ and $[M, D^j] = -jD^{j-1}$: commuting M through $P_k(D)U_t$ contributes $-P_k'(D)U_t + cP_k(D)DU_t$. The degree of P_k is exactly k, with leading coefficient $c^k \neq 0$. Under $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, induction with $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ gives $A_k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; subtracting the lower-degree terms of P_k then gives exactly

$$D^k U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad (k \geq 0). \tag{1198}$$

In particular every derivative of the nonzero $Z_t = U_t \Xi$ lies in N_{ar}. At every point one derivative is nonzero by the entire Taylor theorem, so every $S - \lambda$ is algebraically surjective in this case too. This proof uses only the one forward inclusion. For the stronger equality $U_t N_{\mathrm{ar}} = N_{\mathrm{ar}}$, it also proves $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, by writing each $n \in N_{\mathrm{ar}}$ as $U_t m$ in (1198). None of these exact implications presumes the inclusion or equality for the actual source closure. The explicit obstruction spaces above remain the calculated source question.

32.2 One-sided heat inclusion and the full generalized chain

The already proved commutator identity (1195) and its following heat calculation show that $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ makes every $S - \lambda$ algebraically onto. The single inclusion $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ at any $t \neq 0$ also makes every $S - \lambda$ onto, using every derivative of $U_t \Xi$ and the exact source division map. Here is a stronger consequence that needs no descended derivative.

Lemma 32.1. *Let $T : V \to V$ be a surjective endomorphism of a complex vector space and let $0 \neq v_0 \in \ker T$. Then for every integer $k \geq 1$, $\dim \ker T^k \geq k$. More precisely one can choose $Tv_j = v_{j-1}$ for every $j \geq 1$, and the vectors $v_0, \ldots, v_n$ are linearly independent for every n.*

Proof. Surjectivity gives the successive preimages. It follows inductively that $T^j v_j = v_0$ and $T^{j+1} v_j = 0$. Applying T^n to a relation $\sum_{j=0}^{n} c_j v_j = 0$ gives $c_n v_0 = 0$, hence $c_n = 0$. Repeating with the remaining highest index gives every coefficient zero. The first k vectors lie in $\ker T^k$, proving the bound. □

Applied to $T = S - \lambda$, this proves that either frozen inclusion just stated is incompatible with any nonzero finite-dimensional generalized eigenspace of the actual source operator. The source-independent implication is

$$U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}, \quad t \neq 0, \quad 0 \neq v_0 \in \ker(S - \lambda) \quad \Longrightarrow \quad \dim \ker(S - \lambda)^k \geq k \quad (k \geq 1). \tag{1199}$$

With a descended $d[f] = [Df]$, its more specific ladder is $v_j = (-1)^j d^j v_0 / j!$; indeed $[d, S] = I$ gives $(S - \lambda) d^j v_0 = -j d^{j-1} v_0$. Thus the complete factorial and sign agree with the general preimage construction. This is a proved algebraic incompatibility; it does not assert that the unnamed source quotient has the finite generalized block needed to apply the incompatibility.

32.3 The source Sobolev cokernel and an exact common test-space map

The Sobolev construction cited immediately after Connes–Marcolli Proposition 2.24 is specified in Chapter 2, Section 9. It is a different, explicitly defined quotient, and supplies a useful exact comparison. We retain its weight, its generator, and its limited set of detected jets. The primary locators are Connes–Marcolli, printed pp.407–408 and 416–420 [104], and Connes, Section III, pp.10–15 and Appendix I, pp.62–67 of the author PDF [105]. Here and below $\delta > 1$.

For the rational field, choose the splitting $C_{\mathbb{Q}} = C_{\mathbb{Q},1} \times \mathbb{R}_+^*$ used in the source, give the compact factor Haar mass one, and put $x = e^u$ on the second factor. Its trivial-character component is the Hilbert space

$$B_\delta = L^2\big(\mathbb{R}, (1+u^2)^{\delta/2}\,du\big), \qquad \|f\|_\delta^2 = \int_{\mathbb{R}} |f(u)|^2(1+u^2)^{\delta/2}\,du.$$

The source adelic map is

$$E_{\mathrm{ad}}\psi(g) = |g|^{1/2}\sum_{q\in\mathbb{Q}^*}\psi(qg), \quad \psi \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 = \{\psi : \psi(0) = 0,\ \int \psi\,dx = 0\}.$$

Let P_1 denote averaging over $C_{\mathbb{Q},1}$, and define

$$R_\delta = \overline{P_1 E_{\mathrm{ad}}(\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0)}^{B_\delta}, \qquad H_{\delta,1} = B_\delta/R_\delta.$$

These are exactly the trivial-character relation space and cokernel of the source. Indeed the source completion uses the norm of its image, so the extended isometry has closed range equal to the closure of the uncompleted image. Compact averaging commutes with that equivariant closed range. If b is a limit of image vectors and satisfies $P_1 b = b$, averaging an approximating sequence gives a sequence in the averaged image converging to b. This proves that the displayed closure is the trivial-character part of the full relation space, without replacing any source relation by an orthogonal complement as a definition.

We import explicitly the following result of the source proof: under the bilinear pairing $\langle f, \eta\rangle = \int f(u)\eta(u)\,du$, the annihilator of R_δ in $B_{-\delta} = L^2(\mathbb{R}, (1+u^2)^{-\delta/2}\,du)$ is the closed linear span of

$$u^a e^{itu}, \qquad \zeta(\tfrac{1}{2} + it) = 0, \quad t \in \mathbb{R}, \quad 0 \le a < m_{-t}, \quad a < \frac{\delta-1}{2}. \tag{1200}$$

Here m_{-t} is the original multiplicity of the zero $-t$ of Ξ, equal to that of $\frac{1}{2} + it$ by the retained reflected Mellin coordinate. The source proves the finite combinations after a compact Fourier cutoff and proves that the cutoff vectors are dense by uniformly bounded approximate units. Thus the closed span, not only its finite-dimensional projections, is the imported statement. No such annihilator statement is imported for the unnamed topology τ of the entire-function ring.

The precise cutoff can also be checked directly from the defined norm:

$$\|u^a e^{itu}\|_{-\delta}^2 = \int_{\mathbb{R}} |u|^{2a}(1+u^2)^{-\delta/2}\,du < \infty \quad \iff \quad 2a - \delta < -1.$$

Near zero the integrand is locally integrable for every integer $a \ge 0$; for $|u| \ge 1$ it is bounded above and below by positive constants times $|u|^{2a-\delta}$. Integration of this power proves the strict inequality and the logarithmic divergence at equality. Hence set

$$q_\delta = \left\lceil \frac{\delta-1}{2} \right\rceil, \qquad n_{\delta,\lambda} = \min(m_\lambda, q_\delta). \tag{1201}$$

There are exactly $n_{\delta,\lambda}$ allowed degrees at a real zero λ. The original Connes Theorem 1, Section III, p.13, states the equivalent multiplicity bound $n < (1+\delta)/2$, $n \le m_\lambda$. The inspected Connes–Marcolli printed p.408 instead displays $n < 1 + \delta/2$ in Theorem 2.47; its proof, pp.419–420, equations (2.444) and (2.448), gives precisely the stricter bound proved above. This records the primary discrepancy rather than silently identifying the two printed formulas. No assumption that an actual zeta zero has multiplicity greater than one is used.

Let $\mathcal{E} = C_c^\infty(\mathbb{R};\mathbb{C})$ retain its usual fixed-support smooth-function inductive-limit topology, and let $\mathscr{F}f(s) = \int f(u)e^{-ius}\,du$ as above. Define

$$K_{\text{full}} = \{f \in \mathcal{E} :\ D^a\mathscr{F}f(\lambda) = 0 \text{ for every zero } \lambda \text{ of } \Xi,\ 0 \le a < m_\lambda\}, \tag{1202}$$

$$K_\delta = \{f \in \mathcal{E} :\ D^a\mathscr{F}f(\lambda) = 0 \text{ for every real zero } \lambda \text{ of } \Xi,\ 0 \le a < n_{\delta,\lambda}\}. \tag{1203}$$

The already proved entire division theorem identifies $K_{\text{full}} = \{f : \mathscr{F}f \in \Xi\mathcal{H}_{\le 1}\}$ exactly.

Theorem 32.2 (Exact dense test ingress into the source Sobolev cokernel)**.** *The map $j_\delta : \mathcal{E} \to H_{\delta,1}$ sending a test function to its Hilbert quotient class is continuous, has dense image, and has kernel exactly K_δ. It induces the continuous map*

$$J_\delta : \mathcal{E}/K_{\text{full}} \longrightarrow H_{\delta,1}, \qquad [f] \longmapsto [f], \qquad \ker J_\delta = K_\delta/K_{\text{full}}. \tag{1204}$$

Its image is dense. These statements neither assert surjectivity nor identify the topology of its image with a quotient topology. For the source translation generator D_1, this ingress retains the original spectral coordinate through

$$j_\delta(Tf) = iD_1 j_\delta(f), \qquad T = -i\partial_u, \qquad \mathscr{F}(Tf) = s\mathscr{F}f. \tag{1205}$$

Proof. The inclusion $\mathcal{E} \to B_\delta$ is continuous on every fixed-support space: on $[-L, L]$,

$$\|f\|_\delta^2 \le \left(\int_{-L}^{L} (1+u^2)^{\delta/2}\,du\right) \sup|f|^2.$$

The inductive-limit universal property and the continuous Hilbert quotient map give continuity of j_δ. Smooth compactly supported functions are dense in B_δ: truncate an arbitrary f outside $[-L, L]$, whose squared norm tail tends to zero by integrability; on the enlarged interval $[-L-1, L+1]$, the weight is bounded above and below by positive constants, and convolution with a smooth compactly supported approximate identity approximates the truncation in ordinary L^2 and therefore in the weighted norm. These convolutions have compact support and are smooth. Consequently their quotient images are dense.

The continuous dual of B_δ is exactly $B_{-\delta}$ under the displayed bilinear pairing. To check this, multiply by $(1+u^2)^{\delta/4}$ to identify B_δ isometrically with ordinary L^2; the Riesz representation there, with complex conjugation absorbed in the representing function, gives exactly the inverse-weight dual and its norm. Since R_δ is closed, a vector belongs to it exactly when every member of its annihilator vanishes on that vector. One direction is the definition. For the other, a vector outside a closed Hilbert subspace has a nonzero orthogonal component, and its inner product defines a continuous separating functional. By the explicitly imported closed-span description (1200), it is enough to test the listed log-monomial characters. But

$$D^a\mathscr{F}f(-t) = (-i)^a \int u^a e^{itu} f(u)\,du, \qquad \int u^a e^{itu} f(u)\,du = i^a D^a\mathscr{F}f(-t).$$

Every factor i^a is nonzero. The coordinate $\lambda = -t$ and the proved count (1201) now give $\ker j_\delta = K_\delta$. Every equation defining K_δ appears among the equations defining K_{full}, proving $K_{\text{full}} \subseteq K_\delta$. Thus (1204) is well defined and continuous by the quotient universal property; changing the representative changes its image by zero exactly for a member of K_δ. This proves the kernel and the dense-image assertion.

Finally the source left translation is $(V(e^\varepsilon)f)(u) = f(u-\varepsilon)$, so its infinitesimal generator on $\mathcal{E}$ is $-\partial_u$. The difference quotient converges in B_δ because the supports remain in a fixed compact set and the smooth-function difference quotient converges uniformly there. Passing to the quotient proves that $j_\delta(f)$ belongs to the domain of D_1 and $D_1 j_\delta(f) = j_\delta(-f')$. Multiplication by i gives the first equation of (1205). Integration by parts in the compactly supported Fourier integral gives the second. In particular the source eigenparameter it becomes $-t$ under iD_1, exactly the same s coordinate used in the jet equations; no sign change is absorbed into the source operator. □

For completeness the unspecified source topology also has an exact comparison, requiring no unproved inclusion of its kernel in K_δ. Retain

$$K_\tau = \{f \in \mathcal{E} : \mathscr{F}f \in N_{\mathrm{ar}}\}, \qquad L_{\tau,\delta} = K_\tau \cap K_\delta.$$

Give $\mathcal{E}$ the join of its usual test topology, the inverse-image topology $\mathscr{F}^{-1}\tau$, and the seminorm $\|\cdot\|_\delta$. The last seminorm is already continuous in the test topology, but is displayed to specify every target map. Then K_τ, K_δ and $L_{\tau,\delta}$ are closed, and there are continuous maps

$$Q_{\mathrm{ar}} \longleftarrow \mathcal{E}/L_{\tau,\delta} \longrightarrow H_{\delta,1}, \qquad [f] \longmapsto [\mathscr{F}f], \quad [f] \longmapsto [f]. \tag{1206}$$

Their respective kernels are exactly $K_\tau/L_{\tau,\delta}$ and $K_\delta/L_{\tau,\delta}$; the right image is dense. Indeed each composite on $\mathcal{E}$ is continuous by definition, vanishes on the intersection, and has the asserted kernel before quotienting. The quotient universal property proves the assertions, and density follows from the same set of compact tests as above. The map to the product of the two targets is injective because its kernel is precisely the intersection already divided out. No topological embedding of that image is inferred from injectivity. This is a proved relation to the actual source quotient while its named topology remains retained. The comparison also intertwines the coordinate operators: T preserves K_τ because $\mathscr{F}(Tf) = s\mathscr{F}f$ and $MN_{\mathrm{ar}} \subset N_{\mathrm{ar}}$; it preserves K_δ because

$$D^a(sF)(\lambda) = \lambda D^a F(\lambda) + aD^{a-1}F(\lambda),$$

where the second term is zero for $a = 0$ and otherwise uses an already imposed lower jet. Thus T descends to the common span, whose left map intertwines it with S and whose right map intertwines it with iD_1 on the proved domain of the ingress.

There is also a literal compatibility of the original finite Hermite traces with the adelic map. For the source's even ψ, put $\psi_{\mathrm{ad}} = \psi \otimes \mathbf{1}_{\widehat{\mathbb{Z}}}$. It lies in $\mathcal{S}(\mathbb{A}_\mathbb{Q})_0$ because its value and additive integral are the two original zero constraints. On the idele represented by $(x, 1, 1, \ldots)$, a rational number belongs to every $\mathbb{Z}_p$ exactly when it is an integer: write it in lowest terms and use any prime dividing its denominator for the reverse implication. Therefore

$$E_{\mathrm{ad}}\psi_{\mathrm{ad}}(x) = x^{1/2} \sum_{n \in \mathbb{Z}\setminus\{0\}} \psi(nx) = 2\mathcal{E}\psi(x).$$

Every norm-one rational idele class has a representative consisting of a real sign and finite units. Indeed, for an idele g put $q = \prod_p p^{v_p(g_p)} \in \mathbb{Q}_{>0}$, a finite product. Then $g_p/q \in \mathbb{Z}_p^*$ for every prime, while the norm-one identity gives $|g_\infty| = q$. Dividing by the diagonal rational q gives the asserted representative. The function is unchanged by the finite units; evenness removes the real sign. It therefore lies in the entire trivial-character component. The Fourier transform in $u = \log x$ is thus exactly twice the original source trace, retaining the factor 2. In particular the finite Hermite generators yield $2R_\ell^\pm(s)\Xi(s)$ in this explicit realization. Their logarithmic representatives lie in every B_δ by their proved rapid decay. This proves a concrete arithmetic compatibility as well as the quotient comparison; it does not identify a Sobolev topology with the topology on all of $\mathcal{H}_{\le 1}$.

32.4 The actual finite block in the specified Sobolev source

Retain the source's real frequency t, compact character χ_0, weight $\delta > 1$, and actual zero multiplicity m. The vectors in the dual block of the specified source are

$$\eta_{t,a}(x) = \chi_0(x)|x|^{it}(\log|x|)^a, \qquad a \in \mathbb{Z}_{\geq 0}, \quad a < m, \quad a < (\delta - 1)/2.$$

The last strict bound is exactly integrability of $u^{2a}(1+u^2)^{-\delta/2}$ on $\mathbb{R}$: at infinity its power is $2a - \delta$, integrable precisely when it is less than -1; near zero it is locally integrable for all these a. Let r be the number of permitted integers. The source representation at $\lambda > 0$ acts by

$$\vartheta_m^{\mathrm{tr}}(\lambda)\eta_{t,a} = \lambda^{it} \sum_{b=0}^{a} \binom{a}{b} (\log \lambda)^b \eta_{t,a-b}.$$

Differentiating at $\lambda = e^{\epsilon}$, $\epsilon = 0$, proves

$$(D_{\chi_0}^{\mathrm{tr}} - it)\eta_{t,a} = a\eta_{t,a-1}, \qquad (D_{\chi_0}^{\mathrm{tr}} - it)^k \eta_{t,a} = \frac{a!}{(a-k)!}\eta_{t,a-k} \quad (0 \leq k \leq a). \tag{1207}$$

For $k > a$ the expression is zero. These vectors are independent: divide a proposed relation by the nonzero character and use that $\log|x|$ ranges over $\mathbb{R}$, making the remaining polynomial identically zero. Thus the block has dimension r, one-dimensional ordinary eigenspace, and nilpotence index r. Its multiplicity is generalized multiplicity. The common-test morphism to the actual Hadamard source is (1206), with its exact two kernels; this finite block is not silently identified with a block in that different quotient.

32.5 An explicit spectral model and the inverse at zero

Let $\Lambda = Z(\Xi)$ be the actual zero set. On the complex vector space

$$V_\Lambda = \bigoplus_{\lambda \in \Lambda} \bigoplus_{n \geq 0} \mathbb{C} e_{\lambda,n}$$

every vector has finite support. Give it the finest locally convex topology, defined by all seminorms on this vector space. Coordinate functionals separate points, so it is Hausdorff. Every linear map from it to any locally convex space is continuous: each pullback of a continuous seminorm is among these defining seminorms. With $e_{\lambda,-1} = 0$ set

$$S_\Lambda e_{\lambda,n} = \lambda e_{\lambda,n} + e_{\lambda,n-1}, \qquad d_\Lambda e_{\lambda,n} = -(n+1)e_{\lambda,n+1}.$$

Both are continuous, everywhere-defined, and direct evaluation on the basis proves $[d_\Lambda, S_\Lambda] = I$. For $\mu \notin \Lambda$ the full continuous inverse is

$$(S_\Lambda - \mu)^{-1} e_{\lambda,n} = \sum_{j=0}^{n} \frac{(-1)^j}{(\lambda - \mu)^{j+1}} e_{\lambda,n-j}. \tag{1208}$$

Multiplication in either order cancels adjacent terms of this finite sum and leaves the basis vector. The formula preserves finite support, so it defines an inverse on the whole space, continuous by its topology. For $\mu \in \Lambda$, the μ-summand is a surjective backward shift and every other summand uses the same finite inverse formula. Thus $S_\Lambda - \mu$ is onto, its kernel is exactly $\mathbb{C}e_{\mu,0}$, and

$$\ker(S_\Lambda - \lambda)^k = \operatorname{span}\{e_{\lambda,0}, \ldots, e_{\lambda,k-1}\}, \quad \operatorname{Spec}_{\mathrm{alg}} S_\Lambda = \operatorname{Spec}_{\mathrm{end}} S_\Lambda = \operatorname{Spec}_{\mathrm{point}} S_\Lambda = \Lambda. \tag{1209}$$

Here $\mathrm{Spec}_{\mathrm{end}}$ means failure of a continuous two-sided inverse in the algebra of continuous endomorphisms, not the Hilbert spectral subdivision called continuous spectrum. Since $\Xi(0) \neq 0$, this S_Λ has a continuous inverse at zero.

There is an exact principal-part realization of the model: send $e_{\lambda,n}$ to the class of w^{-n-1} in $\mathbb{C}[w, w^{-1}]/\mathbb{C}[w]$ in its λ-summand. The finite negative Laurent coefficients give its inverse. Multiplication by $\lambda+w$ is S_Λ and differentiation in w is d_Λ, as evaluation on each monomial proves. This complete model shows that the correct zero set as spectrum, geometric multiplicity one, and a continuous inverse at zero can coexist with a continuous Weyl pair. It does not identify this model with Q_{ar}.

For the actual quotient, injectivity of S gives exactly $N_{\mathrm{ar}} \cap s\mathcal{H}_{\leq 1} = sN_{\mathrm{ar}}$: if $sf \in N_{\mathrm{ar}}$, then $S[f] = 0$, so $f \in N_{\mathrm{ar}}$; the converse is polynomial invariance. The original source inverse at zero is explicitly

$$S^{-1}[f] = \left[\frac{f(s) - f(0)\Xi(s)/\Xi(0)}{s}\right]. \tag{1210}$$

The singularity is removable, and division by a degree-one polynomial preserves the original order class. Multiplication by s gives $[f]$, and injectivity gives its unique inverse. If f is changed by $n \in N_{\mathrm{ar}}$, the numerator difference belongs to $N_{\mathrm{ar}} \cap s\mathcal{H}_{\leq 1} = sN_{\mathrm{ar}}$, proving representative independence directly. The usual topological reading of the source resolvent also supplies continuity of this quotient inverse, not continuity of its chosen lift. More generally,

$$S - \lambda \text{ injective} \quad \Longleftrightarrow \quad N_{\mathrm{ar}} \cap (s-\lambda)\mathcal{H}_{\leq 1} = (s-\lambda)N_{\mathrm{ar}}$$

by the same calculation. Saturation at zero does not prove saturation at other points. The missing frozen-descent decision can therefore be approached through an actual finite generalized source block or a range obstruction, as well as through identification of the topology. Neither has been supplied by the primary Hadamard spectrum clauses inspected here.

33 The actual second radial coefficient and its regular arithmetic value

We now evaluate the regular Mellin datum in (1167) for the complete source field, not an auxiliary test function. The calculation works along the inner axis, and retains both powers of the concentration coordinate and the exact force remainder. The source existence theorem is still imported from [7]; the coefficient extraction and its arithmetic image below are consequences proved here. All coefficients are ordinary Taylor coefficients in the physical variable $\sigma = r^2/2$.

33.1 Source axis data, coordinate inverse, and the full local field

Retain the source's fixed choices

$$0 < h < 1/100, \quad A = \tfrac{1}{2}+h, \quad D = \tfrac{1}{2}-h, \quad k = 1+h, \quad C > 1, \quad 0 < j_0 \leq 1/20, \quad \Lambda \geq 1, \quad \sigma_* > 0.$$

These are the constants chosen by the source construction, with its thresholds and dependencies, not independently variable parameters asserted to produce solutions. Here σ_* is the source's positive regularizing parameter, not the physical radial coordinate. At physical axial coordinate z, put $y = z/\sqrt{\nu}$ and retain

$$\tau = 1 - t = q(1-\eta^2), \qquad y = q^D\eta, \qquad X = \frac{\sigma}{\nu q}, \qquad d = 1-\eta^2, \quad L = 1 - 2h\eta^2. \tag{1211}$$

The inverse solves $q - y^2 q^{2h} = \tau$, uniquely on $q > |y|^{1/D}$, and then sets $\eta = yq^{-D}$. The left side starts at zero, tends to infinity, and has derivative $1 - 2hy^2q^{2h-1} = L \geq 1 - 2h > 0$ there. Its inverse is smooth for $\tau > 0$, and $|\eta| < 1$. Thus this is the original coordinate map, including ν, with a proved inverse.

Source (B.1), (B.3), and (B.12), pp. 144–147, specify

$$\begin{aligned} U(\eta) &= 4\eta + j_0, \qquad & H_*(\eta) &= D\eta + dU(\eta), \\ \zeta_*(\eta) &= -\frac{LH_*}{H_*^2 + \sigma_*^2}, \qquad & \varphi(\eta) &= \exp\left(\Lambda \int_0^\eta \zeta_*(w)\,dw\right). \end{aligned} \tag{1212}$$

The function ζ_* is the source auxiliary function, not the Riemann zeta function. In particular $\varphi > 0$ on the real interval and $\varphi' = \Lambda\zeta_*\varphi$.

The actual axis traces, on the inner localized domain, are

$$w_0(z,t) = (u_\nu)_3(0,0,z,t) = \sqrt{\nu}q^{-A}U(\eta), \qquad b_1(z,t) = \partial_\sigma B(0,z,t) = \frac{2}{C}q^{-k}\varphi(\eta). \tag{1213}$$

Here is why these are exact traces of the complete field. Source Lemma 5.1, p. 47, gives $\varphi_n(0,\eta) = U_n(0,\eta) = 0$ at every positive base order. Smoothness factors each such profile by X, and its streamfunction primitive by X^2. The base cutoffs $\chi(c_n q)$ are applied to streamfunctions before taking their curls. Their extra radial term in (5.45), p. 61, is also divisible by X^2 in ru_r; consequently it contributes only $O(r^3)$ to u_r. The axial corrections are $O(r^2)$ and swirl corrections $O(r^3)$. Every fixed z,t derivative of these vanishing axis traces is zero as well. At each preterminal point only finitely many cutoff terms are present locally, since $q > 0$ and the cutoff scales increase to infinity. The annular wave and mean representatives, including finite initialization, vanish in an axis neighborhood by Proposition 9.9, pp. 114–117. Their locally finite sum has the same property. Finally (10.4)–(10.5), p. 118, have localization cutoffs identically one in a fixed neighborhood of the concentrating origin at sufficiently late times. Their derivative terms vanish there. These facts prove (1213) on an open axis neighborhood, not just at one point, and justify its axial and temporal derivatives. The physical map $u_\nu(x,t) = \sqrt{\nu}u_*(x/\sqrt{\nu},t)$, source (10.22), cancels the velocity factor against the radial factor in u_θ/r, giving precisely the second trace above.

This argument does not discard positive-order corrections to b_2. Their radial derivatives need not vanish. We determine their total contribution using the full momentum equation.

33.2 Exact coefficient extraction, with the force torque retained

Locally at the axis, the remaining smooth axisymmetric field has the Cartesian expression

$$u_1 = a(r^2,z,t)x_1 - b(r^2,z,t)x_2, \quad u_2 = b(r^2,z,t)x_1 + a(r^2,z,t)x_2, \quad u_3 = w(r^2,z,t).$$

At $r = 0$, incompressibility gives $2a_0 + \partial_z w_0 = 0$, and $b_1 = 2b_0$. The original angular averages therefore give

$$B = b_1\sigma + b_2\sigma^2 + O(\sigma^3), \quad P = p_2\sigma^2 + O(\sigma^3), \quad Q = q_1\sigma + O(\sigma^2),$$

with the exact coefficients

$$p_2 = 4a_0b_0 = -b_1\partial_z w_0, \qquad q_1 = w_0 b_1. \tag{1214}$$

For example $P = r^2(ar)(br) = 4ab\sigma^2$, which verifies the factor four without changing the radial variable.

Set $r_f(z,t) = (\nabla_x \times f_\nu)_3(0,0,z,t)$. For any smooth Cartesian force, expanding to first order transversely and averaging $x_1 f_2 - x_2 f_1$ gives

$$F(\sigma, z, t) = r_f(z,t)\sigma + O(\sigma^2).$$

Indeed $\langle x_1^2 \rangle = \langle x_2^2 \rangle = \sigma$ and $\langle x_1 x_2 \rangle = 0$; the surviving coefficient is $\partial_1 f_2 - \partial_2 f_1$. This argument does not assume an axisymmetric force away from the axis.

Take the coefficient of σ in the complete equation (1141). Taylor's differentiated remainder and preterminal smoothness justify the operation. Substituting (1214) in (1166) proves

$$\boxed{b_2 = \frac{\partial_t b_1 + w_0 \partial_z b_1 - b_1 \partial_z w_0 - \nu \partial_{zz} b_1 - r_f}{4\nu}.} \tag{1215}$$

Pressure is absent because its full angular torque integrates to zero, as proved in (1140)–(1141), not because the source pressure or force has been removed.

33.3 The complete coefficient along the inner axis

For a function $v = v(\eta)$, define the explicit operators

$$T_b v = \frac{-bv + D\eta v'}{L}, \qquad Z_b v = \frac{2b\eta v + dv'}{L}.$$

Implicit differentiation of (1211) gives

$$q_t = -L^{-1}, \quad \eta_t = D\eta/(qL), \qquad q_z = \frac{2\eta q^{1-D}}{\sqrt{\nu} L}, \quad \eta_z = \frac{dq^{-D}}{\sqrt{\nu} L}.$$

Consequently, with z, t physical and the other physical variable held fixed in each derivative,

$$\partial_t(q^b v) = q^{b-1} T_b v, \quad \partial_z(q^b v) = \nu^{-1/2} q^{b-D} Z_b v, \quad \partial_{zz}(q^b v) = \nu^{-1} q^{b-2D} Z_{b-D} Z_b v. \tag{1216}$$

This is source (4.2), pp. 24–25, at the axis, derived with the physical viscosity map. In particular the order of the last two operators and their different subscripts are part of the formula.

Put

$$\begin{aligned} H(\eta) &= T_{-k}\varphi + U Z_{-k}\varphi - \varphi Z_{-A} U, \\ R(\eta) &= Z_{-k-D} Z_{-k}\varphi. \end{aligned} \tag{1217}$$

These are completely specified functions, not uncomputed source remainders. If desired, the second is a rational expression times the original exponential: writing $V = -2k\eta\varphi + d\varphi'$,

$$R = \frac{2(-k-D)\eta V + dV'}{L^2} - \frac{dVL'}{L^3}, \quad V' = -2k\varphi - 2(k+1)\eta\varphi' + d\varphi'', \quad \varphi'' = (\Lambda\zeta_*' + \Lambda^2\zeta_*^2)\varphi.$$

All denominators are nonzero on $[-1,1]$, and the given expressions specify every derivative appearing in them. Using $A + D = 1$, substitution in (1215) gives

$$\boxed{b_2(z,t) = \frac{q^{-2-h}H(\eta) - q^{-2+h}R(\eta)}{2\nu C} - \frac{r_f(z,t)}{4\nu}.} \tag{1218}$$

This identity holds throughout the inner axis domain just specified. In deriving its first exponent, the two advective terms each have power $q^{-k-A-D} = q^{-k-1}$. The axial diffusion term has power

$q^{-k-2D} = q^{-2+h}$. Neither power has been replaced by the other, and r_f is the actual final force curl.

The first profile has a useful exact sign. Expanding the numerator in (1217), using $U' = 4$, $2(A-k) = -1$, and then $\varphi' = \Lambda\zeta_*\varphi$, proves

$$H(\eta) = \varphi(\eta)\left\{\frac{h-3-j_0\eta}{L} - \frac{\Lambda H_*(\eta)^2}{H_*(\eta)^2+\sigma_*^2}\right\} < 0 \qquad (-1 \le \eta \le 1). \tag{1219}$$

For the expansion before substitution, the coefficient of φ is $k - \eta U - 4d = h - 3 - j_0\eta$, and the coefficient of φ' is $D\eta + dU = H_*$, both divided by L. The strict sign follows from $\varphi > 0$, $L > 0$, $h - 3 + j_0 < 0$, and the nonnegative square in the second term. It does not require $H_* \neq 0$.

33.4 A finite radial closure inside the full cutoff sum

The source recursion gives more than flatness for this particular force coefficient. Denote the actual positive-order base cutoffs by $\chi(c_n q)$, and retain the profiles $\varphi_n(X,\eta)$ from (5.1). The original recurrence (5.3), p. 46, has $b = -k$, $\lambda_n = 2nh$, and its full operators include the $X\partial_X$ terms in source (4.2). At $X = 0$ it gives

$$4\partial_X\varphi_0(0,\eta) = H(\eta), \qquad 4\partial_X\varphi_1(0,\eta) = -R(\eta), \qquad \partial_X\varphi_n(0,\eta) = 0 \quad (n \ge 2). \tag{1220}$$

Here is the complete coefficient argument. At positive order the time operator vanishes on the identically zero axis trace $\varphi_n(0,\eta)$. In each axial transport product $U_i Z_{b+\lambda_j}\varphi_j$, either $i > 0$, in which case $U_i(0,\eta) = 0$, or $j > 0$, in which case both the trace and its η derivative vanish. Its $X\partial_X$ term also vanishes at zero. For radial transport, $V_0 = O(X)$, while $V_i = O(X^2)$ for $i > 0$: the latter follows by integrating $\partial_X V_i = -Z_{-A+\lambda_i}U_i$ and using $U_i = O(X)$. If $i = 0, j > 0$, the factor $\varphi_j' + \varphi_j/X$ is bounded, so its product with V_0 tends to zero. If $i > 0, j = 0$, that factor is $O(X^{-1})$ but $V_i = O(X^2)$, again giving zero. The case with both indices positive also vanishes.

Thus only the negative axial-viscosity term $-Z_{b+\lambda_{n-1}-D}Z_{b+\lambda_{n-1}}\varphi_{n-1}$ can survive. Restriction of each full operator to $X = 0$ equals its axis operator Z used above: applying the second operator does not revive an $X\partial_X$ term, since it still contains a factor X multiplying a smooth function. The term is $-R$ for $n = 1$ and zero for every $n \ge 2$. The left side of (5.3) is $4\varphi_{n,X}$ at zero, proving the positive-order assertions.

At leading order the angular equation with leading residual stress zero, (4.13), has no lower-order axial viscosity. Alternatively its angular transport expression is the same as (5.3) with indices zero and that viscosity term omitted. Incompressibility gives $\lim_{X\downarrow 0} V_0/X = -Z_{-A}U$. Its radial transport limit is therefore $-\varphi Z_{-A}U$. Together with the time and axial terms this is H, proving the first assertion. The radial extensions and moment corrections preserve this axis interval, so these equations concern the actual realized coefficients.

Now differentiate the full swirl sum, not a truncation:

$$B = \frac{2\sigma}{C}q^{-k}\left\{\varphi_0\left(\frac{\sigma}{\nu q},\eta\right) + \sum_{n\ge 1}\chi(c_n q)q^{2nh}\varphi_n\left(\frac{\sigma}{\nu q},\eta\right)\right\}.$$

Local finiteness justifies extracting its coefficient at every preterminal point. The annular contributions vanish near the axis, and the base streamfunction has no azimuthal curl component. Using (1220) proves the second, independent expression for the same full coefficient:

$$\boxed{b_2(z,t) = \frac{q^{-2-h}H(\eta) - \chi(c_1 q)q^{-2+h}R(\eta)}{2\nu C}.} \tag{1221}$$

Equating this with the full PDE identity (1218) determines the force component exactly:

$$r_f(z,t) = \frac{2}{C}\{\chi(c_1 q) - 1\}q^{-2+h}R(\eta). \tag{1222}$$

The cutoff from Lemma 5.4, p. 57, is one on $[0, 1/2]$. Consequently $r_f = 0$ exactly whenever $q \le (2c_1)^{-1}$ in this inner final-cutoff-one region. This is a proved cancellation for the axial curl trace of the force, not a deletion of the actual force and not an assertion of an unforced Navier–Stokes solution. It is consistent with, and stronger for this coefficient than, the flat estimate (10.9). The transition term in (1222) is retained outside that range.

33.5 The fixed origin and a quantitative preterminal sign bound

At $z = 0$, $q = \tau$ and $\eta = 0$. Define the following explicit constants without rescaling any field:

$$\begin{aligned}
S_* &= j_0^2 + \sigma_*^2, \qquad v = \varphi'(0) = -\frac{\Lambda j_0}{S_*},\\
w = \varphi''(0) &= \frac{\Lambda(4+D)(j_0^2 - \sigma_*^2) + \Lambda^2 j_0^2}{S_*^2},\\
K = h - 3 + j_0 v &= h - 3 - \frac{\Lambda j_0^2}{S_*} < 0, \qquad R_0 = w - 2(1+h).
\end{aligned} \tag{1223}$$

To check the second derivative, $H_*(0) = j_0$, $H_*'(0) = 4 + D$, $L(0) = 1$, and $L'(0) = 0$. The quotient rule in (1212) gives $\zeta_*'(0) = (4+D)(j_0^2 - \sigma_*^2)/S_*^2$. Since $\varphi(0) = 1$, its second derivative is $\Lambda\zeta_*'(0) + \Lambda^2\zeta_*(0)^2$, proving w. At zero $Z_b v = v'$, but its second composition includes the derivative of the coefficient $2b\eta$; thus $R(0) = \varphi''(0) - 2k\varphi(0) = R_0$. Equation (1218) becomes the fully evaluated identity

$$\boxed{b_2(0, 1-\tau) = \frac{K\tau^{-2-h} - R_0\tau^{-2+h}}{2\nu C} - \frac{r_f(0, 1-\tau)}{4\nu}.} \tag{1224}$$

In particular the factor ν^{-1} is real coordinate information, not an omitted choice of units.

Here is also a uniform version of the sign statement with a finite threshold. Fix $0 < \eta_c < 1$. For all sufficiently small $q > 0$, the physical axis points

$$z = \sqrt{\nu}q^D\eta, \qquad t = 1 - q(1-\eta^2), \qquad |\eta| \le \eta_c$$

lie in the fixed localization neighborhood above. The profiles and their parameter derivatives are smooth on this compact interval. Let

$$m = (3 - h - j_0)\min_{|\eta|\le\eta_c}\varphi(\eta) > 0, \qquad M = \max_{|\eta|\le\eta_c}|R(\eta)| < \infty.$$

Because $L \le 1$, (1219) gives $H \le -m$. Choose $q_{\text{loc}} > 0$ to enforce both the source domain and the final cutoffs equal to one. Explicitly, it suffices to take

$$q_{\text{loc}} \le \min\{q_{\text{source}}, \tau_0/2, (z_0/2)^{1/D}\},$$

where q_{source} is an inner source-domain bound, and the original final cutoffs are one for $\tau \le \tau_0/2$ and $|y| \le z_0/2$ near the radial axis. Indeed $\tau \le q$ and $|y| = q^D|\eta| \le q^D$. Put

$$q_*^{\text{sign}} = \min\left\{1, q_{\text{loc}}, (2c_1)^{-1}, \left(\frac{m}{2(M+1)}\right)^{1/(2h)}\right\} > 0.$$

For $0 < q < q_*^{\mathrm{sign}}$, multiply (1218) by the positive number $2\nu C q^{2+h}$. The resulting exact quantity is $H - q^{2h}R - (C/2)q^{2+h}r_f$. The last term is exactly zero by (1222); the middle term has absolute value at most $m/2$ by the displayed threshold. Therefore

$$b_2(z,t) \le -\frac{m}{4\nu C}q^{-2-h} < 0 \quad (|\eta| \le \eta_c,\ 0 < q < q_*^{\mathrm{sign}}). \tag{1225}$$

No numerical value for the source-dependent threshold is claimed. The formula retains its potentially small $1/(2h)$ exponent. The full force and every cutoff transition remain specified by (1215) and (1222).

33.6 Exact arithmetic value, diffusion, and the remaining zero map

Retain the full arithmetic transform, multiplier, and their inverse from Section 30. With $\mathfrak{B} = \mathcal{M}\mathcal{A}_h B$, its exact regular value is

$$\boxed{\mathfrak{B}(-2,z,t) = \frac{d_h}{2\nu C}\{q^{-2-h}H(\eta) - q^{-2+h}R(\eta)\} - \frac{d_h}{4\nu}r_f(z,t), \qquad d_h = \frac{7(2^{h+2}-1)\zeta(3)}{32\pi^2} > 0.} \tag{1226}$$

The proof is $\mathfrak{B}(-2) = W_h'(-2)b_2$ from (1161). Locally $\widehat{B}(s) = b_2/(s+2) +$ holomorphic and $W_h(s) = (s+2)(d_h + O(s+2))$; multiplication gives exactly (1226). The classical functional equation [51, (25.4.2)] gives $\zeta'(-2) = -\zeta(3)/(4\pi^2)$, hence the displayed positive constant. This is a regular continued Mellin value, not a convergent unsubtracted integral at $s = -2$. The explicit Taylor-subtracted integral is (1158).

Equations (1225) and (1226) prove a negative divergent regular Mellin value on the actual source axis family. At the origin its leading coefficient is $d_h K/(2\nu C)$, with the exact subleading and force terms in (1224). Yet the arithmetic trace's second derivative is zero: $(\mathcal{A}_h B)''(0) = W_h(-2)B''(0) = 0$. The precise relation between these two statements is the zero-times-pole product just calculated; the coefficient has not been destroyed.

For an explicit balance check at the origin, the other coefficients are

$$b_1 = \frac{2}{C}\tau^{-1-h}, \quad \partial_{zz}b_1 = \frac{2R_0}{\nu C}\tau^{-2+h}, \quad p_2 = -\frac{8}{C}\tau^{-2-h}, \quad \partial_z q_1 = \frac{2}{C}(4 + j_0 v)\tau^{-2-h}, \quad f_1 = r_f.$$

Substitution into (1166) verifies every term. Equivalently the full radial diffusion image has

$$(\mathscr{L}_h\mathcal{A}_h B)'(0) = 4\kappa_h b_2.$$

The arithmetic balance has right side

$$\frac{\kappa_h}{d_h}\{4\nu\mathfrak{B}(-2) - 2\mathfrak{P}(-2)\} = \frac{2\kappa_h}{C}(5 + h + j_0 v)\tau^{-2-h} - \frac{2\kappa_h R_0}{C}\tau^{-2+h} - \kappa_h r_f.$$

Its left side is exactly $\partial_t J_B + \partial_z J_Q - \nu\partial_{zz}J_B - J_F$. Thus both nonlinear fluxes, axial diffusion, and force match the growing residue; there is no scalar heat replacement or hidden compensation.

Finally this sign has a specific scope in the RH investigation. It concerns the linear continued Mellin functional at the trivial zero -2, not the quadratic Weil form. The relation to zeros elsewhere is the exact product $\mathfrak{B}(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s})\widehat{B}(s)$. In $h < \Re s < 1$, $\widehat{B}$ is holomorphic by its original integral, and both dilation factors are nonzero. At any point s_0 there, writing the nonzero analytic factors in their local coordinates proves $\operatorname{ord}_{s_0}\mathfrak{B} = \operatorname{ord}_{s_0}\zeta + \operatorname{ord}_{s_0}\widehat{B}$. The actual B is nonzero by (1213), so Mellin injectivity proves its transform is not identically zero; these orders are well defined. The additional zero multiplicity is exactly the actual fluid factor's order. Its nonvanishing at a proposed nontrivial zeta zero, or a negative full Weil pairing, is not asserted by the sign at -2. This is the proved morphism and its full zero accounting, not an inference of disconnection from different presentations.

34 The complete radial coefficient recursion and its finite source dependence

The preceding second-coefficient cancellation extends to every fixed radial order. We derive the exact recursion from the released source [7, (4.2), (5.2)–(5.6), Lemma 5.1], including the pressure equation and the radial cutoff commutator. The resulting finiteness is in correction order at a fixed Taylor degree. It is not convergence of the uncut formal series, a finite approximation to the whole field, or a replacement of its force by zero.

34.1 Coefficients, source data and the exact derivative operators

Keep $A = 1/2 + h$, $D = 1/2 - h$, $k = 1 + h$, $b = -k$, $c = -A$, $e = -2A$, $\lambda_n = 2nh$, $d = 1 - \eta^2$ and $L = 1 - 2h\eta^2$. The physical variables and their inverse are (1211); in particular $X = \sigma/(\nu q)$, $\sigma = r^2/2$, and $z = \sqrt{\nu} q^D \eta$. For a smooth profile f, let $[X^m]f = \partial_X^m f(0, \eta)/m!$. Define functions of η by

$$\phi_{n,m} = [X^m]\varphi_n, \quad u_{n,m} = [X^m]U_n, \quad \pi_{n,m} = [X^m]\Pi_n, \quad v_{n,l} = [X^l]V_n \quad (l \geq 1).$$

Every negative correction index means zero. The source axis conditions are

$$\phi_{0,0} = \varphi, \quad u_{0,0} = U = 4\eta + j_0, \quad \pi_{0,0} = \Pi_*(\eta), \qquad \phi_{n,0} = u_{n,0} = \pi_{n,0} = 0 \quad (n \geq 1), \quad V_n(0, \eta) = 0.$$

Here φ is exactly (1212). The pressure datum is the fixed source function

$$\Pi_*(\eta) = -\int_0^\infty \frac{E_o(X, \eta)^2}{2X}\, dX$$

from source (4.31), pp. 35–36, with its already chosen exterior profile. It is retained as that datum, not replaced by zero or a freely chosen constant. Its parameter analyticity and the existence of all the source profiles are imported source conclusions; the coefficient identities below are direct consequences proved here.

On functions of η, set

$$t_\beta f = \frac{-\beta f + D\eta f'}{L}, \qquad z_\beta f = \frac{2\beta\eta f + df'}{L},$$

where primes in these two formulas mean η derivatives. The full source operators, which also differentiate X, are

$$T_a = L^{-1}(-a + D\eta\partial_\eta + X\partial_X), \qquad Z_a = L^{-1}(2a\eta + d\partial_\eta - 2\eta X\partial_X).$$

Consequently

$$[X^m]T_a f = t_{a-m}[X^m]f, \qquad [X^m]Z_a f = z_{a-m}[X^m]f, \qquad [X^m]Z_{a-D}Z_a f = z_{a-D-m}z_{a-m}[X^m]f. \tag{1227}$$

Indeed $[X^m]X\partial_X f = m[X^m]f$; parameter differentiation commutes with finite radial differentiation. This proves all three identities, including the order of composition. We abbreviate

$$t_{a,n,m} = t_{a+\lambda_n-m}, \quad z_{a,n,m} = z_{a+\lambda_n-m}, \quad z^{[2]}_{a,n,m} = z_{a+\lambda_n-D-m}z_{a+\lambda_n-m}.$$

The notation names operators, not scalar factors.

34.2 All coupled coefficient equations with their finite sums

All sums below have nonnegative i, j, a_0, b_0 with the displayed constraints; thus they are finite. Put $\omega_{n,l} = [X^l]\Omega_n$. Incompressibility gives

$$v_{n,m+1} = -\frac{z_{c,n,m} u_{n,m}}{m+1}. \tag{1228}$$

For $n, m \geq 0$, the angular, axial and pressure recursions are

$$\begin{aligned} 2(m+1)(m+2)\phi_{n,m+1} &= t_{b,n,m}\phi_{n,m} - z^{[2]}_{b,n-1,m}\phi_{n-1,m} \\ &\quad + \sum_{\substack{i+j=n \\ a_0+b_0=m}} \left((b_0+1)v_{i,a_0+1}\phi_{j,b_0} + u_{i,a_0} z_{b,j,b_0}\phi_{j,b_0}\right), \end{aligned} \tag{1229}$$

$$\begin{aligned} 2(m+1)^2 u_{n,m+1} &= t_{c,n,m}u_{n,m} + z_{e,n,m}\pi_{n,m} - z^{[2]}_{c,n-1,m}u_{n-1,m} \\ &\quad + \sum_{\substack{i+j=n \\ a_0+b_0=m}} \left(b_0 v_{i,a_0+1}u_{j,b_0} + u_{i,a_0} z_{c,j,b_0} u_{j,b_0}\right), \end{aligned} \tag{1230}$$

$$(m+1)\pi_{n,m+1} = C^{-2} \sum_{\substack{i+j=n \\ a_0+b_0=m}} \phi_{i,a_0}\phi_{j,b_0} - \tfrac{1}{2}\omega_{n-1,m+1}. \tag{1231}$$

Every term in the pressure source is specified by

$$\begin{aligned} \omega_{n,m+1} &= t_{0,n,m+1}v_{n,m+1} - 2(m+2)(m+1)v_{n,m+2} - z^{[2]}_{0,n-1,m+1}v_{n-1,m+1} \\ &\quad + \sum_{\substack{i+j=n \\ a_0+b_0=m}} \left((b_0+\tfrac{1}{2})v_{i,a_0+1}v_{j,b_0+1} + u_{i,a_0} z_{0,j,b_0+1} v_{j,b_0+1}\right). \end{aligned} \tag{1232}$$

For a negative first index ω is zero. At $n = 0$, these identities use the source's leading stress-zero equations (4.7), (4.13) on the inner interval, so that no positive-order existence result is being applied to order zero.

Here is a coefficient proof of every factor. Integrating $\partial_X V_n = -Z_{c+\lambda_n} U_n$ with $V_n(0) = 0$ proves (1228). Applying $[X^m]$ to $2(X\varphi_n'' + 2\varphi_n')$ and $2(XU_n'' + U_n')$ gives respectively $2(m+1)(m+2)\phi_{n,m+1}$ and $2(m+1)^2 u_{n,m+1}$, where these primes mean X derivatives. A term $v_{i,a_0+1}X^{a_0+1}$ times $\varphi_j' + \varphi_j/X$ supplies $(b_0+1)v_{i,a_0+1}\phi_{j,b_0}X^{a_0+b_0}$. The same product with U_j' has factor b_0. Equation (1227) supplies the remaining angular and axial operators and all ordered $U_i Z$ products. The coefficient of X^m in Ω_{n-1}/X is $\omega_{n-1,m+1}$, giving (1231). Finally $V_j' - V_j/(2X)$ has coefficient $(b_0 + 1/2)v_{j,b_0+1}$ at degree b_0, and $[X^{m+1}](-2XV_n'') = -2(m+2)(m+1)v_{n,m+2}$. These give precisely (1232), with its shifted axial viscosity. Products of Taylor coefficients here follow from finite Leibniz rules. No infinite Taylor expansion is substituted into a differential equation.

34.3 Finite correction support at every radial degree

Theorem 34.1. *For the full sequence of source profiles and every $m \geq 0$,*

$$\phi_{n,m} = u_{n,m} = \pi_{n,m} = v_{n,m+1} = 0 \quad (n > m).$$

At degree zero the stronger $\pi_{n,0} = 0$ holds for every $n > 0$. The same zero identities hold after every fixed parameter derivative.

Proof. At $m = 0$ the axis data and (1228) prove the assertions. Assume they hold at all degrees at most m. In each product on the right of (1229), a nonzero first factor requires $i \le a_0$, and a nonzero second factor requires $j \le b_0$. Their sum has $n = i + j \le a_0 + b_0 = m$. The time term vanishes for $n > m$, and the shifted viscous term for $n > m + 1$. Thus $\phi_{n,m+1} = 0$ if $n > m + 1$. For (1230) the products have the same bound; its pressure term vanishes when $n > m$ by the degree-m pressure assertion. The other two terms have the just stated bounds. Therefore $u_{n,m+1} = 0$ for $n > m + 1$, and (1228) gives $v_{n,m+2} = 0$ for the same indices.

Now use the newly proved axial coefficient in (1232). The first term and both product sums vanish for $n > m$. The radial viscous term vanishes for $n > m + 1$, and the shifted axial viscous term also vanishes for $n > m+1$. Hence $\omega_{n,m+1} = 0$ for $n > m+1$. The product in (1231) vanishes for $n > m$, whereas $\omega_{n-1,m+1}$ vanishes for $n > m+2$. This already gives $\pi_{n,m+1} = 0$ for $n > m+2$, but the remaining boundary index cancels exactly. Put $\beta = c + \lambda_m - m$. In (1230) at $n = m + 1$, the pressure, time and product terms are zero by the induction assumption. Therefore

$$u_{m+1,m+1} = -\frac{z_{\beta-D} z_\beta u_{m,m}}{2(m+1)^2}, \quad v_{m,m+1} = -\frac{z_\beta u_{m,m}}{m+1}, \quad v_{m+1,m+2} = -\frac{z_{\beta-2D} u_{m+1,m+1}}{m+2}.$$

The last identity uses $\lambda_{m+1} - (m + 1) = \lambda_m - m - 2D$. All terms of $\omega_{m+1,m+1}$ except its two viscosity terms have already vanished. Since $c - D = -1$, its remaining inner axial exponent is $\lambda_m - m - 1 = \beta - D$. Substitution gives

$$\begin{aligned} \omega_{m+1,m+1} &= -2(m+1)(m+2)v_{m+1,m+2} - z_{\beta-2D} z_{\beta-D} v_{m,m+1} \\ &= -\frac{z_{\beta-2D} z_{\beta-D} z_\beta u_{m,m}}{m+1} + \frac{z_{\beta-2D} z_{\beta-D} z_\beta u_{m,m}}{m+1} = 0. \end{aligned}$$

The identical ordered compositions, not any commutation of distinct operators, give this cancellation. Hence $\omega_{n,m+1} = 0$ for all $n > m$. Equation (1231) now gives $\pi_{n,m+1} = 0$ for $n > m+1$, closing the strong induction. The new axial coefficient was computed before the new pressure coefficient, and both viscosity terms were retained. All identities are smooth in η, so their fixed parameter derivatives are zero. □

This is also a constructive finite algorithm. Start with the three specified axis functions and (1228). At degree $m + 1$, compute ϕ, u for $0 \le n \le m + 1$, then v, then ω for $0 \le n \le m + 1$, and finally π for $0 \le n \le m + 1$. The intermediate $\omega_{m+1,m+1}$ is the proved zero just calculated. Only additions, products, parameter derivatives, division by L, and the displayed positive integer denominators occur. The source choice $0 < h < 1/100$ gives $L \ge 1-2h > 0$ on the real interval. For each finite degree the algorithm therefore returns uniquely determined smooth coefficient functions, from the original axis data with no new choice.

34.4 The highest correction is an explicit axial derivative

At $n = j$, $m = j - 1$, the time and product terms in (1229) vanish. Since $b + \lambda_{j-1} - (j - 1) = -k - 2D(j - 1)$, it follows that

$$\phi_{j,j} = -\frac{z_{-k-(2j-1)D} z_{-k-(2j-2)D} \phi_{j-1,j-1}}{2j(j+1)} \quad (j \ge 1). \tag{1233}$$

Iterating this exact equality, with the rightmost operator applied first, proves

$$\phi_{j,j} = \frac{(-1)^j}{2^j j!(j+1)!} z_{-k-(2j-1)D} z_{-k-(2j-2)D} \cdots z_{-k} \varphi. \tag{1234}$$

The denominator is the product of $2\ell(\ell+1)$ for $1 \le \ell \le j$. Empty products at $j=0$ mean the identity. For the physical source trace $b_1 = 2q^{-k}\varphi/C$, repeated use of the physical axial derivative in (1211) gives

$$\partial_z^{2j} b_1 = \frac{2}{C\nu^j} q^{-k-2jD} z_{-k-(2j-1)D} \cdots z_{-k}\varphi.$$

This proves the complete axial meaning of the highest correction, including its sign, amplitude, factorials and viscosity.

The strong pressure assertion also supplies the highest axial coefficient. At $n=j, m=j-1$ in (1230), the pressure term now vanishes along with the time and product terms. The identical induction gives

$$u_{j,j} = \frac{(-1)^j}{2^j (j!)^2} z_{-A-(2j-1)D} \cdots z_{-A} U. \tag{1235}$$

Here the denominator is $\prod_{\ell=1}^{j} 2\ell^2$. Writing $w_0 = \sqrt{\nu} q^{-A} U$, the $n=j$ summand of the physical coefficient $[\sigma^j]u_z$ is consequently $\rho_j(-1)^j \partial_z^{2j} w_0/(2^j(j!)^2)$. This follows by the same $2j$ physical chain rules, retaining the factor $\nu^{1/2-j}$.

34.5 Exact coefficients of the actual cutoff-summed fields

Put $\rho_0(q)=1$ and $\rho_n(q)=\chi(c_n q)$ for $n>0$, with precisely the increasing source scales in Proposition 5.5. The final spatial and temporal cutoffs are one in the central slab near terminal time used in Section 33. All assertions in this subsection concern that slab. At each preterminal axis point the annular wave and mean terms vanish in a neighborhood; the actual field there is the cutoff-summed axisymmetric base. Its locally finite sum and the smooth extensions from the source permit termwise finite radial and parameter differentiation.

Write the actual coefficients as

$$b_m = [\sigma^m]B \ (m \ge 1), \qquad w_m = [\sigma^m]u_z \ (m \ge 0), \qquad a_l = [\sigma^l](ru_r) \ (l \ge 1).$$

These are ordinary coefficients, not derivatives without factorials. Theorem 34.1 and source (5.1), (5.45) give the exact finite formulas

$$b_m = \frac{2}{C\nu^{m-1}} \sum_{n=0}^{m-1} \rho_n q^{-k-m+1+\lambda_n} \phi_{n,m-1}, \tag{1236}$$

$$w_m = \nu^{1/2-m} \sum_{n=0}^{m} \rho_n q^{-A-m+\lambda_n} u_{n,m}, \tag{1237}$$

$$a_l = \nu^{1-l} \sum_{n=0}^{l-1} q^{\lambda_n - l} \left(\rho_n v_{n,l} - \frac{2\eta}{L} q\rho_n'(q) \frac{u_{n,l-1}}{l} \right). \tag{1238}$$

To check the last expression rather than suppress its commutator, the source primitive is $F_n(X) = \int_0^X U_n(x)\,dx$. Its coefficient at X^l is $u_{n,l-1}/l$. Source (5.45) reads $ru_{r,n}^{\mathrm{cut}} = q^{\lambda_n}(\rho_n V_n - 2\eta q \rho_n' F_n/L)$ at viscosity one. The physical viscosity map multiplies ru_r by ν and replaces X by $\sigma/(\nu q)$, giving (1238). Similarly u_z scales by $\sqrt{\nu}$, whereas $B = ru_\theta = 2\sigma q^{-k}\varphi/C$, yielding the other two formulas. Every omitted correction index has a proved zero coefficient; no remainder has been estimated away.

In particular the $n=j$ summand of b_{j+1} is precisely

$$\rho_j(q) \frac{(-1)^j}{2^j j!(j+1)!} \partial_z^{2j} b_1. \tag{1239}$$

The derivative acts on b_1, not on the cutoff. The factors for $j = 1, 2, 3$, respectively, are

$$-\frac{1}{4}, \qquad \frac{1}{48}, \qquad -\frac{1}{1152}.$$

All lower-index contributions remain in (1236).

The nonlinear fluxes retain their full ordered products. Since the actual field is axisymmetric in this neighborhood, $P = (ru_r)B$ and $Q = u_z B$, and therefore

$$p_l = [\sigma^l]P = \sum_{a=1}^{l-1} a_a b_{l-a}, \qquad q_l = [\sigma^l]Q = \sum_{a=0}^{l-1} w_a b_{l-a}. \tag{1240}$$

An empty sum is zero. Outside the neighborhood the original angular averages, not these product expressions, continue to define the full functions. Their axis coefficients agree with (1240).

34.6 Eventual exact disappearance of each fixed angular-force coefficient

Let $f_j = [\sigma^j]F$, with $F = \langle r f_\theta \rangle$. The full physical equation gives, everywhere on the central preterminal slab,

$$f_j = \partial_t b_j + (j+1)p_{j+1} + \partial_z q_j - \nu\{2j(j+1)b_{j+1} + \partial_{zz} b_j\}, \qquad j \geq 1. \tag{1241}$$

Equations (1236)–(1240) make this an exact finite expression, including all cutoff derivatives, with indices at most j. It does not require replacing the force by an auxiliary estimate.

Theorem 34.2. *For each integer $J \geq 1$, every coefficient f_j, $1 \leq j \leq J$, and all its fixed (z,t) derivatives vanish identically on the open part of the central slab where*

$$0 < q < (2c_J)^{-1}.$$

This is a degree-dependent statement about the angular-force trace.

Proof. The source scales are increasing and $\chi = 1$ on $[0, 1/2]$, so on this open region all ρ_n for $1 \leq n \leq J$ are one and all their positive parameter derivatives are zero. Every coefficient in (1241) is consequently the same as for the uncut sum through correction order J.

To verify that no high-order nonlinear term survives this comparison, fix $j \leq J$. At radial degree $j-1$ of the source angular equation, each transport product has correction index at most $j-1$ by Theorem 34.1 and $a_0 + b_0 = j - 1$ in (1229). The time term has that same bound. The radial diffusion term uses $\phi_{n,j}$, and the shifted axial viscosity uses $\phi_{n-1,j-1}$; both have index at most j. Each coefficient at these finitely many correction indices is zero by (1229). Multiplication by the original physical powers and the angular factor r, or equivalently the coefficient derivation of (1241), proves $f_j = 0$. The cutoff commutator in (1238) is zero on this region, and the annular corrections and final-cutoff derivatives give no axis term there. Finally a smooth function identically zero on an open parameter domain has all its parameter derivatives zero on that domain. $\square$

For a compact interval $|\eta| \leq \eta_c < 1$, one can impose the same explicit localization bound $q < q_{\mathrm{loc}}$ as in Section 33; take $q < \min(q_{\mathrm{loc}}, (2c_J)^{-1})$ for the theorem there. No positive bound uniform in J follows: the source has $c_J \longrightarrow \infty$. Nor does a zero Taylor coefficient prove that the full force vanishes off the axis. In particular this does not convert the forced source theorem into an unforced one.

34.7 Exact finite source formulas in all arithmetic residue–value coordinates

Use the unchanged W_h, $c_{h,m}$, and E_m from Section 30, with $\mathfrak{B} = \mathcal{M}\mathcal{A}_h B$. Define $\widetilde{B}_m = E_m(\mathfrak{B})$ and similarly for P, Q, F. The previously proved Mellin continuation and (1236) give

$$\widetilde{B}_m = \frac{2c_{h,m}}{C\nu^{m-1}} \sum_{n=0}^{m-1} \rho_n q^{-k-m+1+2nh} \phi_{n,m-1}. \tag{1242}$$

For odd m this is the residue at $-m$; for even m it is the regular continued value there. In both cases the inverse is the exact division $b_m = \widetilde{B}_m / c_{h,m}$, with the nonzero denominator already proved from the zeta functional equation. Thus a trivial zeta zero retains its even input coefficient through a regular value; it does not erase that coefficient.

The full nonlinear arithmetic evolution is still

$$\partial_t \widetilde{B}_j + \partial_z \widetilde{Q}_j - \nu \partial_{zz} \widetilde{B}_j - \widetilde{F}_j = (j+1) \frac{c_{h,j}}{c_{h,j+1}} (2\nu j \widetilde{B}_{j+1} - \widetilde{P}_{j+1}).$$

Every entry now has the finite source formulas above. On the degree-J domain of Theorem 34.2, $\widetilde{F}_j = 0$ for $j \le J$ by the proved zero coefficient, not by deletion of a force term from the equation. The nonlinear fluxes and axial diffusion persist. There is no assertion that a finite set of $\widetilde{B}_j$ is an autonomous system. The full functions and their flat remainders remain in the invertible arithmetic transform. This supplies exact higher source data for the ongoing arithmetic dynamics; it supplies neither a nontrivial zeta zero nor a sign assertion about the quadratic Weil form.

35 Convergent generating operators for the selected radial diagonals

This section records the exact generating operator for the highest profile-order diagonals of the source radial jets. It is a selected diagonal, not the full Navier–Stokes field. The axis data, viscosity, coordinates, cutoffs and source parameters are retained exactly as in Sections 26 and 34; the source existence theorem remains an imported result [7]. The scalar kernel below is the source's Appendix B function (B.11), and its Bessel provenance is recorded in the NIST DLMF [9, §10.2,§10.9].

35.1 Local analytic axis data and their radius

Use the source parameters

$$A = \tfrac{1}{2} + h, \quad D = \tfrac{1}{2} - h, \quad k = 1 + h, \quad 0 < h < \tfrac{1}{100}, \quad \nu > 0,$$

and put $\tau = 1 - t$, $\sigma = r^2/2$, $y = z/\sqrt{\nu}$,

$$\tau = q(1-\eta^2), \qquad y = q^D \eta.$$

On the central preterminal slab, where the source cutoffs are one near the axis, the retained axis data are

$$w_0(z,t) = \sqrt{\nu}\, q^{-A} U(\eta), \qquad b_1(z,t) = \frac{2}{C} q^{-k} \phi(\eta). \tag{1243}$$

Here U and the holomorphic germ ϕ are the source functions defined in Appendix B; no new amplitude or normalization is introduced.

For a real preterminal point (z_0, t_0), write

$$F(\eta) = \eta(1-\eta^2)^{-D}, \qquad z = \sqrt{\nu}\,\tau^D F(\eta), \qquad q = \frac{\tau}{1-\eta^2}.$$

If η_0 is the corresponding real parameter, then

$$F'(\eta_0) = (1-\eta_0^2)^{-D-1}\big(1-2h\eta_0^2\big) > 0.$$

Choose a disk avoiding ± 1, the roots of $1-2h\eta^2$, and the poles of the rational logarithmic derivative defining ϕ. On that disk the positive real branch of $(1-\eta^2)^{-D}$ has a holomorphic logarithm. The map F has a quantitative inverse there: if $a = F'(\eta_0)$ and M_2 is any bound for $|F''|$ on the disk, then

$$\delta \le \frac{a}{4(M_2+1)}, \qquad R = \frac{\sqrt{\nu}\,\tau_0^D a\delta}{4}$$

give a disk $|z-z_0| < R$ on which the contraction

$$T_Y(\eta) = \eta + \frac{Y - F(\eta)}{a}, \qquad Y = \frac{z}{\sqrt{\nu}\,\tau_0^D},$$

has a unique holomorphic fixed point. Thus both functions in (1243) have a positive local holomorphic radius, with the factor $\sqrt{\nu}\,\tau_0^D$ retained. No uniform radius up to $t=1$ is asserted.

35.2 The two generating kernels

Let f be holomorphic on $|\zeta - z_0| < R$. Whenever $|z-z_0| + |r| < R$, define

$$H_0[f](r,z) = \frac{1}{2\pi}\int_0^{2\pi} f(z + ir\cos\theta)\,d\theta, \tag{1244}$$

$$H_1[f](r,z) = \frac{1}{\pi}\int_0^{2\pi} \sin^2\theta\, f(z + ir\cos\theta)\,d\theta. \tag{1245}$$

The second normalization is $1/\pi$ for the interval $[0,2\pi]$ (equivalently $2/\pi$ on $[0,\pi]$). Both maps are even in r and hence are holomorphic in $\sigma = r^2/2$ near the axis. Taylor expansion and

$$\frac{1}{2\pi}\int_0^{2\pi}\cos^{2j}\theta\,d\theta = \frac{(2j)!}{4^j (j!)^2}, \qquad \frac{1}{\pi}\int_0^{2\pi}\sin^2\theta\cos^{2j}\theta\,d\theta = \frac{(2j)!}{4^j j!(j+1)!}$$

give the absolutely convergent series

$$H_0[f] = \sum_{j\ge 0}\alpha_j[f]\sigma^j, \qquad \alpha_j[f] = \frac{(-1)^j f^{(2j)}}{2^j (j!)^2}, \tag{1246}$$

$$H_1[f] = \sum_{j\ge 0}\beta_j[f]\sigma^j, \qquad \beta_j[f] = \frac{(-1)^j f^{(2j)}}{2^j j!(j+1)!}. \tag{1247}$$

Indeed, Cauchy's estimate $|f^{(2j)}(z)| \le (2j)!M/a^{2j}$ and $\binom{2j}{j} \le 4^j$ bound the jth terms by a geometric series in $|r|/a$. The same argument on a slightly larger compact disk justifies all displayed derivatives. In Bessel notation the identities are

$$H_0 = J_0(r\partial_z), \qquad H_1 = \frac{2J_1(r\partial_z)}{r\partial_z} = f_0(\sigma\partial_z^2), \qquad f_0(s) = \sum_{j\ge 0}\frac{(-s/2)^j}{j!(j+1)!},$$

where the quotient is only notation for the convergent entire series; no inverse of ∂_z is used.

35.3 PDEs, inverse maps, and the selected vector field

Set

$$L_0 = 2\sigma\partial_{\sigma\sigma} + 2\partial_\sigma + \partial_{zz}, \qquad L_1 = 2\sigma\partial_{\sigma\sigma} + 4\partial_\sigma + \partial_{zz}.$$

The coefficient recurrences in (1246)–(1247) prove, on the common analytic domain,

$$L_0 H_0[f] = 0, \qquad L_1 H_1[f] = 0, \qquad \partial_\sigma(\sigma H_1[f]) = H_0[f], \qquad \partial_\sigma H_0[f] = -\tfrac{1}{2} H_1[f'']. \tag{1248}$$

Restriction to $\sigma = 0$ is the inverse of each H_i on its holomorphic solution-germ range. Thus these are explicit linear bijections of analytic germs, not a claim about arbitrary smooth Taylor data.

Define

$$P[f] = \left(-\frac{x_1}{2} H_1[f_z], -\frac{x_2}{2} H_1[f_z], H_0[f]\right), \tag{1249}$$

$$S[g] = \left(-\frac{x_2}{2} H_1[g], \frac{x_1}{2} H_1[g], 0\right), \qquad \sigma = \frac{x_1^2 + x_2^2}{2}. \tag{1250}$$

Then direct differentiation using (1248) gives

$$\operatorname{div} P[f] = \operatorname{div} S[g] = 0, \qquad \operatorname{curl} P[f] = 0, \qquad \operatorname{curl} S[g] = P[g]. \tag{1251}$$

Every Cartesian component is harmonic because

$$\Delta H_0[f] = L_0 H_0[f], \qquad \Delta(x_i G) = x_i L_1 G \quad (i = 1, 2).$$

The selected diagonal of the source is therefore

$$u_{\text{diag}} = P[w_0] + S[b_1], \tag{1252}$$

with

$$u_{z,\text{diag}} = H_0[w_0], \qquad u_{r,\text{diag}} = -\frac{r}{2} H_1[(w_0)_z], \qquad u_{\theta,\text{diag}} = \frac{r}{2} H_1[b_1].$$

The inverse axis map is explicit:

$$w_0 = u_z|_{r=0}, \qquad b_1 = (\operatorname{curl} u)_z|_{r=0} = 2 \lim_{r\to 0} \frac{u_\theta}{r}.$$

This is a bijection only in the stated axisymmetric, axis-regular, componentwise-harmonic analytic-germ category. It is not an identification of the complete source field with a harmonic or irrotational field.

35.4 Restoring the source cutoffs

Let $\rho_0 = 1$ and $\rho_j(z,t) = \chi(c_j q(z,t))$ for $j \geq 1$, exactly as in the source. The correct selected cutoff operators are

$$W_\rho[f] = \sum_{j\geq 0} \rho_j \alpha_j[f] \sigma^j, \qquad G_\rho[f] = \sum_{j\geq 0} \rho_j \beta_j[f] \sigma^j, \qquad B_\rho = \sigma G_\rho[b_1].$$

The sum is locally finite for every preterminal point. The radial component must be reconstructed from incompressibility:

$$R_\rho[f] = r\, u_{r,\rho}[f] = -\int_0^\sigma \partial_z W_\rho[f](s,z)\, ds = -\sum_{j\geq 0} \frac{\partial_z(\rho_j \alpha_j[f])}{j+1} \sigma^{j+1}. \tag{1253}$$

Dropping $\partial_z\rho_j$ would destroy divergence-freeness. For $a=0,1$, Let $d_{0,j}=\alpha_j$, $d_{1,j}=\beta_j$ and $H_{a,\rho}$ for the corresponding cutoff sum. The exact residual is

$$L_a H_{a,\rho}[f]=\sum_{j\geq 0}\sigma^j\left[(\rho_j-\rho_{j+1})\partial_z^2 d_{a,j}[f]+2(\partial_z\rho_j)\partial_z d_{a,j}[f]+(\partial_{zz}\rho_j)d_{a,j}[f]\right]. \tag{1254}$$

The adjacent-index term and both cutoff derivatives are part of the exact formula. For every fixed J, the first $J+1$ coefficients agree with the uncut kernels on the open region where $\rho_0,\ldots,\rho_J=1$; no region uniform in J is asserted. Equation (1253) together with (1254) gives the complete selected cutoff field, not the full lower-profile-order source array.

35.5 Typed handoff to the Mellin hierarchy

Here is a compact representative of the actual local calculation. On a compact preterminal parameter patch choose $\epsilon>0$ inside the common radial neighborhood, and let $\kappa_\epsilon(\sigma)$ be smooth, one for $0\leq\sigma\leq\epsilon$ and zero for $\sigma\geq 2\epsilon$. An explicit such function and the full extension of $\mathcal{A}_h$ to nonzero axis values are proved in Section 36. Set

$$a_W=\kappa_\epsilon W_\rho[w_0],\qquad a_G=\kappa_\epsilon G_\rho[b_1],\qquad a_B=\sigma a_G.$$

The first two inputs are compact and smooth but generally do not vanish at the axis; $a_B\in\mathcal{D}_0$. The original correction cutoffs ρ_j are unchanged. The coefficient maps are exactly

$$\begin{aligned}&[\sigma^j]a_W=\rho_j\alpha_j[w_0],\qquad [\sigma^j]a_G=\rho_j\beta_j[b_1],\quad j\geq 0,\\ &[\sigma^n]a_B=\rho_{n-1}\beta_{n-1}[b_1],\quad n\geq 1.\end{aligned} \tag{1255}$$

The Mellin map acts on the arithmetic transform in the physical variable; $W_h(s)$ multiplies its Mellin transform, not the physical input:

$$\mathfrak{G}_a(s)=\mathcal{M}(\mathcal{A}_h a)(s)=W_h(s)\mathcal{M}a(s).$$

For $n\geq 1$ retain

$$c_{h,n}=\begin{cases}W_h(-n), & n\text{ odd},\\ W_h'(-n), & n\text{ even},\end{cases}\qquad E_n(\mathfrak{G}_a)=\begin{cases}\operatorname{Res}_{s=-n}\mathfrak{G}_a(s), & n\text{ odd},\\ \mathfrak{G}_a(-n), & n\text{ even},\end{cases}$$

The proved continuation gives

$$E_n(\mathfrak{G}_a)=c_{h,n}a_n,\qquad a_n=\frac{E_n(\mathfrak{G}_a)}{c_{h,n}}. \tag{1256}$$

For the two nonvanishing inputs also put $E_0=\operatorname{Res}_{s=0}$ and $c_{h,0}=W_h(0)=(2^h-1)/4$; then $E_0(\mathfrak{G}_a)=c_{h,0}a(0)$. The next section proves this extension, its full inverse, and independence of the auxiliary localization at the level of all residue–value coordinates. The actual angular field uses the shifted index in the second line of (1255), not the unshifted G_ρ index. The full source remainder is carried through the same transform below. A selected coefficient, a regular value at a trivial zero, or a residue is not by itself a value of the complete Weil quadratic form.

35.6 Arithmetic interface and nonclaims

The selected analytic field can be composed with the already proved physical Mellin map in Sections 24–29. That composition is a typed map on the displayed local domain: physical variables (z, σ, t) remain distinct from the arithmetic variable s, and the cutoff residual is retained before applying any Mellin functional. The highest-diagonal identities therefore supply exact input to the existing residue/regular-value hierarchy, but they do not imply a negative Weil quadratic form, an off-critical zero, or a failure of the classical Navier–Stokes theorem. The full source series, its existence theorem and all lower profile orders remain separate dependencies.

36 All source axis derivatives and their exact arithmetic representatives

The actual axial velocity has a nonzero axis value. To carry it alongside the angular momentum we first extend the original arithmetic transform without removing that value. We then compute every axial derivative of both original source traces by Lagrange inversion and insert the results into the radial residue–value maps. The original source choices and cutoff-summed remainder are retained throughout. The imported existence theorem is not strengthened by the calculation.

36.1 The nonzero axis value is a residue at zero

Let $\mathcal{D} = C_c^\infty([0,\infty);\mathbb{C})$, where compact support may meet zero; no vanishing condition is imposed there. Use exactly

$$\phi_a(x) = a(x) - 2^h a(2x), \qquad \psi_a(x) = \phi_a(x) - \tfrac{1}{2}\phi_a(x/2), \qquad G_a(x) = \sum_{n\geq 1} \psi_a(nx), \quad x > 0.$$

The fixed source range is $0 < h < 1/100$. The multiplier remains

$$W_h(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s}).$$

Theorem 36.1. *The formula above extends to an injective linear map $\mathcal{A}_h : \mathcal{D} \to \mathcal{D}$. It agrees with the previous map on $\mathcal{D}_0$. For every $j \geq 0$,*

$$G_a^{(j)}(0) = W_h(-j)a^{(j)}(0), \qquad G_a(0) = \frac{2^h - 1}{4}a(0).$$

On its exact range its inverse is the three series (1157). *Meromorphically on $\mathbb{C}$,*

$$\mathcal{M}G_a(s) = W_h(s)\mathcal{M}a(s), \qquad \operatorname{Res}_{s=0} \mathcal{M}G_a(s) = \frac{2^h - 1}{4}a(0). \tag{1257}$$

Together with the negative-odd residues and negative-even regular values, the residue at zero recovers every input Taylor coefficient.

Proof. If $\operatorname{supp} a \subset [0, R]$, then ψ_a is supported in $[0, 2R]$, so the sum and all its derivatives are locally finite at $x > 0$ and vanish for $x > 2R$. Substitution gives $\int_0^\infty \psi_a = 0$ even when $a(0) \neq 0$. Now

$$\psi_a(0) = \tfrac{1}{2}(1 - 2^h)a(0).$$

Euler–Maclaurin with its integral remainder [9, §2.10(i)], applied to this smooth compact function, gives $G_a(x) = -\psi_a(0)/2 + O(x)$ as $x \downarrow 0$. This yields the stated constant, including its sign. For $j \geq 1$ put $f_j(v) = v^j \psi_a^{(j)}(v)$. Then

$$x^j G_a^{(j)}(x) = \sum_{n\geq 1} f_j(nx), \qquad \int_0^\infty f_j = (-1)^j j! \int_0^\infty \psi_a = 0.$$

Every boundary term in the j integrations by parts contains a positive power of v at zero, regardless of $a(0)$. Moreover $f_j^{(r)}(0) = 0$ for $r < j$ and $f_j^{(r)}(0) = r!\psi_a^{(r)}(0)/(r-j)!$ for $r \geq j$. The proof in Section 29 therefore applies unchanged to every positive derivative: choose $2M - 1 > j$, divide its Euler–Maclaurin remainder by x^j, and bound it by

$$\frac{\|\widetilde{B}_{2M}\|_\infty}{(2M)!} x^{2M-1-j} \|f_j^{(2M)}\|_1.$$

The resulting limits and the fundamental theorem of calculus prove smoothness up to zero, not just an asymptotic Taylor series. The Bernoulli coefficient is exactly $\zeta(-j)(1 - 2^{-j-1})(1 - 2^{h+j})$ for $j \geq 1$. For $j = 0$, $\zeta(0) = -1/2$ gives the same formula.

For completeness the inverse does not require $a(0) = 0$. At each $x > 0$ Möbius inversion is a finite divisor calculation. The second inverse sum telescopes with remainder $2^{-L-1}\phi_a(x/2^{L+1})$; the third telescopes with remainder $2^{-Kh}a(x/2^K)$. Both tend to zero because the recovered functions are bounded at zero and $h > 0$. Their jth derivatives have the extra factors $2^{-(L+1)j}$ and 2^{-Kj}, proving smooth local convergence after recovery of ψ_a. These identities prove both inverse compositions on the actual range and hence injectivity.

Both compact smooth functions a, G_a have their original Mellin integrals on $\Re s > 0$. For $\Re s > 1$, absolute integration and summation are allowed since ψ_a is bounded at zero and compactly supported, and $\sum n^{-\Re s} < \infty$. Scaling in the two differences proves $\mathcal{M}G_a = \zeta(s)(1-2^{s-1})(1-2^{h-s})\mathcal{M}a$ there. Taylor subtraction in (1158), now starting its finite sum at $n = 0$, proves meromorphic continuation of both sides, with only possible simple endpoint poles at $0, -1, -2, \ldots$. The identity theorem extends the product equality. Since $W_h(0) = (2^h - 1)/4 \neq 0$, the residue at zero is precisely $W_h(0)a(0)$. The previously proved functional equation calculation supplies the other nonzero factors $c_{h,n}$. □

Write $E_0 = \mathrm{Res}_{s=0}$ and $c_{h,0} = (2^h - 1)/4$; for $n \geq 1$ retain $E_n, c_{h,n}$ from Section 30. The exact coordinate map is

$$E_n(\mathcal{M}\mathcal{A}_h a) = c_{h,n} a_n, \qquad a_n = \frac{E_n(\mathcal{M}\mathcal{A}_h a)}{c_{h,n}}, \quad n \geq 0. \tag{1258}$$

This recovers the nonzero axial velocity itself at $n = 0$; it does not replace that velocity by its derivative or by an artificially centered one.

36.2 Explicit localization, complete remainders, and the germ map

For $\epsilon > 0$ put

$$e(v) = \begin{cases} e^{-1/v}, & v > 0, \\ 0, & v \leq 0, \end{cases} \qquad \kappa_\epsilon(\sigma) = \frac{e(2 - \sigma/\epsilon)}{e(2 - \sigma/\epsilon) + e(\sigma/\epsilon - 1)}.$$

The denominator is positive for all $\sigma \geq 0$. Each derivative of e from the right at zero vanishes, by exponential decay against every polynomial in $1/v$. Thus κ_ϵ is smooth, equals one on $[0, \epsilon]$, and vanishes on $[2\epsilon, \infty)$. On a compact preterminal parameter patch, choose 2ϵ inside the actual common axis neighborhood. Keep the selected source cutoffs $\rho_j(z,t)$ in W_ρ, G_ρ, and form a_W, a_G, a_B as in Section 35. The sums are locally finite in (z,t) and polynomial in σ, so these inputs belong to $\mathcal{D}, \mathcal{D}, \mathcal{D}_0$, respectively. All their (z,t) derivatives have the same compact support. Hence the differentiated proof of Theorem 36.1 commutes with those derivatives on compact patches.

In particular the exact selected angular coordinates are

$$E_n(\mathcal{M}\mathcal{A}_h a_B) = c_{h,n}\rho_{n-1}\frac{(-1)^{n-1}\partial_z^{2n-2}b_1}{2^{n-1}(n-1)!n!}, \quad n \geq 1. \tag{1259}$$

For the selected axial channel they are

$$E_j(\mathcal{M}\mathcal{A}_h a_W) = c_{h,j}\rho_j\frac{(-1)^j\partial_z^{2j}w_0}{2^j(j!)^2}, \quad j \geq 0. \tag{1260}$$

The shift in (1259) comes from the actual factor $B = \sigma G$, not from a convention for residues.

Let $\mathcal{F} \subset \mathcal{D}$ be the functions flat at zero. The injectivity and full inverse of $\mathcal{A}_h$ induce an exact linear bijection

$$\mathcal{D}/\mathcal{F} \longrightarrow \mathcal{A}_h\mathcal{D}/\mathcal{A}_h\mathcal{F}, \qquad [a] \longmapsto [\mathcal{A}_h a], \tag{1261}$$

with inverse induced by (1157). If two radial localizations are one near zero, their representatives differ by a function zero near zero, so they give the same class in (1261) and the same coordinates (1258). Conversely equality of all those coordinates means that the input difference is flat, since every $c_{h,n}$ is nonzero. This proves exactly the cutoff independence being used. For analytic input germs, flatness implies equality on a neighborhood by convergence of their Taylor series. For smooth inputs the full remainder is instead retained by the original inverse; the quotient is not a replacement for that inverse.

Now let $B_{\rm act}$ be the original compact angular average and $Z_{\rm act} = \langle u_z \rangle$ the compact axial average. Define the latter at the axis by its smooth extension. Indeed angular averaging the Cartesian Taylor polynomial of u_z kills every odd radial degree and produces a polynomial in $r^2 = 2\sigma$. Taylor's differentiated remainder, taken to arbitrary even order, gives the same smooth extension for every finite σ derivative. Compact spatial support gives compact support in σ on each compact parameter patch. Define the full, unestimated differences

$$R_B = B_{\rm act} - a_B, \qquad R_W = Z_{\rm act} - a_W.$$

Both are in $\mathcal{D}$, and $R_B \in \mathcal{D}_0$. Thus

$$\mathcal{M}\mathcal{A}_h B_{\rm act} = \mathcal{M}\mathcal{A}_h a_B + \mathcal{M}\mathcal{A}_h R_B$$

and the identical equality holds for $Z_{\rm act}$. The original full source coefficients of Section 34 remain in the second summand. At the first differing angular coefficient,

$$[\sigma^2]R_B = \frac{q^{-2-h}}{2\nu C}\mathcal{H}(\eta), \quad \mathcal{H} = \phi\left[\frac{h-3-j_0\eta}{L} - \frac{\Lambda H_*^2}{H_*^2+\sigma_*^2}\right] < 0. \tag{1262}$$

Here $\mathcal{H}$ is the function called H in Section 33. Indeed the other term in the complete b_2 is exactly the selected $-\rho_1\partial_{zz}b_1/4$. Positivity of ϕ, L and $h-3-j_0\eta \leq h-3+j_0 < 0$ proves the strict sign on $|\eta| < 1$. Consequently the omitted regular Mellin value is exactly $E_2(\mathcal{M}\mathcal{A}_h R_B) = d_h q^{-2-h}\mathcal{H}/(2\nu C)$. It has not been absorbed into a normalization or silently set to zero.

Localization derivatives also remain in the differential equation. For any smooth B and $L_B = 2\sigma\partial_{\sigma\sigma} + \partial_{zz}$,

$$L_B(\kappa_\epsilon B) - \kappa_\epsilon L_B B = 4\sigma\kappa_\epsilon'\partial_\sigma B + 2\sigma\kappa_\epsilon'' B.$$

This follows by the product rule, with κ_ϵ independent of z, t. The terms are supported away from the axis but are part of the full transformed function. Likewise all nonlinear fluxes use $B_{\text{act}} = a_B + R_B$ and $Z_{\text{act}} = a_W + R_W$ with their cross products; no closed selected-diagonal Navier–Stokes equation is asserted.

36.3 The exact inverse coordinate and all physical axial derivatives

Retain $A = 1/2 + h$, $D = 1/2 - h$, $k = 1 + h$, $\tau = 1 - t > 0$ and

$$x = \frac{z}{\sqrt{\nu}\,\tau^D}, \qquad x = \eta(1-\eta^2)^{-D}, \qquad q = \frac{\tau}{1-\eta^2}.$$

The inverse has its genuine holomorphic germ at zero: the forward map has derivative one, and the chosen logarithm of $1-\eta^2$ is zero there. Its convergence can also be seen directly. For $|x| \leq 1/8$, iteration of $\eta \mapsto x(1-\eta^2)^D$ on $|\eta| \leq 1/4$ stays inside that disk and has derivative bounded by $1/30$. It converges uniformly to the holomorphic inverse. Oddness follows from uniqueness. For a holomorphic function F and $n \geq 1$, the Lagrange formula is

$$[x^n]F(\eta(x)) = \frac{1}{n}[u^{n-1}]F'(u)(1-u^2)^{Dn}. \tag{1263}$$

This is [109, Corollary 5.4.3, (5.65)] in the actual coordinates. To prove it here, change variables in the coefficient residue by $x = f(u) = u(1-u^2)^{-D}$; the residue becomes $\operatorname{Res} F(u)f'(u)f(u)^{-n-1}\,du$. The residue of $(Ff^{-n})'$ is zero, giving $n^{-1}\operatorname{Res} F'f^{-n}\,du$, precisely (1263). For $n = 0$ the answer is separately $F(0)$.

Write the actual axial trace as

$$w_0(z,t) = \sqrt{\nu}\,\tau^{-A}\mathcal{W}(x), \qquad \mathcal{W}(x) = (1-\eta(x)^2)^A(4\eta(x) + j_0) = \sum_{n\geq 0} \mathsf{a}_n x^n.$$

Then $\mathsf{a}_0 = j_0$, $\mathsf{a}_1 = 4$, and for $j \geq 1$

$$\begin{aligned} \mathsf{a}_{2j} &= \frac{j_0(-1)^j A}{j}\binom{(2j-1)D}{j-1}, \\ \mathsf{a}_{2j+1} &= \frac{4(-1)^j}{j}\binom{2jD}{j-1}. \end{aligned} \tag{1264}$$

Every generalized binomial denotes the finite polynomial $\binom{\beta}{m} = \prod_{l=0}^{m-1}(\beta - l)/m!$. For the even coefficients apply (1263) to $j_0(1-u^2)^A$. Its derivative is $-2Aj_0u(1-u^2)^{A-1}$, and $A-1 = -D$ gives the first formula. For the odd part, the identity $A + D = 1$ gives $\eta(1-\eta^2)^A = x(1-\eta^2)$ exactly. Its coefficient at x^{2j+1} is $-4[x^{2j}]\eta^2$; applying (1263) to u^2 gives the second formula. Finally the physical chain rule gives every scale and factorial:

$$\partial_z^n w_0(0,t) = n!\nu^{(1-n)/2}\tau^{-A-nD}\mathsf{a}_n, \qquad n \geq 0. \tag{1265}$$

For the unchanged source angular germ put

$$d(u) = 1-u^2, \quad L(u) = 1-2hu^2, \quad H_*(u) = Du + d(u)(4u+j_0), \quad \zeta_*(u) = -\frac{L(u)H_*(u)}{H_*(u)^2 + \sigma_*^2},$$

$$\phi(u) = \exp\left(\Lambda\int_0^u \zeta_*(v)\,dv\right), \qquad b_1(z,t) = \frac{2}{C}\tau^{-k}d(\eta(x))^k\phi(\eta(x)).$$

These are the selected original axis data, not freely chosen substitutes [7, (B.1)–(B.3), (B.12)]. Their denominator at zero is $j_0^2 + \sigma_*^2 > 0$. Applying (1263) after product differentiation proves

$$\begin{aligned} \partial_z^n b_1(0,t) = {} & \frac{2}{C}\nu^{-n/2}\tau^{-k-nD}(n-1)! \\ & \cdot [u^{n-1}]\phi(u)d(u)^{k-1+nD}\big(\Lambda d(u)\zeta_*(u) - 2ku\big), \qquad n \geq 1. \end{aligned} \tag{1266}$$

The constant is separately $b_1(0,t) = 2\tau^{-k}/C$. The required finite coefficients are constructive: invert the rational denominator as a Taylor series with its nonzero constant and solve

$$\phi_0 = 1, \qquad (m+1)\phi_{m+1} = \Lambda \sum_{l=0}^{m} (\zeta_*)_l \phi_{m-l}.$$

No integration oracle or truncation of the source equation is needed.

36.4 Insertion into the original arithmetic coordinates

At $z = 0$, $q = \tau$. For $j \geq 1$, the exact selected axial coordinate (1260) is therefore

$$\begin{aligned} E_j(\mathcal{M}\mathcal{A}_h a_W)(0,t) = c_{h,j}\rho_j(\tau)\, & \frac{j_0 A(2j)!}{2^j (j!)^2 j}\binom{(2j-1)D}{j-1} \\ & \cdot \nu^{1/2-j}\tau^{-A-2jD}. \end{aligned} \tag{1267}$$

At $j = 0$ the corresponding actual axial residue, with no selected/full difference at the axis, is

$$\operatorname{Res}_{s=0} \mathcal{M}\mathcal{A}_h Z_{\mathrm{act}}(s, 0, 1-\tau) = \frac{2^h - 1}{4}\sqrt{\nu}\, j_0 \tau^{-A}. \tag{1268}$$

Every physical axial derivative of this residue is exactly $c_{h,0}$ times (1265). The angular coordinates are obtained by substituting (1266) into (1259), retaining the shift and the source ρ_j. All lower profile-order contributions remain explicit through (1236)–(1237) and R_B, R_W above. This is an exact source-to-arithmetic propagation. The growing residue in (1268) is at the fixed endpoint $s = 0$, not a newly located zero; no conclusion about a sign of the complete Weil form is drawn from this linear coordinate.

37 Four axial branch singularities and their arithmetic transport

The preceding exact coefficient formulas determine where the source's axial analytic germ first fails to extend holomorphically. We compute its sharp radius, reach four boundary singularities from the original inverse branch, and prove that the zeroth arithmetic residue carries precisely the same singularities. The input is the original axial trace from [7, Appendix B, pp. 144–147]; no source amplitude, physical viscosity, or time exponent is changed.

Retain $0 < h < 1/100$, $A = 1/2 + h$, $D = 1/2 - h$, $0 < j_0 \leq 1/20$, $\nu > 0$, and $\tau = 1 - t > 0$. The source concentration coordinate is still $q = \tau/(1 - \eta^2)$. To avoid confusing it with a new abbreviation, put

$$p_* = 1 - 2h = 2D, \qquad q_* = 2h = 1 - p_*, \qquad B_* = p_*^{p_*} q_*^{q_*}, \qquad R = B_*^{-1/2} = p_*^{-D} q_*^{-h}. \tag{1269}$$

The actual germ is

$$
\begin{aligned}
x &= \frac{z}{\sqrt{\nu}\,\tau^D} = f(\eta) := \eta(1-\eta^2)^{-D},\\
\mathcal{W}(x) &= F(\eta(x)), \qquad F(u) = (1-u^2)^A(4u+j_0),\\
w_0(z,t) &= \sqrt{\nu}\,\tau^{-A}\mathcal{W}(x).
\end{aligned}
\tag{1270}
$$

At zero the logarithm of $1-\eta^2$ is zero, $\eta(0)=0$, and $\eta'(0)=1$. These choices specify the initial inverse branch. Assertions about the actual cutoff-summed field below use the same central cutoff-one axis neighborhood as Sections 33 and 36: the fixed preterminal time is one for which that neighborhood contains the origin. The analytic formula $\mathcal{W}$ itself is defined independently of that localization and has the stated radius for every $\tau > 0$. No identification with a cutoff-modified trace outside the original central patch is made.

Theorem 37.1. *For every parameter in the stated source range, the Taylor series of $\mathcal{W}$ at zero has exact radius R. Four distinct genuine square-root singularities accessible from this disk occur at*

$$
x_{\varepsilon,\varsigma} = \varepsilon R e^{-i\varsigma\pi D}, \qquad \varepsilon,\varsigma \in \{-1,1\}. \tag{1271}
$$

Thus the physical axial Taylor radius is exactly $\sqrt{\nu}\,\tau^D R$. The zeroth Mellin residue of the actual axial average, considered as a function of z, has that same radius and the same four singularities, with the explicit forward and inverse maps given below.

37.1 Exact coefficients, including every exceptional zero

Write $\mathcal{W}(x) = \sum \mathsf{a}_n x^n$. In addition to $\mathsf{a}_0 = j_0$, $\mathsf{a}_1 = 4$, equation (1264) and the gamma recurrence give, for all $j \ge 1$,

$$
\begin{aligned}
\mathsf{a}_{2j} &= j_0 A(-1)^j \frac{\Gamma(p_* j + A)}{\Gamma(j+1)}\,\mathrm{rgamma}(2 - q_* j - D),\\
\mathsf{a}_{2j+1} &= 4(-1)^j \frac{\Gamma(p_* j + 1)}{\Gamma(j+1)}\,\mathrm{rgamma}(2 - q_* j).
\end{aligned}
\tag{1272}
$$

Here $\mathrm{rgamma} = 1/\Gamma$ is entire, including at the gamma poles. To verify the identity, write the finite-product binomial as a gamma quotient away from denominator poles and then continue the entire reciprocal factor. Both numerator arguments are positive. Consequently the exact zero conditions, without an asymptotic exception, are

$$
\begin{aligned}
\mathsf{a}_{2j} = 0 &\iff q_* j + D \in \{2,3,4,\ldots\},\\
\mathsf{a}_{2j+1} = 0 &\iff q_* j \in \{2,3,4,\ldots\}.
\end{aligned}
\tag{1273}
$$

The recurrence and reflection identities used here are [53, §5.5, (5.5.1), (5.5.3)].

For $j > A/q_*$ in the even case and $j > 1/q_*$ in the odd case, reflection gives positive finite envelopes:

$$
\begin{aligned}
\mathsf{a}_{2j} &= (-1)^{j+1}\mathcal{E}_j \cos((2j-1)\pi h),\\
\mathcal{E}_j &= \frac{j_0 A}{\pi}\frac{\Gamma(p_* j + A)\Gamma(q_* j - A)}{\Gamma(j+1)} > 0,\\
\mathsf{a}_{2j+1} &= (-1)^{j+1}\mathcal{O}_j \sin(2\pi h j),\\
\mathcal{O}_j &= \frac{4}{\pi}\frac{\Gamma(p_* j + 1)\Gamma(q_* j - 1)}{\Gamma(j+1)} > 0.
\end{aligned}
\tag{1274}
$$

Indeed rgamma$(2-u) = -\Gamma(u-1)\sin(\pi u)/\pi$ for $u > 1$, including integer $u \geq 2$ by continuity. For the even term use $\sin(\pi(q_* j + D)) = \cos((2j-1)\pi h)$. At $u = 1$ the formal zero times a pole is not a zero; (1272), rather than that undefined product, covers the excluded threshold and every smaller index. For example, $h = 1/200$, $j = 100$ gives the nonzero odd coefficient $1/25$, whereas $j = 200$ gives a genuine zero. No division by a trigonometric factor is used.

37.2 Sharp large-order envelopes and the exact Taylor radius

For fixed h, positive-real Stirling expansion [53, §5.11(i), (5.11.1), (5.11.3)] gives

$$\Gamma(cj+b) = \sqrt{2\pi} e^{-cj}(cj)^{cj+b-1/2}\left(1 + \frac{B_2(b)}{2cj} + O_{b,c}(j^{-2})\right), \quad c > 0, \quad B_2(b) = b^2 - b + \tfrac{1}{6}.$$

This follows by expanding $\log(1 + b/(cj))$ in the logarithmic formula; the constant exponential shift cancels. Therefore the product with shifts (b, c) and slopes (p_*, q_*) is

$$\begin{aligned}\frac{\Gamma(p_* j + b)\Gamma(q_* j + c)}{\Gamma(j+1)} &= \sqrt{2\pi}\, p_*^{b-1/2} q_*^{c-1/2} B_*^j j^{b+c-3/2} \\ &\quad \cdot \left(1 + \frac{1}{j}\left[\frac{B_2(b)}{2p_*} + \frac{B_2(c)}{2q_*} - \frac{1}{12}\right] + O_h(j^{-2})\right),\end{aligned} \tag{1275}$$

where the shifts below are fixed functions of h. The pairs $(A, -A)$ and $(1, -1)$ yield

$$\begin{aligned}\mathcal{E}_j &= C_e B_*^j j^{-3/2}(1 + c_e/j + O_h(j^{-2})), \\ C_e &= j_0 A\sqrt{2/\pi}\, p_*^h q_*^{-1-h}, \qquad c_e = \frac{11 - 2q_* - 7q_*^2}{24 p_* q_*}, \\ \mathcal{O}_j &= C_o B_*^j j^{-3/2}(1 + c_o/j + O_h(j^{-2})), \\ C_o &= 4\sqrt{2/\pi}\, p_*^{1/2} q_*^{-3/2}, \qquad c_o = \frac{13 - 13q_* + q_*^2}{12 p_* q_*}.\end{aligned} \tag{1276}$$

Every constant follows from (1275); none is fitted to numerical data. These are envelope asymptotics, not ratios divided by the possibly zero sine or cosine in (1274). They assert no uniform estimate as $h \downarrow 0$.

The upper bounds give a limsup at most $\sqrt{B_*}$ for each parity. For the reverse bound, the subtraction formula for sine gives

$$|\sin\delta| \leq |\sin\theta| + |\sin(\theta+\delta)|.$$

Set $\delta = \pi q_*$. It lies strictly between zero and $\pi/50$. Each phase sequence in (1274) is a shifted sine with increment δ. Each pair of successive indices therefore has a term of magnitude at least $\sin\delta/2 > 0$. The positive envelopes give infinitely many matching lower bounds for each parity, and hence

$$\limsup_j |\mathsf{a}_{2j}|^{1/(2j)} = \limsup_j |\mathsf{a}_{2j+1}|^{1/(2j+1)} = \sqrt{B_*}. \tag{1277}$$

Cauchy–Hadamard proves the exact radius R. This proof covers every allowed real h, including rational parameters with infinitely many zero coefficients and irrational parameters with very small nonzero subsequences. Since $0 < p_*, q_* < 1$, one has $R > 1$.

37.3 The four singularities belong to the original inverse branch

We first prove that the actual inverse η is holomorphic throughout $|x| < R$, not just on a smaller unquantified neighborhood. The Lagrange formula (1263), applied to $F(u) = u$, gives

$$[x^{2j+1}]\eta(x) = \frac{(-1)^j}{2j+1}\binom{p_*j + D}{j}, \qquad [x^{2j}]\eta(x) = 0.$$

For $j > D/q_*$ reflection rewrites the first expression as

$$\frac{(-1)^j \sin(\pi(q_*j - D))}{\pi(2j+1)} \frac{\Gamma(p_*j + D + 1)\Gamma(q_*j - D)}{\Gamma(j+1)}.$$

The positive gamma-product estimate, with shifts $(D + 1, -D)$ and the extra factor $(2j + 1)^{-1}$, bounds this by $O_h(B_*^j j^{-3/2})$. The inverse power series thus converges on $|x| < R$. Its initial inverse identity continues along paths by local uniqueness wherever f' is nonzero.

Put $r = q_*^{-1/2} > 1$. In the η plane travel from zero to ir on the imaginary axis, then along the upper circle $|\eta| = r$ to either r or $-r$. This path avoids $\eta = \pm 1$. On its first segment,

$$|f(iv)| = v(1 + v^2)^{-D}, \qquad \frac{d}{dv}\big(v(1 + v^2)^{-D}\big) = (1 + v^2)^{-D-1}(1 + q_*v^2) > 0.$$

Even the endpoint modulus is less than $r(r^2 - 1)^{-D} = R$. On each open circular arc,

$$|1 - r^2e^{2i\theta}|^2 = (r^2 - 1)^2 + 4r^2 \sin^2 \theta > (r^2 - 1)^2,$$

so $|f(\eta)| < R$ until its endpoint. The lower-half-plane conjugate paths have the same property. Direct differentiation gives

$$f'(\eta) = (1 - \eta^2)^{-D-1}(1 - q_*\eta^2). \tag{1278}$$

It is nonzero before the endpoints. On every compact initial part of an image path, successive local inverses agree with the disk's $\eta(x)$, starting from their common germ at zero. This proves that the prescribed paths are continuations of that original inverse; it does not presume global injectivity on another logarithm sheet.

At $\eta_c = \varepsilon/\sqrt{q_*}$ let ς record the endpoint logarithm

$$\log(1 - \eta^2) \longrightarrow \log(p_*/q_*) + i\varsigma\pi.$$

The four resulting image endpoints are exactly (1271), distinct because $0 < D < 1/2$. For instance the upper approach to $+r$ has $\varsigma = -1$, and the upper approach to $-r$ has $\varsigma = 1$.

37.4 Convergent square-root expansions with nonzero amplitudes

Let $d_c = 1 - \eta_c^2 = -p_*/q_*$, evaluated using the just specified local logarithm. Differentiating (1278) gives

$$f''(\eta_c) = -2q_*\eta_c d_c^{-D-1}, \qquad \frac{f''(\eta_c)}{x_c} = \frac{2q_*^2}{p_*} \neq 0.$$

It follows that

$$1 - x/x_c = -\frac{q_*^2}{p_*}(\eta - \eta_c)^2 + O((\eta - \eta_c)^3).$$

For an analytic inversion proof, factor the right side as $(\eta - \eta_c)^2 H_c(\eta)$, where $H_c(\eta_c) = -q_*^2/p_* \neq 0$. A local holomorphic square root of H_c exists, and $(\eta - \eta_c)\sqrt{H_c(\eta)}$ has nonzero derivative at η_c. Its holomorphic inverse, composed with $\sqrt{1-x/x_c}$, gives a convergent Puiseux expansion. With that square root asymptotic to the positive one on the circle approach,

$$\eta(x) = \frac{\varepsilon}{\sqrt{q_*}} + \kappa_{\varepsilon,\varsigma}\sqrt{1-x/x_c} + O(1-x/x_c), \qquad \kappa_{\varepsilon,\varsigma} = -i\varepsilon\varsigma\frac{\sqrt{p_*}}{q_*}. \tag{1279}$$

The approach has imaginary part of sign $-\varepsilon\varsigma$; this fixes the displayed sign, while $\kappa^2 = -p_*/q_*^2$ checks its square. Finally

$$F'(\eta_c) = -\frac{2}{q_*}d_c^{-D}(4+\varepsilon A j_0\sqrt{q_*}) \neq 0. \tag{1280}$$

To verify it, differentiate $d^A(4\eta + j_0)$, use $A + D = 1$ and $\eta_c^2 = 1/q_*$. The possible correction to 4 has magnitude less than $51/2000$, since $A < 51/100$, $j_0 \leq 1/20$, $\sqrt{q_*} < 1$. Substitution proves

$$\mathcal{W}(x) = F(\eta_c) + F'(\eta_c)\kappa_{\varepsilon,\varsigma}\sqrt{1-x/x_c} + O(1-x/x_c). \tag{1281}$$

The leading fractional-power coefficient is nonzero at all four points. In particular a holomorphic extension there is impossible. This finishes the analytic part of Theorem 37.1.

37.5 The exact same branch geometry in the arithmetic residue

Define a function of the physical axial coordinate and time by

$$J_0(z,t) = \mathrm{Res}_{s=0}\,\mathcal{M}\mathcal{A}_h Z_{\mathrm{act}}(s,z,t), \qquad c_{h,0} = \frac{2^h-1}{4} > 0.$$

The variables s and z have different explicit roles: s is the Mellin variable, whereas z is the original physical axis coordinate. Theorem 36.1 gives, on the source's central real-axis patch, the exact forward and inverse identities

$$J_0(z,t) = c_{h,0}w_0(z,t), \qquad w_0(z,t) = c_{h,0}^{-1}J_0(z,t). \tag{1282}$$

Indeed evaluation at radial $\sigma = 0$ of the actual axial angular average is w_0, and the zeroth residue multiplies that value by $c_{h,0}$. The full radial remainder has zero zeroth coefficient; it is not discarded from the whole Mellin transform. Therefore (1282) is an equality of the corresponding holomorphic z germs. It is not merely an equality of a finite set of numerical derivatives.

Set $z_c = \sqrt{\nu}\,\tau^D x_c$. Equations (1270), (1281), and (1282) prove the explicit expansion

$$\begin{aligned} J_0(z,1-\tau) = c_{h,0}\sqrt{\nu}\,\tau^{-A}\big(F(\eta_c) + F'(\eta_c)\kappa_{\varepsilon,\varsigma}\sqrt{1-z/z_c} \\ + O(1-z/z_c)\big). \end{aligned} \tag{1283}$$

Every leading coefficient and the inverse scalar in this transport are nonzero. Both physical germs consequently have exact radius $\sqrt{\nu}\,\tau^D R$. Their complete derivative formulas include

$$\begin{aligned} \partial_z^n w_0(0,1-\tau) &= n!\nu^{(1-n)/2}\tau^{-A-nD}\mathsf{a}_n, \\ \partial_z^n J_0(0,1-\tau) &= c_{h,0}n!\nu^{(1-n)/2}\tau^{-A-nD}\mathsf{a}_n, \\ \limsup_n \left|\frac{\partial_z^n J_0(0,1-\tau)}{n!}\right|^{1/n} &= \frac{1}{\sqrt{\nu}\,\tau^D R}. \end{aligned} \tag{1284}$$

The same limsup holds for w_0. The physical singularity locations shrink at the original power τ^D; the original amplitude $\sqrt{\nu}\,\tau^{-A}$ remains visible.

The transferred singularities are in the continued physical variable z of a fixed endpoint residue at $s = 0$. Equations (1282)–(1283) are the precise relation; they do not locate zeros of $\zeta(s)$ or determine the sign of the complete Weil pairing. The full smooth cutoff field has other variables and remainders and has not been asserted globally holomorphic. The original existence theorem remains imported. The standard proofs above cover the whole stated parameter range; the accompanying exact finite and high-precision checks are regressions, not a formal Lean certificate or a numerical substitute for those proofs.

38 Morphisms, interfaces, and obstructions

Every proposed interface must record both original objects and, where meaningful:

domain and codomain,	coordinate formula,
functoriality or compatibility equations,	inverse, section, or retraction,
kernel, image, and fibres,	singular and exceptional loci,
symmetries and equivariance,	information preserved and lost.

Organizational separation never proves mathematical disconnection. A claimed nonexistence of a map must be an explicit obstruction theorem with its quantified class of maps and hypotheses. Candidate Fable/operator/Time/Connes interfaces enter here only after independent literature and local-programme reconstruction; finite calculations do not license unsupported cross-domain meaning.

The literature tranche now contains exact coordinate interfaces in morphism records 0012–0017. They include the Weil logarithmic Mellin bijection, the scaled even Guinand extension, the $k = \mathbb{Q}$ trivial-character finite and pole specialization, symmetric-zero pairing with its central exceptional locus, the residual Guinand–Weil archimedean identity, and the shifted exponential test whose transform is Montgomery's Poisson kernel. Their equations and proofs are in section 2. The final Montgomery gamma/trivial-zero commutator is explicitly open; the partial kernel match is not promoted to a complete formula match.

39 Derived results and global propagation

This satellite is for results derived during the consolidation rather than merely transcribed. Each entry must identify:

a) the exact literature and local inputs;

b) the new statement and proof;

c) novelty-search coverage and the correct attribution tier;

d) every affected definition, theorem, programme, comparison, and obligation;

e) the Lean theorem or executable certificate status;

f) limitations and larger implications not proved.

Refinements of published results are recorded even when they are not novel user results, with the original authors retained as the primary source and the refinement separately attributed.

The first admitted tranche consists of exact coordinate results already isolated during the literature audit. Their complete statements and proofs are in section 40; the corresponding machine records are `RESULT-20260824-0001` through `RESULT-20260824-0007`. They comprise:

1. the exponential/logarithm Haar-coordinate isomorphism, the vertical-line Mellin–Fourier identity, and the multiplicative/additive convolution transport;
2. the continuous Keating–Snaith density coordinate and its explicitly lossy integer projections;
3. the exact transition between Montgomery's finite-height scale and the local-density scale, including collapse and orientation loci;
4. the Fourier-sign involution between Montgomery and Goldston conventions;
5. the weighted pair measure, its complete Fourier–Stieltjes transform, and an explicit signed-measure kernel for restriction to $[-1, 1]$; and
6. the exact affine re-expression of the Keating–Snaith moment parameter.

These are admitted as proved coordinate derivations and literature refinements, not as novelty claims. No larger zeta, random-matrix, spectral, or endpoint implication is inferred.

The second admitted tranche is proved completely in section 41 and machine-recorded in entries `0008-0013` of the results ledger. It contains the Weil logarithmic Mellin test bijection; the exact Guinand–Weil intersection and boundary removal; the $k = \mathbb{Q}$, trivial-character finite, pole, conductor, and real-place coordinates; the symmetric-zero pairing with the central locus checked; the residual archimedean identity with every scalar cancellation displayed; and the Weil test realizing Montgomery's Poisson zero kernel, two prime ranges, logarithmic-derivative coordinate, and pole term. The final gamma/trivial-zero commutator remains explicitly open, so the partial Poisson-kernel match is not promoted to an identity of the two complete explicit formulas.

40 Complete proofs of exact coordinate morphisms

This section gives the complete proofs behind morphism records `0001-0003` and `0006-0011`. The source-dependent coordinates come from Weil's logarithmic Mellin transform, Guinand's explicit logarithmic substitution, Montgomery's pair form, Goldston's opposite Fourier sign, and Keating–Snaith's CUE scaling and moment formulas [26, Collected Papers II, pp. 52–58] [25, p. 115] [24, pp. 181–193] [14, Secs. 3–6] [23, pp. 57–89]. The proofs below are exact coordinate derivations from those retained objects. They are not presently assigned a novelty claim.

40.1 The logarithmic group coordinate

Theorem 40.1 (Exponential/logarithm isomorphism with Haar coordinate)**.** *Let*

$$E : (\mathbb{R}, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \qquad E(u) = e^u.$$

Then E is a real-analytic group isomorphism with real-analytic inverse $L(x) = \log x$. Every fibre of E and L is a singleton. For every nonnegative measurable H, and hence for every absolutely integrable H,

$$\int_0^\infty H(x)\,\frac{dx}{x} = \int_{-\infty}^\infty H(e^u)\,du. \tag{1285}$$

Thus E transports additive Haar measure du exactly to multiplicative Haar measure dx/x.

Proof. For $u, v \in \mathbb{R}$, $E(u+v) = e^{u+v} = e^u e^v = E(u)E(v)$, so E is a homomorphism. Its derivative e^u is positive, its limits at $-\infty$ and $+\infty$ are 0 and $+\infty$, and it is therefore a bijection from $\mathbb{R}$ to $\mathbb{R}_{>0}$. The identities $\log(e^u) = u$ and $e^{\log x} = x$ give the two-sided inverse and singleton fibres. Both functions are real analytic on their stated domains.

For the measure statement, the ordinary one-variable substitution $x = e^u$ has $dx = e^u, du = x, du$, whence $dx/x = du$. Applying the substitution theorem first to nonnegative H proves (1285); applying it to positive and negative, or real and imaginary, parts gives the absolutely integrable case. No statement is made for $x \leq 0$, which lies outside the codomain. □

Theorem 40.2 (Mellin transform on a vertical line is a Fourier transform)**.** *Fix $c \in \mathbb{R}$. Let $f : \mathbb{R}_{>0} \to \mathbb{C}$ be measurable and define*

$$g_c(u) = e^{cu} f(e^u).$$

Then

$$g_c \in L^1(\mathbb{R}, du) \iff \int_0^\infty |f(x)| x^{c-1}\, dx < \infty. \tag{1286}$$

Under these equivalent conditions, using $x^s = e^{s \log x}$ for $x > 0$,

$$\mathcal{M}f(c+it) := \int_0^\infty f(x) x^{c+it-1}\, dx \tag{1287}$$

$$= \int_{\mathbb{R}} g_c(u) e^{itu}\, du = \mathcal{F}_+[g_c](t) = \mathcal{F}_-[g_c](-t), \tag{1288}$$

where $\mathcal{F}_+[g](t) = \int_{\mathbb{R}} g(u) e^{itu} du$ and $\mathcal{F}_-[g](t) = \int_{\mathbb{R}} g(u) e^{-itu} du$. The underlying coordinate map is invertible:

$$f(x) = x^{-c} g_c(\log x), \qquad x > 0. \tag{1289}$$

No Fourier-inversion conclusion is included unless its separate hypotheses hold.

Proof. By theorem 40.1,

$$\int_0^\infty |f(x)| x^{c-1}\, dx = \int_{\mathbb{R}} |f(e^u)| e^{(c-1)u} e^u, du = \int_{\mathbb{R}} |g_c(u)|, du,$$

which proves (1286). The same substitution in the absolutely convergent Mellin integral gives

$$f(e^u) e^{(c+it-1)u} e^u = e^{cu} f(e^u) e^{itu} = g_c(u) e^{itu}.$$

This proves the first two equalities in (1288); the last equality is the literal identity $e^{itu} = e^{-i(-t)u}$. Finally, setting $u = \log x$ in the definition of g_c gives (1289). This is the exact coordinate core of the logarithmic Mellin presentations used by Weil and Guinand; it neither changes their Fourier constants nor erases their test-space hypotheses [26, Collected Papers II, pp. 52–58] [25, p. 115]. □

Theorem 40.3 (Multiplicative–additive convolution conjugacy)**.** *For $f, h \in L^1(\mathbb{R}_{>0}, dx/x)$ define*

$$(f *_\times h)(x) = \int_0^\infty f(y) h(x/y)\, \frac{dy}{y}$$

at every point where the integral is absolutely convergent, and define $U(f)(u) = f(e^u)$. Then U is an isometric linear bijection

$$L^1(\mathbb{R}_{>0}, dx/x) \longrightarrow L^1(\mathbb{R}, du), \qquad U^{-1}(F)(x) = F(\log x),$$

and, almost everywhere,

$$U(f *_\times h) = U(f) *_+ U(h), \qquad (F *_+ H)(u) = \int_{\mathbb{R}} F(v)H(u-v)\,dv. \tag{1290}$$

No normalizing constant occurs.

Proof. The norm identity

$$\int_{\mathbb{R}} |U(f)(u)|, du = \int_0^\infty |f(x)|\,\frac{dx}{x}$$

is (1285). The displayed inverse follows from $e^{\log x} = x$ and $\log(e^u) = u$, so U is an isometric bijection with singleton fibres. For $x = e^u$ and $y = e^v$, $dy/y = dv$ and $x/y = e^{u-v}$. Hence, wherever the integral is absolutely convergent,

$$U(f *_\times h)(u) = \int_{\mathbb{R}} f(e^v)h(e^{u-v}), dv = \int_{\mathbb{R}} U(f)(v)U(h)(u-v)\,dv.$$

Tonelli's theorem applied to $|U(f)(v)U(h)(u-v)|$ shows that this convergence and equality hold for almost every u, proving (1290). This is a transport of the convolution object, not an identification obtained by discarding a measure factor; compare the general LCA-group convolution framework [27, Sec. 1.1]. □

40.2 Height and matrix-size coordinates

Theorem 40.4 (Keating–Snaith continuous density coordinate)**.** *Define*

$$D : (2\pi, \infty) \longrightarrow (0, \infty), \qquad D(T) = \log \frac{T}{2\pi}.$$

Then D is a real-analytic, orientation-preserving bijection with $D^{-1}(n) = 2\pi e^n$, and every fibre is a singleton. If

$$\Delta_\zeta(T) = \frac{2\pi}{\log(T/(2\pi))}, \qquad \Delta_{\mathrm{CUE}}(n) = \frac{2\pi}{n},$$

then $\Delta_{\mathrm{CUE}}(D(T)) = \Delta_\zeta(T)$ exactly.

The integer projections are additional noninjective maps. Explicitly,

$$\lfloor D(T) \rfloor = m \iff T \in [2\pi e^m, 2\pi e^{m+1}) \cap (2\pi, \infty), \quad m \geq 0,$$
$$\lceil D(T) \rceil = m \iff T \in (2\pi e^{m-1}, 2\pi e^m], \quad m \geq 1.$$

For the floor projection the fibre $m = 0$ does not correspond to a positive matrix size. A nearest-integer projection requires an additional tie-breaking rule at half-integers.

Proof. $D'(T) = 1/T > 0$. Moreover, $D(T) \to 0$ as $T \downarrow 2\pi$ and $D(T) \to \infty$ as $T \to \infty$. Thus D is the asserted bijection. Solving $n = \log(T/(2\pi))$ gives $T = 2\pi e^n$, and substitution gives the spacing equality. The floor and ceiling fibres follow by applying the strictly increasing exponential to $m \leq D(T) < m+1$ and $m-1 < D(T) \leq m$, respectively. Keating and Snaith use the retained scale $\log(T/(2\pi))$ in their CUE comparison [23, eqs. (8) and (92)]; the calculation here proves only the scale-coordinate map, not a map from zeros to eigenphases or from zeta to characteristic polynomials. □

Theorem 40.5 (Exact transition between two finite-height pair coordinates)**.** *For $T \geq 2$ and $\Delta \in \mathbb{R}$, retain T and define*

$$u_M(T,\Delta) = \frac{\log T}{2\pi}\Delta, \qquad u_D(T,\Delta) = \frac{\log(T/(2\pi))}{2\pi}\Delta.$$

For $T \neq 2\pi$ both maps in the Δ coordinate are bijective, and

$$u_D = \frac{\log(T/(2\pi))}{\log T}, u_M, \qquad\qquad u_M = \frac{\log T}{\log(T/(2\pi))}, u_D, \tag{1291}$$

$$\Delta = \frac{2\pi}{\log T} u_M, \qquad\qquad \Delta = \frac{2\pi}{\log(T/(2\pi))} u_D. \tag{1292}$$

At $T = 2\pi$, u_D sends every Δ to 0, while u_M remains bijective. The transition reverses orientation for $2 \leq T < 2\pi$ and preserves it for $T > 2\pi$. Its multiplier tends to 1 as $T \to \infty$, but is not equal to 1 at any finite T.

For $T > 2\pi$ the corresponding pointed coordinate

$$U_T(\gamma) = \frac{\log(T/(2\pi))}{2\pi}(\gamma - T)$$

has inverse $U_T^{-1}(u) = T + 2\pi u/\log(T/(2\pi))$, and

$$U_T(\gamma) - U_T(\gamma') = \frac{\log(T/(2\pi))}{2\pi}(\gamma - \gamma').$$

For a labelled finite configuration, retaining all pair differences forgets exactly common translation; forgetting labels may lose additional data, for which no injectivity claim is made.

Proof. Since $T \geq 2$, $\log T > 0$, so $\Delta \mapsto u_M$ is always a positive scalar bijection. If $T \neq 2\pi$, the scalar $\log(T/(2\pi))/(2\pi)$ is nonzero; division gives (1291) and (1292). At $T = 2\pi$ that scalar is zero, proving the stated collapse. Its sign is negative below 2π and positive above 2π, while

$$\frac{\log(T/(2\pi))}{\log T} = 1 - \frac{\log(2\pi)}{\log T} \longrightarrow 1.$$

Because $2\pi \neq 1$, equality with 1 would require $\log(2\pi) = 0$, which never occurs.

The formulas for U_T, its inverse, and its differences are direct scalar algebra. If labelled configurations $(x_i)_{i=1}^m$ and $(y_i)_{i=1}^m$ have $x_i - x_j = y_i - y_j$ for every i, j, then taking $j = 1$ gives $x_i - y_i = x_1 - y_1$ for every i; conversely, a common translation preserves all differences. This proves the precise labelled kernel statement. The two scales are taken respectively from Montgomery and Keating–Snaith [24, pp. 181–182] [23, eqs. (8) and (92)]. No point-process limit follows from the coordinate algebra. ☐

40.3 Fourier signs and the weighted pair measure

Theorem 40.6 (Fourier-sign reflection is an involutive bijection)**.** *For $r \in L^1(\mathbb{R})$ define*

$$\widehat{r}_M(\alpha) = \int_{\mathbb{R}} r(u)e^{-2\pi i\alpha u}\,du, \qquad \widehat{r}_G(\alpha) = \int_{\mathbb{R}} r(u)e^{2\pi i\alpha u}\,du.$$

Let $J(h)(\alpha) = h(-\alpha)$. Then

$$\widehat{r}_G = J(\widehat{r}_M), \qquad J^2 = \mathrm{id}\,.$$

Thus every fibre of the convention change is a singleton. If F is even and the two pairings converge absolutely, then

$$\int_{\mathbb{R}} F(\alpha)\widehat{r}_M(\alpha)\,d\alpha = \int_{\mathbb{R}} F(\alpha)\widehat{r}_G(\alpha)\,d\alpha. \tag{1293}$$

Proof. The identity

$$\widehat{r}_M(-\alpha) = \int_{\mathbb{R}} r(u)e^{2\pi i\alpha u}\,du = \widehat{r}_G(\alpha)$$

proves the first assertion, and $J(J(h))(\alpha) = h(\alpha)$ proves the second. For the pairing, substitute $\beta = -\alpha$ in the left-hand integral:

$$\int_{\mathbb{R}} F(-\beta)\widehat{r}_M(-\beta)\,d\beta = \int_{\mathbb{R}} F(\beta)\widehat{r}_G(\beta)\,d\beta.$$

The retained conventions are Montgomery's and Goldston's [24, pp. 181–182, eqs. (3)–(6)] [14, Sec. 5]; the equality uses evenness and does not normalize either sign away. $\square$

Theorem 40.7 (Weighted pair measure and its complete transform)**.** *Assume the real-ordinate hypothesis used by Montgomery, fix $T \geq 2$, and let γ range with multiplicity over the finite multiset of ordinates with $0 < \gamma \leq T$. Put*

$$w(v) = \frac{4}{4+v^2}, \qquad c_T = \frac{T}{2\pi}\log T,$$

and define the finite positive Borel measure

$$\mu_T = c_T^{-1} \sum_{0<\gamma,\gamma'\leq T} w(\gamma-\gamma')\delta_{(\gamma-\gamma')\log T/(2\pi)}. \tag{1294}$$

Then its plus-sign Fourier–Stieltjes transform is exactly Montgomery's pair form:

$$\widehat{\mu}_{T,+}(\alpha) := \int_{\mathbb{R}} e^{2\pi i\alpha u}\,d\mu_T(u) = c_T^{-1} \sum_{0<\gamma,\gamma'\leq T} T^{i\alpha(\gamma-\gamma')}w(\gamma-\gamma') = F_M(\alpha,T). \tag{1295}$$

If r has the inversion representation $r(u) = \int_{\mathbb{R}} \widehat{r}_M(\alpha)e^{2\pi i\alpha u}d\alpha$ with $\widehat{r}_M \in L^1(\mathbb{R})$, then

$$\int_{\mathbb{R}} r(u)\,d\mu_T(u) = \int_{\mathbb{R}} \widehat{r}_M(\alpha)F_M(\alpha,T)\,d\alpha. \tag{1296}$$

The full transform is injective on finite complex regular Borel measures.

Proof. The sum in (1294) is finite, $c_T > 0$, and $w > 0$, so it defines a finite positive measure. Evaluating its transform gives

$$\begin{aligned}\widehat{\mu}_{T,+}(\alpha) &= c_T^{-1}\sum_{\gamma,\gamma'} w(\gamma-\gamma')\exp\left(2\pi i\alpha\frac{(\gamma-\gamma')\log T}{2\pi}\right)\\ &= c_T^{-1}\sum_{\gamma,\gamma'} w(\gamma-\gamma')e^{i\alpha(\gamma-\gamma')\log T},\end{aligned}$$

which is (1295). Substitute the inversion formula for r into the finite sum defining $\int r, d\mu_T$. Since $\widehat{r}_M \in L^1$ and the sum is finite, Fubini is immediate and yields (1296). These are the exact finite objects in Montgomery's paper [24, pp. 181–182, eqs. (2)–(6)].

For injectivity, if two finite complex regular measures have the same plus-sign transform, their difference σ has $\int e^{2\pi i\alpha u}d\sigma(u) = 0$ for all α. Replacing α by $-\alpha$ gives the zero transform in Rudin's sign convention. Rudin's Fourier–Stieltjes uniqueness theorem therefore gives $\sigma = 0$ [27, p. 29, Thm. 1.7.3(b)]. $\square$

Theorem 40.8 (Explicit kernel of transform restriction)**.** *Let*

$$R : \mathcal{M}(\mathbb{R}) \longrightarrow C([-1,1]), \qquad R(\mu) = \widehat{\mu}_+\big|_{[-1,1]},$$

where $\mathcal{M}(\mathbb{R})$ is the vector space of finite signed Borel measures and $\widehat{\mu}_+(\alpha) = \int e^{2\pi i\alpha u} d\mu(u)$. Then R is not injective. More precisely, there is an explicit nonzero absolutely continuous finite signed measure ν with $R(\nu) = 0$.

Proof. Define the standard smooth bump

$$b(x) = \begin{cases} \exp\bigl(-1/(1-x^2)\bigr), & |x| < 1, \\ 0, & |x| \geq 1, \end{cases}$$

and set

$$\phi(\alpha) = b\bigl(4(\alpha - 5/2)\bigr) + b\bigl(4(\alpha + 5/2)\bigr).$$

Then ϕ is real, even, smooth, nonzero, and its support is contained in $[-11/4, -9/4] \cup [9/4, 11/4]$, which is disjoint from $[-1, 1]$. Put

$$g(u) = \int_{\mathbb{R}} \phi(\alpha) e^{-2\pi i\alpha u}\, d\alpha, \qquad d\nu(u) = g(u)\, du. \tag{1297}$$

Because ϕ is real and even, pairing α with $-\alpha$ shows that g is real. For integers $k, N \geq 0$, differentiation under the compactly supported integral and N integrations by parts give

$$(2\pi i u)^N g^{(k)}(u) = \int_{\mathbb{R}} \frac{d^N}{d\alpha^N}\bigl((-2\pi i\alpha)^k \phi(\alpha)\bigr) e^{-2\pi i\alpha u}\, d\alpha,$$

up to the harmless sign $(-1)^N$. The right side is bounded because its integrand is smooth and compactly supported. Hence g is Schwartz, in particular $g \in L^1(\mathbb{R})$, so ν is a finite signed measure.

Fourier inversion in the displayed 2π convention gives

$$\widehat{\nu}_+(\beta) = \int_{\mathbb{R}} e^{2\pi i\beta u} g(u)\, du = \phi(\beta) \tag{1298}$$

[27, pp. 21–23, Thm. 1.5.1]. Thus $\widehat{\nu}_+ = 0$ on $[-1, 1]$, while $\widehat{\nu}_+ = \phi$ is not the zero function; consequently $\nu \neq 0$. Therefore $R(\nu) = 0$ with $\nu \neq 0$, proving noninjectivity. For any finite signed measure μ, μ and $\mu + \nu$ have the same restricted transform. This proof makes no claim that $\mu + \nu$ remains positive when μ is positive; the obstruction is exactly in the stated signed-measure category. □

40.4 Moment-parameter coordinate

Theorem 40.9 (Keating–Snaith moment parameter is an affine bijection)**.** *The map*

$$s \longmapsto \lambda = s/2$$

is a complex-affine bijection from $\{\Re s > -1\}$ to $\{\Re\lambda > -1/2\}$ with inverse $s = 2\lambda$. Under this map the exact finite-CUE moment formula

$$\mathbb{E}_{U(N)} |Z_N(U;\theta)|^s = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+s)}{\Gamma(j+s/2)^2}$$

becomes exactly

$$\mathbb{E}_{U(N)}|Z_N(U;\theta)|^{2\lambda} = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+2\lambda)}{\Gamma(j+\lambda)^2},$$

and $N^{(s/2)^2} = N^{\lambda^2}$ *for* $N > 0$. *Every parameter fibre is a singleton, and the half-plane boundary and any meromorphic poles are transported rather than removed.*

Proof. The inverse identities $2(s/2) = s$ and $(2\lambda)/2 = \lambda$ prove bijectivity, while $\Re(s/2) > -1/2$ is equivalent to $\Re s > -1$. Substitute $s = 2\lambda$ in every factor: $\Gamma(j+s) = \Gamma(j+2\lambda)$ and $\Gamma(j+s/2) = \Gamma(j+\lambda)$. The power identity is the same substitution, using the real logarithm of $N > 0$ to define complex powers. The starting finite-CUE formula and its domain are Keating–Snaith's theorem [23, pp. 57–64, eqs. (1), (6), and (20)–(26)]. This coordinate equality does not identify a CUE moment with a zeta moment; that transfer remains conjectural in the same source. $\square$

41 Complete proofs of the Guinand–Weil interfaces

This section expands morphism records `0012-0017`. It keeps Guinand's RH hypothesis and half-line cosine transform separate from Weil's general Hecke-character formula until an explicit common subspace and every constant have been written down. The primary inputs are Guinand's Theorems 3 and 3′ and Weil's conditions (A), (B), and formula (11) [25, pp. 112–115] [26, Collected Papers II, pp. 52–58].

41.1 Weil's multiplicative test coordinate

Theorem 41.1 (Exact logarithmic Mellin test-function bijection)**.** *Let* $F : \mathbb{R} \to \mathbb{C}$ *satisfy Weil's condition (A), and for some* $b > 0$ *let*

$$F(x), F'(x) = O\Big(e^{-(1/2+b)|x|}\Big) \qquad (|x| \to \infty) \tag{1299}$$

away from the finitely many derivative-jump points, as in Weil's condition (B). Define

$$\mathcal{B}(F)(v) = v^{-1/2} F(\log v), \qquad v > 0. \tag{1300}$$

Then $\mathcal{B}$ *is injective and has the explicit inverse*

$$\mathcal{B}^{-1}(\phi)(x) = e^{x/2}\phi(e^x). \tag{1301}$$

Consequently it is a bijection from Weil's additive test space onto its explicitly transported multiplicative test space, and every fibre is a singleton. At points away from transported jumps,

$$\phi'(v) = v^{-3/2}\left(F'(\log v) - \frac{1}{2}F(\log v)\right). \tag{1302}$$

If a_j *is a jump location of* F *or* F', *its location becomes* e^{a_j}; *the one-sided values and their averaging convention are transported without merging. The endpoint estimates are*

$$\phi(v) = O(v^b), \qquad \phi'(v) = O(v^{b-1}) \qquad (v \downarrow 0), \tag{1303}$$

$$\phi(v) = O(v^{-1-b}), \qquad \phi'(v) = O(v^{-2-b}) \qquad (v \to \infty). \tag{1304}$$

Finally, wherever Weil's transform is defined,

$$\int_0^\infty \phi(v)v^{s-1}\,dv = \int_{\mathbb{R}} F(x)e^{(s-1/2)x}\,dx = \Phi(s). \tag{1305}$$

Proof. Substituting (1300) into (1301) gives $e^{x/2}e^{-x/2}F(x) = F(x)$; substituting (1301) into (1300) gives $v^{-1/2}v^{1/2}\phi(v) = \phi(v)$. Thus the maps are two-sided inverses. Differentiating $v^{-1/2}F(\log v)$ gives (1302). Since $x \mapsto e^x$ is strictly increasing and bijective, every finite jump location and both of its one-sided germs move bijectively from a_j to e^{a_j}.

For $v \to \infty$, put $x = \log v > 0$. Equation (1299) gives $F(\log v), F'(\log v) = O(v^{-1/2-b})$, which in (1300) and (1302) gives (1304). For $v \downarrow 0$, $x = \log v < 0$ and $e^{-(1/2+b)|x|} = e^{(1/2+b)x} = v^{1/2+b}$, giving (1303). Finally, $v = e^x$ and $dv = e^x dx$ give

$$\phi(v)v^{s-1}dv = e^{-x/2}F(x)e^{(s-1)x}e^x dx = F(x)e^{(s-1/2)x}dx.$$

This proves (1305). The transform and conditions are Weil's literal ones [26, Collected Papers II, pp. 52 and 55–58]; no endpoint decay, jump, or averaging datum has been normalized away. □

41.2 The exact Guinand–Weil intersection

For $1 < p \le 2$, let $\mathcal{G}_p$ denote the half-line functions satisfying Guinand's Theorem 3 hypotheses: f is an integral in Guinand's terminology, $f(x) \to 0$, $xf'(x) \in L^p(0,\infty)$, and its cosine transform is

$$g(y) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x)\cos(xy)\,dx. \tag{1306}$$

Let $\mathcal{G}_p^W$ be the subset for which the scaled even extension below satisfies Weil's (A) and (B), and let $\mathcal{W}_{\text{even}}^{G,p}$ be the even Weil tests whose positive-half restriction, multiplied by $\sqrt{2\pi}$, lies in $\mathcal{G}_p$. These definitions retain the actual intersection; they do not assert equality of the two full test spaces.

Theorem 41.2 (Scaled even-extension bijection and boundary removal)**.** *The map*

$$\mathcal{E} : \mathcal{G}_p^W \longrightarrow \mathcal{W}_{\text{even}}^{G,p}, \qquad \mathcal{E}(f)(x) = \frac{f(|x|)}{\sqrt{2\pi}}, \tag{1307}$$

is a bijection with inverse $f(t) = \sqrt{2\pi}F(t)$ for $t \ge 0$. Its fibres are singletons. If $F = \mathcal{E}(f)$, then

$$\Phi_F(\tfrac{1}{2} + iy) = \int_{\mathbb{R}} F(x)e^{iyx}\,dx = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x)\cos(xy)\,dx = g(y). \tag{1308}$$

Moreover, Weil's condition (B) implies $T^{1/2}f(T) \to 0$, so the boundary term in Guinand's Theorem 3 is removable on this intersection by his Theorem $3'$.

Proof. The image in (1307) is even. Restriction to $[0,\infty)$ followed by multiplication by $\sqrt{2\pi}$ is visibly a two-sided inverse, proving the bijection and singleton fibres. Evenness gives

$$\begin{aligned}\int_{\mathbb{R}} F(x)e^{iyx}\,dx &= \frac{1}{\sqrt{2\pi}} \int_0^\infty f(x)\big(e^{iyx} + e^{-iyx}\big)dx \\ &= \frac{2}{\sqrt{2\pi}} \int_0^\infty f(x)\cos(xy)dx = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x)\cos(xy)dx,\end{aligned}$$

which is (1308). If condition (B) holds with $b > 0$, then for $T > 0$

$$|f(T)| = \sqrt{2\pi}|F(T)| = O\Big(e^{-(1/2+b)T}\Big).$$

Therefore $T^{1/2}|f(T)| \to 0$. Guinand's Theorem $3'$ explicitly lists this limit as a sufficient condition for deleting the boundary term [25, pp. 112–115, Thms. 3 and $3'$]. □

41.3 The rational trivial-character fibre

Theorem 41.3 (Termwise $k = \mathbb{Q}$, $\chi = 1$ specialization)**.** *In Weil's formula (11), choose $k = \mathbb{Q}$ and the trivial character. The retained data are*

$$d = 1, \quad \Delta = 1, \quad N\mathfrak{f} = 1, \quad A = (2\pi)^{-1}, \quad (\eta_1, f_1, \varphi_1) = (1, 0, 0), \quad \delta_\chi = 1. \tag{1309}$$

Let $F = \mathcal{E}(f)$ be in the intersection of theorem 41.2. Then Weil's finite-place term is exactly

$$-\sqrt{\frac{2}{\pi}} \sum_p \sum_{m \geq 1} \frac{\log p}{p^{m/2}} f(m \log p), \tag{1310}$$

his pole integral is

$$\sqrt{\frac{2}{\pi}} \int_0^\infty f(x)(e^{x/2} + e^{-x/2})\, dx, \tag{1311}$$

his conductor constant is

$$-\frac{f(0)}{\sqrt{2\pi}} \log(2\pi), \tag{1312}$$

and the remaining real-place term is

$$-\frac{1}{\sqrt{2\pi}} \operatorname{PF} \int_{\mathbb{R}} f(|x|) \frac{e^{|x|/2}}{|e^x - e^{-x}|}\, dx. \tag{1313}$$

The prime series and pole integrals converge absolutely under condition (B).

Proof. For $\mathbb{Q}$, prime ideals are the ordinary primes, $N(p^m) = p^m$, and the trivial character contributes 1 in both finite-place orientations. Since F is even,

$$F(m \log p) + F(-m \log p) = \frac{2}{\sqrt{2\pi}} f(m \log p) = \sqrt{\frac{2}{\pi}} f(m \log p).$$

Inserting this into Weil's negative finite-place sum proves (1310). Splitting the pole integral at zero, changing x to $-x$ in the negative half, and using evenness gives twice the positive half; $2/\sqrt{2\pi} = \sqrt{2/\pi}$, proving (1311). Since $\log A = -\log(2\pi)$ and $F(0) = f(0)/\sqrt{2\pi}$, (1312) follows. The real-place data in (1309) give Weil's kernel $K_{1,0}(x) = e^{|x|/2}/|e^x - e^{-x}|$; linearity of the PF distribution and $F = f(|\cdot|)/\sqrt{2\pi}$ give (1313).

If condition (B) holds with $b > 0$, then at $m \log p > 0$, $F(\pm m \log p) = O(p^{-m(1/2+b)})$. After the outside factor $p^{-m/2}$, the absolute prime series is bounded by a constant multiple of $\sum_{p,m} (\log p) p^{-m(1+b)}$, which converges. The $e^{x/2}$ pole integrand is $O(e^{-bx})$ and the $e^{-x/2}$ part decays faster. These are literal substitutions into Weil's primary formula [26, Collected Papers II, pp. 57–58, eq. (11)]; other fields, characters, conductors, and complex places remain outside this selected fibre. $\square$

41.4 Zero pairing and the residual archimedean identity

Theorem 41.4 (Symmetric cutoff versus positive ordinates)**.** *Assume RH for ζ, count zeros with multiplicity, and let $F = \mathcal{E}(f)$, so that $\Phi_F(1/2 + iy) = g(y)$ is even. Then, for every cutoff T that does not pass through an ordinate,*

$$\sum_{|\gamma| < T} \Phi_F(\tfrac{1}{2} + i\gamma) = 2 \sum_{0 < \gamma < T} g(\gamma). \tag{1314}$$

There is no omitted central term because $\zeta(1/2) < 0$. For a different self-dual L-function with central-zero multiplicity m_0, the corresponding formula contains the additional term $m_0 g(0)$.

Proof. The functional equation and conjugation symmetry pair every positive ordinate γ with $-\gamma$ with equal multiplicity. Evenness gives $g(-\gamma) = g(\gamma)$, so each noncentral pair contributes $2g(\gamma)$. It remains only to check the central locus.

The alternating eta series at $s = 1/2$ converges, and its even partial sums are

$$\sum_{k=1}^{N} \left((2k-1)^{-1/2} - (2k)^{-1/2}\right) > 0.$$

Their limit is therefore strictly positive. The eta continuation identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ gives

$$\zeta(1/2) = \frac{\eta(1/2)}{1-\sqrt{2}} < 0.$$

Thus $1/2$ is not a zero, proving (1314). Guinand's positive-ordinate convention and Weil's symmetric cutoff are retained from their respective primary formulas [25, pp. 107–115] [26, Collected Papers II, pp. 57–58]. The $m_0 g(0)$ exception is the unpaired central contribution and cannot be removed by evenness. □

Theorem 41.5 (Guinand–Weil residual archimedean identity)**.** *Assume the hypotheses of theorem 41.2, RH, and the literal summation conventions of both primary formulas. Write*

$$\begin{aligned}
P_f &= \sum_p \sum_{m\geq 1} \frac{\log p}{p^{m/2}} f(m \log p),\\
A_+ &= \int_0^\infty f(x) e^{x/2} dx, \qquad A_- = \int_0^\infty f(x) e^{-x/2} dx,\\
H_f &= \frac{1}{2}\int_0^\infty f(x)\left(\frac{1}{x} - \frac{e^{-x/2}}{\sinh x}\right) dx,\\
Z_g(T) &= \sum_{0<\gamma<T} g(\gamma), \qquad L_g(T) = \int_0^T g(t) \log \frac{t}{2\pi} dt,\\
K(x) &= \frac{e^{|x|/2}}{|e^x - e^{-x}|}.
\end{aligned}$$

Then both limits $Z_g = \lim_{T\to\infty} Z_g(T)$ and $L_g = \lim_{T\to\infty} L_g(T)$ exist on this common subspace, and

$$2A_- - \mathrm{PF}\int_{\mathbb{R}} f(|x|)K(x)dx - f(0)\log(2\pi) = 2H_f + \sqrt{\frac{2}{\pi}} L_g. \tag{1315}$$

This is equality of scalar functionals on the stated common test subspace. It does not replace Weil's PF convention by an ordinary integral.

Proof. By theorem 41.4, Weil's symmetric zero limit is $2\lim_T Z_g(T)$, so Weil's formula proves that $\lim_T Z_g(T)$ exists. By theorem 41.2, Guinand's boundary term vanishes, and his Theorem 3′ proves that $\lim_T\{Z_g(T) - L_g(T)/(2\pi)\}$ exists. Therefore

$$L_g(T) = 2\pi\left[Z_g(T) - \left\{Z_g(T) - \frac{L_g(T)}{2\pi}\right\}\right]$$

also has a limit. Denote the two limits by Z_g and L_g. Guinand's theorem now states, in the notation above,

$$P_f - A_+ - H_f = -\sqrt{2\pi}\, Z_g + \frac{1}{\sqrt{2\pi}} L_g. \tag{1316}$$

Indeed, his density term is $(2\pi)^{-1}L_g$, and $\sqrt{2\pi}/(2\pi) = 1/\sqrt{2\pi}$ [25, pp. 114–115, Thms. 3 and 3′].

By theorems 41.3 and 41.4, Weil's formula (11) becomes

$$\begin{aligned} 2Z_g = \sqrt{\frac{2}{\pi}}(A_+ + A_-) - \sqrt{\frac{2}{\pi}}P_f - \frac{f(0)}{\sqrt{2\pi}}\log(2\pi) \\ - \frac{1}{\sqrt{2\pi}}\,\mathrm{PF}\int_{\mathbb{R}} f(|x|)K(x)dx. \end{aligned} \tag{1317}$$

Multiply (1317) by $\sqrt{2\pi}$:

$$2\sqrt{2\pi}Z_g = 2A_+ + 2A_- - 2P_f - f(0)\log(2\pi) - \mathrm{PF}\int_{\mathbb{R}} f(|x|)K(x)dx. \tag{1318}$$

Solving (1316) for $\sqrt{2\pi}Z_g$ and doubling gives

$$2\sqrt{2\pi}Z_g = -2P_f + 2A_+ + 2H_f + \frac{2}{\sqrt{2\pi}}L_g. \tag{1319}$$

Equate (1318) and (1319). The terms $-2P_f + 2A_+$ cancel exactly; moving the rest gives (1315), since $2/\sqrt{2\pi} = \sqrt{2/\pi}$. Every cancellation is displayed, and the PF term remains Weil's literal distribution [26, Collected Papers II, pp. 56–58]. A direct kernel-level proof independent of the two primary formulas would be stronger and remains a separate obligation. □

41.5 Montgomery's Poisson kernel as a Weil test

Theorem 41.6 (Exact Poisson-kernel test and proved term coordinates)**.** *Let $1 < \sigma < 2$, $x \geq 1$, $t \in \mathbb{R}$, and put*

$$a = \sigma - \frac{1}{2}, \qquad u = \log x, \qquad F_{a,t,u}(v) = e^{-a|v-u|}e^{-it(v-u)}.$$

Then $F_{a,t,u}$ satisfies Weil's (A) and (B): it is continuous with one permitted derivative jump at $v = u$, and for every $0 < b < a - 1/2 = \sigma - 1$ it has the required exponential decay. In the strip $|\Re(s) - 1/2| < a$,

$$\Phi_{a,t,u}(s) = \frac{2a\,x^{s-1/2}}{a^2 - (s - 1/2 - it)^2}. \tag{1320}$$

Consequently

$$\Phi_{a,t,u}(\tfrac{1}{2} + i\gamma) = \frac{(2\sigma - 1)x^{i\gamma}}{(\sigma - \frac{1}{2})^2 + (t - \gamma)^2}, \tag{1321}$$

which is Montgomery's exact zero kernel. At a positive integer n,

$$n^{-1/2}F(\log n) = \begin{cases} x^{1/2-\sigma+it}n^{\sigma-1-it}, & n \leq x, \\ x^{\sigma-1/2+it}n^{-\sigma-it}, & n > x, \end{cases} \tag{1322}$$

$$n^{-1/2}F(-\log n) = x^{1/2-\sigma+it}n^{-\sigma+it}. \tag{1323}$$

Thus the positive-log Weil prime term recovers Montgomery's two prime ranges, while the negative-log term equals

$$x^{1/2-\sigma+it}\frac{\zeta'}{\zeta}(\sigma - it) \tag{1324}$$

after Weil's outside minus sign. Finally,

$$\Phi_{a,t,u}(1) = \frac{x^{1/2}(2\sigma - 1)}{(\sigma - 1 + it)(\sigma - it)}, \tag{1325}$$

Montgomery's rational pole term. The remaining gamma/trivial-zero commutator is not claimed proved.

Proof. Away from $v = u$, differentiation multiplies F by $a - it$ on the left and by $-a - it$ on the right, so F is continuous with one first-kind derivative jump. Assigning $F'(u) = -it$ gives the average of the two one-sided derivative values, exactly as required by Weil's convention. Its absolute value is $e^{-a|v-u|}$; choosing $0 < b < a - 1/2$ proves condition (B) for both F and its one-sided derivatives.

Set $w = v - u$ and $z = s - 1/2 - it$. For $|\Re z| < a$, absolute convergence permits the split at $w = 0$:

$$\begin{aligned}
\Phi(s) &= x^{s-1/2} \int_{\mathbb{R}} e^{-a|w|} e^{zw} dw \\
&= x^{s-1/2} \left(\int_0^\infty e^{-(a-z)w} dw + \int_0^\infty e^{-(a+z)w} dw \right) \\
&= x^{s-1/2} \left(\frac{1}{a-z} + \frac{1}{a+z} \right) = \frac{2a\, x^{s-1/2}}{a^2 - z^2},
\end{aligned}$$

proving (1320). At $s = 1/2 + i\gamma$, $z = i(\gamma - t)$ and $2a = 2\sigma - 1$, giving (1321).

If $n \le x$, then $\log n - u \le 0$, so direct substitution gives

$$F(\log n) = e^{-a(u - \log n)} e^{-it(\log n - u)} = x^{-a+it} n^{a-it}.$$

Multiplication by $n^{-1/2}$ and $a = \sigma - 1/2$ gives the first case of (1322). If $n > x$, the same calculation with $|\log n - u| = \log n - u$ gives the second case. Since $u \ge 0$, $-\log n - u \le 0$ for every $n \ge 1$, giving (1323). For $\Re(\sigma - it) > 1$,

$$-\frac{\zeta'}{\zeta}(\sigma - it) = \sum_{n\ge 1} \Lambda(n) n^{-\sigma+it}.$$

Weil's finite-place term has an outside minus sign, so its negative-log part is (1324).

At $s = 1$, factor the denominator:

$$\begin{aligned}
a^2 - (\tfrac{1}{2} - it)^2 &= (a - \tfrac{1}{2} + it)(a + \tfrac{1}{2} - it) \\
&= (\sigma - 1 + it)(\sigma - it),
\end{aligned}$$

which proves (1325). Comparing (1321), (1322), and (1325) with Montgomery's complete formula gives the advertised exact matches [24, pp. 185–187, eqs. (18)–(22)]. Still to be proved is that $\Phi(0)$, the conductor constant, the PF real-place term, and (1324) combine through the functional equation into Montgomery's $-x^{1/2-\sigma+it}\zeta'/\zeta(1 - \sigma + it)$ and his entire trivial-zero series. Until then, the full formulas are not declared identical. □

42 Formalization and open obligations

Every local theorem claim enters a reproducible Lean queue or carries an explicit account of why a different certificate is appropriate. A formal replay never enlarges the statement it checks, and a passing executable calculation never supplies a missing conceptual map.

42.1 Completed finite-coordinate formalization

FORMAL-20260825-0007 is implemented in the source file `formal/lean/ZetaFunctionFoundation/SourceBlockCoordinates.lean`. For maps $\mu, \nu : \mathrm{Fin}(s) \to \mathbb{N}$, it defines the coordinate types

$$J_\mu = \coprod_{i:\mathrm{Fin}(s)} \mathrm{Fin}(\mu_i), \qquad I_{\mu,\nu} = (\mathrm{Fin}(s) \times \mathrm{Fin}(s)) \amalg J_\mu \amalg J_\nu$$

and the full coordinate space $I_{\mu,\nu} \to \mathbb{C}$. The theorems `sourceCoord_card` and `sourceBlock_finrank` prove, without deleting a summand,

$$\#I_{\mu,\nu} = \dim_{\mathbb{C}}(\mathbb{C}^{I_{\mu,\nu}}) = s^2 + \sum_i \mu_i + \sum_j \nu_j.$$

For the first-derivative block, the file retains (A, u, φ, λ) as four separate coordinates. It proves that

$$(A, u, \varphi) \longmapsto (A, u, \varphi, -\operatorname{Tr} A)$$

has the displayed block-extraction inverse, zero kernel, full range, and singleton fibres. It then introduces the literal index decomposition $\mathrm{Fin}(s) \amalg \mathbf{1}$, proves that four-block assembly and extraction are two-sided complex-linear inverses, and proves

$$\operatorname{Tr}\big(\mathrm{blockToMatrix}(A, u, \varphi, \lambda)\big) = \operatorname{Tr}(A) + \lambda.$$

Consequently `sourceSLEquiv` is an explicit linear equivalence with Mathlib's actual special-linear Lie algebra on that index set. The definition `sourceCommutator` pulls the matrix commutator back through this equivalence, and `sourceCommutator_matrix_formula` proves that its image is exactly $XY - YX$. Thus the formal file checks the coordinate map, its inverse, kernel, image, fibres, trace condition, and commutator pullback. It constructs no map from a CUE moment, determinant, Haar law, or limiting object to either side.

The accepted replay uses Lean 4.32.0 and Mathlib commit

`81a5d257c8e410db227a6665ed08f64fea08e997`.

Its source SHA–256 is

`521b647d6449672a7822fc348db680941494eaceff75756d0d3cf761272bce05`,

and the compiled `.olean` SHA–256 is

`79b32538a1f7aaeab42dc2ab402923603b53d1091d857cc06cefa74886697c34`.

The exact command, empty standard streams, exit code zero, and 1,933,365,248-byte aggregate peak are in `formal/lean/receipts/FORMAL-20260825-0007.receipt.json`.

42.2 Completed finite Laurent formalization

FORMAL-20260825-0008 is implemented in the source file `formal/lean/ZetaFunctionFoundation/N3LaurentCertificate.lean`. An exponent is the displayed triple in $\mathbb{Z}^3$, and a Laurent expression is a finite list of exponent/coefficient contributions. Equal exponents are collected only by the explicit function `collectTerms`; `coefficient_insertTerm` and `coefficient_collectTerms` prove that this executable collection preserves every coefficient. The multiplication theorem similarly identifies collected multiplication with the raw finite convolution contribution list. Thus no unchecked preprocessing or silent change of coordinates enters the calculation.

The file records the seven terms of $p = 3 - 2(z_1 + z_2 + z_3) + (z_1z_2 + z_1z_3 + z_2z_3)$, proves coefficientwise that inversion sends p_α to $p_{-\alpha}$, constructs all six factors $(1 - z_i/z_j)$ for $i \neq j$, and extracts the coefficient at $(0, 0, 0)$. Lean verifies the integer numerators

$$84, \quad 2250, \quad 93060, \quad 5083680, \quad 331988544,$$

their exact divisibility by $3! = 6$, and hence the five values in theorem 2.31. The formal theorem is exactly the finite Laurent identity; the analytic passage from normalized Haar measure to the constant-term formula remains the separately cited human proof in theorem 2.31.

The source SHA–256 is

`4a34082b4e60df6511f4e7cca850a102dc2cb858130d2405890247986f766e67`,

and the compiled `.olean` SHA–256 is

`416341a266d0688e7c8082f480d2883bcbea31a5e49842e930f4028084f260e8`.

The accepted replay exits zero with a 1,404,055,552-byte aggregate peak. Its use of `native_decide`, and therefore its compiler trust boundary, is explicit in the receipt and independently backed by the two exact non-Lean implementations.

42.3 Remaining queue

Four successor obligations now bind the exact results added in this tranche. `FORMAL-20260825-0011` stages the compact shifted-Hurwitz work into the finite endpoint-residue equivalence, the all-order coefficient recursion, and the separate analytic Hurwitz/implicit-function layer. The first two are finite coordinate problems; the last requires a located Mathlib meromorphic Hurwitz-zeta and complex implicit-function interface and may not be replaced by axioms.

`FORMAL-20260825-0012` targets theorem 2.32: translation and character operators on an arbitrary finite abelian group, the retained cocycle product, character orthogonality, Hilbert–Schmidt basis, inverse coefficients, kernel, image, fibres, and the exact traceless restriction. No unrecorded enumeration of the finite group may enter the statement.

`FORMAL-20260825-0013` targets the exact $U(2)$ tranche. It separates the finite centre/difference algebra, the two folding branches and boundary fibres, and the exact constant-term moment calculation from the analytic Haar-circle, change-of-variables, beta integral, and pushforward-measure layer. The executable `certificates/u2_exact_law/verify_u2_exact_law.py` already checks, in exact rational arithmetic and by two independently assembled formulas, the first eight moments

$$5,\ 34,\ 285,\ 2766,\ 29666,\ 340740,\ 4108509,\ 51351190.$$

That finite certificate is not represented as a proof of the Jacobian or pushforward measure.

`FORMAL-20260825-0014` targets theorems 2.37 and 2.38. Its finite algebraic stage must formalize the rationalized Z numerator, trace and norm equations, the two discriminant coordinates, the S factorization, the affine (u, v)–(u, X) equivalence, and the lifted inverse modulo $J^2 + S = 0$. Its ordered-real stage must separately prove positivity of D_0, h, g, the exact image Ω_3, surjectivity, two-point and branch fibres, $u < 1$, and bijectivity of the physical lift. The passing SymPy certificate checks the polynomial and rational identities only; it is not to be promoted into an inequality or surjectivity proof.

The separate interface stage `FORMAL-20260826-0015` is complete. Against Lean 4.32.0 and pinned Mathlib v4.32.0 it formalizes the deck-curve equation, the polynomial parametrization and its explicit inverse, the target involution, uniqueness of its fixed point, the general two-fixed-source-point obstruction to equivariant injectivity, the recovered-β_0 carrier parameter, and the exact $J \mapsto -J$ equivariance and coordinates of the sheet map from theorem 2.39. The accepted build contains no `sorry`, added axiom, `admit`, or `unsafe` declaration, and its peak aggregate process-tree working set was 1,452,122,112 bytes under the 4 GB cap. The complete horizontal-interval fibre classification and the ordered-real disc inequalities remain the human-readable proof, not claims made by the finite algebraic checker.

42.4 Completed derivative-coordinate formalizations

`FORMAL-20260826-0016` checks the finite phase-rotated Verblunsky derivative peel: differentiated Szegő transitions, half-plane transport, the retained η-norm identity and strict-ratio interface, the one-step and finite-product telescopes, the triangular coordinate inverse, the terminal derivative identity, and unit-phase norm preservation. Its source is `formal/lean/ZetaFunctionFoundation/VerblunskyPeel.lean`, with SHA–256

`afd9511516c35c993455d24c981acec7b91852e8395d7b0f1d7f86ebf6392e2b`.

The compiled object has SHA–256

`a30409a459eeb959bdd46fe5bad71b7496e209edfaa35add8e653dc88c61e2ac`.

The replay exits zero with empty streams and a 1,499,897,856-byte peak. It does not formalize the probability transport or a large-N theorem.

`FORMAL-20260826-0018` checks the exact identity between the Bourgade–Nikeghbali–Rouault deformed coefficient at one and the retained phase-rotated coordinate, under the displayed nonzero hypotheses. It also proves coordinatewise uniqueness of the recursively lifted derivative state over every retained γ sequence. Its source is `formal/lean/ZetaFunctionFoundation/GammaBetaCrosswalk.lean`, with SHA–256

`0eaaf472571b6b13c8f00d3b32b8797b87d5f5c8b9d37fbd95ee687d2107ae20`,

and the compiled object has SHA–256

`b74b730a1fcd7eb2439c1f890000601215c66fe9e718cff4d4aafa13361fadfe`.

The replay exits zero with empty streams and a 1,414,193,152-byte peak. Exact probability laws, Borel disintegration, and the determinant tilt remain standard human-readable proofs.

`FORMAL-20260826-0019` checks the finite CRS first-jet interface from theorem 2.23. It proves that

$$(V, d) \longmapsto \rho\left(V, d - \frac{N}{2}V\right)$$

and the displayed triangular inverse are two-sided inverses for $\rho \neq 0$. It also proves

$$\frac{\mathrm{i}}{2}V\left[-\mathrm{i}\left(\frac{2NF}{V} - N\right)\right] = N\left(F - \frac{1}{2}V\right) \qquad (V \neq 0)$$

and retains the literal centered value NF at $V = 0$. Its source is

`formal/lean/ZetaFunctionFoundation/CRSJetCrosswalk.lean`.

The source SHA–256 is

`cb738b1de8e530a82ad05e548b2a0fdc351c39f99dd6fd2f7e10c28ada5f7796`, and the compiled object has SHA–256

`645121bfc27878b05fb2152fdc172f84f069840c22b42880f15a7063d10018a3`.

The accepted replay uses Lean 4.32.0 and Mathlib commit

`81a5d257c8e410db227a6665ed08f64fea08e997`,

exits zero with empty streams, contains no `sorry`, added axiom, `admit`, or `unsafe` declaration, and peaks at 1,410,822,144 bytes under the 4 GB cap. It does not formalize the CRS asymptotic analysis or their separate number-theoretic conjecture.

`CERT-DEHAYE-2008-20260826-0001` is the separate exact finite-data certificate for the complete supplemental tranche accompanying [45]. The script `certificates/dehaye_2008/verify_dehaye_supplements.py` parses all 120 supplied numerator polynomials over $\mathbb{Z}[u]$, checks evenness and monicity, replays all sixty denominator-preserving moment-family transforms, reconstructs all sixty denominators Y_r, matches the forty-five embedded table rows, and uses exact isolation after $y = u^2$ for the thirty recorded real-root counts. It also binds the ten supplied artifacts by SHA–256. The script, normalized standard output, and receipt hashes are respectively

`47cc9a04ab90c6e1a1a956c0b650cec8a53d828903c71a794ce7b0089d7e39d6`,
`d53f8a47086a84fc17d1552b9a867f49aa991ae42447dc0ffdfeccdd15e96686`,
`0c17f01996803d852eb1fc41c4cd333fc0d2af0f5d1887732b8533b5196bb705`.

This certificate executes no Magma code and proves no statement in representation theory, no Carlson continuation, and no number-theoretic model transfer.

`FORMAL-20260826-0020` checks the finite Dehaye angular-jet interface in theorem 2.24. It proves both compositions for

$$(V, d) \longmapsto \big(V, \mathrm{i}(d - NV)\big), \qquad (z_0, z_1) \longmapsto \big(z_0, Nz_0 - \mathrm{i}z_1\big),$$

both compositions for the completed diagonal maps $(z_0, z_1) \mapsto (z_0, \mathrm{i}z_1)$ and $(w_0, w_1) \mapsto (w_0, -\mathrm{i}w_1)$, and the exact affine identities

$$\mathrm{i}(u - N) = -\frac{\mathcal{T} + \mathrm{i}N}{2}, \qquad \mathrm{i}\left(u - \frac{N}{2}\right) = -\frac{\mathcal{T}}{2}, \qquad \mathcal{T} = -\mathrm{i}(2u - N).$$

Its source is `formal/lean/ZetaFunctionFoundation/DehayeJetCrosswalk.lean`, with SHA–256

`bd747850358abf964a42368b691b6e898221407b79ed61db7639e9cec73ccc9c`,

and the compiled object has SHA–256

`429318e08f71a8f3107e6d89b9017947e54187db0c61f23ba64f3a8abc8e8279`.

The accepted replay uses Lean 4.32.0 and pinned Mathlib commit

`81a5d257c8e410db227a6665ed08f64fea08e997`.

It exits zero with empty streams, contains no `sorry`, added axiom, `admit`, or `unsafe` declaration, and peaks at 1,401,528,320 bytes under the 4 GB cap. It does not formalize Haar integration, Schur identities, the large-N limit, meromorphic continuation, or the source's number-theoretic proposals.

`FORMAL-20260828-0024` stages the finite algebra of the complete CUE–Mellin reconstruction in section 2.7.1. Its first layer is the exact finite-difference/Hausdorff algebra and Bernstein-weight identities after all measures and support intervals have been stated externally. Its second layer is the terminating $U(2)$ constant-term–${}_3F_2$ equality. Its third layer is the all-order trace recurrence, the descended polynomial formula, and its exact v^s coefficient. Its fourth layer is the fixed-word Parikh-simplex map to the quotient ratio lattice, including kernel and rank. Its fifth layer contains the retained $N = 3$ state substitutions, the affine cubic identity, its $u = 0$ specialization, and the discriminant expansion. The ordered-real support proof, compact-measure convergence, Laplace/Hurwitz/digamma analysis, general lattice asymptotics, and the irreducible-cubic Galois criterion remain complete standard proofs unless and until their required library interfaces are formalized separately. This queue entry records a second checkability layer; it does not turn a proved manuscript theorem into a global proof-status claim. Every future replay is subject to the exact aggregate process-tree ceiling of $4\,\mathrm{GiB} = 4{,}294{,}967{,}296$ bytes and is to be split before elaboration approaches it.

`FORMAL-20260828-0025` stages the finite and algebraic layers of the split-zero and shifted-sheet reconstruction in section 4. Its first layer is the literal coordinate semiring $G(R) \simeq D_R$, both projections, the unique unital Boolean support character, the fibre-product formula for products, and the two-sided multiplicative reconstruction. Its second layer is the complete additive-idempotent skeleton, the R-module fibres and transport maps, and both functors in $G(R)$-SMod $\simeq \mathrm{Diag}_R$, including the natural quasi-inverse equations. Its third layer is the corrected ideal classification: the lifted ring ideals are exactly the nonzero semiring ideals, while $\{\tau\}$ is an additional bottom strictly below $\{\tau, 0_R\}$; lifted ideal products and the supported-zero divisibility criterion are separate targets. Its fourth layer is the rank-two support-cube map to endpoint principal-part subspaces, its inverse residue matrix, fibres, and deck equivariance. Its fifth layer contains the two-point Stone comultiplication, the finite shift-jet and Pochhammer identities, and the prime-power formula for the Dedekind ratio coefficients.

The exact certificate `CERT-GCUE-SPLIT-ZERO-20260828-0001` already checks 26 named finite consequences in two Python/SymPy installations. In particular it independently exposes and verifies the additional bottom ideal in five quotient-ring models. The locally normal shift series, the entire Pochhammer-cancelled Dedekind tail, and the absolute Euler-product expansion remain complete standard proofs in the manuscript; they are not claims made by the finite checker. This Lean queue is an additional certification plan, not a statement that any theorem is unproved outside this formalization.

`FORMAL-20260828-0026` stages the finite categorical and coordinate core of section 5. The targets are the Boolean semimodule identification, the reduced split retraction $\Sigma_m \Delta_m = \mathrm{id}$ at the displayed maps, the terminal-chain inclusion with both adjoints $r_m \dashv i_m \dashv s_m$, the exact fibres of both adjoints, the degree-shift equivalence, the coefficient-support pointed retraction and its cancellation counterexample, the terminal-jet/socle coordinate, and the diagonal fixed-part descent for orbit sizes 1, 2, and 4. The formal statement must preserve the actual object classes: the coefficient-support map is a pointed-set map and is not to be coerced into an additive homomorphism. The product-sheaf and sheafification proofs, analytic local-ring ideal classification, compactified tail stalk and flasqueness, completed-function identities, and equivariance over nonidentity base maps remain the complete standard proofs printed in the chapter until appropriate library interfaces are selected. This queue item concerns only an additional formal certificate; it makes no statement about whether a result is proved in the wider literature.

42.5 Unstabilized semilocal one-zero certification

The certificate with stable ID

`CERT-GNTE-MULTIPLACE-`
`UNSTABILIZED-20260830-0001`

accompanies theorems 9.26 and 9.28 to 9.30. The script

```
certificates/globalization_nte_multiplace_unstabilized/
verify_multiplace_unstabilized.py
```

passes in two Python 3.13/SymPy installations and replays 362 named checks for one through eight finite primes. It checks the exponent and sign cancellations, the finite-boundary screw coordinates, the exact label-collapse matrix and its kernel, the formal and actual root-lattice ranks, explicit clopen sign sections, and the valuation pattern in the nonproperness sequence. Its receipt binds the three content-read source loci and the exact manuscript hash.

`FORMAL-20260830-0037` queues only the finite coordinate and integer layers for a possible Lean replay: generic and boundary exponent identities, the label morphism, its one-dimensional kernel, the codiagonal square, and the root-lattice quotient. The orbit topology, crossed-product exactness, Green imprimitivity, compact-group quotient, six-term naturality, and nonproperness argument remain the complete cited standard proofs in the chapter. No Lean process was launched for this tranche.

42.6 Stable-to-natural extension certification

The certificate with stable ID `CERT-GNTE-STABLE-TO-NATURAL-20260830-0001` accompanies theorems 9.31 and 9.33 to 9.36. Its script

```
certificates/globalization_nte_stable_to_natural/verify_stable_to_natural.py
```

passes 2,361 named checks in Python/SymPy version pairs 3.13.9/1.13.1 and 3.13.1/1.13.3. The exact finite layer covers the orthogonal corner assembly and compression inverse, constant-unit ideal factorization, finite matrix Busby equation, boundary naturality matrix, one-dimensional sign-difference kernel, split-enriched injectivity, evaluation left inverses, nonconstant sampled unit-fibre classes, and the minimal two-coordinate sign carrier. A preflight harness comparison of oppositely oriented basis vectors failed before acceptance; it was replaced by the exact rank-one span test recorded in the receipt. No failed run is presented as a certificate.

`FORMAL-20260830-0038` queues only these finite matrix and integer coordinates for a possible Lean replay. The density and essential-ideal argument, Green modules, linking-algebra stabilization, multiplier/corona extensions, Busby reconstruction, and crossed-product exactness remain the complete cited standard proofs in the chapter. No Lean or Lake process was launched for this tranche. Any future replay must use one guarded worker, `LEAN_NUM_THREADS=1`, `LAKE_JOBS=1`, and the exact aggregate process-tree ceiling of $4\,\mathrm{GiB} = 4{,}294{,}967{,}296$ bytes.

42.7 Nonconstant unit-fibre K-theory certification

The certificate with stable ID `CERT-GNTE-NONCONSTANT-UNIT-K-20260830-0001` accompanies theorems 9.37 to 9.41. Its script

```
certificates/globalization_nte_nonconstant_unit_k/verify_nonconstant_unit_k.py
```

passes 5,315 named checks in Python/SymPy version pairs 3.13.9/1.13.1 and 3.13.1/1.13.3. It checks exact finite orbit lengths and stabilizers, invariant and coinvariant orbit bases, both direct-limit transition maps, the $L_{\mathbf{m}'}/L_{\mathbf{m}}$ coinvariant multiplicity, transition composition and injectivity, finite and archimedean boundary matrices, the global kernel inverse, the odd-difference cokernel coordinate, and the one-prime specialization. An initial harness run negated a value that already represented $-S_h$; it failed before acceptance, was corrected to the manuscript's $h_\infty = -S_h|_{H_P^+}$, and is not represented as a certificate.

`FORMAL-20260830-0039` queues the finite permutation-module and integer-linear-algebra layers for a possible Lean replay. The mapping-torus identification, Pimsner–Voiculescu exact sequence, profinite commutative K-theory, Green imprimitivity, six-term naturality, continuity, and winding arguments remain the complete cited standard proofs in the chapter. No Lean or Lake process was launched for this tranche. Any future replay must use one guarded worker, `LEAN_NUM_THREADS=1`, `LAKE_JOBS=1`, and the current exact aggregate process-tree ceiling of $4\,\mathrm{GiB} = 4{,}294{,}967{,}296$ bytes.

Every future successor build uses the exact $4\,\mathrm{GiB} = 4{,}294{,}967{,}296$-byte aggregate-process-tree ceiling and requires exact toolchain, command, stream, hash, exit, and peak-memory receipts. Earlier receipts retain the historical cap actually enforced when those builds ran.

The unfinished formal obligations are `FORMAL-20260824-0001` through `FORMAL-20260824-0006`. The exp/log, Fourier, convolution, even-extension, finite-place, zero-pairing, finite-height-scale, and shifted-exponential coordinate maps remain to be encoded with every test-space and exceptional-locus hypothesis. Weil's PF distribution, the limiting logarithmic-density functional, and the final gamma/trivial-zero commutator also require an explicit Mathlib integration/distribution interface; none may be introduced as an axiom. Completion of the finite interface certificate above does not discharge any of these analytic items, the full ordered-real stage of `FORMAL-20260825-0014`, or any missing CUE-to-Lie or endpoint interface.

43 Tool and computational resource disclosure

This living corpus is prepared with automated language models, symbolic and numerical software, and—where a bounded theorem is suitable—the Lean proof assistant. The local research-note inputs were produced primarily with ChatGPT 5.4 Pro and ChatGPT 5.5 Pro, with occasional ChatGPT 5.6 Pro input. GPT-5.6 Sol Ultra performs the present consolidation, source crosswalk, proof audit, and document maintenance. Some local notes may contain exploratory conversational input from Claude Opus 4.6 or Fable 5; no particular mathematical claim is attributed to either system without a claim-level audit.

These systems are disclosed as tools, not authors. Credit and responsibility are assigned only to identified humans. Every literature-backed claim must terminate in a human-authored public source; a private local file records local lineage but is not used as scholarly authority for mathematics it may have inherited. A genuinely local result is attributed to its human contributor only after proof, chronology, ownership, and novelty checks. Unresolved provenance is reported as unresolved rather than replaced by model attribution. This policy implements the disclosure, attribution, human-responsibility, review-support, and checkability recommendations of the Leiden Declaration on Artificial Intelligence and Mathematics [93, Recommendations for individual mathematicians].

For each admitted computational or formal certificate, the public tree will retain the source, exact command, dependency and toolchain versions, theorem or check name, input and output hashes, exit status, and a bounded resource receipt. Lean certification supplements the complete human-readable proof; it does not replace it. Every future Lean or Lake session is capped at exactly $4\,\mathrm{GiB} = 4{,}294{,}967{,}296$ aggregate process-tree bytes. The replay wrapper samples the complete spawned process tree, kills the tree when the cap is reached, and removes any pre-existing compiled output before replay so that failure cannot inherit a stale certificate.

The accepted finite-coordinate and Laurent replays use Lean 4.32.0 and Mathlib commit

`81a5d257c8e410db227a6665ed08f64fea08e997`.

Their measured aggregate peaks are 1,933,365,248 and 1,404,055,552 bytes. An earlier exploratory Laurent evaluator imported a larger environment and repeated native evaluation three times; one observed snapshot crossed the cap at 6,050,037,760 aggregate bytes. That run was interrupted, its spawned processes were terminated, and it is not a certificate. The accepted Laurent theorem uses one disclosed `native_decide` evaluation, so the Lean compiler is part of that certificate's trust boundary. The same five integers are independently checked by the sparse-Python and shifted SymPy exact replays recorded in `CERT-N3-CUE-20260825-0001`.

`CERT-U2-CUE-20260825-0001` uses Python 3.13.9 and only exact integer/fraction arithmetic. It independently assembles the normalized $U(2)$ Weyl-density Laurent constant term and the beta-times-circle multinomial moment formula, and checks their agreement for orders one through eight. Its source SHA–256 is

`7d200a1a375d568237a07c51a7eb94d8455a49b5e6b78bacf5cba5d27d2c1610`.

The certificate explicitly excludes the measure pushforward from its scope: the torus map, two inverse branches, boundary fibres, Jacobian, and independence are the written proof in theorem 2.29.

`CERT-N3-PHASE-20260825-0001` uses Python 3.13.9 and SymPy 1.13.1 with exact symbolic polynomial and rational arithmetic. Its twenty-two checks cover the rationalized real and imaginary coordinates of the first peeled $N = 3$ variable Z, the trace and norm identities, the complete base inverse, both discriminant presentations, the factorization of S, the lifted J coordinate and inverse, and the affine (u, v)–(u, X) coordinate change. Its source SHA–256 is

`bca8e81cbc09a2bd2d24f86200daf68b815304f9d6c1a65b731c014ae4328ff4`.

The fresh normalized output agrees byte for byte with the retained output at SHA–256

`62e910773b91a78e9943e9f76d678fe75491f21998afbccd800065dbfe453ed8`.

The executable certificate does not replace the inequality and surjectivity proof in theorems 2.37 and 2.38.

`CERT-N3-DECK-20260826-0001` uses Python 3.13.9 and SymPy 1.13.1 with exact symbolic polynomial and rational arithmetic. Its twenty-one checks cover the affine deck-curve parametrization and inverse, the lossless recovered-β_0 carrier, the exact carrier-equivariance defect, the J-dependent equivariant sheet parameter, both target coordinates, the target curve equation, and the base-coordinate square modulo the physical cover equation. Its source and retained-output SHA–256 values are respectively

`d4299a25508fb5f8505a4346d4aa5e26d8c61c728e2e3448772e406709299a97`,
`15ff919d7bdeddfba1ec130997773df0266fc3deb5204236d62bb8ce6e1adf1e`.

A fresh run parsed to the same twenty-one named checks and the same coordinate/nonclaim record. The companion Lean certificate `FORMAL-20260826-0015` compiled to a 444,184-byte `.olean` at SHA–256

`8bd2c3f79ffe8e4b48220a1679cf3eb143b520896331d773bc6e93415ebf179e`.

Its bounded build exited zero with empty standard streams and a 1,452,122,112-byte peak aggregate working set under the 4,026,531,840-byte cap. Neither executable replaces the written fibre proof or upgrades the interface to a CUE-distribution, critical-line, or zeta-zero statement.

The two exact SymPy certificates discussed here are

`CERT-SIMM-WEI-20260828-0001`,
`CERT-CUE-MELLIN-20260828-0001`.

Both have complete JSON receipts and were replayed under two independently installed Python and SymPy version pairs: 3.13.9/1.13.1 and 3.13.1/1.13.3. The first checks the bounded finite-determinant, $s = 1$, differentiated-pole, Bessel-series, and reciprocal-Gamma truncation calculations listed in section 2.6.2. It does not certify the source's analytic convergence, uniform integrability, Painlevé, or conditional number-theory arguments.

The CUE–Mellin certificate has source SHA–256

`03a5157960c09af23f351bed95159c875d1a22ea01dcbc88c5fedfa22e31576f`

and receipt SHA–256

`6f4501023864e7821e968e20ec65ddf687626a16b1a2f9c07a5d99cac4824b8a`.

Its thirty-nine named checks cover the $U(2)$ Weyl/terminating-${}_3F_2$ identity through order seven, the exact $N = 3$ state-coordinate arrow and norm identity, the descended trace kernel through order eight, the four printed low-order kernels, the Ψ_2 Gram obstruction, the affine cubic and its discriminant, and forty-two terminating Gauss laws. The compact-measure recovery, state-kernel transport, analytic transform identities, multiplier-lattice count, and Galois conclusion retain the complete written proofs in section 2.7.1; no executable finite check is presented as a substitute for them or as an individual-zero statement.

`CERT-GNTE-DIVISOR-20260828-0001` is the exact finite-coordinate certificate for section 5. Its pure-standard-library Python script is retained in the directory `certificates/globalization_nte_divisor_sheaves`, under the filename `verify_chain_cube_packet_core.py`. Fresh runs under Python 3.13.9 and Python 3.13.1 both pass the same twelve named checks. For every $1 \leq m \leq 8$, it exhausts the Boolean cube, verifies the reduced retraction, both chain–cube adjunctions, join preservation, both complete fibre descriptions and their cardinalities, the coefficient-support section and finite-alphabet fibres, the explicit cancellation obstruction, and the degree-shift/socle coordinates. It also exhausts the diagonal fixed parts for packet sizes 1, 2, and 4, and checks the two commuting phase involutions and representative independence on an exact Gaussian-integer grid. The script and receipt SHA–256 values are respectively

`b7192e29d5d558e27cf7b6db6c25960ef6011bbb6f3490833e715c9ff3570b79`,
`26a94d7af47e5d73acaa6d7e752e43df26366bdfaba920fff674aa541371527f`.

The executable deliberately certifies no analytic or sheaf-theoretic claim. Those claims retain their full standard proofs and named human-source dependencies in the chapter.

`CERT-GNTE-STABLE-TO-NATURAL-20260830-0001` is a two-runtime exact SymPy replay of the finite coordinate layer of the stable-to-natural extension morphism. Both Python/SymPy version pairs 3.13.9/1.13.1 and 3.13.1/1.13.3 pass 2,361 named checks. The receipt separately discloses a preflight harness failure caused by comparing oppositely oriented basis-vector lists instead of their rank-one span; the accepted checker uses the exact rank test. The executable does not certify essentiality, Green imprimitivity, Brown–Green–Rieffel stabilization, multiplier or corona extensions, or Busby reconstruction. Those remain the complete cited standard proofs in theorems 9.31, 9.33 and 9.34. It launches neither Lean nor Lake.

`CERT-GNTE-NONCONSTANT-UNIT-K-20260830-0001` is the two-runtime exact SymPy replay of the finite coordinate layer of the complete nonconstant unit-fibre calculation. Both Python/SymPy version pairs 3.13.9/1.13.1 and 3.13.1/1.13.3 pass the same 5,315 named checks, whose ordered-name digest is

`1774e5fe4724618e3e1701568756e13413c5d18c0931bd1eae45193fd7d3c5d2`.

The script and receipt SHA–256 values are respectively

`a312ef288ed67dbe8bf44c37b5c15bcb5699e27ee61d4d494a5c42302954b520`,
`2f22ab6eecc5c12df6f242435091c95333d45c50858542cc737a5c4a109c205f`.

The checker covers finite orbit/stabilizer coordinates, invariant and coinvariant transition matrices including their multiplicities, finite and archimedean boundary matrices, the global kernel/cokernel coordinates, and the one-prime case. It does not certify the mapping-torus theorem, profinite commutative K-theory, Green imprimitivity, six-term exactness, continuity, or winding orientation; those remain the complete cited standard proofs in theorems 9.37 to 9.41. It launches neither Lean nor Lake and makes no finite-set-coherence, endpoint, individual-zero, or global-zeta claim.

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